

For a spherical particle surrounded by an infinite, quiescent fluid, the mass- and heat-transfer coefficients are at their minimum values. Assume an insoluble, solid, spherical particle of radius R_p and diameter $D_p = 2R_p$, suspended in an infinite-fluid medium. The particle is heated so that, at steady state, its surface temperature is constant at T_s . The fluid medium is absolutely quiescent (no free convection) and radiation is ignored so that heat transfer through the fluid is by conduction only. The thermal conductivity k of the fluid is constant, and the temperature far from the particle is T_b . Fourier's second law of heat conduction in the fluid, for spherical coordinates, is

$$\frac{d}{dr} \left(kr^2 \frac{dT}{dr} \right) = 0 \quad (15-50)$$

for $r \geq R_p$, where r is the radial distance from the center of the particle. The boundary conditions are

$$T\{r = R_p\} = T_s \quad (15-51)$$

$$T\{r = \infty\} = T_b \quad (15-52)$$

If (15-50) is integrated twice with respect to r , we obtain:

$$T = -\frac{C_1}{r} + C_2 \quad (15-53)$$

Substitution of the boundary conditions, (15-51) and (15-52), results in an expression for the temperature profile in the fluid:

$$\frac{T - T_b}{T_s - T_b} = \frac{R_p}{r}, \quad r \geq R_p \quad (15-54)$$

The heat flux at the outer surface of the particle is given by Fourier's first law of heat conduction applied to the fluid adjacent to the particle:

$$\frac{q}{A} \Big|_{r=R_p} = -k \frac{dT}{dr} \Big|_{r=R_p} \quad (15-55)$$

From (15-54):

$$\frac{dT}{dr} \Big|_{r=R_p} = -\frac{(T_s - T_b)}{R_p} \quad (15-56)$$

We can also apply Newton's law of cooling for the heat flux at the outer surface of the particle:

$$\frac{q}{A} \Big|_{r=R_p} = h(T_s - T_b) \quad (15-57)$$

Combining (15-55) to (15-57):

$$h = k/R_p \quad (15-58)$$

which rearranges into a Nusselt number form:

$$N_{Nu} = hD_p/k = 2 \quad (15-59)$$

A similar development for convective mass transfer using Fick's laws of diffusion gives

$$N_{Sh_i} = k_{c_i} D_p/D_i = 2 \quad (15-60)$$

where D_i is the diffusivity of component i in the mixture.

When the fluid flows past a single particle, convection increases the convective mass- and heat-transfer coefficients above the values computed from (15-59) and (15-60).

Furthermore, the transport coefficients now vary around the periphery of the particle, with the largest value occurring where the fluid flow first impinges on the particle. Correlations of experimental transport data are usually developed for coefficients averaged over the surface of the particle. Typical correlations are those of Ranz and Marshall [46, 47] for Nusselt numbers as high as 30 and Sherwood numbers to 160:

$$N_{Nu} = 2 + 0.60 N_{Re}^{1/2} N_{Pr}^{1/3} \quad (15-61)$$

$$N_{Sh_i} = 2 + 0.60 N_{Re}^{1/2} N_{Sc_i}^{1/3} \quad (15-62)$$

where

$$N_{Pr} = \text{Prandtl number} = C_p \mu / k$$

$$N_{Sc_i} = \text{Schmidt number} = \mu / \rho D_i$$

$$N_{Re} = \text{Reynolds number} = D_p G / \mu$$

All fluid properties are evaluated at the average temperature of the film or boundary layer. Equations (15-61) and (15-62) reduce to (15-59) and (15-60), respectively, when the fluid mass velocity, G , is zero.

When particles are packed in a bed, the fluid-flow patterns are restricted, and the single-particle correlations of (15-61) and (15-62) cannot be used to estimate the average external-transport coefficients for the particles in the bed. However, Ranz [48] showed that equations of the same form as (15-61) and (15-62) correlate external-transport data for beds packed with spherical particles. Nevertheless, most early investigators, starting with Gamson, Thodos, and Hougen [49], developed correlations in the form of the Chilton and Colburn [50] j -factors:

$$j_D = (N_{StM})(N_{Sc})^{2/3} = f\{N_{Re}\} \quad (15-63)$$

$$j_H = (N_{St})(N_{Pr})^{2/3} = f\{N_{Re}\} \quad (15-64)$$

with

$$N_{StM} = k_c \rho / G \quad \text{and} \quad N_{St} = h / C_p G$$

where different Reynolds-number functions apply to different regions. Various forms of the Reynolds number have been used, including $D_p G / \mu$ and $D_p G / \epsilon_b \mu$, in attempts to account for bed void fraction, ϵ_b , where G is the superficial mass velocity based on the empty-bed cross-sectional area, and G / ϵ_b (a larger value) is the effective mass velocity through the void region of the bed. Notable among correlations of this type are those of Sen Gupta and Thodos [51], Petrovic and Thodos [52], and Dwivedi and Upadhyay [53].

A more-recent study by Wakao and Funazkri [54] reanalyzed 37 sets of previously published mass-transfer data, with Sherwood number corrections for axial dispersion. The resulting correlation, which represents a return to the form of (15-62), is

$$N_{Sh_i} = \frac{k_{c_i} D_p}{D_i} = 2 + 1.1 \left(\frac{D_p G}{\mu} \right)^{0.6} \left(\frac{\mu}{\rho D_i} \right)^{1/3} \quad (15-65)$$

The correlation is compared to 12 sets of gas-phase and 11 sets of liquid-phase data in Figure 15.17. The data cover

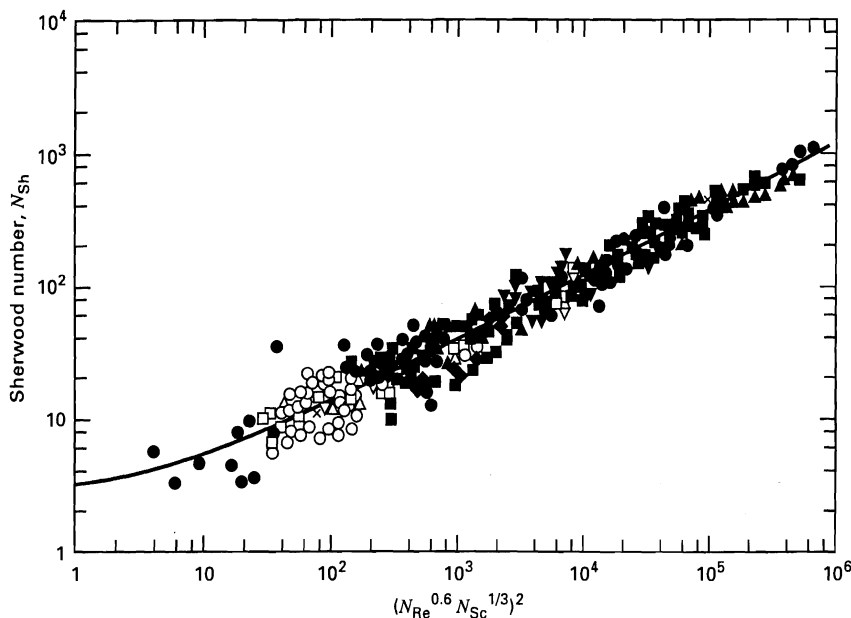


Figure 15.17 Correlation of experimental data for Sherwood number of external mass transfer in a packed bed.

[From N. Wakao and T. Funazkri, *Chem. Eng. Sci.*, 33, 1375 (1978) with permission.]

a Schmidt number range of 0.6 to 70,600, a Reynolds number range of 3 to 10,000, and a particle diameter from 0.6 to 17.1 mm. Particle shapes include spheres, short cylinders, flakes, and granules. By analogy, the corresponding equation for fluid-particle convective heat transfer in packed beds is

$$N_{Nu} = \frac{hD_p}{k} = 2 + 1.1 \left(\frac{D_p G}{\mu} \right)^{0.6} \left(\frac{C_p \mu}{k} \right)^{1/3} \quad (15-66)$$

When (15-65) and (15-66) are used with beds packed with nonspherical particles, D_p is the equivalent diameter of a spherical particle. The following suggestions have been proposed for computing the equivalent diameter from geometric properties of the particle. These suggestions may be compared by considering a short cylinder with diameter, D , equal to the length, L .

1. D_p = diameter of a sphere with the same external surface area:

$$\pi D_p^2 = \pi DL + \pi D^2/2$$

$$\text{and} \quad D_p = (DL + D^2/2)^{0.5} = 1.225D$$

2. D_p = diameter of a sphere with the same volume:

$$\pi D_p^3/6 = \pi D^2 L/4 \text{ and } D_p = (3D^2 L/2)^{1/3} = 1.145D$$

3. D_p = 4 times the hydraulic radius, r_H , where for a packed bed,

$$4r_H = 6/a_v$$

$$a_v = \text{external particle surface area/volume of particle}$$

Thus,

$$a_v = \frac{\pi DL + \pi D^2/2}{\pi D^2 L/4} = \frac{6}{D} \text{ and } D_p = 4r_H = \frac{6D}{6} = 1.0D$$

The use of the hydraulic radius concept is equivalent to replacing D_p in the Reynolds number by $\Psi D'_p$ where Ψ is the sphericity and D'_p is given by Suggestion 2: The sphericity is defined by:

$$\Psi = \frac{\text{Surface area of a sphere of same volume as particle}}{\text{Surface area of particle}}$$

For a cylinder of $D = L$,

$$\Psi = \frac{\pi D_p^2}{\pi DL + 2\left(\frac{\pi D^2}{4}\right)} = \frac{\pi(1.145D)^2}{\frac{3}{2}\pi D^2} = 0.874$$

and $\Psi D'_p = (0.874)(1.145D) = D$, the diameter of the cylinder.

Suggestions 2 and 3 are widely used. Suggestion 3 is conveniently applied to crushed particles of irregular surface, but with no obvious longer or shorter dimension, that is, isotropic in shape. In that case, D'_p is taken as the size of the particle and the sphericity is approximately 0.65, as discussed by Kunii and Levenspiel [55].

EXAMPLE 15.8

Acetone vapor in a nitrogen stream is removed by adsorption in a fixed bed of activated carbon. At a location in the bed where the pressure is 136 kPa, the bulk gas temperature is 297 K, and the bulk mole fraction of acetone is 0.05, estimate the external gas-to-particle mass-transfer coefficient for acetone and the external particle-to-gas heat-transfer coefficient. Additional data are as follows:

$$\begin{aligned} \text{Average particle diameter} &= 0.0040 \text{ m and Gas superficial molar} \\ &\text{velocity} \\ &= 0.00352 \text{ kmol/m}^2\text{-s} \end{aligned}$$

SOLUTION

Because the temperature and composition are known only for the bulk gas and not at the particle external surface, use gas properties at bulk gas conditions. From the ChemCAD process simulation program, relevant properties for use in (15-65) and (15-66) are as follows:

$$\begin{aligned} \text{Viscosity } \mu &= 0.0000165 \text{ Pa-s (kg/m-s); Density } \rho = 1.627 \text{ kg/m}^3 \\ \text{Thermal conductivity } k &= 0.0240 \text{ W/m-K} = 0.024 \times 10^{-3} \text{ kJ/m-K-s} \\ \text{Heat capacity at constant pressure} &= 31.45 \text{ kJ/kmol-K} \\ \text{Molecular weight} &= M = 29.52 \end{aligned}$$

$$\text{Thus, specific heat } C_p = 31.45/29.52 = 1.065 \text{ kJ/kg-K}$$

Other parameters are

$$\begin{aligned}\text{Gas mass velocity } G &= 0.00352(29.52) = 0.1039 \text{ kg/m}^2\text{-s} \\ \text{Assume a sphericity, } \psi, &\text{ of } 0.65; \text{ therefore, } D_p = 0.65(0.004) \\ &= 0.0026 \text{ m}\end{aligned}$$

The diffusivity, D_i , of acetone in nitrogen at 297 K and 136 kPa is independent of the composition and is approximately $0.085 \times 10^{-4} \text{ m}^2/\text{s}$.

$$\begin{aligned}N_{Re} &= D_p G / \mu = 0.0026(0.1039) / (0.0000165) = 16.4 \\ N_{Sc} &= \mu / \rho D_i = 0.0000165 / (1.627)(0.0000085) = 1.19 \\ N_{Pr} &= C_p \mu / k = (1.065)(0.0000165) / (0.000024) = 0.73\end{aligned}$$

From (15-65):

$$N_{Sh} = 2 + 1.1(16.4)^{0.6}(1.19)^{1/3} = 8.24$$

which from Figure 15.17 is well within the data range of the correlation. Thus, the mass-transfer coefficient for acetone is

$$\begin{aligned}k_{ci} &= N_{Sh}(D_i/D_p) = 8.24(0.0000085/0.0026) \\ &= 0.027 \text{ m/s} = 0.088 \text{ ft/s}\end{aligned}$$

From (15-66):

$$\begin{aligned}N_{Nu} &= 2 + 1.1(16.4)^{0.6}(0.73)^{1/3} = 7.31 \\ h &= N_{Nu}(k/D_p) = 7.31(0.0240/0.0026) \\ &= 67.5 \text{ W/m}^2\text{-K} \quad \text{or } 11.9 \text{ Btu/h-ft}^2\text{-}^\circ\text{F}\end{aligned}$$

Internal Transport

Porous-adsorbent particles have a sufficiently high, effective thermal conductivity that temperature gradients within the particle are usually negligible. However, internal (intraparticle) mass transfer in the particle must be considered. Mechanisms for mass transfer in the pores are those described for porous membranes in Section 14.3. However, in membranes, the transport is through the membrane. In sorption applications, transport is only into the interior of the particle during sorption and from the interior of the particle in desorption.

The mathematical model of internal transport in porous particles during adsorption or desorption is very similar to that for catalytic chemical reactions in porous catalyst pellets. The first pore model was that of Thiele [56], who

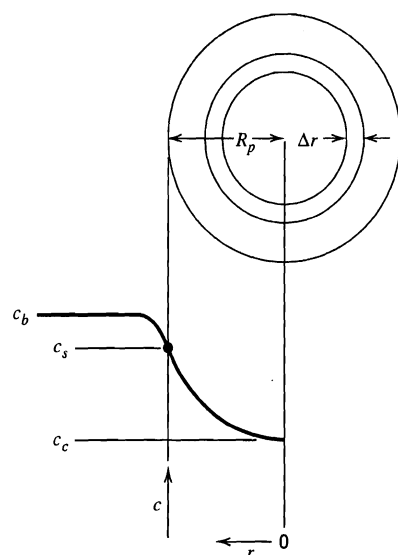


Figure 15.18 Typical solute concentration profile in adsorbent particle.

considered a first-order irreversible reaction taking place isothermally on the surface of a single, straight, cylindrical pore, closed at one end. Thiele's treatment was extended to a porous spherical pellet by Wheeler [57], who utilized an effective diffusivity applied to the case of sorption.

Consider the porous spherical pellet in Figure 15.18, where the fluid concentration, c , refers to the solute. A material balance in moles or mass per unit time over the spherical-shell volume of thickness Δr involves diffusion of the solute into the shell at radius $r + \Delta r$, adsorption within the shell, and diffusion out of the shell at radius r . Using Fick's first law:

$$4\pi(r + \Delta r)^2 D_e \frac{\partial c}{\partial r} \bigg|_{r+\Delta r} = 4\pi r^2 \Delta r \frac{\partial q}{\partial t} + 4\pi r^2 D_e \frac{\partial c}{\partial r} \bigg|_r \quad (15-67)$$

Dividing by $4\pi \Delta r$, taking the limit as $\Delta r \rightarrow 0$, and collecting terms gives

$$D_e \left(\frac{\partial^2 c}{\partial r^2} + \frac{2}{r} \frac{\partial c}{\partial r} \right) = \frac{\partial q}{\partial t} \quad (15-68)$$

The variable q is the amount adsorbed per unit volume of porous pellet. The effective diffusivity, D_e , applies to the entire surface area of the spherical shell, even though only about 50% of it is available as pores for diffusion. For liquid-phase diffusion in the pores, the effective diffusivity is given by (14-14), which involves the volume fraction of pores in the pellet, the solute molecular diffusivity in the fluid within the pore, the pore tortuosity, and a possible restrictive factor for relatively large solute molecules. For gas-phase diffusion in the pores, the effective diffusivity is given by (14-18), which accounts for the possibility of Knudsen diffusion, with diffusivity D_K for very small pore diameters and/or low total pressures. Although (14-14) and (14-18) strictly apply only to equimolar counterdiffusion, they can be used as an approximation for unimolecular diffusion for fluids dilute in the solute molecules. A diffusion mechanism not accounted for directly in (14-18) is that of surface diffusion along the pore wall due to the concentration gradient of the adsorbate (adsorbed solute) along the wall.

Fick's first law for molecular diffusion through a fluid in a pore can be written

$$n_i = -D_i A (dc_i/dx) \quad (15-69)$$

where n is the molar rate of ordinary diffusion of i through the fluid in the x -direction, perpendicular to the cross-sectional area, A , for diffusivity, D_i , and concentration, c_i , in moles/unit volume of fluid. A modified Fick's first law applies to surface diffusion, as suggested by Schneider and Smith [58]. Thus,

$$(n_i)_s = -(D_i)_s b d(c_i)_s / dx \quad (15-70)$$

where b is the perimeter of the surface, $(c_i)_s$ is the surface concentration of adsorbate in moles/unit surface area, and $(D_i)_s$ is the surface diffusivity as defined by (15-70).

For convenience, (15-70) is converted, as follows, to the flux form of (15-69) so that the two mechanisms of diffusion

can be combined in a single transport rate equation. The flux form of (15-69) is

$$N_i = n_i/A = -D_i(dc_i/dx) \quad (15-71)$$

The corresponding flux form of (15-70) is obtained by dividing both sides by the cross-sectional area of the pore and converting the surface concentration, $(c_i)_s$, in moles/unit surface area to the concentration, q , in mol/g of adsorbent, by using the product of the pore surface/pore volume times the reciprocal of the adsorbent particle density times the particle porosity. The result is

$$(N_i)_s = -(D_i)_s \frac{\rho_p}{\epsilon_p} \left(\frac{dq_i}{dx} \right) \quad (15-72)$$

Assuming linear adsorption according to Henry's law:

$$q_i = K_i c_i \quad (15-73)$$

Substituting (15-73) into (15-72) and adding the result to (15-71), the total flux is

$$N_i = - \left[D_i + (D_i)_s \frac{\rho_p K_i}{\epsilon_p} \right] \frac{dc_i}{dx} \quad (15-74)$$

In terms of the effective diffusivity employed in (15-68):

$$D_e = \frac{\epsilon_p}{\tau} \left\{ \left[\frac{1}{(1/D_i) + (1/D_K)} \right] + (D_i)_s \frac{\rho_p K_i}{\epsilon_p} \right\} \quad (15-75)$$

Equation (15-75) needs to be used with caution, because, as discussed by Riekert [59], the tortuosity, τ , for pore-volume diffusion is not necessarily the same as that for surface diffusion.

Based on the study by Sladek, Gilliland, and Baddour [60], values of the surface diffusivity of light gases for physical adsorption are typically in the range of 5×10^{-3} to 10^{-6} cm²/s, with the larger values applying to cases of a low differential heat of adsorption. For nonpolar adsorbates, the surface diffusivity in cm²/s may be estimated from the correlation [60],

$$D_s = 1.6 \times 10^{-2} \exp[-0.45(-\Delta H_{\text{ads}})/mRT] \quad (15-76)$$

where $m = 2$ for conducting adsorbents such as carbon and $m = 1$ for insulating adsorbents.

EXAMPLE 15.9

Porous silica gel of 1.0 mm particle diameter, with a particle density of 1.13 g/cm³, a porosity of 0.486, an average pore radius of 11 Å, and a tortuosity of 3.35 is to be used to adsorb propane from helium. At 100°C, diffusion in the pores is controlled by Knudsen and surface diffusion. Estimate the effective diffusivity. The differential heat of adsorption is -5,900 cal/mol. At 100°C, the adsorption constant (for a linear isotherm) is 19 cm³/g.

SOLUTION

From (14-21), the Knudsen diffusivity for propane is $D_K = 4,850 (22 \times 10^{-8})(373/44.06)^{1/2} = 3.7 \times 10^{-3}$ cm²/s. From (15-76), using $m = 1$, $D_s = 1.6 \times 10^{-2} \exp\{(-0.45)(5,900)/[(1)(1.987)(373)]\} = 4.45 \times 10^{-4}$ cm²/s.

Equation (15-75) reduces to $D_e = \frac{\epsilon_p}{\tau} D_K + \frac{\rho_p K}{\tau} D_s = \frac{0.486}{3.35} (3.17 \times 10^{-3}) + \frac{(1.13)(19)}{3.35} (4.45 \times 10^{-4}) = 0.46 \times 10^{-3} + 2.85 \times 10^{-3} = 3.31 \times 10^{-3}$ cm²/s

Experiments by Schneider and Smith [58] give a value of 1.22×10^{-3} cm²/s for D_e with a value of 0.88×10^{-3} for the contribution of surface diffusion. Thus, the estimated contribution from surface diffusion is high by a factor of about 3. In either case, the fractional contribution due to surface diffusion is large. A detailed review of surface diffusion is given by Kapoor, Yang, and Wong [61].

Mass Transfer in Ion Exchange and Chromatography

As discussed by Helfferich [62], two major mass-transfer resistances occur in ion exchange. The first is the external mass-transfer resistance due to the film or boundary layer surrounding the ion-exchange bead. The second is the internal diffusional resistance due to the resin bead. Either or both resistances can be rate-controlling; in either case, the diameter of the resin bead is an important factor. In general, external mass-transfer film diffusion is rate-controlling at very low exchange-ion concentrations, say below 0.01 N, whereas internal mass transfer (particle diffusion) controls at high concentrations (say above 1.0 N). It has also been observed that a large separation factor, as defined by (15-46), favors external mass-transfer control, and that divalent ions diffuse appreciably more slowly than monovalent ions through the resin bead. Usually, the rate-determining step is not the chemical reaction between the exchanging ions and the resin.

The external mass-transfer coefficient for flow of fluid through a fixed bed of ion-exchange resin beads is estimated from the same relation, (15-65), that is used for applications to adsorption in fixed beds. For internal mass transfer, it is customary to assume that the ion-exchange resin bead is a single quasi-homogeneous phase and that the diffusivity of the diffusing ion is constant at a given temperature. Under these conditions, (15-68) can be applied, where D_e is a diffusivity determined by experiment. In general, such diffusivities depend upon (1) the size and charge of the ion, with the smaller, monovalent ions diffusing the fastest; (2) the degree of cross-linking and resin swelling, with larger diffusivities favored by swelling and a small degree of cross-linking; and (3) temperature.

The most fundamental measurements of diffusivity in ion-exchange resins have been made with isotopes of the ions to obtain self-diffusion coefficients that are independent of ion concentration. Typical data are those of Soldano [63], shown in Figure 15.19 for Na⁺, Zn²⁺, and Y³⁺ in a sulfonated styrene-divinylbenzene cation exchanger at temperatures of 0.2 and 25°C. Recall that typical order-of-magnitude diffusivities for small molecules are as follows:

- 0.1 cm²/s in the gas phase
- 1×10^{-5} cm²/s in the liquid phase
- 1×10^{-7} cm²/s in polymers

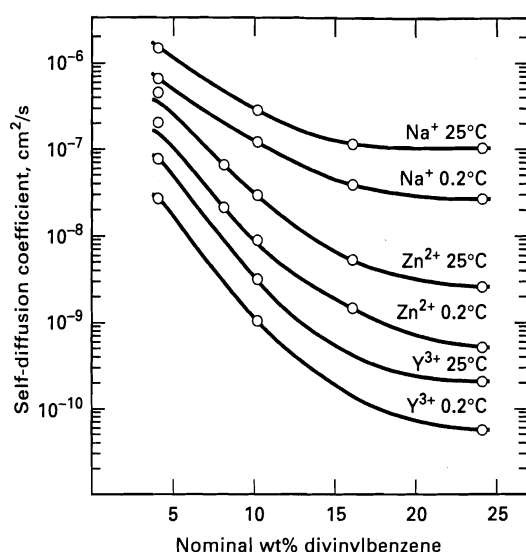


Figure 15.19 Self-diffusion coefficients for cations in a resin as a function of cross-linking with divinylbenzene.

[From B.A. Soldano, *Ann. NY Acad. Sci.*, **57**, 116 (1953) with permission.]

From Figure 15.19, it is seen that diffusivities depend strongly on the degree of cross-linking and the charge on the ion, with values much less than those found in liquids, especially for the divalent and trivalent ions, which have diffusivities even smaller than those observed for small molecules in polymers.

No new fundamental principles are required for formulating mass-transfer relations for chromatography. When packed beds are used, (15-65) and (15-66) are applied to determine external transport coefficients. If a coated flat plate or a tube with a coated inner wall is used, correlations of the type discussed in Chapter 3 are applicable. In some cases, an entry region of finite length exists, particularly for laminar flow, such that the transport coefficients vary with axial location, decrease in value with length, and eventually approach an asymptotic value. For internal diffusion in the sorbent, Fick's second law is applied where the effective diffusivity depends on factors discussed earlier in this section.

15.4 SORPTION SYSTEMS

A variety of equipment configurations and operating procedures are employed for commercial, sorption-separation operations. This variety is due mainly to the wide range of sorbent particle sizes used and the need, in most applications, to regenerate the solid sorbent.

Adsorption

For adsorption, the most widely used equipment configurations and operating procedures are listed in Table 15.7. For analysis purposes, the listed devices may be classified into the three modes of operations, shown schematically in Figure 15.20. An agitated vessel, shown in Figure 15.20a, is used with a batch of liquid to which is added a powdered adsorbent such as activated carbon, of particle diameter typically less than 1 mm, to form a slurry. With good agitation and small particles, the external resistance to mass transfer from the bulk liquid to the external surface of the adsorbent particles is small. For small adsorbent particles, the internal resistance to mass transfer within the pores of the particles is also small. Accordingly, the rate of adsorption is rapid. The required residence time of the slurry in a well-mixed agitated vessel is determined by how fast equilibrium is approached. The main application of this mode of operation is the removal of very small amounts of dissolved, and relatively large molecules, such as coloring agents, from water. Generally, the spent adsorbent, which is removed from the slurry by sedimentation or filtration, is discarded because of the difficulty of desorbing large molecules. The slurry adsorption system, also called *contact filtration*, is also operated continuously.

The cyclic-batch operating mode using a fixed bed, shown schematically in Figure 15.20b, is widely used with both liquid and gas feeds. Adsorbent particle size ranges from 0.05 to 1.2 cm. Bed pressure drop decreases with increasing particle size, but the solute transport rate increases with decreasing particle size. The optimal particle size is determined mainly from these two factors. To avoid jiggling

Table 15.7 Common Commercial Methods for Adsorption Separations

Phase Condition of Feed	Contacting Device	Adsorbent Regeneration Method	Main Application
Liquid	Slurry in an agitated vessel	Adsorbent discarded	Purification
Liquid	Fixed bed	Thermal reactivation	Purification
Liquid	Simulated moving bed	Displacement purge	Bulk separation
Gas	Fixed bed	Thermal swing (TSA)	Purification
Gas	Combined fluidized bed-moving bed	Thermal swing (TSA)	Purification
Gas	Fixed bed	Inert-purge swing	Purification
Gas	Fixed bed	Pressure swing (PSA)	Bulk separation
Gas	Fixed bed	Vacuum swing (VSA)	Bulk separation
Gas	Fixed bed	Displacement purge	Bulk separation

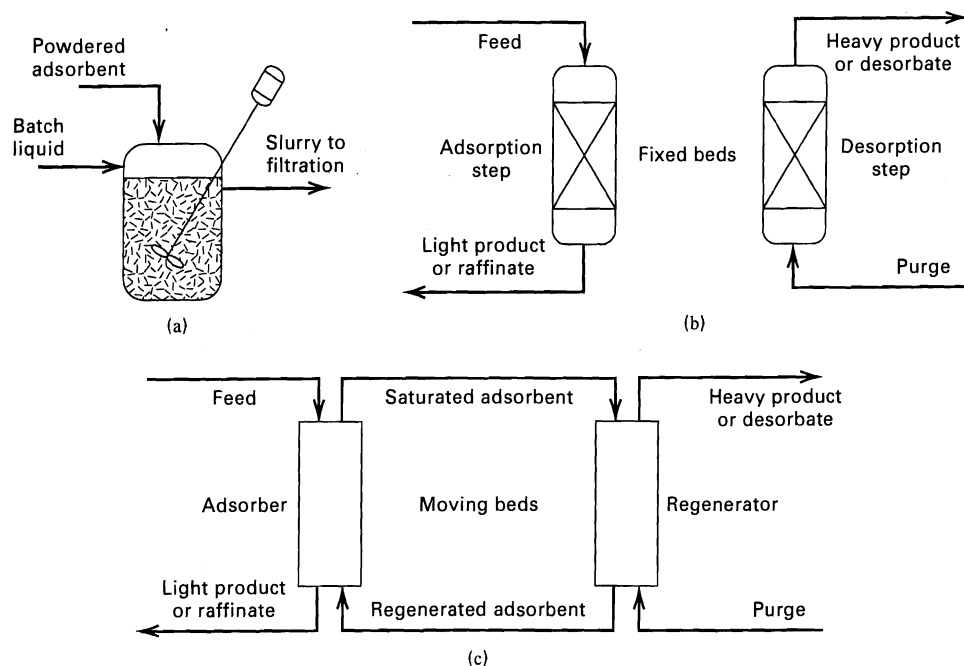


Figure 15.20 Contacting modes for adsorption. (a) Stirred-tank, slurry operation. (b) Cyclic fixed-bed, batch operation. (c) Continuous countercurrent operation.

or fluidizing the bed during adsorption, the flow of the liquid or gas feed is often downward. For removal of small amounts of dissolved hydrocarbons from water, the spent adsorbent is removed from the vessel and reactivated thermally at high temperature or it is discarded. Applications of fixed-bed adsorption, also called *percolation*, include the removal of dissolved organic compounds from water. For purification or bulk separation of gases, the adsorbent is almost always regenerated in place by one of the five methods listed in Table 15.7.

In the *thermal (temperature)-swing-adsorption* (TSA) method, the adsorbent is regenerated by desorption at a temperature higher than that used during the adsorption step of the cycle, as shown in Figure 15.21. The temperature of the bed is increased by (1) heat transfer from heating coils located in the bed followed by pulling a moderate vacuum or (2) more commonly, by heat transfer from an inert, nonadsorbing, hot purge gas, such as steam. Following desorption,

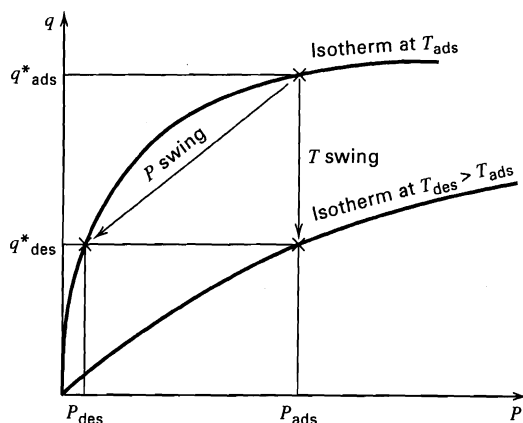


Figure 15.21 Schematic representation of pressure-swing and thermal-swing adsorption.

the bed is cooled before the adsorption step of the cycle is resumed. Because heating and cooling of the bed requires hours, a typical cycle time for TSA is hours to days. Therefore, if the quantity of adsorbent in the bed is to be reasonable, TSA is practical only for purification involving small rates of adsorption. Instead of using a fixed bed, a fluidized bed can be used for adsorption and a moving bed for desorption, as shown in Figure 15.22, provided that the adsorbent particles are attrition-resistant. In the adsorption section, sieve trays are used with the raw gas passing up through the perforations and fluidizing the adsorbent particles. The fluidized solids flow like a liquid across the tray, into the downcomer, and onto the tray below. From the adsorption section, the solids pass to the desorption section, where, as moving beds, they first flow down through preheating tubes and then through desorption tubes. Steam is used for indirect heating in both sets of tubes and for stripping in the desorption tubes. Moving beds, rather than fluidized beds on trays, are used in the desorption section because the stripping-steam flow rate is insufficient for fluidizing the solids. At the bottom of the unit, the regenerated solids are picked up by a carrier gas, which flows up through a gas-lift line to the top, where the solids settle out onto the top tray to repeat the adsorption part of the cycle. According to Keller [64], this configuration, which was announced in 1977, is used in more than 50 units worldwide to remove small amounts of solvents from air. Other applications of TSA include the removal of moisture, CO_2 , and pollutants from gas streams.

In the *inert-purge-swing* method of regeneration, desorption is at the same temperature and pressure as the adsorption step, because the gas used for purging is nonadsorbing (inert) or only weakly adsorbing. This method is used only when the solute is weakly adsorbed, easily desorbed, and of little or no value. The purge gas must be inexpensive so that it does not have to be purified before recycle.

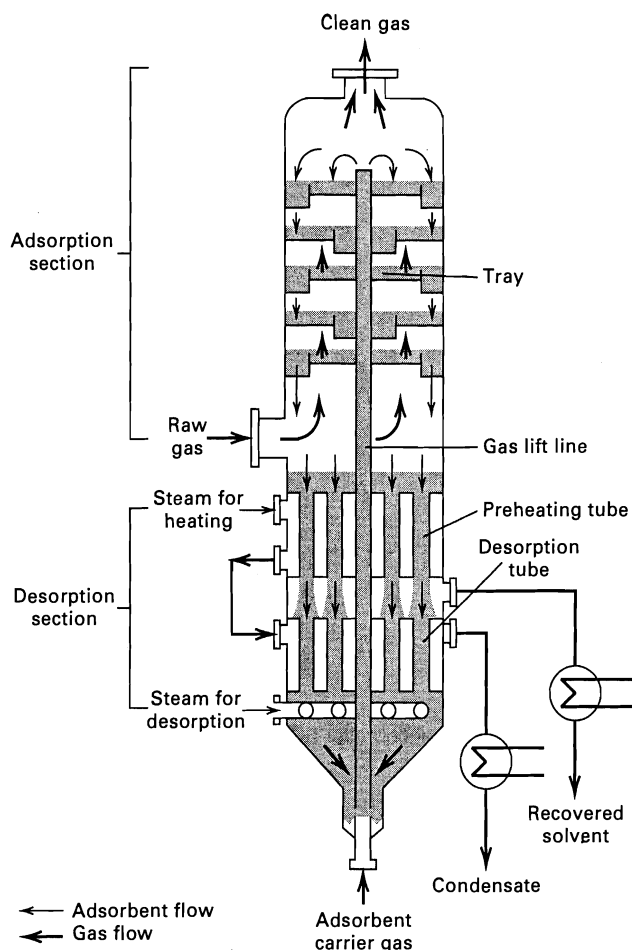


Figure 15.22 Purasiv process with a fluidized bed for adsorption and moving bed for desorption.

[From G.E. Keller, "Separations: New Directions for an Old Field," *AIChE Monograph Series*, 83 (17) (1987) with permission.]

In the *pressure-swing-adsorption* (PSA) cycle, adsorption takes place at an elevated pressure, whereas desorption occurs at near-ambient pressure, as is shown in Figure 15.21. PSA is used for bulk separations because the bed can be depressurized and repressurized rapidly, making it possible to operate at cycle times of seconds to minutes. Because of these short times, the beds need not be large even when a substantial fraction of the feed gas is adsorbed. If adsorption takes place at near-ambient pressure and desorption under vacuum, the cycle is referred to as *vacuum-swing-adsorption* (VSA). PSA and VSA are widely used for the bulk separation of air. If a zeolite adsorbent is used, equilibrium is rapidly established and nitrogen is preferentially adsorbed. The nonadsorbed, high-pressure product gas is a mixture of oxygen and argon with a small amount of nitrogen. If a carbon molecular-sieve adsorbent is used, the particle diffusivity of oxygen is observed to be about 25 times that of nitrogen. As a result, the selectivity of adsorption is controlled by mass transfer, and oxygen is preferentially adsorbed. The resulting high-pressure product gas is nearly pure nitrogen. In both cases, the adsorbed gas, which is desorbed at low pressure, is quite impure. For the separation of air, large plants

use VSA because it is more energy-efficient than PSA. Small plants often use PSA because that cycle is simpler.

In the *displacement-purge* (*displacement desorption*) cycle, a strongly adsorbed purge gas is used in the desorption step to displace the adsorbed species. Another step is required to recover the purge gas. The displacement-purge cycle is considered only where TSA, PSA, and VSA cannot be used because of pressure or temperature limitations. One application is the separation of medium-molecular-weight linear paraffins (C_{10} - C_{18}) from mixtures of branched-chain and cyclic hydrocarbons by adsorption on 5A zeolite. Ammonia, which is easily separated from the paraffins by flash vaporization, is used as the purge.

Most commercial applications of adsorption involve fixed beds that cycle between adsorption and desorption. Thus, compositions, temperature, and/or pressure at a given location in the bed vary with time. Alternatively, a continuous, countercurrent operation, where such variations do not occur, can be envisaged, as shown in Figure 15.20c and discussed in detail by Ruthven and Ching [65]. The main difficulty with such a scheme is the need to circulate the solid adsorbent, as a moving bed, to achieve a steady-state operation. The first commercial application of countercurrent adsorption and desorption was the moving-bed Hypersorption process for the recovery, by adsorption on activated carbon, of light hydrocarbons from various gas streams in petroleum refineries, as discussed by Berg [66]. Only a few units were installed because of problems with attrition of the adsorbent, difficulties in regenerating the adsorbent when heavier hydrocarbons were present in the feed gas, and unfavorable economics compared to those of distillation. Newer adsorbents with a much higher resistance to attrition and possible applications to more difficult separations are reviving interest in moving-bed units.

A successful alternative countercurrent system for commercial application to the separation of liquid mixtures is the simulated moving-bed system, shown in a hybrid system with two distillation columns in Figure 15.23 and known generally as the UOP Sorbex process. As described by Broughton [67], the bed is held stationary in one column, which is equipped with a number (perhaps 12) of liquid feed entry and discharge locations. By shifting, with a rotary valve (RV), the locations of feed entry, desorbent entry, extract (adsorbed) removal, and raffinate (non-adsorbed) removal, a countercurrent movement of solids is simulated by a downward movement of liquid. For the valve positions shown in Figure 15.23, locations 2 (entering desorbent), 5 (exiting extract), 9 (entering feed), and 12 (exiting raffinate) are operational, with all other numbered lines closed. However, liquid is also circulated down through and back up (external to the column) to the top of the column by a pump. Ideally, an infinite number of entry and exit locations on the column would exist and the valve would continuously change the four operational locations. Since this is impractical, a finite number of locations are used and valve changes are made periodically. In Figure 15.23, when the valve is

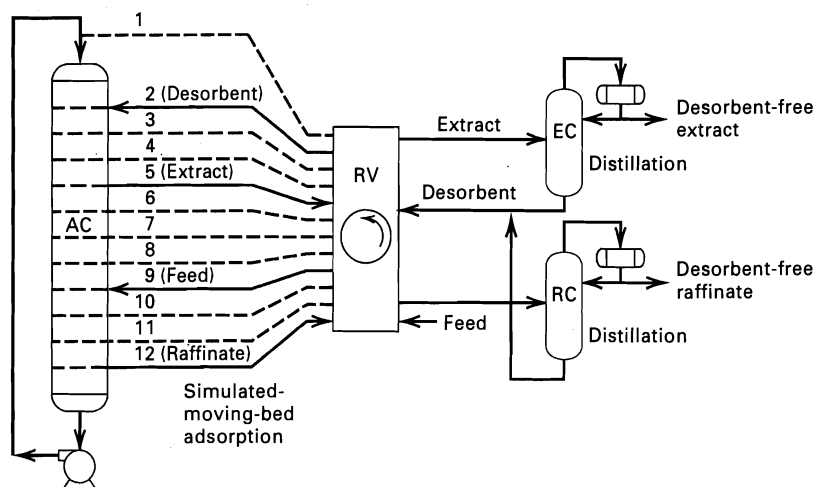


Figure 15.23 Sorbex hybrid simulated moving-bed process for bulk separation. AC, adsorbent chamber; RV, rotary valve; EC, extract column; RC, raffinate column.

[From D.B. Broughton, *Chem. Eng., Progress*, **64** (8), 60–65 (1968) with permission.]

moved to the next position, Lines 3, 6, 10, and 1 become operational. Thus, raffinate removal is relocated from the bottom of the bed to the top of the bed. The result is that the bed has no top or bottom. As discussed by Gembicki et al. [68], 78 Sorbex-type commercial units were installed during 1962–1989 for the bulk separation of *p*-xylene from C_8 aromatics; *n*-paraffins from branched and cyclic hydrocarbons; olefins from paraffins; *p*- or *m*-cymene (or cresol) from cymene (or cresol) isomers; and fructose from dextrose and polysaccharides. Humphrey and Keller [101] cite 100 commercial installations of Sorbex-type units and more than 50 different demonstrated separations.

Ion Exchange

Ion exchange employs the same modes of operation as shown for adsorption in Figure 15.20. Although the use of fixed beds in a cyclic operation is most common, stirred tanks are used for batch contacting, with an attached strainer or filter to separate the resin beads from the solution after

equilibrium conditions are approached. Agitation is mild to avoid resin attrition, but sufficient to achieve complete suspension of the resin. To increase resin utilization and achieve high ion-exchange reaction efficiency, much effort has been expended in the development of continuous, countercurrent contactors, two of which are shown in Figure 15.24. The Higgins contactor [69] operates as a moving, packed bed by using intermittent hydraulic pulses to move incremental portions of the bed from the contacting section, where ion exchange takes place, up, around, and down to the backwash region, down to the regenerating section, and back up through the rinse section to the contacting section to repeat the cycle. Liquid moves countercurrently to the resin. The Himsley contactor [70] has a series of trays, on each of which the resin beads are fluidized by the upward flow of liquid. Periodically, the flow is reversed to move incremental amounts of resin from one stage to the stage below. The batch of resin at the bottom is lifted to the wash column, then to the regeneration column, and then back to the top of the ion-exchange column for reuse.

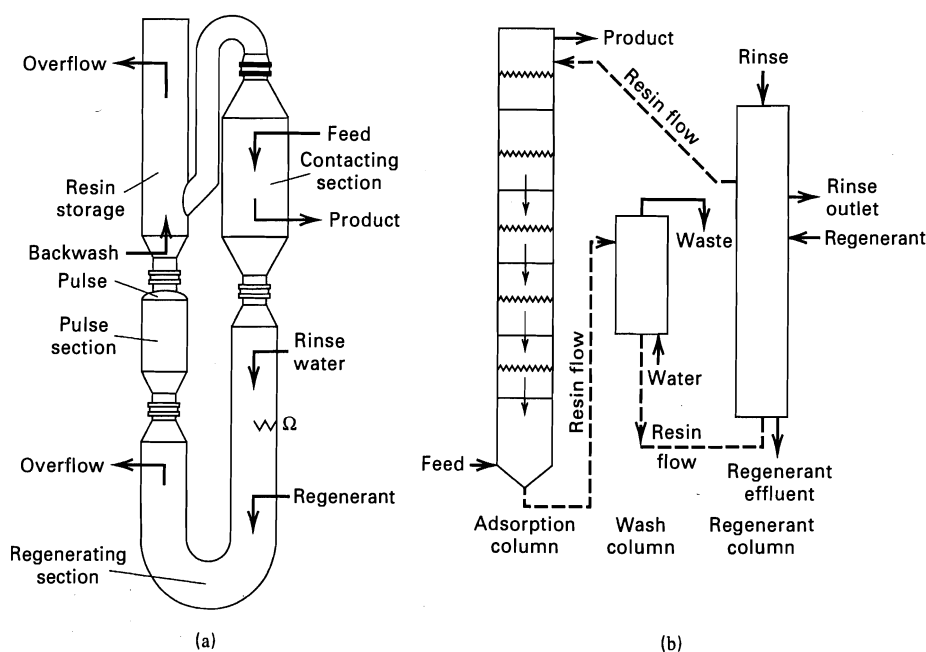


Figure 15.24 Continuous counter-current ion-exchange contactors. (a) Higgins moving packed-bed process. (b) Himsley fluidized-bed process.

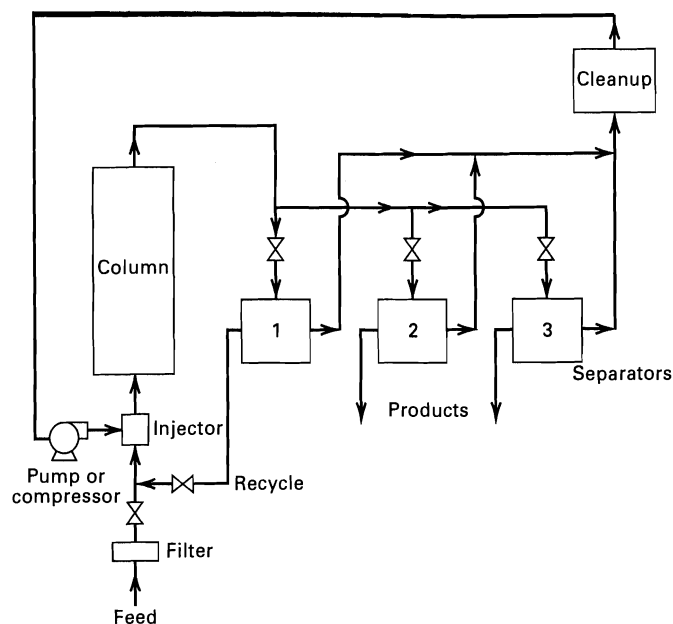


Figure 15.25 Large-scale, batch elution chromatography process.

Chromatography

Operation modes for large-scale, commercial application of chromatography are of two major types, as discussed in a book edited by Ganetsos and Barker [71]. The first, and the most common, is a transient mode that is a scaled-up version of an analytical chromatograph, referred to as large-scale, batch (or elution) chromatography. Packed columns of diameter up to 4.6 m and packed heights to 12 m have been reported. As shown in Figure 15.25 and discussed by Wankat in Chapter 14 of the *Handbook* edited by Rousseau [9], a recycled solvent or carrier gas is fed continuously into the sorbent-packed column. The feed mixture and recycle is pulsed into the column by an injector. A timer or detector (not shown) splits the effluent from the column, sending it to different separators (condensers, evaporators, distillation columns, etc.). Each separator is designed to remove a particular feed component from the carrier fluid. An additional cleanup step is required to purify the carrier fluid before it is recycled to the column. Separator one produces no product because it handles an effluent pulse that contains the carrier fluid and two or more of the feed components, which are recovered and recycled to the column. Thus, if properly designed and operated, the batch chromatograph operates somewhat like a batch-distillation column, producing a nearly pure cut for each component in the feed and slop cuts for recycle. The system shown in Figure 15.25 is designed to separate a binary system. If, say, three more separators are added, the system can separate a five-component feed into five nearly pure products.

The second major type of large-scale chromatograph is the countercurrent flow or simulated-moving-bed mode already discussed for adsorption. This mode is more efficient, but is more complicated and can only separate a mixture into two products. A third mode is the continuous, cross-current

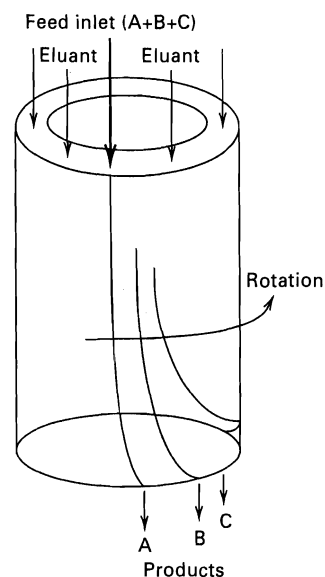


Figure 15.26 Rotating, cross-current, annular chromatograph.

(or rotating) chromatograph, first conceived by Martin [72] and shown schematically in Figure 15.26. The packed annular bed rotates slowly about its axis, past the feed-inlet point. *Eluant* (solvent or carrier gas) enters the top of the bed uniformly over the entire cross-sectional area. Both feed and eluant are fed continuously and are carried downward and around by the rotation of the bed. Because of the different selectivities of the feed components for the sorbent, each component traces a different helical path since each spends a different amount of time in contact with the sorbent. Thus, each component is eluted from the bottom of the packed annulus at a different location. In principle, a multicomponent feed can be separated continuously into nearly pure components following separation of the carrier fluid from each eluted fraction. Units of up to 12 in. in diameter have successfully separated sugars, proteins, and metallic elements.

Slurry Adsorption (Contact Filtration)

Three modes of adsorption from a liquid in an agitated vessel are of interest. The first is the *batch mode* in which a batch of liquid is contacted with a batch of adsorbent for a period of time, followed by discharge of the slurry from the vessel, and filtration to separate the solids from the liquid. The second is the *continuous mode*, in which liquid and adsorbent are continuously added to and removed from the agitated vessel. In the third mode, called the *semibatch* or *semicontinuous mode*, the liquid is continuously fed to and removed from the agitated vessel, where it is contacted with the adsorbent, which is retained in a contacting zone of the vessel until it is nearly spent. Models for each of these three modes are developed next, followed by examples of their application. In all models, the slurry is assumed to be perfectly mixed by agitation in the turbulent regime to produce a fluidized bed of sorbent. Perfect mixing is approached by

using a liquid depth of from one to two vessel diameters, four vertical wall baffles, and one or two marine propellers or pitched-blade turbines on a vertical shaft. With a proper impeller rotation rate, the axial flow achieves complete suspension. For semicontinuous operation, a clear liquid region is maintained above the suspension region for liquid withdrawal.

Because small particles are used in slurry adsorption and because the relative velocity between the particles and the liquid in an agitated slurry is low (small particles tend to move with the liquid), the rate of adsorption is assumed to be controlled by external, rather than internal, mass transfer.

Batch Mode

The rate of adsorption of solute, as controlled by external mass transfer, is

$$-\frac{dc}{dt} = k_L a(c - c^*) \quad (15-77)$$

where c is the concentration of solute in the bulk liquid; c^* is the concentration in equilibrium with the loading on the adsorbent, q ; k_L is the external liquid-phase mass-transfer coefficient; and a is the external surface area of the adsorbent per unit volume of liquid. Starting from feed concentration, c_F , the instantaneous bulk concentration, c , at time t , is related to the instantaneous adsorbent loading, q , by material balance:

$$c_F Q = cQ + qS \quad (15-78)$$

where the adsorbent is assumed to be initially free of adsorbate, Q is the liquid volume (assumed to remain constant for dilute feeds), and S is the mass of adsorbent. The equilibrium concentration, c^* , is given by an appropriate adsorption isotherm: a linear isotherm, the Langmuir isotherm (15-36), or the Freundlich isotherm (15-35). For example, a rearrangement of the latter gives

$$c^* = (q/k)^n \quad (15-79)$$

To solve the system of equations for c and q as a function of time, starting from c_F at $t = 0$, (15-78) is combined with the equilibrium isotherm, for example, (15-79), to eliminate q . The resulting equation is combined with (15-77) to eliminate c^* to give an ODE for c in t , which is integrated analytically or numerically. Corresponding values of q are then obtained from (15-78).

If the equilibrium is represented by a linear isotherm,

$$c^* = q/k \quad (15-80)$$

an analytical integration gives

$$c = \frac{c_F}{\beta} [\exp(-k_L a \beta t) + \alpha] \quad (15-81)$$

where

$$\beta = 1 + \frac{Q}{Sk} \quad (15-82)$$

$$\alpha = \frac{Q}{Sk} \quad (15-83)$$

As the contact time approaches infinity, adsorption equilibrium is approached and for the linear isotherm, from (15-81) or combining (15-78), with $c = c^*$, and (15-80):

$$c\{t = \infty\} = c_F \alpha / \beta \quad (15-84)$$

Continuous Mode

When both liquid and solids flow continuously through a perfectly mixed vessel, (15-77) is converted to an algebraic equation because, as in a perfectly mixed reaction vessel (CSTR), the concentration, c , throughout the vessel, is equal to the exit (outlet) concentration, c_{out} . Thus, in terms of the residence time in the vessel, t_{res} :

$$\frac{c_F - c_{out}}{t_{res}} = k_L a(c_{out} - c^*) \quad (15-85)$$

or, rearranging:

$$c_{out} = \frac{c_F + k_L a t_{res} c^*}{1 + k_L a t_{res}} \quad (15-86)$$

Equation (15-78) becomes

$$c_F Q = c_{out} Q + q_{out} S \quad (15-87)$$

where Q and S are now flow rates. An appropriate adsorption isotherm relates c^* to q_{out} .

For a linear isotherm, (15-80) becomes $c^* = q_{out}/k$, which when combined with (15-87) and (15-86) to eliminate c^* and q_{out} , gives

$$c_{out} = c_F \left(\frac{1 + \gamma \alpha}{1 + \gamma + \gamma \alpha} \right) \quad (15-88)$$

where α is given by (15-83) and

$$\gamma = k_L a t_{res} \quad (15-89)$$

The corresponding q_{out} is given by a rearrangement of (15-87):

$$q_{out} = \frac{Q(c_F - c_{out})}{S} \quad (15-90)$$

For a nonlinear adsorption isotherm, such as (15-35) or (15-36), (15-85) and (15-87) are combined with the isotherm equation, but it may not be possible to express the result explicitly in q_{out} . In that event, a numerical solution is required, as illustrated below in Example 15.10.

Semicontinuous Mode

The most difficult mode to model is the semicontinuous mode, where the adsorbent is retained in the vessel, but the feed liquid enters and exits the vessel at a fixed, continuous flow rate. Both concentration, c , and loading, q , vary with time. With perfect mixing, the outlet concentration is given by (15-86), where t_{res} is the residence time of the liquid in the suspension, and c^* is related to q in the suspension by an appropriate adsorption isotherm. The variation of q in the batch of solids is given by (15-77), rewritten in terms of the

change in q , rather than c :

$$S \frac{dq}{dt} = k_L a (c_{\text{out}} - c^*) t_{\text{res}} Q \quad (15-91)$$

where, for this mode, S is the batch mass of adsorbent in the suspension and Q is the steady, volumetric-liquid flow rate.

Both (15-91) and (15-86) involve c^* , which can be replaced by a function of instantaneous q by selecting an appropriate isotherm. The resulting two equations are then combined to eliminate c_{out} . The resulting ODE is then integrated analytically or numerically to obtain q as a function of time, from which c_{out} as a function of time can be determined from (15-86) and the isotherm. The time-average value of c_{out} is then obtained by integration of c_{out} with respect to time. These steps are illustrated in the following example. For a linear isotherm, the derivation is left as an exercise.

EXAMPLE 15.10

An aqueous solution containing 0.010 mol phenol/L is to be treated at 20°C with activated carbon to reduce the concentration of phenol to 0.00057 mol/L. From Example 4.12, the adsorption equilibrium data are well fitted to the Freundlich equation:

$$q = 2.16c^{1/4.35} \quad (1)$$

or

$$c^* = (q/2.16)^{4.35} \quad (2)$$

where q and c are in mmol/g and mmol/L, respectively. In terms of kmol/kg and kmol/m³, (2) becomes

$$c^* = (q/0.01057)^{4.35} \quad (3)$$

All three modes of slurry adsorption are to be considered. From Example 4.12, the minimum amount of adsorbent is 5 g/L of solution. Laboratory experiments with adsorbent particles 1.5 mm in diameter in a well-agitated vessel have confirmed that the rate of adsorption is controlled by external mass transfer with $k_L = k_c = 5 \times 10^{-5}$ m/s. Particle surface area is 5 m²/kg of particles.

- Using twice the minimum amount of adsorbent in an agitated vessel operated in the batch mode, determine the time in minutes to reduce the phenol content to the desired value.
- For operation in the continuous mode with twice the minimum amount of adsorbent, determine the required residence time in minutes in the agitated vessel. How does this compare to the batch time of part (a)?
- For operation in the semicontinuous mode with 1,000 kg of activated carbon, a liquid feed rate of 10 m³/h, and a liquid residence time equal to 1.5 times the value computed in part (b), determine the run time to obtain a composite liquid product with the desired phenol concentration. Are the results reasonable, or should changes be made to the specifications?

SOLUTION

(a) Batch mode:

$$S/Q = 2(5) = 10 \text{ g/L} = 10 \text{ kg/m}^3$$

$$k_L a = 5 \times 10^{-5}(5)(10) = 2.5 \times 10^{-3} \text{ s}^{-1}$$

$$c_F = 0.010 \text{ mol/L} = 0.010 \text{ kmol/m}^3$$

From (15-78),

$$q = \frac{c_F - c}{S/Q} = \frac{0.010 - c}{10} \quad (4)$$

Substituting (4) into (3),

$$c^* = \left(\frac{0.010 - c}{0.01057} \right)^{4.35} \quad (5)$$

Substituting (5) into (15-77),

$$-\frac{dc}{dt} = 2.5 \times 10^{-3} \left[c - \left(\frac{0.010 - c}{0.01057} \right)^{4.35} \right] \quad (6)$$

where, $c = c_F = 0.010$ kmol/m³ at $t = 0$ and we want t for $c = 0.00057$ kmol/m³. By numerical integration of (6), $t = 1,140$ s = 19 min.

(b) Continuous mode:

Equation (15-85) applies, where all quantities are the same as those determined in part (a) and $c_{\text{out}} = 0.00057$ kmol/m³. Thus,

$$t_{\text{res}} = \frac{c_F - c_{\text{out}}}{k_L a (c_{\text{out}} - c^*)}$$

where c^* is given by (3) with $q = q_{\text{out}}$, and q_{out} is obtained from (15-87). Thus,

$$t_{\text{res}} = \frac{0.010 - 0.00057}{2.5 \times 10^{-3} \left[0.00057 - \left(\frac{0.010 - 0.00057}{0.01057} \right)^{4.35} \right]} = 6,950 \text{ s or } 1.93 \text{ h}$$

This residence time is appreciably longer than the batch time of 1,140 s. In the batch mode, the concentration driving force for external mass transfer is initially $(c - c^*) = c_F = 0.010$ kmol/m³ and gradually declines to a much smaller final value, at 1,140 s, of

$$(c - c^*) = c_{\text{final}} - \left(\frac{0.010 - c_{\text{final}}}{0.01057} \right)^{4.35} = 0.000543 \text{ kmol/m}^3$$

For the continuous mode with perfect mixing in the vessel, the concentration driving force for external mass transfer is always at the final batch value of 0.000543 kmol/m³.

(c) Semicontinuous mode:

Equation (15-91) applies with

$$S = 1,000 \text{ kg}, \quad c_F = 0.010 \text{ kmol/m}^3$$

$$Q = 10 \text{ m}^3/\text{h}, \quad t_{\text{res}} = 10,425 \text{ s}, \quad k_L a = 2.5 \times 10^{-3} \text{ s}^{-1}$$

c^* is given in terms of q by (3) and c_{out} is given by (15-86).

Combining (15-91), (3), and (15-86) to eliminate c^* and c_{out} gives, after simplification,

$$\frac{dq}{dt} = \left(\frac{\gamma}{1 + \gamma} \right) \frac{Q}{S} \left[c_F - \left(\frac{q}{0.01057} \right)^{4.35} \right] \quad (7)$$

where γ is given by (15-89) and the time, t , is the time that the adsorbent remains in the vessel. For values of γ , Q/S , and c_F equal, respectively, to 26.06, 0.01 m³/h·kg, and 0.010 kmol/m³, (7) reduces to

$$\frac{dq}{dt} = 0.00963 \left[0.010 - \left(\frac{q}{0.01057} \right)^{4.35} \right] \quad (8)$$

where t is in hours and q is in kmol. By numerical integration of (8), starting from $q = 0$ at $t = 0$, we obtain q as a function of t as given in Table 15.8. Included are corresponding values of

Table 15.8 Results for Part (c), Semicontinuous Mode, of Example 15.10

Time t , h	q , kmol/kg	kmol/m ³	
		c_{out}	c_{cum}
0.0	0.0	0.000370	0.000370
5.0	0.000481	0.000371	0.000370
10.0	0.000962	0.000398	0.000375
15.0	0.001440	0.000535	0.000401
15.7	0.001506	0.000570	0.000407
20.0	0.001905	0.000928	0.000476
21.0	0.001995	0.001052	0.000501
22.0	0.002084	0.001195	0.000529
23.0	0.002172	0.001356	0.000561
23.2	0.002189	0.001390	0.000568
23.3	0.002197	0.001407	0.000572

c_{out} computed from (15-86) combined with (3) to eliminate c^* :

$$c_{out} = \frac{c_F + \gamma(q/0.01057)^{4.35}}{1 + \gamma} = \frac{0.010 + 26.06(q/0.01057)^{4.35}}{27.06}$$

Also included in Table 15.8 are the cumulative values of c , for all of the liquid effluent that exits the vessel during the period from $t = 0$ to $t = t$, as obtained by integrating c_{out} with respect to time: $c_{cum} = \int_0^t c_{out} dt/t$.

From the results in Table 15.8, it is seen that the loading, q , increases almost linearly during the first 10 h, while the instantaneous phenol concentration c_{out} in the exiting liquid remains almost constant. At 15.7 h, c_{out} has increased to the specified value of 0.00057 kmol/m³, but c_{cum} is only 0.000407 kmol/m³. Therefore, the operation can continue. Finally, at between 23.2 and 23.3 h, c_{cum} reaches 0.00057 kmol/m³ and the operation must be terminated. During operation, the vessel contains 1,000 kg or 2 m³ of adsorbent particles. With a liquid residence time of almost 3 h, the vessel must contain 10(3) = 30 m³. Thus, the vol% solids in the agitated vessel is 6.7. This is reasonable. If the mass of adsorbent in the vessel is increased to 2,000 kg, giving almost 12 vol% solids, the time of operation is doubled to 46.5 h.

Fixed-Bed Adsorption (Percolation)

In the continuous and semicontinuous modes of operation in slurry adsorption, the liquid exiting the vessel always contains unadsorbed solute. If a fixed bed is used, it is possible to obtain a nearly solute-free liquid or gas effluent until the

adsorbent in the bed approaches saturation. A fixed bed is frequently used for gas purification and bulk separation.

Consider the flow, down through a fixed bed of adsorbent, of a fluid containing an adsorbable component (the solute). If (1) external and internal mass-transfer resistances are very small; (2) plug flow is achieved; (3) axial dispersion is negligible; (4) the adsorbent is initially free of adsorbate; and (5) the adsorption isotherm begins at the origin, then local equilibrium between the fluid and the adsorbent is achieved instantaneously, resulting, as shown in Figure 15.27, in a shock like wave, called a *stoichiometric front*, that moves as a sharp concentration front through the bed. This is *ideal (local equilibrium) fixed-bed adsorption*. Upstream of the front, the adsorbent is saturated with adsorbate and the concentration of solute in the fluid is that of the feed, c_F . The loading of adsorbate on the adsorbent is the q_F in equilibrium with c_F . The length (height) and weight of the bed section upstream of the front are LES and WES, respectively, where ES refers to the equilibrium section, called the *equilibrium zone*.

In the upstream region, the adsorbent is spent. Downstream of the stoichiometric front and in the exit fluid, the concentration of the solute in the fluid is zero, and the adsorbent is still adsorbate-free. In this section of the bed, the length and weight are LUB and WUB, respectively, where UB refers to unused bed.

After a period of time, called the *stoichiometric time*, the stoichiometric wave front reaches the end of the bed, the concentration of the solute in the fluid abruptly rises to the inlet value, c_F , no further adsorption is possible, and the adsorption step is terminated. This point is referred to as the *break-point* and the stoichiometric wave front becomes the *ideal breakthrough curve*.

For ideal fixed-bed adsorption, the location of the concentration wave front L , in Figure 15.27, as a function of time, is obtained solely by material balance and adsorption equilibrium considerations. Thus, at equilibrium, the loading in equilibrium with the feed is designated by $q_F = f\{c_F\}$, where $f\{c_F\}$ is given by an appropriate adsorption isotherm. By material balance on the adsorbate before breakthrough occurs: Solute in entering feed = adsorbate. Accordingly:

$$Q_F c_F t_{ideal} = q_F S L_{ideal} / L_B \quad (15-92)$$

where Q_F is the volumetric flow rate of feed, c_F is the concentration of the solute in the feed, t_{ideal} is the time for an

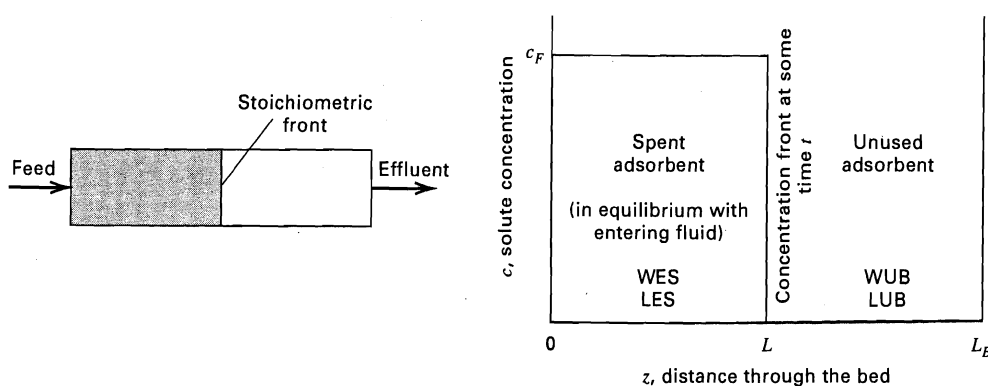


Figure 15.27 Stoichiometric (equilibrium) concentration front for ideal fixed-bed adsorption.

ideal front to reach $L_{\text{ideal}} < L_B$, q_F is the loading per unit mass of adsorbent that is in equilibrium with the feed concentration, S is the total mass of adsorbent in the bed, and L_B is the total bed length.

$$L_{\text{ideal}} = \text{LES} = \left(\frac{Q_F c_F t_{\text{ideal}}}{q_F S} \right) L_B \quad (15-93)$$

$$\text{LUB} = L_B - \text{LES} \quad (15-94)$$

$$\text{WES} = S \left(\frac{\text{LES}}{L_B} \right) \quad (15-95)$$

$$\text{WUB} = S - \text{WES} \quad (15-96)$$

In a real fixed-bed adsorber, the assumptions leading to (15-92) are not valid. Internal transport resistance and, in some cases, external transport resistance are finite. Axial dispersion can also be significant, particularly at low flow rates in shallow beds. These factors contribute to the development of broad concentration fronts like those in Figure 15.28. In Figure 15.28a, typical solute concentration profiles for the fluid are shown as a function of distance through the bed at increasing times t_1 , t_2 , and t_b from the start of flow through the bed. At t_1 , no part of the bed is saturated. At t_2 , the bed is almost saturated for a distance L_s . At L_f , the bed is almost clean. Beyond L_f , little mass transfer occurs at t_2 and the adsorbent is still unused. The region between L_s and L_f is called the mass-transfer zone, MTZ at t_2 , where adsorption takes place. Because it is difficult to determine

where the MTZ zone begins and ends, L_f can be taken where $c/c_F = 0.05$, with L_s at $c/c_F = 0.95$. From time t_2 to time t_b , the S-shaped front moves through the bed. At t_b , the leading point of the MTZ just reaches the end of the bed. This is the breakthrough point. Rather than using $c/c_F = 0.05$, the breakthrough concentration can be taken as the minimum detectable or maximum allowable solute concentration in the effluent fluid.

Figure 15.28b is a typical plot of the ratio of the outlet-to-inlet solute concentration in the fluid as a function of time from the start of flow. The S-shaped curve is called the breakthrough curve. Prior to t_b , the outlet solute concentration is less than some maximum permissible value, say, $c_{\text{out}}/c_F = 0.05$. At t_b , this value is reached, the adsorption step is discontinued, and the regeneration part of the cycle is initiated or the spent adsorbent is discarded. If the adsorption step were to be continued for $t > t_b$, the outlet solute concentration would be observed to rise rapidly, eventually approaching the inlet concentration as the outlet end of the bed became saturated. The time to reach $c_{\text{out}}/c_F = 0.95$ is designated t_e .

The steepness of the breakthrough curve determines the extent to which the capacity of an adsorbent bed can be utilized. Thus, the shape of the curve is very important in determining the length of an adsorption bed. For the ideal case, with a stoichiometric wave front, (15-92) applies and all of the bed is utilized before breakthrough occurs. As the width of the breakthrough curve and the corresponding width of the MTZ for the concentration profiles increase, less and less of the bed capacity can be utilized. The situation is further complicated by the fact that the steepness of the concentration profiles shown in Figure 15.28a increases or decreases with time, depending on the shape of the adsorption isotherm, as shown by DeVault [73], in the following manner.

Assume: (1) plug flow of the fluid through the bed at a constant actual (interstitial) velocity, u ; (2) instantaneous equilibrium of the solute in the bulk fluid with the adsorbate; (3) no axial dispersion; and (4) isothermal conditions. The bed is not initially free of adsorbate and/or the feed to the bed starting at time $t = 0$ is not at constant composition. The superficial fluid velocity is $\epsilon_b u$. A mass balance on the solute for the flow of fluid through a differential adsorption-bed length, dz , over a differential-time duration, dt , gives

$$\epsilon_b u A_b c|_z = \epsilon_b u A_b c|_{z+\Delta z} + \epsilon_b A_b \Delta z \frac{\partial c}{\partial t} + (1 - \epsilon_b) A_b \Delta z \frac{\partial q}{\partial t} \quad (15-97)$$

Dividing by Δz and taking the limit as $\Delta z \rightarrow 0$ gives

$$\frac{\partial c}{\partial t} + u \frac{\partial c}{\partial z} + \frac{(1 - \epsilon_b)}{\epsilon_b} \frac{\partial q}{\partial t} = 0 \quad (15-98)$$

where q is the adsorption loading/unit volume of adsorbent particles, given by an appropriate adsorption isotherm. By the chain rule:

$$\frac{\partial q}{\partial t} = \frac{\partial q}{\partial c} \frac{\partial c}{\partial t} \quad (15-99)$$

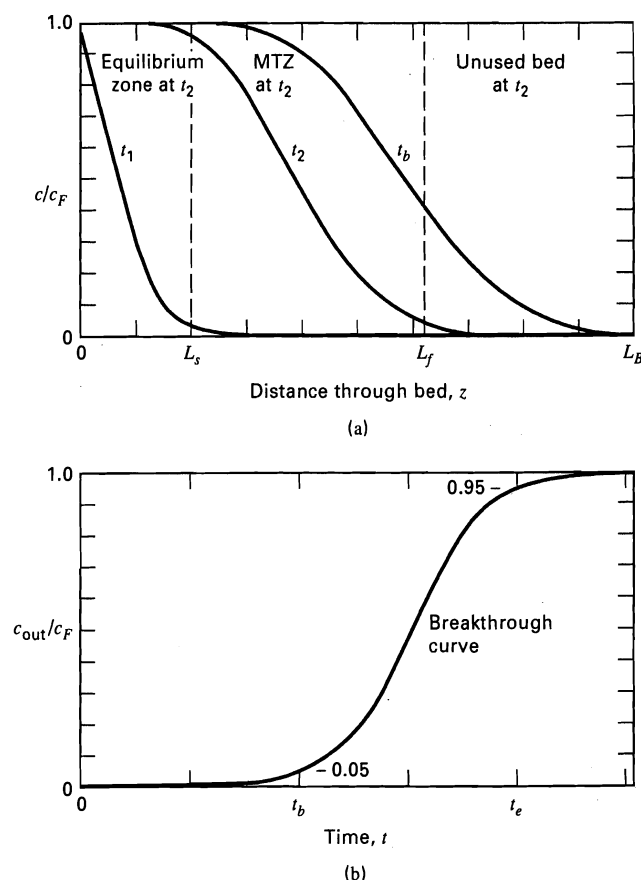


Figure 15.28 Solute wave fronts in a fixed-bed adsorber with mass-transfer effects. (a) Concentration-distance profiles. (b) Breakthrough curve.

This hyperbolic PDE (15-98) gives $c = f\{z, t\}$. Therefore, by the rules of implicit partial differentiation:

$$u_c = \left(\frac{\partial z}{\partial t} \right)_c = - \frac{\left(\frac{\partial c}{\partial t} \right)}{\left(\frac{\partial c}{\partial z} \right)} \quad (15-100)$$

where u_c is the velocity of the concentration wave front, $\partial z / \partial t$ at constant c . Combining (15-98) to (15-100):

$$u_c = \frac{u}{1 + \left(\frac{1 - \epsilon_b}{\epsilon_b} \right) \frac{dq}{dc}} \quad (15-101)$$

This equation gives the velocity of the concentration wave front for the solute in terms of the interstitial fluid velocity and the slope, dq/dc , of the adsorption isotherm. If dq/dc is constant, the wave front moves at a constant value.

In general, the concentration wave front moves through the bed at a velocity, u_c , that is much less than the interstitial fluid velocity. For example, suppose that $\epsilon_b = 0.5$ and the equilibrium adsorption isotherm is given by $q = 5,000c$. Then $dq/dc = 5,000$. Then, from (15-101), $u_c/u = 0.0002$. If the interstitial velocity is 3 ft/s, the velocity of the concentration wave front is only 0.0006 ft/s. If the bed were 6 ft in height, it would take 2.78 h for the concentration wave front to pass through the bed. If the adsorption isotherm is curved, regions of the wave front at a higher concentration move at a velocity different from regions at a lower concentration. Thus, for a linear isotherm (curve A in Figure 15.29a), the width of the MTZ and the wave pattern remain constant. For a favorable isotherm of the Freundlich or Langmuir type (curve B in Figure 15.29a), high-concentration regions move faster than low-concentration regions, and the wavefront steepens with time until a constant pattern front (CPF) is developed, as shown in Figure 15.29b. For the much less common unfavorable type of isotherm (Curve C in Figure 15.29a), low-concentration regions travel faster and the wavefront broadens with time.

For the general case where external and internal mass-transfer resistances are finite and/or axial dispersion is not negligible, methods for predicting concentration profiles and breakthrough curves have been the subject of much study. As will be shown, when mass-transfer resistances are a factor, the concentration fronts develop quite differently from the equilibrium fronts just described. Solutions for a number

of simplified cases are discussed in detail by Ruthven [10]. The PDE for the governing dynamic behavior is a modification of (15-98):

$$-D_L \frac{\partial^2 c}{\partial z^2} + \frac{\partial(u_c c)}{\partial z} + \frac{\partial c}{\partial t} + \frac{(1 - \epsilon_b)}{\epsilon_b} \frac{\partial \bar{q}}{\partial t} = 0 \quad (15-102)$$

where the first term accounts for axial dispersion with eddy diffusivity D_L , the second term permits an axial variation in fluid velocity, and the fourth term is now based on $\bar{q}$, the volume-average adsorbate loading per unit mass. Thus, the latter term accounts for the variation of q throughout the adsorbent particle, due to internal mass-transfer resistance, by averaging the rate of adsorption over the adsorbent particle. The volume-average adsorbate loading for a spherical particle is given by

$$\bar{q} = \left(\frac{3}{R_p^3} \right) \int_0^{R_p} r^2 q dr \quad (15-103)$$

where R_p is the radius of the adsorbent particle.

Equation (15-102) gives the concentration of solute in the bulk fluid as a function of time and location in the bed. Equation (15-68) gives the concentration of the solute in the fluid within the pores of an adsorbent particle. These two equations are coupled together by the continuity condition at the particle surface:

$$D_e \left(\frac{\partial c}{\partial r} \right)_{R_p} = k_c (c - c_{R_p}) \quad (15-104)$$

where k_c is the external mass-transfer coefficient and D_e is the effective diffusivity in the particle, as discussed in Section 15.3. The simultaneous solution of (15-102), (15-103), (15-68), and (15-104) is a formidable task, which can be avoided by using the linear-driving-force (LDF) model formulated by Glueckauf [74, 75] and discussed in detail by Yang [25] and Ruthven [10]. This model, which is widely used to simulate and design fixed-bed adsorption, is based on the following relation, which replaces (15-68) and (15-104):

$$\frac{\partial \bar{q}}{\partial t} = k(q^* - \bar{q}) = kK(c - c^*) \quad (15-105)$$

where q^* is the adsorbate loading in equilibrium with the solute concentration, c , in the bulk fluid; c^* is the concentration in equilibrium with average loading $\bar{q}$; k is the overall mass-transfer coefficient, which includes both external- and internal-transport resistances; and K is the adsorption-

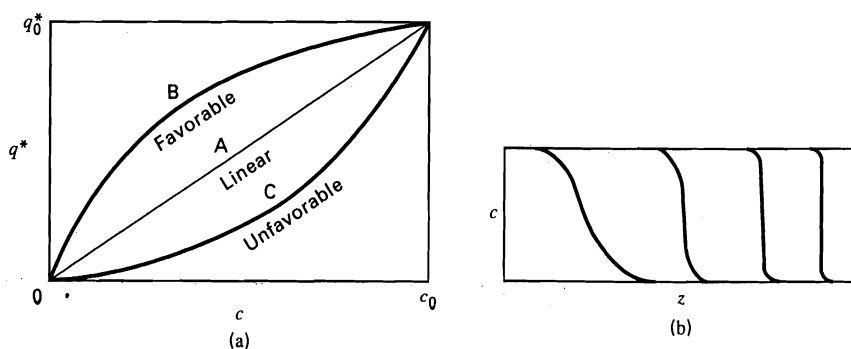


Figure 15.29 Effect of shape of isotherm on sharpness of concentration wavefront. (a) Isotherm shapes. (b) Self-sharpening wavefront caused by a favorable adsorption isotherm.

equilibrium constant for a linear adsorption isotherm of the form $q = Kc$.

A suitable relationship for the factor kK is

$$\frac{1}{kK} = \frac{R_p}{3k_c} + \frac{R_p^2}{15D_e} \quad (15-106)$$

where the first term on the RHS represents the external mass-transfer resistance, $k_c a_v$, since for a sphere, the surface area/unit volume, a_v , is given by

$$4\pi R_p^2 / [(4/3)\pi R_p^3] = 3/R_p$$

The second term in (15-106) represents the internal resistance, which was first developed by Glueckauf [75], but can also be derived by assuming a parabolic adsorbate loading profile, in the particle, as shown by Liaw et al. [76]. Thus, let

$$q = a_0 + a_1 r + a_2 r^2 \quad (15-107)$$

where the constants a_i depend on time and location in the bed, but are independent of r . Because $\partial q / \partial r = 0$ at $r = 0$ (symmetry condition), $a_1 = 0$. Equating Fick's first law for diffusion into the particle at the particle surface, to the rate of accumulation of adsorbate within the particle, assuming that effective diffusivity is independent of concentration, we obtain

$$\frac{\partial \bar{q}}{\partial t} = D_e a_v \frac{\partial q}{\partial r} \Big|_{r=R_p} = \frac{3D_e}{R_p} \frac{\partial q}{\partial r} \Big|_{r=R_p} \quad (15-108)$$

At the particle surface, from (15-107):

$$q_{R_p} = a_0 + a_2 R_p^2 \quad (15-109)$$

Substituting (15-107) with $a_1 = 0$ into (15-103) and integrating gives

$$\bar{q} = a_0 + \frac{3}{5} a_2 R_p^2 \quad (15-110)$$

Combining (15-109) and (15-110) to eliminate a_0 gives

$$a_2 = \frac{5}{2R_p^2} (q_{R_p} - \bar{q}) \quad (15-111)$$

From (15-107):

$$\frac{\partial q}{\partial r} \Big|_{r=R_p} = 2a_2 R_p \quad (15-112)$$

Combining (15-110), (15-111), and (15-108):

$$\frac{\partial \bar{q}}{\partial t} = \frac{15D_e}{R_p^2} (q_{R_p} - \bar{q}) \quad (15-113)$$

Comparing (15-105) with (15-113), we see that the internal resistance is given by the second term in (15-106).

The analytical solution of a simplified form of (15-102), which assumes negligible axial dispersion, constant fluid velocity, u , and the LDF mass-transfer model, is summarized by Ruthven [10] and discussed in detail by Klinkenberg [77]. The solution was first obtained in terms of Bessel functions by Anzelius [78] for the analogous problem of heating or cooling a packed bed of depth z with a fluid. A useful

approximate solution is that of Klinkenberg [79]:

$$\frac{c}{c_F} \approx \frac{1}{2} \left[1 + \operatorname{erf} \left(\sqrt{\tau} - \sqrt{\xi} + \frac{1}{8\sqrt{\tau}} + \frac{1}{8\sqrt{\xi}} \right) \right] \quad (15-114)$$

where

$$\xi = \frac{kKz}{u} \left(\frac{1 - \epsilon_b}{\epsilon_b} \right) = \text{Dimensionless distance coordinate} \quad (15-115)$$

$$\tau = k \left(t - \frac{z}{u} \right) = \text{Dimensionless time coordinate corrected for displacement} \quad (15-116)$$

$$\operatorname{erf}(-x) = -\operatorname{erf}(x) \quad (15-117)$$

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-\eta^2} d\eta \quad (15-118)$$

where ξ and τ are coordinate transformations for z and t , which convert the equations to a much simpler form. The approximation (15-114) is accurate to <0.6% error for $\xi > 2.0$. The $\operatorname{erf}(x)$, which is included as a function in most spreadsheet programs, is 0.0 at $x = 0$ and asymptotically approaches a value of 1.0 for $x > 2.0$, where x is a dummy variable.

Klinkenberg [79] also includes the following approximate solution for profiles of solute concentration in equilibrium with the average sorbent loading:

$$\frac{c^*}{c_F} = \frac{\bar{q}}{q_F^*} \approx \frac{1}{2} \left[1 + \operatorname{erf} \left(\sqrt{\tau} - \sqrt{\xi} - \frac{1}{8\sqrt{\tau}} - \frac{1}{8\sqrt{\xi}} \right) \right] \quad (15-119)$$

where $c^* = \bar{q}/K$ and $c^*/c_F = \bar{q}/q_F^*$, where q_F^* is the loading in equilibrium with c_F .

EXAMPLE 15.11

Air at 70°F and 1 atm, containing 0.9 mol% benzene, enters a fixed-bed adsorption tower at a flow rate of 23.6 lb/min. The tower is 2 ft in inside diameter and is packed to a height of 6 ft with 735 lb of 4 × 6 mesh silica gel (SG) particles having an effective diameter of 0.26 cm and an external void fraction of 0.5. The adsorption isotherm for benzene has been experimentally determined for the conditions of interest and found to be linear over the concentration range of interest, as given by

$$q = Kc^* = 5,120 c^* \quad (1)$$

where

q = lb benzene adsorbed per ft³ of silica gel particles

c^* = equilibrium concentration of benzene in the gas, in lb benzene per ft³ of gas

Mass-transfer experiments, simulating the conditions of the 2-foot-diameter bed, have been carried out and fitted to a linear-driving-force (LDF) model:

$$\frac{\partial \bar{q}}{\partial t} = 0.206 K (c - c^*) \quad (2)$$

where time is in minutes. The constant $k = 0.206 \text{ min}^{-1}$ includes resistances both in the gas film and in the adsorbent pores, with the latter resistance dominant.

Using the approximate concentration-profile equations of Klinkenberg [77], compute a set of breakthrough curves and determine the time when the concentration of benzene in the exiting air rises to 5% of the inlet concentration. Assume isothermal and isobaric operation. Compare the breakthrough time with the time predicted by the equilibrium model.

SOLUTION

For the equilibrium model, the bed becomes completely saturated with benzene at the inlet concentration.

$$\text{MW of entering gas} = 0.009(78) + 0.991(29) = 29.44$$

$$\text{Density of entering gas} = (1)(29.44)/(0.730)(530) = 0.076/\text{lb/ft}^3$$

$$\text{Gas flow rate} = 23.6/0.0761 = 310 \text{ ft}^3/\text{min}$$

$$\begin{aligned} \text{Benzene flow rate in entering gas} &= \frac{(23.6)}{29.44}(0.009)(78) \\ &= 0.562 \text{ lb/min} \end{aligned}$$

or

$$c_F = \frac{0.562}{310} = 0.00181 \text{ lb benzene/ft}^3 \text{ of gas}$$

From (1),

$$q = 5,120(0.00181) = 9.27 \frac{\text{lb benzene}}{\text{ft}^3 \text{ SG}}$$

The total adsorption of benzene at equilibrium

$$= \frac{9.27(3.14)(2)^2(6)(0.5)}{4} = 87.3 \text{ lb}$$

$$\text{Time of operation} = \frac{87.3}{0.562} = 155 \text{ min}$$

For the actual operation, taking into account external and internal mass-transfer resistances, from (15-115) and (15-116),

$$\xi = \frac{(0.206)(5,120)z}{u} \left(\frac{1-0.5}{0.5} \right) = 1,055 z/u$$

$$u = \text{interstitial velocity} = \frac{310}{0.5 \left(\frac{3.14 \times 2^2}{4} \right)} = 197 \text{ ft/min} \quad (3)$$

$$\xi = \frac{1,055}{197} z = 5.36z$$

where z is in feet.

$$\text{When } z = \text{bed height} = 6 \text{ ft}, \xi = 32.2 \text{ and } \tau = 0.206 \left(t - \frac{z}{197} \right) \quad (4)$$

where t is in minutes. For $t = 155$ min (the ideal time), and $z = 6$ ft (the bed height), using (4), $\tau = 32$.

Thus, breakthrough curves should be computed from (15-114) for values of τ and ξ no greater than about 32. For example, when $\xi = 32.2$ (exit end of the bed), and $\tau = 30$, which corresponds to a time $t = 145.7$ minutes, the concentration of benzene in the exiting gas, from (15-114), is

$$\begin{aligned} \frac{c}{c_F} &= \frac{1}{2} \left[1 + \operatorname{erf} \left(30^{0.5} - 32.2^{0.5} + \frac{1}{8(30)^{0.5}} + \frac{1}{8(32.2)^{0.5}} \right) \right] \\ &= \frac{1}{2} [1 + \operatorname{erf}(-0.1524)] = \frac{1}{2} \operatorname{erfc}(0.1524) \\ &= 0.4147 \text{ or } 41.47\% \end{aligned}$$

This far exceeds the specification of $c/c_F = 0.05$ or 5% at the exit. Thus, the time of operation of the bed is considerably less than the

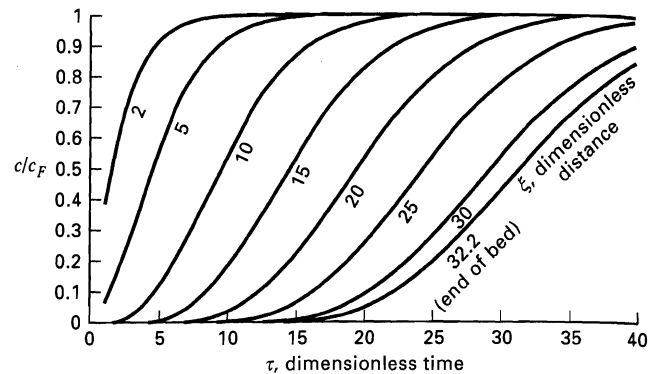


Figure 15.30 Gas concentration breakthrough curves for Example 15.11.

ideal time of 155 min. Figure 15.30 shows breakthrough curves computed from (15-114) over a range of the dimensionless time, τ , for values of the dimensionless distance, ξ , of 2, 5, 10, 15, 20, 25, 30, and 32.2, where the latter corresponds to the exit end of the bed. For $c/c_F = 0.05$ and $\xi = 32.2$, τ is seen to be about 20 (19.9 by calculation).

From (4), with $z = 6$ ft, the time to breakthrough is $t = \frac{20}{0.206} + \frac{6}{197} = 97.1$ min which is 62.3% of the ideal time.

Figure 15.30 or (15-114) can be used to compute the bulk concentration of benzene at various locations in the bed for $\tau = 20$. The results are as follows:

ξ	z , ft	c/c_F
2	0.373	1.00000
5	0.932	0.99948
10	1.863	0.97428
15	2.795	0.82446
20	3.727	0.53151
25	4.658	0.25091
30	5.590	0.08857
32.2	6.000	0.05158

We can also compute, at $\tau = 20$, the adsorbent loading, at various positions in the bed, from (15-119), using $q = 5,120c$. The maximum loading corresponds to c_F . Thus, $q_{\max} = 9.28$ lb benzene/ft³ of SG. Breakthrough curves for the solid loading are plotted in Figure 15.31. As expected, those curves are displaced to the right

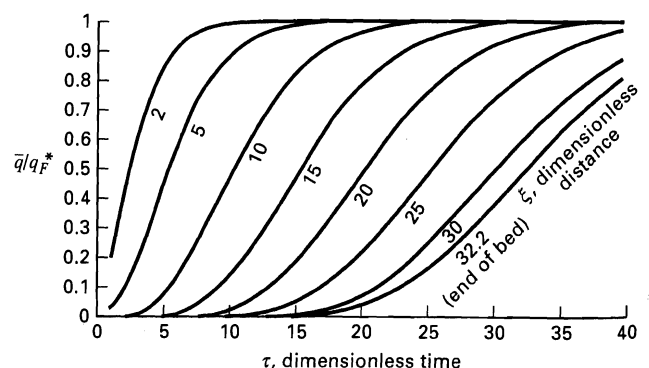


Figure 15.31 Adsorbent loading breakthrough curves for Example 15.11.

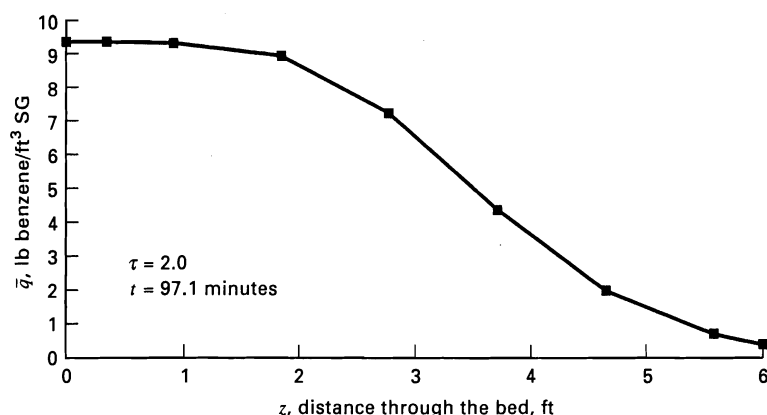


Figure 15.32 Adsorbent loading profile for Example 15.11.

from the curves of Figure 15.30. At $\tau = 20$:

ξ	z , ft	$\frac{c^*}{c_F} = \frac{\bar{q}}{q_F^*}$	$\bar{q}$, lb benzene/ft ³ SG
2	0.373	0.99998	9.28
5	0.932	0.99883	9.27
10	1.863	0.96054	8.91
15	2.795	0.77702	7.21
20	3.727	0.46849	4.35
25	4.658	0.20571	1.909
30	5.590	0.06769	0.628
32.2	6.000	0.03827	0.355

The values of $\bar{q}$ in the preceding table are plotted in Figure 15.32 and integrated over the 6-foot bed length to obtain the average bed loading:

$$\bar{q}_{\text{avg}} = \int_0^6 \bar{q} dz / 6$$

The result is 5.72 lb benzene/ft³ of SG, which is 61.6% of the maximum loading based on the inlet benzene concentration.

If the bed were increased in height by a factor of 5, to 30 ft, $\xi = 161$. The ideal time of operation would be 780 min or 13 h. With mass-transfer effects taken into account, as before, the dimensionless operating time to breakthrough is computed to be $\tau = 132$, or breakthrough time from (4) is

$$t = \frac{132}{0.206} + \frac{30}{197} = 641 \text{ min}$$

which is 82.2% of the ideal time. This represents a substantial increase in bed utilization.

Scale-up for Constant-Pattern Front

In Example 15.11, the wavefront (of the type shown in Figure 15.28a), broadens as it moves through the bed. This is shown in Figure 15.33, where MTZ, the width of the mass-transfer zone, is plotted against the dimensionless time, τ , up to the value of 20 where the front breaks through the 6-foot-long bed. The MTZ in Figure 15.33 is based on a range of c/c_F from 0.95 to 0.05. As seen, MTZ increases from about 2 feet at $\tau = 6$ to about 4 feet at $\tau = 20$. As shown, with

increasing τ , the rate of broadening slows. However, for a deeper bed, it is found that even at $\tau = 100$, the wavefront is still slowly broadening.

The continual broadening of the wavefront determined in Example 15.11 is typical of that obtained with a linear adsorption isotherm (curve A in Figure 15.29a). The wavefront also continues to broaden with an unfavorable isotherm (curve C in Figure 15.29a). But, when the isotherm is of the favorable Langmuir or Freundlich type (curve B in Figure 15.29a), wavefront broadening rapidly diminishes and an asymptotic or *constant-pattern front* (CPF) is developed. For such a front, MTZ becomes constant and curves of c/c_F and $\bar{q}/q^*$ become coincident.

The bed depth at which the CPF is approached depends upon the nonlinearity of the adsorption isotherm and the importance of adsorption kinetics. The mathematical proof of the existence of an asymptotic wavefront solution is given by Cooney and Lightfoot [80], including the case of axial dispersion. Initially, the wavefront broadens because of mass-transfer resistance and/or axial dispersion. Eventually, the opposite influence of a favorable isotherm, as shown in Figure 15.29b, comes into play and an asymptotic wavefront pattern is approached. For a constant-pattern front, Sircar and Kumar [81] present some analytical solutions and Cooney [82] presents a rapid approximate method, illustrated with the Freundlich and Langmuir isotherms, to estimate concentration profiles and breakthrough curves when mass-transfer and equilibrium parameters are available.

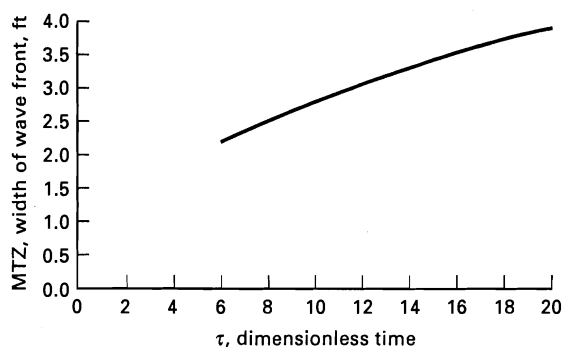


Figure 15.33 Broadening of wavefront in Example 15.11.

When the constant-pattern-front assumption is valid, it can be used to determine the length of a full-scale adsorbent bed from breakthrough curves obtained in small-scale laboratory experiments. This widely used technique is described by Collins [83] for purification applications. The adsorbent bed is considered to be the sum of two sections, analogous to those mentioned for ideal, fixed-bed adsorption. Thus, the total bed length is estimated to be the sum of the length, LES, of the ideal, fixed-bed adsorber plus an additional length, called the LUB, that depends on the observed width of the MTZ and the shape of the c/c_F profile within that zone. The total required bed length is

$$L_B = \text{LES} + \text{LUB} \quad (15-120)$$

For the ideal, fixed-bed adsorber, with $\text{MTZ} = 0$, LUB is not necessary, but if $L_B > \text{LES}$, then LUB is the length of unused bed. However, when an MTZ is present, then an LUB is necessary and is referred to as the equivalent length of unused bed. To determine LUB from an experimental breakthrough curve, for the same feed composition and superficial velocity to be used in the commercial adsorber, and for a CPF, the front is located such that in Figure 15.34, area A is equal to area B. Then:

$$\text{LUB} = \text{Ideal wavefront velocity} \times (t_s - t_b) = \frac{L_e}{t_s} (t_s - t_b) \quad (15-121)$$

where L_e is the length of the experimental bed. For the ideal case, a solute mass balance for a cylindrical bed of diameter D gives

$$c_F Q_F t = q_F \rho_b \pi \frac{D^2}{4} (\text{LES}) \quad (15-122)$$

where t is the time to breakthrough, from which the LES can be determined.

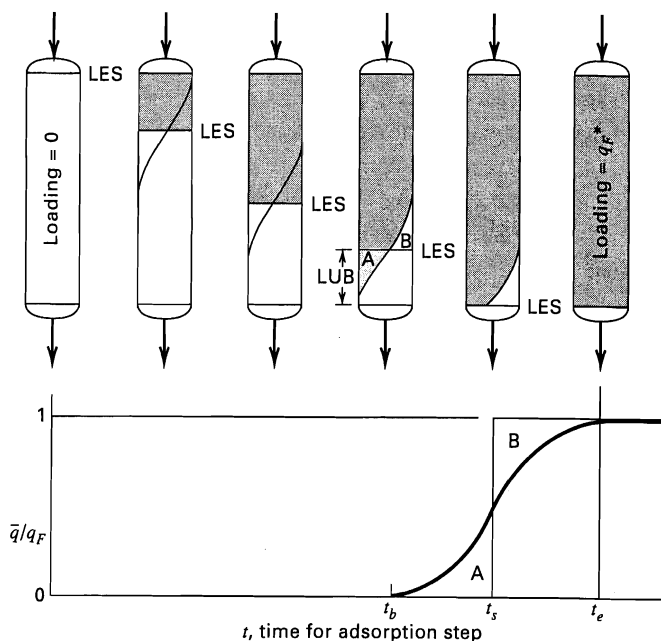


Figure 15.34 Determination of bed length from laboratory measurements.

Instead of positioning the stoichiometric front for equal areas in Figure 15.34, the LUB can be determined from the experimental breakthrough-curve data by computing t_s from

$$t_s = \int_0^{t_e} \left(1 - \frac{c}{c_F}\right) dt \quad (15-123)$$

If, in Figure 15.34, t_s is located midway between t_b and t_e , such that the shape of the experimental breakthrough curve below area B is equivalent to the curve above area A, then $\text{LUB} = \text{MTZ}/2$, i.e., one-half of the width of the mass-transfer zone. In the absence of experimental breakthrough data, a conservative estimate of MTZ is 4 ft.

EXAMPLE 15.12

Collins [83] presents the following experimental data for the adsorption of water vapor from nitrogen in a fixed bed of 4A molecular sieves:

Bed depth = 0.88 ft, temperature = 83°F (negligible temperature change), pressure = 86 psia (negligible pressure drop), G = entering gas molar velocity = 29.6 lbmol/h-ft², entering water content = 1,440 ppm (by volume), initial adsorbent loading = 1 lb/100 lb sieves, and bulk density of bed = 44.5 lb/ft³. For the entering gas moisture content, c_F , the equilibrium loading, q_F , = 0.186 lb H₂O/lb solid.

c_{exit} , ppm (by volume)	Time, h	c_{exit} , ppm (by volume)	Time, h
<1	0–9.0	650	10.8
1	9.0	808	11.0
4	9.2	980	11.25
9	9.4	1,115	11.5
33	9.6	1,235	11.75
80	9.8	1,330	12.0
142	10.0	1,410	12.5
238	10.2	1,440	12.8
365	10.4	1,440	13.0
498	10.6		

Determine the bed height required for a commercial unit to be operated at the same temperature, pressure, and entering gas mass velocity and water content to obtain an exiting gas with no more than 9 ppm (by volume) of water vapor with a breakthrough time of 20 h.

SOLUTION

$$c_F = \frac{1,440(18)}{106} = 0.02592 \text{ lb H}_2\text{O/lbmol N}_2$$

$$G = \frac{Q_F}{\pi D^2/4} = 29.6 \text{ lbmol N}_2/\text{h-ft}^2 \text{ of bed cross-section}$$

Initial moisture content of bed = 0.01 lb H₂O/lb solid

From (15-122),

$$\text{LES} = \frac{(0.02592)(29.6)(20)}{(0.186 - 0.01)(44.5)} = 1.96 \text{ ft}$$

Use the integration method to obtain LUB. From the data:

Take $t_e = 12.8 \text{ h}$ (1,440 ppm) and $t_b = 9.4 \text{ h}$ (9 ppm)

By numerical integration of the breakthrough-curve data, using (15-123): $t_s = 10.93$ h

From (15-121),

$$LUB = \left(\frac{10.93 - 9.40}{10.93} \right) (0.88) = 0.12 \text{ ft}$$

From (15-120):

$L_B = 1.96 + 0.12 = 2.08$ ft or a bed utilization of $\frac{1.96}{2.08} \times 100\% = 94.2\%$.

Alternatively, the following approximate calculation can be made. Let t_b , the beginning of breakthrough, be 5% of the final ppm or $0.05(1440) = 72$ ppm. Using the experimental data, this corresponds to $t_b = 9.76$ h. Let t_e , the end of breakthrough, be 95% of the final ppm or $0.95(1440) = 1370$ ppm, corresponding to $t_e = 12.25$ h. Let t_s = the midpoint or $(9.76 + 12.25)/2 = 11$ h. The ideal wave-front velocity $= L_e/t_s = 0.88/11 = 0.08$ ft/h. From (15-121), $LUB = 0.08(11 - 9.76) = 0.1$ ft. That is, the $MTZ = 0.2$ ft and $L_B = 1.96 + 0.1 = 2.06$ ft.

Thermal-Swing Adsorption

Thermal (temperature)-swing adsorption (TSA), in its simplest configuration, is carried out with two fixed beds in parallel, operating cyclically, as in Figure 15.20b. While one bed is adsorbing solute at near-ambient temperature, $T_1 = T_{\text{ads}}$, the other bed is regenerated by desorbing adsorbate at a higher temperature, $T_2 = T_{\text{des}}$, at which the equilibrium adsorbate loading is much less for a given concentration of solute in the fluid, as illustrated in Figure 15.21. Although the desorption step might be accomplished in the absence of a purge fluid by simply vaporizing the adsorbate, readsorption of some solute vapor would occur upon cooling the bed. Thus, it is best to remove the desorbed adsorbate with a purge. The desorption temperature is high, but not so high as to cause deterioration of the adsorbent. TSA is best applied to the removal of contaminants present at low concentrations in the feed fluid. In that case, nearly isothermal adsorption and desorption is achieved. An ideal cycle involves four steps:

(1) adsorption at T_1 to breakthrough, (2) heating of the bed to T_2 , (3) desorption at T_2 to a low adsorbate loading, and (4) cooling of the bed to T_1 . Practical cycles do not operate with isothermal steps. Instead, Steps 2 and 3 are combined for the regeneration part of the cycle, with the bed being simultaneously heated and desorbed with preheated purge gas until the temperature of the effluent approaches that of the inlet purge. Steps 1 and 4 may also be combined because, as discussed in detail by Ruthven [10], the thermal wave precedes the MTZ front. Thus, adsorption takes place at essentially the feed-fluid temperature.

The heating and cooling steps cannot be accomplished instantaneously because of the relatively low bed thermal conductivity. Although heat transfer can be done indirectly from jackets surrounding the beds or from coils located within the beds, bed temperature changes are more readily achieved by preheating or precooling a purge fluid, as shown in Figure 15.35. The purge fluid can be a portion of the feed or effluent, or some other fluid. The purge fluid can also be used in the desorption step. When the adsorbate is valuable and easily condensed, the purge fluid might be a noncondensable gas. When the adsorbate is valuable, but not easily condensed, and is essentially insoluble in water, steam may be used as the purge fluid, followed by condensation of the steam to separate it from the desorbed adsorbate. When the adsorbate is not valuable, fuel and/or air can be used as the purge fluid, followed by incineration. Often the amount of purge used in the regeneration step is much less than the amount of feed sent to the bed in the adsorption step. In Figure 15.35, the feed fluid is a gas. The spent bed is heated and regenerated with preheated feed gas, which is then cooled to condense the desorbed adsorbate.

Because of the time to heat and cool a fixed bed, cycle times for TSA are long, usually extending over periods of hours or days. The longer the cycle time, the longer the required bed length, and the greater the percent utilization of the bed during adsorption. However, for a given cycle time,

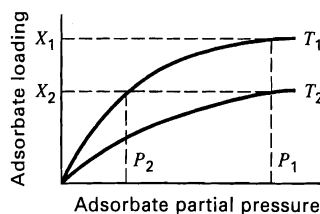
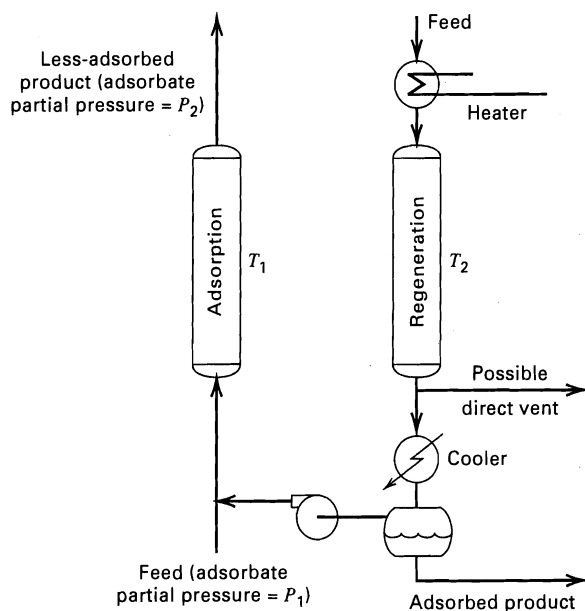


Figure 15.35 Temperature-swing adsorption cycle.

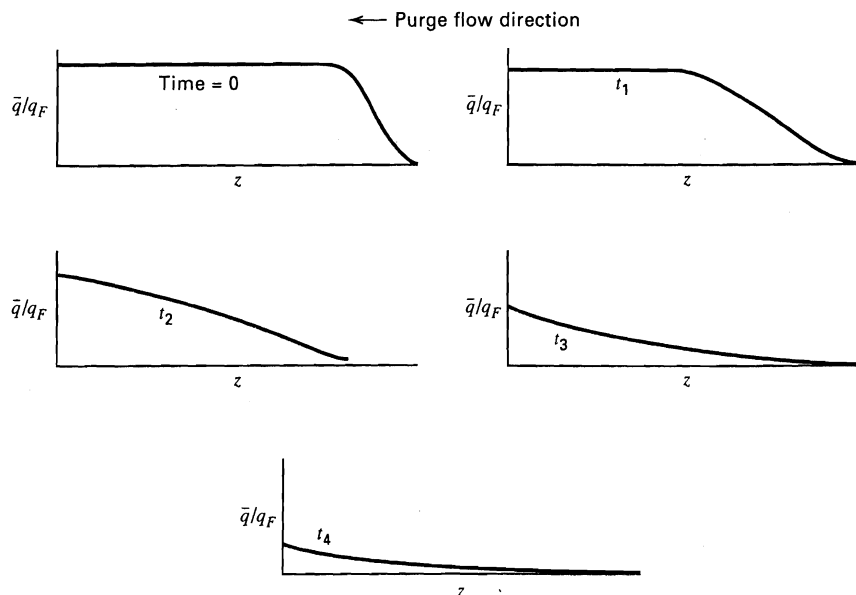


Figure 15.36 Sequence of loading profiles during countercurrent regeneration.

when the width of the MTZ is an appreciable fraction of the bed height, such that the capacity of the bed is poorly utilized, consideration should be given to a *lead-trim-bed* arrangement of two beds in series for the adsorption step. When the lead bed is spent, it is switched to regeneration. At this point in time, the trim bed has an MTZ occupying a considerable portion of the bed, and that bed becomes the lead bed, with a regenerated bed becoming the trim bed. In this manner only a fully spent bed is switched to regeneration. Thus, a total of three beds is used. If the flow rate of the feed stream is very high, beds in parallel may be required for both adsorption and desorption.

The adsorption step is usually conducted with the feed fluid flowing downward through the bed. The flow direction for desorption can be either downward or upward, but the upward, countercurrent direction is usually preferred because it is more efficient. Consider the sequence of loading fronts shown in Figure 15.36, for regeneration countercurrent to adsorption. Although the bed is shown in a horizontal position, it must be positioned vertically. The feed fluid flows downward, entering at the left and leaving at the right. At time $t = 0$, breakthrough has occurred, with a loading profile as shown at the top, where the MTZ is seen to be about 25% of the bed. If the purge fluid for regeneration also flows downward (entering at the left), all of the adsorbate will have to move through the unused portion of the bed. Thus, some

of the desorbed adsorbate will be readsorbed in the unused section and then desorbed a second time. If countercurrent regeneration is used, the unused portion of the bed is never contacted with desorbed adsorbate. During a countercurrent regeneration step, the loading profile changes progressively, as shown for a series of times in Figure 15.36. The right-side end of the bed, where the purge enters, is desorbed first. At the end of regeneration, the residual loading may be uniformly zero or, more likely, finite and nonuniform as shown at the bottom of Figure 15.36. If the latter, then the useful cyclic capacity, called the *delta loading*, is as shown in Figure 15.37.

Calculations of the concentration and loading profiles during desorption are only approximated by (15-114) and (15-119) because the loading is not uniform at the beginning of desorption. A numerical solution for the desorption step can be obtained in the following fashion using a procedure discussed by Wong and Niedzwiecki [84]. Although their method was developed for the adsorption step, it is readily applied to desorption. In the absence of axial dispersion and for constant fluid velocity, (15-102) and (15-105) are rewritten as

$$u \frac{\partial \phi}{\partial z} + \frac{\partial \phi}{\partial t} + \left(\frac{1 - \epsilon_b}{\epsilon_b} \right) kK(\phi - \psi) = 0 \quad (15-123)$$

$$\frac{\partial \psi}{\partial t} = k(\phi - \psi) \quad (15-124)$$

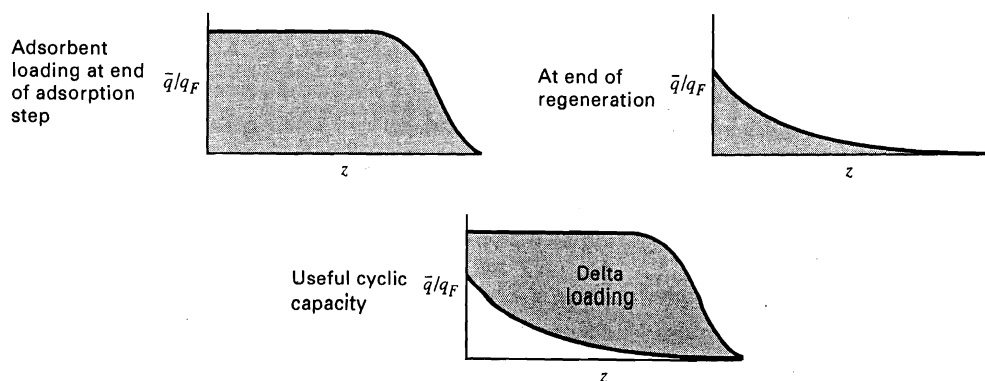


Figure 15.37 Delta loading for regeneration step.

where

$$\phi = c/c_F \quad (15-125)$$

$$\psi = \bar{q}/q_F^* \quad (15-126)$$

and c_F and q_F^* are the values at the beginning of the adsorption step. The boundary conditions are as follows:

At $t = 0$: $\phi = \phi\{z\}$ at the end of the adsorption step

$\psi = \psi\{z\}$ at the end of the adsorption step

where, for countercurrent desorption, it is best to let z start from the bottom of the bed (called z') and increase in the direction of purge-gas flow. Thus, u in (15-123) is positive.

At $z' = 0$: $\phi = 0$ (no solute in the entering purge gas)

$\psi = 0$

Partial differential equations (15-123) and (15-124) in independent variables z and t can be converted to a set of ordinary differential equations (ODEs) in independent variable t by the method of lines (MOL), which was first applied to parabolic PDEs by Rothe in 1930, as discussed by Liskovets [85], and subsequently to elliptic and hyperbolic PDEs. The MOL is developed in detail by Schiesser [86]. The lines refer to the z' -locations of the ODEs. To obtain the set of ODEs, the z' -coordinate is divided into N increments or $N + 1$ grid points that are usually evenly spaced. For many problems, 20 increments are sufficient. Letting i be the index for each grid point in z' , starting from the end where the purge gas enters, and discretizing $\partial\phi/\partial z'$, (15-123) and (15-124) become

$$\frac{d\phi_i}{dt} = -u \left(\frac{\Delta\phi}{\Delta z'} \right)_i - \left(\frac{1 - \epsilon_b}{\epsilon_b} \right) kK(\phi_i - \psi_i), \quad i = 1, N + 1 \quad (15-127)$$

$$\frac{d\psi_i}{dt} = k(\phi_i - \psi_i), \quad i = 1, N + 1 \quad (15-128)$$

where the initial conditions ($t = 0$) for ϕ_i and ψ_i are as given above. Before we can integrate (15-127) and (15-128), we must provide a suitable approximation for $(\Delta\phi/\Delta z')_i$. In general, for a moving-front problem of the hyperbolic type here for adsorption and desorption, the simple central difference

$$\left(\frac{\Delta\phi}{\Delta z'} \right)_i \approx \frac{\phi_{i+1} - \phi_{i-1}}{2\Delta z'}$$

is not adequate. Instead, Wong and Niedzwiecki [84] found that a five-point, biased, upwind, finite-difference approximation, discussed by Schiesser [87], is very effective. This approximation, which is derived from a Taylor's series analysis, places emphasis on conditions upwind of the moving front. At an interior grid point:

$$\left(\frac{\Delta\phi}{\Delta z'} \right)_i \approx \frac{1}{12\Delta z'} [-\phi_{i-3} + 6\phi_{i-2} - 18\phi_{i-1} + 10\phi_i + 3\phi_{i+1}] \quad (15-129)$$

Note that the coefficients of the ϕ -factors, inside the square brackets, sum to 0. At the last grid point, $N + 1$, where the

purge gas exits, (15-129) is replaced by

$$\left(\frac{\Delta\phi}{\Delta z'} \right)_{N+1} \approx \frac{1}{12\Delta z'} [3\phi_{N-3} - 16\phi_{N-2} + 36\phi_{N-1} - 48\phi_N + 25\phi_{N+1}] \quad (15-130)$$

For the first three node points, the following approximations replace (15-129):

$$\left(\frac{\Delta\phi}{\Delta z'} \right)_1 \approx \frac{1}{12\Delta z'} [-25\phi_1 + 48\phi_2 - 36\phi_3 + 16\phi_4 - 3\phi_5] \quad (15-131)$$

$$\left(\frac{\Delta\phi}{\Delta z'} \right)_2 \approx \frac{1}{12\Delta z'} [-3\phi_1 - 10\phi_2 + 18\phi_3 - 6\phi_4 + \phi_5] \quad (15-132)$$

$$\left(\frac{\Delta\phi}{\Delta z'} \right)_3 \approx \frac{1}{12\Delta z'} [\phi_1 - 8\phi_2 + 0\phi_3 + 8\phi_4 - \phi_5] \quad (15-133)$$

However, because values of ϕ_1 (at $z' = 1$) are given as a boundary condition, (15-131) is not needed.

Equations (15-127) to (15-133) with boundary conditions for ϕ_1 and ψ_1 constitute a set of $2N$ ODEs as an initial-value problem, with time as the independent variable. However, the values of ϕ_i and ψ_i at the different axial locations can change with time at vastly different rates. For example, in Figure 15.36 for desorption fronts, if we divide the bed length, L , into 20 equal-width increments, starting from the right-hand side where the purge gas enters, we see that initially ψ_{21} , where the purge gas exits, is not changing at all, while ψ_5 is changing rapidly. Near the end of the desorption step, ψ_{21} is changing rapidly, while ψ_5 is not. The same observations hold for ϕ_i . This type of response is referred to as *stiffness*, as described by Schiesser [87] and in *Numerical Recipes* by Press et al. [88]. If we attempt to integrate the set of ODEs with simple Euler or Runge-Kutta methods, not only do we encounter truncation error, but, with time, the computed values of ϕ_i and ψ_i go through enormous instability, characterized by wild swings between large and impossible positive and negative values. Even if the length is divided into many more than 20 increments and very small time steps are used, instability is still often encountered.

The integration of a stiff set of ODEs is most efficiently carried out by variable-order/variable-step-size implicit methods of the type first developed by Gear [89]. These methods are included in a widely available software package called ODEPACK, described by Byrne and Hindmarsh [90]. The subject of stiffness is also discussed in Chapter 13.

EXAMPLE 15.13

In Example 15.11, benzene is adsorbed from air at 70°F and 1 atm onto silica gel in a fixed-bed adsorber, 6 ft in length. Breakthrough occurs at close to 97.1 min for $\phi = 0.05$. At that time, values of $\phi = c/c_F$ and $\psi = \bar{q}/q_F^*$ in the bed are distributed as follows, where z' is measured backwards from the exit of the bed for the adsorption step. These results were obtained by the numerical

method just described, as applied to the adsorption step, and are in close agreement with the approximate, analytical Klinkenberg solution given in Example 15.11.

$z', \text{ ft}$	$\phi = c/c_F$	$\psi = \bar{q}/q_F^*$
0	0.05227	0.03891
0.3	0.07785	0.05913
0.6	0.11314	0.08776
0.9	0.16008	0.12690
1.2	0.22017	0.17850
1.5	0.29394	0.24387
1.8	0.38042	0.32310
2.1	0.47678	0.41459
2.4	0.57825	0.51469
2.7	0.67861	0.61786
3.0	0.77108	0.71728
3.3	0.84969	0.80603
3.6	0.91057	0.87858
3.9	0.95281	0.93207
4.2	0.97848	0.96690
4.5	0.99172	0.98636
4.8	0.99731	0.99531
5.1	0.99921	0.99857
5.4	0.99987	0.99960
5.7	1.00000	1.00000
6.0	1.00000	1.00000

If the bed is regenerated isothermally with pure air at 1 atm and 145°F, and the desorption of benzene during the heat-up period is neglected, determine the loading, $\bar{q}$, profile at times of 15, 30, and 60 min for pure stripping air interstitial velocities of: (a) 197 ft/min, and (b) 98.5 ft/min. At 145°F and 1 atm, the adsorption isotherm, in the same units as in Example 15.11, is

$$q = 1,000c^* \quad (1)$$

giving an equilibrium loading of about 20% of that at 70°F. Assume that k is unchanged from the value of 0.206 in Example 15.11.

SOLUTION

This problem is solved by the MOL with 20 increments in z' , using the subroutine LSODE in ODEPACK to integrate the set of ODEs. The user supplies the FORTRAN MAIN program and the subroutine FEX, shown in Table 15.9, for the derivative functions given by (15-127) to (15-130) and (15-132) to (15-133). The program LSODE includes detailed instructions for writing these two routines. Note that the program in Table 15.9 actually includes both the adsorption and desorption steps for desorption conditions of 30 min at 197 ft/min.

The computed loading profiles for all conditions are plotted in Figures 15.38a and b, for desorption interstitial velocities of 197 and 98.5 ft/min, respectively, where z is the distance from the feed gas inlet end of the bed for the adsorption step. The curves are similar to those shown in Figure 15.30. For the 197 ft/min case, desorption is almost complete at 60 min with less than 1% of the bed still loaded with benzene. If this velocity were used, this would allow $97.1 - 60 = 37.1$ min for heating and cooling the bed before and

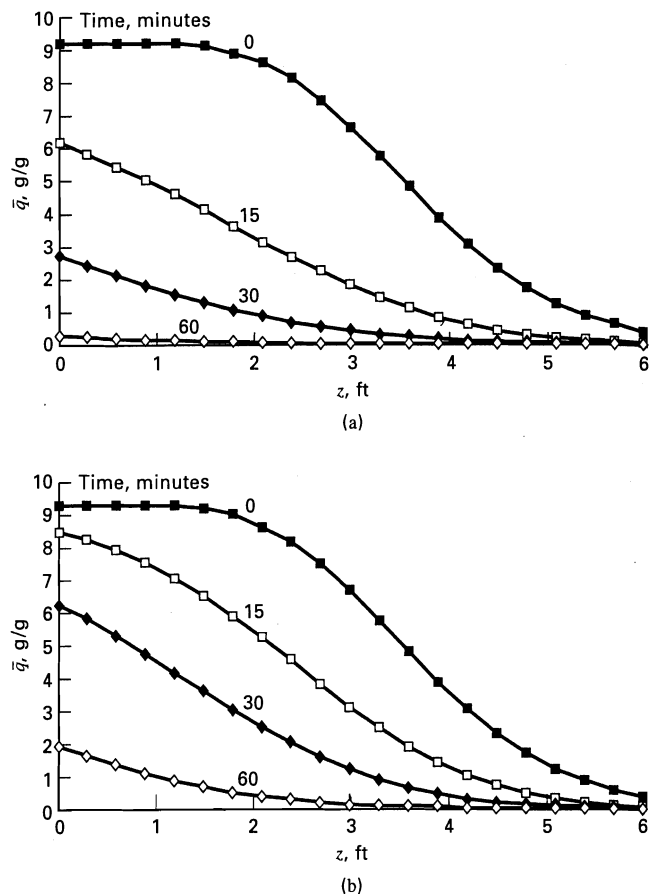


Figure 15.38 Regeneration loading profiles for Example 15.13.

- (a) Regeneration air interstitial velocity = 197 ft/min.
(b) Regeneration air interstitial velocity = 98.5 ft/min.

after desorption. For the 98.5 ft/min case at 60 min, about 5% of the bed is still loaded with benzene. This may be acceptable, but the resulting adsorption step would take a little longer because initially the bed would not be clean. Several cycles are required to establish a cyclic steady state, whose development is considered in the next section on pressure-swing adsorption.

Pressure-Swing Adsorption

Pressure-swing adsorption (PSA) and vacuum-swing adsorption (VSA), in their simplest configurations, are carried out with two fixed beds in parallel, operating in a cycle, as in Figure 15.39. Unlike TSA, where thermal means is used to effect the separation, PSA and VSA use mechanical work to increase the pressure or create a vacuum. While one bed is adsorbing at one pressure, the other bed is desorbing at a lower pressure, as was illustrated in Figure 15.21. Unlike TSA, which can be used to purify gases or liquids, PSA and VSA are used only with gases, because a change in pressure has little or no effect on the equilibrium loading for liquid adsorption. PSA was originally used only for purification, as in the removal of moisture from air by the "heatless drier," which was invented by C.W. Skarstrom in 1960 to compete with TSA. However, by the early 1970s, PSA was being applied to bulk separations such as the partial separation of air

Table 15.9 FORTRAN Computer Program for Example 15.13

```

PROGRAM tsa
IMPLICIT DOUBLE PRECISION(A-H, O-Z)
EXTERNAL FEX
DIMENSION C(40),ATOL(60),RWORK(4162),IWORK(90),CH(40),DL(40)
COMMON CF,VEL,AK,A(20)
open (unit=3,file='n1.out')
write(3,*)'desorption velocity=197ft/min, desorption time=30min'
NEQ=60
CF0=0.00181
CF1=0.0
TCA=97.1
NUMCYCLE=1
MXSTEP=2000
DO 55 I=1,20
55 C(I)=0.0

DO 56 I=21,40
56 C(I)=0.0
CONTINUE

T=0.D0
TOUT=0.0
ITOL=2
RTOL=1.D-6
DO 57 I=1, 60
57 ATOL(I)=1.0 D-12
CONTINUE

ITASK=1
ISTATE=1
IOPT=1
IWORK(6)=2000
LRW=4162
LIW=90
MF=22

CF=CF0
C0=CF
AK=5120.
Q0=AK*C0

1 CALL LSODE(FEX,NEQ,C,T,TOUT,ITOL,RTOL,ATOL,ITASK,ISTATE,
IOPT,RWORK,LRW,IWORK,LIW,JEX,MF)
WRITE(3,*)'CONDITIONS AT THE BEGINNING '
WRITE(3,*)'TIME(SEC)=',TOUT
WRITE(3,*)'CONC. GAS PHASE'
WRITE(3,*)C0, (C(I),I=1,20)
WRITE(3,*)'LOADING gm/gm'
WRITE(3,*)Q0, (C(I),I=21,40)
write(3,128)
128 format(///)
C0DL=1.0
Q0DL=1.0
DO 989 I=1,20
DL(I)=C(I)/C0
DL(I+20)=C(I+20)/Q0
989 CONTINUE
WRITE(3,*)'DIMENSIONLESS CONDITIONS AT THE BEGINNING'
WRITE(3,*)'TIME(SEC)=',TOUT
WRITE(3,*)'DIMENSIONLESS GAS CONCENTRATION C/CF'
WRITE(3,*)C0DL, (DL(I),I=1,20)
WRITE(3,*)'DIMENSIONLESS LOADING Q/Q0'
WRITE(3,*)Q0DL, (DL(I),I=21,40)
write(3,129)

```

(continued)

Table 15.9 (Continued)

```

129      format(/////////)
        DO 1000 KK=1,NUMCYCLE
C-----
C-----ADSORPTION STEP-----
C-----
        T=0.0

        CF=CF0
        C0=CF
        AK=5120.
        Q0=AK*C0

        VEL=197.0
        ISTATE=1
        TOUT= 97.1

        CALL LSODE(FEX,NEQ,C,T,TOUT,ITOL,RTOL,ATOL,ITASK,ISTATE,
1         IOPT,RWORK,LRW,IWORK,LIW,JEX,MF)
        IF(KK.EQ.1)GOTO18
        IF((KK/25)*25.NE.KK)GOTO81
18      WRITE(3,*) 'CONDITIONS AT THE END OF ADSORPTION STEP'
        WRITE(3,*) 'STEP TIME(SEC)=',TOUT
        WRITE(3,*) 'CONC. OF GAS PHASE'
        WRITE(3,*)C0,(C(I),I=1,20)
        WRITE(3,*) 'LOADING gm/gm'
        WRITE(3,*)Q0,(C(I),I=21,40)
        WRITE(3,741)
741     FORMAT(///)
        C0DL=1.0
        Q0DL=1.0
        DO 990 I=1,20
        DL(I)=C(I)/C0
        DL(I+20)=C(I+20)/Q0
990     CONTINUE
        WRITE(3,*) 'DIMENSIONLESS CONDITIONS AT THE END OF ADSORPTION'
        WRITE(3,*) 'STEP TIME(SEC)=',TOUT
        WRITE(3,*) 'DIMENSIONLESS GAS CONCENTRATION C/CF'
        WRITE(3,*)C0DL,(DL(I),I=1,20)
        WRITE(3,*) 'DIMENSIONLESS LOADING Q/Q0'
        WRITE(3,*)Q0DL,(DL(I),I=21,40)
        WRITE(3,238)
238     FORMAT(/////////)
C-----
C-----DESORPTION BY TEMPERATURE SWING-----
C-----
81      T=0.0
        VEL=197.0
        TOUT=30.0
        ISTATE=1

        CF=CF1
        C0=CF
        AK=1000.0
        Q0=AK*C0

        DO 91 I=1,40
        CH(I)=C(I)
91      CONTINUE

        DO 92 I=1,19
        C(I)=CH(20-I)
92      CONTINUE
        C(20)=CF0

        DO 95 I=1,19

```

Table 15.9 (Continued)

```

95      C(20+I)=CH(40-I)
        C(40)=CF0*5120.
        CALL LODE (FEX,NEQ,C,T,TOUT,ITOL,RTOL,ATOL,ITASK,
1       ISTATE,IOPTR,WORK,LRW,IWORK,LIW,JEX,MF)

        DO 93 I=1,40
          CH(I)=C(I)
93      CONTINUE

        DO 94 I=1,19
          C(I)=CH(20-I)
          C(20+I)=CH(40-I)
94      CONTINUE
          C0=C(20)
          Q0=C(40)
          C(20)=CF1
          C(40)=1000.*CF1
          IF(KK.EQ.1)GOTO38
          IF((KK/25)*25.NE.KK)GOTO1000
38      WRITE(3,*)'CONDITIONS AT THE END OF DESORPTION '
          WRITE(3,*)'STEP TIME(SEC)=',TOUT
          WRITE(3,*)'CONC. OF GAS PHASE '
          WRITE(3,*)C0,(C(I),I=1,20)
          WRITE(3,*)'LOADING gm/gm'
          WRITE(3,*)Q0,(C(I),I=21,40)
          WRITE(3,264)
264     FORMAT (///)
          C0DL=C0/CF0
          Q0DL=Q0/(CF0*5120.)
          DO 991 I=1,20
            DL(I)=C(I)/CF0
            DL(I+20)=C(I+20)/(CF0*5120.)
991     CONTINUE
          WRITE(3,*)'DIMENSIONLESS CONDITIONS AT THE END OF DESORPTION'
          WRITE(3,*)'STEP TIME(SEC)=',TOUT
          WRITE(3,*)'DIMENSIONLESS GAS CONCENTRATION C/CF'
          WRITE(3,*)C0DL,(DL(I),I=1,20)
          WRITE(3,*)'DIMENSIONLESS LOADING Q/Q0'
          WRITE(3,*)Q0DL,(DL(I),I=21,40)
          WRITE(3,365)
365     FORMAT(/////////)
1000    CONTINUE

C-----
C-----
C-----
        WRITE(3,60)IWORK(11),IWORK(12),IWORK(13)
60      FORMAT(/12H NO. STEPS =,I4,11H NO. F-S =,I4,11H NO. J-S =,I4)
        STOP
80      WRITE(3,90)ISTATE
90      FORMAT(///22H ERROR HALT.. ISTATE =,I3)
        close(unit=3)
        STOP
        END

C-----
        SUBROUTINE FEX (NEQ,T,C,CDOT)
        IMPLICIT DOUBLE PRECISION(A-H,O-Z)
        DIMENSION C(40), CDOT(40)
        COMMON CF,VEL,AK,A(20)
        E=0.5
        C0=CF
        Q0=AK*C0
        DZ=6.0/20.0      !FT

```

(continued)

Table 15.9 (Continued)

	AA=-VEL
	BB=-(1.0-E)/E
	R4FDX=1./(12.*DZ)
	A(1)=R4FDX*
1	(-3.*C0-10.*C(1)+18.*C(2)-6.*C(3)+1.*C(4))
	A(2)=R4FDX*
1	(1.*C0-8.*C(1)+0.*C(2)+8.*C(3)-1.*C(4))
	A(3)=R4FDX*
1	(-1.*C0+6.*C(1)-18.*C(2)+10.*C(3)+3.0*C(4))
	DO 455 I=4,19
	A(I)=R4FDX*
1	(-1.*C(I-3)+6.*C(I-2)-18.*C(I-1)+10.*C(I)+3.*C(I+1))
455	CONTINUE
	A(20)=R4FDX*
1	(3.*C(16)-16.*C(17)+36.*C(18)-48.*C(19)+25.*C(20))
	DO 676 I=1,20
	CDOT(20+I)=0.206*(AK*C(I)-C(20+I))
	CDOT(I)=AA*A(I)+BB*CDOT(20+I)
676	CONTINUE
	RETURN
	END

to produce either nitrogen or oxygen and to the removal of impurities and pollutants from other gas streams. PSA can also be used for vapor recovery, as discussed and illustrated by Ritter and Yang [91].

A typical sequence of steps in the Skarstrom cycle, operating with two beds, is shown in Figure 15.40. Each bed operates alternately in two half-cycles of equal duration: (1) pressurization followed by adsorption, and (2) depressurization (blowdown) followed by a purge. The feed gas is used for pressurization, while a portion of the effluent product gas is used for purge. Thus, in Figure 15.40, while adsorption is taking place in bed 1, part of the gas leaving bed 1

is routed to bed 2 to purge that bed in a direction countercurrent to the direction of flow of the feed gas during the adsorption step. When moisture is to be removed from air, the dry-air product is produced during the adsorption step in each of the two beds. In Figure 15.40, the adsorption and purge steps represent less than 50% of the total cycle time. In many commercial applications of PSA, these two steps consume a much greater fraction of the cycle time because pressurization and blowdown can be completed rapidly. Therefore, cycle times for PSA and VSA are short, typically seconds to minutes. Thus, small beds have relatively large throughputs. With the valving shown in Figure 15.39, the

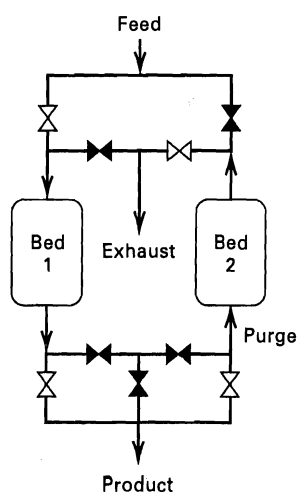


Figure 15.39 Pressure-swing-adsorption cycle.

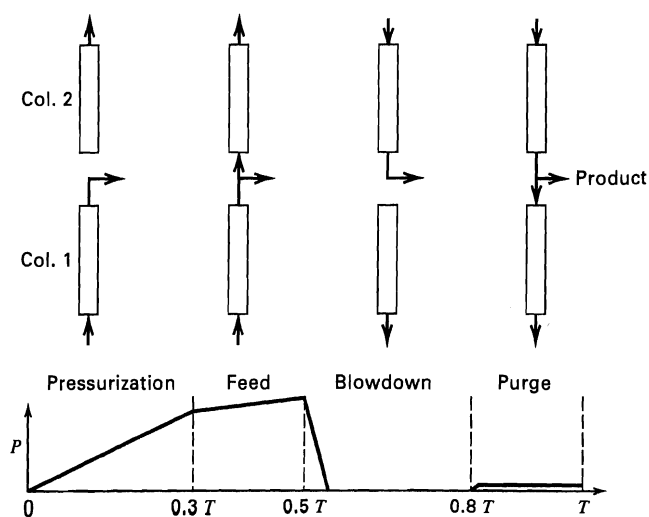


Figure 15.40 Sequence of cycle steps in PSA.

entire cyclic sequence can be programmed to operate automatically. With some valves open and others closed, as in Figure 15.39, adsorption takes place in Bed 1 and purge takes place in Bed 2. During the second half of the cycle, the valve openings and beds are switched.

Since the introduction of the Skarstrom cycle, numerous improvements have been made to increase product purity, product recovery, adsorbent productivity, and energy efficiency, as discussed by Yang [25] and by Ruthven, Farooq, and Knaebel [92]. Among these modifications are the use of (1) three, four, or more beds; (2) a pressure-equalization step in which both beds are equalized in pressure following purge of one bed and adsorption in the other; (3) pretreatment or guard beds to remove strongly adsorbed components that might interfere with the separation of other components; (4) purge with a strongly adsorbing gas; and (5) the use of an extremely short cycle time to approach isothermal operation, if a longer cycle causes an undesirable increase in temperature during adsorption and an undesirable decrease in temperature during desorption.

Separations by PSA and VSA are controlled by adsorption equilibrium or adsorption kinetics, where the latter refers to mass transfer external and/or internal to the adsorbent particle. Both types of control are important commercially. For the separation of air with zeolites, adsorption equilibrium is the controlling factor, with nitrogen more strongly adsorbed than oxygen and argon. For air with 21% oxygen and 1% argon, oxygen of about 96% purity can be produced. When carbon molecular sieves are used, oxygen and nitrogen have almost the same adsorption isotherms, but the effective diffusivity of oxygen is much larger than that of nitrogen. Consequently, a nitrogen product of very high purity (>99%) can be produced.

PSA and VSA cycles have been modeled successfully for both equilibrium and kinetic-controlled cases. The models and computational procedures are similar to those used for

TSA. The models are particularly useful for optimizing cycles. Of particular importance in PSA and TSA is the determination of the cyclic steady state. In TSA, following the desorption step, the regenerated bed is usually clean. Thus, a cyclic steady state is closely approached in one cycle. In PSA and VSA, this is often not the case, and complete regeneration is seldom achieved or necessary. It is only required to attain a cyclic steady state whereby the product obtained during the adsorption step has the desired purity and at cyclic steady state, the difference between the loading profiles after adsorption and desorption is equal to the solute entering in the feed. Starting with a clean bed, the attainment of a cyclic steady state for a fixed cycle time may require tens or hundreds of cycles. Consider an example from a study by Mutasim and Bowen [93] on the removal of ethane and carbon dioxide from nitrogen with 5A zeolite, at ambient temperature with adsorption and desorption for 3 min each at 4 bar and 1 bar, respectively, in beds 0.25 m in length. Figures 15.41a and b show the computed development of the loading and gas concentration profiles at the end of each adsorption step for ethane, starting from a clean bed. At the end of the first cycle, the bed is still clean beyond about 0.11 m. By the end of the 10th cycle, a cyclic steady state has almost been attained, with a clean bed existing only near the very end of the bed. Experimental data points for ethane loading at the end of 10 cycles agree reasonably well with the computed profile from a mathematical model.

Modeling of PSA and VSA cycles is carried out with the same equations as for TSA. However, the assumptions of negligible axial diffusion and isothermal operation may be relaxed. For each cycle, the pressurization and blowdown steps are often ignored and the initial conditions for adsorption and desorption are the final conditions for the desorption and adsorption steps, respectively, of the previous cycle. This is illustrated in the following example. Calculations can also be made with Aspen Adsorb of the Aspen Engineering Suite.

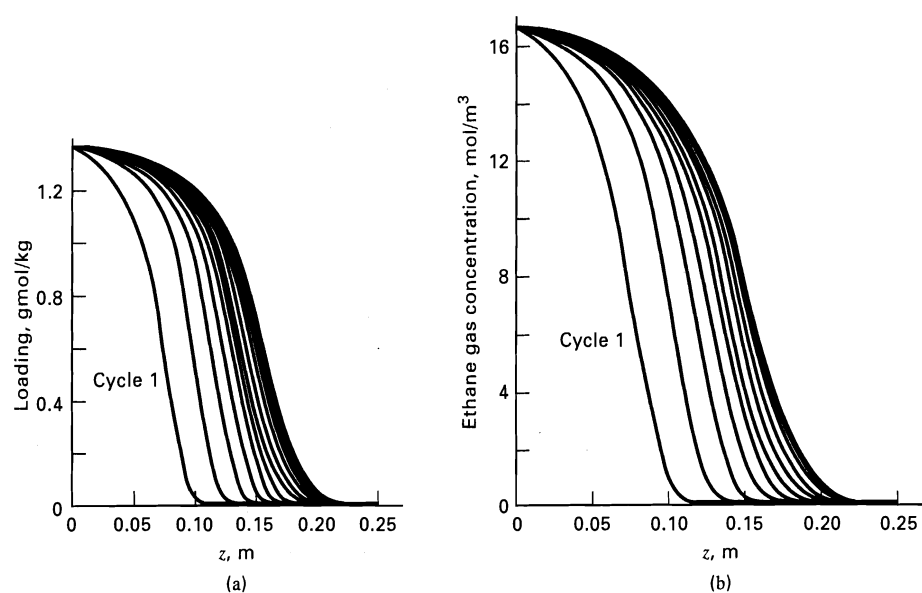


Figure 15.41 Development of cyclic steady-state profiles. (a) Loading profiles for first 11 cycles. (b) Ethane gas concentration profiles for first 16 cycles.

EXAMPLE 15.14

Ritter and Yang [91] conducted an experimental and theoretical study of the use of PSA to recover dimethyl methylphosphonate (DMMP) vapor from air. For the following data and operating conditions, starting with a clean bed, use the method of lines with a stiff integrator to estimate the concentration and loading profiles for the beds, the percent of the feed gas recovered as essentially pure air, and the average mole fraction of DMMP in the effluent gas leaving the desorption step during the third cycle.

Feed-Gas Conditions:

236 ppm by volume of DMMP in dry air at 294 K and 3.06 atm

Adsorbent:

BPL activated carbon, 5.25 g in each bed, 0.07 cm average particle diameter, and 0.43 bed porosity.

Bed dimensions: 1.1 cm i.d. by 12.8 cm each

$$\text{Langmuir adsorption isotherm: } q = \frac{48,360 p_{\text{DMMP}}}{1 + 98,700 p_{\text{DMMP}}},$$

where q is in g/g and p is in atm.

Overall mass-transfer coefficient: $k = 5 \times 10^{-3} \text{ s}^{-1}$

Cycle conditions (all at 294 K):

1. Pressurization with pure air from p_L to p_H in negligible time.
2. Adsorption at $p_H = 3.06 \text{ atm}$ with feed gas for 20 minutes.
 $u = \text{interstitial velocity} = 10.465 \text{ cm/s}$.
3. Blowdown from p_H to p_L with no loss of DMMP from the adsorbent or gas in the voids of the bed in negligible time.
4. Desorption at $p_L = 1.07 \text{ atm}$ with product gas (pure air) for 20 minutes. Interstitial velocity, u , corresponding to use of 41.6% of the product gas leaving the adsorption step.

SOLUTION

This example can be solved using the same equations and numerical techniques employed in Example 15.13, but noting that the units of q are different and a Langmuir isotherm replaces Henry's law. If the bed is not clean following the first desorption step, the results for the second and third cycles will differ from the first. The results are not presented here, but the calculations are required in Exercise 15.30.

Continuous, Countercurrent Adsorption Systems

In previous subsections, slurry and fixed-bed modes of adsorption, shown in Figures 15.20a and b, were considered. While these are traditional modes of adsorber operation, the third mode of operation in Figure 15.20c, continuous, countercurrent operation, has an important advantage because, as in a heat exchanger, an adsorber, and other separation cascades, countercurrent flow maximizes the average driving force for transport. In adsorption, this increases the efficiency of adsorbent use.

In Figure 15.20c, both liquid or gas mixtures undergoing separation and the solid adsorbent particles move through the system. However, as discussed in detail by Ruthven and Ching (65) and Wankat (97), the advantage of countercurrent operation can also be achieved by a simulated-moving-bed (SMB) operation, with one widely used implementation shown in Figure 15.23, wherein adsorbent particles remain fixed in a bed. In this subsection, the continuous, countercurrent system shown in Figure 15.20c is considered, while the next subsection covers the SMB. Both types of operation can be applied to purification or bulk separation.

McCabe-Thiele and Kremser Methods for Purification

Consider a binary mixture, dilute in a solute that is to be removed by adsorption in a continuous, countercurrent system of the type shown in Figure 15.42a. Only the solute is adsorbed. Feed F , with solute concentration c_F , enters the adsorption section, ADS, at Plane P_1 , from which adsorbent S leaves with a solute loading q_F . Purified feed called the raffinate, with solute concentration c_R , leaves the adsorption section at Plane P_2 , countercurrent to adsorbent of loading q_R , which enters at the top of the bed. At Plane P_3 , a purge called the desorbent, D , with solute concentration c_D , enters at the bottom of the desorption section, DES, from which the adsorbent leaves to enter the adsorption section. We assume that the desorbent does not adsorb and exits from the desorption

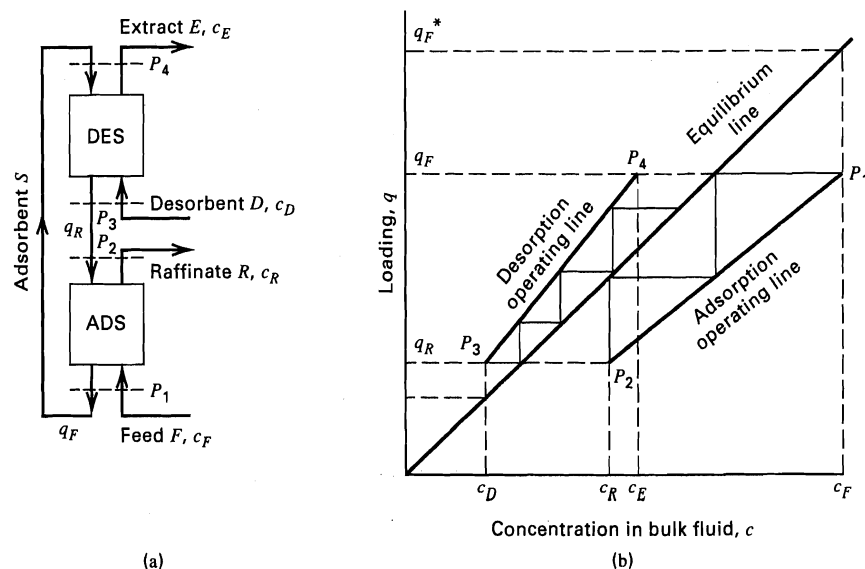


Figure 15.42 Continuous, countercurrent adsorption-desorption system. (a) System sections and flow conditions. (b) McCabe-Thiele diagram.

section as extract E , with solute concentration c_E , at Plane 4, where recycled adsorbent enters the desorption bed to complete the cycle.

If the system is dilute in the solute, if the solute adsorption isotherms for the feed solvent and the purge fluid are identical, and the system operates at constant temperature and pressure, the McCabe–Thiele diagram for the solute resembles that shown in Figure 15.42b, where the operating and equilibrium lines are straight because of the dilute condition. Note that the proper directions for mass transfer require that the adsorption and desorption operating lines lie below and above, respectively, the equilibrium line. These three lines are represented by the following equations:

Adsorption Operating Line:

$$q = \frac{F}{S}(c - c_F) + q_F \quad (15-134)$$

Desorption Operating Line:

$$q = \frac{D}{S}(c - c_D) + q_R \quad (15-135)$$

Equilibrium Line:

$$q = Kc \quad (15-136)$$

where F , S , and D are solute-free mass flow rates, and all solute concentrations are per solute-free carrier.

In Figure 15.42b, as the concentration of solute in the entering desorbent (purge), c_D , approaches zero, and solute concentration in the exiting raffinate, c_R , approaches zero, in order to avoid a large number of stages, it is necessary to select the adsorbent and desorbent flow rates so that

$$\frac{F}{S} < K < \frac{D}{S}$$

Thus, because more purge, D , than feed, F , is required, this system is only economical when the purge fluid is inexpensive. From the equilibrium and operating lines in Figure 15.42b, 2 and 3.3 equilibrium stages are determined for the adsorption and desorption sections, respectively, by stepping off stages in the McCabe–Thiele diagram. When the equilibrium and operating lines are straight, as in Figure 15.42b,

the algebraic Kremser method, rather than the graphical McCabe–Thiele method, can be employed. The Kremser equation, discussed in Section 6.4, is written in the following end-point form for the adsorption or desorption section:

$$N_t = \frac{\ln \left[\frac{c_1 - q_1/K}{c_2 - q_2/K} \right]}{\ln \left[\frac{c_1 - c_2}{q_1/K - q_2/K} \right]} \quad (15-137)$$

where 1 and 2 refer to opposite ends of the section, such as Planes 1 and 2 in Figure 15.42a, which are chosen so that $q_1 > q_2$.

If the operating conditions, e.g., temperature, for the two sections can be altered so as to place the equilibrium line for desorption below that for adsorption, it becomes possible to use a portion of the raffinate for desorption. This situation, shown in Figure 15.43, is achieved by desorbing at elevated temperature or, in the case of gas adsorption, at reduced pressure. Now, as shown in Figure 15.43, F/S can be greater than D/S . With a portion of raffinate used in Bed 2 (DES), the net raffinate product is $F - D$. Note that in this case, the two operating lines must intersect at the point (q_R, c_R) . By adjusting D/F , this point can be moved closer and closer to the origin so as to achieve any raffinate purity, c_R , desired, but at the expense of more theoretical stages and, therefore, deeper beds.

For a number of theoretical stages, N_t , in either the adsorption or desorption sections, bed height L can be determined from

$$L = N_t(\text{HETP}) \quad (15-138)$$

Values of HETP, which depend on mass-transfer resistances and axial dispersion, must be determined from experimental measurements. For large-diameter beds, typical values of HETP are in the range of 0.5–1.5 ft [97, 98].

McCabe–Thiele Method for Bulk Separation

Figure 15.44 shows a continuous, countercurrent adsorption–desorption process for bulk separation of a binary mixture.

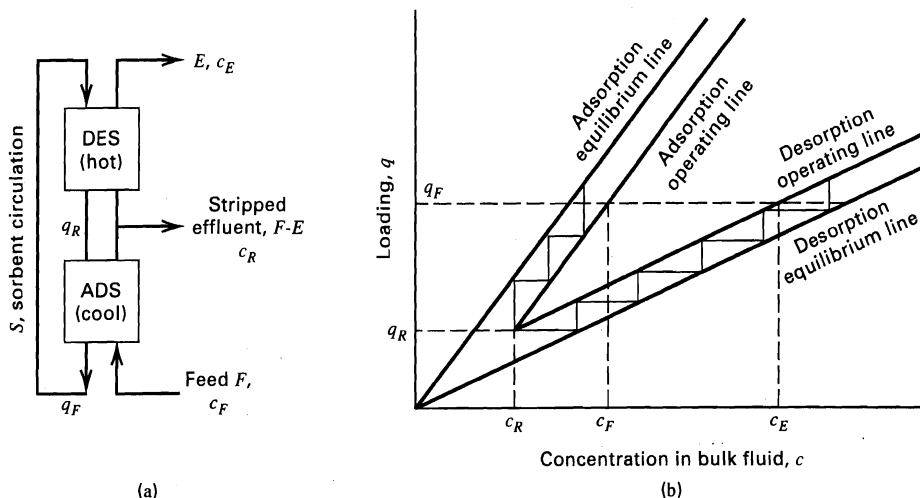


Figure 15.43 Continuous, countercurrent system with a temperature swing. (a) System sections and flow conditions. (b) McCabe–Thiele diagram.

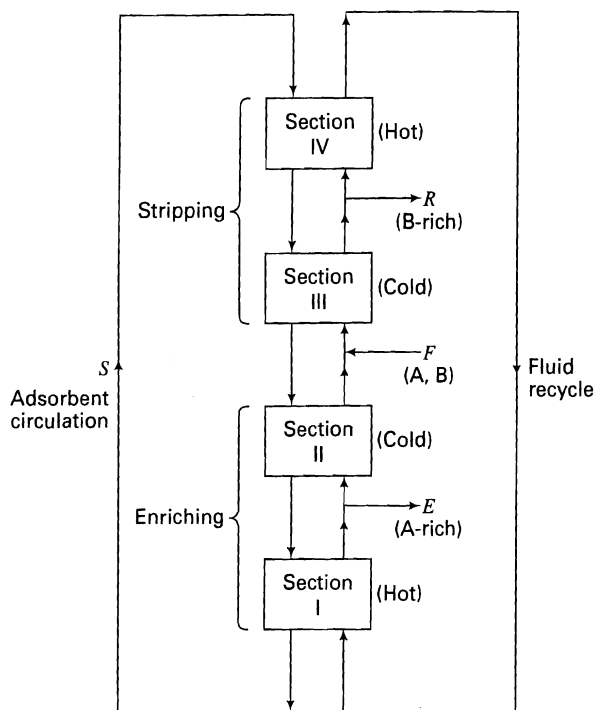


Figure 15.44 Continuous, countercurrent system for bulk separation.

The feed consists of component A, which is more strongly adsorbed, and B, which is less strongly adsorbed. The process consists of four sections (also called zones), numbered from the bottom up. Adsorbent S is circulated through the system, passing downward through the four sections, preferentially adsorbing A, leaving B to preferentially pass upward. To provide flexibility, a thermal swing is used, with Sections II and III operating at low or ambient temperature, while Sections I and IV operate at elevated temperature. The feed, F , enters between Sections II and III, passing up through Section III, where A is preferentially adsorbed at a relatively cold temperature. Product R, rich in B, is removed between Sections III and IV. At the higher temperature in Section IV, residual A and B is desorbed, with the fluid leaving from Section IV recycled to Section I. Adsorbent with mainly adsorbed A passes downward from Section III to Section II and then to Section I, where component A is desorbed to produce product E, rich in A, which is removed between Sections I and II. The system resembles an inverted distillation column, with the top two sections (III and IV) providing a stripping action to produce a product rich in the less strongly adsorbed component, while the bottom two sections provide an enriching action to produce a product rich in the more strongly adsorbed component. An equipment arrangement similar to that in Figure 15.44 was used in the Hypersorption moving bed process [66] for separating hydrogen and methane from ethane and heavier hydrocarbons, except that Section IV was a cooler, Section I was a steam stripper, and gas leaving Section IV was used to lift the adsorbent from Section I to Section IV. Additional flexibility can be achieved for the system in Figure 15.44 by providing separate adsorbent-circulation loops for the top two and bottom two sections.

EXAMPLE 15.15

One hundred pounds per minute (dry basis) of air at 80°F and 1 atm with 65% relative humidity is dehumidified isothermally and isobarically to 10% relative humidity in a continuous, countercurrent moving-bed adsorption unit. The adsorbent is dry silica gel (SG) having a particle-diameter range of 1.42 to 2.0 mm. Over the water-partial-pressure range of interest, the adsorption isotherm is given by measurements of Eagleton and Bliss [94] as

$$q_{\text{H}_2\text{O}} = 29c_{\text{H}_2\text{O}} \quad (1)$$

with concentration in lb $\text{H}_2\text{O}/\text{lb}$ dry air and loading in lb $\text{H}_2\text{O}/\text{lb}$ dry SG. If 1.5 times the minimum flow rate of silica gel is used, determine the number of equilibrium stages required.

SOLUTION

For relative humidities of 65% and 10%, the corresponding moisture contents are, from a humidity chart, 0.0143 and 0.0022 lb $\text{H}_2\text{O}/\text{lb}$ dry air, respectively.

In this case, Figure 15.42b applies for the adsorption section. Using the nomenclature in that figure:

$$F = 100 \text{ lb/min}, \quad c_F = 0.0143 \text{ lb H}_2\text{O}/\text{lb dry air}, \\ c_R = 0.0022 \text{ lb H}_2\text{O}/\text{lb dry air}, \quad \text{and } q_R = 0.$$

The value of q_F depends on adsorbent flow rate, S , which is 1.5 times the minimum value. At minimum-adsorbent rate, exiting adsorbent is in equilibrium with the entering gas. Therefore, from (1): $q_F^* = 29(0.0143) = 0.415$ lb $\text{H}_2\text{O}/\text{lb}$ dry SG. The amount of water vapor adsorbed is $F(c_F - c_R) = 100(0.0143 - 0.0022) = 1.21$ lb/min. Therefore: $S_{\min} = \frac{1.21}{0.415} = 2.92$ lb dry SG/min. If 1.5 times the minimum amount of silica gel is used: $S = 1.5 S_{\min} = 1.5(2.92) = 4.38$ lb dry SG/min. By material balance: $q_F = \frac{1.21}{4.38} = 0.276$ lb $\text{H}_2\text{O}/\text{lb}$ dry SG. From (15-137), with $K = 29$ from (1) and letting F be at plane 1 and R at plane 2:

$$N_t = \frac{\ln \left[\frac{0.0143 - 0.276/29}{0.0022 - 0} \right]}{\ln \left[\frac{0.0143 - 0.0022}{0.276/29 - 0} \right]} = 3.2 \text{ stages}$$

Simulated-Moving-Bed Systems

Continuous, countercurrent, moving-bed systems, often referred to as “true-moving-bed” (TMB) systems, encounter operating difficulties, including abrasion of adsorbent particles, failure to approach a plug flow of the particles as they move downward, and channeling of fluid through the moving bed. Alternatively, as shown in one implementation in Figure 15.23, a continuous countercurrent operation can be simulated by using a column containing a series of fixed beds and periodically moving the locations at which streams enter and leave the column. “Simulated-moving-bed” (SMB) systems have found widespread commercial application for liquid separations in the petrochemical, food, biochemical, pharmaceutical, and fine chemical industries, say of components A and B, when employing a circulating desorbent D (also called a diluent or eluent) to aid in the separation. In some cases the properties of D are such that,

like A and B, it can be adsorbed. Then, D can displace A and/or B from the sorbent pores, while A and/or B can displace D. In that case, a hybrid process of SMB adsorption and distillation, as shown in Figure 15.23, is often utilized, where following the SMB, a D-free extract of A and a D-free raffinate of B are obtained by distillation, with recovered D recycled to the SMB. In other cases, D is a component of the feed and is not adsorbed, but simply acts as a carrier and stripping agent for separation of A from B. For example, an aqueous solution of glucose and fructose is often separated by an SMB into an extract of aqueous glucose and a raffinate of aqueous fructose, which may be the final products. In the literature, simulated-moving-bed operations are often referred to as chromatographic, rather than adsorptive, separations.

An SMB can be treated as a countercurrent cascade of sections (or zones), rather than stages, where stream entry or withdrawal points bound the sections. As discussed by Zang

and Wankat [99], two-, three-, and four-section systems for producing two products, and a nine-section system for three products are described in the literature, with the four-section system, shown in Figure 15.45a, being the most common commercial design. More recently, Kim and Wankat [100] proposed SMB designs with from 12 to 32 sections for separation of quaternary mixtures.

Operation of a simulated moving bed is best understood by studying the two representations of a four-section system and the accompanying fluid composition profile in Figure 15.45. The schematic representation in Figure 15.45a shows a TMB, with circulation of solid adsorbent S down through four dense-bed sections in a closed cycle, while Figure 15.45b represents an actual SMB system, comprised of four sections divided into 12 fixed-bed subsections, shown as rectangles, with periodic movement of fluid inlet and outlet ports, shown as circles. The sections in Figure 15.45a are sometimes

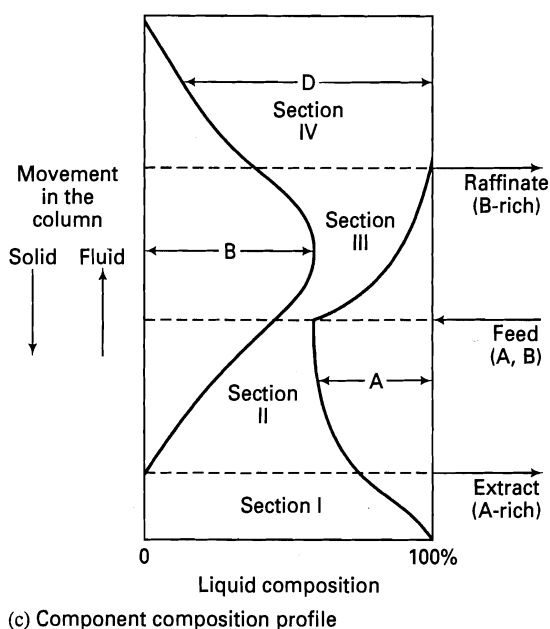
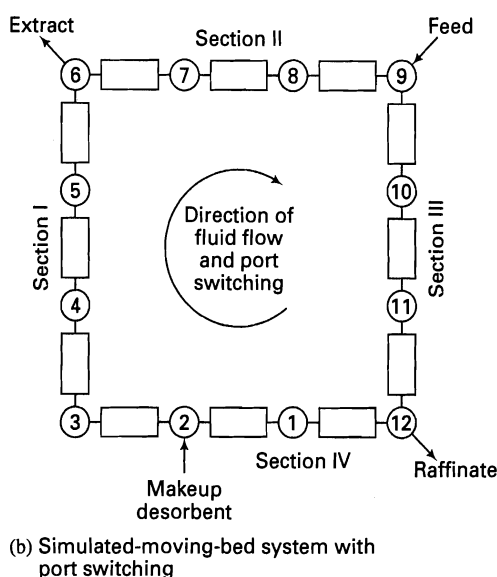
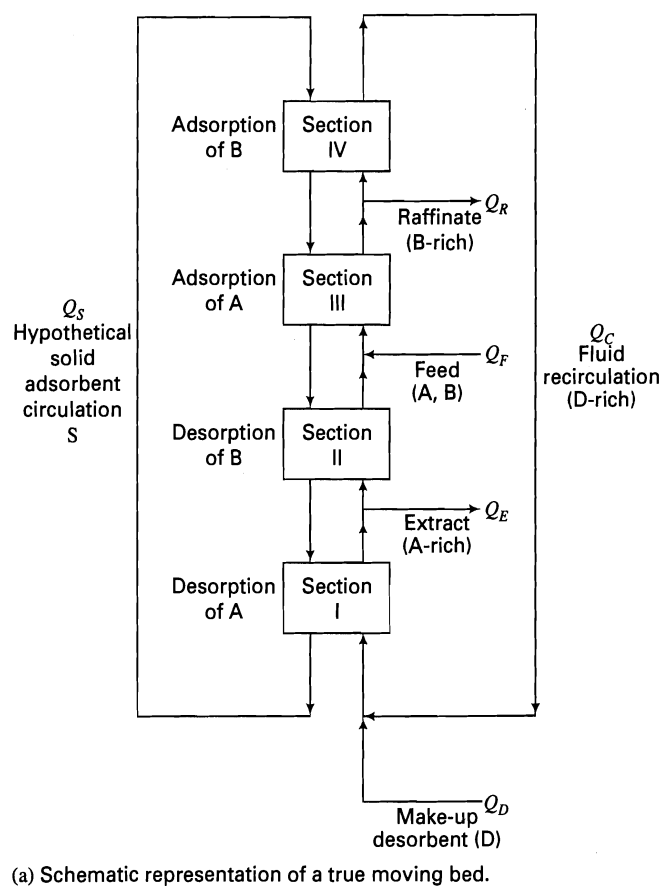


Figure 15.45 Four-section system.

referred to as zones, and the fixed-bed subsections in Figure 15.45b are often referred to as beds and sometimes columns. In the equivalent TMB case of Figure 15.45a, fluid of changing composition with respect to feed components A and B, and desorbent D, flows upward through the downward-flowing adsorbent beds. From the top of Section IV, fluid rich in D is recirculated to Section I. Fluid feed is shown as a binary mixture of A and B, which enters between Sections II and III. Component A is more strongly adsorbed than D, which is more strongly adsorbed than B. The desired result is that A is almost completely separated from B. However, appreciable amounts of D may appear in both the B-rich raffinate and A-rich extract. Thus, makeup D is added to the recirculated fluid.

Each of the four sections in Figure 15.45a performs a different primary function, listed in Figure 15.45a. More detail follows for the case where D, as well as A and B, are adsorbed. A typical component composition profile is shown in Figure 15.45c.

- Section I:** Desorb A. Entering S contains adsorbed A and D. Ideally, entering fluid is nearly pure D. Exiting S contains adsorbed D. Exiting fluid is A and D, part of which is withdrawn as A-rich extract.
- Section II:** Desorb B. Entering S contains adsorbed A, B, and D. Entering fluid is A and D. Exiting S contains adsorbed A and D. Exiting fluid is A, B, and D.
- Section III:** Adsorb A. Entering S contains adsorbed B and D. Entering fluid is A, B, and D from section II and fresh feed of A and B. Exiting S contains adsorbed A, B, and D. Exiting fluid is B and D, part of which is withdrawn as B-rich raffinate.
- Section IV:** Adsorb B. Entering S contains adsorbed D. Entering fluid is B and D. Exiting S contains adsorbed B and D. Ideally, exiting fluid is nearly pure D.

The steady-state separation achieved by the TMB in Figure 15.45a can be a close approximation to that achieved by the SMB, shown for a commercial Sorbex system in Figure 15.23 and by a simpler representation in Figure 15.45b. In both figures, it is seen that four sections are provided with a total of 12 ports for fluid feeds to enter, or fluid products to exit. In Figure 15.45b, it is clear that ports divide each section into subsections, four for Section I, three for Section II, three for Section III, and two for Section IV. As each section is divided into more subsections (thereby adding more ports), the SMB system more closely approaches the separation achieved in the corresponding TMB. In Figure 15.45b, only ports 2, 6, 9, and 12 are open. After an increment of time (called the switching time or port switching interval, t^*), those ports are closed and 3, 7, 10, and 1 are opened. In this manner, the ports are closed and opened in sequence in the direction shown. By periodically shifting feed and

product positions by one port position in the direction of fluid flow, movement of solid adsorbent in the opposite direction within the sections is simulated. Because of stream additions and withdrawals between sections, flow rates in each of the four sections are different. Figure 15.23 shows a pump for controlling the fluid flow rate at the bottom of the SMB. Although sections are switched, the pump is not. Therefore, the pump must be programmed for four different flow rates depending on the section to which the pump is currently connected.

A number of models have been developed for designing and analyzing SMBs. These include: (1) TMB equilibrium-stage model using a McCabe–Thiele-type analysis, (2) TMB local adsorption-equilibrium model, (3) TMB rate-based model, and (4) SMB rate-based model. The first three assume steady-state conditions with continuous, countercurrent flows of fluid and solid adsorbent, approximating SMB operation with a TMB. The SMB rate-based model applies to transient operation for start-up, approach to a cyclic steady state, and shut-down. The simplest of the four approaches is the TMB equilibrium-stage model, but it is difficult to apply to multicomponent systems with nonlinear adsorption-equilibrium isotherms. The TMB local adsorption-equilibrium model, although ignoring the effects of axial dispersion and fluid-particle mass transfer, has proved useful for establishing reasonable operating flow rates in multiple sections of an SMB because, for many applications, behavior of an SMB is determined largely by adsorption equilibria. For a linear adsorption isotherm, Wankat [102] has successfully applied this method to SMBs with up to 32 sections for feeds dilute in the solutes. Methods for solving the TMB local adsorption equilibrium model for multicomponent systems, including concentrated mixtures, with nonlinear adsorption isotherms, have been presented by a number of investigators, including Storti et al. [103], who extended the pioneering work of Rhee, Aris, and Amundson [104] for a single section to the commonly used four-section unit, and Mazzotti et al. [105] for multicomponent systems. For a final design, rate-based models are preferred. These models, which account for axial dispersion in the bed, particle-fluid mass-transfer resistances, and nonlinear adsorption isotherms, are available in the program Aspen Chromatography, of the Aspen Engineering Suite, for both TMB steady-state operation and SMB dynamic operation. The local adsorption equilibrium and rate-based models are described next, followed by illustrative examples, two of which are solved using Aspen Chromatography. Equations are presented for four-section units, but are readily extended to more sections.

Steady-state Local Adsorption Equilibrium TMB Model

The TMB model describes continuous, steady-state, multicomponent adsorption with countercurrent flow of the fluid and solid adsorbent, as shown in Figure 15.46 for a single section of height Z of a multisection system, subject to these

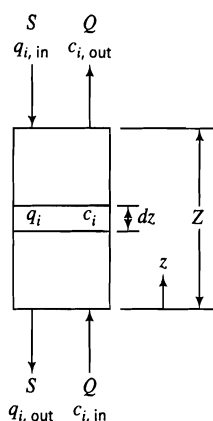


Figure 15.46 TMB local-adsorption-equilibrium model for a single section.

assumptions:

- One-dimensional plug flow of both phases with no channeling.
- Constant volumetric flow rates, of Q for the liquid and Q_s for the solid.
- Constant external void fraction, ϵ_b , of the solids bed.
- Negligible axial dispersion and particle-fluid mass-transfer resistances.
- Local adsorption equilibrium between solute concentrations, c_i , in the bulk liquid and adsorption loading, q_i , on the solid.
- Isothermal and isochoric conditions.

For a differential-bed thickness, dz , where component i undergoes mass transfer between the two phases, this mass balance applies:

$$Q \frac{dc_i}{dz} - S \frac{dq_i}{dz} = 0 \quad (15-138)$$

Boundary conditions are

$$z = 0, c_i = c_{i,in} \quad \text{and} \quad z = Z, q_i = q_{i,in}$$

The solution to (15-138) depends on the equilibrium adsorption isotherm. Typically, when the fluid is a liquid dilute in

solutes, a linear (Henry's law) isotherm, $q_i = K_i c_i$, is used, where q_i is on a particle volume basis so that K_i is dimensionless. For the bulk separation of liquid mixtures, where concentrations of the feed components and desorbent are not small, a nonlinear, extended-Langmuir-equilibrium-adsorption isotherm of the constant-selectivity form, from Example (15.6) applies

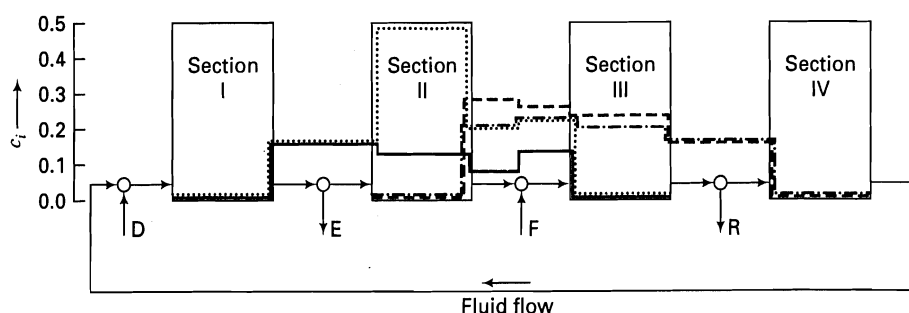
$$q_i = \frac{(q_i)_m K_i c_i}{1 + \sum_j K_j c_j} \quad (15-139)$$

In either case, the solution of Rhee, Aris, and Amundson [104], when extended to multiple (e.g., four) sections, as by Storti et al. [103], predicts constant component concentrations in each section, but with discontinuities at either one or both section boundaries. Typical concentration profiles are shown in Figure 15.47 for a four-solute system (1, 2, 3, and 4), where a set of stationary rectangular (shock-like) waves of constant concentration exists in the fluid phase in each section. The concentration profile for the desorbent (component 5) is not shown. Note that the concentrations of the four solutes for this local equilibrium assumption are negligible in Sections I and IV, where only desorbent is present.

The usefulness of local equilibrium theory is in approximate determinations of required solid adsorbent and fluid flow rates in each section of a TMB in order to achieve a perfect separation of two solutes. The description of the method, first developed by Ruthven and Ching [106] and extended by Zhong and Guiochon [107], is facilitated by applying local adsorption-equilibrium theory to the simple case of a feed dilute in binary solutes, A and B, that are to be completely separated. Assume diluent, D, does not adsorb and Henry's law governs adsorption equilibrium, with $K_A > K_B$ (i.e., A is more strongly adsorbed). First, we define a set of flow rate ratios, m_j , one for each section, j , as

$$m_j = \frac{Q_j}{Q_s} = \frac{\text{volumetric fluid phase flow rate}}{\text{volumetric solid particle phase flow rate}} \quad (15-140)$$

For conditions of local adsorption equilibrium, the following necessary and sufficient conditions apply to each section for



Component	Relative adsorption selectivity
1	1.00
2	1.12
3	2.86
4	5.71
5 (not shown)	1.90

Figure 15.47 Typical solute-concentration profiles for local adsorption equilibrium in a four-section unit.

complete separation:

$$K_A < m_I < \infty \quad (15-141)$$

$$K_B < m_{II} < K_A \quad (15-142)$$

$$K_B < m_{III} < K_A \quad (15-143)$$

$$0 < m_{IV} < K_B \quad (15-144)$$

Constraint (15-142) ensures that net flow rates of components A and B will be positive (upward) in section I. Constraint (15-144) ensures that the net flow rates of components A and B will be negative (downward) in Section IV. Constraints (15-142) and (15-143) are most important because they ensure sharpness of the separation. They cause net flow rates of A and B to be negative (downward) and positive (upward), respectively, in the two central sections II and III. Inequality constraints (15-141) to (15-144) may be converted to equality constraints, where β , the safety margin, is discussed shortly.

$$Q_I/Q_S = K_A\beta \quad (15-145)$$

$$(Q_I - Q_E)/Q_S = K_B\beta \quad (15-146)$$

$$(Q_I - Q_E + Q_F)/Q_S = K_A/\beta \quad (15-147)$$

$$(Q_I - Q_E + Q_F - Q_R)/Q_S = K_B/\beta \quad (15-148)$$

Solving (15-145) to (15-148) by eliminating Q_I gives

$$Q_S = \frac{Q_F}{K_A/\beta - K_B\beta} \quad (15-149)$$

$$Q_E = Q_S(K_A - K_B)\beta \quad (15-150)$$

$$Q_R = Q_S(K_A - K_B)/\beta \quad (15-151)$$

Then, using (15-145),

$$Q_I = Q_C + Q_D = Q_S K_A \beta \quad (15-152)$$

Therefore,

$$Q_C = Q_S K_A \beta - Q_D \quad (15-153)$$

where Q_C = fluid recirculation rate before adding makeup desorbent. By an overall material balance,

$$Q_D = Q_E + Q_R - Q_F \quad (15-154)$$

Restrictions on flow rate ratios, m_{II} and m_{III} in inequality constraints (15-142) and (15-143), are conveniently represented by the *triangle method* of Storti et al. [107], as shown in Figure 15.48. If values of m_{II} and m_{III} within the triangular region are selected, a perfect separation is possible. However, if $m_{II} < K_B$, some B will appear in the extract; if $m_{III} > K_A$, some A will appear in the raffinate. If $m_{II} < K_B$ and $m_{III} > K_A$, extract will contain some B and raffinate will contain some A.

The permissible range for safety margin, β , in (15-145) to (15-151) is determined from inequality constraints (15-142) and (15-143). Let

$$\gamma_{i,j} = \frac{m_j}{K_i} = \frac{Q_j}{Q_S K_i} \quad (15-155)$$

In Section II, we require that $\gamma_{A,II} > 1$ and $\gamma_{B,II} < 1$. In terms of safety margin, β , we can apply (15-155) to give corresponding equalities, $Q_{II}/Q_S = K_A/\beta$ and $Q_{II}/Q_S = K_B\beta$, assuming equal β in all four sections. Equating these two equalities for the same safety margin gives $\beta = \sqrt{K_A/K_B}$, which is the maximum value of β for a perfect separation, the minimum value being 1.0. Above the maximum value of β , some sections will encounter negative fluid flow rates. Below a β value of 1.0, a perfect separation will not be achieved. As the value of β increases from minimum to maximum, fluid flow rates in the sections increase, often exponentially. Thus, estimation of operating flow rates is generally carried out using a value of β close to, but above, 1.0, e.g., 1.05 (unless it exceeds the maximum value of β). Note that as the separation factor, K_A/K_B , approaches 1.0, not only does the separation become more difficult, but also, the permissible range of β becomes smaller. In the triangle method, illustrated in Figure 15.48, the upper left corner of the triangle corresponds to $\beta = 1$, while the maximum value of β occurs when $m_{II} = m_{III}$, which falls on the 45° line between the values K_A and K_B . Extensions of the above binary procedures for estimating operating flow rates to cases of both constant selectivity Langmuir adsorption isotherms and to more complex nonlinear isotherms are given by Mazzotti et al. [109] and for multicomponent systems by Mazzotti et al. [105]. With nonlinear adsorption isotherms, the right triangle of Figure 15.48 is distorted to a shape with one or more curved sides.

EXAMPLE 15.16

Fructose (A) is separated from glucose (B) in a four-section SMB unit. The aqueous feed of 1.667 mL/min contains 0.467 g/min of A, 0.583 g/min of B, and 0.994 g/min of water. For the adsorbent and

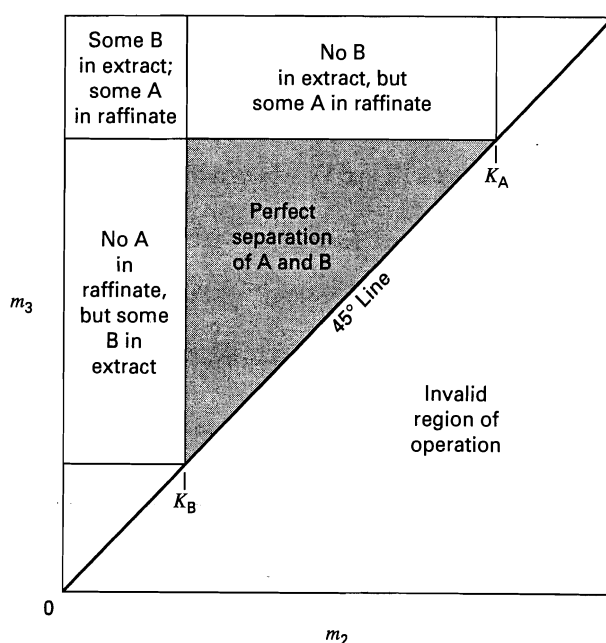


Figure 15.48 Triangle method for determining necessary values of flow rate ratios.

expected concentrations and temperature of the operation, Henry's law holds, with constants of $K_A = 0.610$ and $K_B = 0.351$ for fluid concentrations in g/mL and loadings in g/mL of adsorbent particles. Water is assumed not to adsorb. Estimate operating flow rates in mL/min to achieve a perfect separation of fructose from glucose for a TMB. Note that the extract will contain fructose, while raffinate will contain glucose. Conversion of the results to SMB operation will be made in Example 15.17.

SOLUTION

Equations (15-149) to (15-154) apply. The minimum value of β is 1.0, while the maximum value is $\sqrt{K_A/K_B} = \sqrt{0.610/0.351} = 1.32$. Calculations are most conveniently carried out with a spreadsheet. With reference to Figure 15.45 for the case of a TMB, the results for values of $\beta = 1.0, 1.05, 1.20$ are:

	Volumetric Flow Rates, mL/min		
	$\beta = 1.0$	$\beta = 1.05$	$\beta = 1.20$
Feed, Q_F	1.667	1.667	1.667
Solid particles, Q_S	6.436	7.848	19.132
Extract, Q_E	1.667	2.134	5.946
Raffinate, Q_R	1.667	1.936	4.129
Recirculation, Q_C	2.259	2.624	5.596
Make-up desorbent, Q_D	1.667	2.403	8.408
Q_I	3.926	5.027	14.004
Q_{II}	2.259	2.893	8.058
Q_{III}	3.926	4.560	9.725
Q_{IV}	2.259	2.624	5.596

Note that the lowest section fluid flow rates, Q_I – Q_{IV} , correspond to $\beta = 1.0$. At $\beta = 1.2$, section fluid flow rates, as well as the adsorbent particles flow rate, become significantly higher. The most concentrated products (extract and raffinate) and the smallest flow rate of make-up desorbent are also achieved with the lowest β value.

Steady-state TMB Model

This model, which assumes plug flow at isothermal, isobaric, and constant-fluid-velocity conditions in each section, j , ($j = 1$ to 4) requires for each component, i ($i = 1$ to C) the following equations, where each section begins at $z = 0$, where the fluid enters, and ends at $z = L_j$. Unlike the previous local adsorption-equilibrium model, axial dispersion and fluid-particle mass transfer are taken into account.

- (1) Mass-balance equation for the bulk fluid phase, f , [similar to (15-102)]:

$$-D_{L_j} \frac{d^2 c_{i,j}}{dz^2} + u_{fj} \frac{dc_{i,j}}{dz} + \frac{(1 - \epsilon_b)}{\epsilon_b} J_{i,j} = 0 \quad (15-156)$$

where the first term accounts for axial dispersion, J_i is the mass-transfer flux between the bulk fluid phase and the sorbate in the pores of the solid, and u_f is the interstitial fluid velocity, where for an adsorbent bed of cross-sectional area, A_b ,

$$u_{fj} = \frac{Q_j}{\epsilon_b A_b} \quad (15-157)$$

- (2) Mass-balance equation for the sorbate, s , on the solid phase:

$$u_s \frac{d\bar{q}_{i,j}}{dz} - J_{i,j} = 0 \quad (15-158)$$

where u_s is the true moving-solid velocity, where

$$u_s = \frac{Q_S}{(1 - \epsilon_b) A_b} \quad (15-159)$$

- (3) Fluid-to-solid mass transfer [similar to (15-105)]:

$$J_{i,j} = k_{i,j}(q_{i,j}^* - \bar{q}_{i,j}) \quad (15-160)$$

- (4) Adsorption isotherm [e.g., the multicomponent, extended-Langmuir equation of (15-139)]:

$$q_{i,j}^* = f\{\text{all } c_{i,j}\} \quad (15-161)$$

This system of 4C second-order ODEs and 4C first-order ODEs, together with the algebraic equations for mass-transfer rates and adsorption equilibria, requires 12C boundary conditions, i.e., 3C for each section.

At the entrance, $z = 0$, to each section, we require a boundary condition that accounts for axial dispersion. This has been discussed extensively in the literature, e.g., Danckwerts [110]. Most often used is

$$u_{fj}(c_{i,j,0} - c_{i,j}) = -\epsilon_b D_{L_j} \frac{dc_{i,j}}{dz} \quad (15-162)$$

where $c_{i,j,0}$ is the concentration of component i entering ($z = 0$) section j .

For continuity of bulk fluid concentrations and sorbate loadings in moving from one section to another, the following boundary conditions apply at boundaries of adjacent sections:

At Sections I and II where extract is withdrawn:

$$c_{i,I,z=L_j} = c_{i,II,z=0} \quad (15-163)$$

$$q_{i,I,z=L_j} = q_{i,II,z=0} \quad (15-164)$$

At Sections III and IV where raffinate is withdrawn:

$$c_{i,III,z=L_j} = c_{i,IV,z=0} \quad (15-165)$$

$$q_{i,III,z=L_j} = q_{i,IV,z=0} \quad (15-166)$$

At Sections II and III where the feed enters:

$$c_{i,III,z=0} = \frac{Q_{II}}{Q_{III}} c_{i,II,z=L_{II}} + \frac{Q_F}{Q_{III}} c_{i,F} \quad (15-167)$$

$$q_{i,II,z=L_j} = q_{i,III,z=0} \quad (15-168)$$

At Sections IV and I where make-up desorbent enters:

$$c_{i,I,z=0} = \frac{Q_{IV}}{Q_I} c_{i,IV,z=L_{IV}} + \frac{Q_D}{Q_I} c_{i,D} \quad (15-169)$$

$$q_{i,IV,z=L_j} = q_{i,I,z=0} \quad (15-170)$$

where the volumetric fluid flow rates, which change from section to section, are subject to

$$Q_I = Q_{IV} + Q_D \quad (15-171)$$

$$Q_{II} = Q_I - Q_E \quad (15-172)$$

$$Q_{III} = Q_{II} + Q_F \quad (15-173)$$

$$Q_{IV} = Q_{III} - Q_E \quad (15-174)$$

It is important to note that for an SMB, solid particles do not flow down through the unit, but are retained in stationary beds in each section. To obtain the same true velocity difference between the fluid and solid particle phase, the upward fluid velocity in the SMB must be the sum of the absolute true velocities in the upward-moving fluid and downward-moving solid particle phases in the TMB. Thus, using (15-157) and (15-159),

$$(Q_j)_{SMB} = (Q_j)_{TMB} + \left(\frac{\epsilon_b}{1 - \epsilon_b} \right) (Q_s)_{TMB} \quad (15-175)$$

The TMB model can be solved by any of a number of techniques, as discussed by Constantinides and Mostoufi [111], with the Newton shooting method being preferred. An example of the application of the steady-state TMB model is given after the next subsection that treats dynamic SMB models.

Dynamic SMB Model

The equations are subject to the same assumptions as the steady-state TMB model. Changes in the equations permit the model to take into account time of operation, t , and to use a fluid velocity relative to the stationary solid particles. In addition, equations now must be written for each bed subsection (also referred to as a column), k , between adjacent ports, as shown in Figure 15.45b. The revised equations are

- (1) Mass-balance equation for the bulk fluid phase, f [similar to (15-102)]:

$$\frac{\partial c_{i,k}}{\partial t} - D_{L_j} \frac{\partial^2 c_{i,k}}{\partial z^2} + u_{f,k} \frac{\partial c_{i,k}}{\partial z} + \frac{(1 - \epsilon_b)}{\epsilon_b} J_{i,k} = 0 \quad (15-176)$$

- (2) Mass-balance equation for sorbate on the solid phase:

$$\frac{\partial \bar{q}_{i,k}}{\partial t} - J_{i,k} = 0 \quad (15-177)$$

where the interstitial fluid velocity for SMB operation is related to that for TMB operation at a particular location by

$$(u_f)_{SMB} = (u_f)_{TMB} + |(u_s)|_{TMB} \quad (15-178)$$

SMB and TMB models are further connected by an equation that relates solid velocity in the TMB model to a port-switching time, t^* , and bed height between adjacent ports, L_k , for use in this SMB model:

$$u_s = \frac{L_k}{t^*} \quad (15-179)$$

The boundary conditions for the TMB model apply to SMB models. In addition, initial conditions are needed for fluid

concentrations, $c_{i,j}$, and sorbate loadings, $\bar{q}_{i,j}$, throughout the adsorbent beds; e.g., at $t = 0$, $c_{i,k} = 0$ and $\bar{q}_{i,k} = 0$.

The SMB model, which involves PDEs, rather than ODEs, is much more difficult to solve than the steady-state TMB, because it involves moving concentration fronts. In Aspen Chromatography, the dynamic SMB equations are solved by discretizing the first- and second-order spatial terms of the PDEs to obtain a large set of ODEs and algebraic equations, which constitute a DAE (differential algebraic equations) system. A number of discretization or differencing methods are provided. Each complete cycle of the SMB model provides a different result, which ultimately leads to a cyclic steady state. Studies have shown that if the number of bed subsections per section is at least four and the number of cycles is 10 or more, the steady-state TMB result closely approximates the SMB result. Therefore, if only steady-state results are of interest, the simpler steady-state TMB model is best employed.

All four models can be applied to a gas or liquid mixture, with the latter being the most widely applied to industrial separations. Regardless of the model used for design of an SMB (dynamic SMB or steady-state TMB), the basic information required is:

1. Flow rate and composition of the feed (binary of A and B, or multicomponent).
2. Selection of a suitable adsorbent, S, and desorbent, D.
3. Nominal bed operating temperature, T , and pressure, P .
4. A suitable adsorption isotherm for all components, with known constants at the bed operating conditions.
5. Desired separation, which may be purity (on a desorbent-free basis) and desired recovery of the most strongly adsorbed component in the extract.

Not initially known, but required before calculations can be made, are:

6. Total bed height and inside diameter of the adsorption column.
7. Amount of adsorbent in the column.
8. Desorbent recirculation rate.
9. Flow rates of extract and raffinate.
10. Overall mass-transfer coefficients for transport of solutes between bulk fluid and sorbate layer on the adsorbent.
11. Eddy diffusivity for axial dispersion.
12. Spacing of inlet and outlet ports.

Some guidance on initial values for items 6, 10, and 11 is sometimes provided in patents for similar separations. For example, for the separation of xylene mixtures using para diethylbenzene as desorbent, Minceva and Rodrigues [112] suggest:

- Molecular-sieve zeolite adsorbent with a spherical particle diameter, d_p , between 0.25 and 1.00 mm and a particle density, ρ_p , of 1.39 g/cm³.

- Operating temperature between 140°C and 185°C with an operating pressure sufficient to maintain a liquid phase.
- Liquid interstitial velocity, u_f , between 0.4 and 1.2 cm/s.
- Four sections with eight to 24 subsections (beds).

For a commercial-size unit, the following are suggested:

- Bed height, L_k , in each subsection from 40 to 120 cm.
- Equation (15-106) for estimating overall mass-transfer coefficient, $k_{i,j}$, for solute transport between bulk fluid and sorbate layer on the adsorbent.
- An axial diffusivity, $D_{L,j}$, defined in terms of a Peclet number, where

$$N_{Pe} = \frac{u_f(\text{characteristic length})}{D_L} \quad (15-180)$$

Characteristic lengths equal to bed depth or particle diameter have been used. Most common for TMB and SMB is bed depth, with Peclet numbers in the 1000–2000 range.

EXAMPLE 15.17

Use the results of the fructose-glucose separation of Example 15.16, for $\beta = 1.05$, with the steady-state TMB model of Aspen Chromatography to estimate product compositions obtained with the following laboratory-size SMB unit:

Number of sections = 4

Number of subsections (beds) in each section (column) = 2

All bed diameters = 2.54 cm

All bed heights = 10 cm

Bed void fraction = 0.40

Particle diameter = 500 microns (0.5 mm)

Overall mass-transfer coefficient for A and B = 10 min^{-1}

Peclet number high enough that axial dispersion is negligible

SOLUTION

To use Aspen Chromatography, the recirculating liquid flow rate for a TMB must be converted to a SMB using (15-175), and solid-particle flow rate must be converted to a port switching time given by (15-179). From (15-175), using the results for $\beta = 1.05$ in Example 15.16,

$$(Q_C)_{\text{SMB}} = 2.624 + \left(\frac{0.40}{1 - 0.40} \right) 7.848 = 7.856 \text{ mL/min}$$

The total liquid rate in Section I of the SMB is

$$(Q_I)_{\text{SMB}} = (Q_C)_{\text{SMB}} + Q_D = 7.856 + 2.403 = 10.259 \text{ mL/min}$$

This is the maximum volumetric flow rate in the SMB and it is of interest to calculate the corresponding interstitial fluid velocity. From (15-157),

$$\begin{aligned} (u_f)_{\text{SMB}} &= \frac{(Q_I)_{\text{SMB}}}{\epsilon_b A_b} = \frac{10.259}{0.40 \left[\frac{3.14(2.54)^2}{4} \right]} \\ &= 5.06 \text{ cm/min} = 0.0844 \text{ cm/s} \end{aligned}$$

This fluid velocity is low, but it corresponds to a desirable bed diameter-to-particle diameter ratio of $2.54/0.05 = 49$. To increase fluid velocity to, say, 0.4 cm/s, the bed diameter would be decreased to 1.17 cm, giving a bed diameter-to-particle diameter of 23, which would still be acceptable.

From (15-159), the true velocity of the solid particles in each bed is

$$u_s = \frac{Q_s}{(1 - \epsilon_b) A_b} = \frac{7.848}{(1 - 0.40) \left[\frac{3.14(2.54)^2}{4} \right]} = 2.58 \text{ cm/min}$$

From (15-179), port-switching time for subsection bed height, L , of 10 cm is,

$$t^* = \frac{L}{u_s} = \frac{10}{2.58} = 3.88 \text{ min}$$

The following results were obtained from Aspen Chromatography for a steady-state TMB:

	Feed	Desorbent	Extract	Raffinate
Flow rate, mL/min	1.667	2.403	2.134	1.936
Concentrations, g/L:				
Fructose	280.0	0.0	211.6	12.7
Glucose	350.0	0.0	8.4	295.3
Water	596.0	996.0	861.7	795.8
Mass fraction on water-free basis:				
Fructose	0.444		0.962	0.040
Glucose	0.556		0.038	0.960

As seen in the table, a reasonably sharp separation between fructose and glucose is achieved. In Exercise 15.39, modifications to the input data are studied in an attempt to improve separation sharpness.

EXAMPLE 15.18

Minceva and Rodrigues [112] consider the industrial-scale separation of paraxylene from a liquid mixture of C_8 aromatics in a four-section SMB. Feed to the unit is 1,450 L/min with the composition shown in a table below, which also contains results for this example. The adsorbent is a molecular-sieve zeolite with a particle density of 1.39 g/cm³ and a particle diameter of 0.092 cm that packs a bed with an external void fraction of 0.39. The desorbent is paradiethylbenzene (PDEB). With reference to Figure 15.45, the number of subsections is 6, 9, 6, and 3, respectively, in Sections I to IV. The height of each bed subsection is 1.135 m, with a bed diameter of 4.117 m. The operation takes place at 180°C and a pressure above 12 bar, sufficient to prevent vaporization. At these conditions, the extended Langmuir adsorption isotherm (15-139) correlates adsorption equilibrium, yielding the following constants. Note that this is a constant-selectivity isotherm; therefore, the selectivity relative to paradiethylbenzene is tabulated.

Component	q_m , mg/g	K , cm ³ /mg	Selectivity
Paraxylene	130.3	1.0658	0.9969
Paradiethylbenzene	107.7	1.2935	1.0000
Ethylbenzene	130.3	0.3067	0.2689
Metaxylene	130.3	0.2299	0.2150
Orthoxylene	130.3	0.1884	0.1762

Note that the desorbent does not have the most desirable equilibrium adsorption property because its selectivity does not lie between that of paraxylene and the C_8 components of the feed. Take the overall mass-transfer coefficient between sorbate and bulk fluid, in (15-160), as 2 min^{-1} for each component. For axial dispersion, assume a Peclet number of 700 in (15-180) with a characteristic length of bed height.

Using Aspen Chromatography with the TMB model as an approximation of the SMB, determine steady-state flow rates and compositions of extract and raffinate, together with the composition profiles in the four sections for the following operating conditions:

Extract flow rate = 1,650 L/min

Raffinate flow rate = 2,690 L/min

Circulation flow rate, $(Q_C)_{\text{SMB}}$, before adding makeup
DPEB = 5,395 L/min

Port-switching interval, $t^* = 1.15 \text{ min}$

SOLUTION

By an overall material balance, the DPEB makeup flow rate is

$$Q_D = Q_E + Q_R - Q_F = 1,650 + 2,690 - 1,450 = 2,890 \text{ L/min}$$

From the switching time, using (15-179), with a 1.135-m bed height,

$$u_s = 1.135/1.15 = 0.987 \text{ m/min} = 98.7 \text{ cm/min}$$

The adsorbent bed cross-sectional area, $A_b = 3.14(4.117)^2/4 = 13.31 \text{ m}^2$.

From (15-159), the volumetric flow rate of the solid particles in a TMB is

$$\begin{aligned} Q_S &= u_s(1 - \epsilon)A_b = 0.987(1 - 0.39)(13.31) \\ &= 8.014 \text{ m}^3/\text{min} = 8,014 \text{ L/min} \end{aligned}$$

Liquid flow rates in the four sections are as follows, where both $(Q_j)_{\text{SMB}}$ and $(Q_j)_{\text{TMB}}$ flow rates are included, where the former are computed by material balance and the latter from (15-175). For example,

$$(Q_I)_{\text{SMB}} = (Q_C)_{\text{SMB}} + Q_D = 5,395 + 2,890 = 8,285 \text{ L/min}$$

$$\begin{aligned} (Q_I)_{\text{TMB}} &= (Q_I)_{\text{SMB}} - [0.39/(1 - 0.39)]Q_S \\ &= 8,285 - 0.639(8,014) = 3,164 \text{ L/min} \end{aligned}$$

Section in Figure 15.45	$(Q_j)_{\text{SMB}}$, L/min	$(Q_j)_{\text{TMB}}$, L/min
I	8,285	3,164
II	6,635	1,514
III	8,085	2,964
IV	5,395	274

Results of the Aspen Chromatography calculations for the steady-state TMB model, but on an SMB basis are:

Wt% of component	Feed	Desorbent	Extract	Raffinate
Ethylbenzene	14.0	0.0	0.00	7.63
Metaxylene	49.7	0.0	0.00	27.09
Orthoxylene	12.7	0.0	0.00	6.92
PDEB	0.0	100.0	80.79	57.85
Paraxylene	23.6	0.0	19.21	0.51

Note that an excellent separation between paraxylene and the other feed components of the feed is achieved. However, both the extract and raffinate contain a substantial fraction of desorbent, PDEB. The desorbent in both products is recovered by the hybrid SMB-distillation process shown in Figure 15.23. Component concentration profiles in the four sections, as computed by Aspen Chromatography, are shown in Figure 15.49. In Sections I and III particularly, they differ considerably from the flat profile predictions of the simple, local-equilibrium TMB model. The circulating desorbent is predicted to be essentially pure PDEB.

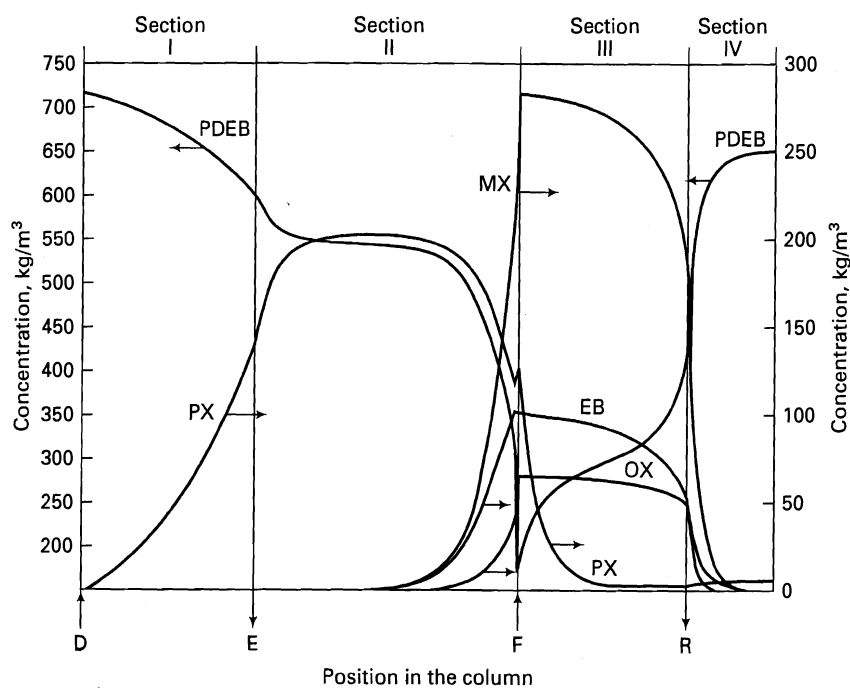


Figure 15.49 Concentration profiles in the liquid for SMB of Example 15.18.

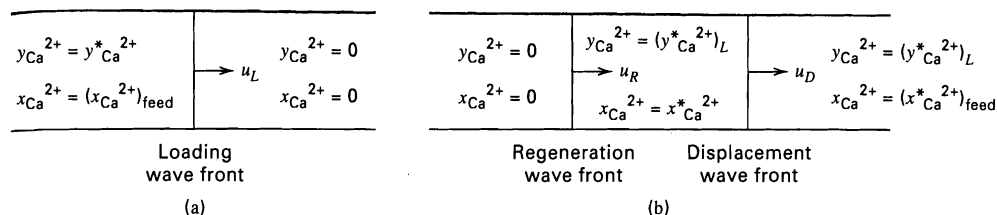


Figure 15.50 Ion exchange in a cyclic operation with a fixed bed. (a) Loading step. (b) Displacement and regeneration steps.

Ion-Exchange Cycle

Although ion exchange has a wide range of applications, water softening with gel resins continues to be the major one. Usually a fixed bed is used, which is operated in a cycle of four steps: (1) loading, (2) displacement, (3) regeneration, and (4) washing. The solute ions removed from water in the loading step are mainly Ca^{2+} and Mg^{2+} , which are absorbed by resin while an equivalent amount of Na^+ is transferred from resin to water as feed solution flows down through the bed. If mass transfer is rapid, the solution and resin are at equilibrium at all points in the bed. With a divalent ion (e.g., Ca^{2+}) replacing a monovalent ion (e.g., Na^+), the equilibrium expression is given by (15-44), where A is the divalent ion. If $(Q/C)^{n-1} K_{A,B} \gg 1$, equilibrium for the divalent ion is very favorable (see Figure 15.29a) and a self-sharpening front of the type shown in Figure 15.29b develops. In that case, which is common, ion exchange is well approximated using simple stoichiometric or shock-wave front theory for adsorption, assuming plug flow. As the front moves down through the bed, the resin behind or upstream of the front is in equilibrium with the feed composition. Ahead or downstream of the front, water is essentially free of the divalent ion(s). Breakthrough occurs when the front reaches the end of the bed.

Suppose the only cations in the feed are Na^+ and Ca^{2+} . Then, from (15-44):

$$K_{Ca^{2+},Na^+} \left(\frac{Q}{C} \right) = \frac{y_{Ca^{2+}}(1 - x_{Ca^{2+}})}{x_{Ca^{2+}}(1 - y_{Ca^{2+}})} \quad (15-181)$$

where Q is total concentration of the two cations in the resin, in eq/L of bed of wet resin, and C is total concentration of the two ions in the solution, in eq/L of solution. One mole of Na^+ is 1 equivalent, while 1 mole of Ca^{2+} is 2 equivalents. The quantities y_i and x_i are equivalent (rather than mole) fractions. From Table 15.5, using (15-45), the molar selectivity factor is

$$K_{Ca^{2+},Na^+} = 5.2/2.0 = 2.6$$

For a given loading step during water softening, values of Q and C remain constant. Thus, for a given equivalent fraction, $x_{Ca^{2+}}$ in the feed, (15-181) is solved for the equilibrium $y_{Ca^{2+}}$. By material balance, for a given bed volume, the time t_L for the loading step is computed. The loading wavefront velocity is $u_L = L/t_L$ where L is the height of the bed. Equivalent fractions ahead of and behind the loading front are shown in Figure 15.50a. Typically, feed-solution superficial mass velocities are about 15 gal/h-ft², but can be much higher at the expense of larger pressure drops.

At the end of the loading step, the bed voids are filled with feed solution, which must be displaced from the bed. This is best done with a regeneration solution, which is usually a concentrated salt solution that flows upwards through the bed. Thus, the displacement and regeneration steps are combined. Following displacement, mass transfer of Ca^{2+} from the resin beads to the regenerating solution takes place while an equivalent amount of Na^+ is transferred from the solution to the resin. In order for equilibrium to be favorable for regeneration with Na^+ , it is necessary for $(Q/C)K_{Ca^{2+},Na^+} \ll 1$. In that case, which is just the opposite for loading, the wavefront during regeneration sharpens quickly into a shock-like wave. This criterion can be satisfied by using a saturated salt solution to give a large value for C .

During displacement and regeneration, two concentration waves move through the bed. The first is the displacement front; the second, the regeneration front. For plug flow and negligible mass-transfer resistance, the resin and solution are in equilibrium at all locations in the bed. Again (15-181) is used to solve for the equilibrium equivalent fractions, which are shown for the displacement and regeneration steps in Figure 15.50b. The displacement time, t_D , is determined from the interstitial velocity, u_D , of the fluid during displacement:

$$t_D = L/u_D \quad (15-182)$$

The regeneration time, t_R , is determined by material balance, from which the regeneration wavefront velocity is $u_R = L/t_R$. In general, the mass velocity of the regeneration solution is less than that of the feed solution. The cycle is completed by displacing, with water, the salt solution in the bed voids. The cycle calculations are illustrated by the following example.

EXAMPLE 15.19

Hard water, containing 500 ppm (by weight) of magnesium carbonate and 50 ppm of NaCl, is to be softened at 25°C in an existing fixed bed of gel resin of a cation capacity of 2.3 eq/L of bed volume. The bed is 8.5 ft in diameter and packed to a height of 10 ft, with a wetted-resin void fraction of 0.38. During the loading step, the recommended throughput is 15 gal/min-ft². During displacement, regeneration, and washing, the flow rate is reduced to 1.5 gal/min-ft². The displacement and regeneration solutions are water saturated with NaCl (26 wt%). Determine: (a) flow rate of feed solution, L/min, (b) loading time to breakthrough, h, (c) loading wavefront velocity, cm/min, (d) flow rate of regeneration solution, L/min, (e) displacement time, h, (f) additional time for regeneration, h, (g) regeneration wavefront velocity, cm/min, (h) amount of regeneration solution for one cycle, L, and (i) Washing time, h.

SOLUTION

Molecular weight, M , of $\text{MgCO}_3 = 83.43$

$$\begin{aligned}\text{Concentration of } \text{MgCO}_3 \text{ in feed} &= \frac{500(1,000)}{83.43(1,000,000)} \\ &= 0.006 \text{ mol/L or } 0.012 \text{ eq/L}\end{aligned}$$

M of $\text{NaCl} = 58.45$

$$\begin{aligned}\text{Concentration of } \text{NaCl} \text{ in feed} &= \frac{50(1,000)}{58.45(1,000,000)} \\ &= 0.000855 \text{ mol/L or eq/L}\end{aligned}$$

$$(a) \text{ Bed cross-section area} = 3.14(8.5)^2/4 = 56.7 \text{ ft}^2.$$

$$\begin{aligned}\text{Feed-solution flow rate} &= 15(56.7) \\ &= 851 \text{ gpm or } 3,219 \text{ L/min}\end{aligned}$$

(b) Behind the loading wavefront:

$$x_{\text{Ca}^{2+}} = \frac{0.012}{0.012 + 0.000855} = 0.9335$$

Since no NaCl in the feed is exchanged: $C = 0.012 \text{ eq/L}$ and $Q = 2.3 \text{ eq/L}$

From Table 15.5,

$$K_{\text{Mg}^{2+}, \text{Na}^+} = 3.3/2 = 1.65$$

From (15-181), for Mg^{2+} instead of Ca^{2+} as the exchanging ion, with $x_{\text{Mg}^{2+}} =$ that of the feed from Figure 15.50a:

$$1.65 \left(\frac{2.3}{0.012} \right) = \frac{(y_{\text{Mg}^{2+}}^*)(1 - 0.9335)}{(0.9335)(1 - y_{\text{Mg}^{2+}}^*)}$$

Solving: $y_{\text{Mg}^{2+}}^* = 0.9998$. Thus, sodium ion is displaced from the resin almost completely.

$$\text{Bed volume} = (56.7)(10) = 567 \text{ ft}^3 \text{ or } 16,060 \text{ L}$$

$$\text{Total bed capacity} = 2.3(16,060) = 36,940 \text{ eq}$$

$$\text{Mg}^{2+} \text{ absorbed by resin} = 0.9998(36,940) = 36,930 \text{ eq}$$

$$\text{Mg}^{2+} \text{ entering bed in feed solution} = 0.012(3,219) = 38.63 \text{ eq/min}$$

$$t_L = \frac{36,930}{38.63} = 956 \text{ min or } 15.9 \text{ h}$$

$$(c) u_L = L/t_L = 10/956 = 0.01046 \text{ ft/min or } 0.319 \text{ cm/min.}$$

$$(d) \text{ Flow rate of regeneration solution} = (1.5/15)(3,219) = 321.9 \text{ L/min.}$$

(e) Displacement time = time for 321.9 L/min to displace liquid in the voids.

$$\text{Void volume} = 0.38(16,060) = 6,103 \text{ L and } t_D = \frac{6103}{321.9} = 19 \text{ min}$$

(f) For a 26 wt% NaCl solution at 25°C , density from *Perry's Chemical Engineers' Handbook* = 1.19443 g/cm^3 .

Flow rate of Na^+ in regeneration solution

$$= \frac{321.9(1,000)(1.19443)(0.26)}{58.45} = 1,710 \text{ eq/min}$$

$$\begin{aligned}\text{NaCl concentration in regenerating solution} &= \frac{1,710}{321.9} \\ &= 5.31 \text{ eq/L} = c_R\end{aligned}$$

From (15-181), noting conditions in Figure 15.49:

$$\frac{Q}{c_R} K_{\text{Mg}^{2+}, \text{Na}^+} = 1.65 \left(\frac{2.3}{5.31} \right) = 0.715$$

This is less than 1, but not much less than 1. Therefore, the regeneration wavefront may not sharpen rapidly. Assume a shock-wave-like front anyway.

From (15-181),

$$0.715 = \frac{(0.99998)(1 - x_{\text{Mg}^{2+}}^*)}{x_{\text{Mg}^{2+}}^*(1 - 0.99998)}$$

Solving: $x_{\text{Mg}^{2+}}^* = 0.9998$

So downstream of the regeneration wavefront, but upstream of the displacement wavefront, the liquid contains very few sodium ions.

Chromatographic Separations

The separation of multicomponent mixtures into more than two products usually requires more than one separation device. For example, if a four-component mixture (A, B, C, D) is to be separated by distillation into pure products, a sequence of three trayed columns is almost always used. If the order of decreasing volatility is A, B, C, and D, the first column might produce a distillate of nearly pure A; the second column a distillate of nearly pure B; and the third column a distillate of nearly pure C and a bottoms of nearly pure D. Four other sequences are possible, depending upon the selection of the split for each column.

Chromatography is one of the few separation techniques that can separate a multicomponent mixture into nearly pure components in a single device, generally a column packed with a suitable sorbent. The degree of separation depends upon column length and differences in component affinities for the sorbent.

As an example, consider a mixture of three components, A, B, and C, in order of decreasing affinity for the sorbent, S. If the separation is achieved by adsorption, then A is the most strongly adsorbed. A feed mixture, insufficient to load the sorbent, is introduced as a pulse into one end (feed end) of the packed chromatographic column. The resulting initial concentrations for the three components are shown in Figure 15.51a, where most of the bed remains clean of adsorbates. An *elutant*, such as a carrier gas or solvent that has little or no affinity for the sorbent, is now introduced continuously into the feed end of the bed, causing the three components to desorb, with C desorbing most readily. However, as the desorbed components are carried down the bed by the elutant into cleaner regions of the bed, the components are successively readsorbed and then redesorbed to produce three waves, as shown in Figure 15.51b. Because of differences in affinities for the sorbent, the three waves, which initially overlap considerably, gradually overlap less (Figure 15.51c), and finally, if the column is long enough, become completely separated, as in Figure 15.51d and e. In that case, the components are eluted from the column, one at a time. In Figure 15.51e, all components but A have been eluted. As the separated waves elute, the area under each component wave is proportional to the mass of the component moving through the packed column.

Chromatography, as discussed here, is limited to batch processes in which feed is introduced as pulses into an elutant carrier gas or solvent, which then contacts the sorbent. All feed components have affinities for the sorbent, but the elutant does not. This type of chromatography is sometimes

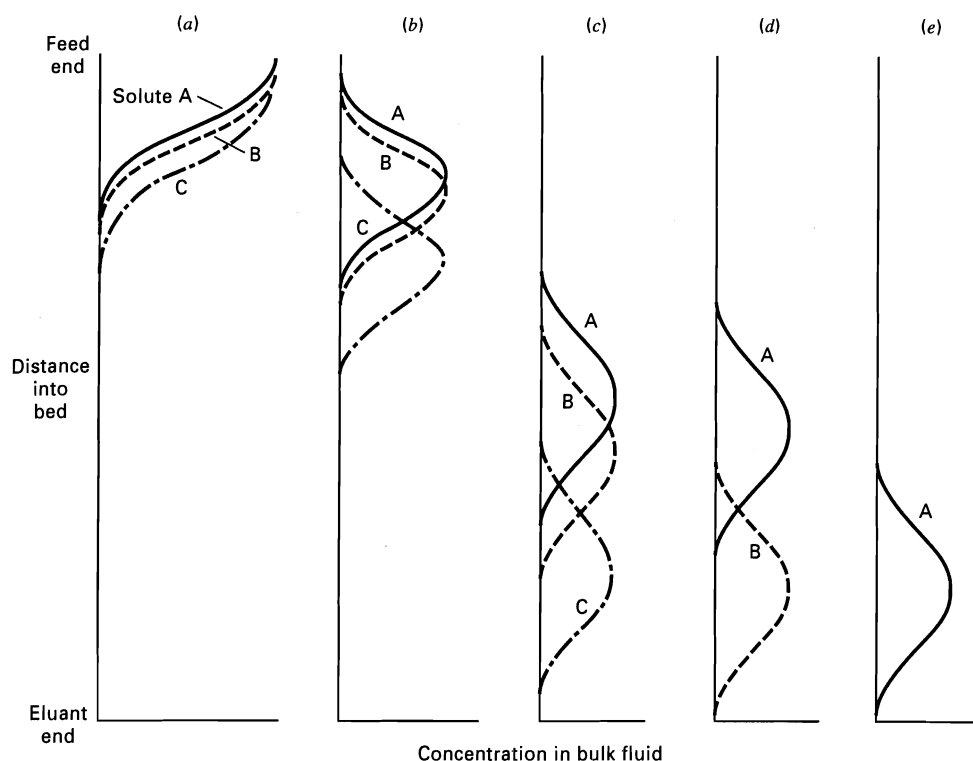


Figure 15.51 Movement of concentration waves during separation in a chromatographic column.

referred to as batch or elution chromatography. Chromatography in the broader sense, as mentioned by Ruthven [10], refers to any separation process involving partitioning of components between a flowing fluid and a solid adsorbent (or a solid-supported liquid absorbent). The previously presented simulated-moving-bed (SMB) system, which uses a circulating desorbent that may also partition, may be viewed as a chromatographic process.

Equilibrium Wave Pulse Theory for Linear Isotherm

A simple and useful wave theory for chromatography is based on isothermal, plug flow, negligible axial dispersion, and local equilibrium everywhere. This theory, when developed for adsorption, results in the stoichiometric wavefront that was shown in Figure 15.27. For chromatography, where solutes are pulsed into the column, a wave pulse rather than a wavefront results, as shown in Figure 15.52a. For a stoichiometric (equilibrium) wave, the pulse is a square wave rather than a Gaussian-distribution-like wave of the type shown in Figure 15.51c. The latter type of wave results when axial dispersion occurs, mass-transfer resistances are important, radial variations of the fluid velocity occur, and/or the pulse of solutes is not a square wave.

If the sorbent is nonporous, such as a gel, and the adsorption isotherm is linear for each solute i ($q = Kc$), then (15-101) applies and each solute wave velocity, u_i , in terms of the interstitial fluid velocity is given by

$$u_i = \frac{u}{1 + \frac{1 - \epsilon_b}{\epsilon_b} K_i} \quad (15-183)$$

This equation applies to both the leading and trailing edges of the feed pulse to produce solute movement diagrams, used extensively by Wankat [95]. When the eluant is dilute

in the solutes, such that the sorption equilibrium constants do not depend on composition, but only on temperature, (15-183) applies independently to each solute in a multi-component mixture. For a strongly sorbed solute, such as B in Figure 15.52b, K is large, and the second term in the

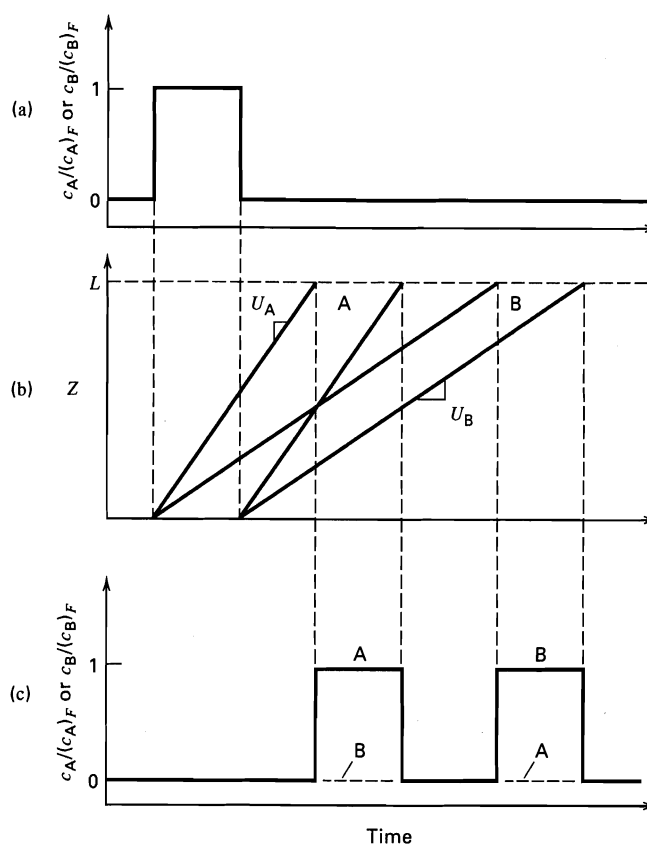


Figure 15.52 Ideal solute wave pulses in a chromatographic column.

denominator dominates, so that solute moves much slower through the bed than does the elutant. For a weakly sorbed solute, such as A in Figure 15.52b, K is small and the denominator is not much greater than 1. For such a solute, its velocity may not be significantly smaller than the elutant velocity.

In Figure 15.52b, the wave velocities of both the leading and trailing edges of the wave pulses are constant, with $u_A > u_B$. For each solute, the pulse time to move through a column of length L is $t_i = L/u_i$. The wave pulse of A reaches the end of the column in less time than the B wave pulse. In Figure 15.52c, the product-concentration ratios, c_A/c_{A_F} and c_B/c_{B_F} , are shown at the end of the bed as a function of time. The widths of these product waves are identical to the widths of the feed pulses, as illustrated in the following example. This simple, equilibrium-wave pulse theory for linear isotherms can be used to obtain an approximate estimate of the separation achievable in a chromatographic column. Unfortunately, the estimate is not conservative when computing the necessary column length, because the pulses broaden, as shown after the following example.

EXAMPLE 15.20

An aqueous solution of 3 g/cm³ each of glucose (G), sucrose (S), and fructose (F) is to be separated in a chromatographic column, packed with an ion-exchange resin of the calcium form. In the range of expected solute concentrations, the sorption isotherms are linear and independent, with $q_i = K_i c_i$, where q_i is in grams sorbent per 100 cm³ resin and c_i is in grams solute per 100 cm³ solution. From experiment:

Solute	K
Glucose	0.26
Sucrose	0.40
Fructose	0.66

The superficial solution velocity is 0.031 cm/s and bed void fraction is 0.39. If a 500-second pulse, t_p , of feed is followed by elution with pure water, what length of column packing is needed to separate the three solutes if sorption equilibrium is assumed? How soon after the first pulse begins can a second 500-second pulse begin?

SOLUTION

Interstitial solution velocity = $0.031/0.39 = 0.0795$ cm/s.

Wave velocity for glucose from (15-183):

$$u_G = \frac{0.0795}{1 + \left(\frac{1 - 0.39}{0.39}\right)(0.26)} = 0.0565 \text{ cm/s}$$

Similarly,

$$u_S = 0.0489 \text{ cm/s} \quad \text{and} \quad u_F = 0.0391 \text{ cm/s}$$

The smallest difference in wave velocities is between glucose and sucrose. Therefore, the separation between these two waves determines the column length. The minimum column length, assuming equilibrium, corresponds to the time at which the trailing edge of the glucose wave pulse, together with the leading edge of the

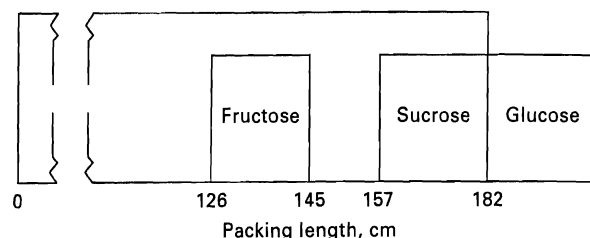


Figure 15.53 Locations of solute waves of first pulse for Example 15.20 at 3,718 s.

sucrose wave pulse, leaves the column. Thus, if t_p is the duration of the first pulse and L is the length of the packing:

$$t_p + \frac{L}{u_G} = \frac{L}{u_S} \quad (1)$$

Thus,

$$500 + \frac{L}{0.0565} = \frac{L}{0.0489}$$

Solving, length of packing, $L = 182$ cm. The glucose just leaves the column at

$$500 + \frac{182}{0.0565} = 3,718 \text{ s}$$

The locations of the three wave fronts in the column at 3,718 s are shown in Figure 15.53.

The time at which the second pulse begins is determined so that the trailing edge of the first fructose wave pulse just leaves the column as the second pulse of glucose begins to leave the column. This time, based on the fructose, is $500 + 182/0.0391 = 5,155$ s. It takes the leading edge of a glucose wave $182/0.0565 = 3,220$ s to pass through the column. Therefore, the second pulse can begin at $5,155 - 3,220 = 1,935$ s. This establishes the following ideal cycle: pulse: 500 s, elute: 1,435 s, pulse: 500 s, elute: 1,435 s, etc. In the real case, where we account for mass-transfer resistance, as shown in the next example, the column will have to be longer.

Analytical Solution for Rate-based Chromatography with Linear Isotherm

A plot of solute concentrations in the elutant as a function of time is a chromatogram. When mass-transfer resistances, axial dispersion, and other non-ideal phenomena are not negligible, the solute concentrations in a chromatogram will not appear as square waves, but will exhibit the wave shapes in Figure 15.51. Carta [96] developed analytical solutions for chromatographic response to periodic injections of rectangular feed pulses, taking into account mass-transfer resistances for solute mixtures having linear, independent adsorption isotherms. Carta's solution for the LDF approximation is readily applied to the determination of the necessary length of packing and frequency of feed pulses for the chromatographic separation of a feed mixture.

For each solute in the feed, (15-102) is simplified by neglecting axial dispersion and assuming a constant interstitial fluid velocity, u , through the packing:

$$\frac{\partial c_i}{\partial t} + u \frac{\partial c_i}{\partial z} + \frac{(1 - \epsilon_b)}{\epsilon_b} \frac{\partial \bar{q}_i}{\partial t} = 0 \quad (15-184)$$

The linear driving force approximation (15-105), the linear isotherm, and (15-106) for the overall mass-transfer resistance are assumed to apply for each solute. For periodic, rectangular feed pulses, the boundary conditions for feed pulses of duration, t_F , each followed by an elution period of duration, t_E , are for each solute concentration, $c_i\{z, t\}$:

Initial condition:

$$\text{At } t = 0, c_i\{z, 0\} = 0 \quad (15-185)$$

Feed pulse:

$$\begin{aligned} \text{At } z = 0, c_i\{0, t\} &= (c_i)_F \\ \text{for } (j-1)(t_F + t_E) < t < j(t_F + t_E) - t_E \end{aligned} \quad (15-186)$$

Elution period:

$$\begin{aligned} \text{At } z = 0, c_i\{0, t\} &= 0 \\ \text{for } j(t_F + t_E) - t_E < t < j(t_F + t_E) \end{aligned} \quad (15-187)$$

where $j = 1, 2, 3, \dots$ is an index that accounts for the periodic nature of the feed and elution pulses. Thus, with $j = 1$, the feed pulse takes place from $t = 0$ to $t = t_F$ and the elution pulse is from t_F to $t_F + t_E$.

Carta solved the linear system of (15-184), (15-105), and (15-106) for conditions (15-185) to (15-187) by the Laplace transform method to obtain the following series solution, in terms of dimensionless parameters, which is applied to each solute in the feed pulse:

$$\begin{aligned} X &= \frac{r_F}{2r} + \frac{2}{\pi} \sum_{m=1}^{\infty} \left[\frac{1}{m} \exp\left(-\frac{m^2 n_f}{m^2 + r^2}\right) \sin\left(\frac{m\pi r_F}{2r}\right) \right. \\ &\quad \left. \times \cos\left(\frac{m\theta_f}{r} - \frac{m\pi r_F}{2r} - \frac{m\beta n_f}{r} - \frac{mr n_f}{m^2 + r^2}\right) \right] \end{aligned} \quad (15-188)$$

where

$$X = c/c_F \quad (15-189)$$

$$r = \frac{k}{2\pi K}(t_F + t_E) \quad (15-190)$$

$$r_F = \frac{k}{\pi K} t_F \quad (15-191)$$

$$n_f = \frac{(1 - \epsilon_b)kz}{\epsilon_b u} \quad (15-192)$$

$$\theta_f = \frac{kt}{K} \quad (15-193)$$

$$\beta = \frac{\epsilon_b}{(1 - \epsilon_b)K} \quad (15-194)$$

$$K = q/c \quad (15-195)$$

and where

$$k = \frac{1}{\frac{R_p}{3k_c} + \frac{R_p^2}{15D_e}} \quad (15-196)$$

When nonlinear adsorption isotherms such as the Freundlich equation (15-35), the Langmuir equation (15-36), or extensions thereof to multicomponent mixtures [e.g., concentration forms of (15-32) or (15-33)] are necessary, the analytical solution of Carta is not applicable. However, the method of lines, using five-point, biased, upwind, finite-difference approximations, as described earlier in the section on thermal-swing adsorption, can be applied to obtain a numerical solution.

EXAMPLE 15.21

Use Carta's equation with the following properties to compute the chromatogram for the conditions of Example 15.20 with a packing length of 182 cm. Does a significant overlap of peaks result?

Property	Glucose	Sucrose	Fructose
K	0.26	0.40	0.66
D_e , cm ² /s	1.1×10^{-8}	1.8×10^{-8}	2.8×10^{-8}
k_c , cm/s	5.0×10^{-3}	5.0×10^{-3}	5.0×10^{-3}

$$\begin{aligned} \epsilon_b &= 0.39, R_p = 0.0025 \text{ cm}, u = 0.0795 \text{ cm/s}, \\ z &= 182 \text{ cm}, t_E = 2,000 \text{ s}, \text{ and } t_F = 500 \text{ s} \end{aligned}$$

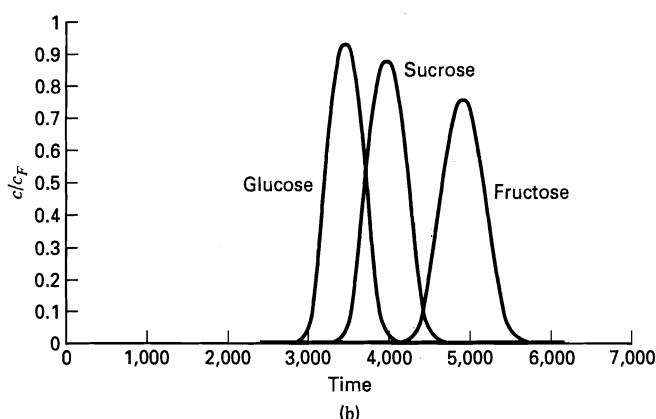
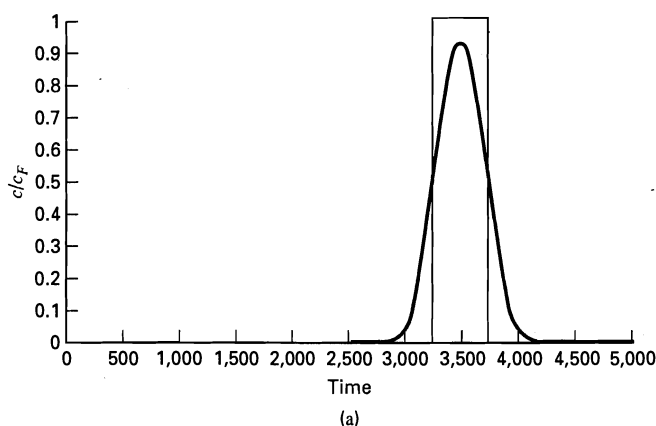


Figure 15.54 Computed chromatograms for Example 15.21. (a) Comparison of ideal to nonideal wave for fructose. (b) Computed chromatogram for nonideal eluant.

SOLUTION

Values of k and the computed dimensionless parameters from (15-148) to (15-152) are as follows:

	Glucose	Sucrose	Fructose
r	40.22	42.66	40.06
r_F	16.09	17.07	16.03
n_f	94.13	153.6	238.0
θ_f	$0.1011 t$	$0.1072 t$	$0.1007 t$
β	2.459	1.598	0.9687
k, s^{-1}	0.0263	0.0429	0.0665

where t is in seconds.

SUMMARY

1. Sorption is a generic term for the selective transfer of a solute from the bulk of a liquid or gas to the surface and/or into the bulk of a solid or liquid. Thus, sorption includes adsorption and absorption. The sorbed solute is commonly called the sorbate.
2. For commercial applications, a sorbent should have high selectivity, high capacity, rapid solute transport rates, stability, strength, and ability to be regenerated. An adsorbent should have small pores so as to give a large surface area per unit volume.
3. Physical adsorption of pure gases and gas mixtures is easily measured. Adsorption of pure liquids and liquid mixtures is not easily measured.
4. The most widely used commercial adsorbents are carbon (activated and molecular-sieve), molecular-sieve zeolites, silica gel, and activated alumina.
5. The most widely used ion exchangers are water-swallowable, solid gel resins based on the copolymerization of styrene and a cross-linking agent, such as divinylbenzene. They can be cation or anion exchangers. Ions are exchanged stoichiometrically on an equivalent basis. Thus, Ca^{2+} is exchanged for $2 Na^+$.
6. Sorbents for chromatographic separations are typically solid adsorbents, liquid adsorbents supported on or bonded to an inert solid, or a gel.
7. The most commonly used adsorption isotherms for gases and liquids are Henry's law (linear isotherm), the Freundlich isotherm, and the Langmuir isotherm. The latter asymptotically approaches the linear isotherm at low concentrations and an asymptotic value, representing maximum surface coverage, at high concentrations. For mixtures, extended versions of the Freundlich and Langmuir isotherms are often used.
8. Ion-exchange equilibrium is most commonly represented by an equilibrium constant based on the law of mass action. Because of the dilute conditions common in chromatography, a linear equilibrium isotherm is commonly employed.
9. For physical adsorption, the rate of adsorption is almost instantaneous after the solute reaches the sorbing surface. Thus, only external and internal mass-transfer resistances need be considered. External mass-transfer coefficients are generally obtained from empirical correlations of the Chilton-Colburn j -factor type. Internal mass transfer is generally based on a modified Fick's first law using an effective diffusivity that depends on particle porosity, pore

Values of $X = c/c_F$ are computed with these parameters using (15-188) for values of time, t , in the neighborhood of times for the equilibrium-based waves. The resulting chromatogram for glucose is shown in Figure 15.54a, compared to the equilibrium rectangular wave (shown as a dashed line) determined in Example 15.20 using (15-183). The areas under the two curves should be identical. The equilibrium-based wave appears to be centered in time within the mass-transfer-based wave.

In Figure 15.54b, the complete computed chromatogram is plotted for the three carbohydrates. It is seen that the effect of mass transfer is to cause the peaks to overlap significantly. To obtain a sharp separation, it is necessary to lengthen the column or reduce the feed pulse time, t_F .

tortuosity, bulk molecular diffusivity, and surface diffusivity. Diffusivities in ion-exchange resin gels depend strongly on the degree of cross-linking.

10. A wide variety of sorption systems are used, including slurry adsorption in various modes of operation, fixed-bed adsorption, and simulated, continuous, countercurrent adsorption. When sorbent regeneration is necessary, the system must be operated on a cycle. For fixed beds, the most common cycles are temperature-swing adsorption (TSA) and pressure-swing adsorption (PSA). Ion exchange almost always includes a regeneration step, using a displacement fluid. In a chromatographic separation, adsorption and regeneration take place in the same column.

11. For the design and operation of all sorption systems, the adsorption isotherm is of great importance because it relates, at equilibrium, the concentration of the solute in the fluid to its loading as a sorbate in and/or on the sorbent. Most commonly, the overall rate of adsorption is expressed in the form of a linear driving force (LDF) model, where the driving force is the difference between bulk concentration and concentration in equilibrium with the loading. The coefficient in the LDF equation is a combined overall mass-transfer coefficient and area for sorption.

12. In ideal, fixed-bed operation, solute-sorbate equilibrium between the flowing fluid and the static bed is assumed everywhere. For plug flow and negligible axial dispersion, the result is a sharp concentration front that moves like a shock wave (stoichiometric front) through the bed. Upstream of the front, the sorbent is spent and in equilibrium with the feed mixture. Downstream of the front, the sorbent is clean of sorbate. Typically, the stoichiometric front travels through the bed at a much slower velocity than the interstitial velocity of the fluid feed. The time for the concentration front to reach the end of the bed is the breakthrough time.

13. When mass-transfer effects are taken into account, the concentration front broadens into an S-shaped curve such that at breakthrough only a portion of the sorbent is fully loaded. When mass-transfer coefficients and sorption isotherms are known, these curves can be readily computed with the Klinkenberg equations. Alternatively, when the shapes of experimental concentration fronts appear to exhibit a constant pattern, because of favorable adsorption equilibrium, commercial-size adsorption beds can be scaled-up directly from experimental breakthrough data by the method of Collins.

14. Thermal-swing adsorption (TSA) can be used to remove small concentrations of solutes from gas and liquid mixtures. Typically, adsorption is carried out at ambient temperature and desorption at an elevated temperature. Because bed heating and cooling between the adsorption and desorption steps is not instantaneous, TSA cycles are long, typically hours or days. The desorption step, starting with a partially loaded bed, can be computed numerically by the method of lines, using a stiff integrator.

15. Pressure-swing adsorption (PSA) is used to separate air and enrich hydrogen-containing streams. Adsorption is carried out at an elevated or ambient pressure, whereas desorption occurs at ambient pressure or in a vacuum; the latter is often referred to as vacuum-swing adsorption (VSA). Because pressure swings can be made rapidly, PSA cycles are short, typically seconds or minutes. Usually, it is not necessary to regenerate the bed completely. When that circumstance exists and the cycle is fixed, a number of cycles may be needed to approach a cyclic steady-state operation.

16. Although continuous, countercurrent adsorption with a moving bed is difficult to achieve successfully in practice, an SMB system is becoming popular, particularly for separation of solutes

in dilute aqueous solutions and for bulk-liquid separations. Design procedures for SMB systems, which require solution of systems of differential-algebraic equations, are highly developed.

17. Design calculations for ion-exchange operations are based on the equilibrium assumption for both the loading and regeneration steps.

18. In chromatography, the feed is periodically pulsed into a column packed with sorbent. Between feed pulses, an elutant is passed through the column, causing the less strongly sorbed solutes to move through the column more rapidly than other solutes. If the column is long enough, a multicomponent feed can be completely separated, with solutes eluted one by one from the column. In the absence of mass-transfer resistances, a rectangular feed pulse is separated into individual solute rectangular pulses, whose position-time curves are readily established. When mass-transfer effects are important, the rectangular pulses take on a Gaussian distribution that can be predicted by the analytical solution of Carta, provided that a linear adsorption isotherm applies and axial dispersion is negligible.

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EXERCISES

Section 15.1

15.1 Porous particles of activated alumina have a BET surface area of 310 m²/g, a particle porosity of 0.48, and a particle density of 1.30 g/cm³. Determine: (a) specific pore volume in cm³/g, (b) true solid density, g/cm³, and (c) approximate pore diameter in angstroms from (15-2).

15.2 Carbon molecular sieves are available in two forms from a Japanese manufacturer:

	Form A	Form B
Pore volume, cm ³ /g	0.18	0.38
Average pore diameter	5 Å	2.0 μm

Estimate the surface area of each form.

15.3 Representative properties of small-pore silica gel are as follows: pore diameter = 24 Å; particle porosity = 0.47; particle density = 1.09 g/cm³; and specific surface area = 800 m²/g

(a) Are these values reasonably consistent? (b) If the adsorption capacity for water vapor at 25°C and 6 mmHg partial pressure is 18% by weight, what fraction of a monolayer is adsorbed?

15.4 The following data were obtained in a BET apparatus for adsorption equilibrium of nitrogen on silica gel (SG) at −195.8°C. Estimate the specific surface area in m²/g of silica gel. How does your value compare with that in Table 15.2?

N ₂ Partial Pressure, torr	Volume of N ₂ Adsorbed in cm ³ (0°C, 1 atm) per gram SG
6.0	6.1
24.8	12.7
140.3	17.0
230.3	19.7
285.1	21.5
320.3	23.0
430	27.7
505	33.5

15.5 Estimate the maximum ion-exchange capacity in meq/g resin for an ion-exchange resin made from 8 wt% divinylbenzene and 92 wt% styrene.

Section 15.2

15.6 Shen and Smith [*Ind. Eng. Chem. Fundam.*, **7**, 100–105 (1968)] measured equilibrium-adsorption isotherms at four different temperatures for pure benzene vapor on silica gel, having the following properties: surface area = 832 m²/g, pore volume = 0.43 cm³/g, particle density = 1.13 g/cm³, and average pore diameter = 22 Å.

The adsorption data are as follows:

Partial Pressure of Benzene, atm	Moles Adsorbed/g Gel × 10 ⁵			
	70°C	90°C	110°C	130°C
5.0 × 10 ^{−4}	14.0	6.7	2.6	1.13
1.0 × 10 ^{−3}	22.0	11.2	4.5	2.0
2.0 × 10 ^{−3}	34.0	18.0	7.8	3.9
5.0 × 10 ^{−3}	68.0	33.0	17.0	8.6
1.0 × 10 ^{−2}	88.0	51.0	27.0	16.0
2.0 × 10 ^{−2}	—	78.0	42.0	26.0

(a) For each temperature, obtain a best fit of the data to (1) linear, (2) Freundlich, and (3) Langmuir isotherms. Which isotherm(s), if any, fit the data reasonably well?

(b) Do the data represent less than a monolayer of adsorption?

(c) From the data, estimate the heat of adsorption. How does this value compare to the heat of vaporization (condensation) of benzene?

15.7 The separation of propane and propylene is accomplished by distillation, but at the expense of more than 100 trays and a reflux ratio of greater than 10. Consequently, the use of adsorption has been investigated in a number of studies. Jarvelin and Fair [*Ind. Eng. Chem. Research*, **32**, 2201–2207 (1993)] measured adsorption-equilibrium data at 25°C for three different zeolite molecular sieves (ZMSs) and activated carbon. The data were fitted to the Langmuir isotherm with the following results:

Adsorbent	Sorbate	q_m	K
ZMS 4A	C ₃	0.226	9.770
	C ₃ [−]	2.092	95.096
ZMS 5A	C ₃	1.919	100.223
	C ₃ [−]	2.436	147.260
ZMS 13X	C ₃	2.130	55.412
	C ₃ [−]	2.680	100.000
Activated carbon	C ₃	4.239	58.458
	C ₃ [−]	4.889	34.915

where q and q_m are in mmol/g and p is in bar.

(a) Which component is most strongly adsorbed by each of the adsorbents? (b) Which adsorbent has the greatest adsorption capacity? (c) Which adsorbent has the greatest selectivity? (d) Based on equilibrium considerations, which adsorbent is best for the separation?

15.8 Ruthven and Kaul [*Ind. Eng. Chem. Res.*, **32**, 2047–2052 (1993)] measured adsorption isotherms for a series of gaseous aromatic hydrocarbons on well-defined crystals of NaX zeolite over ranges of temperature and pressure. For 1,2,3,5-tetramethylbenzene

at 547 K, the following equilibrium data were obtained with a vacuum microbalance:

q , wt%	7.0	9.1	10.3	10.8	11.1	11.5
p , torr	0.012	0.027	0.043	0.070	0.094	0.147

Obtain a best fit of the data to the linear, Freundlich, and Langmuir isotherms, with q in mol/g and pressure in atm. Which isotherm gives the best fit?

15.9 Lewis, Gilliland, Chertow, and Hoffman [*J. Am. Chem. Soc.*, **72**, 1153–1157 (1950)] measured adsorption equilibria for pure propane, pure propylene, and binary mixtures thereof, on activated carbon and silica gel. Adsorbate capacity was high on carbon, but selectivity was poor. Selectivity was high on silica gel, but capacity was low. For silica gel (751 m²/g), the following pure component data were obtained at 25°C:

Propane		Propylene	
P , torr	q , mmol/g	P , torr	q , mmol/g
11.1	0.0564	34.2	0.3738
25.0	0.1252	71.4	0.7227
43.5	0.1980	91.6	0.7472
71.4	0.2986	194.3	1.129
100.0	0.3850	198.3	1.168
158.9	0.5441	271.5	1.401
227.5	0.7020	353.2	1.562
304.2	0.843	550.7	1.918
387.0	1.010	555.2	1.928
468.0	1.138	760.6	2.184
569.0	1.288		
677.8	1.434		
775.0	1.562		

The following mixture data were measured at 25°C, over a pressure range of 752–773 torr:

Total Pressure, torr	Millimoles of Mixture Adsorbed/g	y_{C_3} , Mole Fraction in Gas Phase	x_{C_3} , Mole Fraction in Adsorbate
769.2	2.197	0.2445	0.1078
760.9	2.013	0.299	0.2576
767.8	2.052	0.4040	0.2956
761.0	2.041	0.530	0.2816
753.6	1.963	0.5333	0.3655
766.3	1.967	0.5356	0.3120
754.0	1.974	0.6140	0.3591
753.6	1.851	0.6220	0.5550
754.0	1.701	0.6252	0.7007
760.0	1.686	0.7480	0.723
—	2.180	0.671	0.096
760.0	1.993	0.8964	0.253
760.0	1.426	0.921	0.401

(a) Fit the pure component data to Freundlich and Langmuir isotherms. Which gives the best fit? Which component is most strongly adsorbed?

(b) Use the results of the Langmuir fits in part (a) to predict binary-mixture adsorption using the extended Langmuir equation, (15-32). Are the predictions adequate?

(c) Ignoring the pure-component data, fit the binary-mixture data to the extended Langmuir equation, (15-32). Is the fit better than that obtained in part (b)?

(d) Ignoring the pure-component data, fit the binary mixture data to the extended Langmuir–Freundlich equation, (15-33). Is the fit adequate? Is the fit better than that in part (c)?

(e) For the binary-mixture data, compute the relative selectivity,

$$\alpha_{C_3, C_2} = y_{C_3}(1 - x_{C_3})/[x_{C_3}(1 - y_{C_3})]$$

for each condition. Does α vary widely or is the assumption of constant α reasonable?

15.10 In Example 15.6, pure-component, liquid-phase adsorption data are used with the extended-Langmuir isotherm to predict a binary-solute data point. Use the following mixture data to obtain the best fit to an extended Langmuir–Freundlich isotherm of the form

$$q_i = \frac{(q_0)_i k_i c_i^{1/n_i}}{1 + \sum_j k_j c_j^{1/n_j}} \quad (1)$$

Data for binary-mixture adsorption on activated carbon (1000 m²/g) at 25°C for acetone (1) and propionitrile (2) are as follows:

Solution Concentration, mol/L		Loading, mmol/g	
c_1	c_2	q_1	q_2
5.52E – 5	7.46E – 5	0.0192	0.0199
6.14E – 5	7.71E – 5	0.0191	0.0198
1.06E – 4	1.35E – 4	0.0308	0.0320
1.12E – 4	1.46E – 4	0.0307	0.0319
3.03E – 4	2.32E – 3	0.0378	0.263
3.17E – 4	2.34E – 3	0.0378	0.264
3.25E – 4	3.89E – 4	0.0644	0.0672
1.42E – 3	1.58E – 3	0.161	0.169
1.42E – 3	1.61E – 3	0.161	0.169
1.43E – 3	1.60E – 3	0.161	0.169
2.09E – 3	3.84E – 4	0.250	0.0390
2.17E – 3	3.85E – 4	0.251	0.0392
4.99E – 3	5.24E – 3	0.291	0.307
5.06E – 3	5.31E – 3	0.288	0.305
7.41E – 3	2.42E – 2	0.237	0.900
7.52E – 3	2.47E – 2	0.236	0.896
2.79E – 2	7.59E – 3	0.802	0.251
4.00E – 2	3.44E – 2	0.715	0.822
4.02E – 2	3.42E – 2	0.717	0.834

15.11 Sircar and Myers [*J. Phys. Chem.*, **74**, 2828–2835 (1970)] measured liquid-phase adsorption at 30°C for a binary mixture of cyclohexane (1) and ethyl alcohol (2) on activated carbon. Assuming no adsorption of ethyl alcohol, they used (15-34) to obtain the

following results:

x_1	q_1^e , mmol/g	x_1	q_1^e , mmol/g
0.042	0.295	0.440	0.065
0.051	0.485	0.470	0.000
0.072	0.517	0.521	-0.129
0.148	0.586	0.537	-0.362
0.160	0.669	0.610	-0.643
0.213	0.661	0.756	-1.230
0.216	0.583	0.848	-1.310
0.249	0.595	0.893	-1.180
0.286	0.532	0.920	-1.230
0.341	0.383	0.953	-0.996
0.391	0.192	0.974	-0.470

- (a) Plot the data as q_1^e against x_1 . Explain the shape of the curve.
 (b) In what regions of concentration could the Freundlich isotherm be fitted to the data? Make the fits.

15.12 Both the adsorptive removal of small amounts of toluene from water and small amounts of water from toluene are important in the process industries. Activated carbon is particularly effective for removing soluble organic compounds (SOCs) from water. Activated alumina is effective for removing soluble water from toluene. Fit each of the following two sets of equilibrium data for 25°C to both the Langmuir and Freundlich isotherms. For each case, which isotherm provides the better fit? Could a linear isotherm be used?

Toluene (in Water) Activated Carbon		Water (in Toluene) Activated Alumina	
c , mg/L	q , mg/g	c , ppm (by Weight)	q , g/100g
0.01	12.5	25	1.9
0.02	17.1	50	3.1
0.05	23.5	75	4.2
0.1	30.3	100	5.1
0.2	39.2	150	6.5
0.5	54.5	200	8.2
1	90.1	250	9.5
2	70.2	300	10.9
5	125.5	350	12.1
10	165	400	13.3

15.13 Derive (15-44). Use this equation to solve the following problem. Sulfate ion is to be removed from 60 L of water by exchanging it with chloride ion on 1 L of a strong-base resin with relative molar selectivities as listed in Table 15.6 and an ion-exchange capacity of 1.2 eq/L of resin. The water to be treated has a sulfate-ion concentration of 0.018 eq/L and a chloride-ion concentration of 0.002 eq/L. Following the attainment of equilibrium ion exchange, the treated water will be removed and the resin will be regenerated with 30 L of 10 wt% aqueous NaCl.

- (a) Write the ion-exchange reaction.
 (b) Determine the value of $K_{\text{SO}_4^{2-}, \text{Cl}^-}$.
 (c) Calculate equilibrium concentrations $c_{\text{SO}_4^{2-}}$, c_{Cl^-} , $q_{\text{SO}_4^{2-}}$, and q_{Cl^-} in eq/L for the initial ion-exchange step.
 (d) Calculate the concentration of Cl^- in eq/L for the regenerating solution.

(e) Calculate $c_{\text{SO}_4^{2-}}$, c_{Cl^-} , $q_{\text{SO}_4^{2-}}$, and q_{Cl^-} upon reaching equilibrium in the regeneration step.

(f) Are the separations sufficiently selective?

15.14 Silver ion in methanol was exchanged with sodium ion using Dowex 50 cross-linked with 8% divinylbenzene by Gable and Stroebel [*J. Phys. Chem.*, **60**, 513–517 (1956)]. The molar selectivity coefficient was found to vary somewhat with the equivalent fraction of Na^+ in the resin as follows:

x_{Na^+}	0.1	0.3	0.5	0.7	0.9
$K_{\text{Ag}^+, \text{Na}^+}$	11.2	11.9	12.3	14.1	17.0

If the wet capacity of the resin is 2.5 eq/L and the resin is initially saturated with Na^+ , calculate the equilibrium equivalent fractions if 50 L of 0.05-M Ag^+ in methanol is treated with 1 L of wet resin.

15.15 Ion exclusion is a process that uses ion-exchange resins to separate nonionic organic compounds from ionic species contained in a polar solvent, usually water. The resin is presaturated with the same ions as in the solution, thus eliminating ion exchange. However, in the presence of the polar solvent, resins undergo considerable swelling by absorbing the solvent. Experiments have shown that a nonionic solute will distribute between the solution outside the resin and the solution within the resin, while the ions can only exchange.

A feed solution of 1,000 kg contains 6 wt% NaCl, 35 wt% glycerol, and 47 wt% water. This solution is to be treated with Dowex-50 ion-exchange resin in the sodium form, after prewetting with water, to recover 75% of the glycerol. The following data for the glycerol distribution coefficient,

$$K_d = \frac{\text{mass fraction in solution inside resin}}{\text{mass fraction in solution outside resin}}$$

were reported by Asher and Simpson [*J. Phys. Chem.*, **60**, 518–521 (1956)]:

Mass Fraction Glycerol in Solution Outside Resin	K_d	
	6 wt% NaCl	12 wt% NaCl
0.10	0.75	0.91
0.20	0.80	0.93
0.30	0.83	0.95
0.40	0.85	0.97

If the prewetted resin contains 40 wt% water, determine the kilograms of resin (dry basis) required.

Section 15.3

15.16 Benzene vapor in an air stream is adsorbed in a fixed bed of 4×6 mesh silica gel packed to an external void fraction of 0.5. The bed is 2 feet in inside diameter and the air flow rate is 25 lb/min (benzene-free basis). At a location in the bed where the pressure is 1 atm, the temperature is 70°F, and the bulk mole fraction of benzene is 0.005, estimate the external, gas-to-particle mass-transfer and heat-transfer coefficients.

15.17 Water vapor in an air stream is to be adsorbed in a 12.06-cm-inside-diameter column packed with 3.3-mm-diameter Alcoa F-200 activated alumina beads with an external porosity of 0.442. At a location in the bed where the pressure is 653.3 kPa, the temperature is 21°C, the gas flow rate is 1.327 kg/min, and the

dew-point temperature is 11.2°C, estimate the external, gas-to-particle mass-transfer and heat-transfer coefficients.

15.18 For the conditions of Example 15.8 estimate the effective diffusivity of acetone vapor in the pores of activated carbon with the following properties: particle density = 0.85 g/cm³, particle porosity = 0.48, average pore diameter = 25 Å, and tortuosity = 2.75.

Consider both bulk and Knudsen diffusion, but ignore surface diffusion.

15.19 For the conditions of Exercise 15.16, estimate the effective diffusivity of benzene vapor in the pores of silica gel with the following properties: particle density = 1.15 g/cm³, particle porosity = 0.48, average pore diameter = 30 Å, and tortuosity = 3.2.

Consider all mechanisms of diffusion. The adsorption equilibrium constant is given in Example 15.11, and the differential heat of adsorption is -11,000 cal/mol.

15.20 For the conditions of Exercise 15.17, estimate the effective diffusivity of water vapor in the pores of activated alumina with the following properties: particle density = 1.38 g/cm³, particle porosity = 0.52, average pore diameter = 60 Å, and tortuosity = 2.3.

Consider all mechanisms of diffusion except surface diffusion.

Section 15.4

15.21 Adsorption with activated carbon, made from bituminous coal, of soluble organic compounds (SOCs) to purify surface and ground water is a proven technology, as discussed by Stenzel [*Chem. Eng. Prog.*, **89** (4), 36-43 (1993)]. The less-soluble organic compounds, such as chlorinated organic solvents and aromatic solvents, are the more strongly adsorbed. Water containing 3.3 mg/L of trichloroethylene (TCE) is to be treated with activated carbon to obtain an effluent with only 0.01 mg TCE/L. At 25°C, adsorption equilibrium data for TCE on activated carbon are correlated with the following Freundlich equation:

$$q = 67 c^{0.564} \quad (1)$$

where

q = mg TCE/g carbon and c = mg TCE/L solution

The TCE is to be removed by slurry adsorption using a powdered form of the activated carbon, with an average particle diameter of 1.5 mm. In the absence of any laboratory data on mass-transfer rates, assume that the rate of adsorption for the small particles is controlled by external mass transfer with a Sherwood number of 30. Particle surface area is 5 m²/kg. The molecular diffusivity of TCE in low concentrations in water at 25°C may be determined from the Wilke-Chang equation.

- Determine the minimum amount of adsorbent needed.
- For operation in the batch mode with twice the minimum amount of adsorbent, determine the time to reduce the TCE content to the desired value.
- For operation in the continuous mode using twice the minimum amount of adsorbent, determine the required residence time.
- For operation in the semicontinuous mode at a feed rate of 50 gpm and for a liquid residence time equal to 1.5 times that computed in part (c), determine the amount of activated carbon to give a reasonable vol% solids in the tank and a run time of not less than 10 times the liquid residence time.

15.22 Repeat Exercise 15.21 for water containing 0.324 mg/L of benzene (B) and 0.630 mg/L of m-xylene (X).

Adsorption isotherms at 25°C for these low concentrations are essentially independent and are given by

$$q_B = 32 c_B^{0.428} \quad (1)$$

$$q_X = 125 c_X^{0.333} \quad (2)$$

The feed concentrations of the SOC's in the feed are to be reduced to 0.002 mg/L each.

15.23 Repeat Exercise 15.21 for water containing 0.223 mg/L chloroform, whose concentration is to be reduced to 0.010 mg/L. The adsorption isotherm at 25°C is given by

$$q = 10 c^{0.564} \quad (1)$$

15.24 Three fixed-bed adsorbers containing 10,000 lb of granules of activated carbon ($\rho_b = 30 \text{ lb/ft}^3$) each are to be used to treat 250 gpm of water containing 4.6 mg/L of 1,2-dichloroethane (D) to reduce the concentration to less than 0.001 mg/L. Each carbon bed has a height equal to twice the diameter. Two beds are to be placed in series so that when Bed 1 (the lead bed) becomes saturated with D at the feed concentration, that bed is removed. Bed 2 (the trailing bed), which is partially saturated at this point, depending upon the width of the MTZ, becomes the lead bed, and previously idle Bed 3 takes the place of Bed 2. While Bed 1 is off-line, its spent carbon is removed and replaced with fresh carbon. The spent carbon is incinerated. The equilibrium adsorption isotherm for D is given by $q = 8 c^{0.57}$, where q is in mg/g and c is in mg/L. Once the cycle is established, how often must the carbon in a bed be replaced? What is the maximum width of the MTZ that will allow saturated loading of the lead bed?

15.25 The fixed-bed adsorber series arrangement of Exercise 15.24 is to be used to treat 250 gpm of water containing 0.185 mg/L of benzene (B) and 0.583 mg/L of m-xylene (X). However, because the two solutes may have considerably different breakthrough times, more than two operating beds in series may be needed. The adsorption isotherms are given in Exercise 15.22, where q is in mg/g and c is in mg/L. From laboratory measurements, the widths of the mass-transfer zones are estimated to be $\text{MTZ}_B = 2.5 \text{ ft}$ and $\text{MTZ}_X = 4.8 \text{ ft}$. Once the cycle is established, how often must the carbon in the bed be replaced?

15.26 Air at 80°F, 1 atm, 80% relative humidity, and a superficial velocity of 100 ft/min passes through a 5-ft-high bed of 2.8-mm-diameter spherical particles of silica gel ($\rho_b = 39 \text{ lb/ft}^3$). The adsorption equilibrium isotherm at 80°F is given by

$$q_{\text{H}_2\text{O}} = 15.9 p_{\text{H}_2\text{O}} \quad (1)$$

where q is in lb H₂O/lb gel and p is in atm. The overall mass-transfer coefficient can be estimated from (15-106), using an effective diffusivity of 0.05 cm²/s and with k_c estimated from (15-65). Using the approximate concentration-profile equations of Klinkenberg, compute a set of breakthrough curves and determine the time when the humidity of the exiting air reaches 0.0009 lb H₂O/lb dry air. Assume isothermal and isobaric operation. Compare the time to breakthrough with the time for the equilibrium model. At breakthrough, what is the approximate width of the mass-transfer zone. What is the average loading of the bed at breakthrough?

15.27 A train of four 55-gallon canisters of activated carbon is to be used to reduce the nitroglycerine (NG) content of 400 gph of

wastewater from 2,000 ppm by weight to less than 1 ppm. Each cannister has a diameter of 2 ft and holds 200 lb activated carbon ($\rho_b = 32 \text{ lb/ft}^3$). Each cannister is equipped with a liquid-flow distributor to promote plug flow through the bed of carbon. The effluent from the first cannister is monitored so that when a 1 ppm threshold of NG is reached, that cannister is removed from the train and a fresh cannister is added to the end of the train. The spent carbon is mixed with coal for use as a fuel in a coal-fired power plant at the process site. Using the following pilot-plant data, estimate how many cannisters are needed each month and the monthly cannister cost at \$700 per cannister.

Pilot-plant data:

Tests with the same 55-gallon cannister to be used in the commercial process; water flow rate = 10 gpm; NG content in feed = 1,020 ppm by weight.

Breakthrough correlation:

$t_B = 3.90L - 2.05$, where t_B = time, h, at breakthrough of the 1 ppm threshold and L = bed depth of carbon in feet.

15.28 Air at a flow rate of 12,000 scfm (60°F, 1 atm) and containing 0.5 mol% ethyl acetate (EA) and no water vapor is to be treated with activated carbon (C) ($\rho_b = 30 \text{ lb/ft}^3$) with an equivalent particle diameter of 0.011 ft in a fixed-bed adsorber to remove the ethyl acetate, which will be subsequently stripped from the carbon by steam at 230°F. Based on the following data, determine the diameter and height of the carbon bed, assuming adsorption at 100°F and 1 atm and a time-to-breakthrough of 8 h with a superficial gas velocity of 60 ft/min. If the bed height-to-diameter is unreasonable, what change in design basis would you suggest?

Adsorption isotherm data (100°F) for EA:

p^{EA} , atm	q , lb EA/lb C	p^{EA} , atm	q , lb EA/lb C
0.0002	0.125	0.0020	0.227
0.0005	0.164	0.0050	0.270
0.0010	0.195	0.0100	0.304

Breakthrough data at 100°F and 1 atm for EA in air at a gas superficial velocity of 60 ft/min in a 2-ft dry bed:

Mole Fraction EA in Effluent	Time, Min	Mole Fraction EA in Effluent	Time, Min
0.00005	60	0.00100	95
0.00010	66	0.00250	120
0.00025	75	0.00475	160
0.00050	84		

15.29 In Examples 15.11 and 15.13, benzene is adsorbed from air at 70°F in a 6-ft-high bed of silica gel and then stripped with air at 145°F. If the bed height is changed to 30 ft, the following data are obtained for breakthrough at 641 minutes for the adsorption step:

z , ft	$\phi = c/c_F$	$\psi = \bar{q}/q_F^*$	z , ft	$\phi = c/c_F$	$\psi = \bar{q}/q_F^*$
0–12	1.000	1.000	14	1.000	1.000
13	1.000	1.000	15	1.000	1.000

z , ft	$\phi = c/c_F$	$\psi = \bar{q}/q_F^*$	z , ft	$\phi = c/c_F$	$\psi = \bar{q}/q_F^*$
16	0.999	0.999	24	0.599	0.575
17	0.997	0.997	25	0.468	0.444
18	0.992	0.990	26	0.343	0.321
19	0.978	0.975	27	0.235	0.217
20	0.951	0.944	28	0.150	0.137
21	0.901	0.890	29	0.090	0.081
22	0.825	0.808	30	0.050	0.044
23	0.722	0.701			

If the bed is regenerated isothermally with pure air at 1 atm and 145°F, and the desorption of benzene during the heatup period is neglected, determine the loading, $\bar{q}$, profile at a time sufficient to remove 90% of the benzene from the bed if an interstitial pure air velocity of 98.5 ft/min is used. Values of k and K at 145°F are given in Example 15.13.

15.30 Use the method of lines with a five-point, biased, upwind finite-difference approximation and a stiff integrator to perform PSA cycle calculations that approach the cyclic steady state for the data and design basis in Example 15.14, starting from: (a) a clean bed, and (b) a bed saturated with the feed. Are the two cyclic steady states essentially the same?

15.31 Solve Example 15.14 for $P_L = 0.12$ atm and an interstitial velocity during desorption that corresponds to the use of 44.5% of the product gas from the adsorption step.

15.32 For the separation of air by PSA, adsorption of both O_2 and N_2 must be considered. Develop a model for this case taking into account two species mass balances, overall mass balance, two species mass-transfer rates, and two extended-Langmuir isotherms. Each of the two main steps can be isothermal and isobaric. Can your PDE equations still be solved by the method of lines with a stiff integrator? If so, outline a procedure for doing it.

15.33 Two adsorption-based separation processes not considered in this chapter because of lack of significant commercial application are (1) parametric pumping, first conceived by R.H. Wilhelm in the early 1960s, and (2) cycling-zone adsorption, invented by R.L. Pigford and co-workers in the late 1960s. The status of and future for these two processes was assessed by Sweed in 1984 [*AIChE Symp. Series*, **80** (233), 44–53 (1984)]. Describe in detail each of these processes. Can either be used for both gas-phase and liquid-phase adsorption?

15.34 A gas mixture containing 55 mol% propane and 45 mol% propylene is to be separated into products containing 10 and 90 mol% propane by adsorption in a continuous, countercurrent adsorption system operating at 25°C and 1 atm. The adsorbent is silica gel, for which equilibrium data are given in Exercise 15.9. Determine by the McCabe–Thiele method: (a) the adsorbent flow rate per 1,000 m^3 of feed gas at 25°C and 1 atm if 1.2 times the minimum rate is used, and (b) the number of theoretical stages required.

15.35 Repeat Example 15.19, except for a feed containing 400 ppm (by weight) of CaCl_2 and 50 ppm of NaCl .

15.36 An aqueous solution, buffered to a pH of 3.4 by sodium citrate and containing 20 mol/m^3 each of glutamic acid, glycine, and valine, is separated in a chromatographic column, packed with Dowex 50W-X8 in the sodium form to a depth of 470 mm. The resin is 0.07 mm in diameter and packs to a bed void fraction

Part 4

Separations That Involve a Solid Phase

Chapters 16, 17, and 18 cover separation operations in which one or more components in a solid phase undergo mass transfer to or from a fluid phase. Chapter 16 covers selective leaching of material from a solid to a liquid solvent. This operation is widely used in the food industry. Crystallization from a liquid and desublimation from a vapor are discussed in Chapter 17,

where evaporation, which often precedes crystallization, is included. Both solution crystallization to produce inorganic crystals and melt crystallization to produce organic crystals are considered. Chapter 18 is concerned with the drying of solids and the myriad types of equipment used industrially. A section on psychrometry is included.

Chapter 16

Leaching and Washing

Leaching, sometimes called solid-liquid (or liquid-solid) extraction, involves the removal of a soluble fraction (the solute or *leachant*) of a solid material by a liquid solvent. The solute diffuses from the solid into the surrounding solvent. Either the extracted solid fraction or the insoluble solids, or both may be the valuable product. Leaching is widely used in the metallurgical, natural product, and food industries. In the former, leaching may involve oxidation or reduction reactions of the solid with the solvent. Equipment is available to conduct leaching under batch, semicontinuous, or continuous operating conditions. Effluents from a leaching stage are essentially solids-free liquid, called the *overflow*, and wet solids, called the *underflow*. To reduce the concentration of solute in the liquid portion of the underflow, leaching is often accompanied by two or more countercurrent-flow

washing stages. The combined process produces a final overflow, called the *extract*, which contains some of the solvent and most of the solute; and a final underflow, called the *extracted* or *leached solids*, which are wet with almost-pure, remaining solvent. Ideally, the soluble solids are perfectly separated from the insoluble solids, but solvent is distributed to both products. Additional processing of the extract and the leached solids is necessary to recover the solvent for recycle.

Some industrial applications of leaching include: (1) removal of copper from ore with sulfuric acid, (2) recovery of gold from ore with sodium-cyanide solution, (3) extraction of sugar from sugar beets with hot water, (4) extraction of tannin from tree bark with water, and (5) the removal of caffeine from green coffee beans with supercritical CO₂.

16.0 INSTRUCTIONAL OBJECTIVES

After completing this chapter, you should be able to:

- Describe equipment used for batch and continuous leaching.
- Explain differences between leaching and washing.
- List assumptions for an ideal, equilibrium leaching or washing stage.
- Calculate recovery of a solute for a continuous, countercurrent leaching and washing system for a constant ratio of liquid to solids in the underflow or for a variable-underflow ratio.
- Apply rate-based leaching to food-processing applications.
- Develop and apply the rate-based, shrinking-core model to reactive mineral processing.

Industrial Example

As an example of leaching, consider the extraction of vegetable oil from soybeans with a commercial hexane solvent in a pilot-plant size countercurrent-flow, multistage leaching unit, as described by Othmer and Agarwal [1]. Although edible oils can be extracted from a number of different field and tree crops, including coconuts, cottonseeds, palm, peanuts, grapeseed, soybeans, and sunflower seeds, the highest percentage of edible oil is from soybeans. In 1979–1980, world production of soybean oil was 14 million metric tons. Oil from soybeans is high in polyunsaturated fats and, thus, is less threatening from a cholesterol standpoint. When the oil content of seeds and beans is high, some of the oil can be removed by compressing the solids in a process known as

expression, as discussed in Perry and Green [2]. For soybeans, whose oil content is typically less than 0.30 lb per lb of dry and oil-free solids, leaching is a more desirable technique than expression because a higher yield of oil can be achieved by leaching.

The soybeans are dehulled, cleaned, cracked, and flaked before being fed to the extractor. Typically, the cleaned soybeans contain 8 wt% moisture and 20 wt% oil. The dry and oil-free soybeans have a particle specific gravity of about 1.425 and the oil has a specific gravity of 0.907 with a viscosity at 25°C of 50 cP. Approximately 50% of the flake volume is taken up by oil, moisture, and air. It might be expected that whole soybeans, rather than flakes, might be fed to the extractor, with leaching taking place by molecular diffusion of the

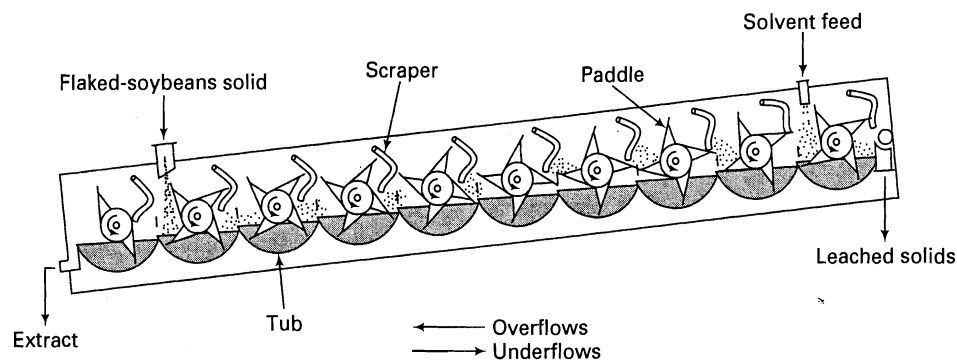


Figure 16.1 Kennedy extractor for leaching of oil from soybeans.

solvent uniformly into the seed, followed by diffusion of the oil through the solvent and out of the seed. If so, the mass-transfer process within the seed could be modeled with Fick's second law. However, experiments by King, Katz, and Brier [3] found that although extraction of oil with solvent in uniformly porous inorganic solids, like porous clay plates, obeyed Fick's law of molecular diffusion, extraction of oil from soybeans did not, presumably because of the complex internal structure of soybeans. Furthermore, Othmer and Agarwal [1], using whole and carefully cut half soybeans, found that diffusion was extremely slow. After 168 hours in contact with hexane, less than 0.08% of the original oil in the whole beans and less than 0.19% of the oil in the half beans was extracted. Such a slow diffusion rate for particles that are approximately 5 mm in diameter is probably due to the location of the oil within the insoluble cell walls, requiring that the oil pass through the walls by slow osmotic pressure differences.

The extent and rate of extraction of the oil is greatly enhanced by flaking the soybeans to thicknesses in the 0.005–0.02-in. range. The flaking process ruptures the cell walls, greatly facilitating contacting of the oil with the solvent. Using trichloroethylene [3] or *n*-hexane [1] as the solvent, with flakes of diameters ranging from 0.04 to 0.24 in., approximately 90% of the oil can be extracted in 100 minutes. The ideal solvent for commercial applications of the leaching of soybeans should have a high solubility of the oil to minimize the amount of solvent needed, a high volatility to facilitate the recovery of solvent from the oil by evaporation or distillation, nonflammability to eliminate the chance of fire and explosion, low cost, ready availability, chemical stability, low toxicity, and compatibility with inexpensive materials of construction

to minimize or eliminate corrosion. Although in a number of respects, especially nonflammability, trichloroethylene is an ideal solvent, it is now considered to be a very hazardous chemical because of its high toxicity. The favored solvent is commercial hexane (mostly *n*-hexane), which presents a fire hazard but has a low toxicity.

The pilot-plant leaching unit used by Othmer and Agarwal, and known as the Kennedy extractor, is shown in Figure 16.1. The soybeans enter continuously at the low end and are leached in a countercurrent cascade of tubs by hexane solvent, which enters at the upper end. The flakes and solvent are agitated and the underflows are pushed uphill from one tub to the next by slowly rotating paddles and scrapers, while the overflows move downhill from tub to tub. The paddles are perforated to drain the solids when they are lifted above the liquid level in the tub by the paddle. The cascade can contain as many tubs as required; Othmer and Agarwal used 15.

Soybean flakes of 0.012 in. average thickness and containing, on the average, 10.67 wt% moisture and 0.2675 g oil/g dry oil-free flakes were fed to the Kennedy extractor at a continuous flow rate of 6.375 lb/h. The solvent flow rate was 10.844 lb/h. Leaching took place at ambient conditions. After 11 hours of operation, a steady state was achieved that delivered an extract, called a *miscella*, of 7.313 lb/hr, containing 15.35 wt% oil. The leached solids contained 0.0151 g oil/g dry oil-free flakes. Thus, 94.4% of the oil was extracted. The residence time in each tub was 3 minutes, giving a total residence time of 45 minutes. From these data, a mass-balance check can be made for oil and solvent, and the liquid-to-solids ratio in the leached solids can be estimated. These calculations are left as an exercise.

16.1 EQUIPMENT FOR LEACHING

Often, the solids to be leached must undergo pretreatment before being fed to the extraction equipment, in order to obtain reasonable leaching times. For example, seeds and beans are dehulled, cracked, and flaked, as described above

for soybeans. When vegetable and animal material cannot be flaked, it may be possible to cut it into thin slices, as is done with sugar beets prior to the leaching of the sugar with water. In this case, the cell walls are left largely intact to minimize the leaching of undesirable material, such as colloids and

albumens. Metallurgical ores are crushed and ground to small particles because the small regions of leachable material may be surrounded by relatively impermeable insoluble material. When the insoluble material is quartzite, leaching may be extremely slow. Van Arsdale [4] cites the very important effect of particle size on the time required for

effective leaching by aqueous sulfuric acid of a copper ore. The times for particle diameters of 150 mm, 6 mm, and less than 0.25 mm are approximately 5 years, 5 days, and 5 hours, respectively. When the solid to be leached contains a high percentage of solute, pretreatment may not be necessary because disintegration of the remaining skeleton of insoluble material may take place at the surface of the particles as leaching takes place. When the entire solid is soluble, leaching may be rapid, such that only one stage of extraction, called *dissolution*, may be required.

Industrial equipment for solid-liquid extraction is designed for either batchwise or continuous processing. The method of contacting of the solid with the solvent is either by *percolation* of the solvent through a bed of solids or by *immersion* of the solid in the solvent with agitation of the solid-liquid mixture. When immersion is used, countercurrent, multistage operation is common. With percolation, either a stagewise or differential contacting device can be used. An extractor must be efficient for its particular application so as to minimize the need for solvent because of the high cost of solvent recovery.

Batch Extractors

When the solids to be leached are in the form of fine particles, perhaps smaller than 0.1 mm in diameter, batch leaching is conveniently conducted in an agitated vessel. A simple configuration is the Pachuca tank [5], shown in Figure 16.2a and used extensively in the metallurgical industry. The tank is a tall, cylindrical vessel constructed of wood, concrete, or a metal-like steel, which can be lined with an inert,

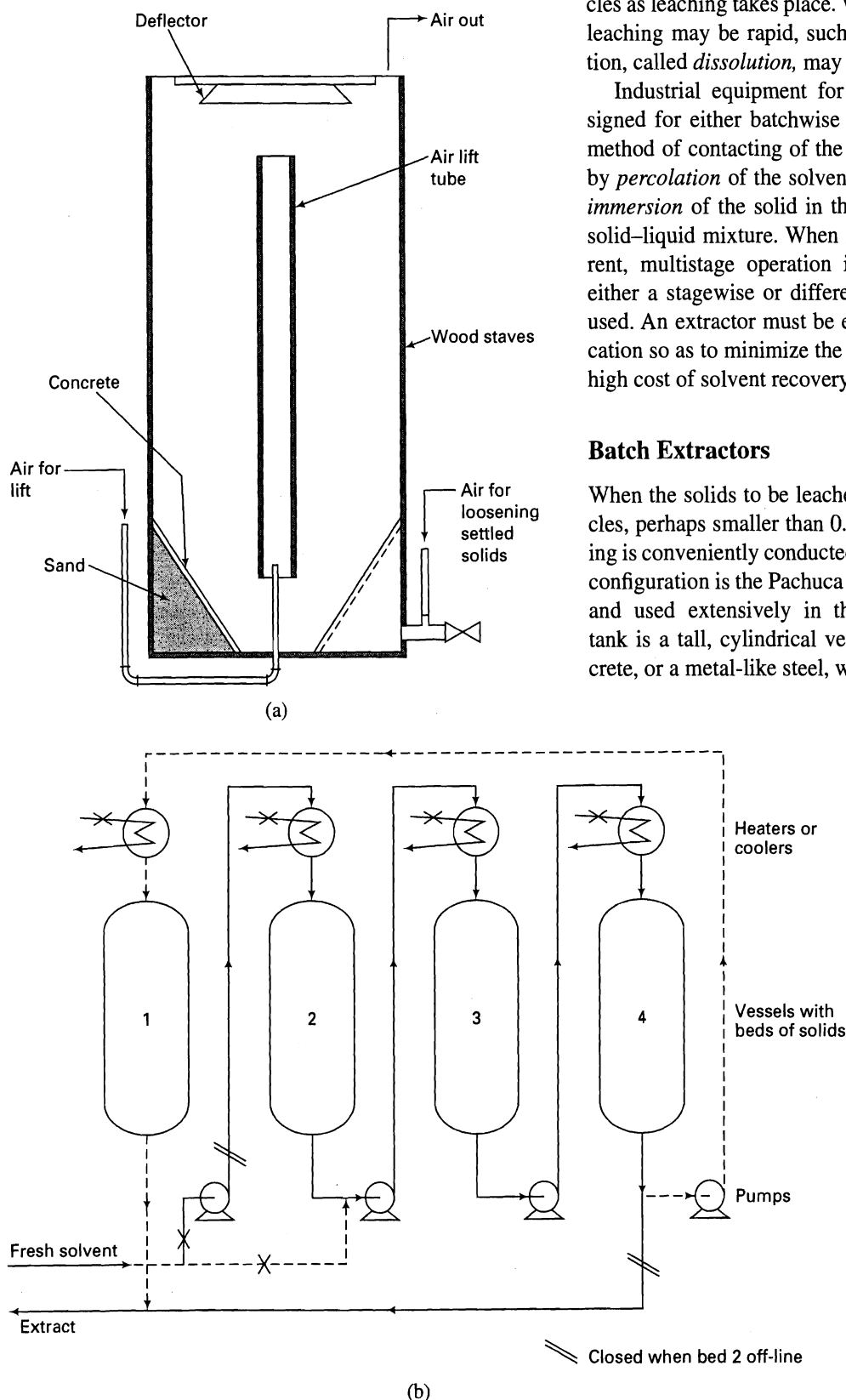


Figure 16.2 (a) Pachuca tank for batch leaching of small particles.

[From *Handbook of Separation Techniques for Chemical Engineers*, 2nd edition, P.A. Schweitzer, Editor-in-chief, McGraw-Hill, New York (1988) with permission.]

(b) Shanks countercurrent multibatch battery system for leaching of large particles by percolation.

[From C.J. King, *Separation Processes*, 2nd edition, McGraw-Hill, New York (1980) with permission.]

noncorrosive material such as lead. The solvent and solids are placed in the tank and agitation is achieved by an air lift, whereby air bubbles entering at the bottom of a circular tube, concentric with the tank, cause upward flow and subsequent circulation of the solid-liquid suspension. During agitation, air continuously enters and leaves the vessel. When the desired degree of leaching has been accomplished, agitation is stopped and the solids are allowed to settle into a sludge at the bottom, from where the sludge is removed with the assistance of air. The supernatant extract is removed by siphoning from the top of the tank. Agitation can also be achieved by a paddle stirrer or by the use of a propeller mounted in a draft tube in a manner to provide upward flow and subsequent circulation of the solid-liquid suspension, much like that in the Pachuca tank.

When the solids are too coarse to be easily suspended by immersion in the solvent with agitation, the percolation technique can be used. Again a tall, cylindrical vessel is employed. The solids to be leached are dumped into the vessel, followed by percolation of the solvent down through the bed of solids, much like fixed-bed adsorption. To achieve a high concentration of the solute in the solvent, a series of vessels is arranged in a multibatch, countercurrent-leaching technique developed in 1841 by James Shanks and often called an extraction battery using the Shanks system. This technique can be used for such applications as the batch removal of tannin from wood or bark, sugar from sugar beets, and water-soluble substances from coffee, tea, and spices. A typical arrangement of vessels is shown in Figure 16.2b, where Vessel 1 is shown off-line for emptying and refilling of solids. The solvent enters and percolates down through the solids in Vessel 2, and then percolates through Vessels 3 and 4, leaving as final extract from Vessel 4. The extraction of solids in Vessel 2 is completed first. When that occurs, Vessel 2 is taken off-line for emptying and refilling of solids and Vessel 1 is placed on-line. Now the fresh solvent first enters Vessel 3, followed by Vessels 4 and 1. In this manner, fresh solvent always contacts the solids that have been leached for the longest time, thus realizing the benefits of countercurrent contacting. Heat exchangers are provided between vessels to

adjust the temperature of the liquid phase, and pumps can be used to move the liquid from vessel to vessel. Any number of vessels can be included in the battery. Although the Shanks system is batchwise with respect to the solids, the system is continuous with respect to the solvent and extract.

Espresso Machine

A widely used batch-leaching machine is the espresso coffee maker. Although a beverage from coffee beans was first made about 1100 BC on the Arabian peninsula, it was not until many centuries later that a method and a machine was devised to produce a high-quality coffee, called espresso (a term that connotes a cup of coffee expressly for you, made quickly and individually, and intended to be drunk right away). The prototype of the espresso machine was created in France in 1822, and the first commercial espresso machine was manufactured in Italy in 1905. By the 1990s, espresso machines were in common use in many countries of the world, producing billions of cups annually. Today, coffee, and, particularly, espresso, is a world commodity that is second only to oil.

A photograph of a typical consumer espresso machine is shown in Figure 16.3a, while a simpler diagram to help understand its operation is presented in Figure 16.3b. In the machine, 7 to 9 grams of coffee beans are ground to a powder of particle size ranging from 250 to 750 microns by a special burr grinder that minimizes the temperature increase of the grounds. The bed of powder, contained as a thin layer in a filter housing, is tamped to increase its uniformity. Water is pumped to a pressure of 9–15 atmospheres and heated rapidly to from 88 to 92°C. The high pressure is required for pressure infusion of hot water into the particles so that extraction can proceed rapidly. During a period of from 20 to 30 sec, the hot water is percolated through the bed of coffee powder to produce a 45-ml shot. The shot has a viscosity of warm honey, and is topped by a thick, dark, golden-cream foam (“crema”). A typical machine produces two shots, which can be added to water, milk, or other liquids or blends to produce various beverages, including

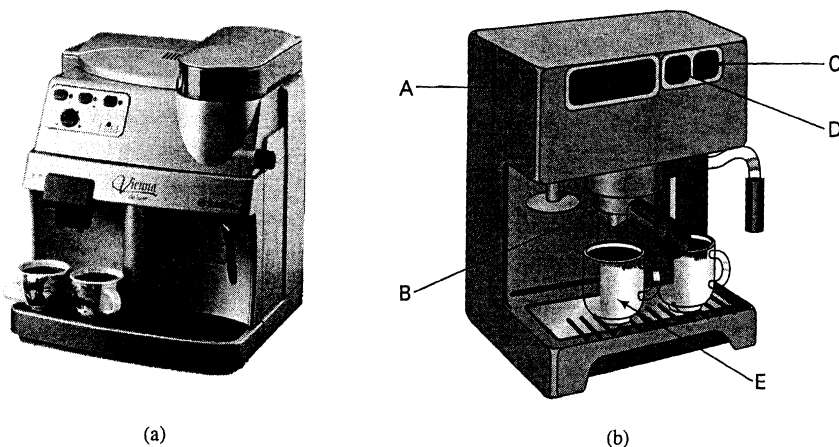
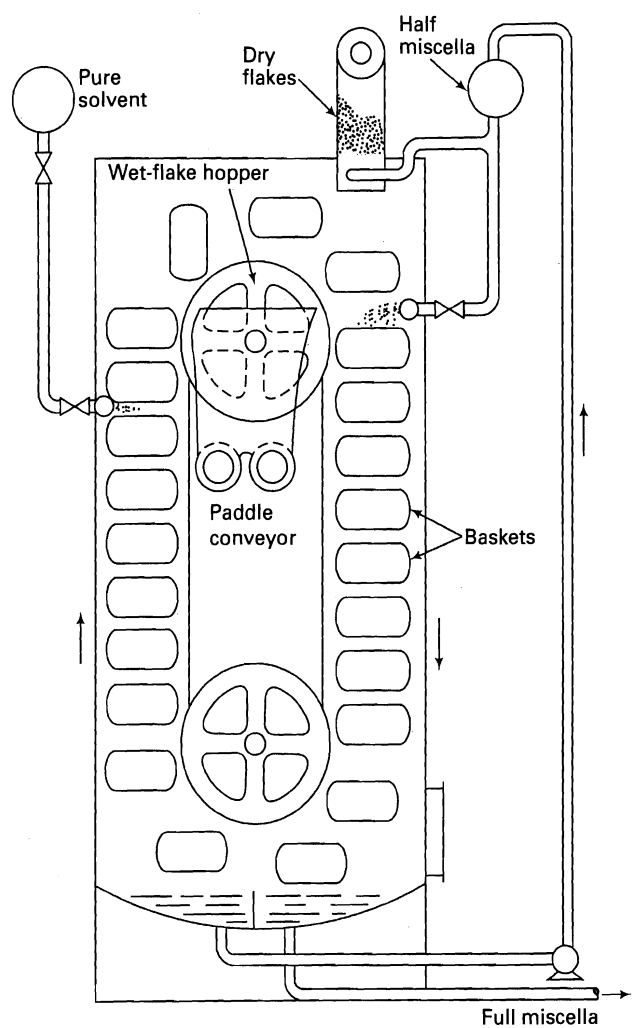
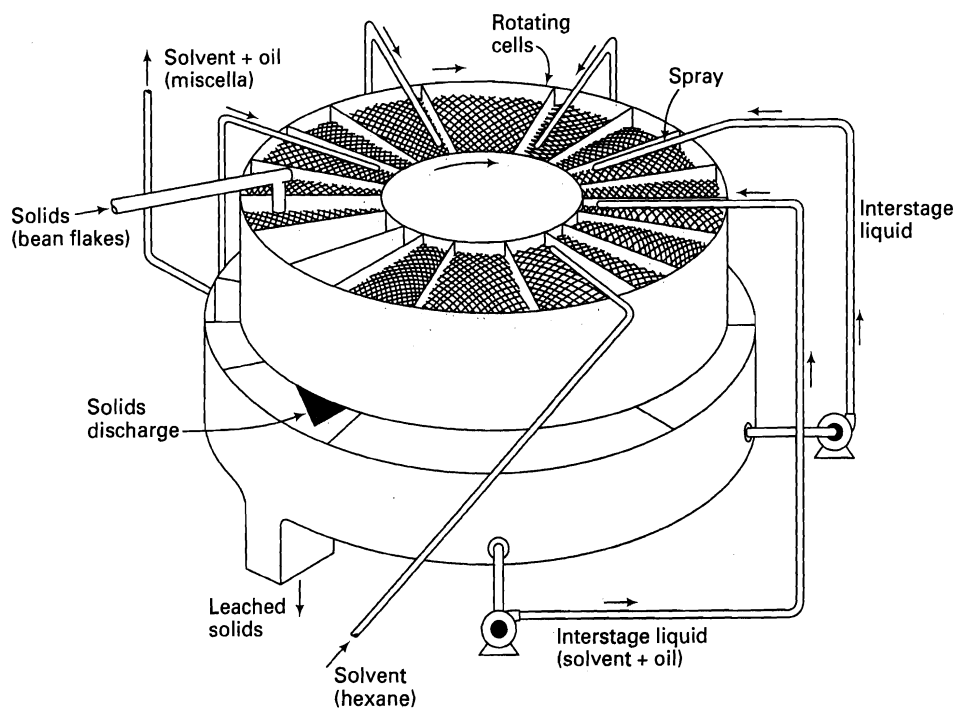


Figure 16.3 (a) Consumer espresso machine. (b) Operation: (A) pressure vessel, (B) portafilter holding ground coffee, (C) on/off switch, with built-in pressure indicator, (D) solenoid valve for espresso coffee, (E) shot cup holding leached, espresso coffee.



(a) Bollman vertical, moving-basket, conveyor extractor



(b) Rotocel extractor

Americano, Breve, Cappuccino, Latte, Lungo, Macchiato, Mocha, or Ristretto.

The short-time interval between grinding and leaching, the short residence time of the leaching process, the very small particle size of the coffee powder, and the controlled temperature and pressure of the water during leaching combine to maximize the extraction of the favorable flavor-and-aroma chemicals and minimize the extraction of chemicals associated with bitterness, such as quinine and caffeine. Furthermore, the crema traps the aroma, preventing its escape into the surrounding air. At the pull of a lever or the push of a button, espresso coffee is produced that is concentrated, full in body, and rich in flavor and aroma. A number of experimental studies of espresso production are reported in the literature, e.g., Andueza et al. [16, 17, 18].

Continuous Extractors

When leaching is to be carried out on a large scale, it is preferable to use an extraction device that can be operated with continuous flow of both solids and liquid. Many such patented devices are commercially available, especially for applications in the food industry, as discussed by Schwartzberg [6]. Some of the more widely discussed extractors are shown schematically in Figure 16.4. These devices differ mainly with respect to the manner in which the solids are transported through and the degree to which agitation of the solid-liquid mixture is provided in the equipment.

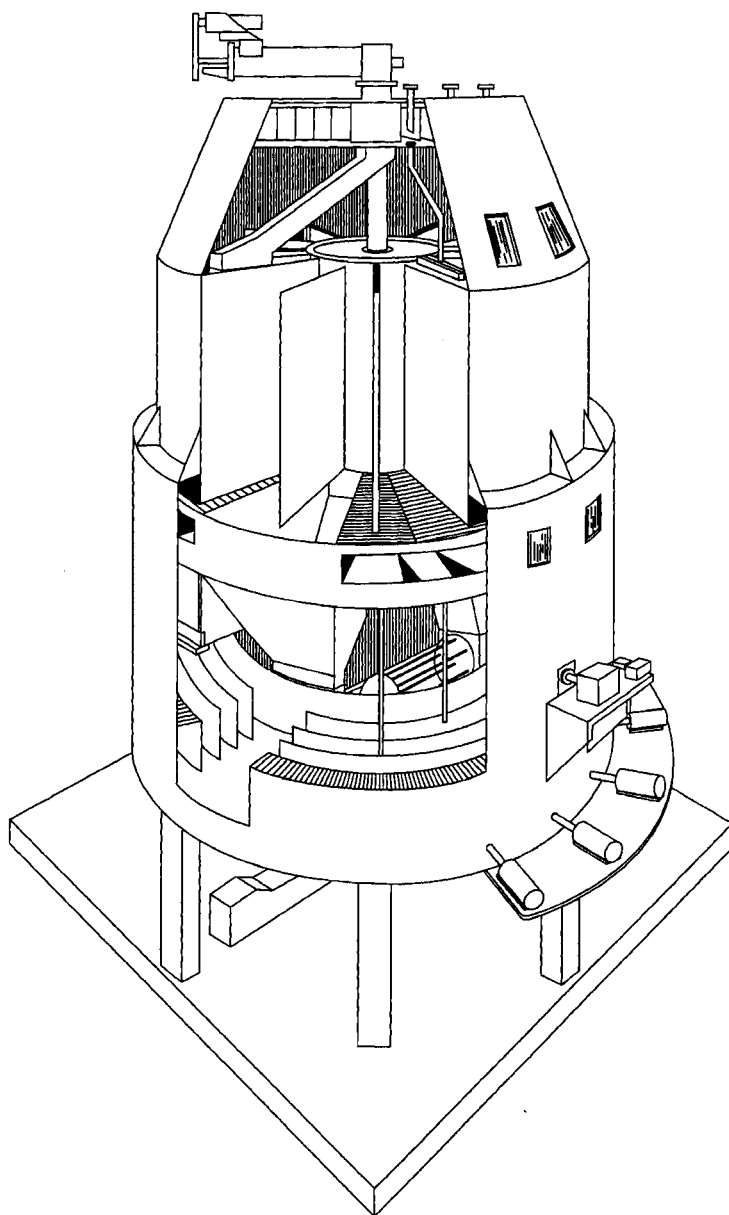
According to Schwartzberg, several extractors that have been described in the literature are now either obsolete or infrequently used, because of various limitations, including ineffective contacting of solid and liquid phases, bypassing,

Figure 16.4 Equipment for continuous leaching.

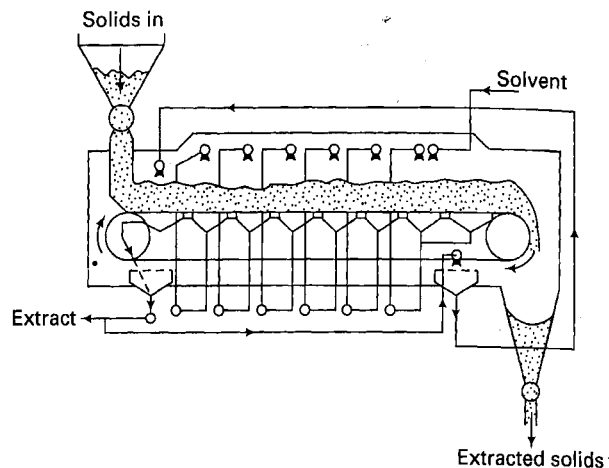
[From *Handbook of Separation Techniques for Chemical Engineers*, 2nd edition, P.A. Schweitzer, Editor-in-chief, McGraw-Hill, New York (1988) with permission.]

[From R.E. Treybal, *Mass-Transfer Operations*, 3rd edition, McGraw-Hill, New York (1980) with permission.]

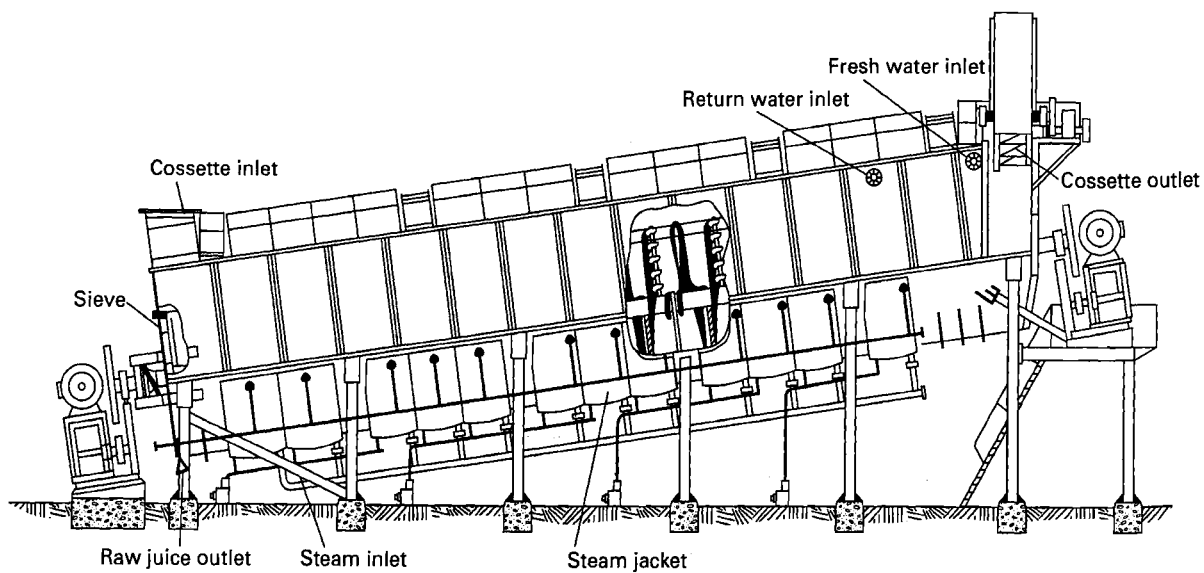
(continued)



(c) French stationary-basket extractor



(d) Continuous, perforated-belt extractor



(e) D.D.S. double-screw, slope extractor

Figure 16.4 (Continued).

and fines entrainment. These include the Hildebrand, Detrex, Anderson, Allis Chalmers, and Bonotto extractors. The Kennedy extractor described above for application to oil extraction from soybeans may also have a low efficiency in some applications, but is still used and still available.

The Bollman vertical moving-basket conveyor extractor, shown in Figure 16.4a, has been widely used to extract oil from flaked seeds and beans. Baskets with perforated bottoms are moved around a vertical loop by a motor-driven chain drive. The solvent percolates down, from basket to basket, through the solids. When a basket reaches the top of the extractor, the basket is inverted to dump the extracted solids and then filled with fresh solids. The flow of liquid is countercurrent to the solids contained in ascending baskets and cocurrent in descending baskets. Fresh solvent enters near the top of the ascending leg and collects as "half miscella" in the left-hand part of the sump at the bottom. From there, the half miscella is pumped to the top of the descending leg, from which it flows down to the right-hand part of the sump and is withdrawn as final extract, called "full miscella." A typical Bollman extractor is 14 m high, with each basket filled with solids to a depth of about 0.5 m. According to Coulson et al. [7], the baskets are rotated very slowly at about one revolution per hour so as to give solids residence times of about 60 minutes. Each basket contains about 350 kg of flaked solids. Thus, for the 23 baskets shown in Figure 16.4a, almost 200,000 kg of solids can be extracted per day. About equal mass flows of flaked solids and solvent are fed to the extractor, and the full miscella is essentially solids-free with about 25 wt% oil.

Another widely used continuous extractor for flaked seeds and beans is the Rotocel extractor, shown in Figure 16.4b. In this device, which resembles a carousel and conveniently simulates the Shanks system, walled, annular sectors, called cells, on a horizontal plane, are slowly rotated by a motor. The cells, which hold the solids and are perforated on the bottom for solvent drainage, successively pass a solids feed area, a series of solvent sprays, a final spray and drainage area, and a solids-discharge area. Fresh solvent is supplied to the cell located just below the final spray and drainage area, from where the drained liquid is collected and pumped to the preceding cell location. The drainage from that cell is collected and pumped to the cell preceding that cell and so on. In this manner, a countercurrent flow of solids and liquid is achieved. The extracted solids typically contain 25–30 wt% liquid. Rotocel extractors are typically 3.4–11.3 m in diameter, 6.4–7.3 m in height, with bed depths of 1.8–3.0 m. They have processed up to three-million kg/day of flaked soybeans. The number of cells can be varied and residence time can be varied by varying the rate of rotation. A widely used variation of the Rotocel extractor is the French stationary-basket extractor, which is shown in Figure 16.4c and has about the same size and capacity as the Rotocel extractor. The sector cells do not move. Instead, the solids feed spout and solids discharge zone rotate, with periodic switching of the solvent feed and discharge connections. Thus, the weight of moving parts is reduced.

Continuous, perforated-belt extractors of the type shown in Figure 16.4d are widely used to process sugar cane, sugar beets, oilseeds, and apples (for apple juice). The feed solids are fed from a hopper to a slow-moving, continuous and non-partitioned, perforated belt driven by motorized sprockets at either end. The height of the solids on the belt can be controlled by a damper at the outlet of the feed hopper. The belt speed is automatically adjusted to maintain the desired depth of solids. Extracted solids are discharged into an outlet hopper at the end of the belt by a scraper. Side walls prevent the solids from falling off the sides of the belt. Below the belt are compartments for collecting solvent. Fresh solvent is sprayed over the solids and above the compartments in a countercurrent fashion, starting from the discharge-end of the belt, in as many as 17 passes. Bed depths may range from 0.8 to 2.6 m. Units from 7 to 37 m long and with belts from 0.5 to 9.5 m wide have processed as much as 7,000,000 kg/day of sugar cane or sugar beets.

The D.D.S. (De Danske Sukker-fabriker) double-screw, slope extractor, shown in Figure 16.4e, is a very versatile unit. Although used mainly for extraction of sugar beets, the device has been applied successfully to a very wide range of other feed materials, including sugar cane, flaked seeds and beans, apples, pears, grapes, cherries, ginger, licorice, red beets, carrots, fishmeal, coffee, and tea. The opposite-turning screws of the metal ribbons are pitched so that they both move the solids uphill in parallel, cylindrical troughs. Extract flows through the screw surface downhill to achieve a differential, countercurrent flow with the solids. A novel feature is the ability to turn one screw slightly faster and then slightly slower than the other screw, causing the solids to be periodically squeezed. Units range in size from 2 to 3.7 m in diameter and 21–27 m in length and have been used to process as much as 3,000,000 kg/day of sugar beets in the form of cossettes (long, thin strips).

Continuous, Countercurrent Washing

When leaching is very rapid, as with small particles containing very soluble solutes, or when leaching has already been completed, or when solids are formed by chemical reactions in a solution, it is common to countercurrently wash the solids to reduce the concentration of the solute in the liquid adhering to the solids. This can be accomplished in a series of *gravity thickeners* or centrifugal thickeners, called *hydroclones*, arranged for countercurrent flow of the underflows and overflows as shown in Figure 16.5, and sometimes called a continuous, countercurrent decantation system. A typical continuous gravity thickener is shown in detail in Figure 16.6a. The combined feed to the thickener consists of feed solids or underflow from an adjacent thickener, together with fresh solvent or overflow from an adjacent thickener. The thickener has a threefold function. First, it must thoroughly mix the liquid and solids to obtain a uniform concentration of solute in the liquid. The thickener must also produce an overflow free of solids and an underflow with as

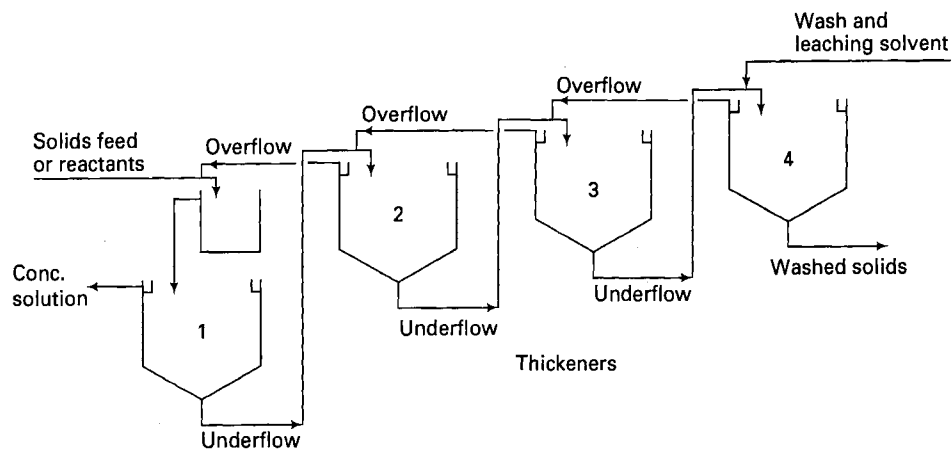
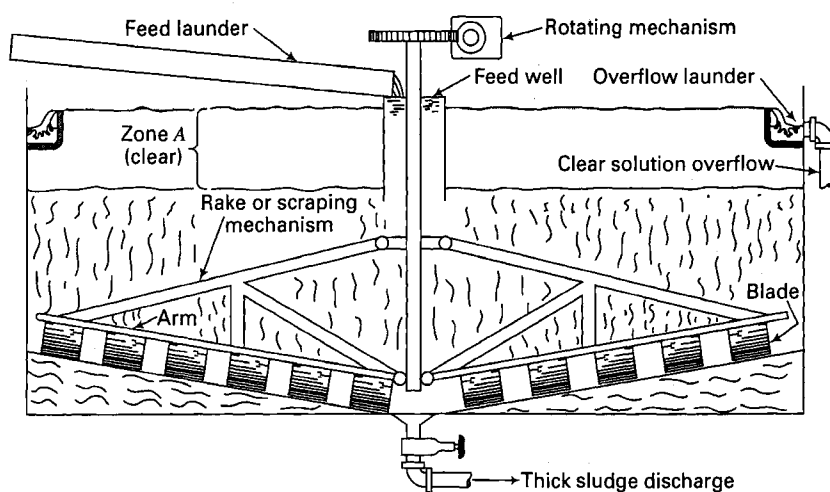
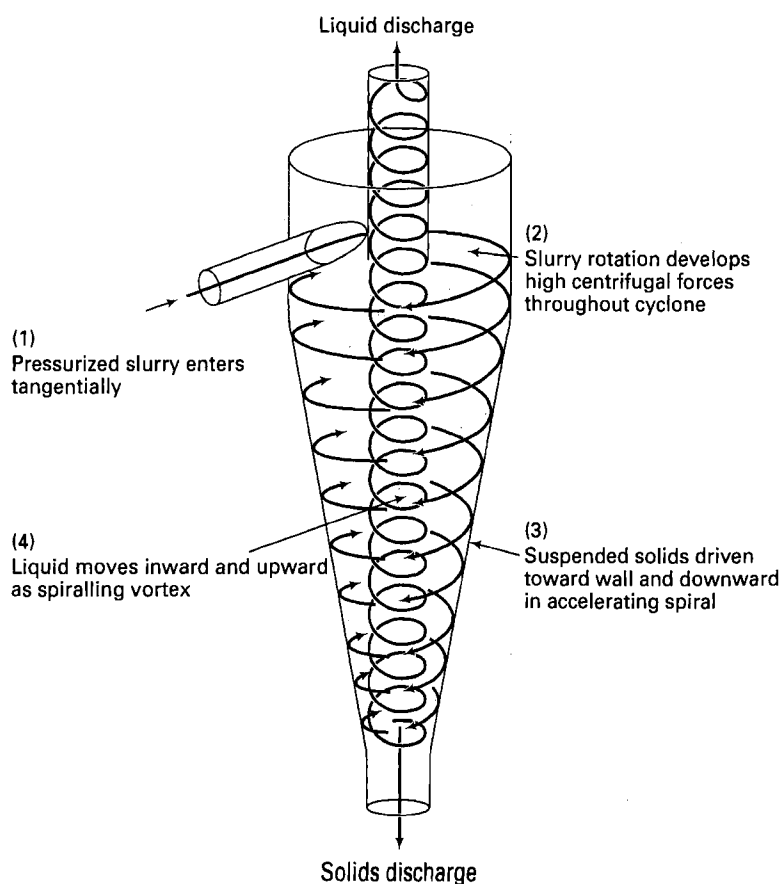


Figure 16.5 Continuous, countercurrent washing system using thickeners.



(a) Gravity thickener



(b) Hydroclone, centrifugal thickener

Figure 16.6 Thickeners.

[From *Handbook of Separation Techniques for Chemical Engineers*, 2nd edition, P.A. Schweitzer, Editor-in-chief, McGraw-Hill, New York (1988) with permission.]

high a fraction of solids as possible. The thickener consists of a large-diameter shallow tank with a flat or slightly conical bottom. The combined feed enters the tank near the center by a means of a feed launder that discharges into a feed well. Settling and sedimentation of the solid particles occur by gravity due to a solid particle density that is greater than the liquid density. In essence, solids flow downward and liquid flows upward. Around the upper, inner periphery of the tank is an overflow launder or weir for continuously removing clarified liquid. Solids settling to the bottom of the tank are moved inward toward a thick sludge discharge by a slowly rotating motor-driven rake. Thickeners as large as 100 m in diameter and 3.5 m high have been constructed. In large thickeners, the rake revolves at about 2 rpm.

The residence times of solids and liquids in a gravity thickener are often large (minutes or hours) and, as such, are sufficient to provide adequate residence time for mass transfer and mixing when small particles are involved. When long residence times are not needed and the overflow need not be perfectly clear of solids, the hydroclone, shown in Figure 16.6b, should be considered. The pressurized feed slurry enters the device tangentially to create, by centrifugal force, a downward spiraling motion. The higher-density, suspended solids are, by preference, driven to the wall, which becomes conical as it extends downward, and discharged as a thickened slurry at the bottom of the hydroclone. The liquid, which is forced to move inward and upward as a spiraling vortex, exits from a vortex-finder pipe that extends downward from the closed top of the hydroclone to a location just below the feed entry.

16.2 EQUILIBRIUM-STAGE MODEL FOR LEACHING AND WASHING

The simplest model for a continuous, countercurrent leaching and washing system, as shown in Figure 16.7, assumes that the solid feed consists of a solute, which is completely soluble in the solvent, and an inert substance or carrier that is not soluble in the solvent. Furthermore, the rate of leaching is rapid such that it is completed in a single leaching stage, which is followed by a series of one or more washing stages to reduce the concentration of solute in the liquid adhering to the solids in the underflow streams. All overflow

streams are assumed to be free of solids. In Figure 16.7:

S = mass flow rate of inert solids, which is constant from stage to stage.

V = mass flow rate of entering solvent or overflow liquid (solvent plus solute), which varies from stage to stage.

L = mass flow rate of underflow liquid (solvent plus solute), which varies from stage to stage.

y = mass fraction of solute in the overflow liquid.

x = mass fraction of solute in the underflow liquid.

Alternatively, V and L can refer to mass flow rates of solvent on a solute-free basis and the symbols Y and X can be used to refer to mass ratios of solute to solvent in the overflow liquid and underflow liquid, respectively. Mole or volume flow rates can also be used.

An ideal leaching or washing stage is defined by Baker [8] as one where:

1. Any entering solid solute is completely dissolved into the liquid in the stage (assuming that the liquid contains sufficient solvent).
2. The composition of the liquid in the stage is uniform throughout, including any liquid within pores of the inert solid.
3. Solute is not adsorbed on the surfaces of the inert solid.
4. The inert solids leaving in the underflow from each stage are wet with liquid, such that mass ratio of solvent in that liquid (or the total liquid) to inert solids is constant from stage to stage.
5. Because of (2), the concentration of solute in the overflow is equal to that in the liquid portion of the underflow. This is equivalent to an equilibrium assumption.
6. Overflows contain no solids.
7. Solvent is not vaporized, adsorbed, or crystallized in a stage.

For the continuous, countercurrent system of ideal leaching stages in Figure 16.7, solute and total-liquid material balances can be used to solve various types of problems, including (1) the determination of the number of ideal stages required to achieve a specified degree of washing and (2) the

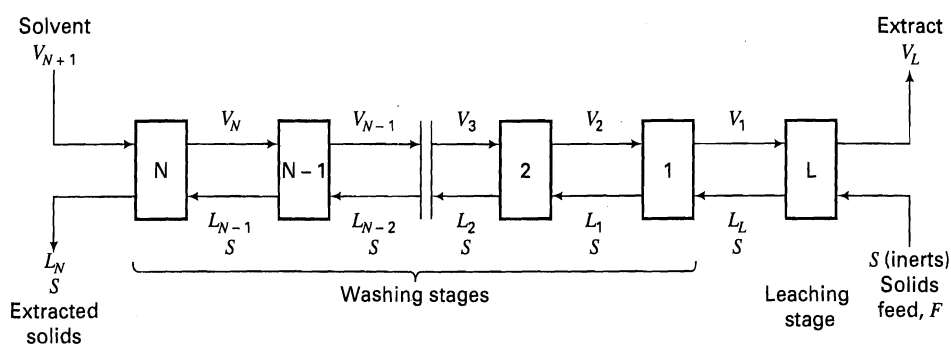


Figure 16.7 Continuous, countercurrent, ideal-stage leaching and washing system.

determination of the effect of washing for a specified degree of washing for a specified number of ideal stages. Depending on the nature of the problem, either an algebraic or a graphical method may be preferred. For most problems, it is usually best to consider the leaching stage separate from the washing stages, as illustrated in the following examples.

EXAMPLE 16.1

A finely divided solids feed, F , of 150 kg/h, containing 1/3 water-soluble Na_2CO_3 and 2/3 insoluble ash is to be leached and washed at 30°C in a two-stage, countercurrent system with 400 kg/h of water. The leaching stage consists of an agitated vessel that discharges the slurry into a thickener. The washing stage consists of a second thickener. Experiments show that the sludge underflow from each thickener will contain 2 kg of liquid (water and carbonate) per kg of insoluble ash. Assume ideal stages.

- (a) Calculate the percent (%) recovery of carbonate in the final extract.
 (b) If a third stage is added, calculate the amount of additional carbonate that will be recovered.

SOLUTION

At 30°C, the solubility of the carbonate in water is 38.8 kg/100 kg of water. Therefore, there is sufficient water to dissolve all the carbonate. Referring to Figure 16.7:

- (a) $N = 1$ (L is the leaching stage)

$$S = \frac{2}{3}(150) = 100 \text{ kg/h of insoluble ash}$$

$$\text{Na}_2\text{CO}_3 \text{ in entering solids} = \frac{1}{3}(150) = 50 \text{ kg/h}$$

$$V_2 = \text{entering solvent} = 400 \text{ kg/h}$$

$$L_L = L_1 = 2S = 200 \text{ kg/h}$$

By total liquid material balances on Stage 1 and Stage L ,

$$V_1 = V_2 + L_L - L_1 = 400 + 200 - 200 = 400 \text{ kg/h}$$

$$V_L = V_1 + 50 - L_L = 400 + 50 - 200 = 250 \text{ kg/h}$$

Na_2CO_3 material balance around washing Stage 1:

$$x_L L_L = y_1 V_1 + x_1 L_1 \quad (1)$$

$$200x_L = 400y_1 + 200x_1 \quad (2)$$

But, $y_1 = x_1$ from item (5) above for an ideal stage.

Combining (1) and (2),

$$x_L = 3x_1 \quad (3)$$

Na_2CO_3 material balance around leaching Stage L :

$$y_1 V_1 + 50 = x_L L_L + y_L V_L \quad (4)$$

$$400y_1 + 50 = 200x_L + 250y_L \quad (5)$$

But, again $y_1 = x_1$ and $y_L = x_L$ (6)

Combining (4), (5), and (6),

$$x_1 = 1.125x_L - 0.125 \quad (7)$$

Combining (3) and (7) and solving,

$$x_L = 0.158$$

Therefore, $y_L = 0.158$

From (7),

$$x_1 = 0.0526$$

$$\begin{aligned} \text{Recovery of Na}_2\text{CO}_3 &= \frac{y_L V_L}{50} = \frac{(0.158)(250)}{50} \\ &= 0.79 \text{ or } 79\% \end{aligned}$$

- (b) $N = 2$ washing stages

$$V_3 = 400 \text{ kg/h}$$

$$L_L = L_1 = L_2 = 2S = 200 \text{ kg/h}$$

$$V_2 = V_1 = 400 \text{ kg/h}$$

$$V_L = 250 \text{ kg/h}$$

Na_2CO_3 material balance around Stage 2:

$$\begin{aligned} x_1 L_1 &= y_2 V_2 + x_2 L_2 \\ 200x_1 &= 400y_2 + 200x_2 \end{aligned} \quad (8)$$

$$\text{But, } y_2 = x_2 \quad (9)$$

Combining (8) and (9),

$$x_1 = 3x_2 \quad (10)$$

Na_2CO_3 material balance around Stage 1:

$$\begin{aligned} y_2 V_2 + x_L L_L &= x_1 L_1 + y_1 V_1 \\ 400y_2 + 200x_L &= 200x_1 + 400y_1 \end{aligned} \quad (11)$$

$$\text{But, } y_2 = x_2 \quad (12)$$

$$y_1 = x_1 \quad (13)$$

Combining (11), (12), and (13),

$$x_L = 3x_1 - 2x_2 \quad (14)$$

Na_2CO_3 material balance around Stage L :

This is the same as (7) in Part (a).

Solving (10), (14), and (7), which are all linear,

$$x_L = 0.1795$$

$$y_L = 0.1795$$

$$x_1 = 0.0769$$

$$x_2 = 0.0256$$

$$\text{Recovery of Na}_2\text{CO}_3 = \frac{y_L V_L}{50} = \frac{(0.1795)(250)}{50} = 0.898 \text{ or } 89.8\%$$

From Part (a), for two stages, recovery of Na_2CO_3 is $0.158(250) = 39.5 \text{ kg/h}$

For three stages, recovery is $0.1795(250) = 44.9 \text{ kg/h}$

Recover $44.9 - 39.5 = 5.4 \text{ kg/h}$ more Na_2CO_3 with three stages.

For this example, it is difficult to apply a graphical McCabe–Thiele-type method because only the slope of the operating line is known and not either end point.

EXAMPLE 16.2

Baker [8] presents the following problem, for which a McCabe–Thiele graphical method can be applied. Two tons (4,000 lb) per day of waxed paper containing 25 wt% soluble wax and 75 wt% insoluble pulp are to be dewaxed by leaching with kerosene in a continuous, countercurrent contacting system of the type shown in Figure 16.7. The wax will be completely dissolved by the kerosene in the leaching stage, L. Subsequent washing stages will be used to reduce the wax content in the liquid adhering to the pulp leaving the last stage, N , to 0.2 lb wax/100 lb pulp. The kerosene entering the system is recycled from a solvent-recovery system and contains 0.05 lb wax/100 lb kerosene. The final extract is to contain 5 lb wax/100 lb kerosene. Experiments show that the underflow from each stage will contain 2 lb kerosene/lb insoluble pulp. Determine the number of washing stages required, exclusive of the leaching stage.

SOLUTION

Referring to Figure 16.7 and using the nomenclature defined at the beginning of this section for the case of concentrations on a mass-ratio basis and flow rates on a solute-free basis, the following material balance equations apply:

Overall mass balance on solute (wax):

$$0.25(4,000) + \frac{0.05}{100} V_{N+1} = \frac{5}{100} V_L + \frac{0.2}{100} (0.75)(4,000)$$

or

$$0.05V_L - 0.0005V_{N+1} = 994 \quad (1)$$

Overall mass balance on solvent (kerosene):

$$V_{N+1} = V_L + 2(0.75)(4,000)$$

or

$$V_{N+1} = V_L + 6,000 \quad (2)$$

Solving (1) and (2)

$$V_{N+1} = \text{kerosene in entering solvent} = 26,140 \text{ lb/day}$$

$$V_L = \text{kerosene in exiting extract} = 20,140 \text{ lb/day}$$

Thus, the final underflow contains $26,140 - 20,140 = 6,000$ lb/day of kerosene. Also,

$$X_N = \left[\frac{0.2}{100} (0.75)(4,000) \right] / 6,000 \\ = 0.001 \text{ lb wax/lb kerosene in the final underflow.}$$

Material balances can now be made around the leaching stage.

Mass balance on kerosene:

$$V_1 = V_L + 2(0.75)(4,000)$$

Thus,

$$V_1 = 20,140 + 6,000 = 26,140 \text{ lb/day.}$$

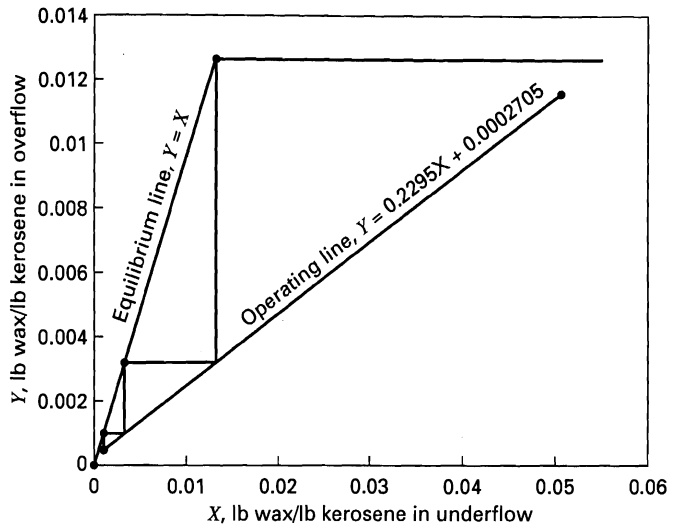


Figure 16.8 McCabe–Thiele diagram for Example 16.2.

Mass balance on wax:

$$26,140 Y_1 + (0.25)(4,000) = \frac{5}{100} (20,140) + X_L (2)(0.75)(4,000) \quad (3)$$

But

$$X_L = Y_L = 0.05 \text{ lb wax/lb kerosene} \quad (4)$$

Substituting (4) into (3) and solving,

$$Y_1 = 0.01174 \text{ lb wax/lb kerosene}$$

We now have the end points (solute compositions) at the two ends of the washing cascade. Referring to Figure 16.7,

$$Y_{N+1} = 0.0005 \quad \text{and} \quad X_N = 0.001 \text{ in lb wax/lb kerosene}$$

$$Y_1 = 0.01174 \quad \text{and} \quad X_L = 0.05 \text{ in lb wax/lb kerosene}$$

Furthermore, for the washing section, the mass ratio of kerosene in the underflow to kerosene in the overflow is constant at a value of

$$\frac{L_{n-1}}{V_n} = \frac{2(0.75)(4,000)}{26,140} = 0.2295$$

Therefore, the operating line on a McCabe–Thiele plot of Y versus X will be a straight line through the two end points, with a slope of 0.2295. This corresponds to a line given by the equation, $Y = 0.2295X + 0.0002705$. The equilibrium line is simply $Y = X$. The operating and equilibrium lines are plotted in Figure 16.8. The ideal washing stages, which can be stepped off starting from either end of the operating line, are stepped off here from stage N . It is seen that slightly less than three ideal washing stages are needed.

McCabe–Smith Algebraic Method

It is seen in Figure 16.8 that it can be difficult to accurately step off the number of washing stages. When the above ideal-stage model of Baker [8] applies, a more accurate algebraic method, developed by McCabe and Smith [9] from the Kremser equation of Chapter 5, can be applied. The

method is developed here using solute concentrations in mass fractions, but the final equations can also be applied with mass ratios.

Combining (5-48), (5-50), (5-51), and (5-54),

$$y_{N+1} = y_1 \left(\frac{1 - A^{N+1}}{1 - A} \right) - y_0^* A \left(\frac{1 - A^N}{1 - A} \right) \quad (16-1)$$

where N = number of ideal washing stages and

$$y_0^* = Kx_0 = Kx_L = y_L \quad (16-2)$$

For washing,

$$K = y/x = 1 \text{ and, therefore, } A = L/KV = L/V$$

Equation (16-1) can be written as follows by collecting terms in A^{N+1} and A :

$$A^{N+1}(y_1 - y_L) = A(y_{N+1} - y_L) + (y_1 - y_{N+1}) \quad (16-3)$$

This equation can be simplified by writing an overall solute balance around all washing stages:

$$y_{N+1}V_{N+1} + x_L L_L = y_1 V_1 + x_N L_N \quad (16-4)$$

But, for ideal washing stages, since

$$V_{N+1} = V_1, \quad L_L = L_N, \quad \text{and} \quad A = L/V,$$

(16-4) can be simplified to

$$y_{N+1} = y_1 + Ax_N - Ax_L \quad (16-5)$$

But $y_L = x_L$. Therefore, (16-5) can be written:

$$(y_1 - y_{N+1}) = A(y_L - x_N) \quad (16-6)$$

Combining (16-3) and (16-6) and rearranging gives

$$A^N = \left(\frac{y_{N+1} - x_N}{y_1 - y_L} \right) \quad (16-7)$$

Solving (16-7) for N with $A = L/V$, gives

$$N = \frac{\log \left(\frac{x_N - y_{N+1}}{y_L - y_1} \right)}{\log(L/V)} \quad (16-8)$$

The argument of the log term of the denominator can be written in terms of the end points to give

$$\frac{L}{V} = \left(\frac{y_1 - y_{N+1}}{x_L - x_N} \right) = \left(\frac{y_1 - y_{N+1}}{y_L - x_N} \right) \quad (16-9)$$

Combining (16-8) and (16-9),

$$N = \frac{\log \left(\frac{x_N - y_{N+1}}{y_L - y_1} \right)}{\log \left(\frac{y_1 - y_{N+1}}{y_L - x_N} \right)} \quad (16-10)$$

When the constant underflow liquid is given in terms of total (solvent plus solute) liquid, (16-8) or (16-9) is used directly, where L and V are total liquid flow rates. When the underflow liquid is given in terms of just the solvent, the y and x solute mass fractions in (16-8) and (16-9) are replaced by Y and X solute mass ratios, and V and L are liquid flow rates of solute-free solvent.

EXAMPLE 16.3

Solve for the number of ideal, continuous, countercurrent washing stages for the conditions of Example 16.2 using the McCabe-Smith equations.

SOLUTION

From the problem statement and the overall material balance and leaching stage calculations for Example 16.2,

$$Y_{N+1} = 0.0005$$

$$X_N = 0.001$$

$$Y_2 = 0.01174$$

$$Y_L = X_L = 0.05$$

$$L/V = 0.2295$$

From the mass ratio form of (16-8), the number of ideal leaching stages is:

$$N = \frac{\log \left(\frac{0.001 - 0.0005}{0.05 - 0.01174} \right)}{\log(0.2295)} = 2.95$$

The same result is obtained if (16-10) is used.

When the rate of leaching is slow, several countercurrent stages may be required, during which the effect of washing will be diminished. This is illustrated in the following example.

EXAMPLE 16.4

In Example 16.1, part (b), leaching was assumed to be completed in one stage, with two additional stages provided for washing. The recovery of the solute, Na_2CO_3 , in the extract was 89.8%. Recalculate this example, assuming that 1/2 of the carbonate is leached in the first stage and the remaining 1/2 in the second stage, leaving only the last stage as a true washing stage. For convenience, number the stages as in Figure 16.9, which includes flow rates that are given or easily computed.

SOLUTION

Na_2CO_3 material balance around Stage 3:

$$\begin{aligned} x_2 L_2 &= y_3 V_3 + x_3 L_3 \\ 200x_2 &= 400y_3 + 200x_3 \end{aligned} \quad (1)$$

$$\text{But,} \quad y_3 = x_3 \quad (2)$$

$$\text{Combining (1) and (2),} \quad x_2 = 3x_3 \quad (3)$$

Na_2CO_3 material balance around Stage 2:

$$\begin{aligned} y_3 V_3 + x_1 L_1 &= x_2 L_2 + y_2 V_2 \\ 400y_3 + 200x_1 + 25 &= 200x_2 + 425y_2 \end{aligned} \quad (4)$$

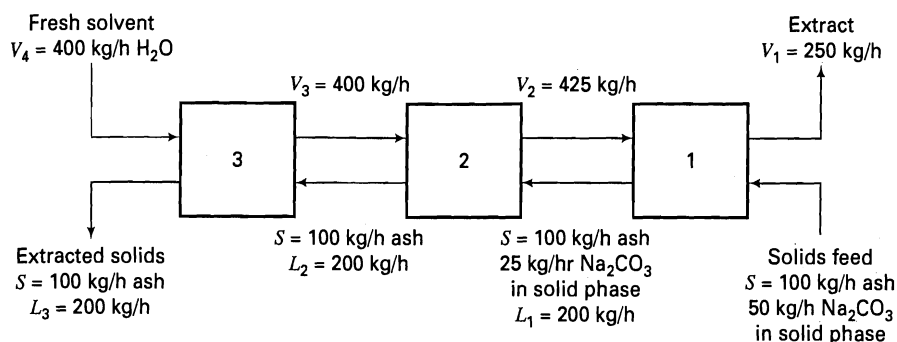


Figure 16.9 Leaching and washing system for Example 16.4.

But, $y_2 = x_2$ (5)

Combining (2), (4), and (5),

$$x_1 = 3.125x_2 - 2x_3 - 0.125 \quad (6)$$

Na_2CO_3 material balance around Stage 1:

$$y_2 V_2 + 50 = x_1 L_1 + y_1 V_1 + 25 \quad (7)$$

$$425y_2 + 50 = 200x_1 + 250y_1 + 25$$

But, $y_1 = x_1$ (8)

Combining (5), (7), and (8)

$$x_2 = 1.059x_1 - 0.0588 \quad (9)$$

Solving (3), (6), and (9),

$$x_1 = y_1 = 0.1681$$

$$x_2 = 0.1192$$

$$x_3 = 0.0397$$

Recovery of

$$\text{Na}_2\text{CO}_3 = \frac{y_1 V_1}{50} = \frac{(0.1681)(250)}{50} = 0.841 \text{ or } 84.1\%$$

which is almost 6% less than in Part (b) of Example 16.1, where leaching was completed in one stage.

Variable Underflow

In previous Examples 16.1–16.4, the ratio of liquid to solids in the underflow from stage to stage was assumed to be constant. Experiments by Ravenscroft [10] on the extraction of oil from granulated halibut livers by diethylether showed that the ratio of liquid to solids in the underflow increased significantly with increasing concentration of oil in the liquid in the range of 0.04–0.64 gal oil/gal liquid. In the leaching experiments, equilibrium was achieved in 2–3 minutes of agitation, but 10 minutes was used. Thirty minutes was allowed for settling of the extracted livers, after which the free solution (overflow) was decanted, leaving the underflow. Ravenscroft ascribed the variable-underflow effect to an appreciable increase in viscosity and density of the liquid as the concentration of the oil was increased. When neither density nor viscosity vary significantly, experimental data of Othmer and Agarwal [1] show that the main variable affecting the liquid-to-solid ratio in the underflow is surface area-to-volume ratio of the solids. When the flow rate of underflow varies, the operating line on a McCabe–Thiele diagram will be curved instead of straight. The curvature can be established by computing two arbitrary, intermediate points, as illustrated in the following example.

EXAMPLE 16.5

Oil is to be extracted from 10,000 lb/h of granulated halibut livers, based on oil-free livers, which contain 0.043 gal of extractable oil per pound of oil-free livers. It is desired to extract 95% of the oil in a countercurrent extraction system using oil-free diethylether as the solvent. The final extract is to contain 0.65 gal oil per gal of extract. Assume that the volumes of oil and ether are additive, and that

leaching will be completed in one stage. Although the experimental underflow data of Ravenscroft show some scatter, use the following smoothed data to predict underflow rates:

Gal oil per gal liquid	Gal liquid retained per lb oil-free livers
0.00	0.035
0.10	0.042
0.20	0.049
0.30	0.058
0.40	0.069
0.50	0.083
0.60	0.100
0.70	0.132

Determine the number of ideal stages required.

SOLUTION

First, determine the flow rate of solvent.

Oil in liver feed = $10,000(0.043) = 430$ gal liquid/h

Oil in extract = $430(0.95) = 408.5$ gal/h

Oil in underflow of extracted livers = $430 - 408.5 = 21.5$ gal/h

By an iterative procedure, determine the gal/h of ether in the final underflow of extracted livers:

Assume 0.10 gal oil/gal liquid:

From the above data table, have 0.042 gal liquid/lb oil-free livers

Therefore, have $0.042(10,000) = 420$ gal liquid/h and $0.10(420) = 42$ gal/h oil, which is higher than the required value of 21.5 gal/h from above.

Assume 0.05 gal oil/gal liquid:

Linear interpolation of the above data table gives 0.0385 gal liquid/lb oil-free livers.

Therefore, have $0.0385(10,000) = 385$ gal liquid/h

$0.05(385) = 19.3$ gal oil/h, which is too low

By interpolation, have 0.055 gal oil/gal liquid and 0.0389 gal liquid/lb oil-free livers

Therefore, final underflow contains $0.0389(10,000) - 21.5 = 367.5$ gal ether/h

The final extract contains 408.5 gal oil/h, with given value of 65 vol % oil.

Therefore, final extract contains

$$\frac{0.35}{0.65}(408.5) = 220 \text{ gal ether/h}$$

Ether in solvent feed = $220 + 367.5 = 587.5$ gal/h.

Overall material balance is

	Solvent feed	Livers feed	Final extract	Final underflow
Oil-free livers, lb/h	0	10,000	0	10,000
ether, gal/h	587.5	0	220	367.5
oil, gal/h	0	430	408.5	21.5

Ideal Leaching Stage:

Referring to Figure 16.7, the underflow leaving the leaching stage, L , will have the same concentration of oil as in the final extract. That concentration is the specification of 0.65 gal oil/gal liquid. From the above data table, by linear interpolation, gal liquid retained/lb oil-free livers = 0.116.

Therefore, letting x_j and y_j = volume fractions,

$$L_L = 0.116(10,000) = 1,160 \text{ gal liquid}$$

$$x_L = 0.65 \text{ gal oil/gal liquid}$$

Oil material balance around leaching stage:

$$y_1 V_1 + 430 = x_L L_L + 408.5$$

$$\text{or } y_1 V_1 = 0.65(1,160) + 408.5 - 430 = 732.5 \text{ gal oil}$$

Ether material balance around leaching stage

$$(1 - y_1) V_1 = 0.35(1,160) + 220 = 626 \text{ gal ether/h}$$

$$y_1 = \frac{732.5}{732.5 + 626} = 0.539 \text{ gal oil/gal liquid}$$

We now have the end points of the operating line for the washing stages:

Stage N	Stage 1
$y_{N+1} = 0$	$y_1 = 0.539$
$x_N = 0.055$	$x_L = 0.65$

These two pairs of points together with the straight equilibrium line, $y = x$ are plotted on the McCabe-Thiele diagram of Figure 16.10. Because the underflow is variable, a straight line does not connect the end points of the operating line.

Calculation of Intermediate Points on the Operating Line

Determine by oil and total liquid material balances, values of y_{n+1} for $x_n = 0.30$ and 0.50 . The material balance can be made from the solvent feed end to an arbitrary stage n .

For $x_n = 0.3$, with above data table giving 0.058 gal liquid/lb oil-free livers

Oil mass balance:

$$0.3(10,000)(0.058) = y_{n+1} V_{n+1} + 21.5 \quad (1)$$

Total liquid mass balance:

$$10,000(0.058) + 587.5 = V_{n+1} + (367.5 + 21.5) \quad (2)$$

Solving (1) and (2),

$$V_{n+1} = 778.5 \text{ gal/h}$$

$$y_{n+1} = 0.196 \text{ gal oil/gal liquid}$$

For $x_n = 0.50$, with 0.083 gal liquid/lb oil-free livers,

Oil mass balance:

$$0.5(10,000)(0.083) = y_{n+1} V_{n+1} + 21.5 \quad (3)$$

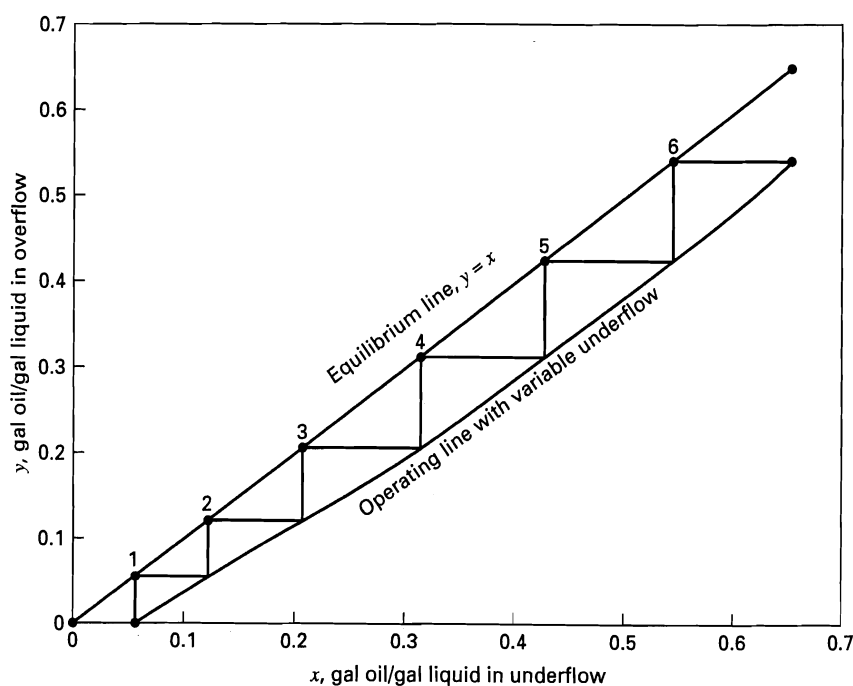


Figure 16.10 McCabe-Thiele diagram for Example 16.5.

Total liquid mass balance:

$$10,000(0.083) + 587.5 = V_{n+1} + (367.5 + 21.5) \quad (4)$$

Solving (3) and (4),

$$V_{n+1} = 1,028.5 \text{ gal/h}$$

$$y_{n+1} = 0.383 \text{ gal oil/gal liquid}$$

When these two sets of x_n , y_{n+1} points are plotted, a curved operating line is obtained as shown in Figure 16.10. A total of almost exactly six washing stages can be stepped off. Adding the leaching stage to this gives a total of seven ideal stages.

16.3 RATE-BASED MODEL FOR LEACHING

Leaching involves the transfer of a solute from the interior of a solid into the bulk of a liquid solvent or extract. The process can be modeled by considering two steps (in series): (1) molecular diffusion of the solute through the solid, and (2) convection and eddy diffusion of the solute through the solvent or extract that is exterior to the solid. Practical rates of molecular diffusion through the solid are only achieved after the solvent penetrates the solid to become occluded liquid, unless the solvent is initially present in the solid. The solute then dissolves into that liquid and diffuses at a reasonable rate to the surface of the solid, leaving behind the insoluble solids and any sparingly soluble materials in the form of a framework. If relative motion exists between the solids and the exterior solvent solution, the resistance to mass transfer in the fluid phase may be negligible compared to that in the solid, and the entire leaching process can be modeled by diffusion through the solid.

Food Processing

Schwartzberg and Chao [11] present a summary of published experimental and theoretical studies of solute diffusion in the leaching of food materials in the form of slices, near-cylinders, and nearly spherical particles, including a compilation of effective diffusivities. As with diffusion of liquids and gases in porous-solid adsorbents, the diffusivity can be expressed as a true molecular diffusivity in the occluded fluid phase, or as an effective diffusivity through the entire solid, including the insoluble-solid framework, sometimes called the *marc*, and the occluded liquid. When an effective diffusivity, D_e , is used, and Fick's laws are applied, the concentration driving force is taken to be the concentration of solute, X_i in mass per unit volume of solid particle. Thus, if r is the direction of diffusion, Fick's first law for the solute, i , is

$$n_i = -D_e A \left(\frac{\partial X_i}{\partial r} \right) \quad (16-11)$$

Fick's second law for constant effective diffusivity in the direction r is

$$\frac{\partial X_i}{\partial t} = \frac{D_e}{r^{v-1}} \frac{\partial}{\partial r} \left[r^{v-1} D_e \frac{\partial X_i}{\partial r} \right] \quad (16-12)$$

where by comparison to (3-69), (3-70), and (3-71), $v = 1, 2$, and 3 for rectangular, cylindrical, and spherical coordinates, respectively.

The rate of mass transfer through the solvent external to the solid can be written in terms of a mass-transfer coefficient and the concentration of the solute in the solvent or extract Y_i , as in (3-105), where subscripts s and b refer to the solid-liquid interface and the bulk liquid, respectively,

$$n_i = k_c A [(Y_i)_s - (Y_i)_b] \quad (16-13)$$

At the solid-liquid interface at the exterior surface of the solid, (16-11) and (16-13) can be equated:

$$-D_e \left(\frac{\partial X_i}{\partial r} \right)_s = k_c [(Y_i)_s - (Y_i)_b] \quad (16-14)$$

Let

$$m = Y_i / X_i$$

$$a = \text{characteristic dimension of the solid,} \quad (16-15)$$

e.g., the radius of a cylinder or spherical particle, or the half-thickness of a slice.

Combining (16-14) and (16-15) and expressing the result in the form of dimensionless groups,

$$-\left[\frac{\partial \left(\frac{Y_i}{(Y_i)_s - (Y_i)_b} \right)}{\partial (r/a)} \right]_s = \left(\frac{mk_c a}{D_e} \right) \quad (16-16)$$

The dimensionless group on the right-hand side of (16-16) is called the Biot number for mass transfer:

$$(N_{Bi})_M = \frac{mk_c a}{D_e} \quad (16-17)$$

which is analogous to the more common Biot number for heat transfer,

$$(N_{Bi}) = \frac{ha}{k} \quad (16-18)$$

Biot numbers are quantitative measures of the ratio of internal (solid) resistance to external (fluid) resistance to transport. In Section 3.3, transient solutions are given to (16-12) for different geometries for the case of an initial uniform concentration, $X_o = c_o$, of the solute in the solid. At time $t = 0$, the solute concentration in the solid phase at the solid-fluid interface is suddenly brought to and then held at $X_s = c_s$. The solutions given are

Geometry	Concentration Profile	Rate of Mass Transfer at Interface	Average Concentration in Solid
Semi-infinite	(3-75)	(3-78)	—
Slab of finite thickness	(3-80), (3-81), and Figure 3.8	(3-82)	(3-85) and Figure 3.9
Infinite cylinder	Figure 3.10	—	Figure 3.9
Sphere	Figure 3.11	—	Figure 3.9

These solutions apply to the case in which the Biot number for mass transfer is infinite, such that the resistance in the fluid phase is negligible and $(Y_i)_b = (Y_i)_s$.

For an infinite Biot number, as indicated above, the solute concentration profile as a function of time is given in Figure 3.8 while the average solute concentration in the solid is given in Figure 3.9. In these plots of the solutions, a dimensionless time, the Fourier number for mass transfer, is used, where

$$(N_{Fo})_M = \frac{D_e t}{a^2} \quad (16-19)$$

and a = flake or slice half thickness. When the internal resistance to mass transfer is negligible, which is almost never the case in the leaching of foods, the solution for the uniform concentration of solute in the solid is given by the following equation, whose derivation is left as an exercise:

$$\frac{X_i - \frac{(Y_i)_b}{m}}{(X_i)_o - \frac{(Y_i)_b}{m}} = \exp\left(\frac{-k_c t m}{a}\right) \quad (16-20)$$

When $(N_{Bi})_M > 200$, the external (fluid) mass-transfer resistance is negligible and Figures 3.8 and 3.9 can be applied. When $(N_{Bi})_M < 0.001$, the internal (solid) mass-transfer resistance is negligible and (16-20) applies. When $(N_{Bi})_M$ lies between these two extremes, both resistances must be taken into account. Solutions for this general case are given by Schwartzberg and Chao [11].

Effective diffusivities for solutes in solids are complicated because they depend on the volume fraction of and concentration of solute in the occluded solvent, temperature, tortuosity of the diffusion path, and extent of adsorption of the solute by the marc. Values of solute effective diffusivities in a variety of foods, with water as the solvent, are tabulated by Schwartzberg and Chao [11]. Typical values for sucrose, when the cell walls are hard (e.g., sugar cane and coffee), range from 0.5 to $1.0 \times 10^{-6} \text{ cm}^2/\text{s}$. When cells walls are soft (e.g., sugar beets, potatoes, apples, celery, and onions), values are higher, ranging from 1.5 to $4.5 \times 10^{-6} \text{ cm}^2/\text{s}$. When the solute is a salt (e.g., NaCl and KCl), effective diffusivities are about four times higher. As mentioned above, the diffusion of oil from flaked oil seeds does not follow Fick's law. If, nevertheless, Fick's law is applied to determine the effective diffusivity, the values are found to decrease significantly with time. For example, data of Karnofsky [12], who leached oil from soybeans, cottonseeds, and flaxseeds, with hexane, give values of effective diffusivity that decrease over the course of extraction by about one order-of-magnitude. Other foods exhibit the same trend under certain conditions. Frequently, the diffusivity is not a constant, but varies with flake or slice thickness and solute concentration. Schwartzberg [11] discusses possible reasons for these effects.

A thin slice or flake of solid can be treated as a slab of finite thickness, with mass transfer from the thin edges ignored. For this case, (3-85) or Figure 3.9 can be used to determine the effective diffusivity from experimental leaching data or predict the rate of leaching. In (3-85), E_{avgslab}

is the fractional unaccomplished approach to equilibrium for extraction, which decreases with time. As seen in Figure 3.9 and as can be demonstrated with (3-85), when $(N_{Fo})_M > 0.10$, the series solution is converged to less than a 2% error with only one term of the infinite series, given by

$$E_{\text{avgslab}} = \frac{8}{\pi^2} \exp\left(-\frac{(N_{Fo})_M \pi^2}{4}\right)$$

or

$$\ln E_{\text{avgslab}} = \ln(8/\pi^2) - \frac{\pi^2}{4} (N_{Fo})_M \quad (16-21)$$

Thus, if Fick's law holds, and the diffusivity is constant, a plot of experimental data as $\log E_{\text{avgslab}}$ against time should yield a straight line with a negative slope from which the effective diffusivity can be determined, as illustrated in the following example.

EXAMPLE 16.6

In the commercial extraction of sugar (sucrose) from sugar beets with water, the process is controlled by diffusion through the sugar beet. Yang and Brier [13] conducted diffusion experiments with beets that were sliced into cosettes that were 0.0383 in. thick $\times$ 0.25 in. wide and 0.5–1.0 in. long. Typically, the cosettes contained 16 wt% sucrose, 74 wt% water, and 10 wt% insoluble fiber. Experiments were conducted at temperatures ranging from 65 to 80°C, with solvent water rates from 1.0 to 1.2 lb/lb fresh cosettes. For a temperature of 80°C and a solvent water rate of 1.2 lb/lb fresh cosettes, the following smoothed data were obtained:

E_{avg}	t , min.
1.0	0
0.39	10
0.19	20
0.10	30
0.050	40
0.025	50
0.0135	60

These data are plotted in Figure 16.11, where it is seen that a straight line can be passed through the data in the range of time from 10 to 60 minutes. From the slope of this line, using (16-19) and (16-21),

$$\frac{\pi^2}{4} \left(\frac{D_e}{a^2}\right) = 0.00113 \text{ sec}^{-1}$$

$$\text{Since } a = \text{half thickness} = \frac{0.0383}{2} (2.54) = 4.86 \times 10^{-2} \text{ cm}$$

Therefore,

$$D_e = \frac{0.00113(4.86 \times 10^{-2})^2(4)}{(3.14)^2} = 1.1 \times 10^{-6} \text{ cm}^2/\text{s}$$

For a continuous, countercurrent extractor, (16-21) can be used to determine the approximate time required for leaching the solids. The time is given in terms of $E = E_{\text{avg}}$ by

$$t = \int_{E_{\text{in}}}^{E_{\text{out}}} \frac{dE}{\left(\frac{dE}{dt}\right)} \quad (16-22)$$

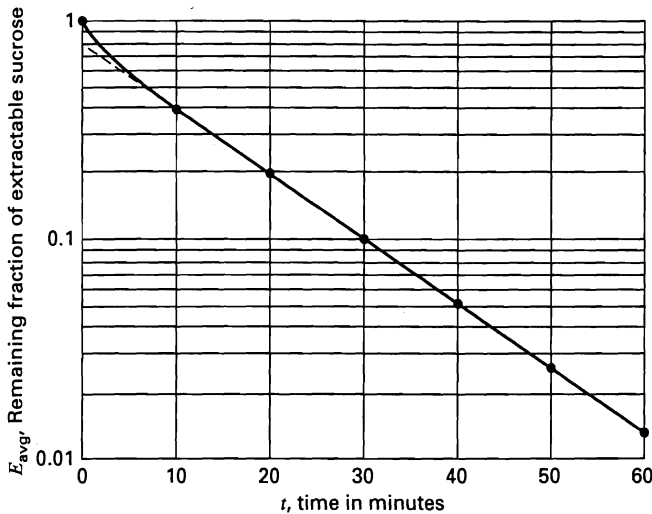


Figure 16.11 Experimental data for leaching of sucrose from sugar beets with water for Example 16.6.

If the solute diffusivity is constant, (dE/dt) , except for small values of time, can be obtained by differentiating (16-21) and combining the result to eliminate time, t , to give

$$\frac{dE}{dt} = -\frac{\pi^2 D_e E}{4a^2} \quad (16-23)$$

Substitution of (16-23) into (16-22), followed by integration, gives

$$t = \frac{4a^2}{\pi^2 D_e} \ln \left(\frac{E_{in}}{E_{out}} \right) \quad (16-24)$$

When the solute diffusivity is not constant, which is more common, experimental plots of E as a function of time can be used directly to obtain values of (dE/dt) for use in (16-22), which can be graphically or numerically integrated, as shown by Yang and Brier [13].

EXAMPLE 16.7

The sucrose in 10,000 lb/h of sugar beets containing 16 wt% sucrose, 74 wt% water, and 10 wt% insoluble fiber is extracted in a continuous, countercurrent extractor at 80°C with 12,000 lb/h of water. If 98% of the sucrose is extracted and no net mass transfer of water occurs, determine the residence time in minutes for the beets. Assume the beets are sliced to 1 mm in thickness and that the effective sucrose diffusivity is that computed in Example 16.6.

SOLUTION

By material balance, the extracted beets contain

$$\begin{aligned} 0.02(0.16)(10,000) &= 32 \text{ lb/h sucrose} \\ 0.74(10,000) &= 7,400 \text{ lb/h water} \\ 0.10(10,000) &= 1,000 \text{ lb/h insoluble fiber} \\ \text{Total} &= 8,432 \text{ lb/h} \end{aligned}$$

Thus,

$$\begin{aligned} X_{out} &= 32/8,432 = 0.0038 \text{ lb/lb} \\ X_{in} &= 1,600/10,000 = 0.160 \text{ lb/lb} \end{aligned}$$

where X is expressed on a weight fraction basis.

At the beet inlet end,

$$E_{in} = \frac{0.16 - (Y/m)_{\text{extract out}}}{0.16 - (Y/m)_{\text{extract out}}} = 1.0$$

At the beet outlet end, $(Y/m)_{\text{solvent in}} = 0$

Therefore,

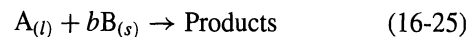
$$E_{out} = \frac{0.0038}{1.0} = 0.0038$$

From (16-24),

$$t = \frac{4 \left(\frac{0.1}{2} \right)^2}{(3.14)^2 (1.1 \times 10^{-6})} \ln \left(\frac{1.0}{0.0038} \right) = 5,140 \text{ s} = 85.6 \text{ min}$$

Mineral Processing

Leaching can be used to recover valuable metals from low-grade ores. The leaching process is accomplished by reacting part of the ore with a constituent of the leach liquor, to produce ions of the metal, which are soluble in the liquid. In general, the reaction can be written as



The removal of reactant B from the ore leaves pores in the solid particle for reactant A to diffuse through to reach reactant B in the interior of the particle.

Figure 16.12 shows a spherical mineral particle undergoing leaching. As the process proceeds, an outer porous leached shell develops, leaving an unleached core. The steps involved are:

1. Mass transfer of reactant A from the bulk liquid to the outer surface of the particle.
2. Pore diffusion of reactant A through the leached shell.
3. Chemical reaction at the interface between the leached shell and the unleached core.
4. Pore diffusion of the reaction products back through the leached shell.
5. Mass transfer of the reaction products back into the bulk liquid surrounding the particle.

Because the diameter of the unleached core shrinks with time, a mathematical model for the process, first conceived for application to gas-solid combustion reactions by Yagi and Kunii [14] in 1955 and extended to liquid-solid leaching by Roman, Benner, and Becker [15] in 1974 is referred to as the *shrinking-core model*. Although any one or more of the above five steps can control the process, the rate of leaching is often controlled by Step 2. Therefore, although the general model has been developed for all possibilities, the leaching model presented here is derived on the assumption that Step 2 is controlling.

Referring to Figure 16.12, assume that dr_c/dt , the rate of movement of the reaction interface at r_c , is small with respect to the diffusion velocity of reactant A, in (16-25), through the porous, leached layer. This is referred to as the

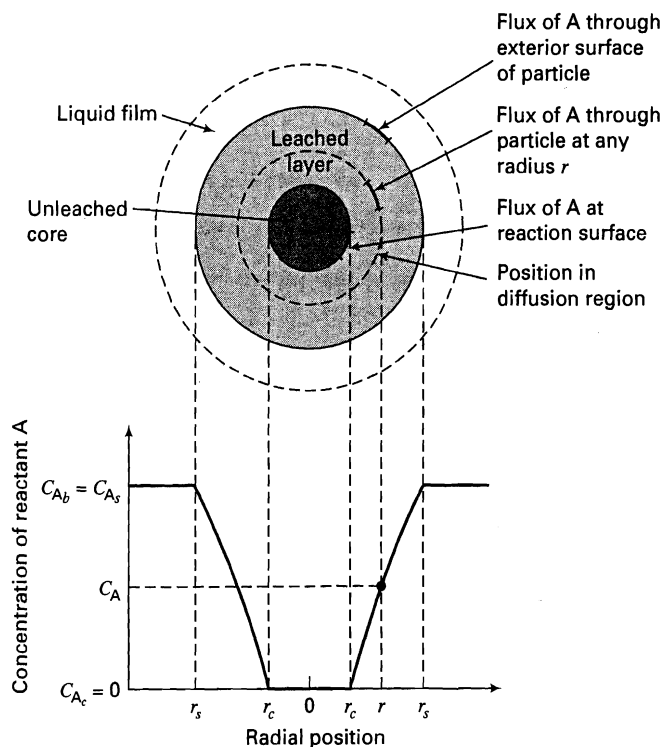


Figure 16.12 Shrinking-core model when diffusion through the leached shell is controlling.

pseudo-steady-state assumption. Although it is valid for the gas–solid case, it is less satisfactory for the liquid–solid case here. The importance of this assumption is that it allows us to neglect the accumulation of reactant A as a function of time in the leached layer as that layer increases in thickness, with the result that the model can be formulated as an ordinary differential equation rather than as a partial differential equation. Thus, the rate of diffusion of reactant A through the porous, leached layer is given by Fick's second law, (3-71), ignoring the term on the left-hand side and replacing the molecular diffusivity with an effective diffusivity:

$$\frac{D_e}{r^2} \frac{d}{dr} \left(r^2 \frac{dc_A}{dr} \right) = 0 \quad (16-26)$$

with boundary conditions:

$$\begin{aligned} c_A &= c_{A_s} = c_{A_b} \quad \text{at} \quad r = r_s \\ c_A &= 0 \quad \text{at} \quad r = r_c \end{aligned}$$

These boundary conditions hold because the mass-transfer resistance in the liquid film or boundary layer is assumed negligible and the interface reaction is assumed to be instantaneous and complete, respectively.

If (16-26) is integrated twice and the boundary conditions are applied, the result after simplification is

$$c_A = c_{A_b} \left[\frac{1 - \frac{r_c}{r}}{1 - \frac{r_c}{r_s}} \right] \quad (16-27)$$

To obtain a relationship between r_c and time, t , differentiate (16-27) with respect to r and evaluate the differential at

$r = r_c$:

$$\left. \frac{dc_A}{dr} \right|_{r=r_c} = \frac{c_{A_b}}{r_c \left(1 - \frac{r_c}{r_s} \right)} \quad (16-28)$$

The rate of diffusion at $r = r_c$ is given by Fick's first law:

$$n_A = \frac{dN_A}{dt} = -4\pi r_c^2 D_e \left(\frac{dc_A}{dr} \right)_{r=r_c} \quad (16-29)$$

where N_A = moles of A

Combining (16-28) and (16-29),

$$-\frac{dN_A}{dt} = \frac{4\pi r_c^2 D_e c_{A_b}}{\left(1 - \frac{r_c}{r_s} \right)} \quad (16-30)$$

By stoichiometry, from (16-25),

$$\frac{dN_A}{dt} = \frac{1}{b} \frac{dN_B}{dt} \quad (16-31)$$

By material balance,

$$\frac{dN_B}{dt} = \frac{\rho_B}{M_B} \frac{d}{dt} \left(\frac{4}{3} \pi r_c^3 \right) = \frac{4\pi r_c^2 \rho_B}{M_B} \frac{dr_c}{dt} \quad (16-32)$$

where

ρ_B = initial mass of reactant B per unit volume of solid particle

M_B = molecular weight of B

Combining (16-30) with (16-32),

$$\frac{-\rho_B}{M_B} \left(\frac{1}{r_c} - \frac{1}{r_s} \right) r_c^2 dr_c = b D_e c_{A_b} dt \quad (16-33)$$

Integration of (16-33) and application of the boundary condition,

$$r_c = r_s \quad \text{at} \quad t = 0$$

gives

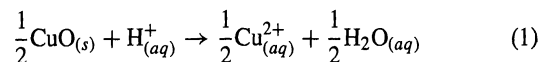
$$t = \frac{\rho_B r_s^2}{6 D_e b M_B c_{A_b}} \left[1 - 3 \left(\frac{r_c}{r_s} \right)^2 + 2 \left(\frac{r_c}{r_s} \right)^3 \right] \quad (16-34)$$

For complete leaching, $r_s = 0$, and (16-34) becomes

$$t = \frac{\rho_B r_s^2}{6 D_e b M_B c_{A_b}} \quad (16-35)$$

EXAMPLE 16.8

A copper ore containing 2 wt% CuO is to be leached with 0.5-M H_2SO_4 . The reaction is



The leaching process is controlled by the diffusion of hydrogen ions through the leached layer. The effective diffusivity, D_e , of the hydrogen ion has been determined by laboratory tests to be $0.6 \times 10^{-6} \text{ cm}^2/\text{s}$. The specific gravity of the ore is 2.7. For ore particles of diameter equal to 10 mm, estimate the time required to leach 98% of the copper, assuming that the CuO is uniformly distributed throughout the particles. Also, check the validity of the

pseudo-steady-state assumption by comparing the amount of hydrogen ions held up in the liquid in the pores with the amount reacted with CuO.

SOLUTION

If 98% of the cupric oxide is leached, then r_c corresponds to 2% of the particle volume. Thus,

$$\frac{4}{3}\pi r_c^3 = (0.02)\frac{4}{3}\pi r_s^3$$

or

$$r_c = (0.02)^{1/3} r_s = (0.02)^{1/3} (0.5) = 0.136 \text{ cm}$$

$$\rho_B = 0.02(2.7) = 0.054 \text{ g/cm}^3 = \text{density of CuO in the ore}$$

$$M_B = 79.6 = \text{molecular weight of CuO}$$

From (16-25) and (1), $b = 0.5$.

$$\text{For } 0.5\text{-MH}_2\text{SO}_4, c_{H_b}^+ = \frac{2(0.5)}{1000} = 0.001 \text{ mol/cm}^3.$$

From (16-34), with $r_c/r_s = 0.136/0.500 = 0.272$,

$$t = \frac{(0.054)(0.5)^2}{6(0.6 \times 10^{-6})(0.5)(79.6)(0.001)} [1 - 3(0.272)^2 + 2(0.272)^3] \\ = 77,000 \text{ sec} = 21.4 \text{ h}$$

Now, check the validity of the pseudo-steady-state assumption.

The specific gravity of CuO is 6.4 g/cm³.

$$100 \text{ g of ore occupies } 100/2.7 = 37.0 \text{ cm}^3.$$

The CuO in this amount of ore occupies

$$0.02(100)/6.4 = 0.313 \text{ cm}^3 \text{ or } 0.845\% \text{ of the particle volume}$$

$$\text{Volume of one particle} = \frac{4}{3}\pi r_s^3 = \frac{4}{3}(3.14)(0.5)^3 = 0.523 \text{ cm}^3.$$

$$\text{Volume of CuO as pores in one particle} = 0.00845(0.523) = 0.0044 \text{ cm}^3.$$

Mols of H⁺ in pores, based on the bulk concentration to be conservative:

$$0.001(0.0044) = 4.4 \times 10^{-6} \text{ mol}$$

98% of CuO leached in a particle, in mol units:

$$\frac{0.98(0.02)(2.7)(0.523)}{79.6} = 3.5 \times 10^{-4} \text{ mol}$$

which requires $7.0 \times 10^{-4} \text{ mol H}^+$ for reaction.

Because this value is approximately two orders or magnitude larger than the conservative estimate of H⁺ in the pores, the pseudo-steady-state assumption is reasonable.

SUMMARY

1. Leaching is similar to liquid-liquid extraction, except that the solute initially resides in a solid. Leaching is widely used to remove solutes from foods and minerals.
2. When leaching is rapid, it can be accomplished in one stage. However, the leached solid will retain surface liquid that contains the solute. To recover most of the solute in the extract, it is desirable to add one or more washing stages in a countercurrent arrangement.
3. Leaching of large solids can be very slow because of very small values of diffusivities in solids. Therefore, it is common to reduce the size of the solids by crushing, grinding, flaking, slicing, etc.
4. Industrial leaching equipment is available for batch or continuous processing. The solids are contacted with the solvent by either percolation or immersion. Large, continuous, countercurrent extractors can process up to 7,000,000 kg/day of food solids.
5. Washing of large flow rates of leached solids is commonly carried out in thickeners that can be designed to produce a clear liquid overflow and a concentrated solids underflow. When a clear overflow is not critical, hydroclones can replace thickeners.

6. An equilibrium-stage model is widely used for continuous, countercurrent systems when leaching is rapid and washing is desirable for high solute recovery. The model assumes that the concentration of the solute in the overflow leaving a stage equals that in the liquid retained on the solid leaving the stage in the underflow.
7. When the ratio of liquid to solids in the underflow is constant from stage to stage, the equilibrium-stage model can be applied algebraically by a modified Kremser method or graphically by a modified McCabe-Thiele method. If the underflow is variable, the graphical method with a curved operating line is applied.
8. When leaching is slow, as with food solids or low-grade ores, leaching calculations must be done on a rate basis. In some cases, the diffusion of solutes in food solids does not obey Fick's law, because of complex membrane and fiber structures.
9. The leaching of low-grade ores by reactive-leaching is conveniently carried out with a shrinking-core diffusion model, using a pseudo-steady-state assumption.

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EXERCISES

Section 16.1

16.1 Using experimental data from pilot-plant tests of soybean extraction by Othmer and Agarwal, summarized at the beginning of this chapter, check the mass balances for oil and hexane around the extractor, assuming the moisture is retained in the flakes, and compute the mass ratio of liquid oil to flakes in the leached solids leaving the extractor.

Section 16.2

16.2 Barium carbonate, which is essentially water insoluble, is to be made by precipitation from an aqueous solution containing 120,000 kg/day of water and 40,000 kg/day of barium sulfide, with the stoichiometric amount of solid sodium carbonate. The reaction also produces a by-product of water-soluble sodium sulfide. The process will be carried out in a continuous, countercurrent system of five thickeners. The reaction will take place completely in the first thickener to which will be fed the solid sodium carbonate, the aqueous solution of barium sulfide, and the overflow from the second thickener. Sufficient fresh water will enter the last thickener so that the overflow from the first thickener will be 10 wt% sodium sulfide, assuming that the underflow from each thickener contains two parts of water per one part of barium carbonate by weight.

- (a) Draw a schematic diagram of the process and label it with all the given information.
- (b) Determine the kg/day of sodium carbonate required and the kg/day of barium carbonate and sodium sulfide produced by the reaction.
- (c) Determine the kg/day of fresh water needed, the wt% of sodium sulfide in the liquid portion of the underflow that leaves each thickener, and the kg/day of sodium sulfide that will remain with the barium carbonate product after it is dried.

16.3 Calcium-carbonate precipitate can be produced by the reaction of an aqueous solution of sodium carbonate and calcium oxide. The by-product is aqueous sodium hydroxide. Following decantation, the slurry leaving the precipitation tank is 5 wt% calcium carbonate, 0.1 wt% sodium hydroxide, and the balance water. One hundred thousand lb/h of this slurry is fed to a two-stage, continuous, countercurrent washing system to be washed with 20,000 lb/h of fresh water. The underflow from each thickener will contain 20 wt% solids. Determine the percent recovery of sodium hydroxide in the extract and wt% sodium hydroxide in the dried, calcium-carbonate product. Based on calculations, is it worthwhile to add a third stage?

16.4 Zinc is to be recovered from an ore containing zinc sulfide. The ore is first roasted with oxygen to produce zinc oxide, which is then leached with aqueous sulfuric acid to produce water-soluble zinc sulfate and an insoluble, worthless residue called gangue. The

decanted sludge of 20,000 kg/h contains 5 wt% water, 10 wt% zinc sulfate, and the balance as gangue. This sludge is to be washed with water in a continuous, countercurrent washing system to produce an extract, called a strong solution, of 10 wt% zinc sulfate in water, with a 98% recovery of the zinc sulfate. Assume that the underflow from each washing stage contains, by weight, two parts of water (sulfate-free basis) per part of gangue. Determine the number of stages required.

16.5 Fifty-thousand kg/h of flaked soybeans, containing 20 wt% oil, is to be leached of the oil with the same flow rate of *n*-hexane in a countercurrent-flow system consisting of an ideal leaching stage and three ideal washing stages. Experiments show that the underflow from each stage will contain 0.8 kg liquid/kg soybeans (oil-free basis).

- (a) Determine the % recovery of oil in the final extract.
- (b) If leaching requires three of the four stages, such that one-third of the leaching occurs in each of these three stages, followed by just one true washing stage, determine the % recovery of oil in the final extract.

16.6 One hundred tons per hour of a feed containing 20 wt% Na_2CO_3 and the balance insoluble solids is to be leached and washed with water in a continuous, countercurrent system. Assume that leaching will be completed in one ideal stage. It is desired to obtain a final extract containing 15 wt% solute, with a 98% recovery of solute. The underflow from each stage will contain 0.5 lb solution/lb insoluble solids. Determine the number of ideal washing stages required.

16.7 Titanium dioxide, which is the most common white pigment in paint, can be produced from the titanium mineral, rutile, by chlorination to TiCl_4 , followed by oxidation to TiO_2 . To purify the insoluble titanium dioxide, it is washed free of soluble impurities in a continuous, countercurrent system of thickeners with water. Two hundred thousand kg/h of 99.9 wt% titanium dioxide pigment is to be produced by washing, followed by filtering and drying. The feed contains 50 wt% TiO_2 , 20 wt% soluble salts, and 30 wt% water. The wash liquid is pure water at a flow rate equal to that of the feed on a mass-flow basis.

- (a) Determine the number of washing stages required if the underflow from each stage is 0.4 kg solution/kg TiO_2 .
- (b) Determine the number of washing stages required if the underflow is variable as follows:

Concentration of solute, kg/solute/kg solution	Retention of solution, kg solution/kg TiO_2
0.0	0.30
0.2	0.34
0.4	0.38
0.6	0.42

Section 16.3

16.8 Derive (16-20), assuming that $(Y_i)_b$, k_c , m , and a are constants and that $(X_i)_o$ is uniform through the solid.

16.9 Derive (16-24).

16.10 Data of Othmer and Agarwal [1] for the batch extraction of oil from soybeans by oil-free *n*-hexane at 80°F are as follows:

Time, min	Oil content of Soybeans, g/g Dry, Oil-free Soybeans
0	0.203
0.5	0.1559
1	0.1359
2	0.1190
4	0.0981
7	0.0775
12	0.0591
20	0.04197
35	0.03055
60	0.02388
120	0.02107

Determine whether these data are consistent with a constant effective diffusivity of oil in soybeans.

16.11 Estimate the molecular diffusivity of sucrose in water at infinite dilution at 80°C, noting that the value is $0.54 \times 10^{-5} \text{ cm}^2/\text{s}$ at 25°C. Give reasons for the difference between the value you obtain and the value for effective diffusivity in Example 16.6.

16.12 The sucrose in ground coffee particles of an average diameter of 2 mm is to be extracted with water in a continuous, counter-current extractor at 25°C. The diffusivity of the sucrose in the particles has been determined to be about $1.0 \times 10^{-6} \text{ cm}^2/\text{s}$. Estimate the time in minutes to leach 95% of the sucrose. For a sphere, with $N_{\text{FOM}} > 0.10$,

$$E_{\text{ave}} = \frac{6}{\pi^2} \exp\left(\frac{-\pi^2 D_e t}{a^2}\right)$$

16.13 For the conditions of Example 16.8, determine the effect on leaching time of particle size over the range of 0.5 mm to 50 mm.

16.14 For the conditions of Example 16.8, determine the effect of % recovery of copper over the range of 50–100%.

16.15 Repeat Example 16.8, except that the ore contains 3 wt% Cu_2O .

16.16 For the shrinking-core model, if the rate of leaching is controlled by an interface chemical reaction that is first order in the concentration of reactant A, derive the expression,

$$t = \frac{\rho_B r_s}{b M_B k C_{A_b}} \left(1 - \frac{r_c}{r_s}\right)$$

where k = first-order rate constant.

Crystallization, Desublimation, and Evaporation

Crystallization is a solid–fluid separation operation in which crystalline particles are formed from a homogeneous fluid phase. Ideally, the crystals are a pure chemical, obtained in a high yield with a desirable shape and a reasonably uniform and desirable size. Crystallization is one of the oldest known separation operations, with the recovery of sodium chloride as salt crystals from water by evaporation dating back to antiquity. Even today, the most common applications are the crystallization from aqueous solution of various inorganic salts, a short list of which is given in Table 17.1. All these cases are referred to as *solution crystallization* because the inorganic salt is clearly the solute, which is crystallized, and water is the solvent, which remains in the liquid phase. The phase diagram for systems suitable for solution crystallization is a solubility curve, such as shown in Figure 17.1a and described earlier in Chapter 4.

For the formation of organic crystals, organic solvents such as acetic acid, ethyl acetate, methanol, ethanol, acetone, ethyl ether, chlorinated hydrocarbons, benzene, and petroleum fractions may be preferred choices, but they must be used with great care when they are toxic or flammable with a low flash point and a wide range of explosive limits.

For either aqueous or organic solutions, crystallization is effected by cooling the solution, evaporating the solvent, or a combination of the two. In some cases, a mixture of two or more solvents may be best, examples of which include water with the lower alcohols, and normal paraffins with chlorinated solvents. Also, the addition of a second solvent is sometimes used to reduce the solubility of the solute. When water is the additional solvent, the process is called *watering-out*; when an organic solvent is added to an aqueous salt solution, the process is called *salting-out*. For both of these cases of solvent addition, fast crystallization called *precipitation* can occur, resulting in large numbers of very small crystals. Precipitation also occurs when one product of two

reacting solutions is a solid with low solubility. For example, when aqueous solutions of silver nitrate and sodium chloride are mixed together, insoluble silver chloride is precipitated leaving an aqueous solution of mainly soluble sodium nitrate.

When both components of a homogeneous, binary solution have melting (freezing) points not far removed from each other, the solution is referred to as a *melt*. If, as in Figure 17.1b, the phase diagram for the melt exhibits a eutectic point, it is possible to obtain, in one step called *melt crystallization*, pure crystals of one component or the other, depending on whether the composition of the melt is to the left or right of the eutectic composition. If, however, solid solutions form, as shown in Figure 17.1c, a process of repeated melting and freezing steps, called *fractional melt crystallization*, is required to obtain nearly pure crystalline products. A higher degree of purity can be achieved by a technique called *zone melting* or *refining*. Examples of binary organic systems that form eutectics include metaxylene-paraxylene and benzene-naphthalene. Binary systems of naphthalene-beta naphthol and naphthalene- β naphthylamine, which form solid solutions, are not as common.

Crystallization can also occur from a vapor mixture by a process more properly called *desublimation*. A number of pure compounds, including phthalic anhydride and benzoic acid, are produced in this manner. When two or more compounds tend to desublime, a fractional desublimation process can be employed to obtain near-pure products.

Crystallization of a compound from a dilute, aqueous solution is often preceded by *evaporation* in one or more vessels, called *effects*, to concentrate the solution, and followed by partial separation and washing of the crystals from the resulting slurry, called the *magma*, by centrifugation or filtration. The process is completed by drying the crystals to a specified moisture content.

17.0 INSTRUCTIONAL OBJECTIVES

After completing this chapter, you should be able to:

- Describe different types of crystallization.
- Explain how crystals grow.
- Explain how crystal-size distribution can be measured, tabulated, and plotted.

- Explain the importance of supersaturation in crystallization.
- Differentiate between primary and secondary nucleation of crystals.
- Use mass-transfer theory to determine rate of crystal growth.
- Describe major types of batch and continuous solution-crystallization equipment.
- Apply the MSMPR model to design of a continuous, vacuum, evaporating crystallizer of the draft-tube baffled (DTB) type.
- Understand precipitation.
- Describe equipment for melt crystallization.
- Apply mass-transfer theory to a falling-film melt crystallizer.
- Apply the ideal zone-melting model.
- Differentiate between crystallization and desublimation.
- Describe evaporation equipment.
- Derive and apply the ideal evaporator model.
- Design multiple-effect evaporation systems.

Industrial Example

Consider the crystallization of $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ (Epsom salt) from an aqueous solution. The solid-liquid phase diagram for the $\text{MgSO}_4 \cdot \text{H}_2\text{O}$ system at 1 atm is shown in Figure 17.2. Depending on the temperature, four different hydrated forms of MgSO_4 are possible: $\text{MgSO}_4 \cdot \text{H}_2\text{O}$, $\text{MgSO}_4 \cdot 6\text{H}_2\text{O}$, $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$, and $\text{MgSO}_4 \cdot 12\text{H}_2\text{O}$. Furthermore, a eutectic of the latter hydrate with ice is possible. To obtain the usually desired heptahydrate, crystallization must occur in the temperature range from 36°F to 118°F (Point *b* to Point *c*). Within this range, the solubility of MgSO_4 (anhydrous or hydrate-free basis) increases almost linearly from about 21 to 33 wt%.

A representative commercial process for producing 4,205 lb/hr (dry basis) of $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ crystals from a 10 wt% aqueous solution at 1 atm and 70°F is shown in Figure 17.3. This solution is first concentrated in a double-effect evaporation system with forward feed and then mixed with recycled mother liquors from the hydroclone and centrifuge. The combined feed of 14,326 lb/h containing 31.0 wt% MgSO_4 at 120°F and 1 atm enters an evaporative, vacuum crystallizer constructed of 316 stainless steel and shown in more detail in Figure 17.4.

The crystallizer utilizes internal circulation of 6,000 gpm of magma up through a draft tube equipped with a 3-Hp marine-propeller agitator to obtain near-perfect mixing of

Table 17.1 Some Inorganic Salts Recovered from Aqueous Solutions

Chemical Name	Formula	Common Name	Crystal System
Ammonium chloride	NH_4Cl	sal-ammoniac	cubic
Ammonium sulfate	$(\text{NH}_4)_2\text{SO}_4$	mascagnite	orthorhombic
Barium chloride	$\text{BaCl}_2 \cdot 2\text{H}_2\text{O}$		monoclinic
Calcium carbonate	CaCO_3	calcite	rhombohedral
Copper sulfate	$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$	blue vitriol	triclinic
Magnesium sulfate	$\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$	Epsom salt	orthorhombic
Magnesium chloride	$\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$	bischofite	monoclinic
Nickel sulfate	$\text{NiSO}_4 \cdot 6\text{H}_2\text{O}$	single nickel salt	tetragonal
Potassium chloride	KCl	muriate of potash	cubic
Potassium nitrate	KNO_3	nitre	hexagonal
Potassium sulfate	K_2SO_4	arcanite	orthorhombic
Silver nitrate	AgNO_3	lunar caustic	orthorhombic
Sodium chlorate	NaClO_3		cubic
Sodium chloride	NaCl	salt, halite	cubic
Sodium nitrate	NaNO_3	chile salt petre	rhombohedral
Sodium sulfate	$\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$	glauber's salt	monoclinic
Sodium thiosulfate	$\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$	hypo	monoclinic
Zinc sulfate	$\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$	white vitriol	orthorhombic

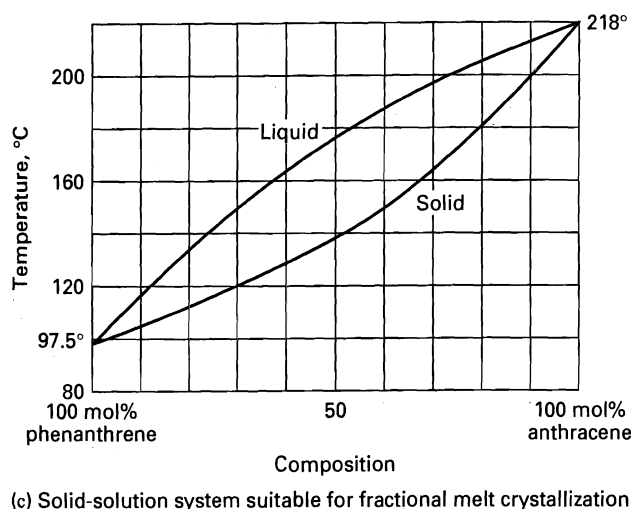
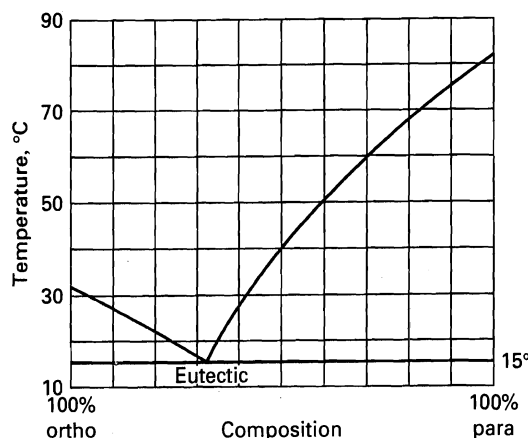
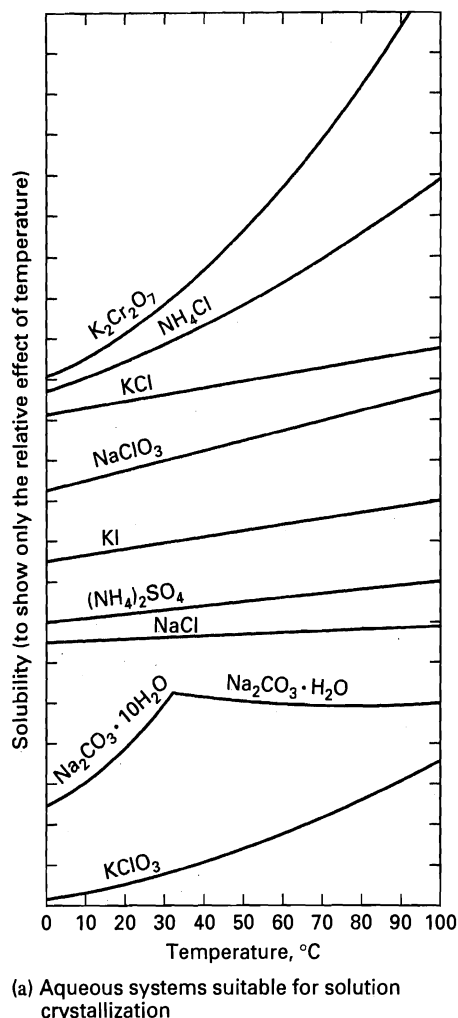


Figure 17.1 Different types of solubility curves.

[From *Handbook of Separation Techniques for Chemical Engineers*, 2nd ed., P.A. Schweitzer, Editor-in-chief, McGraw-Hill, New York (1988) with permission.]

the magma. Mother liquor, which is separated from crystals during upward flow outside of the skirt baffle, is circulated externally at the rate of 625 gpm, by a 10-Hp stainless-steel pump, through a 300-ft² stainless-steel, plate-and-frame heat exchanger, where 2,052,000 Btu/hr of heat is transferred to the solution from 2,185 lb/h of condensing 20 psig steam to provide supersaturation and energy to evaporate 2,311 lb/h of water in the crystallizer.

The vapor leaving the top of the crystallizer is condensed by direct contact with cooling water in a barometric condenser, attached to which are ejectors to pull a vacuum of 0.867 psia in the vapor space of the crystallizer. The product magma, at a temperature of 105°F, consists of 7,810 lb/h of mother liquor saturated with 30.6 wt% MgSO₄ and 4,205 lb/h of crystals. This corresponds to a magma containing 35% crystals by weight or 30.2% crystals by volume, based on a crystal density of 1.68 g/cm³ and a mother liquor density of 1.35 g/cm³. The boiling-point elevation of the saturated

mother liquor at 105°F is 8°F. Thus, the vapor leaving the crystallizer is superheated by the same 8°F. The magma residence time in the crystallizer is 4 hours, which is sufficient to produce the following crystal-size distribution:

35 wt% on 20 mesh U.S. screen
80 wt% on 40 mesh U.S. screen
99 wt% on 100 mesh U.S. screen

The crystallizer for the representative process is 30 ft high with a vapor-space diameter of 5-1/2 ft and a magma-space diameter of 10 ft. The magma is thickened to 50 wt% crystals in a hydroclone, from which the mother-liquor overflow is recycled to the crystallizer and the underflow slurry is sent to a continuous centrifuge, where the slurry is further thickened to 65 wt% crystals and washed. Filtrate mother liquor from the centrifuge is also recycled to the crystallizer. The centrifuge cake is fed to a continuous direct-heat rotary dryer to reduce the moisture content of the crystals to 1.5 wt%.

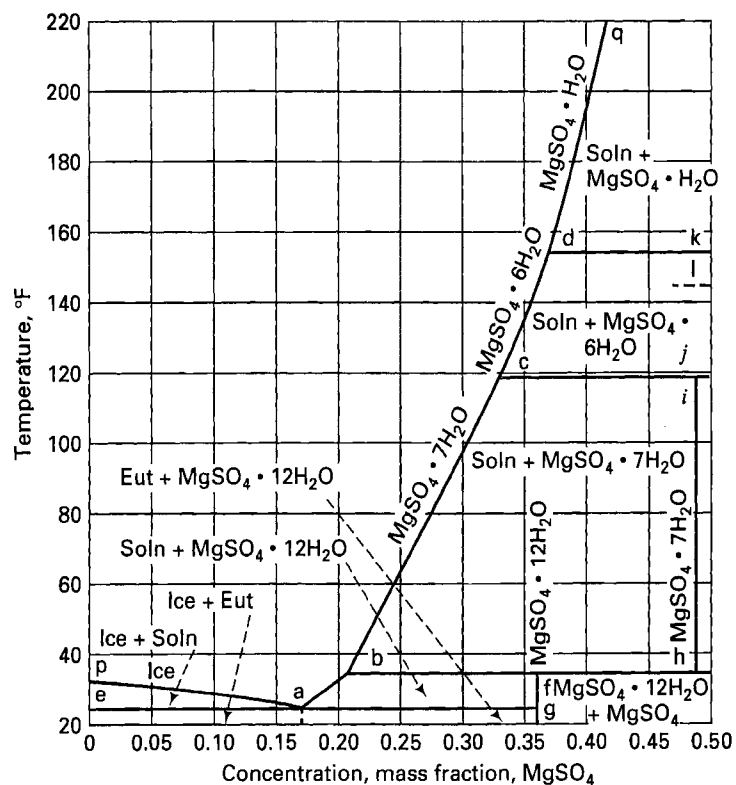


Figure 17.2 Solid-liquid phase diagram for the MgSO_4 - $n\text{H}_2\text{O}$ system at 1 atm.

[From W.L. McCabe, J.C. Smith, and P. Harriott, *Unit Operations of Chemical Engineering*, 5th ed., McGraw-Hill, New York (1993) with permission.]

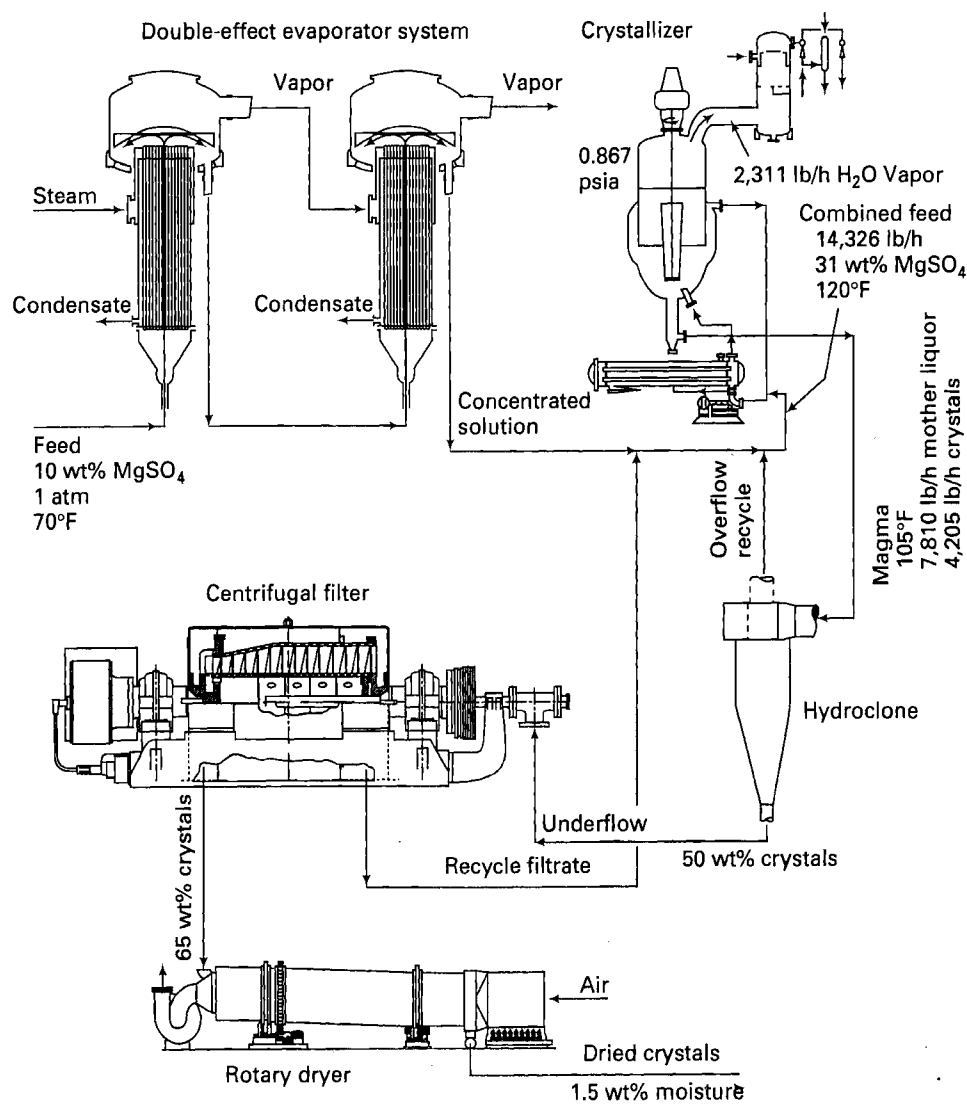


Figure 17.3 Process for production of $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$.

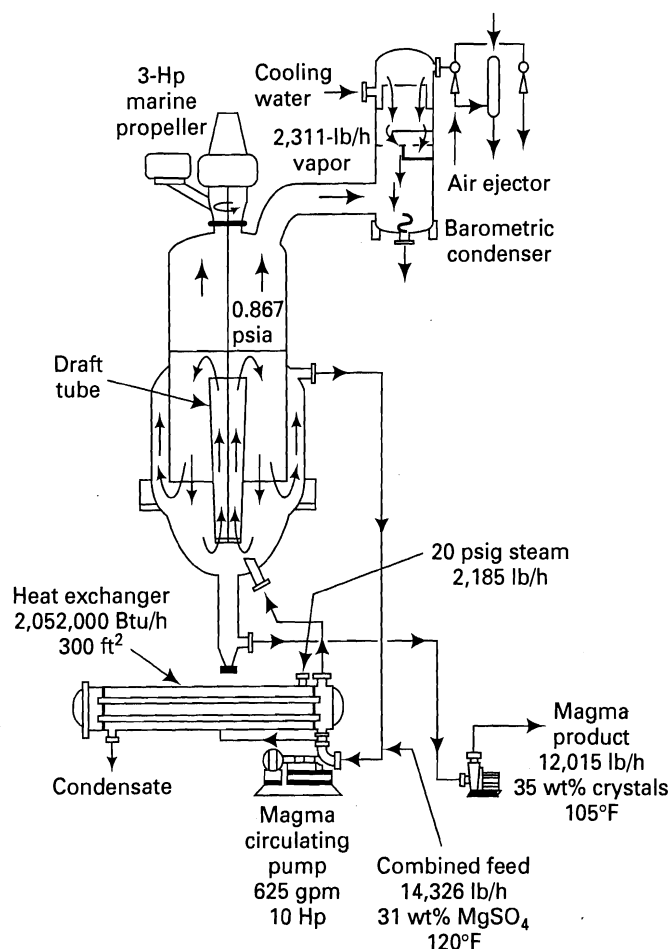


Figure 17.4 Crystallizer for production of $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ crystals.

17.1 CRYSTAL GEOMETRY

In a solid, the motion of molecules, atoms, or ions is restricted largely to oscillations about fixed positions. If the solid is amorphous, these positions are not arranged in a regular or lattice pattern; if the solid is crystalline, they are. Amorphous solids are isotropic, such that physical properties are independent of the direction of measurement; crystalline solids are anisotropic, unless the crystals are cubic in structure.

When crystals grow, unhindered by other surfaces such as container walls and other crystals, they form polyhedrons with flat sides and sharp corners. Crystals are never spherical in shape. Although two crystals of a given chemical may appear quite different in size and shape, they always have something in common, known as the Law of Constant Interfacial Angles, proposed by Hauy in 1784. This law states that the angles between corresponding faces of all crystals of a given substance are constant, even though the crystals vary in size and in the development of the various faces (called the *crystal habit*). The interfacial angles and lattice dimensions can be measured accurately by x-ray crystallography.

As discussed by Mullin [1], early investigators found that crystals consist of many units, each shaped like the larger

crystal. This led to the concept of a space lattice as a regular arrangement of points (molecules, atoms, or ions) such that if a line is drawn between any two points and then extended in both directions, the line will pass through other lattice points with an identical spacing. In 1848, Bravais showed that only the 14 space lattices shown in Figure 17.5 are possible. Based on the symmetry of the three mutually perpendicular axes with respect to their relative lengths (a, b, c) and the angles (α, β, γ) between the axes, the 14 lattices can be classified into the seven crystal systems listed in Table 17.2. For example, the cubic (regular) system includes the simple cubic lattice, the body-centered cubic lattice, and the face-centered lattice. Examples of the seven crystal systems are included in Table 17.1. The five sodium salts included in that table form three of the seven crystal systems.

Actual crystals of a given substance and a given crystal system can exhibit markedly different appearances when the faces grow at different rates, particularly when these rates vary markedly from stunted growth in one direction, so as to give plates, or by exaggerated growth in another direction, to give needles. For example, potassium sulfate, which belongs to the orthorhombic-crystal system, can take on any of the shapes (crystal habits) shown in Figure 17.6, including plates, needles, and prisms. When product crystals of a particular crystal habit are desired, experimental research may be required to find the necessary processing conditions. Modifications of crystal habit are most often accomplished by deliberate addition of impurities to the solution.

Crystal-Size Distributions

Typical magmas from a crystallizer contain a distribution of crystal sizes and shapes. It is highly desirable to characterize a batch of crystals (or particles in general) by an average crystal size and a crystal-size distribution. This is often accomplished by defining a characteristic crystal dimension. However, as shown in Figure 17.6, some crystal shapes might require two characteristic dimensions, while one might suffice for others. One solution to this problem, which is particularly applicable to the correlation of transport rates involving particles, is to relate the irregular-shaped particle to a sphere by the sphericity, ψ , defined as

$$\psi = \frac{\text{surface area of a sphere with the same volume as the particle}}{\text{surface area of the particle}} \quad (17-1)$$

For a sphere, $\psi = 1$, while for all other particles, $\psi < 1$. For a spherical particle of diameter, D_p , the surface area, s_p , to volume, v_p , ratio is

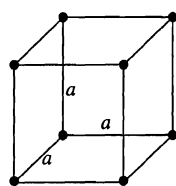
$$(s_p/v_p)_{\text{sphere}} = (\pi D_p^2)/(\pi D_p^3/6) = 6/D_p$$

Therefore, (17-1) becomes

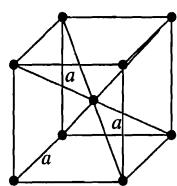
$$\psi = \frac{6}{D_p} \left(\frac{v_p}{s_p} \right)_{\text{particle}} \quad (17-2)$$

Table 17.2 The Seven Crystal Systems

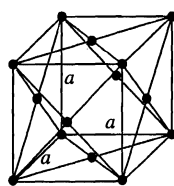
Crystal System	Space Lattices	Length of Axes	Angles between Axes
Cubic (regular)	Simple cubic	$a = b = c$	$\alpha = \beta = \gamma = 90^\circ$
	Body-centered cubic		
	Face-centered cubic		
Tetragonal	Square prism	$a = b < c$	$\alpha = \beta = \gamma = 90^\circ$
	Body-centered square prism		
Orthorhombic	Simple orthorhombic	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90^\circ$
	Body-centered orthorhombic		
	Base-centered orthorhombic		
	Face-centered orthorhombic		
Monoclinic	Simple monoclinic	$a \neq b \neq c$	$\alpha = \beta = 90^\circ$
	Base-centered monoclinic		$\gamma \neq 90^\circ$
Rhombohedral (trigonal)	Rhombohedral	$a = b = c$	$\alpha = \beta = \gamma \neq 90^\circ$
Hexagonal	Hexagonal	$a = b \neq c$	$\alpha = \beta = 90^\circ$
			$\gamma \neq 120^\circ$
Triclinic	Triclinic	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90^\circ$



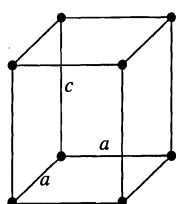
Simple cubic



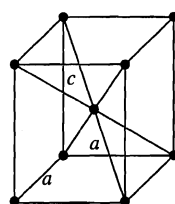
Body-centered cubic



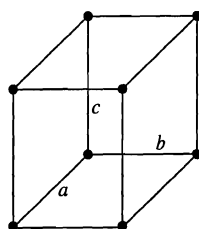
Face-centered cubic



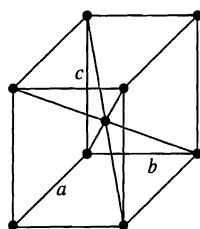
Simple tetragonal



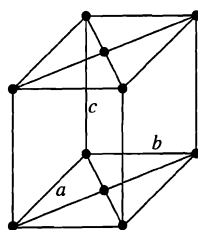
Body-centered tetragonal



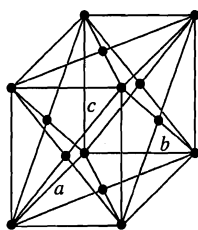
Simple orthorhombic



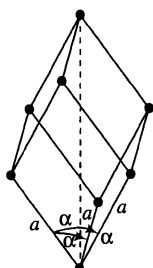
Body-centered orthorhombic



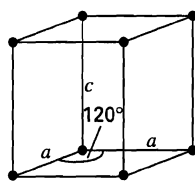
Base-centered orthorhombic



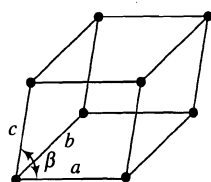
Face-centered orthorhombic



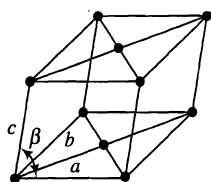
Rhombohedral



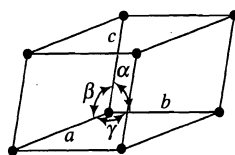
Hexagonal



Simple monoclinic



Base-centered monoclinic



Triclinic

Figure 17.5 The 14 space lattices of Bravais.

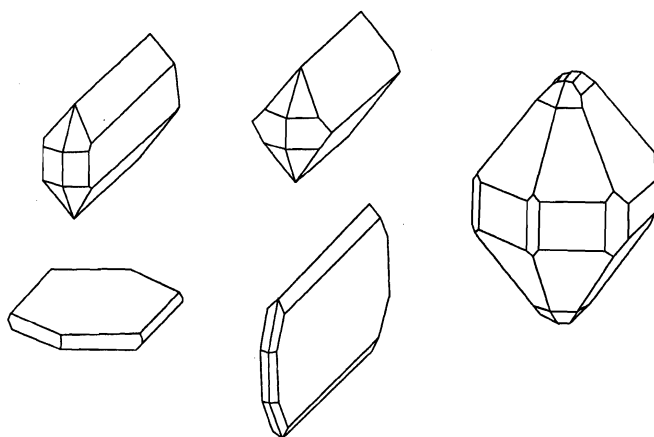


Figure 17.6 Some crystal habits of orthorhombic, potassium-sulfate crystals.

EXAMPLE 17.1

Estimate the sphericity of a cube of dimension a on each side.

SOLUTION

$$v_{\text{cube}} = a^3$$

$$s_{\text{cube}} = 6a^2$$

From (17-2),

$$\psi = \frac{6}{D_p} \left(\frac{a^3}{6a^2} \right) = \frac{a}{D_p}$$

Because the volumes of the sphere and the cube must be equal,

$$\pi D_p^3 / 6 = a^3$$

Solving,

$$D_p = 1.241 a$$

Then,

$$\psi = a / (1.241 a) = 0.806$$

The most common methods for measuring particle size are listed in Table 17.3 together with their useful particle-size ranges. Because of the irregular shapes of crystals, it should not be surprising that the different methods can give results that may differ by as much as 50%. Crystal-size distributions are most often determined with U.S. (or British)

Table 17.3 Methods of Measuring Particle Size

Method	Size Range, Microns
Woven-wire screen	32–5600
Coulter electrical sensor	1–200
Gravity sedimentation	1–50
Optical microscopy	0.5–150
Laser-light scattering	0.04–2000
Centrifugal sedimentation	0.01–5
Electron microscopy	0.001–5

Table 17.4 U.S. Standard Screens ASTM E11

Mesh Number	Opening of Square Aperture		
	in.	mm	μm
3-1/2	0.220	5.60	5600
4	0.187	4.75	4750
5	0.157	4.00	4000
6	0.132	3.35	3350
7	0.110	2.80	2800
8	0.0929	2.36	2360
10	0.0787	2.00	2000
12	0.0669	1.70	1700
14	0.0551	1.40	1400
16	0.0465	1.18	1180
18	0.0394	1.000	1000
20	0.0335	0.850	850
25	0.0280	0.710	710
30	0.0236	0.600	600
35	0.0197	0.500	500
40	0.0167	0.425	425
45	0.0140	0.355	355
50	0.0118	0.300	300
60	0.00984	0.250	250
70	0.00835	0.212	212
80	0.00709	0.180	180
100	0.00591	0.150	150
120	0.00492	0.125	125
140	0.00417	0.106	106
170	0.00354	0.090	90
200	0.00295	0.075	75
230	0.00248	0.063	63
270	0.00209	0.053	53
325	0.00177	0.045	45
400	0.00150	0.038	38
450	0.00126	0.032	32

standard wire-mesh screens [ASTM E11 (1989)] derived from the earlier Tyler standard screens. The U.S. standard is based on a 1-mm (1000- μm)-square aperture-opening screen called Mesh No. 18 because there are 18 apertures per inch. The standard Mesh numbers are listed in Table 17.4, where each successively smaller aperture differs from the preceding aperture by a factor of approximately $(2)^{1/4}$. Mechanical shaking is applied to conduct the sieving operation, using a stack of ordered screens.

When wire-mesh screens are used to determine crystal-size distribution, the crystal size is taken to correspond to the screen aperture through which the crystal just passes. However, because of the irregularity of particle shape, this should be considered as a nominal value only. This is particularly true for plates and needles, as illustrated in Figure 17.7.

Particle-size-distribution data, called a *screen analysis*, are presented in the form of a table, from which differential and cumulative plots can be made, usually on a mass-fraction

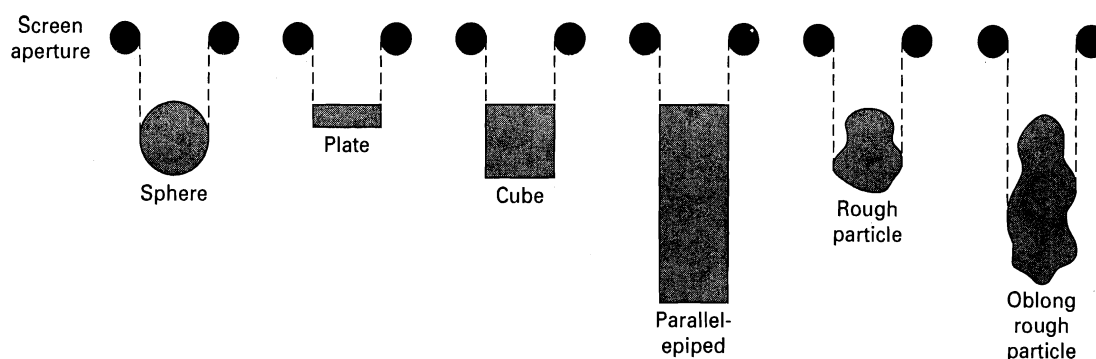


Figure 17.7 Different particle shapes that just pass through the same screen.

basis. Consider the following laboratory screen-analysis data presented by Graber and Taboada [2] for crystals of $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$ (Glauber's salt) grown at about 18°C during an average residence time of 37.2 min in a well-mixed, laboratory, cooling crystallizer. The smallest screen used was 140 mesh, with particles passing through that screen being retained on a pan.

Mesh Number	Aperture, D_p , mm	Mass Retained on Screen, Grams	% Mass Retained
14	1.400	0.00	0.00
16	1.180	9.12	1.86
18	1.000	32.12	6.54
20	0.850	39.82	8.11
30	0.600	235.42	47.95
40	0.425	89.14	18.15
50	0.300	54.42	11.08
70	0.212	22.02	4.48
100	0.150	7.22	1.47
140	0.106	1.22	0.25
Pan	—	0.50	0.11
		491.00	100.00

Differential Screen Analysis

A *differential screen analysis* is made by determining the arithmetic-average aperture for each mass fraction that passes through one screen but not the next screen. Thus, from the above table, a mass fraction of 0.0186 passes through a screen of 1.400-mm aperture, but does not pass through a screen of 1.180-mm aperture. The average of these two apertures is 1.290 mm, which is taken to be the nominal particle size for that mass fraction. The following differential analysis is computed in this manner, where the designation $-14 + 16$ refers to those particles passing through a 14-mesh screen and retained on a 16-mesh screen.

Mesh Range	$\bar{D}_p$, Average Particle Size, mm	Mass Fraction, x_i
-14 + 16	1.290	0.0186
-16 + 18	1.090	0.0654
-18 + 20	0.925	0.0811
-20 + 30	0.725	0.4795
-30 + 40	0.513	0.1815
-40 + 50	0.363	0.1108

Mesh Range	$\bar{D}_p$, Average Particle Size, mm	Mass Fraction, x_i
-50 + 70	0.256	0.0448
-70 + 100	0.181	0.0147
-100 + 140	0.128	0.0025
-140 + (170)	0.098	0.0011
		1.0000

A plot of the differential screen analysis is shown in Figure 17.8 both as (a) an x - y plot and as (b) a histogram. If a wide range of screen aperture is covered, it is best to use a log scale for that variable.

Cumulative Screen Analysis

Screen analysis data can also be plotted as cumulative-weight-percent oversize or (which is more common) undersize as a function of screen aperture. For the above data of Graber and Taboada [2], the two types of *cumulative screen analysis* are as follows:

Aperture, D_p , mm	Cumulative wt% Undersize	Cumulative wt% Oversize
1.400	100.00	0.00
1.180	98.14	1.86
1.000	91.60	8.40
0.850	83.49	16.51
0.600	35.54	64.46
0.425	17.39	82.61
0.300	6.31	93.69
0.212	1.83	98.17
0.150	0.36	99.64
0.106	0.11	99.89

Because 0.11 wt% passed through a 0.106-mm aperture but was retained on a pan with no indication of just how small these retained particles were, the cumulative wt% undersize and oversize cannot be taken to 0 and 100%, respectively.

The above cumulative screen analyses are plotted in Figure 17.8c. The two curves, which are mirror images of each other, cross at a median size where 50 wt% is larger in size and 50 wt% is smaller. As with differential plots, a log scale is preferred if a large range of screen aperture is covered. A log scale for the cumulative wt% may also be preferred if an appreciable fraction of the data points lie below 10%.

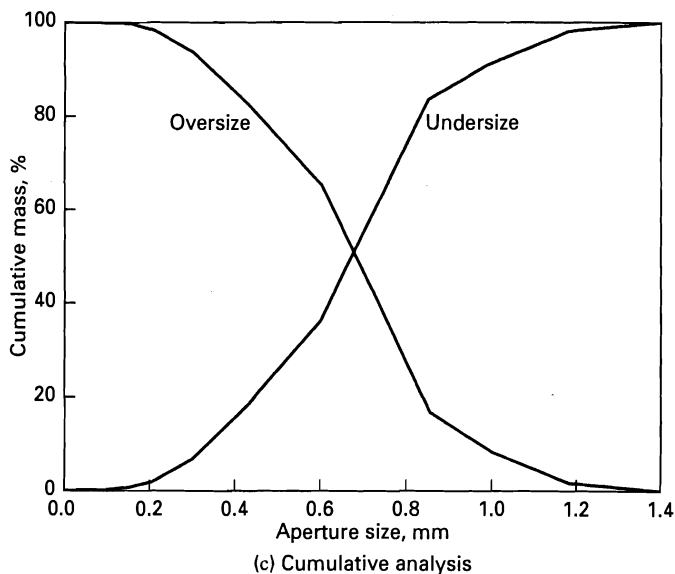
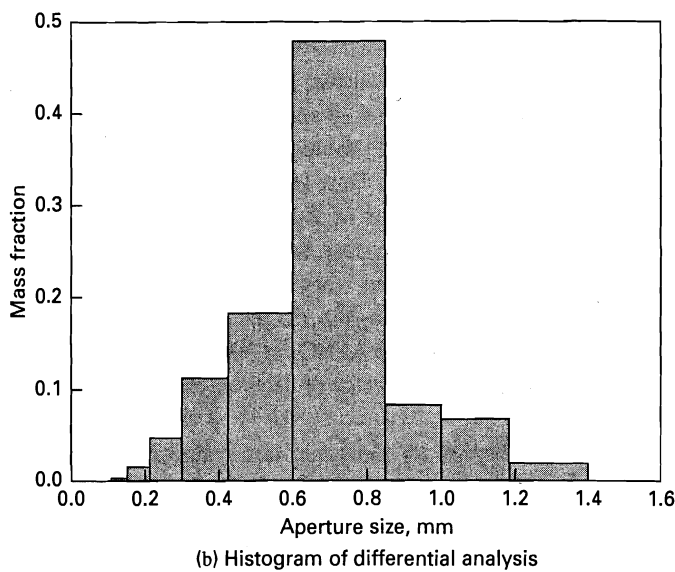
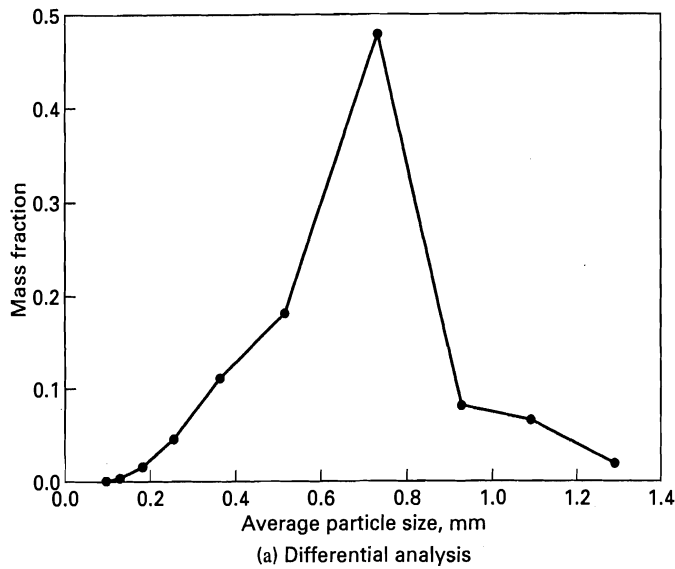


Figure 17.8 Screen analyses for data of Graber and Taboada [2].

A number of different mean particle sizes that are derived from screen analysis are used in practice, depending upon the application. Of these, the most useful are: (1) surface-mean diameter, (2) mass-mean diameter, (3) arithmetic-mean diameter, and (4) volume-mean diameter.

Surface-Mean Diameter

The specific surface area (area/mass) of a particle of spherical or other shape is defined by

$$A_w = s_p/m_p = s_p/v_p\rho_p \quad (17-3)$$

Combining (17-2) and (17-3)

$$A_w = 6/\psi\rho_p\bar{D}_p \quad (17-4)$$

For n mass fractions, x_i , each of average aperture $\bar{D}_{pi}$, from a screen analysis, the overall specific surface area is given by

$$A_w = \sum_{i=1}^n \frac{6x_i}{\psi\rho_p\bar{D}_{pi}} = \frac{6}{\psi\rho_p} \sum \frac{x_i}{\bar{D}_{pi}} \quad (17-5)$$

The surface-mean diameter is defined by

$$A_w = \frac{6}{\psi\rho_p\bar{D}_S} \quad (17-6)$$

Combining (17-5) and (17-6),

$$\bar{D}_S = \frac{1}{\sum_{i=1}^n \frac{x_i}{\bar{D}_{pi}}} \quad (17-7)$$

which can be used to determine $\bar{D}_S$ from a screen analysis. This mean diameter is sometimes referred to as the Sauter mean diameter and as the volume-surface-mean diameter. It is often used for skin friction, heat-transfer, and mass-transfer calculations involving particles.

Mass-Mean Diameter

The mass-mean diameter is defined by

$$\bar{D}_W = \sum_{i=1}^n x_i \bar{D}_{pi} \quad (17-8)$$

Arithmetic-Mean Diameter

The arithmetic-mean diameter is defined in terms of the number of particles, N_i , in each size range:

$$\bar{D}_N = \frac{\sum_{i=1}^n N_i \bar{D}_{pi}}{\sum N_i} \quad (17-9)$$

The number of particles is related to the mass fraction of particles by

$$\begin{aligned} x_i &= \frac{\text{mass of particles of average size } \bar{D}_{pi}}{\text{total mass}} \\ &= \frac{N_i f_v (\bar{D}_{pi})^3 \rho_p}{M_t} \end{aligned} \quad (17-10)$$

where

$$f_v = \text{volume shape factor defined by } v_p = f_v \bar{D}_{p_i}^3$$

$$M_t = \text{total mass} \quad (17-11)$$

For spherical particles, $f_v = \pi/6$.

If (17-10) is solved for N_i , substituted into (17-9), and simplified, we obtain

$$\bar{D}_N = \frac{\sum_{i=1}^n \left(\frac{x_i}{\bar{D}_{p_i}^2} \right)}{\sum_{i=1}^n \left(\frac{x_i}{\bar{D}_{p_i}^3} \right)} \quad (17-12)$$

Volume-Mean Diameter

The volume-mean diameter, $\bar{D}_V$, is defined by

$$(f_v \bar{D}_V^3) \sum_{i=1}^n N_i = \sum_{i=1}^n (f_v \bar{D}_{p_i}^3) N_i \quad (17-13)$$

Solving (17-13) for $\bar{D}_V$ for a constant value of f_v gives

$$\bar{D}_V = \left(\frac{\sum_{i=1}^n N_i \bar{D}_{p_i}^3}{\sum_{i=1}^n N_i} \right)^{1/3} \quad (17-14)$$

The corresponding relation in terms of x_i rather than N_i is obtained by combining (17-14) with (17-10), giving

$$\bar{D}_V = \left(\frac{1}{\sum \frac{x_i}{\bar{D}_{p_i}^3}} \right)^{1/3} \quad (17-15)$$

EXAMPLE 17.2

Using the screen analysis data of Graber and Taboada given above, compute all four mean diameters.

SOLUTION

Since the data are given in weight (mass) fractions, use (17-7), (17-8), (17-12), and (17-15).

$\bar{D}_p$, mm	x	$x/\bar{D}_p$	$x\bar{D}_p$	$x/\bar{D}_p^2$	$x/\bar{D}_p^3$
1.290	0.0186	0.0144	0.0240	0.0112	0.0087
1.090	0.0654	0.0600	0.0713	0.0550	0.0505
0.925	0.0811	0.0877	0.0750	0.0948	0.1025
0.725	0.4795	0.6614	0.3476	0.9122	1.2583
0.513	0.1815	0.3538	0.0931	0.6897	1.3444
0.363	0.1108	0.3052	0.0402	0.8409	2.3164
0.256	0.0448	0.1750	0.0115	0.6836	2.6703
0.181	0.0147	0.0812	0.0027	0.4487	2.4790
0.128	0.0025	0.0195	0.0003	0.1526	1.1921
0.098	0.0011	0.0112	0.0001	0.1145	1.1687
	1.0000	1.7695	0.6658	4.0032	12.5909

From (17-7),

$$\bar{D}_S = \frac{1}{1.7695} = 0.565 \text{ mm}$$

From (17-8),

$$\bar{D}_W = 0.666 \text{ mm}$$

From (17-12),

$$\bar{D}_N = \frac{4.0032}{12.5909} = 0.318 \text{ mm}$$

From (17-15),

$$\bar{D}_V = \left(\frac{1}{12.5909} \right)^{1/3} = 0.430 \text{ mm}$$

Thus, the mean diameters vary significantly.

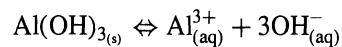
17.2 THERMODYNAMIC CONSIDERATIONS

Solubility and Material Balances

Important thermodynamic properties for crystallization operations include melting point, heat of fusion, solubility, heat of crystallization, heat of solution, heat of transition, and supersaturation. For binary systems of water and soluble inorganic and organic chemicals, Mullin [1] presents extensive tables of solubility, as a function of temperature, and heat of solution at infinite dilution and room temperature (approximately 18–25°C). Data in water are listed in Table 17.5 for the inorganic salts of Table 17.1, where solubility data are given on a hydrate-free basis.

Solubilities are seen to vary widely from as low as 4.8 g/100 g of water for Na_2SO_4 (as the decahydrate) at 0°C to 952 g/100 g of water for AgNO_3 at 100°C. For KNO_3 , the solubility increases by a factor of 18.6 for the same temperature increase.

The solubility of an inorganic compound can be even much lower than that shown for Na_2SO_4 . Such compounds are generally considered to be just slightly or sparingly soluble or almost insoluble. The solubility of such compounds is usually expressed as a solubility product, K_c , in terms of ion concentrations. Data for several compounds are given in Table 17.6. For example, consider $\text{Al}(\text{OH})_3$, which is sparingly soluble with a solubility product of $K_c = 1.1 \times 10^{-15}$ at 18°C and dissolves according to the equation



By the law of mass action, the equilibrium constant, called the solubility product for dissolution, is given by

$$K_c = \frac{(c_{\text{Al}^{3+}})(c_{\text{OH}^{-}})^3}{a_{\text{Al}(\text{OH})_3}} = (c_{\text{Al}^{3+}})(c_{\text{OH}^{-}})^3 = 1.1 \times 10^{-15}$$

where the activity of $\text{Al}(\text{OH})_3$ solid is taken as 1.0. Since, by stoichiometry,

$$(c_{\text{OH}^{-}}) = 3(c_{\text{Al}^{3+}}) \quad \text{and} \quad K_c = (c_{\text{Al}^{3+}})^4(3)^3 = 1.1 \times 10^{-15}$$

then,

$$(c_{\text{Al}^{3+}}) = c_{\text{dissolved Al}(\text{OH})_3} = 8 \times 10^{-5} \text{ gmoles/L}$$

which is a very small concentration.

Table 17.5 Solubility and Heat of Solution at Infinite Dilution of Some Inorganic Compounds in Water (A Positive Heat of Solution Is Endothermic)

Compound	Heat of Solution of Stable Hydrate (at Room Temperature) kcal/mole Compound	Solubility (Hydrate-free Basis) g/100 g H ₂ O at T, °C								Stable Hydrate at Room Temperature
		0	10	20	30	40	60	80	100	
NH ₄ Cl	+3.8	29.7	33.4	37.2	41.4	45.8	55.2	65.6	77.3	0
(NH ₄) ₂ SO ₄	+1.5	71.0	73.0	75.4	78.0	81.0	88.0	95.3	103.3	0
BaCl ₂	+4.5	31.6	33.2	35.7	38.2	40.7	46.4	52.4	58.3	2
CuSO ₄	+2.86	14.3	17.4	20.7	25.0	28.5	40.0	55.0	75.4	5
MgSO ₄	+3.18	22.3	27.8	33.5	39.6	44.8	55.3	56.0	50.0	7
MgCl ₂	-3.1	52.8	53.5	54.5	56.0	57.5	61.0	66.0	73.0	6
NiSO ₄	+4.2	26	32	37	43	47	55	63	—	7
KCl	+4.4	27.6	31.0	34.0	37.0	40.0	45.5	51.1	56.7	0
KNO ₃	+8.6	13.3	20.9	31.6	45.8	63.9	110	169	247	0
K ₂ SO ₄	+6.3	7.4	9.3	11.1	13.1	14.9	18.3	21.4	24.2	0
AgNO ₃	+5.4	122	170	222	300	376	525	669	952	0
NaClO ₃	+5.4	80	89	101	113	126	155	189	233	0
NaCl	+0.93	35.6	35.7	35.8	36.1	36.4	37.1	38.1	39.8	0
NaNO ₃	+5.0	72	78	85	92	98	—	133	163	0
Na ₂ SO ₄	+18.7	4.8	9.0	19.4	40.8	48.8	45.3	43.7	42.5	10
Na ₂ S ₂ O ₃	+11.4	52	61	70	84	103	207	250	266	5
Na ₃ PO ₄	+15.0	1.5	4	11	20	31	55	81	108	12

Table 17.6 Concentration Solubility Products of Some Sparingly Soluble Inorganic Compounds

Compound	T, °C	K _c
Ag ₂ CO ₃	25	6.15 × 10 ⁻¹²
AgCl	25	1.56 × 10 ⁻¹⁰
Al(OH) ₃	15	4 × 10 ⁻¹³
Al(OH) ₃	18	1.1 × 10 ⁻¹⁵
BaSO ₄	18	0.87 × 10 ⁻¹⁰
CaCO ₃	15	0.99 × 10 ⁻⁸
CaSO ₄	10	1.95 × 10 ⁻⁴
CuSO ₄	16-18	2 × 10 ⁻⁴⁷
Fe(OH) ₃	18	1.1 × 10 ⁻³⁶
MgCO ₃	12	2.6 × 10 ⁻⁵
ZnS	18	1.2 × 10 ⁻²³

For less sparingly soluble compounds, the equilibrium constant, called K_a , is more rigorously expressed in terms of ionic activities or activity coefficients:

$$K_a = \frac{(a_{\text{Al}^{3+}})(a_{\text{OH}^{-}})^3}{a_{\text{Al(OH)}_3}} = (\gamma_{\text{Al}^{3+}})(c_{\text{Al}^{3+}})(\gamma_{\text{OH}^{-}})^3(c_{\text{OH}^{-}})^3$$

In general, $\gamma \approx 1.0$ for $c < 1 \times 10^{-3}$ gmoles/L. As c rises above 1×10^{-3} gmoles/L, γ decreases, but may pass through a minimum and then increase. Mullin [1] presents activity-coefficient data at 25°C for soluble inorganic compounds over a wide range of concentration.

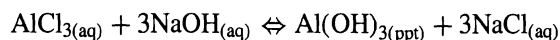
Although the solubility of most inorganic compounds increases with increasing temperature, a few common compounds exhibit a so-called negative or inverted solubility, in certain ranges of temperature, where solubility decreases with increasing temperature. These compounds are the so-called hard salts, which include anhydrous Na₂SO₄ and CaSO₄.

A considerable change in the solubility curve can occur when a phase transition from one stable hydrate to another takes place. For example, in Table 17.5, Na₂SO₄ · 10H₂O is the stable form from 0°C to about 32.4°C. In that temperature range, the solubility increases rapidly from 4.8 to 49.5 g (hydrate-free basis)/100 g H₂O. From 32.4°C to 100°C, the stable form is Na₂SO₄, whose solubility decreases slowly from 49.5 to 42.5 g/100 g H₂O. In the phase diagram of Figure 17.2 for the MgSO₄-water system, the solubility-temperature curves of each of the four hydrated forms has a distinctive slope.

The solubility characteristic of a solute in a particular solvent is, by far, the most important property for determining: (1) the best method for causing crystallization, and (2) the ease or difficulty in growing crystals. Crystallization by cooling is only attractive for compounds having a solubility that decreases rapidly with decreasing temperature above ambient temperature. Such is not the case for most of the compounds in Table 17.5. For NaCl, crystallization by cooling would be undesirable because the solubility decreases

only by about 10% when the temperature decreases from 100 to 0°C. For most soluble inorganic compounds, cooling by evaporation is the preferred technique.

Solid compounds with a very low solubility can be produced by reacting two soluble compounds. For example, in Table 17.6, solid $\text{Al}(\text{OH})_3$ can be formed by the reaction



However, the reaction is so fast that only very fine crystals, called a precipitate, are produced, with no simple method to cause them to grow to large crystals.

EXAMPLE 17.3

The concentrate from an evaporation system is 4,466 lb/h of 37.75 wt% MgSO_4 at 170°F and 20 psia. It is mixed with 9,860 lb/h of a saturated aqueous recycle filtrate of MgSO_4 at 85°F and 20 psia and sent to a vacuum crystallizer, operating at 85°F and 0.58 psia in the vapor space, to produce water vapor and a magma of 20.8 wt% crystals and 79.2 wt% saturated solution. The magma is sent to a filter, from which filtrate is recycled as mentioned above. Determine the lb/h of water evaporated and the maximum production rate of crystals in tons/day (dry basis for 2000 lb/ton).

SOLUTION

For the saturated filtrate at 85°F, the weight fraction of MgSO_4 , from Figure 17.2, is 28 wt%. Therefore, MgSO_4 in the recycle filtrate is $9,860(0.28) = 2,760$ lb/h. By material balance around the mixing step,

	lb/h		
Component	Feed	Recycle Filtrate	Crystallizer Feed
MgSO_4	1,686	2,760	4,446
H_2O	2,780	7,100	9,880
	4,466	9,860	14,326

The material balance around the crystallizer is conveniently made by a balance on MgSO_4 . At 85°F, from Figure 17.2, the magma is 20.8 wt% $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ crystals and 79.2 wt% of 28 wt% aqueous MgSO_4 liquid. Because the MW of MgSO_4 and $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ are 120.4 and 246.4, respectively, the crystals are $120.4/246.4 = 0.4886$ mass fraction MgSO_4 . Therefore, by a MgSO_4

balance,

$$4,446 = 0.28L + 0.4886S \quad (1)$$

where

L = lb/h of liquid

S = lb/h of crystals

Also,

$$S = 0.208(S + L) \quad (2)$$

Solving (1) and (2) simultaneously,

$$S = 2,856 \text{ lb/h}$$

$$L = 10,876 \text{ lb/h}$$

By a total material balance around the crystallizer,

$$F = V + L + S$$

where F = total feed rate and V = evaporation rate. Therefore,

$$14,326 = V + 10,876 + 2,856 \quad (3)$$

Solving,

$$V = 594 \text{ lb/h}$$

The results in tabular form are:

	lb/h for crystallizer			
Component	Feed	Vapor	Liquid	Crystals
$\text{MgSO}_{4(\text{aq})}$	4,446	0	3,045	0
$\text{MgSO}_4 \cdot 7\text{H}_2\text{O}_{(\text{s})}$	0	0	0	2,856
H_2O	9,880	594	7,831	0
	14,326	594	10,876	2,856

The maximum production rate of crystals is

$$\frac{2,856}{2,000}(24) = 34.3 \text{ tons/day}$$

A large number of organic compounds, particularly organic acids with relatively moderate melting points (125–225°C), are also soluble in water. Some data are given in Table 17.7, where it is seen that the solubility often increases significantly with increasing temperature. For example, the solubility of o-phthalic acid increases from a very low value of 0.56 to 18.0 g/100 g H_2O when the temperature increases from 20 to 100°C.

Table 17.7 Solubility and Melting Point of Some Organic Compounds in Water

Compound	Melting Point, °C	Solubility, g/100 g H_2O at T , °C							
		0	10	20	30	40	60	80	100
Adipic acid	153	0.8	1.0	1.9	3.0	5.0	18	70	160
Benzoic acid	122	0.17	0.20	0.29	0.40	0.56	1.16	2.72	5.88
Fumaric acid (trans)	287	0.23	0.35	0.50	0.72	1.1	2.3	5.2	9.8
Maleic acid	130	39.3	50	70	90	115	178	283	—
Oxalic acid	189	3.5	6.0	9.5	14.5	21.6	44.3	84.4	—
o-phthalic acid	208	0.23	0.36	0.56	0.8	1.2	2.8	6.3	18.0
Succinic acid	183	2.8	4.4	6.9	10.5	16.2	35.8	70.8	127
Sucrose	d	179	190	204	219	238	287	362	487
Urea	133	67	85	105	135	165	250	400	730
Uric acid	d	0.002	0.004	0.006	0.009	0.012	0.023	0.039	0.062

EXAMPLE 17.4

Oxalic acid is to be crystallized from a saturated aqueous solution initially at 100°C. To what temperature does the solution have to be cooled to crystallize 95% of the acid as the dihydrate?

SOLUTION

Assume a basis of 100 g of water. From Table 17.7, the amount of dissolved oxalic acid at 100°C is 84.4 g.

Amount to be crystallized = $0.95(84.4) = 80.2$ g.

Amount of oxalic acid left in solution = $84.4 - 80.2 = 4.2$ g.

MW of oxalic acid = 90.0.

MW of water = 18.0.

Water of hydration for $2\text{H}_2\text{O} = \frac{2(18.0)}{90.0} = 0.4 \frac{\text{g H}_2\text{O}}{\text{g oxalic acid}}$.

Therefore water of crystallization = $0.4(80.2) = 32.1$ g H_2O .

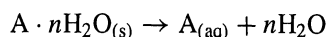
Liquid water remaining = $100 - 32.1 = 67.9$ g.

Final solubility must be $\frac{4.2}{67.9} \times 100 = 6.19 \frac{\text{g}}{100 \text{ g H}_2\text{O}}$.

From Table 17.7, by linear interpolation, temperature = 10.6°C.

Enthalpy Balances

When an anhydrous solid compound, whose solubility increases with increasing temperature, dissolves isothermally in water or some other solvent, heat must be absorbed by the solution. This amount of heat per mole of compound in an infinite amount of solvent varies with temperature and is referred to as the *heat of solution at infinite dilution* ($\Delta H_{\text{sol}}^{\infty}$). For example, in Table 17.5, the solubility of anhydrous NaCl is seen to increase slowly with increasing temperature from 10 to 100°C. Correspondingly, the heat of solution at infinite dilution in Table 17.5 is modestly endothermic (+) at room temperature. In contrast, the solubility of anhydrous KNO_3 increases more rapidly with increasing temperature, resulting in a higher endothermic heat of solution. For compounds that form hydrates, the heat of solution at infinite dilution may be negative (exothermic) for the anhydrous form, but becomes less negative and often positive as higher hydrates are formed by the reaction:



For example, the following heats of solution at infinite dilution in kJ/mol of compound at 18°C for four hydrates of MgSO_4 clearly show this effect:

MgSO_4	-88.3
$\text{MgSO}_4 \cdot \text{H}_2\text{O}$	-58.6
$\text{MgSO}_4 \cdot 6\text{H}_2\text{O}$	-2.3
$\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$	+13.3

Heats of a solution for a number of hydrated and anhydrous compounds are listed in Table 17.5.

As a solid compound continues to dissolve in a solvent, the heat of solution, which is now referred to as the integral

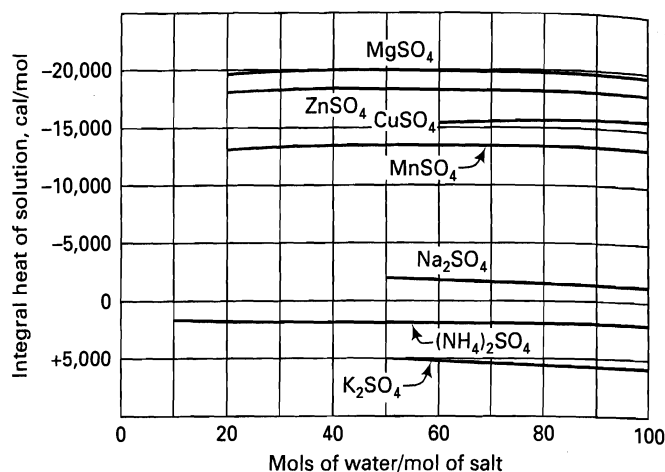


Figure 17.9 Integral heats of solution for sulfates in water at 25°C.

heat of solution, varies somewhat, as shown in Figure 17.9 for several compounds as a function of concentration. The integral heat of solution at saturation is numerically equal, but opposite in sign, to the heat of crystallization. The difference between the integral heat of solution at saturation and the heat of solution at infinite dilution is the heat of dilution:

$$\Delta H_{\text{sol}}^{\text{sat}} - \Delta H_{\text{sol}}^{\infty} = \Delta H_{\text{dil}}$$

with

$$\Delta H_{\text{sol}}^{\text{sat}} = -\Delta H_{\text{crys}}$$

As indicated in Figure 17.9, the heat of dilution is relatively small; therefore, it is common to use:

$$\Delta H_{\text{crys}} \approx -\Delta H_{\text{sol}}^{\infty}$$

An energy-balance calculation around a crystallizer is complex because it can involve not only the integral heat of solution and/or heat of crystallization, but also the specific heats of the solute and solvent and the heat of vaporization of the solvent. The calculation is readily made if an enthalpy-mass fraction diagram is available for the system, including solubility and phase-equilibria data. Mullin [1] lists 11 aqueous binary systems for which such a diagram has been constructed. A diagram for the MgSO_4 - H_2O system is shown in Figure 17.10. The enthalpy datum is pure liquid water at 32°F (consistent with steam tables in American Engineering Units) at Point *p* and solid MgSO_4 at 32°F (not shown in Figure 17.10).

Points *a* to *l*, *n*, *p*, and *q* in the enthalpy-mass fraction diagram of Figure 17.10 correspond to the same points in the phase diagram of Figure 17.2. In Figure 17.10, the isotherms in the region above Curve *pabcdq* pertain to enthalpies of unsaturated solutions of MgSO_4 . The straightness of these isotherms indicates that the heat of dilution is almost negligible. In this solid-free region, a 30 wt% aqueous solution of MgSO_4 has a specific enthalpy at 110°F of -31 Btu/lb solution.

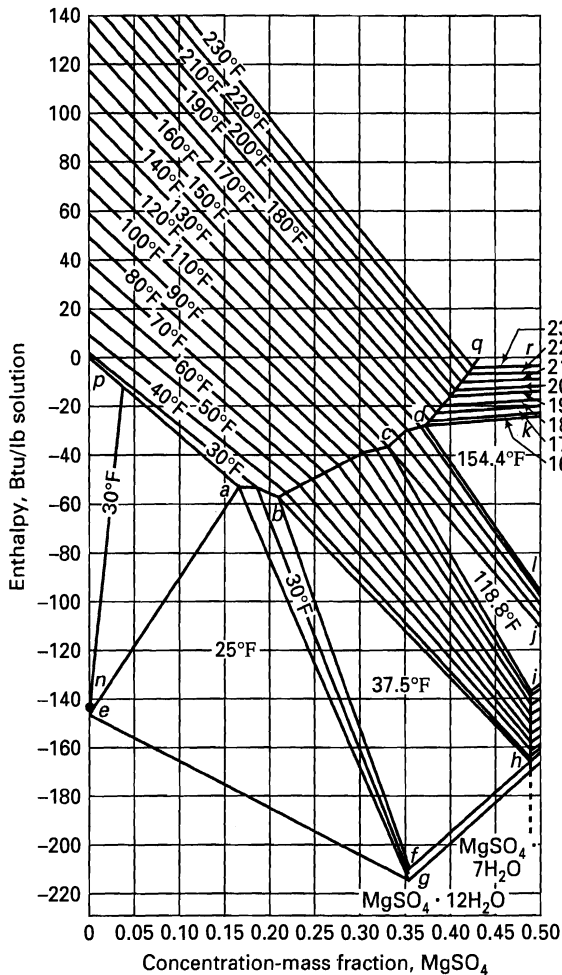


Figure 17.10 Enthalpy-concentration diagram for the $\text{MgSO}_4\text{-H}_2\text{O}$ system at 1 atm.

In the region below the solubility curve $pabcdq$, in both Figures 17.2 and 17.10, the following phases exist at equilibrium:

Region	Temperature Range, °F	Phases
pae	25–32	ice and aqueous solution of MgSO_4
ea	25	ice and eutectic mixture
ag	25	eutectic and $\text{MgSO}_4 \cdot 12\text{H}_2\text{O}$
$abfg$	25–37.5	saturated solution and $\text{MgSO}_4 \cdot 12\text{H}_2\text{O}$
$bcih$	37.5–118.8	saturated solution and $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$
$cdlj$	118.8–154.4	saturated solution and $\text{MgSO}_4 \cdot 6\text{H}_2\text{O}$
$dqrk$	154.4–	saturated solution and $\text{MgSO}_4 \cdot \text{H}_2\text{O}$

Pure ice exists at Point e , where in Figure 17.10 the specific enthalpy is -147 Btu/lb, which is the heat of crystallization of water at 32°F . If a 30 wt% aqueous solution of MgSO_4 is cooled from 110°F to 70°F , the equilibrium magma will consist of a saturated solution of 26 wt% MgSO_4 and crystals of $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ (49 wt% MgSO_4) as determined from the ends of the 70°F tie line that extends

from solubility curve bc to $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ solid line ih . The relative amounts of the two equilibrium phases can be computed from a MgSO_4 balance. For a basis of 100 lb of mixture, let S be the pounds of crystals and A be the pounds of saturated aqueous solution. Thus, the MgSO_4 balance is

$$0.30(100) = 0.49S + 0.26A$$

where

$$100 = S + A$$

Solving, $S = 17.4$ lb and $A = 82.6$ lb. The enthalpy of the mixture at 70°F is -65 Btu/lb, which is equivalent to enthalpies of -46 and -155 Btu/lb, respectively, for the solution and crystals.

EXAMPLE 17.5

For the conditions of Example 17.3, calculate the Btu/h of heat addition for the crystallizer.

SOLUTION

An overall energy balance around the crystallizer gives

$$m_{\text{feed}} H_{\text{feed}} + Q_{\text{in}} = m_{\text{vapor}} H_{\text{vapor}} + m_{\text{liquid}} H_{\text{liquid}} + m_{\text{crystals}} H_{\text{crystals}} \quad (1)$$

where liquid refers to the saturated-liquid portion of the magma.

From the solution to Example 17.3, the feed consists of two streams:

$$m_{\text{feed}1} = 4,466 \text{ lb/hr of } 37.75 \text{ wt\% } \text{MgSO}_4 \text{ at } 170^\circ\text{F}$$

$$m_{\text{feed}2} = 9,860 \text{ lb/hr of } 28.0 \text{ wt\% } \text{MgSO}_4 \text{ at } 85^\circ\text{F}$$

From Figure 17.10,

$$H_{\text{feed}1} = -20 \text{ Btu/lb}$$

$$H_{\text{feed}2} = -43 \text{ Btu/lb}$$

Therefore,

$$m_{\text{feed}} H_{\text{feed}} = 4,466(-20) + 9,860(-43) = -513,000 \text{ Btu/h}$$

$$m_{\text{vapor}} = 594 \text{ lb/h at } 85^\circ\text{F and } 0.58 \text{ psia}$$

The vapor enthalpy does not appear on Figure 17.10, but enthalpy tables for steam can be used since they are based on the same datum (i.e., liquid water at 32°F).

Therefore, $H_{\text{vapor}} = 1099$ Btu/lb from steam tables and

$$m_{\text{vapor}} H_{\text{vapor}} = 594(1099) = 653,000 \text{ Btu/h}$$

The liquid plus crystals can be treated together as the magma. From the solution to Example 17.3,

$$\begin{aligned} m_{\text{liquid}} + m_{\text{crystals}} &= m_{\text{magma}} = 10,876 + 2,856 \\ &= 13,732 \text{ lb/h of } 32.4 \text{ wt\% } \text{MgSO}_4 \text{ at } 85^\circ\text{F} \end{aligned}$$

From Figure 17.10,

$$H_{\text{magma}} = -67 \text{ Btu/lb}$$

and

$$m_{\text{magma}} H_{\text{magma}} = 13,732(-67) = -920,000 \text{ Btu/lb}$$

From (1),

$$Q_{\text{in}} = 653,000 - 920,000 - (-513,000) = 246,000 \text{ Btu/h}$$

In the absence of an enthalpy-mass fraction diagram, a reasonably accurate energy balance can be made if data for heat of crystallization and specific heats of the solutions are available or can be estimated and the heat of dilution is neglected, as shown in the next example.

EXAMPLE 17.6

The feed to a cooling crystallizer is 1,000 lb/h of 32.5 wt% MgSO_4 in water at 120°F. This solution is cooled to 70°F to form crystals of the heptahydrate. Estimate the heat removal rate in Btu/h.

SOLUTION

Material balance

From Figure 17.2, the feed at 120°F contains no crystals, but the magma at 70°F consists of crystals of the heptahydrate and a mother liquor of 26 wt% MgSO_4 . By material balance in the manner of Example 17.3, the following results are obtained:

	lb/hr		
	Feed	Mother Liquor	Crystals
H_2O	675	530	0
MgSO_4	325	186	0
$\text{Mg}_2\text{SO}_4 \cdot 7\text{H}_2\text{O}$	0	0	284
	1,000	716	284

Take a thermodynamic path consisting of cooling the feed from 120°F to 70°F followed by crystallization at 70°F. From Hougen, Watson, and Ragatz [3], the specific heat of the feed is approximately constant over the temperature range at 0.72 Btu/lb-°F. Therefore, the heat that must be removed to cool the feed to 70°F is

$$1,000(0.72)(120 - 70) = 36,000 \text{ Btu/h}$$

For data presented earlier in this section, the heat of crystallization can be taken as the negative of the heat of solution at infinite dilution:

$$-13.3 \text{ kJ/mole of heptahydrate}$$

$$\text{or} \quad -23.2 \text{ Btu/lb of heptahydrate}$$

Therefore, the heat that must be removed during crystallization of the heptahydrate is

$$284(23.2) = 6,600 \text{ Btu/h}$$

The total heat removal is $36,000 + 6,600 = 42,600 \text{ Btu/h}$.

If this example is solved with Figure 17.10, in the manner of Example 17.5, the result is 44,900 Btu/h, which is 5.4% higher.

17.3 KINETIC AND TRANSPORT CONSIDERATIONS

Crystallization is a complex phenomenon that involves three steps: *nucleation*, mass transfer of the solute to the crystal surface, and incorporation of the solute into the crystal

lattice structure. Collectively, these phenomena are referred to as *crystallization kinetics*. Experimental data show that the driving force for all three steps is *supersaturation*.

Supersaturation

The solubility property discussed in the previous section refers to relatively large crystals of the size that can be seen by the naked eye, i.e., larger than 20 μm in diameter. As crystal size decreases below this diameter, solubility noticeably increases, making it possible to supersaturate a solution if it is cooled slowly without agitation. This phenomenon, based on the work of Miers and Isaac [4] in 1907, is represented in Figure 17.11, where the normal solubility curve, c_s , is represented as a solid line. The solubility of very small crystals can fall in the metastable region which is shown to have a metastable limiting solubility, c_m , given by the dashed line.

Consider a solution at a temperature, T_1 , given by the vertical line in Figure 17.11. If the concentration is given by Point *a*, the solution is undersaturated and crystals of all sizes dissolve. At Point *b*, equilibrium exists between a saturated solution and crystals that can be seen by the naked eye. In the metastable region at Point *c*, crystals can grow but cannot nucleate. If no crystals are present, none can form. For that concentration, the difference between the temperature at Point *e* on the solubility curve and Point *c* in the metastable region is the supersaturation temperature difference, which may be about 2°F. The supersaturation, $\Delta c = c - c_s$, is the difference in concentration between Points *c* and *b*.

At Point *d*, spontaneous nucleation of very small crystals, invisible to the naked eye, occurs. The difference in temperature between Points *f* and *d* is the limit of the supersaturation temperature difference. The limiting supersaturation is $\Delta c_{\text{limit}} = c_m - c_s$.

The relationship between solubility and crystal size is given quantitatively by the Kelvin equation (also known as

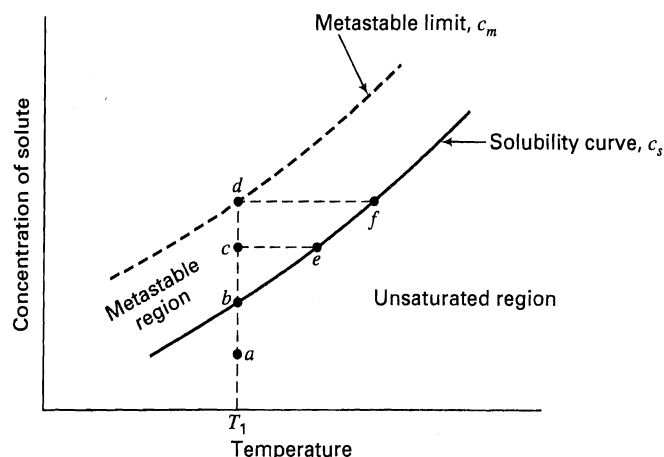


Figure 17.11 Representative solubility-supersolubility diagram.

the Gibbs–Thompson and Ostwald equations):

$$\ln\left(\frac{c}{c_s}\right) = \left(\frac{4v_s\sigma_{s,L}}{vRTD_p}\right) \quad (17-16)$$

where

v_s = molar volume of the crystals

$\sigma_{s,L}$ = interfacial tension

v = number of ions/molecule of solute

c/c_s = supersaturation ratio = S

Measured values of interfacial tension (also called surface energy) range from as low as 0.001 J/m² for very soluble compounds to 0.170 for compounds of low solubility.

As might be expected, (17-16), in a more general form, applies to the effect of droplet diameter on vapor pressure and solubility in another liquid phase.

It is common to define a relative supersaturation, s , by

$$s = \frac{c - c_s}{c_s} = \frac{c}{c_s} - 1 = S - 1 \quad (17-17)$$

In practice, s is usually less than 0.02 or 2%. For such small values, $\ln(c/c_s)$ can be approximated by s with no more than a 1% error.

EXAMPLE 17.7

Determine the effect of crystal diameter on the solubility of KCl in water at 25°C.

SOLUTION

From Table 17.5, by interpolation, $c_s = 35.5$ g/100 g H₂O.

Because KCl dissociates into K⁺ and Cl⁻, $v = 2$.

MW of KCl = 74.6.

Density of KCl crystals = 1980 kg/m³.

$$v_s = \frac{74.6}{1980} = 0.0376 \text{ m}^3/\text{kmol}$$

$$T = 298 \text{ K}$$

$$R = 8314 \text{ J/kmol-K}$$

From Mullin [1], page 200,

$$\sigma_{s,L} = 0.028 \text{ J/m}^2$$

From (17-16),

$$\begin{aligned} c &= c_s \exp\left(\frac{4v_s\sigma_{s,L}}{vRTD_p}\right) \\ &= 35.5 \exp\left[\frac{4(0.0376)(0.028)}{2(8315)(298)D_p}\right] \text{ for } D_p \text{ in m, or} \\ &= 35.5 \exp(0.00085/D_p, \mu\text{m}) \end{aligned} \quad (1)$$

$D_p, \mu\text{m}$	c/c_s	$c, \text{g KCl/100 g H}_2\text{O}$
0.01	1.0887	38.65
0.10	1.0085	35.80
1.00	1.00085	35.53
10.00	1.000085	35.50
100.00	1.0000085	35.50

Nucleation

To determine the volume or residence time of the magma in a crystallizer, the rate of *nucleation* (birth) of crystals and their rate of growth must be known or estimated. The relative rates of nucleation and growth are very important because they determine crystal size and size distribution. Nucleation may be *primary* or *secondary* depending on whether the supersaturated solution is free of crystalline surfaces or contains crystals, respectively. Primary nucleation requires a high degree of supersaturation and is the principal mechanism occurring in precipitation. The theory of primary nucleation is well developed and applies as well to the condensation of liquid droplets from a supersaturated vapor and the formation of droplets of a second liquid phase from an initial liquid phase. However, secondary nucleation is the principal mechanism in commercial crystallizers, where crystalline surfaces are present and large crystals are desirable.

Primary Nucleation

Primary nucleation can be homogeneous or heterogeneous. The former occurs within the supersaturated solution in the absence of any foreign matter, such as dust. Molecules in the solution first associate to form a *cluster*, which may dissociate or grow. If a cluster gets large enough to take on the appearance of a lattice structure, it becomes an *embryo*. Further growth can result in a stable crystalline *nucleus* whose size exceeds that given by (17-16) for the prevailing degree of supersaturation.

The rate of homogeneous nucleation is given by classical chemical kinetics in conjunction with (17-16), as discussed by Nielsen [5]. The resulting expression is

$$B^o = A \exp\left[\frac{-16\pi v_s^2 \sigma_{s,L}^3 N_a}{3v^2(RT)^3 \left[\ln\left(\frac{c}{c_s}\right)\right]^2}\right] \quad (17-18)$$

where

B^o = rate of homogeneous primary nucleation, number of nuclei/cm³-s

A = frequency factor

N_a = Avogadro's number = 6.022×10^{26} molecules/kmol

Theoretically, $A = 10^{30}$ nuclei/cm³-s; however, observed values are generally different due to the unavoidable presence of foreign matter. Thus, (17-18) can also be applied to heterogeneous primary nucleation, where A is determined experimentally and may be many orders of magnitude different from the theoretical value. A value of 10^{25} is often quoted.

The rate of primary nucleation is extremely sensitive to the supersaturation ratio, S , defined by (17-17), as illustrated in the following example.

EXAMPLE 17.8

Using the data in Example 17.7, estimate the effects of relative supersaturation on the primary homogeneous nucleation of KCl from an aqueous solution at 25°C. Use values of s corresponding to values of c/c_s of 2.0, 1.5, and 1.1.

SOLUTION

For $c/c_s = 2.0$, $\ln(c/c_s) = 0.693$. From (17-18), using data in Example 17.7,

$$B^0 = 10^{30} \exp \left[\frac{-16(3.14)(0.0376)^2(0.028)^3(6.022 \times 10^{26})}{3(2)^2(8315)^3(298)^3(0.693)^2} \right]$$

$$= 2.23 \times 10^{25} \text{ nuclei/cm}^3\text{-s}$$

Calculations for the other values of c/c_s are obtained in the same manner with the following results:

c/c_s	B^0 , nuclei/cm ³ -s
2.0	2.23×10^{25}
1.5	2.60×10^{16}
1.1	0

Since large values of the supersaturation ratio ($c/c_s > 1.02$) are essentially impossible for crystallization of solutes of moderate to high solubility (e.g., solutes listed in Tables 17.5 and 17.7), primary nucleation for these solutes never occurs. However, for relatively insoluble solutes (e.g., solutes listed in Table 17.6), large values of c/c_s can be generated rapidly from ionic reactions causing rapid precipitation of very fine particles. If $A = 10^{25}$ is used, B^0 is divided by 10^5 .

Secondary Nucleation

Nucleation in industrial crystallizers occurs mainly by secondary nucleation caused by the presence of existing crystals in the supersaturated solution. Secondary nucleation can occur by: (1) fluid shear past crystal surfaces that sweeps away nuclei, (2) collisions of crystals with each other, and (3) collisions of crystals with metal surfaces such as the crystallizer vessel wall or agitator blades. The latter two mechanisms, which are referred to as *contact nucleation* are the most common types since they can occur at the low values of relative supersaturation that are typically encountered in industrial applications.

In the absence of a theory for the complex phenomena of secondary nucleation, the following empirical power-law function, which correlates much of the experimental data, is widely used:

$$B^0 = k_N s^b M_T^j N^r \quad (17-19)$$

where M_T = mass of crystals per volume of magma and N = agitation rate (e.g., rpm of an impeller). The constants, k_N , b , j , and r , are determined from experimental data, on the system of interest, as discussed below in the section on a crystallizer model.

Crystal Growth

In 1897, Noyes and Whitney [6] presented a mass-transfer theory of crystal growth based on equilibrium at the crystal-solution interface. Thus, they wrote

$$dm/dt = k_c A(c - c_s) \quad (17-20)$$

where dm/dt = rate of mass deposited on the crystal surface, A = surface area of the crystal, k_c = mass-transfer coefficient, c = mass solute concentration in the bulk supersaturated solution, and c_s = mass solute concentration in the solution at saturation. Nernst [7] proposed the existence of a thin stagnant film of solution adjacent to the crystal face through which molecular diffusion of the solute took place. Thus, $k_c = D/\delta$, where D = diffusivity and δ = film thickness, where the latter was assumed to depend on velocity of the solution past the crystal as determined by the degree of agitation.

The theory of Noyes and Whitney was challenged by Miers [8], who showed experimentally that an aqueous solution in contact with crystals of sodium chlorate was not saturated at the crystal-solution interface, but was supersaturated. This finding led to a two-step theory of crystal growth, referred to as the diffusion-reaction theory, as described by Valetton [9]. Mass transfer of solute from the bulk of the solution to the crystal-solution interface occurs in the first step, as given by a modification of (17-20):

$$dm/dt = k_c A(c - c_i) \quad (17-21)$$

where c_i is the supersaturated concentration at the interface. In the second step, a first-order reaction is assumed to occur at the crystal-solution interface, in which solute molecules are integrated into the crystal-lattice structure. Thus, for this kinetic step,

$$dm/dt = k_i A(c_i - c_s) \quad (17-22)$$

If (17-21) and (17-22) are combined to eliminate c_i , we obtain

$$dm/dt = \frac{A(c - c_s)}{1/k_c + 1/k_i} \quad (17-23a)$$

Typically, k_c will depend on the velocity of the solution as shown in Figure 17.12. At low velocities, the growth rate

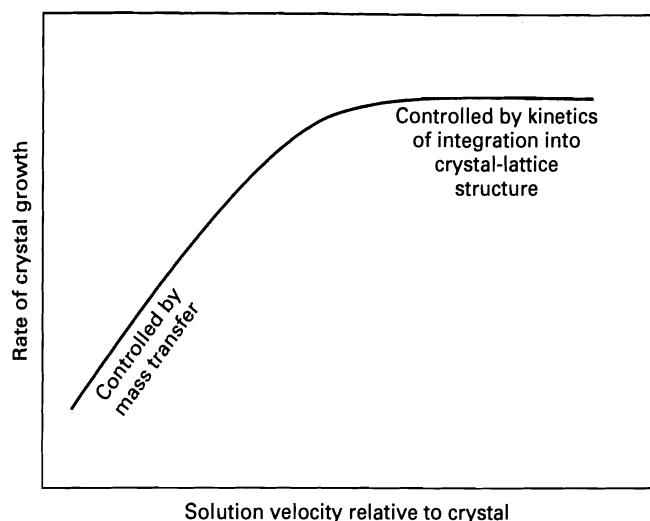


Figure 17.12 Effect of solution velocity past crystal on the rate of crystal growth.

may be controlled by the first step. The second step can be important, especially when the solution velocity past the crystal surface is high, such that k_c is large compared to k_i . In adsorption, the kinetic step is rarely important. It is also unimportant in dissolution, the reverse of crystallization.

The mass-transfer coefficient, k_c , for the first step, is independent of the crystallization process and can be estimated from general fluid-solid particle mass-transfer-coefficient correlations described in Chapters 3 and 15. The kinetic coefficient, k_i , is peculiar to the crystallization process. A number of theories have been advanced for the kinetic step, as discussed in Myerson [10]. One prominent theory is that of Burton, Cabrera, and Frank [11], which is based on a growth spiral starting from a screw dislocation, as shown in Figure 17.13 and verified in some experimental studies using scanning-electron microscopy. A dislocation is an imperfection in the crystal structure. The screw-dislocation theory predicts a growth rate proportional to: $(c_i - c_s)^2$ at low supersaturation and to $(c_i - c_s)$ at high supersaturation. Unfortunately, the theory does not provide a means to predict k_i . Accordingly, (17-23a) is generally applied with k_c estimated from available correlations and k_i back-calculated from experimental data.

Although crystals do not grow as spheres, let us develop an equation for the rate of increase of the diameter of a

spherical crystal. Rewriting (17-23a) in terms of an overall coefficient,

$$dm/dt = K_c A(c - c_s) \quad (17-23b)$$

Since $A = \pi D_p^2$ and $m = \frac{\pi D_p^3}{6} \rho$

Equation (17-23b) becomes

$$\frac{dD_p}{dt} = \frac{2K_c(c - c_s)}{\rho} = \frac{2K_c(\Delta c)}{\rho} \quad (17-24)$$

If the rate of growth is controlled by k_i , which is assumed to be independent of D_p , then

$$\frac{\Delta D_p}{\Delta t} = \frac{2k_i \Delta c}{\rho} \quad (17-25)$$

and the rate of increase of crystal size is linear in time for a constant supersaturation. If the rate of growth is controlled by k_c at a low velocity, then, from (15-60),

$$K_c = k_c = 2\mathcal{D}/D_p \quad (17-26)$$

where $\mathcal{D}$ is solute diffusivity.

Substitution of (17-26) into (17-24) gives

$$\frac{dD_p}{dt} = \frac{4\mathcal{D}(\Delta c)}{D_p \rho} \quad (17-27)$$

Integration from D_{p_0} to D_p gives

$$\frac{D_p^2 - D_{p_0}^2}{2} = \frac{4\mathcal{D}(\Delta c)}{\rho} t \quad (17-28)$$

If $D_{p_0} \ll D_p$, (17-28) reduces to

$$D_p = \left(\frac{8\mathcal{D}(\Delta c)t}{\rho} \right)^{1/2} \quad (17-29)$$

In this case, the increase in crystal diameter slows with time.

At higher solution velocities where k_c still controls, the use of (15-62) results in

$$K_c = k_c = C_1/D_p^{1/2} \quad (17-30)$$

For this case, the increase in crystal diameter also slows with time, but not as rapidly as predicted by (17-29). It is

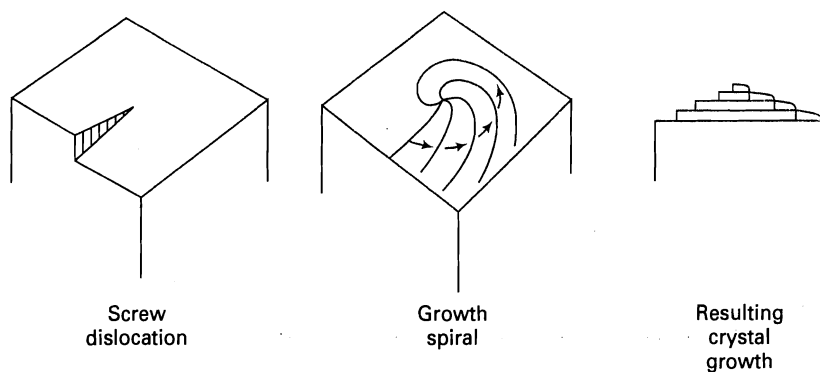


Figure 17.13 Screw-dislocation mechanism of crystal growth.

common to assume that the rate of crystal growth is controlled by k_i and, thus, is not dependent on crystal size and is, therefore, invariant with time. This assumption will be utilized in a latter section of this chapter to develop a useful crystallizer model.

EXAMPLE 17.9

The heptahydrate of MgSO_4 is to be crystallized batchwise from a seeded aqueous solution. A low supersaturation is to be used to avoid primary nucleation and a mild agitation is to be applied to minimize secondary nucleation. The temperature in the crystallizer will be maintained at 35°C , at which the solubility of MgSO_4 in water is 30 wt%. The crystallizer will be charged with 3,000 lb of a saturated solution at that temperature. To this solution will be added 2 lb of heptahydrate seed crystals of $50\ \mu\text{m}$ in diameter. A supersaturation of 0.01 gm of heptahydrate per gm of solution at 35°C will be maintained during crystallization by operating the crystallizer at vacuum and using heat exchange and the heat of crystallization to gradually evaporate water from the solution. Based on the assumptions and data listed below, determine the following if the final crystal size is to be $400\ \mu\text{m}$.

- Yield of crystals of heptahydrate in pounds.
- Number of crystals.
- Amount of water in pounds that will have to be evaporated.
- Product magma density in pounds of crystals per pound of solution.
- Crystallizer volume in gallons if the volume occupied by the magma during operation is at the most 50% of the crystallizer volume.
- Crystallizer pressure in psia and the boiling-point elevation in $^\circ\text{F}$.
- Time in minutes to grow the crystals to the final size.
- Amount of heat transfer in Btu.
- Also, determine whether crystal growth is controlled by mass transfer, by surface reaction (incorporation into the lattice), or by both.

Assumptions and Data:

- No primary or secondary nucleation.
- Properties of aqueous 30 wt% MgSO_4 at 35°C :
 - density = $1.34\ \text{g/cm}^3$
 - viscosity = 8 cP
 - diffusivity of $\text{MgSO}_4 = 1.10 \times 10^{-5}\ \text{cm}^2/\text{s}$
- Density of the crystals = $1.68\ \text{g/cm}^3$.
- Crystal shape can be approximated as a sphere.
- Average crystal-face growth rate, including effects of both mass transfer and surface reaction = $0.005\ \text{mm/min}$.
- Solution velocity past crystal face = $5\ \text{cm/s}$.

SOLUTION

Molecular weights: MgSO_4	120.5
$\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$	246.6
H_2O	18.02

Initial charge:

$$\begin{aligned} &900\ \text{lb aq MgSO}_4 \\ &2,100\ \text{lb H}_2\text{O} \\ &3,000\ \text{lb} \\ &+2\ \text{lb MgSO}_4 \cdot 7\text{H}_2\text{O crystal seeds} \end{aligned}$$

- (a) Crystals grow from $50\ \mu\text{m}$ to $400\ \mu\text{m}$ in diameter:

$$\text{yield} = 2 \left(\frac{400}{50} \right)^3 = 1,024\ \text{lb MgSO}_4 \cdot 7\text{H}_2\text{O}$$

- (b) Number of crystals, based on the crystal seeds,

$$\begin{aligned} &= \frac{\text{mass of seeds}}{\text{mass/seed}} \\ &= \frac{2(454)}{1.68 \left(\frac{3.14}{6} \right) (50 \times 10^{-4})^3} \\ &= 8.26 \times 10^9, \text{ where mass/seed} = \rho_p V_p = \rho \frac{\pi D_p^3}{6} \end{aligned}$$

- (c) Pounds of heptahydrate crystallized = $1,024 - 2 = 1,022\ \text{lb}$

$$\text{H}_2\text{O of hydration} = 1,022 \left(\frac{246.6 - 120.5}{246.6} \right) = 523\ \text{lb}$$

$$\text{MgSO}_4 \text{ in crystals} = 1,022 - 523 = 499\ \text{lb}$$

$$\text{MgSO}_4 \text{ left in solution} = 900 - 499 = 401\ \text{lb}$$

$$\text{H}_2\text{O left in solution} = 401 \left(\frac{70}{30} \right) = 936\ \text{lb}$$

$$\text{H}_2\text{O evaporated} = 2,100 - 523 - 936 = 641\ \text{lb}$$

- (d) Final mother liquor = $936 + 401 = 1,337\ \text{lb}$

$$\text{Magma density} = \left(\frac{1,024}{1,337} \right) = 0.766 \left(\frac{\text{lb crystals}}{\text{lb mother liquor}} \right)$$

- (e) Initially, using the factor of $8.33\ \text{lb/gal}$ for $1.0\ \text{g/cm}^3$,

$$\text{Solution volume} = \frac{3,000}{8.33(1.34)} = 269\ \text{gal}$$

Finally,

$$\text{Solution volume} = \frac{1,337}{8.33(1.34)} = 120\ \text{gal}$$

$$\text{Crystal volume} = \frac{1,024}{8.33(1.68)} = 73\ \text{gal}$$

$$\text{Total volume} = 193\ \text{gal}$$

Therefore, initial conditions before crystallization will control the volume.

$$\text{Make crystallizer volume} = \frac{269}{0.5} = 538\ \text{gal}$$

- (f) Calculate H_2O partial pressure by Raoult's law applied to mother liquor:

$$\text{lbmol MgSO}_4 = \frac{401}{120.5} = 3.33$$

$$\text{lbmol H}_2\text{O} = \frac{936}{18.02} = 51.95$$

$$\text{Total} = 55.28\ \text{lbmol}$$

$$x_{\text{H}_2\text{O}} = \frac{51.95}{55.28} = 0.94$$

$$\text{At } 35^\circ\text{C} = 95^\circ\text{F}, P_{\text{H}_2\text{O}}^s = 0.8153\ \text{psia.}$$

Therefore, $p_{H_2O} = x_{H_2O} P_{H_2O}^s = 0.94(0.8153) = 0.766$ psia.

This corresponds to a saturation temperature of 93°F.

Therefore, boiling-point elevation = 2°F.

(g) Growth rate = 0.005 mm/min.

Therefore, diameter grows at 0.01 mm/min.

Must grow from 50 μm = 0.05-mm diameter to 0.40-mm diameter.

$$\text{Time} = \frac{0.40 - 0.05}{0.01} = 35 \text{ min}$$

(h) Enthalpy balance:

$$F H_F + Q = V H_V + M H_M \quad (1)$$

Assume charge is at 35°C = 95°F; then, from Figure 17.10, $H_F = -40$ Btu/lb.

From the steam tables, $H_V = 1,102.2 + 0.9$ (for the boiling-point elevation) = 1,103.1 Btu/lb.

$$\begin{aligned} \text{Wt\% MgSO}_4 \text{ in magma} &= \frac{(499 + 401 + 1)}{(3,000 + 2 - 641)} = \frac{901}{2,361} \\ &= 0.38. \end{aligned}$$

From Figure 17.10, $H_M = -90$ Btu/lb.

From (1),

$$\begin{aligned} Q &= 641(1,103.1) + 2,361(-90) - 3,002(-40) \\ &= 615,000 \text{ Btu} = \text{heat transferred to crystallizer} \end{aligned}$$

(i) Assume mass-transfer controlled, using molar amount of crystals, n , and molar concentrations, c .

The molar form of (17-21) is

$$\frac{dn}{dt} = 4\pi r^2 k_c (\Delta c), \quad (2)$$

where r is the crystal radius

$$n = \frac{4}{3}\pi r^3 \rho_M, \text{ using a molar density.}$$

Therefore,

$$\frac{dn}{dt} = 4\pi r^2 \rho_M \frac{dr}{dt} \quad (3)$$

Equating (2) and (3),

$$\begin{aligned} 4\pi r^2 k_c (\Delta c) &= 4\pi r^2 \rho_M \frac{dr}{dt} \\ \frac{dr}{dt} &= \frac{k_c \Delta c}{\rho_M} \quad (4) \end{aligned}$$

From (15-62),

$$N_{Sh} = \frac{k_c D_p}{\mathcal{D}} = \frac{2k_c r}{\mathcal{D}} = 2 + 0.6(N_{Re})^{1/2}(N_{Sc})^{1/3}$$

$$\text{For } r = \frac{50}{2} = 25 \mu\text{m},$$

$$N_{Re} = \frac{D_p v \rho}{\mu} = \frac{(50 \times 10^{-6})(100)(5)(1.34)}{0.08} = 0.42$$

$$N_{Sc} = \frac{\mu}{\rho \mathcal{D}} = \frac{0.08}{1.34(1.10 \times 10^{-5})} = 5,430$$

$$N_{Sh} = 2 + 0.6(0.42)^{1/2}(5,430)^{1/3}$$

$$= 2 + 6.8 = 8.8 = \frac{2k_c r}{\mathcal{D}}$$

$$k_c = \frac{(8.8)(1.1 \times 10^{-5})}{2(25 \times 10^{-4})} = 0.019 \text{ cm/s}$$

$$\text{For } r = \frac{400}{2} = 200 \mu\text{m},$$

$$N_{Re} = 0.42 \left(\frac{400}{50} \right) = 3.36$$

$$\begin{aligned} N_{Sh} &= 2 + 0.6(3.36)^{1/2}(5,430)^{1/3} \\ &= 2 + 19.3 = 21.3 \end{aligned}$$

$$k_c = \frac{(21.3)(1.1 \times 10^{-5})}{2(200 \times 10^{-4})} = 0.006 \text{ cm/s}$$

Thus, k_c changes by a factor of 3 as crystal size changes from 25 to 200 μm .

Assume a Δc based on the given supersaturation of 0.01 g crystal per g solution.

$$\Delta c = \frac{0.01(1.34)}{246.6} = 54.3 \times 10^{-6} \frac{\text{mol}}{\text{cm}^3}$$

$$\rho_M \text{ of crystals} = \frac{1.68}{246.6} = 6.81 \times 10^{-3} \frac{\text{mol}}{\text{cm}^3}$$

From (4),

$$\frac{dr}{dt} = k_c \frac{54.3 \times 10^{-6}}{6.81 \times 10^{-3}} = 0.008 k_c$$

for the lowest value of $k_c = 0.006$ cm/s,

$$\frac{dr}{dt} = 0.008(0.006) = 48 \times 10^{-6} \text{ cm/s}$$

or

$$480 \times 10^{-6}(60) = 0.029 \text{ mm/min.}$$

But this is six times faster than the growth rate. Therefore, crystal growth is largely controlled by surface reaction.

17.4 EQUIPMENT FOR SOLUTION CRYSTALLIZATION

Before developing a crystallizer model in Section 17.5, it is useful to describe the most widely used equipment for solution crystallization. Such equipment may be classified according to the three schemes shown in Table 17.8.

Table 17.8 Classification of Equipment for Solution Crystallization

Operation Modes	Methods for Achieving Supersaturation	Crystallizer Features for Achieving Desired Crystal Growth
Batch	Cooling	Agitated or nonagitated
Continuous	Evaporation	Baffled or unbaffled
		Circulating liquor or circulating magma
		Classifying or nonclassifying
		Controlled or uncontrolled
		Cooling jacket or cooling coils

Although industrial equipment is available for batch or continuous operation, the latter is generally preferred. The choice of method for achieving supersaturation depends strongly on the effect of temperature on solubility. From the data in Table 17.5, it is seen that for many inorganic compounds in the near-ambient temperature range, e.g., 10–40°C, the change in solubility is small and may be insufficient to utilize the cooling method. This is certainly the case for MgCl_2 and NaCl . For KNO_3 and Na_2SO_4 , crystallization by cooling may be feasible. The majority of industrial crystallizers use the evaporation method or a combination of cooling and evaporation. Direct-contact cooling with agitation and evaporation can be achieved by bubbling air through the magma.

To produce crystals of a desired size distribution, considerable attention must be paid to the selection of features of the design of the crystallizer. The use of mechanical agitation can result in smaller and more uniformly sized crystals of a higher purity that are produced in less time. The use of vertical baffles can promote more uniform mixing. Supersaturation and uniformity can be controlled by circulation between a crystallizing zone and a supersaturation zone. In a circulating-liquor design, only the mother liquor is circulated, while in the circulating-magma design the mother liquor and crystals are circulated together. The circulation may be limited to the crystallizer vessel or may include pumping through an external heat exchanger. In a classifying crystallizer, the smaller crystals are separated from the larger and retained in the crystallizing zone for further growth or removed from the zone and redissolved. In a controlled design, one or more techniques are used to control the degree or supersaturation to avoid undesirable nucleation.

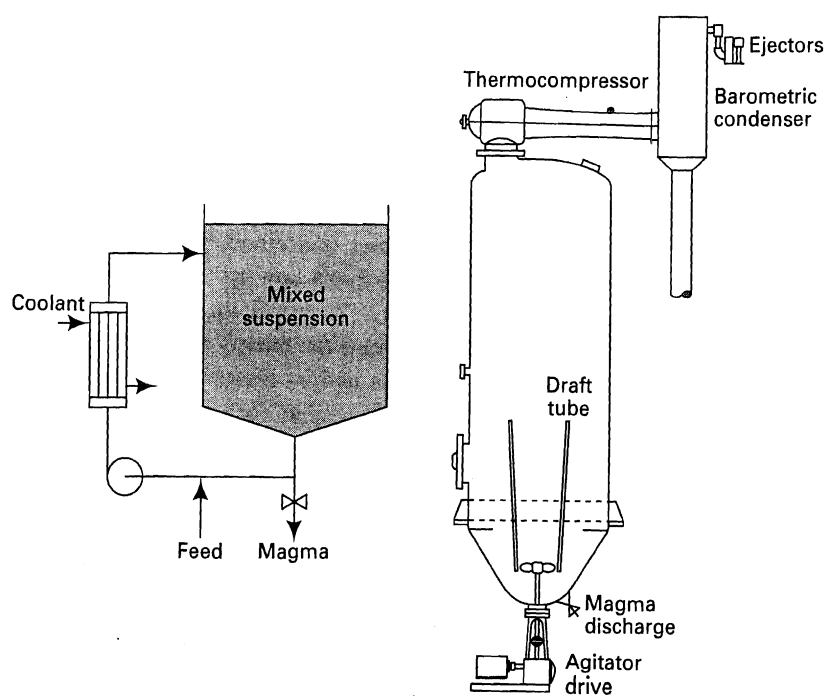
Cooling crystallizers may use a vessel jacket or internal coils, with the latter preferred because of the ease of wiping the crystals off the cooling surface.

A large number of patented commercial crystallizer designs have been developed. Many of them are described in Myerson [10] and by Mullin [1]. Only four of the more common solution crystallizers are described and illustrated here. The designs suffice to illustrate most of the features found in the many other designs.

Circulating, Batch Crystallizers

Although batch crystallizers can be operated without agitation or circulation by simply charging a hot solution to an open vessel and allowing the solution to stand as it cools by natural convection, the resulting crystals may be undesirably large, interlocked, impure because of entrapment of mother liquor, and difficult to remove from the vessel. Therefore, if the desired product of the crystallization operation is the crystals, it is preferable to use a more elaborate crystallizer configuration similar to either of the two batch crystallizers shown in Figure 17.14. In the design with external circulation, a high magma velocity is used for the flow of magma through the tubes of the heat exchanger to obtain a reasonable heat-transfer rate with a small temperature-driving force and minimization of crystal formation on the heat exchanger tube surfaces. This design can also be used for continuous crystallization, when the solubility-temperature curve dictates cooling crystallization.

In Figure 17.14b, crystallization is accomplished by evaporation under a vacuum pulled by a two-stage, steam-jet-ejector system though a water-cooled condenser. The



(a) Circulation of magma through an external, cooling heat exchanger

(b) Internal circulation with a draft tube

Figure 17.14 Circulating, batch cooling crystallizers.

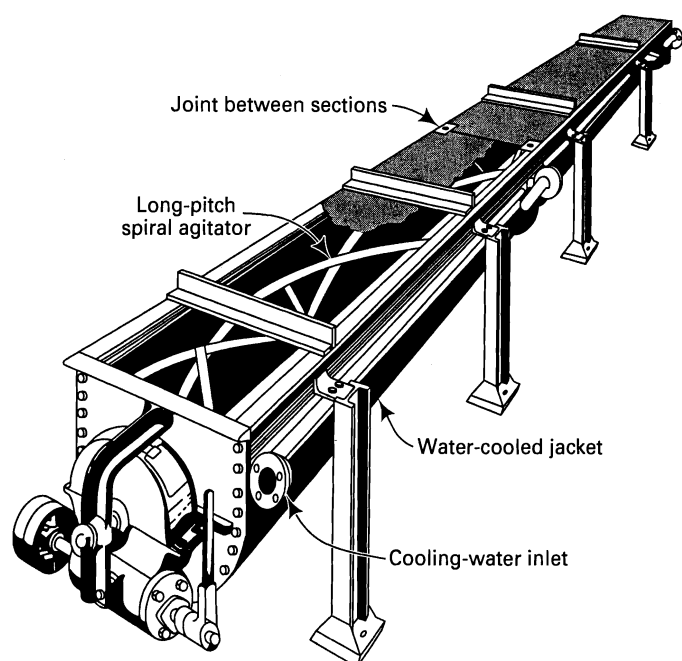


Figure 17.15 Swenson-Walker continuous, cooling crystallizer.

magma is circulated internally through a draft tube by a propeller. The energy for evaporation is supplied from the hot feed. A typical cycle, which includes charging the feed to the vessel, crystallization, and removal of the magma, may range from 2 to 8 h.

Continuous, Cooling Crystallizers

Figure 17.15 is a schematic diagram of a typical scraped-surface crystallizer known as the Swenson-Walker continuous, cooling crystallizer, described in detail by Seavoy and Caldwell [12]. The feed flows through a semicylindrical trough, typically 1 m wide $\times$ 3–12 m long. The trough has a water-cooled jacket and is provided with a low-speed (3–10 rpm) helical agitator-conveyor that scrapes the wall and prevents growth of crystals on the trough wall and promotes crystal growth by gentle agitation. Standard-size units can be linked together. The crystallization process may be controlled by the rather slow rate of heat transfer, with the major resistance due to the magma on the inside. Overall heat-transfer coefficients of only 10–25 Btu/h-ft²-°F (57–142 W/m²-K) are typically observed, based on a log-mean temperature difference between the magma and the coolant. Production rates of up to 20 tons per day of crystals of Na₃PO₄ · 12 H₂O and Na₂SO₄ · 10 H₂O of moderate size and uniformity have been achieved. Both salts show a significant decrease in solubility with decreasing temperature making the cooling crystallizer a viable choice.

EXAMPLE 17.10

The cooling crystallizer of Example 17.6 is to be a scraped-surface unit with 3 ft² of cooling surface per foot of running length of crystallizer. Cooling will be provided by a countercurrent flow of chilled water entering the cooling jacket at 60°F and leaving at

85°F. The overall heat-transfer coefficient, U , is expected to be 20 Btu/hr-ft²-°F. What length of crystallizer is needed?

SOLUTION

From Example 17.6, using Figure 17.10, the required rate of heat transfer is 44,900 Btu/h. The log mean temperature driving force for heat transfer is

$$\Delta T_{LM} = \frac{(120 - 85) - (70 - 60)}{\ln \left(\frac{120 - 85}{70 - 60} \right)} = 20^\circ\text{F}$$

The area for heat transfer is

$$A = \frac{Q}{U \Delta T_{LM}} = \frac{44,900}{20(20)} = 112 \text{ ft}^2$$

The crystallizer length = 112/3 = 37 ft.

Continuous, Vacuum, Evaporating Crystallizers

A large number of designs for continuous, vacuum, evaporating crystallizers have been developed. A particularly successful and widely used design of this type is the Swenson draft-tube, baffled (DTB) crystallizer, described by Newman and Bennett [13] and shown in one of several variations in Figure 17.16. In the main body of the crystallizer, evaporation occurs, under vacuum, at the boiling surface, which is located several inches above a draft tube that extends down to within several inches of the bottom of the main body of the crystallizer vessel. Near the bottom and inside of the draft tube is a low-rpm propeller that directs the magma upward through the draft tube toward the boiling surface under conditions of a small degree of supercooling and in the absence of any violent flashing action. Thus, nucleation and buildup of crystals on the walls are minimized. Surrounding the draft tube is an annular space where the magma flows back downward for re-entry into the draft tube. The outer wall of the annular space is a skirt baffle, surrounded by an annular settling zone, whose outer wall is the wall of the crystallizer. A portion (perhaps 10%) of the magma flowing downward through the first annular space turns around and flows outward and upward through the settling zone where larger crystals can settle, leaving a mother liquor containing fine crystals, which flows to a circulating pipe where it is joined by the feed and flows upward through a pump and then a heat exchanger. The circulating solution is heated several degrees to provide the energy for feed preheat and subsequent evaporation and to dissolve the finer crystals. The circulating magma re-enters the main body of the crystallizer just below the bottom of the draft tube. Further classification of crystals by size can be accomplished by providing an elutriation leg, as shown at the bottom of the main body of the crystallizer in Figure 17.16. In that case, product magma is withdrawn through a pipe from a nozzle located near the bottom of the elutriation leg where the largest crystals are present. Otherwise, the product magma may be withdrawn from the lower part of the annular region surrounding the draft tube.

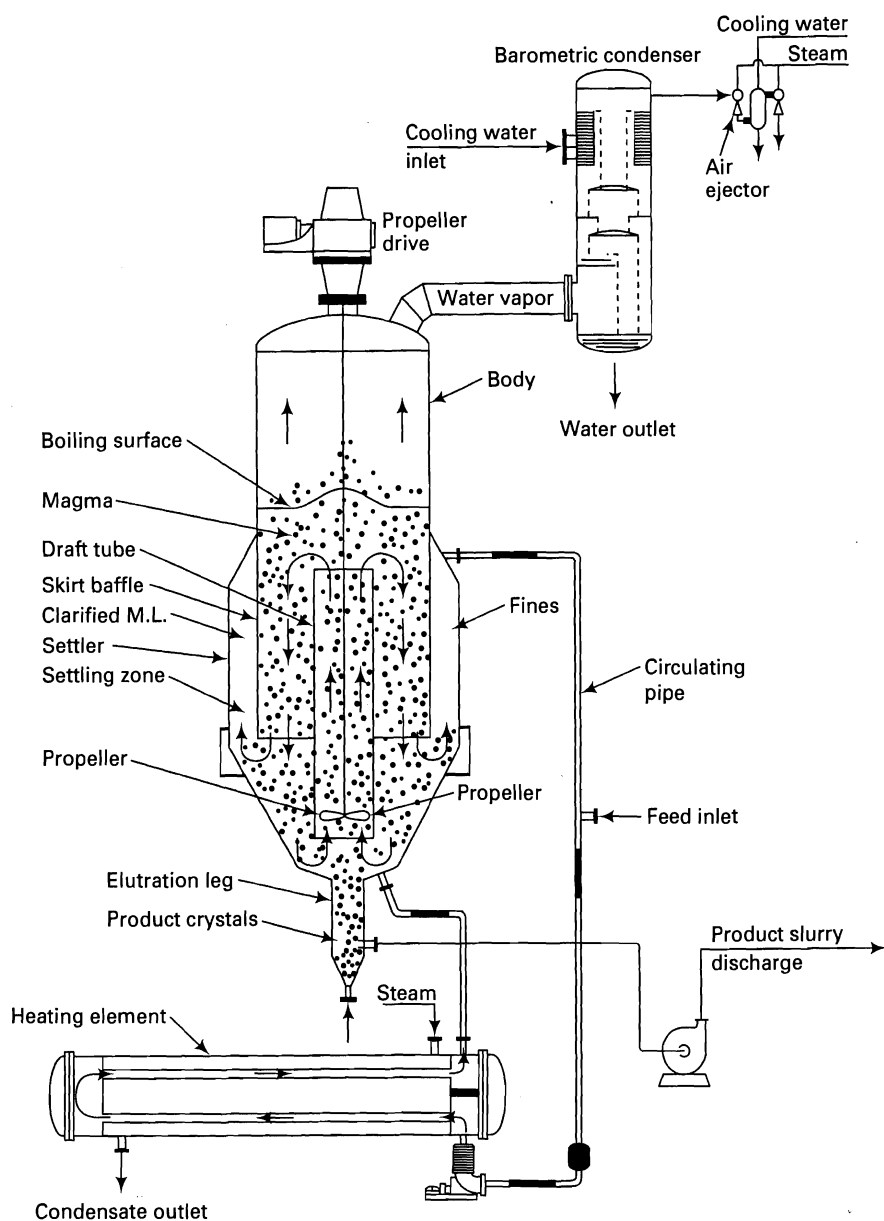


Figure 17.16 Swenson draft-tube, baffled (DTB) crystallizer.

17.5 THE MSMPR CRYSTALLIZATION MODEL

Because of the popularity of the DTB crystallizer, a mathematical model, due to Randolph [14], for its design and analysis is quite useful and is now found in the process equipment model libraries of a few continuous, steady-state, computer-aided simulation programs. This model is referred to as the *Mixed-Suspension, Mixed-Product-Removal (MSMPR)* model and is based on the following assumptions:

1. Continuous, steady-flow, steady-state operation.
2. Perfect mixing of the magma.
3. No classification of crystals.
4. Uniform degree of supersaturation throughout the magma.
5. Crystal growth rate independent of crystal size.
6. No crystals in the feed, but seeds are added initially.

7. No crystal breakage.
8. Uniform temperature.
9. Mother liquor in product magma in equilibrium with the crystals.
10. Nucleation rate constant and uniform and due to secondary nucleation by crystal contact.
11. Crystal-size distribution (CSD) uniform in the crystallizer and equal to that in the magma.
12. All crystals have the same shape.

Modifications to the model to account for: (1) classification of crystals due to settling, elutriation, and fines dissolving, and (2) variable growth rate are discussed by Randolph and Larson [15]. The central part of the MSMPR model is the estimation, by a crystal-population balance, of the crystal-size distribution (CSD), which is determined in practice by the rpm of the propeller in the draft tube and the external circulation rate. It is relatively easy to conduct experiments

in a small laboratory crystallizer that approximates the MSMPR model and can provide some of the necessary crystal nucleation rate and growth rate data to design a large-scale, industrial crystallizer.

Crystal-Population Balance

The crystal-population balance accounts for all crystals in the magma and, together with the mass balance, makes possible the determination of the CSD. Let L = a characteristic crystal size (e.g., from a screen analysis), N = cumulative number of crystals of size L and smaller in the magma in the crystallizer, and V_{ML} = volume of mother liquor in the magma in the crystallizer. A typical cumulative-numbers undersize plot based on these variables is shown in Figure 17.17, where the slope of the curve, n , at a given value of L is the number of crystals per unit size per unit volume, given by

$$n = \frac{d(N/V_{ML})}{dL} = \frac{1}{V_{ML}} \frac{dN}{dL} \quad (17-31)$$

The limits of n , as shown in Figure 17.17, vary from n^0 at $L = 0$ to 0 at $L = L_T$, the largest crystal size. In the MSMPR model, the cumulative plot of Figure 17.17 is independent of time and location in the magma. The plot is, in fact, the numbers-cumulative CSD for the crystals in the product magma.

For a constant, crystal-size growth rate, independent of crystal size, $G = dL/dt$, or

$$\Delta L = G \Delta t \quad (17-32)$$

and

$$L = G t_L \quad (17-33)$$

where t_L = residence time in the magma in the crystallizer for crystals of size L . Equation (17-32) is referred to as the ΔL law of McCabe [16], who found that the law correlated experimental data on the growth of crystals of KCl and $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ surprisingly well. Although McCabe's experiments were conducted in a small, laboratory crystallizer under ideal conditions, his resulting ΔL law is applied to commercial crystallization even though conditions may be far from ideal.

From (17-31), $dN = n V_{ML} dL$ = number of crystals in the size range dL . Now withdraw $(\Delta n dL)$ crystals per unit volume in time increment Δt . Because of the perfect mixing assumption for the magma,

$$\begin{aligned} & \frac{\text{number of crystals withdrawn}}{\text{mother-liquor volume withdrawn}} \\ &= \frac{\text{number of crystals}}{\text{mother-liquor volume in the crystallizer}} \end{aligned}$$

Therefore,

$$\begin{aligned} & \frac{\text{number of crystals withdrawn}}{\text{number of crystals in crystallizer}} \\ &= \frac{\text{mother-liquor volume withdrawn}}{\text{mother-liquor volume in the crystallizer}} \end{aligned}$$

or

$$\frac{\Delta n dL}{n dL} = - \frac{\Delta n}{n} = \frac{Q_{ML} \Delta t}{V_{ML}} \quad (17-34)$$

where Q_{ML} = volumetric flow rate of mother liquor in the withdrawn product magma and V_{ML} = volume of mother liquor in the crystallizer. Combining (17-32) and (17-34) and taking the limit,

$$- \frac{dn}{dL} = \frac{Q_{ML} n}{G V_{ML}} \quad (17-35)$$

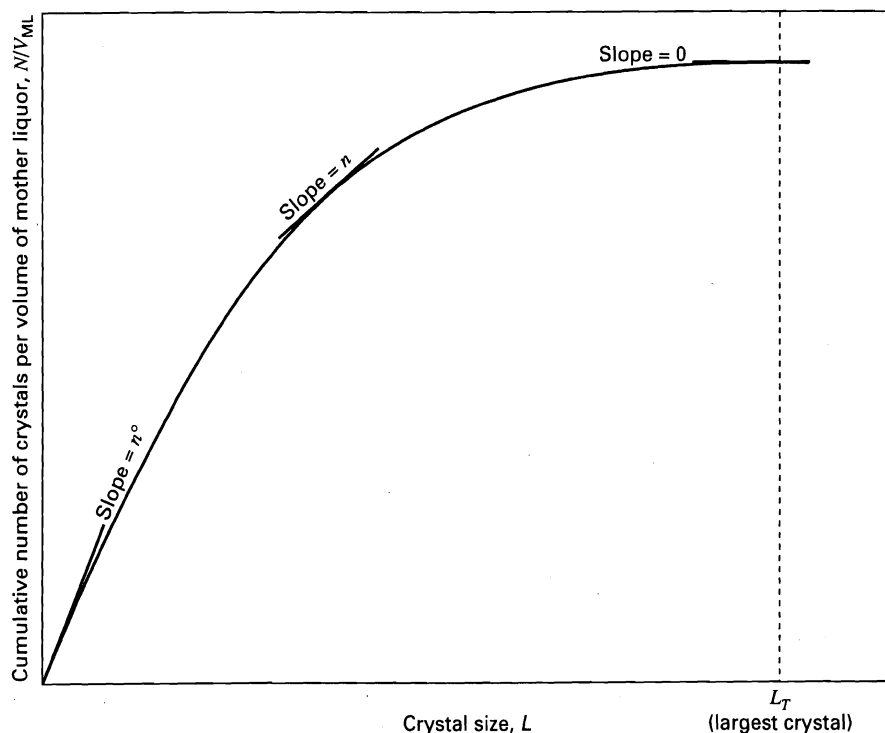


Figure 17.17 Typical cumulative-numbers-undersize distribution.

which is a simplified version of the following more-general, transient-population balance equation that allows for crystals in the feed and a nonuniform growth rate, but does assume a constant volume of mother liquor in the crystallizer:

$$\frac{\partial n}{\partial t} + \frac{\partial(nG)}{\partial L} + \frac{(Q_{ML})_{out} n}{V_{ML}} - \frac{(Q_{ML})_{in} n_{in}}{V_{ML}} = 0 \quad (17-36)$$

The retention time of mother liquor in the crystallizer is $\tau = V_{ML}/Q_{ML}$. Therefore, (17-35) can be rewritten as

$$-\frac{dn}{n} = \frac{dL}{G\tau} \quad (17-37)$$

If (17-37) is integrated for a constant growth rate and residence time, we obtain

$$n = n^0 \exp(-L/G\tau) \quad (17-38)$$

This equation is the starting point for determining distribution curves for crystal population, crystal size or length, crystal surface area, and crystal volume or mass.

For example, to obtain crystal population, the number of crystals per unit volume of mother liquor below size L is

$$N/V_{ML} = \int_0^L n dL \quad (17-39)$$

The total number of crystals per unit volume of mother liquor is

$$N_T/V_{ML} = \int_0^\infty n dL \quad (17-40)$$

Combining (17-38) to (17-40), the cumulative number of crystals of size smaller than L , as a fraction of the total is

$$x_n = \frac{\int_0^L n^0 e^{-L/G\tau} dL}{\int_0^\infty n^0 e^{-L/G\tau} dL} = 1 - \exp(-L/G\tau) \quad (17-41)$$

Or if we define a dimensionless crystal size as

$$z = L/G\tau \quad (17-42)$$

then,

$$x_n = 1 - e^{-z} \quad (17-43)$$

A plot of (17-43), shown in Figure 17.18a, is referred to as the cumulative distribution or cumulative crystal population. For a given value of z , x_n is the fraction of crystals having a smaller value of z . Also of interest is the corresponding differential plot of dx_n/dz , shown in Figure 17.18b, where from (17-43),

$$\frac{dx_n}{dz} = e^{-z} \quad (17-44)$$

The differential plot gives the fraction of the crystals in a given interval of z . At small values of z , the fraction is seen to be large, while at large values of z , the fraction is small.

In Figures 17.18a and b, four different distribution plots are shown. From statistics, (17-41) is one of a number of *moment equations*, which for a relation $n = f(z)$, are

given by

$$x_k = \frac{\int_0^z n z^k dz}{\int_0^\infty n z^k dz} \quad (17-45)$$

where k is the order of the moment. Thus, (17-43) is obtained by setting $k = 0$ and thus, is the zeroth moment of the distribution. Results for this moment and the corresponding first (length or size), second (area), and third (volume or mass) moments are summarized in Table 17.9. Corresponding cumulative and differential plots are included in Figure 17.18.

Of particular interest in the design and operation of a crystallizer is the predominant crystal size, L_{pd} , in terms of the volume or mass distribution. This size corresponds to the peak of the differential-mass distribution and is derived as follows: From Table 17.9,

$$dx_m/dz = (z^3/6)e^{-z} \quad (17-46)$$

At the peak,

$$\frac{d\left(\frac{dx_m}{dz}\right)}{dz} = 0 = \frac{3z^2 e^{-z}}{6} - \frac{z^3 e^{-z}}{6} \quad (17-47)$$

Solving (17-47) for z ,

$$z = 3 = \frac{L}{G\tau} \quad (17-48)$$

Therefore,

$$L_{pd} = 3G\tau \quad (17-49)$$

Similar developments using the differential expressions in Table 17.9 show that the most populous sizes in terms of the number of crystals, the size of crystals, and the surface area of crystals are 0, $G\tau$, and $2G\tau$, respectively. If L_{pd} is selected, (17-41) and (17-43) can be used to estimate cumulative and differential screen analyses.

To utilize the distributions of Table 17.9, values of G and τ are needed. The growth rate depends on the supersaturation and the degree of agitation. The residence time depends on the design and operation of the crystallizer. It is also useful to know B^0 and n^0 , which are related as follows:

$$\frac{1}{V_{ML}} \frac{dN}{dt} = \frac{1}{V_{ML}} \frac{dN}{dL} \left(\frac{dL}{dt} \right) \quad (17-50)$$

$$\lim_{L \rightarrow 0} \frac{1}{V_{ML}} \frac{dN}{dt} = B^0$$

$$\frac{dL}{dt} = G$$

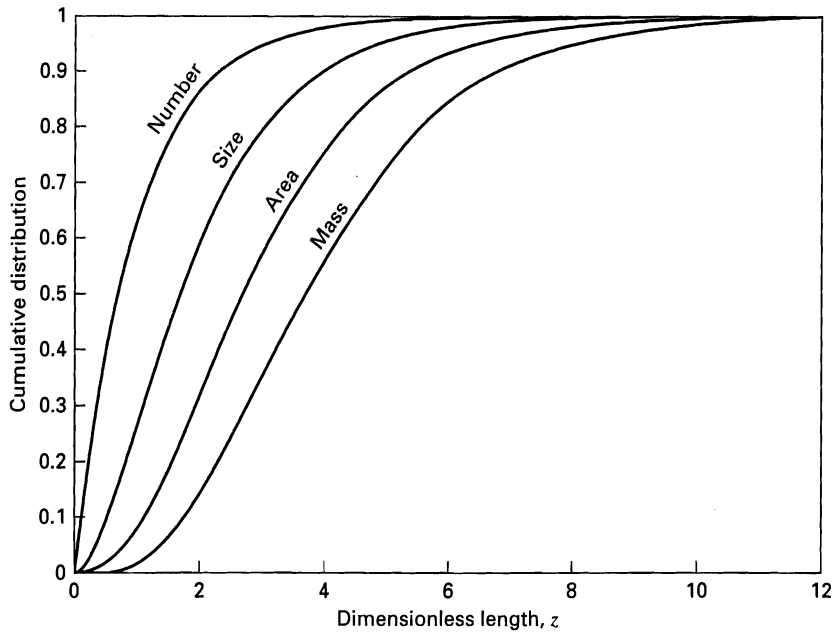
$$\lim_{L \rightarrow 0} \frac{1}{V_{ML}} \frac{dN}{dL} = n^0$$

Therefore,

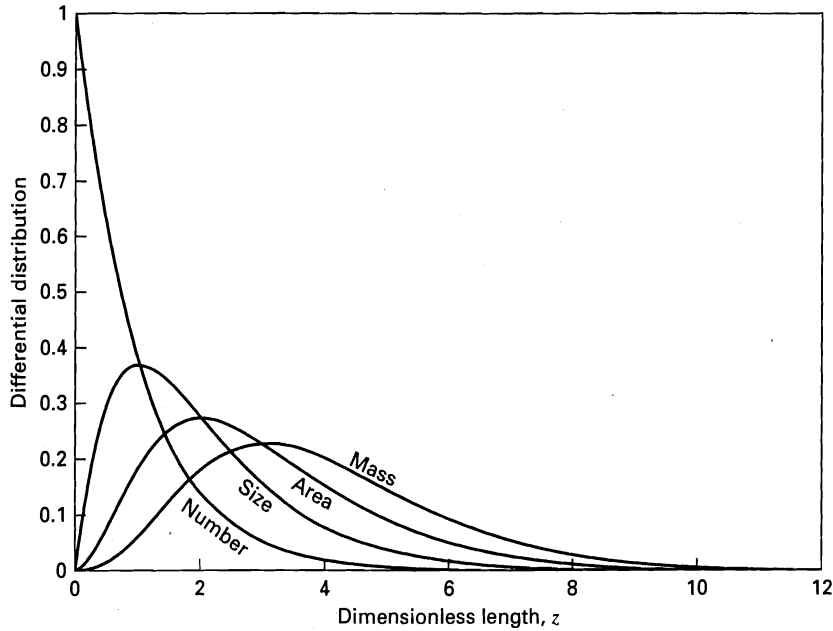
$$B^0 = Gn^0 \quad (17-51)$$

Combining (17-51) with (17-38),

$$n = \frac{B^0}{G} \exp(-L/G\tau) \quad (17-52)$$



(a) Cumulative distributions



(b) Differential distributions

Figure 17.18 Crystal-size distributions predicted by the MSMPR model.**Table 17.9** Cumulative and Differential Plots for Moments of Crystal Distribution for Constant Growth Rate

Moment	Distribution Basis	Cumulative	Differential
Zeroth	Number	$x_n = 1 - e^{-z}$	$dx_n/dz = e^{-z}$
First	Size or length	$x_L = 1 - (1 + z)e^{-z}$	$dx_L/dz = ze^{-z}$
Second	Area	$x_a = 1 - \left(1 + z + \frac{z^2}{2}\right)e^{-z}$	$dx_a/dz = \frac{z^2}{2}e^{-z}$
Third	Volume or mass	$x_m = 1 - \left(1 + z + \frac{z^2}{2} + \frac{z^3}{6}\right)e^{-z}$	$dx_m/dz = \frac{z^3}{6}e^{-z}$

$$z = L/G\tau.$$

This equation can be used with experimental data, as shown in the following example, to obtain values of nucleation and growth rates for a given set of operating conditions when the assumptions of the MSMPR model hold. The effect of operating conditions on B^0 can be expressed as an empirical, power-law function of the form given by (17-19). However, since the growth rate can be proportional to the relative supersaturation, s , raised to an exponent, (17-19) can be rewritten as

$$B^0 = k'_N G^i M_T^j N^r \quad (17-53)$$

Unfortunately, k'_N can be sensitive to the size of the equipment and is, therefore, best determined from data for a commercial crystallizer as discussed by Zumstein and Rousseau [17]. The exponents i , j , and r can be determined from small-scale experiments.

The necessary nucleation rate for a crystallization operation is related to the predominant crystal size by the MSMPR model. From (17-40) and (17-38),

$$n_c = N_T / V_{ML} = \int_0^\infty n dL = n^0 \tau G \int_0^\infty e^{-z} dz = n^0 \tau G \quad (17-54)$$

The mass of crystals per unit volume of mother liquor is

$$m_c = \int_0^\infty m_p n dL \quad (17-55)$$

where m_p = mass of particle, given by

$$m_p = f_v L^3 \rho_p \quad (17-56)$$

where f_v is defined in (17-11).

Combining (17-55), (17-56), and (17-38), followed by integration gives

$$m_c = 6 f_v \rho_p n^0 (G \tau)^4 \quad (17-57)$$

Combining (17-54) and (17-57), the number of crystals per unit mass of crystals is

$$\frac{n_c}{m_c} = \frac{1}{6 f_v \rho_p (G \tau)^3} \quad (17-58)$$

Combining (17-48) with (17-58),

$$\frac{n_c}{m_c} = \frac{9}{2 f_v \rho_p L_{pd}^3} \quad (17-59)$$

or the corresponding required nucleation rate is

$$B^0 = \frac{n_c C}{m_c V_{ML}} = \frac{9C}{2 f_v \rho_p V_{ML} L_{pd}^3} \quad (17-60)$$

where C = mass rate of production of crystals.

EXAMPLE 17.11

A continuous, vacuum-evaporating crystallizer of the DTB type is to be used to produce 2,000 lb/hr of $\text{Al}_2(\text{SO}_4)_3 \cdot 18\text{H}_2\text{O}$ ($\rho_p = 105 \text{ lb/ft}^3$). The magma contains 0.15 ft^3 crystals/ ft^3 magma and the residence time of the magma in the crystallizer is 2 h. The desired predominant crystal size on a mass basis is 0.417 mm.

Estimate with the MSMPR model:

- Required crystal growth rate in ft/h
- Necessary nucleation rate in nuclei/h-ft³ of mother liquor in the crystallizer
- Number of crystals produced per hour
- Tables and plots of estimated cumulative and differential screen analyses of the product crystals on a mass or volume basis

Also explain how the required growth and nucleation rates for the operating crystallizer might be achieved.

SOLUTION

- (a) From (17-48),

$$G = \frac{L_{pd}}{3\tau} = \frac{(0.417/304.8)}{3(2)} = 2.28 \times 10^{-4} \text{ ft/h}$$

- (b) Need volume of mother liquor in the crystallizer.

$$\text{Volume of crystals produced} = \frac{2,000}{105} = 19.1 \text{ ft}^3 \text{ h}$$

$$\text{Volume of crystals in crystallizer} = 19.1(2) = 38.2 \text{ ft}^3$$

$$\text{Volume of magma in crystallizer} = \frac{38.2}{0.15} = 255 \text{ ft}^3$$

$$\text{Volume of mother liquor in crystallizer} = V_{ML} = 255 - 38.2 = 217 \text{ ft}^3$$

From (17-60), assuming $f_v = 0.5$,

$$B^0 = \frac{9(2,000)}{2(0.5)(105)(217)(0.417/304.8)^3} = 3.1 \times 10^8 \frac{\text{nuclei}}{\text{h-ft}^3}$$

- (c) Number of crystals produced = $3.1 \times 10^8(217) = 6.7 \times 10^{10}/\text{h}$.

$$\begin{aligned} \text{(d)} \quad z &= \frac{L}{G\tau} = \frac{L}{(2.28 \times 10^{-4})(2)} \\ &= 2.2 \times 10^3 L, \text{ ft} \quad \text{or} \quad 7.19 L, \text{ mm} \end{aligned}$$

From Table 17.9,

$$x_m = 1 - \left(1 + z + \frac{z^2}{2} + \frac{z^3}{6}\right) e^{-z} \quad (1)$$

$$\frac{dx_m}{dz} = \frac{z^3}{6} e^{-z} \quad (2)$$

Using (1) and (2) with a spreadsheet for the older Tyler mesh sizes, the following results are obtained for the cumulative and differential screen analyses:

Tyler Mesh	Opening, mm	Dimensionless Length, z	Cumulative Screen Analysis, %	Differential Screen Analysis %
8	2.357	16.96	100.00	0.00
10	1.667	11.99	99.77	0.18
14	1.179	8.48	96.95	2.11
20	0.833	5.99	84.82	8.95
28	0.589	4.24	61.16	18.31
35	0.417	3.00	35.20	22.40
48	0.295	2.12	16.50	19.05
65	0.208	1.50	6.54	12.53
100	0.147	1.06	2.29	6.87
150	0.104	0.75	0.73	3.31
200	0.074	0.53	0.22	1.46

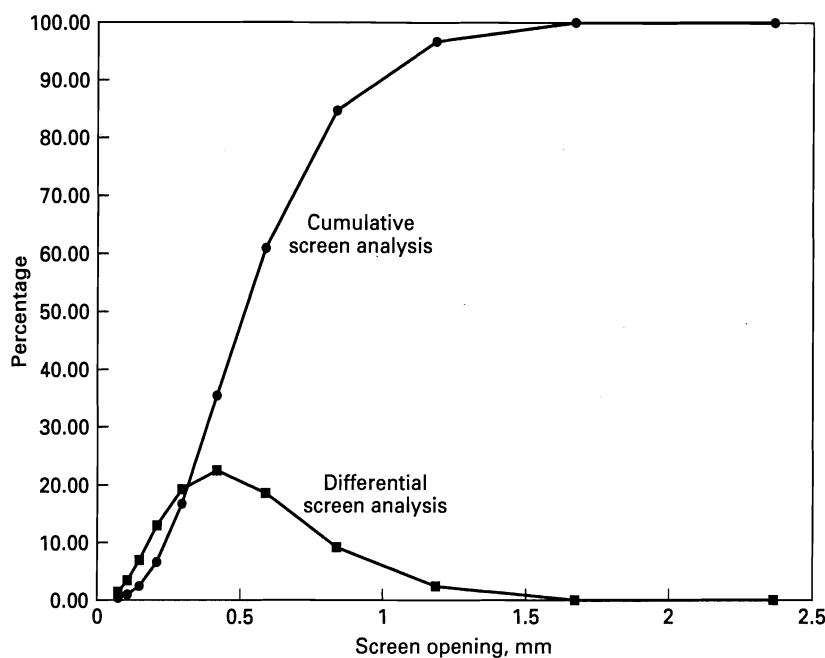


Figure 17.19 Predicted cumulative and differential screen analyses for Example 17.11.

A plot of the cumulative and differential screen analyses is shown in Figure 17.19.

Both growth and nucleation rates depend on the supersaturation. The growth may also depend on the relative velocity between the

crystals and the mother liquor. The nucleation rate depends upon the degree of agitation. With a DTB crystallizer, the agitator rpm and the magma circulation rate through the external heat exchanger can be adjusted to achieve the required growth and nucleation rates.

17.6 PRECIPITATION

Solution crystallization, as discussed above, occurs when a solution containing a moderately to strongly soluble solute is cooled or partially evaporated, causing the solute to exceed its solubility sufficiently to partially crystallize. The degree of supersaturation is small, primary nucleation is avoided, and crystal growth is slow. If the process is carried out under controlled conditions, reasonably large and desirable crystals can be grown. In many respects, the process of precipitation is just the opposite of solution crystallization. Precipitation, as discussed by Nielsen [18], involves solutes that are only sparingly soluble. The precipitate is formed by changing solution pH, solvent concentration, solution temperature, or most commonly by the addition of a reagent to a solution, resulting in a reaction to produce another chemical that is almost insoluble in the resulting solution. The latter is commonly referred to as *reactive crystallization*. The degree of supersaturation produced by the reaction is very large, causing a high degree of primary nucleation. Although some growth occurs as the supersaturation is depleted, precipitates generally consist of very small particles that may be crystalline in nature and are frequently *aggregates* and *agglomerates*. Aggregates are masses of crystallites that are weakly bonded together. Agglomeration can follow, cementing aggregates together. Typical chemicals that are produced by precipitation are the sparingly soluble compounds listed in Table 17.6. Precipitation may be thought of as *fast crystallization*.

In precipitation, particle size is related to the solubility by (17-16), the Kelvin equation. However, for precipitates formed from ionic reactions in solution, the supersaturation ratio, $S = c/c_s$, is replaced by $(\pi/K_c)^{1/v}$, where π = the ionic concentration product for the reaction, K_c = (equilibrium) solubility product, and v = sum of the number of cations and anions that form the precipitated compound. Thus, for aluminum hydroxide at 15°C,

$$\pi = (c_{\text{Al}}^{+3})(c_{\text{OH}}^{-})^3$$

with K_c given in Table 17.6 and $v = 1 + 3 = 4$. The Kelvin relation is of great importance in precipitation because nucleation can be due to homogeneous primary nucleation.

The extent of supersaturation is a very important factor in precipitation. When a reagent is added to a solution to form a sparingly soluble compound, a very high supersaturation can be developed, depending upon the ionic concentrations in the reagent and solution before mixing. For example, consider the formation of a precipitate of BaSO_4 from an aqueous reagent solution containing Ba^{++} ions and an aqueous solution of, e.g., sulfuric acid.

From Table 17.6, the solubility product of BaSO_4 at 18°C is 0.87×10^{-10} . Figure 17.20a, taken from Nielsen [19], is a plot of sulfate ion concentration versus barium ion concentration, both in the solution just after mixing and before precipitation, with contours of constant supersaturation ratio, S . The dashed lines refer to an ideal situation where activity

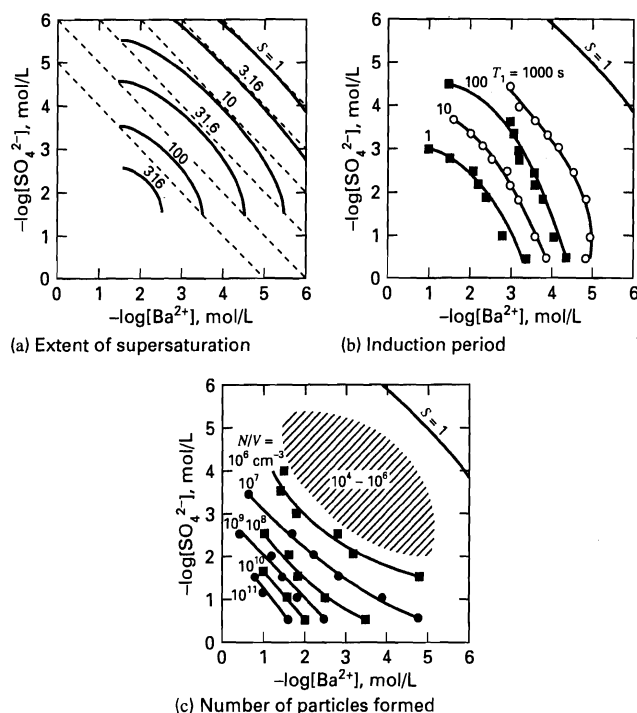


Figure 17.20 Precipitation diagrams for BaSO₄.

coefficients are 1.0 (concentrations = activities) and no complexes of Ba⁺⁺ and SO₄⁼, are formed. The solid, curved contours take these deviations from the ideal situation into account. Note the very large supersaturation ratios that can occur. For example, if the SO₄⁼ concentration in the solution is 0.01 mol/L and the Ba⁺⁺ concentration is 4.7×10^{-4} mol/L, the supersaturation ratio = 100.

Typically, precipitation does not take place immediately upon development of a large supersaturation because of the slow growth of very small particles. However, after a certain period of time, called the induction period, T_1 visible precipitation begins. As shown in Figure 17.20b, this period is highly dependent on the initial-ion concentrations and, by superposition of Figure 17.20a, on the supersaturation ratio. For example, at a supersaturation ratio of about 300, the induction period is only one second, while at a supersaturation ratio of about 10, the induction period is more than one minute.

As shown in Figure 17.20c, the number of particles formed per unit volume of solution, N/V , also depends strongly on the initial concentrations of the anion and cation and, therefore, on the supersaturation at high ion concentrations. A minimum number of particles per unit volume or a corresponding maximum particle size is frequently observed. For BaSO₄, this maximum size occurs for a supersaturation ratio of about 300. The number of particles formed also depends on particle growth rate, which may be controlled by mass transfer of the ions to the particle surface and/or integration of ions into the particle crystalline structure (surface reaction). Particle growth rates during

precipitation generally follow one of the following laws, which exhibit different dependencies on the relative supersaturation, $s = S - 1$:

Linear

$$G = k_1 s \quad (17-61)$$

Parabolic

$$G = k_2 s^2 \quad (17-62)$$

Exponential

$$G = k_3 f\{s\} \quad (17-63)$$

The linear rate law, which often applies for $G > 10$ nm/s, may be due to mass-transfer control or surface-adsorption control, where the latter depends more strongly on temperature. The parabolic rate law, for which G may be < 10 nm/s, applies to screw-dislocation-controlled growth. The exponential law, where $f\{s\}$ can involve a complex exponential, log, and/or power-law dependency corresponds to growth control by surface nucleation. The latter two mechanisms can occur in parallel. Rate constants for many electrolytes for the three different rate laws are given by Nielsen [20]. When growth is rapid, coprecipitation of soluble electrolytes may occur by entrapment, thus making it difficult to obtain a pure precipitate product.

Because a precipitate is formed at considerable supersaturation, the resulting particle shapes may be far from the shape corresponding to a minimum Gibbs energy, which depends on the particle surface area and interfacial tension. However, if the precipitate and mother liquor are allowed to age, then, as discussed by Nielsen, [20], the precipitate particle sizes and shapes can tend toward equilibrium values by: (1) flocculation and sintering of fine particles (2) transport of ions over the surface, and (3) ripening by dissolution and redeposition. Ripening can result in the release of coprecipitates, thus increasing precipitate purity.

The small particles produced in abundance during precipitation have a tendency to cluster together by interparticle-collision phenomena, variously referred to as agglomeration, aggregation, and flocculation. Such clusters, often called agglomerates, are common when the number of particles/cm³ of solution exceeds 10^7 . For BaSO₄, Figures 17.20a and c show that agglomeration may require a very large initial supersaturation ratio. Agglomeration is important mainly when particles sizes are between 1 μ m and 50 μ m.

EXAMPLE 17.12

Fitchett and Tarbell [21] studied the effect of impeller rpm on the crystal-size distribution obtained for the continuous precipitation of barium sulfate when mixing solutions of sodium sulfate and barium chloride. The contents of the 1.8-L crystallizer were assumed to be perfectly mixed. In their Run 15, they used feeds with dissolved salts in stoichiometric ratio to give a sodium-chloride concentration of 0.15 mol/L and an average residence time of 38 s.

Results for two different impeller speeds were as follows:

Size, μm	ln n , Number Density of Crystals, ln ($\mu\text{m}^{-1} \text{L}^{-1}$)	
	950 rpm, 0.361 J/s · kg Feeds	400 rpm, 0.028 J/s · kg Feeds
7	22.07	23.45
9	21.66	22.75
11	21.35	22.08
13	20.97	21.36
15	20.77	20.75
17	20.41	20.14
19	20.04	19.57
21	19.77	18.94
23	19.48	18.43
25	19.09	17.75
27	18.85	
29	18.49	
31	18.11	
33	17.87	

Using the MSMPR model, determine from the two sets of data:

- n^0 , number density of nuclei, nuclei/ $\mu\text{m}\cdot\text{L}$
- G , linear growth rate, $\mu\text{m}/\text{s}$
- B^0 , nucleation rate, nuclei/L·s
- mean crystal length, μm
- n_c , number of crystals/volume of mother liquor, crystals/ m^3

SOLUTION

- (a) From (17-38), if each set of data is fitted to

$$\ln n = \ln n^0 - L/G\tau \quad (1)$$

the best straight line yields an intercept of $\ln n^0$ and a slope of $-1/G\tau$, from which n^0 and G can be determined for $\tau = 38$ s.

Using a spreadsheet, the results for 950 rpm are

intercept = 23.13, slope = -0.1601

Therefore, $n^0 = \exp(23.13) = 1.11 \times 10^{10}$ crystals/ $\mu\text{m} \cdot \text{L}$

- (b) $G = 1/[(38)(0.1601)] = 0.164 \mu\text{m}/\text{s}$

- (c) The nucleation rate is given by (17-51):

$$B^0 = Gn^0 = 0.164(1.11 \times 10^{10}) = 1.82 \times 10^9 \text{ nuclei/L} \cdot \text{s}$$

- (d) The mean particle length is determined from the value of z for the maximum value of dx_L/dz , which, from Table 17.9 is ze^{-z} .

$$d \frac{(ze^{-z})}{dz} = e^{-z} - ze^{-z} = 0$$

Therefore, $z = 1 = L_{\text{mean}}/G\tau$ from (17-42)

$$L_{\text{mean}} = G\tau = 0.164(38) = 6.23 \mu\text{m}$$

- (e) The number of crystals per unit volume of mother liquor is given by (17-54).

$$\begin{aligned} n_c &= n^0 \tau G = B^0 \tau = 1.82 \times 10^9 (38) \\ &= 6.92 \times 10^{10} \text{ crystals/L} = 6.92 \times 10^7 \text{ crystals/m}^3 \end{aligned}$$

- (f) In a similar manner, the following results are obtained for 400 rpm:

Intercept = 25.53, slope = -0.313

$$n^0 = 1.22 \times 10^{11} \text{ crystals}/\mu\text{m} \cdot \text{L}$$

$$G = 0.0841 \mu\text{m}/\text{s}$$

$$B^0 = 1.03 \times 10^{10} \text{ nuclei/L} \cdot \text{s}$$

$$L_{\text{mean}} = 3.20 \mu\text{m}$$

$$n_c = 3.91 \times 10^8 \text{ crystals/m}^3$$

Comparing the two sets of results, we see that a higher agitator speed gives a larger mean crystal size, a larger growth rate, and a lower nucleation rate.

17.7 MELT CRYSTALLIZATION

Solution crystallization refers to crystallization of a solute from a solvent. It is most commonly conducted with aqueous solutions of dissolved inorganic salts. The phase equilibrium diagram for a water-salt system, e.g., Figure 17.2, typically includes temperatures ranging from the eutectic temperature (below 0°C) to a temperature exceeding the melting point of ice, but not greater than 200°C . Since the melting points of inorganic salts greatly exceed 200°C , the pure-salt melting point is, accordingly, not included on the diagram.

For mixtures of organic compounds, the situation is quite different. An analysis by Matsuoka et al. [22] found that more than 70% of the common organic compounds had melting points between 0 and 200°C . For binary mixtures of such compounds, the phase-equilibrium diagrams will typically include the melting points of both compounds. Typical diagrams are shown in Figure 17.1. In (b), crystals of ortho-chloronitrobenzene can be formed if the feed composition is less than the eutectic composition based on para-chloronitrobenzene; otherwise, pure para-chloronitrobenzene can be formed. The exception is the eutectic composition. *Eutectic-forming* systems consist of compounds that can not substitute for each other in the crystal lattice. Thus, the eutectic mixture consists of two different solid phases. The two solubility curves separating the liquid-phase region from the two solid-liquid regions are sometimes referred to as freezing-point curves, and mixtures at conditions in the liquid-phase region are referred to as *melts*.

Much less common are *solid-solution-forming* systems of the type shown in Figure 17.1c for the phenanthrene-anthracene system. These systems consist of compounds so nearly alike in structure that they can substitute for each other in the lattice structure to form a single solid phase of the two compounds over a wide range of composition. With this type of system, a mixture is crystallized. The liquid-solid phase diagram resembles that for vapor-liquid equilibrium, where the freezing-point and melting-point curves replace the dew-point and bubble-point curves, respectively. Mixtures in the liquid-phase region above the freezing-point curve are also referred to as *melts*. A mixture in the region

between the two curves separates into a liquid phase and a solid phase, neither of which is pure.

Crystallization of melts from either eutectic-forming or solid-solution-forming mixtures is called melt crystallization. Although theoretically, melt crystallization of eutectic-forming systems, like solution crystallization of such systems, can produce pure crystals, the crystalline product from commercial single-stage crystallizers may not meet purity specifications for reasons discussed in detail by Wilcox [23]. In that case, repeated stages of melting and crystallization may be necessary. With eutectic-forming systems, multiple stages are mandatory to produce crystals of high purity.

The separation of organic mixtures is most commonly achieved by distillation. However, if distillation: (1) requires more than 100 theoretical stages, (2) cannot produce products that meet specifications (e.g., purity and color), (3) causes decomposition of feed components, or (4) requires extreme conditions of temperature or pressure (e.g., vacuum), other separation methods should be considered. According to Wynn [24], if the compounds to be separated are: (1) disubstituted benzenes, diphenyl alkyls, phenones, secondary or tertiary aromatic or aliphatic amines, isocyanates, fused-ring compounds, heterocyclic compounds, and carboxylic acids of MW < 150; (2) have a melting-temperature range from 0 to 160°C; (3) required to be very high-purity products; or if (4) a laboratory test produces a clearly defined solid phase from which the liquid phase drains freely, then melt crystallization should be considered as an alternative or supplement, as in a hybrid process, to distillation.

Equipment for Melt Crystallization

As with solution crystallization, a large number of crystallizer designs have been proposed for melt crystallization.

Only some of the more widely used commercial units are discussed here. Myerson [10] gives a detailed discussion. In all cases, crystallization is caused by cooling the mixture.

The two major methods used in melt crystallizers are *suspension crystallization* and *layer crystallization* by progressive freezing. In the former, crystals of a desired size distribution are grown slowly in a suspension by subcooling a seeded-feed melt. In the latter, crystals of uncontrolled size are grown rapidly on a cooled surface, wherein subcooling is supplied through the crystallized layer. In suspension crystallization, the remaining melt must be separated from the crystals by centrifugation, filtration, and/or settling. In layer crystallization, the remaining melt or residual liquid is drained from the solid layer, followed by melting of that layer.

Figure 17.21 shows a two-stage, scraped-wall-crystallizer system used for suspension crystallization. A cooling medium is used to control the surface temperatures of the two scraped-wall units, causing crystals to grow, which are subsequently scraped off by screws. The melt-crystal mixture is circulated through a ripening vessel. The two scraped-wall units of a typical system are 3.6 m long with 3.85 m² of total heat-transfer surface area. The screws are driven by a 10-kW motor.

Of more importance in commercial applications of melt crystallization is the falling-film crystallizer developed by Sulzer Brothers Limited and shown in Figure 17.22. This equipment is particularly useful for producing high-purity crystals (>99.9%) at high capacity (>10,000 tons/y). A large pair of units, each 4 m in diameter and containing 1,100 12-m-high tubes can produce 100,000 tons/yr of very pure crystals, with typical layer growth rates of 1 in./h. The feed melt flows as a film down the inside of the tubes, over a crystal layer that forms and grows by progressive freezing because the wall of the tube is cooled from the outside by a coolant. When a predetermined crystal layer thickness, typically 5–20 mm, is

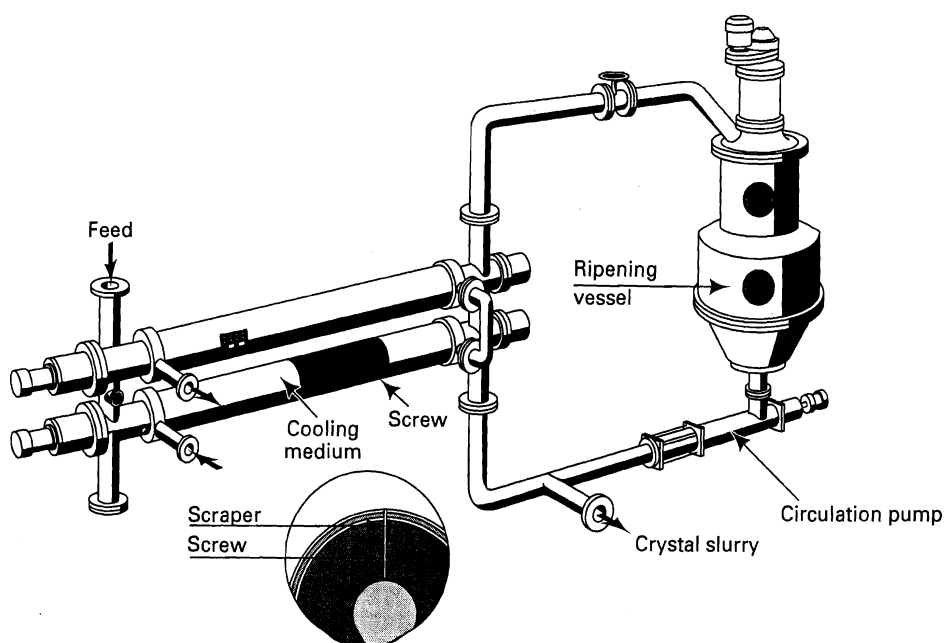


Figure 17.21 Two-stage, scraped-wall, melt crystallizer.

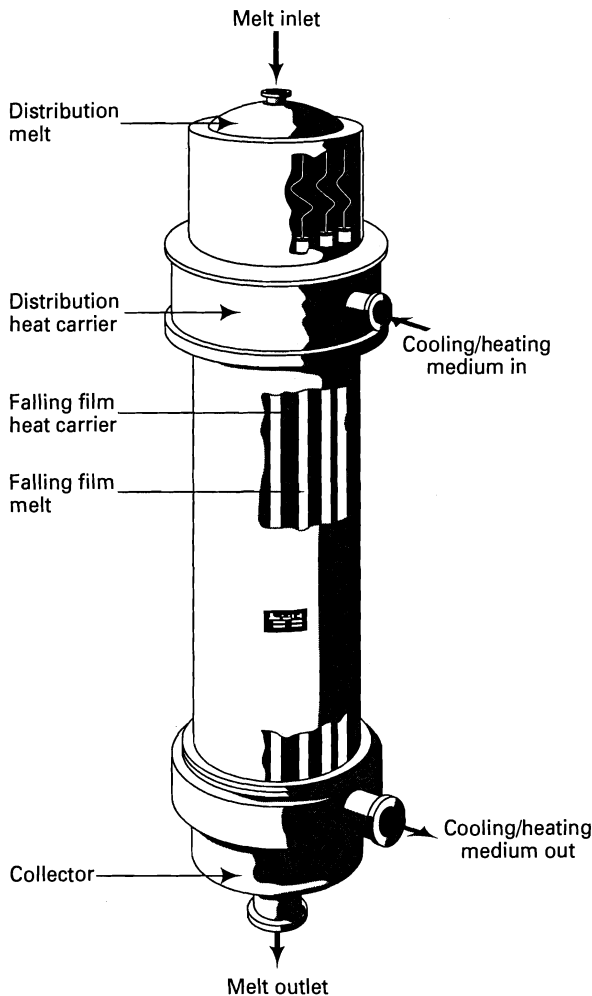


Figure 17.22 Sulzer falling-film melt crystallizer.

reached, the feed melt flow rate is stopped. The tubes are now slightly warmed to cause partial melting, called sweating, to remove impurities that may be stuck or bonded to the crystal layer. This is followed by complete melting of the remaining layer, which is of high purity. During the initial crystallization phase, melt is circulated at a high rate compared to the crystallization rate so that a uniform temperature and melt composition is approached down the length of the tube. The coolant also flows as a film down along the outside surface of the tubes.

Consider the freezing step in a falling-film crystallizer. A typical temperature profile is shown in Figure 17.23. The melt enters at the top of the tube and flows as a film down the inside wall. A coolant, at a temperature below the freezing point of the melt, also enters at the top of the tube and flows as a film down the outside wall. Heat is transferred from the melt to the coolant, causing the melt to form a crystal layer at the inside wall of the tube. As the melt and coolant flow down the tube, their temperatures decrease and increase, respectively. The thickness of the crystal layer increases with time, with the latent heat of fusion transferred from the crystal-melt interface to the coolant.

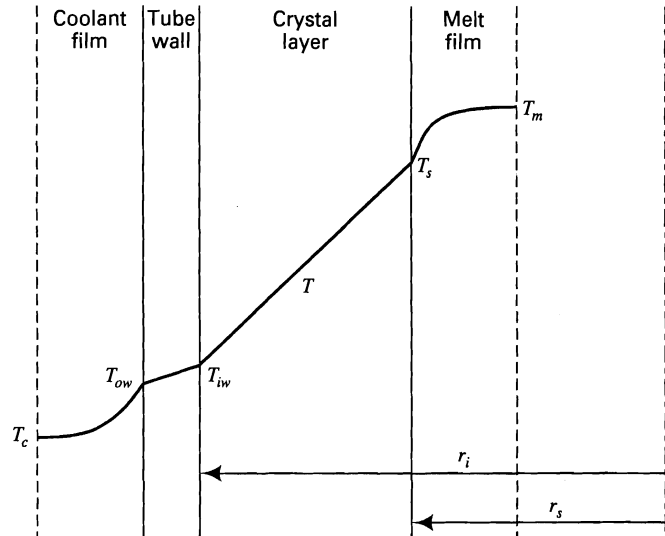


Figure 17.23 Temperature profile for melt crystallization from a falling film at a vertical location along the tube.

For a eutectic-forming, binary melt, only one component will crystallize, although small amounts of the other component may be trapped in the crystal layer, particularly if the rate of crystal formation is too fast. The temperature at the interface of the crystal layer and the melt can be assumed to be the equilibrium temperature corresponding to the saturated melt composition on the solubility curve. If mass transfer of the crystallizing component from the melt film to the phase interface is rapid, the interface temperature will correspond to the melt-film composition at that vertical location. Furthermore, if the thermal resistances of the coolant film, tube wall, and melt film are negligible compared to that of the crystal layer, and if the heat capacity of the crystal layer and metal wall are negligible, then the following simple model for the rate of increase of thickness of the crystal layer with time can be constructed.

At a particular vertical location, $T_s \approx T_m$, the melting point, and $T_{iw} \approx T_c$, the coolant bulk temperature. The rate of heat released by freezing, ΔH_f , is equal to the rate of heat conduction through the crystal layer. Thus, for a planar wall, referring to Figure 17.23, the heat evolved from fusion is equated to the rate of heat conduction through the crystal layer to give

$$\Delta H_f \frac{dm}{dt} = -A \rho_c (\Delta H_f) \frac{dr_s}{dt} = \frac{k_c A (T_m - T_c)}{r_i - r_s} \quad (17-64)$$

with an initial condition $r_s = r_i$ at $t = 0$. Integration of (17-64) gives

$$\frac{(r_i - r_s)^2}{2} = \frac{k_c (T_m - T_c) t}{\rho_c \Delta H_f} \quad (17-65)$$

or crystal layer thickness is

$$(r_i - r_s) = \sqrt{\frac{2k_c(T_m - T_c)t}{\rho_c \Delta H_f}} \quad (17-66)$$

A similar derivation for a cylindrical tube wall, where r_i = inside radius of the tube, gives

$$\frac{1}{4}(r_i^2 - r_s^2) - \frac{r_s^2}{2} \ln\left(\frac{r_i}{r_s}\right) = \frac{k_c(T_m - T_c)t}{\rho_c \Delta H_f} \quad (17-67)$$

As would be expected, the value of the left-hand side of (17-67) approaches the value of the left-hand side of (17-65) as the value of r_s approaches the value of r_i . For the planar wall, (17-66) shows that the crystal layer grows as the square root of time. Thus, for a given period of time, the growth during the first half of the time period is $(1/\sqrt{2}) = 70.7\%$ of the total growth. For the cylindrical-tube wall, if the growth is from the wall to half of the radius, 67.9% of that growth occurs during the first half of the time period to produce

EXAMPLE 17.13

A melt of 80 wt% naphthalene and 20 wt% benzene at the saturation temperature is fed to a falling-film crystallizer, where coolant enters at 15°C. Estimate the time required for the crystal-layer thickness near the top of the 8-cm i.d. tubes to reach 2 cm.

SOLUTION

From Figure 17.24, by extrapolation the saturation temperature of the melt is 62°C. Therefore, naphthalene will crystallize, with

$$T_m - T_c = (62 - 15)1.8 = 84.6^\circ\text{F}$$

$$r_i = 8/2 = 4 \text{ cm} = 0.131 \text{ ft}$$

$$r_s = 4 - 2 = 2 \text{ cm} = 0.0655 \text{ ft}$$

75.2% of the crystal layer. The time required for this thickness of growth for the cylindrical tube is only 80% of the time for the planar wall. Thus, a conservative result is obtained by using the much simpler planar wall case of (17-66).

During operation of the falling-film crystallizer, the temperature of the melt film decreases as it flows down the tube because of heat transfer from the melt to the colder crystal layer. Based on the earlier assumptions, the melt temperature at any elevation will be the temperature corresponding to the solubility curve for the melt composition at that elevation. If we: (1) assume that the sensible heat from the cooling of the melt layer is negligible compared to the latent heat of fusion, (2) neglect any sensible-heat storage in the crystal layer and tube wall, (3) neglect vertical conduction in the tube wall, crystal layer, and melt layer, and (4) assume a constant coolant temperature, we can couple (17-66) with a material balance for the depletion of the crystallizing component as it flows down the inside tube wall. The result, which is left as an exercise, gives the crystal-layer thickness as a function of time and vertical location.

The estimate can be made with (17-67), based on the following properties for naphthalene:

$$\rho_c = 71.4 \text{ lb/ft}^3$$

$$k_c = 0.17 \text{ Btu/h-ft-}^\circ\text{F}$$

$$\Delta H_f = 63.9 \text{ Btu/lb}$$

From (17-67), solving for time, t ,

$$t = \frac{(71.4)(63.9)}{(0.17)(84.6)} \times \left[\frac{1}{4}(0.131^2 - 0.0655^2) - \frac{0.0655^2}{2} \ln\left(\frac{0.131}{0.0655}\right) \right] = 0.549 \text{ h}$$

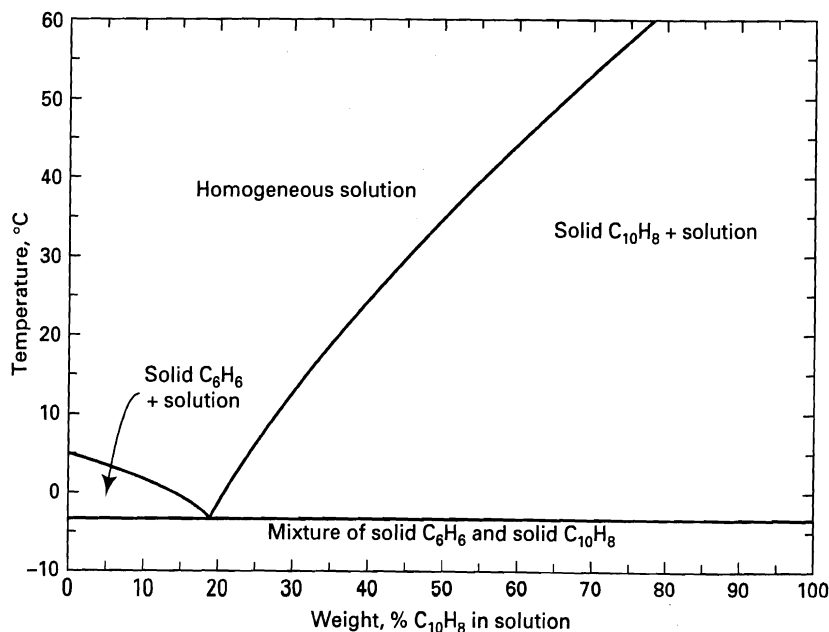


Figure 17.24 Solid-liquid phase diagram for the naphthalene-benzene system.

17.8 ZONE MELTING

When the melt consists of two components that form a solid solution, such as shown in Figure 17.1c, the liquid and solid phases at equilibrium contain amounts of both components, similar to the behavior previously discussed for vapor-liquid equilibrium. Accordingly, multiple stages of crystallization are required to obtain products of high purity. A particularly useful technique for doing this, especially when impurities are to be removed to achieve a very high-purity product, is zone melting, as developed by Pfann [25, 26] and discussed further by Zief and Wilcox [27].

The zone-melting technique, when carried out batchwise, is illustrated in Figure 17.25. Starting with an impure crystal slab, a melt zone, which is thin compared to the length of the slab, is passed slowly (typically at 1 cm/h for organic mixtures) through the slab from one end to the other by fixing the slab and using a moving heat source or, less commonly, by moving the slab through a fixed heat source. Radiofrequency (RF) induction heating is particularly convenient and causes mixing in the melt. The slab can be arranged horizontally or vertically, with the latter orientation preferred because it can take advantage of any density difference be-

tween the crystals and the melt. Zone melting can be applied to the composition region near either end of the phase-equilibrium diagram. For example, for the binary mixture of phenanthrene (P) and anthracene (A) in Figure 17.1c, zone melting could be used to remove small amounts of A from P or small amounts of P from A.

A solid-liquid equilibrium distribution coefficient, K , can be defined in a manner analogous to that for liquid-liquid equilibrium. Thus,

$$K_i = \frac{\text{concentration of impurity, } i, \text{ in the solid phase}}{\text{concentration of impurity, } i, \text{ in the melt phase}} \quad (17-68)$$

For example, if the concentration is expressed in mole fractions and the impurity is anthracene in the system of Figure 17.1c, then at 120°C, $K_i = 0.30/0.12 = 2.50$. At 200°C, with phenanthrene as the impurity $K_i = 0.11/0.28 = 0.393$. Thus, K_i can be greater than or less than one. When greater than one, the impurity raises the melting point and the impurity concentrates in the solid phase; when less than one, the impurity lowers the melting point and concentrates in the melt phase. However, when K_i approaches a value of 1, purification by zone melting becomes very difficult.

Near either end of the composition range, the equilibrium curves for the solid and liquid phases approach straight lines, and therefore, the values of K_i become constants. In these composition ranges, an equation is readily developed to predict concentration of the impurity in the solid phase upstream of the moving melt zone as a function of distance down the crystal layer in the direction of movement of the melt zone. Figure 17.26 shows the position of the melt zone during zone melting. The melt zone is shown to move a distance dz . Assume that: (1) the melt zone of width, ℓ , is perfectly mixed with impurity concentration, w_L , in weight fraction; (2) no diffusion of impurity in the solid phases occurs; (3) the initial concentration of the impurity is uniform at w_o ; and (4) the concentration of the impurity in the melt zone is in equilibrium with that in the solid phase upstream of the melt zone. A mass balance on the impurity for a dz movement of the melt

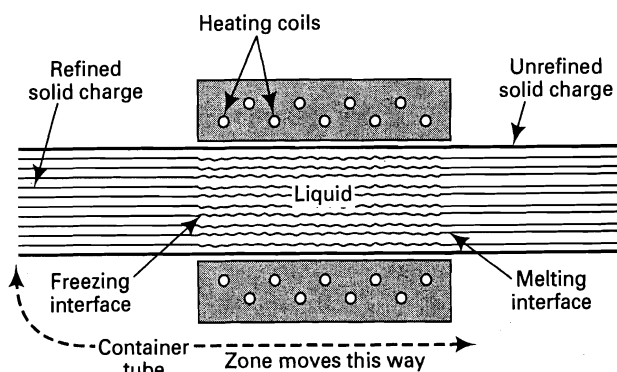


Figure 17.25 Zone melting.

[From Perry's *Chemical Engineers' Handbook*, 6th ed., R.H. Perry, D.W. Green, and J.O. Maloney, eds., McGraw-Hill, New York (1984) with permission.]

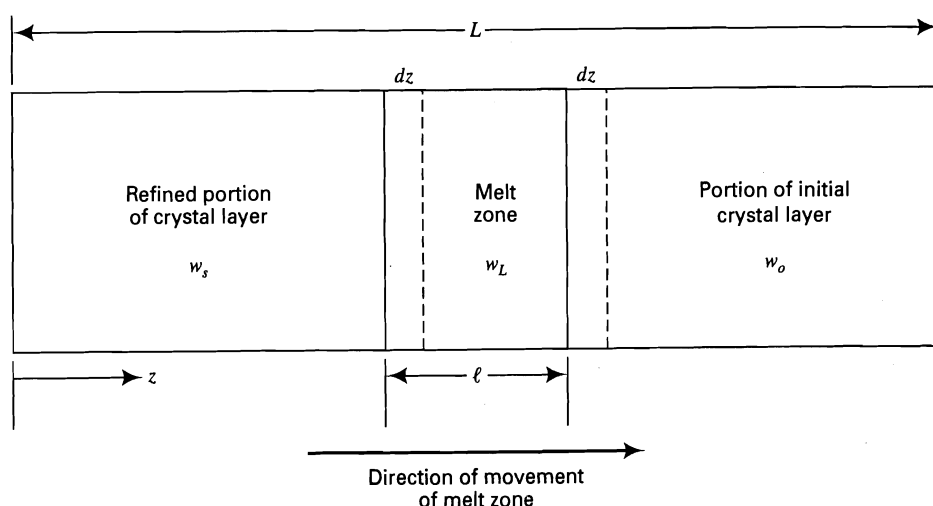


Figure 17.26 Ideal, zone-melting model.

zone is given by

$$\begin{aligned} & (\text{Mass of impurity added to melt zone}) \\ & - (\text{mass of impurity removed from melt zone}) \\ & = (\text{increase in the mass of impurity in the melt zone}) \end{aligned}$$

Thus,

$$w_o \rho_c A_c dz - w_s \rho_c A_c dz = \rho_L A_c \ell dw_L \quad (17-69)$$

From (17-68), let

$$K = w_s/w_L \quad (17-70)$$

Therefore,

$$dw_L = dw_s/K \quad (17-71)$$

Combining (17-69) to (17-71) to eliminate w_L and assuming $\rho_L = \rho_c$, we obtain

$$\frac{K}{\ell} \int_0^z dz = \int_{Kw_o}^{w_s} \frac{dw_s}{w_o - w_s} \quad (17-72)$$

Integration gives

$$\frac{w_s}{w_o} = 1 - (1 - K) \exp(-Kz/\ell) \quad (17-73)$$

for

$$z/\ell = 0 \quad \text{to} \quad z/\ell = \frac{L}{\ell} - 1$$

EXAMPLE 17.14

A crystal layer of 1 wt% phenanthrene and 99 wt% anthracene is subjected to zone melting with a melt-zone width equal to 0.1 of the length of the crystal layer ($\ell/L = 0.1$). The distribution coefficient for phenanthrene in the dilute composition region is 0.36. Determine the phenanthrene concentration profile in weight fractions at the conclusion of zone melting when the melt reaches the last 10% of the crystal layer length.

SOLUTION

From Figure 17.1c and the value of K , the phenanthrene favors distribution to the melt phase. Therefore, as the melt zone moves down the crystal layer, the phenanthrene will migrate from the upstream portion of the crystal layer into the melt zone. When the leading edge of the melt zone reaches the end of the crystal layer, all of the phenanthrene that has migrated will be in the melt layer at a uniform concentration equal to $w_s\{z/\ell = 0.9\}/K$. Assume instant freezing of this melt zone so as to maintain the uniform composition. From (17-73), with $K = 0.36$ and $w_o = 0.01$,

$$w_s = 0.01[1 - 0.64 \exp(-0.36z/\ell)] \quad (1)$$

Solving (1) for values of $z/\ell = 0$ to 9 gives

z/ℓ	w_s
0	0.0036
1	0.0055
2	0.0069
3	0.0078

z/ℓ	w_s
4	0.0085
5	0.0089
6	0.0093
7	0.0095
8	0.0096
9	0.0097

In the melt zone, $w_L = w_s/K = 0.0097/0.36 = 0.0269$. A plot of the predicted profile is shown in Figure 17.27. To further refine the anthracene, additional passes of zone melting can be made to move more of the impurity into the melt zone. However, for each pass after the first, (17-73) is not valid because at the beginning of each of the additional passes, w_o is not uniform. For example, at the beginning of the second pass, w_o becomes w_s in Figure 17.27. For the additional passes, it is necessary to numerically solve the following differential-equation form of (17-72).

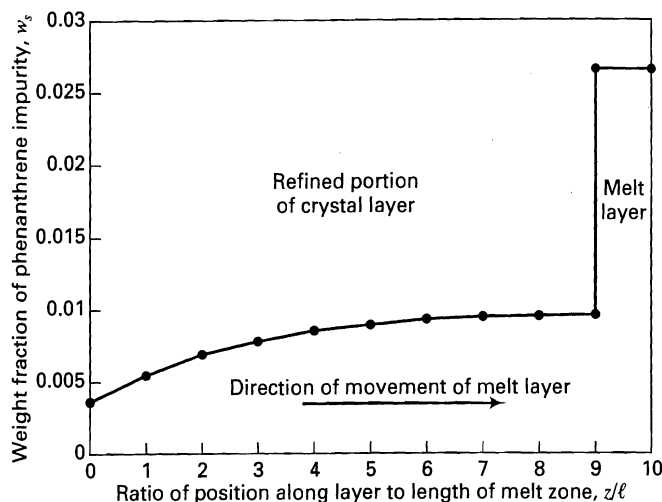


Figure 17.27 Predicted impurity profile for Example 17.14.

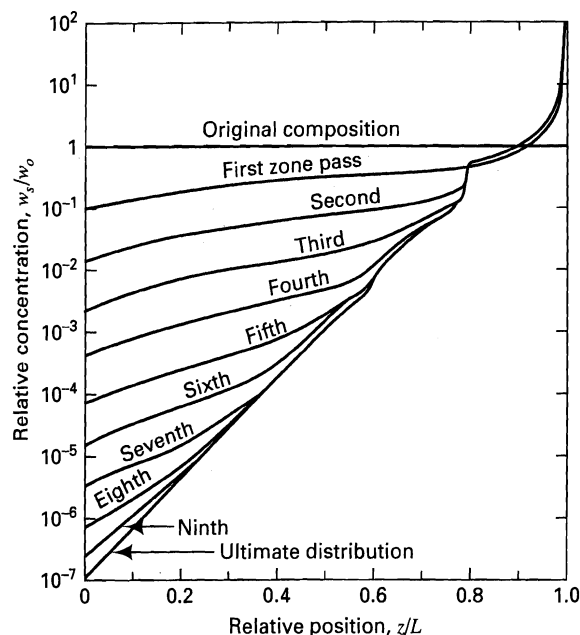


Figure 17.28 Concentration profiles for multiple zone-melting passes.

$$\frac{dw_s}{dz} = \frac{K}{\ell}(w_o - w_s) \quad (17-74)$$

where $w_o = w_o\{z = z\}$ from the results for w_s for the results from the previous pass, and $w_s\{z = 0\} = Kw_o\{z = 0\}$.

Typical impurity-concentration profiles for multiple melt-zone passes are shown in Figure 17.28 from calculations by Burris, Stockman, and Dillon [28] for the case of $K = 0.1$ and a melt-zone width of 20% of the crystal-layer length ($\ell/L = 0.2$). However, in their calculations, the melt zone was allowed to diminish to zero as the heater moved away from the crystal layer, resulting in a steep gradient from $z/L = 0.8$ to 1.0. It is seen that an ultimate distribution of the impurity is approached after 9 passes. The number of passes required to approach the ultimate distribution is given approximately by $2(L/\ell) + 1$, as observed by Herington [29]. In this example, a highly pure crystalline product can be achieved if a portion of the more pure end of the final crystal layer is taken.

17.9 DESUBLIMATION

When crystallization occurs from a vapor phase, rather than from a liquid solution or melt, the operation is referred to as *desublimation*. The reverse operation, i.e., the vaporization of a solid directly to a vapor, is *sublimation*. To understand how such a phase change can occur without going through the liquid phase, consider the phase diagram for naphthalene shown in Figure 17.29. As with most chemicals that have a vapor pressure of much less than 1 atm at the melting temperature, that temperature coincides, within a fraction of a degree Celsius, with the triple point. Below the triple-point temperature of 80.2°C, corresponding to a vapor pressure of 7.8 torr, naphthalene cannot exist as a liquid, regardless of the pressure. Below 80.2°C, naphthalene exists as a solid provided that the pressure is greater than the vapor pressure of naphthalene at the prevailing temperature. However, if the pressure falls below the vapor pressure, solid naphthalene will sublime directly to a vapor.

If naphthalene is present in an ideal gas mixture with a noncondensable inert gas, sometimes called an entrainer, at a temperature below the triple point of 80.2°C, desublimation of the naphthalene to a crystalline solid will occur if the partial pressure of naphthalene in the gas is increased to a value that exceeds its vapor pressure. For example, consider a vapor mixture of 5 mol% naphthalene in nitrogen at 70°C and a total pressure of 40 torr. The partial pressure of naphthalene is $0.05(40) = 2$ torr. From Figure 17.29, this partial pressure is less than the vapor pressure of 3.8 torr. If the total pressure is increased, desublimation will begin when the total pressure is increased to $3.8/0.05 = 76$ torr. Alternatively, if the pressure is maintained at 40 torr, but the temperature is reduced, desublimation will begin to take place at 61°C, where the vapor pressure of naphthalene is 2 torr. Unless nitrogen is trapped, the naphthalene crystals will be pure.

The most common occurrence of desublimation is the formation of snow crystals by condensation of water vapor from air at temperatures below 0°C. At temperatures above -40°C, the snow crystals form by heterogeneous nucleation on fine mineral particles of perhaps 10^{-5} – 10^{-2} mm in diameter. At temperatures below -40°C, homogeneous nucleation can take place. As the snow crystals fall through the atmosphere, they cluster together into snowflakes. The difference between the melting point of ice and the triple point of water is less than 0.01°C, with a corresponding vapor pressure of only 4.6 torr.

As discussed by Nord [30] and Kudela and Sampson [31], a number of chemicals are amenable to purification by desublimation, preceded perhaps by sublimation. In general, they are chemicals that are solids at ambient temperature and pressure and have solid vapor pressures greater than 5 $\mu\text{m Hg}$ at moderate operating temperatures. These chemicals are listed in Table 17.10. Major applications of desublimation, to obtain near-pure solid chemicals or pure-chemical solid films on substrates, include crystallization of water-insoluble

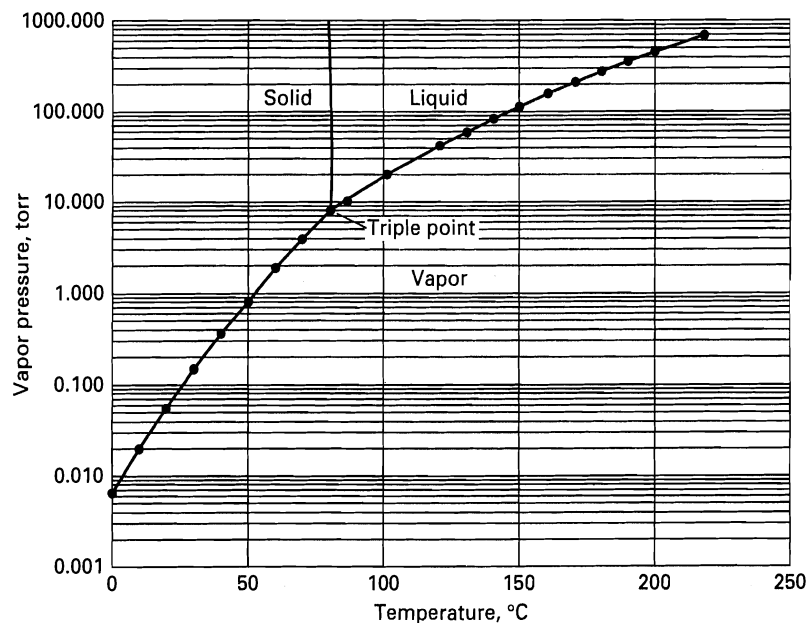


Figure 17.29 Vapor pressure of naphthalene.

Table 17.10 Chemicals Amenable to Purification by Desublimation

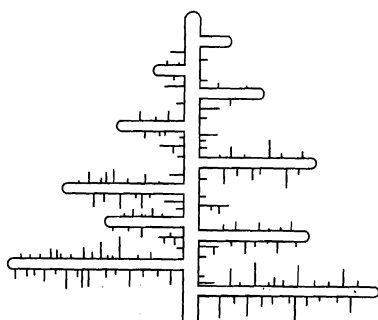
Aluminum chloride	Molybdenum Trioxide
Anthracene	Naphthalene
Anthranilic acid	β -Naphthol
Anthraquinone	Phthalic anhydride
Benzanthrone	o-Phthalimide
Benzoic acid	Pyrogallol
Calcium	Salicylic acid
Camphor	Sulfur
Chromium chloride	Terephthalic acid
Cyanuric chloride	Titanium tetrachloride
Ferric chloride	Thymol
Hafnium tetrachloride	Uranium hexafluoride
Iodine	Zirconium tetrachloride
Magnesium	

organic chemicals from mixtures with inert, nonvolatile gases and vapor-deposition of metals.

Desublimation is almost always effected by cooling the gas mixture at nearly constant pressure. As discussed by Holden and Bryant [32], cooling may be achieved by any of the following techniques:

1. Heat transfer from the gas through a solid surface, on which the crystals form.
2. Quenching by addition of a vaporizable liquid.
3. Quenching by addition of a cold, noncondensable gas.
4. Expansion of the gas mixture through a nozzle.

The first technique, which is widely used, has the disadvantage that the crystals must be removed by scraping of or by melting from the heat-transfer surface. The second technique, which is less common, must use a liquid in which the sublimate is not soluble. If excess water is used, this technique will produce a slurry of the crystals and nonvaporized liquid. The third technique, which is also common, produces dry crystals, sometimes called a snow. The fourth technique is much less common, because the degree of cooling necessary may require either a high pressure upstream of the nozzle or a vacuum downstream.

**Figure 17.30** Dendritic growth of a crystal.

The number, size, and shape of the crystals produced in the snow of the third technique depend on the relative rates of nucleation, crystal growth, and crystal agglomeration. Frequently, as commonly observed with snow crystals, dendritic growth occurs, producing undesirable crystals of the shape shown in Figure 17.30. The main stem grows rapidly, followed by slower rates of growth of primary branches, and much slower growth of secondary branches. If the dendritic crystals can be suspended in the vapor long enough, the spaces between the branches can fill in to produce a more-desirable dense shape.

Desublimation in a Heat Exchanger

This technique is used in industry to recover, from a gas mixture, a number of organic chemicals, including anthracene, maleic anhydride, naphthalene, phthalic anhydride, and salicylic acid. The crystals are deposited on the outside of the tubes while a coolant flows through the inside of the tubes. If the rate of desublimation is controlled by the rate of conduction of heat through the deposited crystal layer, then a relationship for the time to deposit a crystal layer of given thickness is derived in a manner similar to that for the falling-film melt crystallizer in Section 17.8. The result, which is similar to (17-67), except that the crystal layer grows outward from the outside radius of the cylindrical tube, r_o , instead of inward from radius, r_i in (17-67), is

$$t = \frac{\rho_c \Delta H_s}{k_c(T_g - T_c)} \left[\frac{r_s^2}{2} \ln \left(\frac{r_s}{r_o} \right) - \frac{1}{4}(r_s^2 - r_o^2) \right] \quad (17-75)$$

where r_s is the radius to the crystal layer-gas interface. Equation (17-75) ignores any sensible heat associated with cooling of the gas. If the sensible heat is not negligible, then it must be added in (17-75) to the heat of sublimation, ΔH_s . If the heat-transfer resistance in the gas phase is negligible, the temperature, T_g , in (17-75) corresponds to the temperature where the partial pressure of the desubliming solute equals its vapor pressure. The derivation of (17-75) including the sensible-heat correction is given in detail by Singh and Tawney [33].

EXAMPLE 17.15

A desublimation unit of the heat-exchanger type is to be sized for the recovery of 100 kg/h of naphthalene (N) from a gas stream, where the other components are noncondensable under the conditions of operation. The heat-exchanger tubes are 1 m long with an outside diameter of 2.5 cm. Tube spacing is such that N can build up to a maximum thickness of 1.25 cm. The gas enters the unit at 800 torr and 80°C with a mole fraction for N of 0.0095. The coolant is cooling water, which flows through the inside of the tubes, countercurrently to the flow of gas. The inlet and outlet temperatures of the cooling water are 25°C and 45°C, respectively. Determine the number of tubes needed and the time required to reach the maximum thickness, if the gas leaves at 60°C.

SOLUTION

Assume the properties

$$C_P \text{ of gas} = 0.26 \text{ cal/g-}^\circ\text{C}$$

$$MW \text{ of naphthalene} = 128.2$$

$$k_c \text{ of solid naphthalene} = 1.5 \text{ cal/h-cm-}^\circ\text{C}$$

$$\Delta H_s \text{ of naphthalene} = 115 \text{ cal/g}$$

$$\rho_c \text{ of solid naphthalene} = 1.025 \text{ g/cm}^3$$

When the maximum thickness of N is achieved, the amount of sublimate per tube, if uniform, is

$$\begin{aligned} m &= \pi(r_s^2 - r_o^2)L\rho_c \\ &= 3.14[(1.25 + 1.25)^2 - 1.25^2]100(1.025) \\ &= 1,510 \text{ g} = 1.51 \text{ kg} \end{aligned}$$

The entering gas has a partial pressure for N of $0.0095(800) = 7.6$ torr. From Figure 17.29, the saturation temperature for this partial pressure is 79.7°C . This is just slightly below the entering-gas temperature of 80°C , which is less than the triple-point temperature. Therefore, N will condense as a solid. At the exit temperature of the gas, 60°C , the vapor pressure of N is 1.8 torr. Assuming saturation at the exit and no pressure drop, the mole fraction of N in the exit gas will be $1.8/800 = 0.00225$. Thus, per mole of entering gas, 0.0073 moles of N will be condensed. If the gas is assumed to be nitrogen and N, then M_r , the ratio of the mass of gas mixture to the mass of N condensed, will be

$$M_r = \frac{28(0.9905) + 128.2(0.0095)}{(128.2)(0.0073)} = 30.94$$

The sensible heat plus the heat of fusion is

$$\begin{aligned} M_r C_{P_g}(T_{\text{in}} - T_{\text{out}})_g + \Delta H_s &= 30.94(0.26)(80 - 60) + 115 \\ &= 276 \text{ cal/g} \end{aligned}$$

Thus, for this example, with a very low mole fraction of the chemical undergoing desublimation, the sensible heat effect is large. From (17-75), the time to reach the maximum thickness of N, using the temperature driving force across the solid N of 35°C , is

$$\begin{aligned} t &= \frac{(1.025)(276)}{(1.5)(35)} \left[\frac{(2.5)^2}{2} \ln \left(\frac{2.5}{1.25} \right) - \frac{1}{4}(2.5^2 - 1.25^2) \right] \\ &= 5.37 \text{ h} \end{aligned}$$

The number of tubes required is

$$\frac{(100)(5.37)}{1.51} = 356$$

17.10 EVAPORATION

Before crystallization of a solute from an aqueous solution takes place, it is customary to bring the concentration of the solute close to the solubility curve. This is accomplished by evaporation of the solvent in an evaporator. Such a device is also used to concentrate solutions even when the solute is not subsequently crystallized, e.g., solutions of sodium hydroxide. When the vapor formed is essentially pure, there is no mass-transfer resistance in the vapor phase. When the liquid phase is agitated, mass-transfer in the liquid phase is sufficiently rapid that the rate of evaporation of solvent can be determined by the rate of heat transfer from the heating medium, usually condensing steam, to the solution.

Evaporators differ in configuration and the degree of agitation of the liquid phase. The five most widely used continuous-flow evaporators are shown schematically in Figure 17.31. Their main characteristics are as follows:

- (a) **Horizontal-tube evaporator.** This unit consists of a horizontal cylindrical vessel, equipped in the lower section with a horizontal bundle of tubes, inside of which steam condenses and outside of which the solution to be concentrated boils. The vapor produced leaves the surface of the liquid solution. Agitation is provided only by the movement of the bubbles formed. Therefore, this type of unit is only suitable for low-viscosity solutions that do not deposit scale on the heat-transfer surfaces.
- (b) **Short-vertical-tube evaporator.** This unit differs significantly from the horizontal-tube evaporator. The tube bundle is arranged vertically, with the solution

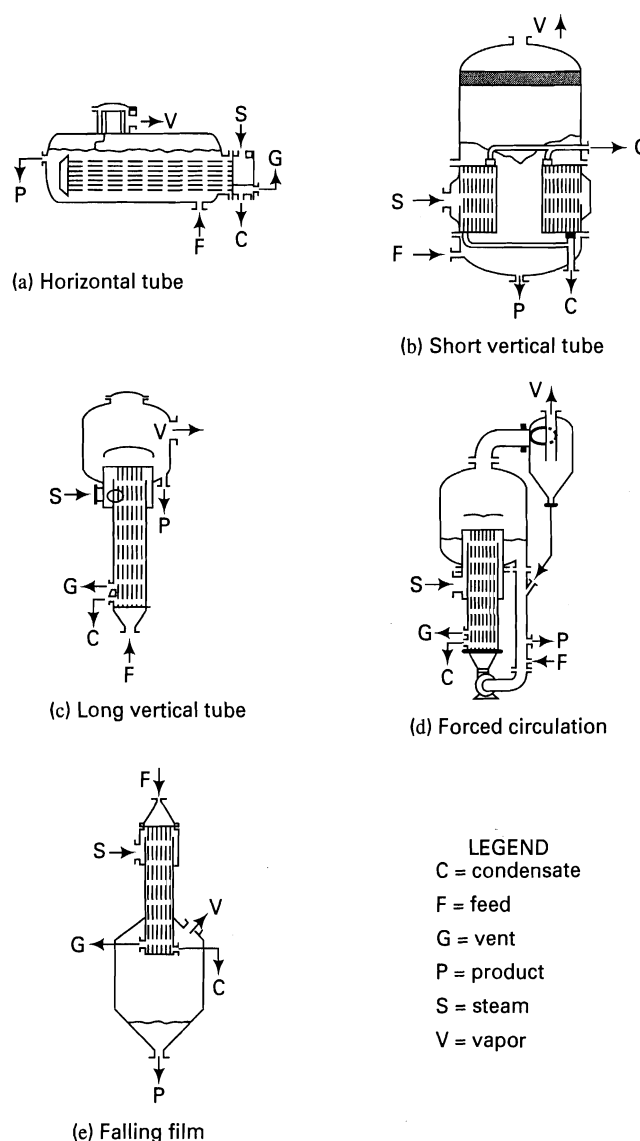


Figure 17.31 Common types of evaporators.

[From *Perry's Chemical Engineers' Handbook*, 6th edition, R.H. Perry, D.W. Green, and J.O. Maloney, eds., McGraw-Hill, New York (1984) with permission.]

inside the tubes and steam condensing outside. Boiling inside the tube causes the solution to circulate, thus providing additional agitation and, therefore, higher heat-transfer coefficients. Nevertheless, this type of evaporator is not suitable for very viscous solutions.

- (c) **Long-vertical-tube evaporator.** By lengthening the vertical tubes and providing a separate vapor-liquid disengagement chamber, as shown in Figure 17.31c, a higher tube-entering liquid velocity can be achieved and, thus, an even higher heat-transfer coefficient.
- (d) **Forced-circulation evaporator.** To handle very viscous solutions, a pump is used to force the solution upward through relatively short tubes.
- (e) **Falling-film evaporator.** This unit is widely used to concentrate heat-sensitive solutions such as fruit juices. The solution enters at the top and flows as a film down the inside walls of the tubes. The concentrate and the vapor produced are separated at the bottom.

For a given pressure in the vapor space of an evaporator, the boiling temperature of an aqueous solution will be equal to that of pure water if the solute is not dissolved in the water, but rather consists of small, insoluble, colloidal material. If the solute is soluble, the boiling temperature will be greater than that of pure water by an amount known as the

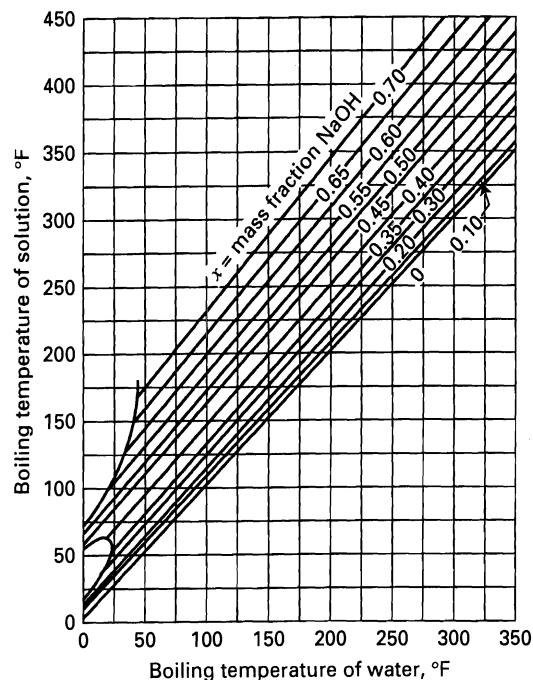


Figure 17.32 Dühring chart for aqueous solutions of sodium hydroxide.

[From W.L. McCabe, J.C. Smith, and P. Harriott, *Unit Operations of Chemical Engineering*, 5th ed., McGraw-Hill, New York (1993) with permission.]

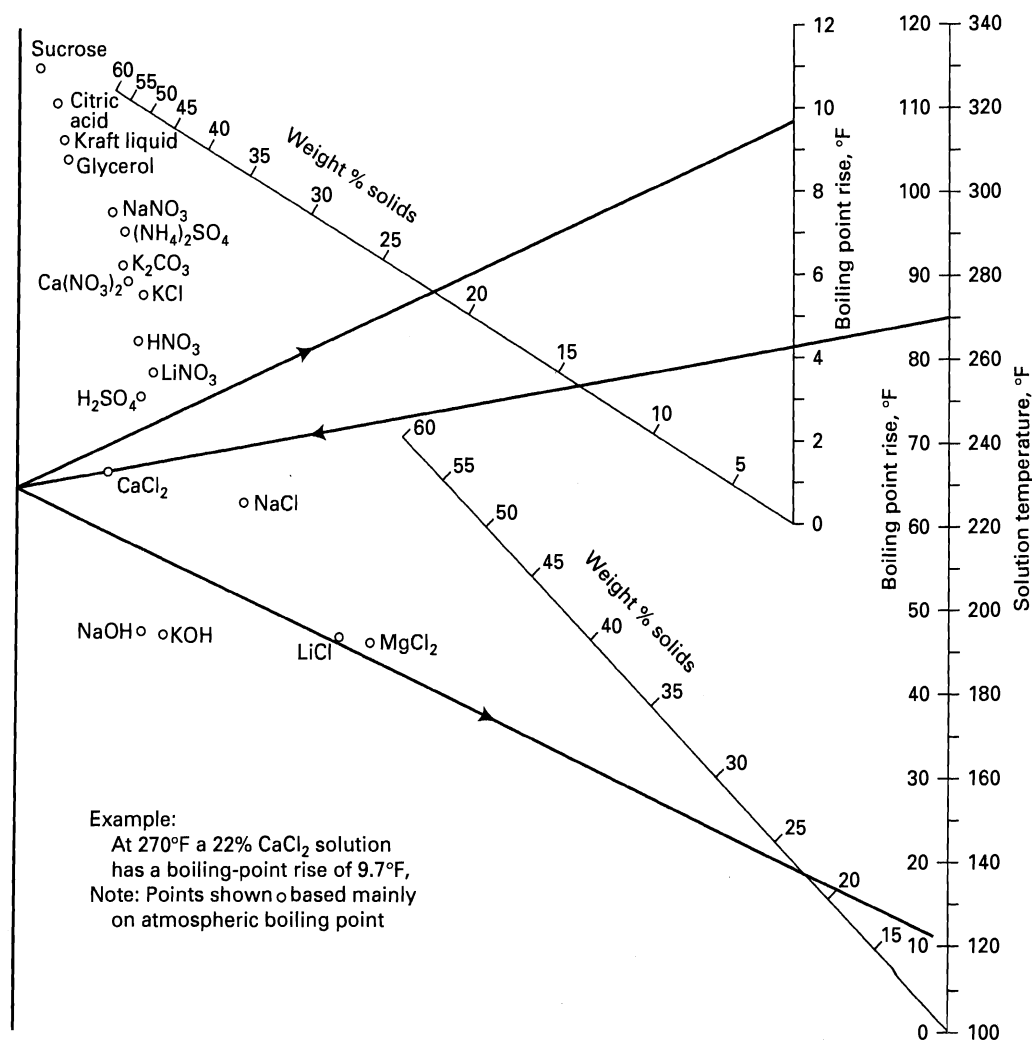


Figure 17.33 Nomograph for boiling-point elevation of aqueous solutions.

[From Perry's *Chemical Engineers' Handbook*, 6th ed., R.H. Perry, D.W. Green, and J.O. Maloney, eds., McGraw-Hill, New York (1984) with permission.]

boiling-point elevation of the solution. If, as is usually the case, the solute has little or no vapor pressure, the evaporator pressure is equal to the partial pressure of the water in the solution. Then, by a modified Raoult's law,

$$P = p_{\text{H}_2\text{O}} = \gamma_{\text{H}_2\text{O}} x_{\text{H}_2\text{O}} P_{\text{H}_2\text{O}}^s \quad (17-76)$$

Thus, for a given P and $x_{\text{H}_2\text{O}}$, a temperature can be determined to give the required $P_{\text{H}_2\text{O}}^s$ provided that $\gamma_{\text{H}_2\text{O}}$ can be estimated. For solutions dilute in the solute, $\gamma_{\text{H}_2\text{O}}$ approaches a value of 1.0. For concentrated solutions, the value of $\gamma_{\text{H}_2\text{O}}$ can be estimated from correlations for electrolyte solutions as discussed by Reid, Prausnitz, and Poling [34].

Alternatively, the boiling temperature of the solution can be estimated by using a *Dühring-line chart* if it is available for the particular solute. Such a chart is shown in Figure 17.32 for the sodium hydroxide-water solution. The straight lines on this chart for different mass fractions of NaOH obey *Dühring's rule*, which states that as the pressure is increased, the boiling temperature of the solution increases linearly with the boiling temperature of the pure solvent. A nomograph for other solutes in water is given in Figure 17.33. The use of Figures 17.32 and 17.33, and (17-76) is illustrated in the following example.

EXAMPLE 17.16

An aqueous, NaOH solution is being evaporated at 6 psia. If the solution is 35-wt% NaOH, determine:

- The boiling temperature of the solution.
- The boiling-point elevation.
- The activity coefficient for water from (17-76).

SOLUTION

From steam tables, pure water has a vapor pressure of 6 psia at 170°F.

- From the Dühring-line chart of Figure 17.32, the boiling temperature of the solution is 207°F.
- The boiling-point elevation is:

$$207 - 170 = 37^\circ\text{F}$$

Alternatively, the nomograph of Figure 17.33 may be used by drawing a straight line through the point for NaOH and a solution (boiling-point) temperature of 207°F. That line is extended to the left to the intersection with the leftmost vertical line. A straight line is then drawn from that intersection point through the lower, inclined line labeled weight percent solids at 35 (wt% NaOH). This second line is extended so as to intersect the right, vertical line. The value of the boiling-point elevation at this point of interaction is read as 36°F, which is close to the value of 37°F from the Dühring chart.

- The mole fraction of water in the solution is, for complete ionization of NaOH:

$$x_{\text{H}_2\text{O}} = \frac{0.65/18}{0.65/18 + 2(0.35)/40} = 0.674$$

Vapor pressure of pure water at 207°F = 13.3 psia. From (17-76),

$$\gamma_{\text{H}_2\text{O}} = \frac{6}{0.674(13.3)} = 0.67$$

Evaporator Model

The following mathematical model is widely used to make material balance, energy balance, and heat-transfer rate calculations to size evaporators operating under continuous-flow, steady-state conditions. The model is based on the schematic diagram in Figure 17.34. A so-called thin liquor at temperature, T_f , with weight-fraction solute, w_f , is fed to the evaporator at mass flow rate, m_f . A heating medium, e.g., saturated steam, is fed to the heat-exchanger tubes at temperature and pressure, T_s and P_s , and mass flow rate, m_s . Saturated condensate leaves the heat exchanger at the same temperature and pressure. Heat-transfer rate, Q , to the solution in the evaporator causes the solution in the evaporator at temperature, T_e , to partially evaporate to produce a vapor at temperature, T_v , with flow rate, m_v . The so-called thick-liquor concentrate leaves at temperature, T_p , with weight-fraction solute, w_p , at mass flow rate, m_p . The heat exchanger has a heat-transfer area, A , and overall heat-transfer coefficient, U , based on that area.

Key assumptions in formulating the mathematical model are:

- The thin-liquor feed has only one volatile component, e.g., water.
- Only the latent heat of the heating steam at T_s is available for heating and vaporizing the solution in the evaporator.

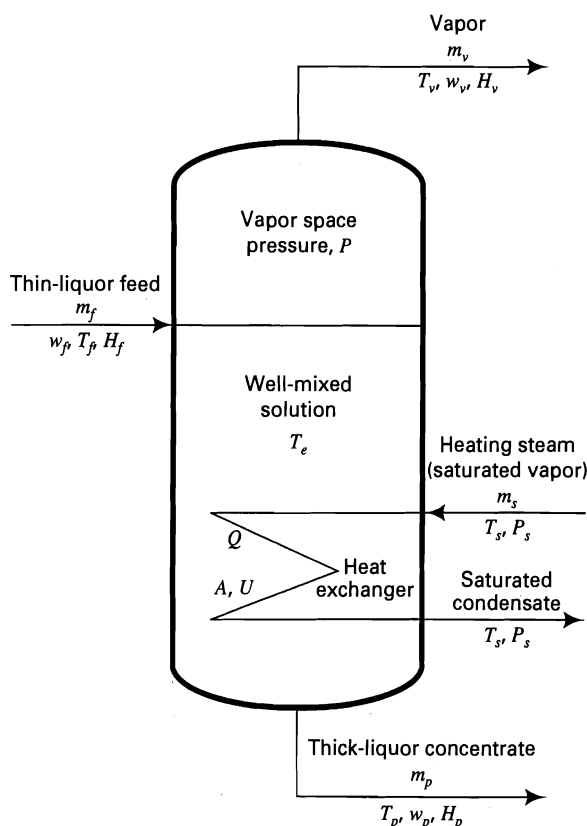


Figure 17.34 Continuous-flow, steady-state model for an evaporator.

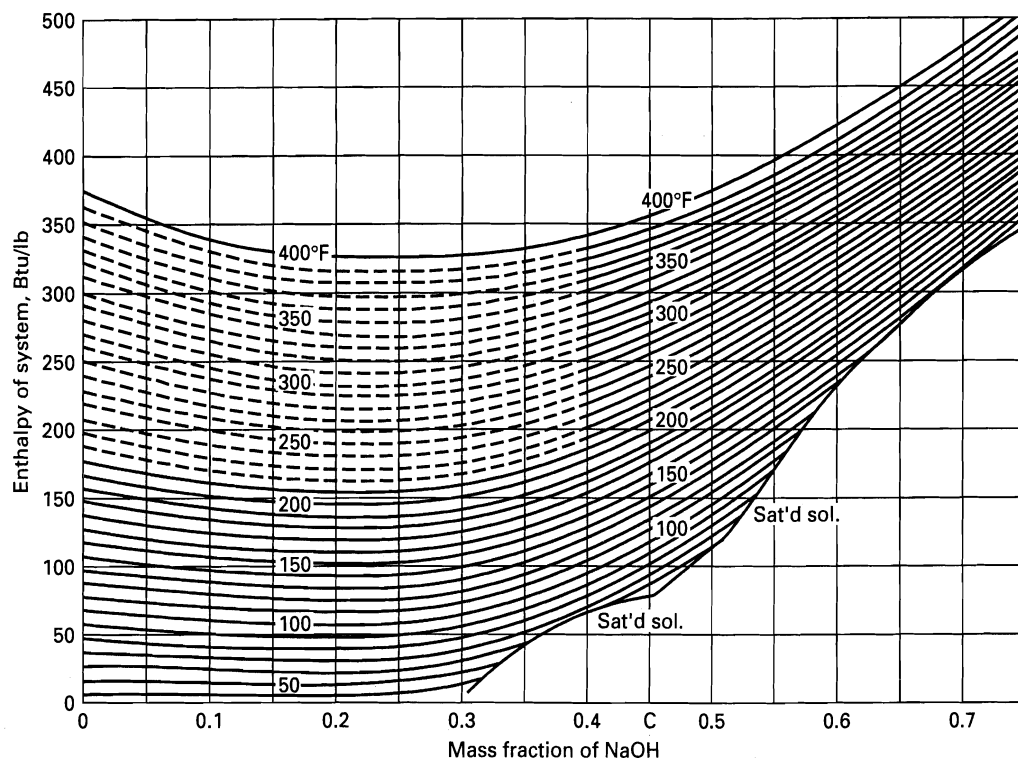


Figure 17.35 Enthalpy-concentration diagram for sodium hydroxide-water system.

[From W.L. McCabe, J.C. Smith, and P. Harriott, *Unit Operations of Chemical Engineering*, 5th ed., McGraw-Hill, New York (1993) with permission.]

3. The boiling action on the heat-exchanger surfaces agitates the solution, in the evaporator, sufficiently to achieve perfect mixing. Therefore, the temperature of the solution in the evaporator equals the exiting temperature of the thick-liquor concentrate. Thus, $T_e = T_p$. Also, $T_v = T_p$.
4. Because of Assumptions 2 and 3, the overall temperature driving force for heat transfer $= \Delta T = T_s - T_p$.
5. The ΔT is high enough to achieve nucleate boiling and not so high as to cause film boiling.
6. The exiting vapor temperature, $T_v = T_p = T_e$, corresponds to evaporator vapor-space pressure, P , taking into account the boiling-point elevation of the solution unless the solute is small, insoluble particles, such as colloidal matter.
7. No heat loss from the evaporator.

Based on the above assumptions, the mathematical model is as follows:

Total mass balance:

$$m_f = m_p + m_v \quad (17-77)$$

Mass balance on the solute:

$$w_f m_f = w_p m_p \quad (17-78)$$

Energy (enthalpy) balance on the solution:

$$Q = m_v H_v + m_p H_p - m_f H_f \quad (17-79)$$

where H = enthalpy per unit mass

Energy (enthalpy) balance on the heating steam:

$$Q = m_s \Delta H^{\text{vap}} \quad (17-80)$$

Heat-transfer rate:

$$Q = U A (T_s - T_p) \quad (17-81)$$

The procedure used to solve these five equations depends upon the specifications of the problem. The following example is just one illustration. The solution of the energy-balance equations is greatly facilitated if an enthalpy-concentration diagram is available for the solute-solvent system. A diagram of this type for the NaOH-water system is given in Figure 17.35. In this diagram, the enthalpy datum for water is the pure liquid at 32°F, so that the diagram has the same datum as the steam tables. For NaOH, the datum is NaOH at infinite dilution in water at 20°C (68°F). Figure 17.35, together with the Dühring chart of Figure 17.32, were first prepared by McCabe [35].

EXAMPLE 17.17

An existing forced-circulation evaporator is to be used to concentrate 44 wt% NaOH to 65 wt% NaOH using steam at 3 atm gage (barometer reads 1 atm). The feed temperature will be 40°C and the pressure in the vapor space of the evaporator will be 2.0 psia. The evaporator has a heat-transfer area of 232 m². The overall heat-transfer coefficient for the given conditions is estimated to be 2,000 W/m²·°C. The density of the feed solution is 1,450 kg/m³. Neglect heat losses from the evaporator. Determine:

- (a) The temperature of the solution in the evaporator in °C.
- (b) The heating steam requirement in kg/h.
- (c) The m³/h of feed solution that can be sent to the evaporator.
- (d) The kg/h of concentrated NaOH solution leaving the evaporator.
- (e) The rate of evaporation in kg/h.

SOLUTION

(a) At 2.0 psia, the boiling temperature of pure water is 126°F. From Figure 17.32, for 65 wt% NaOH, the solution boiling point is 240°F or 116°C. This is a considerable boiling-point elevation of 114°F.

(b) In American engineering units,

$$A = 232/0.0929 = 2,497 \text{ ft}^2$$

$$U = 2,000/5.674 = 352.5 \text{ Btu/h-ft}^2\text{-}^\circ\text{F}$$

$$T_s = 291^\circ\text{F for 4 atm saturated steam}$$

The driving force for heat transfer is

$$\Delta T = T_s - T_p = 291 - 240 = 51^\circ\text{F}$$

From (17-81),

$$Q = UA \Delta T = 352.5(2,497)(51) = 44,900,000 \text{ Btu/h}$$

The heat of vaporization of steam at 291°F is 917 Btu/lb.

From (17-80),

$$m_s = Q/\Delta H^{\text{vap}} = 44,900,000/917 = 48,950 \text{ lb/h}$$

or

$$Q = 48,950(0.4536) = 22,200 \text{ kg/h}$$

It should be noted that the heat flux is $44,900,000/2,497 = 18,000 \text{ Btu/h-ft}^2$. This heat flux is safely in the nucleate-boiling region.

(c) From (17-79), using Figure 17.35 for NaOH solutions and the steam tables for water vapor, the energy balance on the solution is as follows, where the evaporated water is superheated steam at 240°F and 2.0 psia., and $m_v = m_f - m_p$:

$$44,900,000 = (m_f - m_p)(1168) + m_p(340) - m_f(115) \quad (1)$$

From (17-78),

$$0.44m_f = 0.65m_p \quad (2)$$

Substituting (2) into (1) to eliminate m_p , we obtain

$$m_f = \frac{44,900,000}{\left(1 - \frac{0.44}{0.65}\right)(1,168) + \left[\frac{0.44}{0.65}(340) - 115\right]} = 91,170 \text{ lb/h}$$

or

$$m_f = (91,170)(0.4536) = 41,350 \text{ kg/h}$$

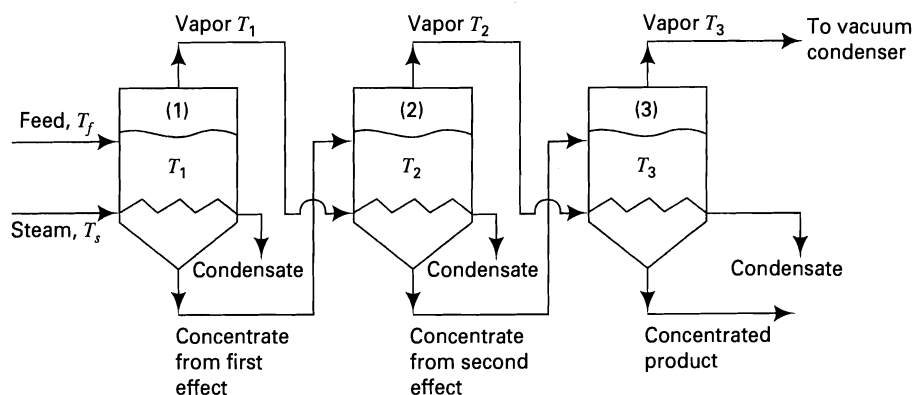
(d) From (2),

$$m_p = \frac{0.44}{0.65}(41,350) = 27,990 \text{ kg/h}$$

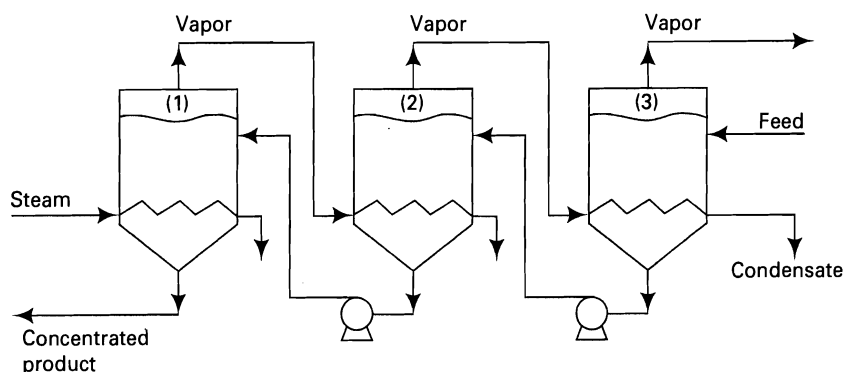
(e) $m_v = m_f - m_p = 41,350 - 27,990 = 13,360 \text{ kg/h}$

Multiple-Effect Evaporator Systems

When condensing steam is used to evaporate water from an aqueous solution, the heat of condensation of the higher-temperature condensing steam is less than the heat of vaporization of the lower-temperature boiling water. Consequently, less than 1 kilogram of vapor is produced per kilogram condensation of heating steam. This ratio is called the *economy*. In Example 17.17, the economy is $13,360/22,200 = 0.602$ or 60.2%. To reduce the amount of steam required and, thereby, increase the economy, a series of evaporators, called *effects*, can be used as shown in Figure 17.36. The increased economy is achieved by operating the effects



(a) Forward-feed, triple-effect



(b) Backward-feed, triple-effect

Figure 17.36 Multiple-effect evaporator systems.

at different pressures, and thus at different boiling temperatures, so that vapor produced in one effect can be condensed to supply the heat in another effect.

In Figure 17.36a, referred to as a *forward-feed, triple-effect-evaporator* system, approximately one-third of the total evaporation occurs in each effect. The fresh feed solution and steam both enter the first effect, which operates at pressure P_1 . The concentrate from the first effect is sent to the second effect. The vapor produced in the first effect is also sent to the second effect where it condenses, giving up its heat of condensation to cause additional evaporation of the solution. To achieve a temperature-driving force for heat transfer in the second effect, the pressure of the second effect, P_2 , is lower than that of the first effect. This procedure is repeated in the third effect. For three effects, the flow rate of steam entering the first is only about one-third of the amount of steam that would be required if only one effect were used. However, the temperature-driving force in each of the three effects is only about one-third of that in a single effect. Therefore, the heat transfer area of each of the three evaporators in a triple-effect system is approximately the same as for the one evaporator in a single-effect unit. Therefore, the savings in annual heating-steam cost must offset the additional capital investment for equipment.

When the temperature of the fresh feed is significantly below its saturation temperature corresponding to the pressure in the first effect, *backward-feed* operation is desirable, as shown in Figure 17.36b. The cold fresh feed is sent to the third effect, which operates at the lowest pressure and, therefore, the lowest temperature. Unlike the forward-feed system, pumps are required to move the concentrate from one effect to the next because $P_1 > P_2 > P_3$. However, unlike gas compressors, liquid pumps are not high-cost items.

Calculations for multieffect evaporator systems involve the same types of mass-balance, energy-balance, and heat-transfer equations as for a single-effect system. These equations are usually solved by an iterative method, especially when boiling-point elevations occur. The particular iterative procedure depends on the problem specifications. The following example illustrates a typical procedure.

EXAMPLE 17.18

A feed of 44,090 lb/h of an aqueous solution containing 8 wt% colloids is to be concentrated to 45 wt% colloids in a triple-effect-evaporator system using forward feed. The feed enters the first effect at 125°F, and the third effect operates at 1.94 psia in the vapor space. Fresh saturated steam at 29.3 psia is used for heating in the first effect. The specific heat of the colloids can be assumed constant at 0.48 Btu/lb-°F. Overall heat-transfer coefficients are estimated to be:

Effect	U , Btu/h-ft ² -°F
1	350
2	420
3	490

If the heat-transfer areas of each of the three effects are to be equal, determine:

Evaporation temperatures, T_1 and T_2 , in the first two effects.

Heating steam flow rate, m_s .

Solution flow rates, m_1 , m_2 , and m_3 leaving the three effects.

SOLUTION

The unknowns, which number seven, are A ($= A_1 = A_2 = A_3$), T_1 , T_2 , m_s , m_1 , m_2 , and m_3 . Therefore, seven independent equations are needed. Because the solute is colloids (insolubles), boiling-point elevations do not occur.

It is convenient to add three additional unknowns and, therefore, three additional equations, making a total of 10 equations. The added unknowns are the heat-transfer rates, Q_1 , Q_2 , and Q_3 in the three effects. The 10 equations, which are similar to (17-78) to (17-81), are

Overall colloid mass balance

$$w_f m_f = w_3 m_3 \quad (1)$$

Energy balances on the solutions

$$Q_1 = (m_f - m_1)H_{v1} + m_1 H_1 - m_f H_f \quad (2)$$

$$Q_2 = (m_1 - m_2)H_{v2} + m_2 H_2 - m_1 H_1 \quad (3)$$

$$Q_3 = (m_2 - m_3)H_{v3} + m_3 H_3 - m_2 H_2 \quad (4)$$

Energy balances on steam and water vapors

$$Q_1 = m_s \Delta H_s^{\text{vap}} \quad (5)$$

$$Q_2 = (m_f - m_1) \Delta H_2^{\text{vap}} \quad (6)$$

$$Q_3 = (m_1 - m_2) \Delta H_3^{\text{vap}} \quad (7)$$

Heat-transfer rates

$$Q_1 = U_1 A_1 (T_s - T_1) = U_1 A_1 \Delta T_1 \quad (8)$$

$$Q_2 = U_2 A_2 (T_1 - T_2) = U_2 A_2 \Delta T_2 \quad (9)$$

$$Q_3 = U_3 A_3 (T_2 - T_3) = U_3 A_3 \Delta T_3 \quad (10)$$

From (1),

$$m_3 = (w_f/w_3)m_f = (0.08/0.45)(44,090) = 7,838 \text{ lb/h}$$

Also, the flow rate of colloids in the feed is $(0.08)(44,090) = 3,527 \text{ lb/h}$.

Initial estimates of solution temperature in each effect:

With no boiling-point elevation, the temperature of the solution in the third effect is the saturation temperature of water at the specified pressure of 1.94 psia or 125°F. The temperature of the heating steam entering the first effect is the saturation temperature of 249°F at 29.3 psia. Thus, if only one effect were used, the temperature driving force for heat transfer, ΔT , would be $249 - 125 = 124^\circ\text{F}$. With no boiling-point elevations, the ΔT s for the three effects must sum to the value for one effect. Thus,

$$\Delta T_1 + \Delta T_2 + \Delta T_3 = 124^\circ\text{F} \quad (11)$$

As a first approximation, assume that the ΔT s for the three effects are, using (8)–(10), inversely proportional to the given values of U_1 , U_2 , and U_3 . Thus,

$$\Delta T_1 = \frac{U_3}{U_1} \Delta T_3 \quad (12)$$

$$\Delta T_2 = \frac{U_3}{U_2} \Delta T_3 \quad (13)$$

Solving (11), (12), and (13), we obtain

$$\Delta T_1 = 48.6^\circ\text{F}$$

$$\Delta T_2 = 40.6^\circ\text{F}$$

$$\Delta T_3 = 34.8^\circ\text{F}$$

$$T_1 = T_s - \Delta T_1 = 249 - 48.6 = 200.4^\circ\text{F}$$

$$T_2 = T_1 - \Delta T_2 = 200.4 - 40.6 = 159.8^\circ\text{F}$$

$$T_3 = T_2 - \Delta T_3 = 159.8 - 34.8 = 125^\circ\text{F}$$

Initial Estimates of m_1 and m_2 :

The total evaporation rate for the three effects is $m_f - m_3 = 44,090 - 7,838 = 36,252$ lb/h. Assume, as a first approximation, equal amounts of vapor produced in each effect. Then,

$$m_f - m_1 = 36,252/3 = 12,084 \text{ lb/h}$$

$$m_1 = 44,090 - 12,084 = 32,006 \text{ lb/h}$$

$$m_2 = 32,006 - 12,084 = 19,922 \text{ lb/h}$$

$$m_3 = 19,922 - 12,084 = 7,838 \text{ lb/h}$$

Corresponding estimates of the mass fractions of colloids are

$$w_1 = 3,527/32,006 = 0.110$$

$$w_2 = 3,527/19,922 = 0.177$$

$$w_3 = 3,527/7,838 = 0.450 \text{ (given)}$$

The remaining calculations are iterative in nature to obtain corrected values of T_1 , T_2 , m_1 , and m_2 , as well as values of A , m_s , Q_1 , Q_2 , and Q_3 . These calculations are best carried out on a spreadsheet. Each iteration consists of the following steps:

Step 1

Combine (2) through (7) to eliminate Q_1 , Q_2 , and Q_3 . Using the approximations for T_1 , T_2 , T_3 , w_1 , and w_2 , the specific enthalpies for the resulting equations are calculated and the equations are solved for new approximations of m_s , m_1 , and m_2 . Corresponding approximations for w_1 and w_2 are computed.

For the first iteration, the enthalpy values are

$$\Delta H_s^{\text{vap}} = 946.2 \text{ Btu/lb}$$

$$\Delta H_2^{\text{vap}} = 977.6 \text{ Btu/lb}$$

$$\Delta H_3^{\text{vap}} = 1,002.6 \text{ Btu/lb}$$

$$H_{v1} = 1,146 \text{ Btu/lb}$$

$$H_{v2} = 1,130 \text{ Btu/lb}$$

$$H_{v3} = 1,116 \text{ Btu/lb}$$

$$H_f = 0.92(92.9) + 0.08(0.48)(125 - 32) = 89.0 \text{ Btu/lb}$$

$$H_1 = 0.89(168.4) + 0.110(0.48)(200.4 - 32) = 158.8 \text{ Btu/lb}$$

$$H_2 = 0.823(127.7) + 0.177(0.48)(159.8 - 32) = 116.0 \text{ Btu/lb}$$

$$H_3 = 0.55(92.9) + 0.45(0.48)(125 - 32) = 71.2 \text{ Btu/lb}$$

When these enthalpy values are substituted into the combined energy balances, the following equations are obtained:

$$m_s = 49,250 - 1.043 m_1 \quad (14)$$

$$44,090 = 1.994 m_1 - 1.037 m_2 \quad (15)$$

$$8,168 = 1.997 m_2 - m_1 \quad (16)$$

Solving (14), (15), and (16),

$$m_s = 15,070 \text{ lb/h}$$

$$m_1 = 32,770 \text{ lb/h}$$

$$m_2 = 20,500 \text{ lb/h}$$

Corresponding values of colloid mass fractions are

$$w_1 = 0.108$$

$$w_2 = 0.172$$

It may be noted that these values of m_1 , m_2 , w_1 , and w_2 are close to the first approximations. This is often the case.

Step 2

Using the values computed in Step 1, values of Q are determined from (5), (6), and (7); and values of A are determined from (8), (9), and (10).

$$Q_1 = 15,070(946.2) = 14,260,000 \text{ Btu/h}$$

$$Q_2 = (44,090 - 32,770)(977.6) = 11,070,000 \text{ Btu/h}$$

$$Q_3 = (32,770 - 20,500)(1,002.6) = 12,400,000 \text{ Btu/h}$$

$$A_1 = \frac{14,260,000}{(350)(48.6)} = 838 \text{ ft}^2$$

$$A_2 = \frac{11,070,000}{(420)(40.6)} = 649 \text{ ft}^2$$

$$A_3 = \frac{12,400,000}{(490)(34.8)} = 727 \text{ ft}^2$$

Step 3

Because the three areas are not equal, calculate the arithmetic-average, heat-transfer area and a new set of ΔT driving forces from (8), (9), and (10). Normalize these values so they sum to the overall ΔT (124°F in this example). From the ΔT values, compute T_1 and T_2 .

$$A_{\text{avg}} = \frac{838 + 649 + 727}{3} = 738 \text{ ft}^2$$

$$\Delta T_1 = \frac{14,260,000}{(350)(738)} = 55.2^\circ\text{F}$$

$$\Delta T_2 = \frac{11,070,000}{(420)(738)} = 35.7^\circ\text{F}$$

$$\Delta T_3 = \frac{12,400,000}{(490)(738)} = 34.3^\circ\text{F}$$

These values sum to 125.2°F . Therefore, they are normalized to

$$\Delta T_1 = 55.2(124/125.2) = 54.7^\circ\text{F}$$

$$\Delta T_2 = 35.7(124/125.2) = 35.3^\circ\text{F}$$

$$\Delta T_3 = 34.3(124/125.2) = 34.0^\circ\text{F}$$

$$T_1 = 249 - 54.7 = 194.3^\circ\text{F}$$

$$T_2 = 194.3 - 35.3 = 159.0^\circ\text{F}$$

Steps 1 through 3 are now repeated using the new values of T_1 and T_2 from Step 3 and the new values of w_1 and w_2 from Step 1. The iterations are continued until the values of the unknowns no longer change significantly and $A_1 = A_2 = A_3$. The subsequent iterations for this example are left as an exercise. Based on the results of the first iteration, the economy of the three-effect system is

$$\frac{44,090 - 7,838}{15,070} = 2.41 \text{ or } 241\%$$

Overall Heat-Transfer Coefficients in Evaporators

In an evaporator, the overall heat-transfer coefficient, U , depends mainly on the steamside, condensing coefficient, the solution-side coefficient, and a scale or fouling resistance on the solution side. The conduction resistance of the metal wall of the heat-exchanger tubes is usually negligible. Steam condensation is generally of the film, rather than dropwise, type. When boiling occurs on the surfaces of the heat-exchanger tubes, it is of the nucleate-boiling, rather than film-boiling, regime. In the absence of boiling on the tube surfaces, heat transfer is by forced convection to the solution. Local coefficients for film condensation, nucleate boiling, and forced convection of aqueous solutions are all relatively large, of the order of 1,000 Btu/h-ft²-°F (5,700 W/m²-K). Thus, the overall coefficient would be about one-half of this. However, when fouling occurs, the overall coefficient can be substantially

Table 17.11 Typical Heat-Transfer Coefficients in Evaporators

Type Evaporator	U	
	Btu/h-ft ² -°F	W/m ² -K
Horizontal-tube	200–500	1,100–2,800
Short-tube vertical	200–500	1,100–2,800
Long-tube vertical	200–700	1,100–3,900
Forced circulation	400–2,000	2,300–11,300

less. Table 17.11, taken from Geankoplis [36], lists ranges of overall heat-transfer coefficients for different types of evaporators. The higher coefficients in forced-circulation evaporators are mainly a consequence of the greatly reduced fouling due to the high liquid velocity in the tubes.

SUMMARY

1. Crystallization involves the formation of solid crystalline particles from a homogeneous fluid phase. However, if the fluid is a gas, the process is usually referred to as desublimation.
2. In crystalline solids, as opposed to amorphous solids, the molecules, atoms, and/or ions are arranged in a regular lattice pattern. When crystals grow unhindered, they form polyhedrons with flat sides and sharp corners. Although the faces of a crystal may grow at different rates, referred to as crystal habit, the Law of Constant Interfacial Angles restricts the angles between corresponding faces to be constant. Crystals can form only seven different crystal systems, which include 14 different space lattices. Because of crystal habit, a given crystal system can take on different shapes, e.g., plates, needles, and prisms, but not spheres.
3. Crystal-size distributions can be determined or formulated in terms of differential or cumulative analyses, which are convertible, one from the other. A number of different, mean-particle sizes can be derived from crystal-size distribution data.
4. The most important thermodynamic properties for crystallization calculations are melting point, solubility, heat of fusion, heat of crystallization, heat of solution, heat of transition, and supersaturation. Solubilities of inorganic salts in water can vary widely from a negligible value to a concentration of greater than 50 wt%. Many salts crystallize in hydrated forms, with the number of waters of crystallization of the stable hydrate depending upon temperature. The solubility of sparingly soluble compounds is usually expressed in terms of a solubility product. When available, phase diagrams and enthalpy-concentration diagrams are extremely useful for making material- and energy-balance calculations.
5. Crystals smaller in size than can be seen by the naked eye (<20 mm) are more soluble than the normally listed solubility. Supersaturation ratio for a given crystal size is the ratio of the actual solubility of a small-size crystal to the solubility of larger crystals that can be seen by the naked eye. The driving force for nucleation and growth of crystals is supersaturation.
6. Primary nucleation, which requires a high degree of supersaturation, occurs in systems free of crystalline surfaces, and can be homogeneous or heterogeneous. Secondary nucleation occurs

when crystalline surfaces are present. Crystal growth involves the mass transfer of the solute up to the crystal surface followed by incorporation of the solute into the crystal-lattice structure.

7. Equipment for solution crystallization can be classified according to operation mode (batch or continuous), method for achieving supersaturation (cooling or evaporation), and features for achieving desired crystal growth (e.g., agitation, baffles, circulation, and classification). Of primary importance is the effect of temperature on solubility. Three of the most widely used types of equipment for solution crystallization are: (1) batch crystallizer with external or internal circulation, (2) continuous, cooling crystallizer, and (3) continuous, vacuum, evaporating crystallizer.

8. The MSMPR crystallization model is widely used to simulate the often used continuous, vacuum, evaporating draft-tube, baffled crystallizer. Some of the assumptions of this model are perfect mixing of the magma, no classification of crystals, uniform degree of supersaturation throughout the magma, crystal growth rate independent of crystal size, no crystals in the feed, no crystal breakage, uniform temperature, equilibrium in product magma between mother liquor and crystals, constant and uniform nucleation rate due to secondary nucleation by crystal contact; uniform crystal-size distribution, and all crystals with the same shape.

9. For a specified crystallizer feed, magma density, magma residence time, and predominant crystal size, the MSMPR model can predict the required nucleation rate and crystal-growth rate, number of crystals produced per unit time, and crystal-size distribution.

10. Precipitation, leading to very small crystals, occurs with solutes that are only sparingly soluble. The precipitate is often produced by reactive crystallization from the addition of two soluble salt solutions, producing one soluble and one insoluble salt. Unlike solution crystallization, which takes place at a low degree of supersaturation, precipitation occurs at a very high supersaturation that results in very small crystals.

11. When both components of a mixture can be melted at reasonable temperatures (e.g., certain mixtures of organic compounds), melt crystallization can be used to separate the components. If the

components form a eutectic mixture, pure crystals of one of the components can be formed. If the components form a solid solution, repeated stages of melting and crystallization are required to achieve high purity.

12. A large number of crystallizer designs have been proposed for melt crystallization. The two major methods are suspension crystallization and layer crystallization. Of particular importance is the falling-film crystallizer, which can be designed and operated for high production rates when the components form eutectic mixtures. For components that form solid solutions, the zone-melting technique developed by Pfann can be employed to produce nearly pure compounds.

13. A number of chemicals are amenable to purification by desublimation, preceded perhaps by sublimation. Desublimation is almost always achieved by cooling the gas mixture at constant pressure. The cooling can be accomplished by heat transfer, quenching with a vaporizable liquid, or quenching with a cold, non-condensable gas.

14. Evaporation can be used to concentrate a solute prior to solution crystallization. Common evaporators include the horizontal-

tube unit, short-vertical-tube unit, long-vertical-tube unit, forced-circulation unit, and the falling-film unit. For a given evaporation pressure, the presence of the solution can cause a boiling-point elevation.

15. The most widely used evaporator model assumes that the liquor being evaporated is well-mixed such that the temperature and solute concentration are uniform and at exiting conditions.

16. The economy of an evaporator is defined as the mass ratio of water evaporated to heating steam required. The economy can be increased by using multiple evaporator effects that operate at different pressures such that vapor produced in one effect can be used as heating steam in a subsequent effect. The solution being evaporated can progress through the effects in forward, backward, or mixed directions.

17. Evaporators typically operate so that solutions are in the nucleate-boiling regime. Overall, heat-transfer coefficients are generally high because boiling occurs on one side and condensation on the other side.

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EXERCISES

Section 17.1

17.1 Estimate the sphericities of the following simple particle shapes:

(a) a cylindrical needle with a height, H , equal to 5 times the diameter, D

(b) a rectangular prism of sides a , $2a$, and $3a$

17.2 A certain circular plate of diameter, D , and thickness, t , has a sphericity of 0.594. What is the ratio of t to D ?

17.3 A laboratory screen analysis for a batch of crystals of hypo (sodium thiosulfate) is as follows. Prepare both differential and cumulative-undersize plots of the data, using a spreadsheet.

U.S. Screen	Mass Retained, gm
6	0.0
8	8.8
12	21.3
16	138.2
20	211.6
30	161.7
40	81.6
50	44.1
70	28.7
100	13.2
140	9.6
170	8.8
230	7.4
	735.0

In preparing your plots, determine whether arithmetic, semilog, or log-log plots are preferred.

17.4 Derive expressions for the surface-mean and mass-mean diameter from a particle-size analysis based on counting, rather than weighing, particles in given size ranges, letting N_i be the number of particles in a given size range of average diameter, $\bar{D}_{pi}$.

17.5 Using the screen analysis of Exercise 17.3, calculate, with a spreadsheet, the surface-mean, mass-mean, arithmetic-mean, and volume-mean crystal diameters, assuming that all particles have the same sphericity and volume shape factor.

17.6 A precipitation process for producing perfect spheres of silica has been developed. The individual particles are so small that most cannot be discerned by the naked eye. Using optical microscopy, the particle size distribution has been measured, with results given in the table below. Using these data on a spreadsheet program:

(a) Produce plots of the differential and cumulative particle-size analyses

(b) Determine:

- (1) surface-mean diameter
- (2) arithmetic-mean diameter
- (3) mass-mean diameter
- (4) volume-mean diameter

PSD of Silica Spheres

Particle-Size Interval, μm	Number of Particles
1.0–1.4	2
1.4–2.0	5
2.0–2.8	14

PSD of Silica Spheres

Particle-Size Interval, μm	Number of Particles
2.8–4.0	60
4.0–5.6	100
5.6–8.0	190
8.0–12.0	250
12.0–16.0	160
16.0–22.0	110
22.0–30.0	70
30.0–42.0	28
42.0–60.0	10
60.0–84.0	1
Total	1,000

17.7 A screen analysis for a sample of glauber's salt from a commercial crystallizer is as follows, where the crystals can be assumed to have a uniform sphericity and volume shape factor.

U.S. Screen	Mass Retained, gm
14	0.0
16	0.9
18	25.4
20	111.2
25	113.9
30	225.9
35	171.7
40	116.5
45	55.1
50	31.5
60	8.7
70	10.5
80	4.4
	875.7

Use a spreadsheet to determine in microns:

- (a) a plot of the differential analysis
- (b) a plot of the cumulative oversize analysis
- (c) a plot of the cumulative undersize analysis
- (d) the surface-mean diameter
- (e) the mass-mean diameter
- (f) the arithmetic-mean diameter
- (g) the volume-mean diameter

Section 17.2

17.8 1,000 grams of water is mixed with 50 grams of Ag_2CO_3 and 100 grams of AgCl . At equilibrium at 25°C , calculate the concentrations in moles/liter of Ag^+ , Cl^- , and CO_3^{2-} ions and the grams of Ag_2CO_3 and AgCl in the solid phases.

17.9 5,000 lb/h of a saturated aqueous solution of $(\text{NH}_4)_2\text{SO}_4$ at 80°C is cooled to 30°C . At equilibrium, what is the amount of crystals formed in lb/h. If during the cooling process, 50% of the water is evaporated, what is the amount of crystals formed in lb/h?

17.10 7,500 lb/h of a 50 wt% aqueous solution of FeCl_3 at 100°C is cooled to 20°C . At 100°C , the solubility of the FeCl_3 is 540 g/100 g of water. At 20°C , the solubility is 91.8 g/100 g water and crystals of FeCl_3 are the hexahydrate. At equilibrium at 20°C , determine the lb/h of crystals formed.

17.11 The concentrate from an evaporation system is 5,870 lb/h of 35 wt% MgSO_4 at 180°F and 25 psia. It is mixed with 10,500 lb/h of saturated aqueous recycle filtrate of MgSO_4 at 80°F and 25 psia. The mixture is sent to a vacuum crystallizer, operating at 85°F and 0.58 psia in the vapor space, to produce steam and a magma of 25 wt% crystals and 75 wt% saturated solution. Determine the lb/h of water evaporated and the maximum production rate of crystals in tons/day (dry basis for 2000 lb/ton).

17.12 Urea is to be crystallized from an aqueous solution that is 90% saturated at 100°C. If 90% of the urea is to be crystallized in the anhydrous form and the final solution temperature is to be 30°C, what fraction of the water must be evaporated?

17.13 In Examples 17.3 and 17.5, heat addition to the crystallizer is by an external heat exchanger through which magma is circulated, as shown in Figure 17.16. If instead the heat is added to the feed, determine the new feed temperature. Which is the preferable way to add the heat?

17.14 For the conditions of Exercise 17.11, determine the rate at which heat must be added to the system.

17.15 For the conditions of Example 17.4, calculate the amount of heat in calories/100 grams of water that must be removed to cool the solution from 100 to 10.6°C.

Section 17.3

17.16 Based on the following data, compare the effect of crystal size on solubility in water at 25°C for (1) KCl (see Example 17.7), a soluble inorganic salt, with that for (2) BaSO_4 , an almost insoluble inorganic salt, and (3) sucrose, a very soluble organic compound.

$$\sigma_{s,L} \text{ for barium sulfate} = 0.13 \text{ J/m}^2$$

$$\sigma_{s,L} \text{ for sucrose} = 0.01 \text{ J/m}^2$$

What conclusions can you draw from the results?

17.17 Determine the supersaturation ratio, S , required to permit 0.5- μm -diameter crystals of sucrose (MW = 342 and $\rho_c = 1,590 \text{ kg/m}^3$) to grow if $\sigma_{s,L} = 0.01 \text{ J/m}^2$.

17.18 The Kelvin equation, (17-16), predicts that solubility increases to infinity as the crystal diameter decreases to zero. However, measurements by L. Harbury [*J. Phys. Chem.*, **50**, 190–199 (1946)] for several inorganic salts in water show a maximum in the solubility curve and a solubility that approaches zero as crystal size is reduced to zero. Harbury's explanation is that the surface energy of the crystals depends not only on interfacial tension, but also on surface electrical charge, given by

$$2q^2 v_s / \pi \kappa R T D_p^4$$

where

q = electrical charge on the crystal

κ = dielectric constant

Modify (17-16) to take into account electrical charge. Make sure your equation predicts a maximum.

17.19 Using the following data, compare the effect of supersaturation ratio over the range of 1.005 to 1.02 on the primary homogeneous nucleation of AgNO_3 , NaNO_3 , and KNO_3 from aqueous solutions at 25°C:

	AgNO_3	NaNO_3	KNO_3
Crystal density, g/cm^3	4.35	2.26	2.11
Interfacial tension, J/m^2	0.0025	0.0015	0.0030

17.20 Estimate the effect of relative supersaturation on the primary, homogeneous nucleation of BaSO_4 from an aqueous solution at 25°C, if

$$\text{Crystal density} = 4.50 \text{ g/cm}^3$$

$$\text{Interfacial tension} = 0.12 \text{ J/m}^2$$

17.21 Repeat parts (g) and (i) of Example 17.9 if the solution velocity past the crystal face is reduced from 5 cm/s to 1 cm/s.

Section 17.4

17.22 The feed to a cooling crystallizer is 2,000 kg/h of 30 wt% Na_2SO_4 in water at 40°C. This solution is to be cooled to a temperature at which 50% of the solute will be crystallized as the decahydrate. Estimate the required heat-transfer area in m^2 if an overall heat-transfer coefficient of 15 $\text{Btu/h-ft}^2\text{-}^\circ\text{F}$ can be achieved. Assume a constant specific heat for the aqueous solution of 0.80 $\text{cal/g-}^\circ\text{C}$. Chilled cooling water will flow countercurrently to the crystallizing solution, entering the crystallizer at 10°C, and exiting at a temperature sufficient to give a log-mean driving force of at least 10°C.

17.23 Two tons per hour of the dodecahydrate of sodium phosphate ($\text{Na}_3\text{PO}_4 \cdot 12\text{H}_2\text{O}$) is to be crystallized by cooling, in a cooling crystallizer, an aqueous solution that enters saturated at 40°C and leaves at 20°C. Chilled cooling water flows countercurrently, entering at 10°C and exiting at 25°C. The expected overall heat-transfer coefficient is 20 $\text{Btu/h-ft}^2\text{-}^\circ\text{F}$. The average specific heat of the solution is 0.80 $\text{cal/g-}^\circ\text{C}$. Estimate:

(a) The tons (2,000 lb) per hour of feed solution.

(b) The heat-transfer area in ft^2 .

(c) The number of crystallizer units required if each 10-ft-long unit contains 30 ft^2 of heat-transfer surface.

Section 17.5

17.24 An aqueous feed of 10,000 kg/h, saturated with BaCl_2 at 100°C, enters a crystallizer that can be simulated with the MSMRP model. However, crystallization is achieved with negligible evaporation. The magma leaves the crystallizer at 20°C with crystals of the dihydrate. The crystallizer has a volume (vapor-space free basis) of 2.0 m^3 . From laboratory experiments, the crystal growth rate is essentially constant at $4.0 \times 10^{-7} \text{ m/s}$. Using the data below, determine:

(a) The kg/h of crystals in the magma product.

(b) The predominant crystal size in mm.

(c) The mass fraction of crystals in the size range from U.S. Standard 20 mesh to 25 mesh.

Data:

Density of the dihydrate crystals = 3.097 g/cm^3

Density of an aqueous, saturated solution of barium chloride at 20°C = 1.29 g/cm^3

17.25 The feed to a continuous crystallizer that can be simulated with the MSMRP model is 5,000 kg/h of 40 wt% sodium acetate in water. Monoclinic crystals of the trihydrate will be formed. The pressure in the crystallizer and the heat-transfer rate in the associated heat exchanger are such that 20% of the water in the feed will be evaporated at a crystallizer temperature of 40°C. The crystal growth rate, G , is 0.0002 m/h and a predominant crystal size, L_{pd} , of 20 mesh is desired.

Determine:

- The kg/h of crystals in the exiting magma.
- The kg/h of mother liquor in the exiting magma.
- The volume in m^3 of magma in the crystallizer if
density of the crystals = 1.45 g/cm^3
density of the mother liquor = 1.20 g/cm^3

Solubility data:

$T, ^\circ\text{C}$	Solubility, g sodium acetate/100 g H_2O
30	54.5
40	65.5
60	139

17.26 An MSMPR-type crystallizer is to be designed to produce 2,000 lb/h of crystals of the heptahydrate of magnesium sulfate with a predominant crystal size of 35 mesh. The magma will be 15 vol% crystals. The temperature in the crystallizer will be 50°C and the residence time will be 2 h. The densities of the crystals and mother liquor are 1.68 and 1.32 g/cm^3 , respectively. Determine:

- The exiting flow rates in cubic feet per hour of
Crystals
Mother liquor
Magma
- The crystallizer volume in gallons, if the vapor space equals the magma space.
- The approximate dimensions in feet of the crystallizer, if the body is cylindrical with a height equal to twice the diameter.
- The required crystal growth rate in feet per hour.
- The necessary nucleation rate in nuclei per hour per cubic feet of mother liquor in the crystallizer.
- The number of crystals produced per hour.
- A screen analysis table covering a U.S. mesh range of 3-1/2 to 200, giving the predicted % cumulative and % differential screen analyses of the product crystals.
- Plots of the screen analyses predicted in part (g).

Section 17.6

17.27 Refer to Example 17.12. In Run 15, Fitchett and Tarbell also made measurements of number density of crystals at 200 rpm, for which the data can be fitted well by the equation

$$\ln n = 26.3 - 0.407 L \quad (1)$$

where

n = number density of crystals

L = crystal size, μm

Using the MSMPR model, determine in the same units as for Example 17.12:

- n°
- G
- B°
- mean crystal length
- n_c
- Are your results consistent with the trends found in Example 17.12?
- Using your results and those in Example 17.12, predict the growth rate and mean-crystal length if no agitation is used.

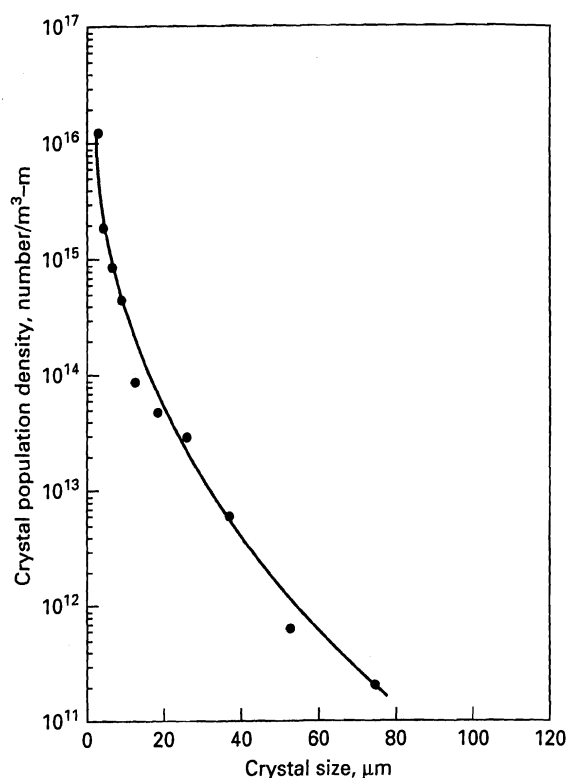


Figure 17.37 Population density of CaCO_3 for Exercise 17.28.

17.28 Tai and Chen [*AIChE J.*, **41**, 68–77 (1995)] studied the precipitation of calcium carbonate by mixing aqueous solutions of sodium carbonate and calcium chloride in an MSMPR crystallizer with pH control, such that the form of CaCO_3 was calcite rather than aragonite or vaterite. In Run S-2, which was conducted at 30°C , a pH of 8.65, and 800 rpm, with a residence time of 100 min, the crystal population density data were as shown in Figure 17.37.

Because the data do not plot as a straight line, they do not fit (17-38).

- Develop an empirical equation that will fit the data and determine, by regression, the constants.
- Can nucleation rate and growth rate be determined from the data? If so, how?

17.29 Tsuge and Matsuo ["Crystallization as a Separation Process," *ACS Symposium Series 438*, edited by Myerson and Toyokura, ACS, Washington, DC (1990), pp. 344–354] studied the precipitation of $\text{Mg}(\text{OH})_2$ by reacting aqueous solutions of MgCl_2

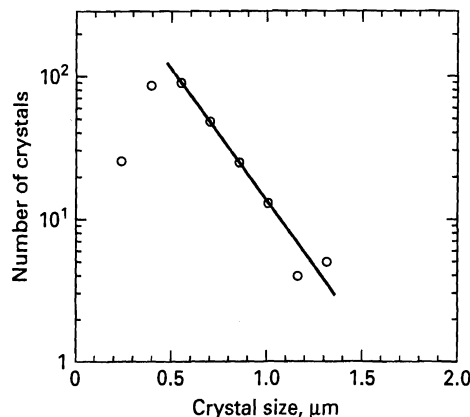


Figure 17.38 Crystal-size distribution of $\text{Mg}(\text{OH})_2$ for Exercise 17.29.

and $\text{Ca}(\text{OH})_2$ in a 1-liter MSMR crystallizer operating at 450 rpm and 25°C . Crystal sizes were measured by a scanning electron microscope (SEM) and analyzed by a digitizer. Crystal size was taken to be the maximum length. A typical plot of the crystal-size distribution is given in Figure 17.38 for an assumed residence time of 5 min. Assuming that the number of crystals is proportional to $\exp(-L/G\tau)$, as in (17-38), determine:

- Growth rate
- Nucleation rate
- Predominant crystal size

Section 17.7

17.30 The feed to the top of a falling-film crystallizer is a melt of 60 wt% naphthalene and 40 wt% benzene at saturation conditions. If the coolant enters at the top at 10°C , determine the crystal-layer thickness for up to 2 cm, as a function of time. Necessary physical-property data are given in Example 17.13.

17.31 Paradichlorobenzene melts at 53°C , while orthodichlorobenzene melts at -17.6°C . They form a eutectic of 87.5 wt% of the ortho isomer at -23°C . The normal boiling points of these two isomers differ by about 5°C . A mixture of 80 wt% of the para isomer at the saturation temperature of 43°C is fed to the top of a falling-film crystallizer, where coolant enters at 15°C . If 8-cm i.d. tubes are used, determine the time for the crystal-layer thickness at the top of the tube to reach 2 cm. Which isomer will crystallize? Necessary physical properties are given in Perry's Chemical Engineers' Handbook, except for crystal thermal conductivity, for which we assume a value of $0.15 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F}$ for either isomer.

17.32 Derive (17-67).

Section 17.8

17.33 Derive the following expression for the average impurity concentration over a particular length of crystal layer, $z_2 - z_1$, after one pass or partial pass of zone melting.

$$w_{\text{avg}} = w_o \left\{ \frac{\ell(1-K)}{K(z_2 - z_1)} [\exp(-z_2 K/\ell) - \exp(-z_1 K/\ell)] + 1 \right\} \quad (1)$$

Using the results of Example 17.14, calculate w_{avg} for $z_1 = 0$ and $z_2/\ell = 9$.

17.34 In Example 17.14, let the last 20% of the crystal layer be removed, following the first pass, to $z/\ell = 9$. Calculate from (1), in Exercise 17.33, the average impurity concentration in the remaining crystal layer.

17.35 A bar of 98 wt% Al with 2 wt% of Fe impurity is to be subjected to one pass of zone refining. The solid-liquid equilibrium distribution coefficient for the impurity is 0.29. If $z/\ell = 10$ and the resulting bar is cut off at $z_2 = 0.75z$, calculate the concentration profile for Fe and the average concentration from (1) in Exercise 17.33.

Section 17.9

17.36 A desublimation unit of the heat-exchanger type is to be sized for the recovery of 200 kg/h of benzoic acid (BA) from a gas stream containing 0.8 mol% BA and 99.2 mol% N_2 . The gas enters the unit at 780 torr at 130°C and leaves without pressure drop at 80°C . The coolant is pressurized cooling water that enters at 40°C

and leaves at 90°C , in countercurrent flow to the gas. The heat-exchanger tubes are of the type in Example 17.15.

Some properties of benzoic acid are given in Exercise 17.37. In addition,

$$k_c \text{ of solid benzoic acid} = 1.4 \text{ cal/h}\cdot\text{cm}\cdot^\circ\text{C}$$

$$\rho_c \text{ of solid benzoic acid} = 1.316 \text{ g/cm}^3$$

Determine the number of tubes needed and the time required to reach the maximum thickness of benzoic acid of 1.25 cm.

17.37 Benzoic acid is to be crystallized by bulk-phase desublimation from N_2 using a novel method described by Vitovec, Smolik, and Kugler [*Coll. Czech. Chem. Commun.*, **42**, 1108–1117 (1977)]. The gas, containing 6.4 mol% benzoic acid and the balance N_2 , flows at $3 \text{ m}^3/\text{h}$ at 1 atm and a temperature of 10°C above the dew point. The gas is directly cooled by the vaporization of $150 \text{ cm}^3/\text{h}$ of a water spray at 25°C . The gas is further cooled in two steps by nitrogen quench gas at 1 atm as follows:

Step	Quench Gas Flow Rate, m^3/h	Quench Gas Temp., $^\circ\text{C}$
1	1.5	105
2	2.0	25

The quench gases enter through porous walls of the vessel so as to prevent crystallization on the vessel wall. Based on the following data for benzoic acid, determine the final gas temperature and the fractional yield of benzoic-acid crystals, assuming equilibrium in the exiting gas.

Melting point = 122.4°C

Specific heat of solid and vapor = $0.32 \text{ cal/g}\cdot^\circ\text{C}$

Heat of sublimation = 134 cal/g

Vapor pressure:

$T, ^\circ\text{C}$	Vapor Pressure, torr
96	1
105	1.7
119.5	5
132.1	10
146.7	20
162.6	40
172.8	60

The vapor pressure data can be extrapolated to lower temperatures by the Antoine equation.

17.38 Derive (17-75).

Section 17.10

17.39 Fifty-thousand pounds per hour of a 20 wt% aqueous solution of NaOH at 120°F is to be fed to an evaporator operating at 3.7 psia, where the solution is concentrated to 40 wt% NaOH. The heating medium is saturated steam at a temperature 40°F higher than the exiting temperature of the caustic solution. Determine:

- Boiling-point elevation of the solution
- Saturated-heating-steam temperature and pressure
- Evaporation rate
- Heat-transfer rate
- Heating-steam flow rate
- Economy
- Heat-transfer area if $U = 300 \text{ Btu/h}\cdot\text{ft}^2\cdot^\circ\text{F}$

17.40 A 10 wt% aqueous solution of NaOH at 100°F and a flow rate of 30,000 lb/h is to be concentrated to 50 wt% by evaporation using saturated steam at 115 psia.

(a) If a single-effect evaporator is used with $U = 400 \text{ Btu/h-ft}^2\text{-}^\circ\text{F}$ and a vapor-space pressure of 4 in. Hg, determine the heat-transfer area and the economy.

(b) If a double-effect evaporator system is used with forward feed and $U_1 = 450 \text{ Btu/h-ft}^2\text{-}^\circ\text{F}$ and $U_2 = 350 \text{ Btu/h-ft}^2\text{-}^\circ\text{F}$, and a vapor-space pressure of 4 in. Hg in the second effect, determine the heat-transfer area of each effect, assuming equal areas, and the overall economy.

17.41 A 10 wt% aqueous solution of MgSO_4 at 14.7 psia and 70°F is sent to a double-effect evaporator system with forward feed

at a flow rate of 16,860 lb/h, to be concentrated to 30 wt% MgSO_4 . The pressure in the second effect is 2.20 psia. The heating medium is saturated steam at 230°F. Estimated heat-transfer coefficients in $\text{Btu/h-ft}^2\text{-}^\circ\text{F}$ are 400 for the first effect and 350 for the second effect. If the heat-transfer areas of the two effects are to be the same, and boiling-point elevations are neglected, determine:

(a) The pressure in the first effect

(b) The percent of the total evaporation occurring in the first effect.

(c) The heat-transfer area of each effect.

(d) The economy.

Chapter 18

Drying of Solids

Drying involves the removal of moisture (either water or other volatile compounds) from solids, solutions, slurries, and pastes to give solid products, which often, after drying, are final products ready to be packaged. In the feed to a dryer, the moisture may be a liquid, a solute in a solution, or a solid. In the first two cases, the moisture is evaporated; in the latter case, the moisture is sublimed. The term drying is also applied to a gas mixture in which a condensable vapor is removed from a noncondensable gas by cooling, as discussed in Chapter 4; and to the removal of moisture from a liquid or gas by sorption, as discussed in Chapters 6 and 16. This chapter deals only with drying operations that give solid products in various sizes and shapes.

Drying is widely used in industrial processes. Applications include the removal of moisture from: (1) crystalline particles of inorganic salts and organic compounds to cause them to be free-flowing; (2) biological materials, including foods, to prevent spoilage and decay from micro-organisms that cannot live without water; (3) pharmaceuticals; (4) detergents; (5) lumber, paper, and fiber products; (6) dyestuffs; (7) solid catalysts; (8) milk; and (9) films and coatings.

Drying can be expensive, especially when large amounts of water, with its high heat of vaporization, must be evaporated. Therefore, it is important, before drying, to remove as much moisture as possible by mechanical means such as expression; gravity, vacuum, or pressure filtration; settling; and by centrifugal means.

Because drying involves vaporization or sublimation of the moisture, heat must be transferred to the material being

dried. The most commonly employed modes of heat transfer for drying are: (1) convection from a hot gas in contact with the material, (2) conduction from a hot, solid surface in contact with the material, (3) radiation from a hot gas or hot surface in view of the material, and (4) heat generation within the material by dielectric, or microwave heating. These different modes can sometimes be used to advantage, depending on whether the moisture to be removed is on the surface of the solid and/or inside the solid.

Of importance in the drying of solids is the temperature at which the moisture evaporates. When the first mode is employed and the moisture is a continuous liquid film or is rapidly supplied to the surface from the interior of the solid, the rate of evaporation is independent of the properties of the solid and can be determined by the rate of convective heat transfer from the gas to the surface. Then, the temperature of the evaporating surface is the wet-bulb temperature of the gas provided that the dryer operates adiabatically. If the convective heat transfer is supplemented by radiation, the temperature of the evaporating surface will be higher than the wet-bulb temperature of the gas. In the absence of contact with a convective-heating gas, as in the latter three modes, and when a sweep gas is not present, such that the dryer operates nonadiabatically, the temperature of the evaporating moisture is its boiling-point temperature at the pressure in the dryer. In evaporators, if the moisture contains dissolved, nonvolatile substances, the boiling-point temperature will be elevated.

18.0 INSTRUCTIONAL OBJECTIVES

After completing this chapter, you should be able to:

- Describe two common modes of drying.
- Discuss industrial drying equipment.
- Use a psychrometric chart to determine drying temperature.
- Differentiate between the adiabatic-saturation and wet-bulb temperatures.
- Explain equilibrium-moisture content of solids.
- Explain types of moisture content used in making dryer calculations.
- Describe the four different periods in direct-heat drying.
- Calculate drying rates for different periods.
- Apply models for a few common types of dryers.

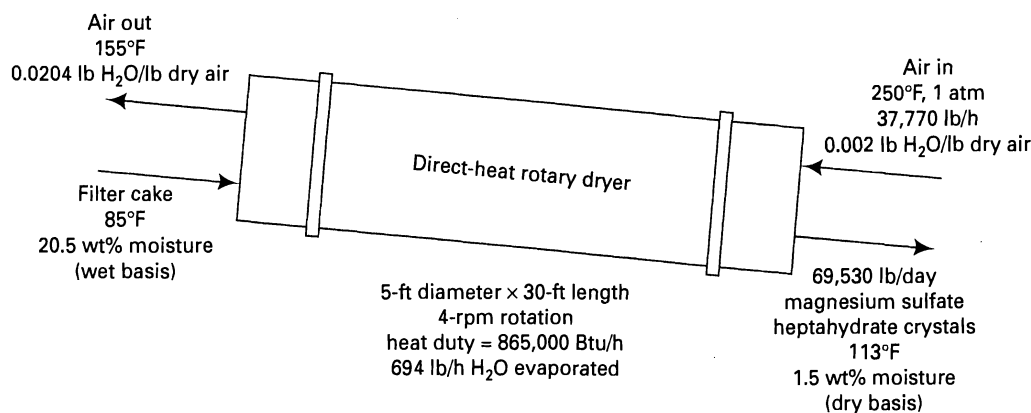


Figure 18.1 Process for drying magnesium-sulfate-heptahydrate filter cake.

Industrial Example

As an example of an industrial application of drying, consider the continuous production of 69,530 lb/day of $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ crystalline solids containing 0.015 lb H_2O /lb dry solid. The feed to the dryer, shown in Figure 18.1, consists of a filter cake from a continuous, rotary-drum, vacuum filter. The cake is at 85°F and contains 20.5 wt% moisture on the wet basis. Because the crystals are relatively coarse and free-flowing when partially dry, and nonsticking, a direct-heat rotary dryer consisting of a slightly inclined, rotating, cylindrical shell is used. The filter-cake feed enters the high end of the dryer from an inclined, vibrated chute. Heated air at 250°F and atmospheric pressure, with an absolute humidity of 0.002 lb H_2O /lb dry air enters at the other end of the dryer, at a flow rate of 37,770 lb/h. In order to obtain good contact between the wet crystals and the hot air, the dryer is provided with internal, longitudinal flights that extend the entire length of the shell. As the shell rotates, the flights lift the solids until they reach their angle of repose and then shower down through the hot air, which is in countercurrent flow to the net longitudinal direction of flow of the solids. The dry solids discharge at 113°F through a rotating valve into a screw conveyor. The air, which has been cooled to 155°F and humidified to 0.0204 lb H_2O /lb dry air, by contact with the wet solids, exits from the opposite end of the dryer, where the moist air is drawn through a fan and exhausted to the surrounding atmosphere.

The hot air causes the evaporation of 694 lb/h of water. Most of this evaporation takes place at a temperature of approximately 94.5°F, which is the average of the entering

and exiting gas wet-bulb temperatures of 95.5 and 93.5°F, respectively. In addition, the hot air must heat the solids from 85°F to 113°F and the evaporated moisture to 155°F. The total rate of convective heat transfer, Q , from the gas to the solids is 865,000 Btu/h. This value ignores heat loss from the shell to the surroundings and thermal radiation to the solids from the hot gas or the inside surfaces of the flighted shell.

Of the total heat load, approximately 83% is required to evaporate the moisture, with the balance of only 17% supplying sensible heat. Therefore, a reasonably accurate log-mean, temperature-driving force can be based on the assumption of a constant temperature at the gas–wet solids interface equal to the average air wet-bulb temperature of 94.5°F. Thus,

$$\Delta T_{LM} = \frac{(250 - 94.5) - (155 - 94.5)}{\ln \left(\frac{250 - 94.5}{155 - 94.5} \right)} = 100.6^\circ\text{F}$$

For a direct-heat, rotary dryer, it is convenient to characterize the convective heat transfer by an overall, volumetric heat-transfer coefficient, Ua , which for this example is 14.6 Btu/h-ft³-°F. The required cylindrical shell volume, V , from $Q = UaV\Delta T_{LM}$, is 590 ft³. The dryer diameter is 5 ft, which gives an entering, superficial hot-air velocity of 9.56 ft/s, which is sufficiently low to prevent entrainment of solid particles in the exiting air. The length of the cylindrical shell is 30 ft. While moving through the dryer, the bulk solids, with a bulk density of 62 lb dry solids/ft³, occupy 8 vol% of the dryer, and have a residence time of one hour. The shell rotates at 4 rpm.

18.1 DRYING EQUIPMENT

Many different forms of materials are sent to drying equipment, including granular solids, pastes, slabs, films, slurries, and liquids. No one device can handle efficiently such a wide variety of materials. Accordingly, a large number of different types of commercial dryers have been developed. These dryers can be classified in a number of ways. Perhaps

most important is the mode of operation with respect to the material being dried. Batch operation is generally indicated when the production rate is less than 500 lb/h of dried solid, while continuous operation is preferred for a production rate of more than 2,000 lb/h. In the example above, the production rate is 2,900 lb/h and a continuous drying operation was selected.

A second method of classification is the mode used to supply heat to evaporate the moisture. As mentioned above, direct-heat (also called adiabatic or convection) dryers contact material with a hot gas, which not only provides the required energy to heat the material and evaporate the moisture, but also sweeps away the moisture. When the continuous mode of operation is used, the hot gas can flow countercurrently, cocurrently, or in crossflow to the material being dried. Countercurrent flow is the most efficient configuration, but cocurrent flow may be required if the material being dried is temperature-sensitive.

Indirect-heat (also called nonadiabatic) dryers provide the heat to the material indirectly by conduction and/or radiation from a hot surface. The energy may also be generated within the material by dielectric, radio frequency, or microwave heating. Indirect-heat dryers may also be operated under vacuum to reduce the temperature at which the moisture is evaporated. A sweep gas is not necessary, but can be provided to help remove the moisture. In general, indirect-heat dryers are more expensive than direct-heat dryers. Therefore, the former type is generally used only when the material is either temperature-sensitive or subject to breakage of crystals, with dust or fines formation.

A third method for classifying dryers is the degree to which the material to be dried is agitated. In some dryers, the material is stationary. At the opposite extreme is the fluidized-bed dryer. Agitation increases the rate of heat transfer to the material but, if too severe, can cause crystal breakage and dust formation. Agitation in a continuous dryer may be necessary if the material is sticky.

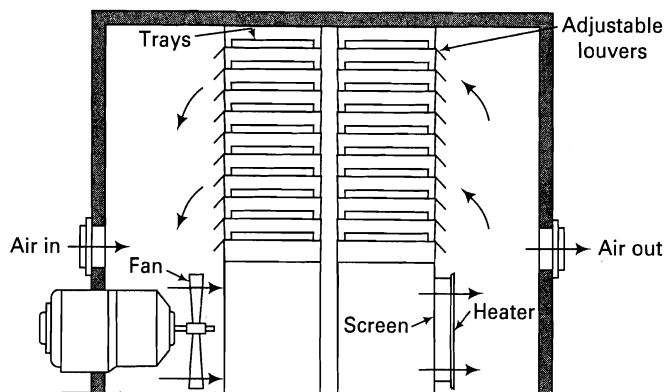
In the following, only some of the more widely used commercial dryers are described. A more complete coverage is given in the *Handbook of Industrial Drying* [1]. Extensive performance data for many of the types of dryers are given in *Perry's Chemical Engineers' Handbook* [2] and by Walas [3]. Batch dryers are discussed first, followed by continuous dryers.

Batch Operation

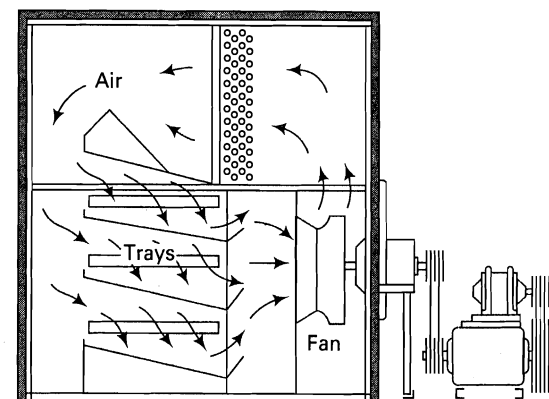
Equipment for drying batches includes: (1) tray (also called cabinet, compartment, or shelf) dryers; and (2) agitated dryers. Together, these two types cover many of the modes of heat transfer and agitation discussed above and can handle a wide variety of wet-solid feeds, such as slurries, filter cake, and particulate materials.

Tray Dryers

The oldest and simplest batch dryer is the tray dryer, which is shown schematically in Figure 18.2 and is particularly useful when low production rates of multiple products are involved and where drying times vary from hours to days. The material to be dried is loaded to a depth of typically 0.5–4 in. in removable trays that may measure $30 \times 30 \times 3$ in. and are stacked on shelves about 3 in. apart in a cabinet or on a truck that is wheeled into a chamber. If the wet solids are



(a) Cross-circulation



(b) Through-circulation

Figure 18.2 Tray dryers.

granular or shaped into briquettes, noodles, or pellets, with appreciable voids, the tray bottom can be perforated or can be a screen so that the heating gas can be passed down through the material (called through-circulation) as shown in Figure 18.2b. Otherwise, the tray bottom is solid and the hot gas is passed at velocities of typically 3–30 ft/s over the open surface of the tray (called cross-circulation), as shown in Figure 18.2a.

Although fresh hot gas might be used for each pass through or across the material, it is almost always more economical to recirculate the gas, providing venting and makeup fresh gas at rates of 5–50% of the circulation rate to maintain the humidity at an acceptable level. Typically the gas is heated with an annular, finned-tube heat exchanger by steam condensing inside the tubes. If the moisture being evaporated is water, steam requirements can range from 1.5 to 7.5 lb steam/lb water evaporated. It is important to provide baffles in tray dryers to promote uniform distribution of hot gas across or through the trays and thus, achieve uniform drying.

Tray dryers are also available for operation under vacuum and with indirect heating. In one configuration, the trays are placed on hollow shelves that carry condensing steam and, thus, act as heat exchangers. Heat is transferred by conduction to a tray from the top of the shelf supporting it and by radiation from the bottom of the shelf located directly above the tray. Typical performance data for direct-heat, crossflow-circulation tray dryers are given in Table 18.1.

Table 18.1 Performance Data for Direct-Heat, Crossflow-Circulation Tray Dryers

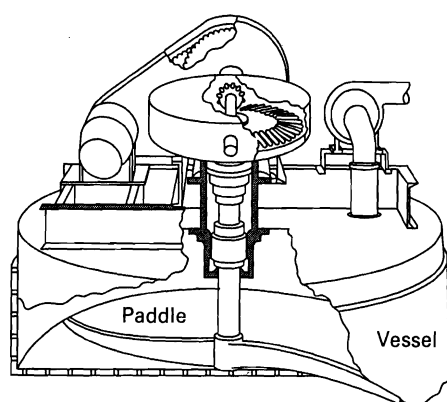
Material	Aspirin-Base Granules	Chalk	Filter Cake
Number of trays	20	72	80
Area/tray, ft ²	3.5	15.7	3.5
Total loading, lb wet	56	1,800	2,800
Depth of loading, in.	0.5	2.0	1.0
% Initial moisture	15	46	70
% Final moisture	0.5	2	1
Maximum air temp., °F	122	180	200
Drying time, h	14	4.5	45

Agitated Dryers

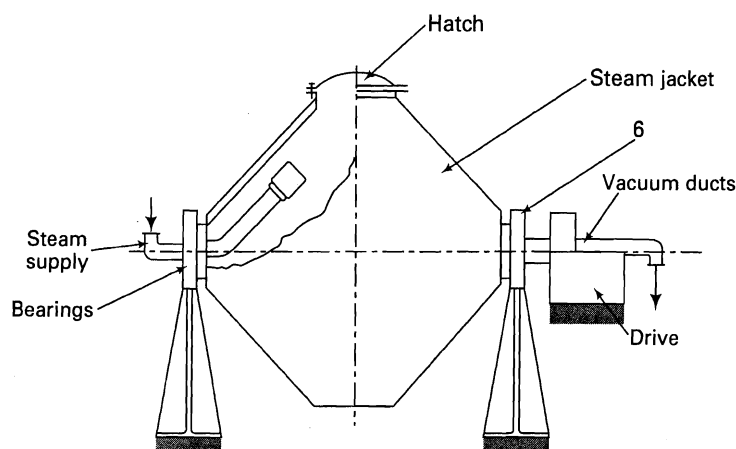
As discussed by van't Land [4] and Uhl and Root [5], indirect heat with agitation and, perhaps, under vacuum, may be desirable for batch drying when any of the following conditions exist: (1) material oxidizes, becomes explosive, or becomes

dusty during drying; (2) moisture is valuable, toxic, flammable, or explosive; (3) material tends to agglomerate or set up if not agitated; and (4) maximum product temperature is less than about 30°C. In most applications, the rate of heat transfer is controlled by contact resistance at the inner wall of the jacketed vessel and conduction into the material being dried. A wide variety of heating fluids can be used, including hot water, steam, Dowtherm, hot oil, and molten salt.

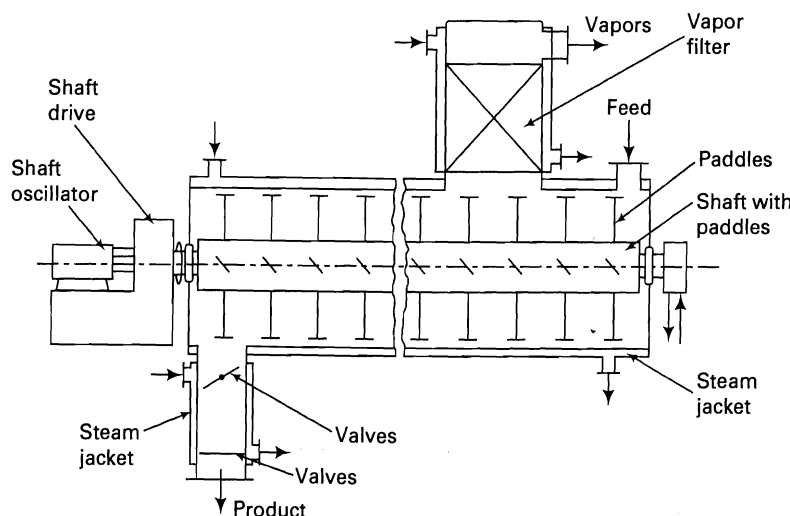
When only Condition 3 applies, the atmospheric, agitated-pan dryer, shown in Figure 18.3a, may be employed, particularly when the feed is a liquid, slurry, or paste. This dryer consists of a shallow (2–3-ft high), jacketed, flat-bottomed vessel, equipped with a paddle agitator that rotates at 2–20 rpm and scrapes the inner wall to help prevent cake buildup. Typical units range in size from 3 to 10 ft in diameter, with a capacity of up to 1,000 gallons and from 15 to 300 ft² of heat-transfer surface. When using steam as the heating medium in the jacket, overall heat-transfer coefficients may vary from 5 to 75 Btu/h-ft²-°F. The material to be dried occupies about two-thirds of the volume of the vessel. The



(a) Atmospheric pan dryer



(b) Rotating, double-cone vacuum dryer



(c) Paddle-agitated cylinder dryer

Figure 18.3 Agitated dryers.

[From Perry's *Chemical Engineers' Handbook*, 6th ed., R.H. Perry, D.W. Green, and J.O. Maloney, Eds., McGraw-Hill, New York (1984) with permission.]

degree of agitation can be varied during the drying cycle. For example, with a thin-liquid feed, agitation may vary from very low initially to very high if a sticky paste forms, followed by a moderate degree of agitation when the final product of a granular solid begins to form. Typically, several hours are required for drying. Units that can be operated under vacuum are also available.

When any or all of the above four conditions apply, but only a mild degree of agitation is required, the jacketed, rotating, double-cone (also called tumbler) vacuum dryer, shown schematically in Figure 18.3b, can be used. V-shaped tumblers are also available. The conical shape facilitates discharge of the dried product, but no means is provided to prevent cake buildup on the inner walls. Double-cone volumes range from 0.13 to 16 m³, with heat-transfer surface areas of 1 to 56 m². Additional heat-transfer surface can be provided by internal tubes or plates. Up to 70% of the volume can be occupied by the feed. A typical H₂O evaporation rate when operating at 10 torr with heating steam at 2 atm is 1 lb/h-ft² of heat-transfer surface.

A more widely used agitated dryer that is applicable when any or all of the above four conditions are relevant is the ribbon- or paddle-agitated, horizontal-cylinder dryer, shown in the paddle form in Figure 18.3c. The cylinder is jacketed and stationary. The ribbons or paddles provide agitation and scrape the inner walls to prevent solids buildup. As discussed by Uhl and Root [5], cylinder dimensions range up to diameters of 6 ft and lengths up to 40 ft. The agitator can be rotated over a wide range of rates, with values of 4–140 rpm having been reported, resulting in overall heat-transfer coefficients of 5–35 Btu/h-ft²·°F. Typically from 20 to 70% of the cylinder volume is filled with feed and drying times vary from 4 to 16 hours. In more advanced versions of this type of dryer, as discussed by McCormick [6], one or two parallel rotating shafts can be provided that intermesh with stationary, lump-breaking bars to increase the range of application. The paddles can also be hollow to provide additional heat-transfer

surface. This type of dryer can also be operated in a continuous mode.

Continuous Operation

A wide variety of industrial drying equipment for continuous operation is available. The following descriptions cover most of the widely used types, organized by the nature of the wet feed: (1) granular, crystalline, and fibrous solids, cakes, extrusions, and pastes; (2) liquids and slurries; and (3) sheets and films. In addition, infrared, microwave, and freeze drying is described.

Tunnel Dryers

The simplest, most widely applicable, and perhaps oldest continuous dryers are the tunnel dryers, which are suitable for any material that can be placed into trays and is not subject to dust formation. The trays are stacked onto wheeled trucks, which are conveyed progressively in series through a tunnel where the material in the trays is contacted by cross-circulation of hot gases. As shown in Figure 18.4, the hot gases can flow countercurrently or cocurrently to the movement of the trucks. More complex flow configurations are also possible. As a truck of dried material is removed from the discharge end of the tunnel, a truck of wet material enters at the feed end. The overall drying operation is not truly continuous because wet material must be loaded into the trays and dried material removed from the trays manually outside the tunnel. Tray spacings and dimensions, as well as hot-gas velocities, are the same as for batch tray dryers. A typical tunnel might be 100 ft long and able to house 15 trucks.

Belt or Band Dryers

A truly continuous operation can be achieved by carrying the solids as a layer on a belt conveyor, with hot gases passing over the material. The endless belt is constructed of hinged,

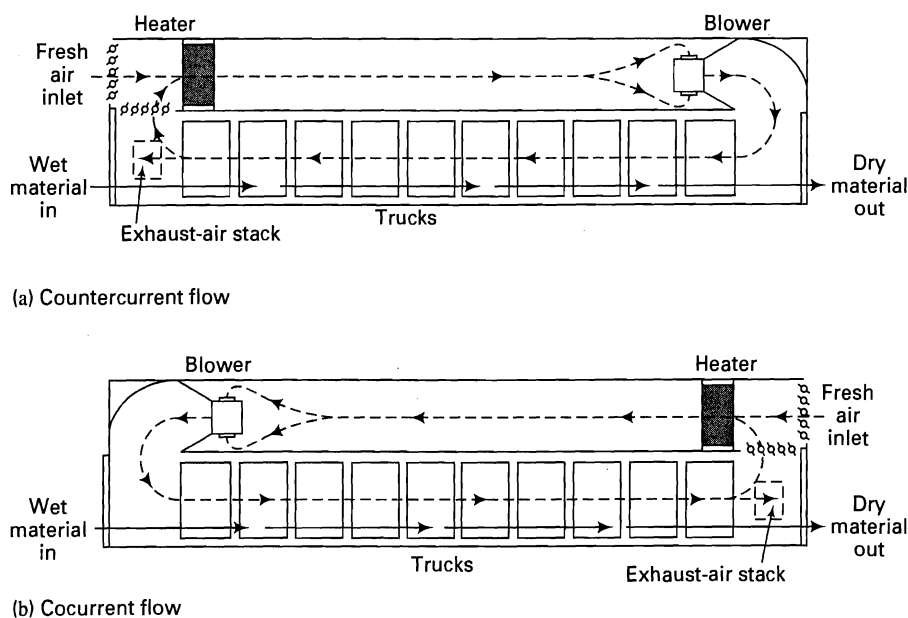


Figure 18.4 Tunnel dryer.

slotted-metal plates, or, preferably a thin metal band, which is ideal for slurries, pastes, and sticky materials. The bands are up to 1.5 m wide $\times$ 1 mm thick.

Much more common are screen or perforated-belt or band-conveyor dryers, which, as shown in Figure 18.5a, use circulation of heated gases upward and/or downward through a moving, permeable, layered, bed of wet material from 1 to 6 in. in depth. Multiple sections shown in Figure 18.5b, each with a fan and set of gas-heating coils, can be arranged in series to provide a dryer, with a single belt, as long as 150 ft with a 6-ft width, giving drying times up to 2 h with a belt speed of about 1 ft/min. To be permeable, the wet material must be of a granular-like form. If not, the material can be preformed by scoring, granulation, extrusion, pelletization, flaking, or briquetting. Particle sizes typically range from 30 mesh to 2 in. Hot-gas superficial velocities through the bed typically range from 0.5 to 1.5 m/s, with maximum bed pressure drops of 50-mm head of water. Heating gases are usually provided by heat transfer from condensing steam in finned-tube heat exchangers to temperatures in the range of 50–180°C, but temperatures up to 325°C are feasible by other means. Continuous, through-circulation conveyor dryers have been used to remove moisture from a wide variety of materials, some of which are listed in Table 18.2, which includes, in parenthesis, the method of preforming, if necessary.

In a typical application with a perforated-band-conveyor dryer, 50-ft long $\times$ 75-in. wide, 1,800 lb/h of calcium carbonate with a moisture content of 0.005 lb H₂O/lb carbonate is produced in a residence time of 40 minutes from 6-mm-diameter carbonate extrusions with a moisture content of 1.5 lb H₂O/lb carbonate, using air heated to 320°F by 160 psig steam and passing through the extrusions at a superficial

Table 18.2 Materials Dried in Through-circulation Conveyor Dryers

Aluminum hydrate (scored on filter)
Aluminum stearate (extruded)
Asbestos fiber
Breakfast food
Calcium carbonate (extruded)
Cellulose acetate (granulated)
Charcoal (briquetted)
Cornstarch
Cotton linters
Cryolite (granulated)
Dye intermediates (granulated)
Fluorspar
Gelatin (extruded)
Kaolin (granulated)
Lead arsenate (granulated)
Lithopone (extruded)
Magnesium carbonate (extruded)
Mercuric oxide (extruded)
Nickel hydroxide (extruded)
Polyacrylic nitrile (extruded)
Rayon staple and waste
Sawdust
Scoured wool
Silica gel
Soap flakes
Soda ash
Starch (scored on filter)
Sulfur (extruded)
Synthetic rubber (briquetted)
Tapioca
Titanium dioxide (extruded)
Zinc stearate (extruded)

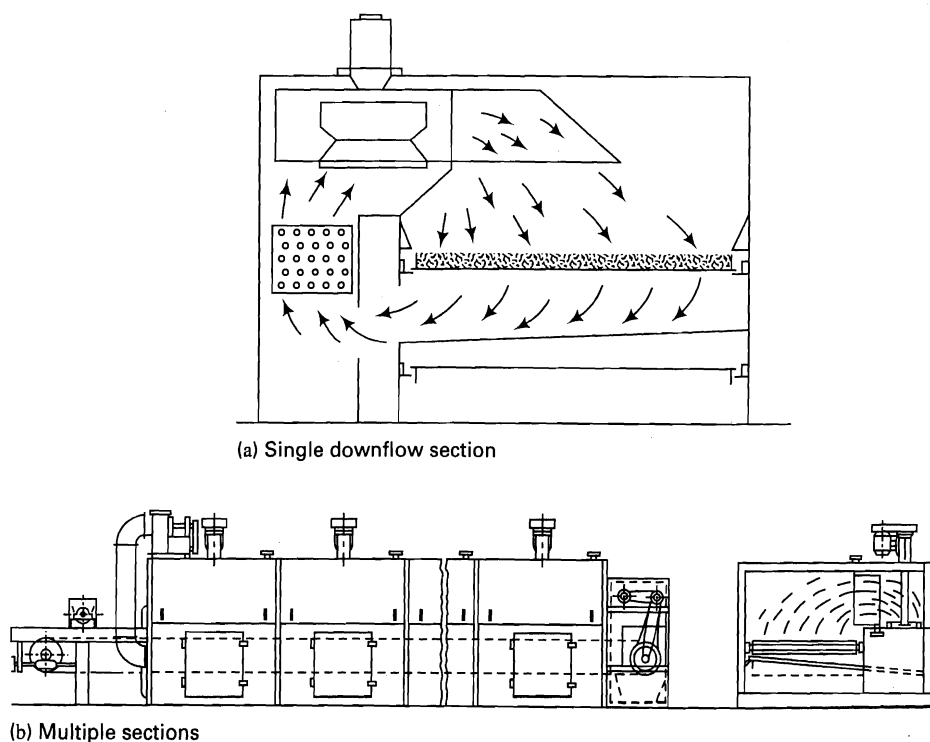


Figure 18.5 Perforated-belt or band-conveyor dryer.

velocity of 2.7 ft/s. The heating steam consumption is 1.75 lb/lb H_2O evaporated.

Turbo-Tray Tower Dryers

When floor space is limited, but head-room is available, the turbo-tray or rotating-shelf dryer, shown in Figure 18.6, may be a good choice for rapid drying of free-flowing, nondusting, granular solids. Annular shelves, mounted one above the other, are slowly rotated at up to one rpm by a central shaft. Wet feed enters through the roof onto the top shelf as it rotates under the feed opening. At the end of one revolution, a stationary wiper causes the material to fall through a radial slot onto the shelf below, where it is spread into a pile of uniform thickness by a stationary leveler. This action is repeated on each shelf until the dried material is discharged from the bottom of the unit. Also mounted on the central

shaft are turbo-type fans that provide cross-circulation of hot gases at velocities of 2 to 8 ft/s across the shelves, and heating elements located at the outer periphery of the unit. The bottom section of shelves can be used as a solids-cooling zone. Because the solids are showered through the hot gases and redistributed from shelf to shelf, drying time is reduced from that for a cross-circulation, stationary-tray dryer. Typical turbo-tray dryers range in size from 2 to 20 m in height and 2 to 11 m in diameter with as many as 58 shelves, giving shelf areas from 5.5 to 1,675 m^2 .

Overall heat-transfer coefficients, based on shelf area, of 30–120 $J/m^2 \cdot s \cdot K$ have been observed, giving, in some cases, moisture-evaporation rates comparable to through-circulation, belt- or band-conveyor dryers. Materials successfully handled in turbo-tray dryers include calcium hypochlorite, urea, calcium chloride, sodium chloride, antibiotics, antioxidants, and water-soluble polymers. The unit is particularly useful when product contamination must be avoided and the wet solids contain volatiles besides water. Capacities of up to 24,000 lb/h of dried product are quoted.

Direct-Heat Rotary Dryers

A widely used dryer for the evaporation of water from free-flowing granular, crystalline, and flaked solids of relatively small size, especially when breakage of the solids can be tolerated, is the direct-heat rotary dryer. As shown in Figure 18.7a, the dryer consists of a rotating, cylindrical shell that is slightly inclined from the horizontal with a slope of less than 8 cm/m. Wet solids enter the cylinder through a chute at the high end and dry solids discharge from the low end. Hot gases (heated air, flue gas, or superheated steam) generally flow countercurrent to the solids, but cocurrent flow can also be employed for temperature-sensitive solids. With cocurrent flow, the cylinder may not need to be inclined because the gas will help move the solids to the discharge end. To enhance the rate of heat transfer from the gas to the solids, longitudinal lifting flights, available in several different designs, two of which are shown in Figure 18.7b, are mounted on the inside of the rotating shell, causing the solids to be lifted and then showered through the hot gas during each revolution of the cylinder. Typically the bulk solids occupy 10–18% of the cylinder volume, with residence times from 5 min to 2 h. Resulting water-evaporation rates are 5–50 $kg/h \cdot m^3$ of dryer volume. The gas blower can be located so as to push or pull the gas through the dryer, with the latter location favored if the material tends to form appreciable dust. Knockers, located on the outside wall of the shell, are sometimes used to prevent solids from sticking to the inside wall of the shell. Rotary dryers are available in a wide range of sizes, typically from 1 to 20 ft in diameter and 4–150 ft long. Superficial gas velocities may be limited by dust entrainment and are usually in the range of 0.5–10 ft/s. The peripheral velocity of the shell is typically 1 ft/s. A wide variety of materials, some of which are listed in Table 18.3, are dried in direct-heat rotary dryers.

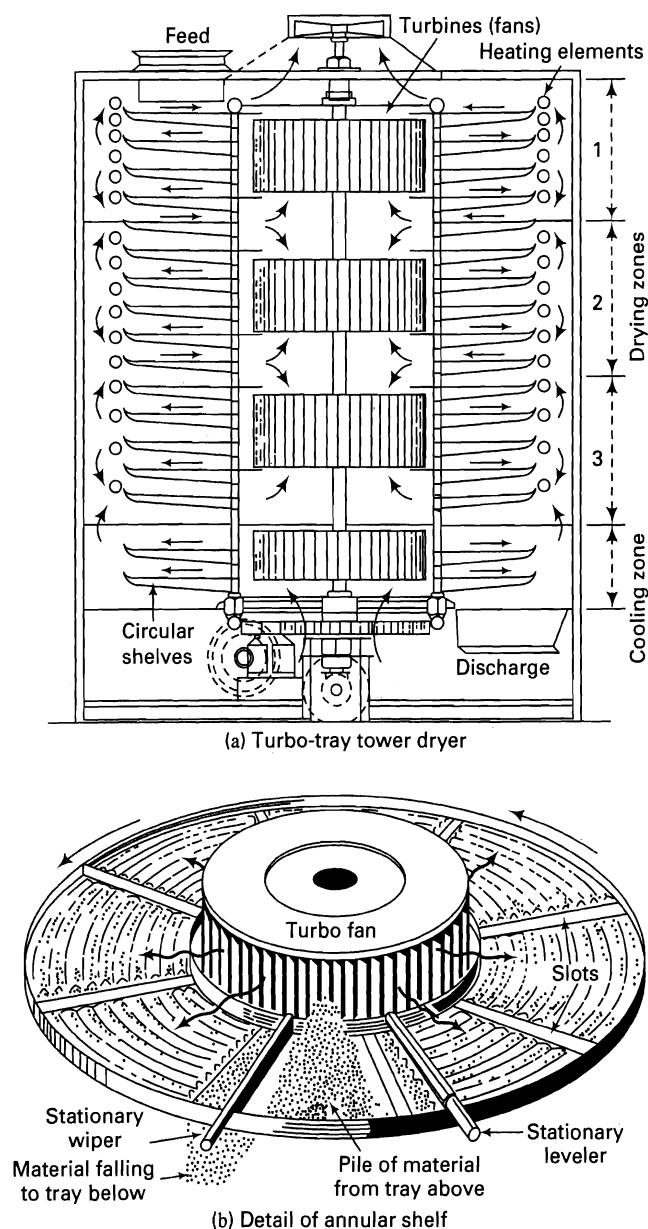
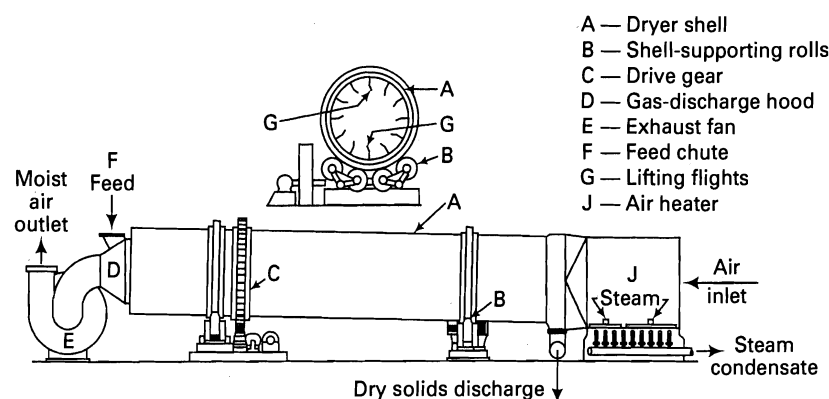
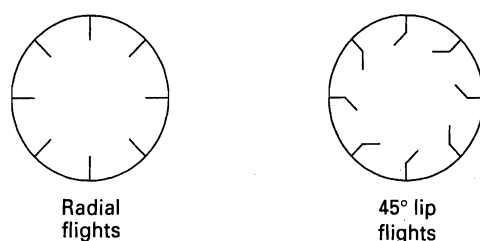


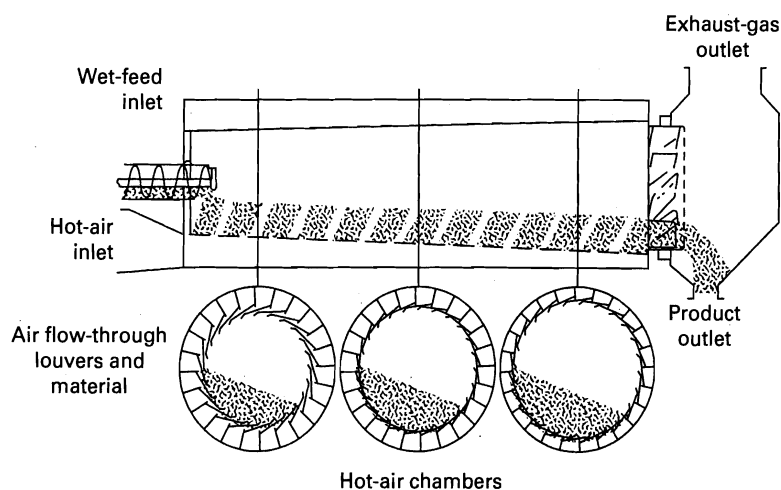
Figure 18.6 Rotating-shelf dryer.



(a) Rotary dryer



(b) Lifting flights



(c) Roto-louvre dryer

Figure 18.7 Direct-heat rotary dryer.

[From W.L. McCabe, J.C. Smith, and P. Harriott, *Unit Operations of Chemical Engineering*, 5th ed., McGraw-Hill, New York (1993) with permission.]

[From *Perry's Chemical Engineers' Handbook*, 6th edition, R.H. Perry, D.W. Green, and J.O. Maloney, Eds., McGraw-Hill, New York (1984) with permission.]

Table 18.3 Materials Dried in Direct-heat Rotary Dryers

Ammonium nitrate prills	Sand
Ammonium sulfate	Sodium chloride
Blast furnace slag	Sodium sulfate
Calcium carbonate	Stone
Cast-iron borings	Styrene
Cellulose acetate	Sugar beet pulp
Coppers	Urea crystals
Fluorspar	Urea prills
Ilmenite ore	Vinyl resins
Oxalic acid	Zinc concentrate

Roto-Louvre Dryer

A further improvement in the rate of heat transfer from hot gas to solids in a rotating cylinder is the through-circulation

action achieved in the Roto-Louvre dryer, shown in Figure 18.7c. A double wall provides an annular passage for hot gas, which passes through louvers and then through the rotating bed of solids. Because the gas pressure drop through the bed may be significant, both inlet and outlet gas blowers are often provided to maintain an internal pressure close to atmospheric. These dryers range in size from 3 to 12 ft in diameter and 9–36 ft long, with water-evaporation rates reported as high as 12,300 lb/hr. Roto-Louvre dryers are useful for processing coarse, free-flowing, dust-free granular solids.

Indirect-Heat, Steam-Tube, Rotary Dryers

When materials are: (1) free-flowing and granular, crystalline, or flaked; (2) wet with water or organic solvents;

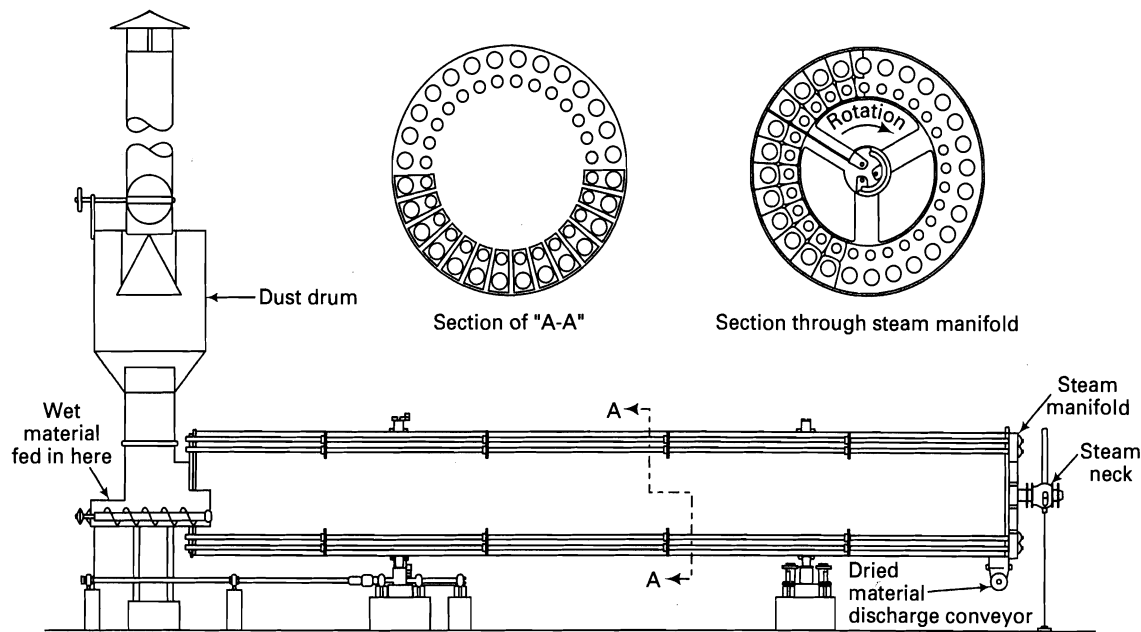


Figure 18.8 Indirect-heat, steam-tube, rotary dryer.

[From *Perry's Chemical Engineers' Handbook*, 6th ed., R.H. Perry, D.W. Green, and J.O. Maloney, Eds., McGraw-Hill, New York (1984) with permission.]

and/or (3) subject to undesirable breakage, dust formation, or contamination by air or flue gases, an indirect-heat, steam-tube, rotary dryer is often selected. A version of this dryer, shown in Figure 18.8, consists of a rotating cylinder that houses two concentric rows of longitudinal finned or unfinned tubes that carry condensing steam and rotate with the cylinder. Wet solids are fed into one end of the cylinder through a chute or by a screw conveyor. A gentle solids-lifting action is provided by the tubes. The dried product discharges from the other end after suitable contact with the hot-tube surfaces. The moisture (water or solvent) evaporates at about the boiling temperature, but can be swept out by a small purge of inert gas. Steam enters the tubes through a central revolving inlet manifold. Condensate is discharged into a collection ring. With unfinned tubes, overall heat-transfer coefficients based on the surface area of the tubes

range from 30 to 85 J/m²-s-K, when the solids occupy 10–20% of the dryer volume. Steam-tube rotary dryers range in size from 3 to 8 ft in diameter by 15–80 ft long, with one or two rows of 14–90 tubes each, of 2.5–4.5 in. in diameter. The largest-size dryers contain a single row of 90 tubes. Rotation rates range from 3 to 6 rpm. Materials successfully dried include inorganic crystals, silica, mica, flotation concentrates, pigment filter cakes, precipitated calcium carbonate, distillers' grains, brewers' grains, citrus pulp, cellulose acetate, starch, and high-moisture organic compounds.

Screw-Conveyor Dryers

Used less often than rotary dryers is the screw-conveyor dryer, as shown in Figure 18.9, consisting of a trough or cylinder that carries a hollow screw inside of which steam

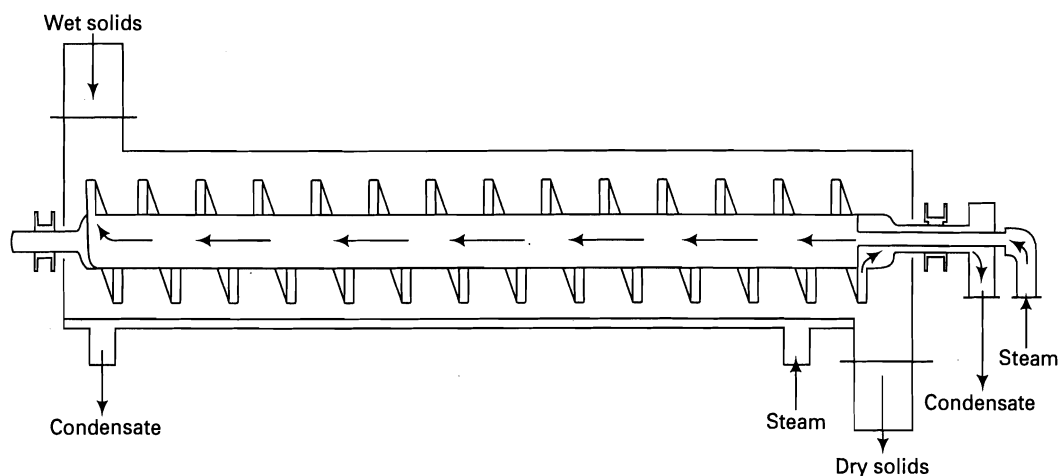


Figure 18.9 Screw-conveyor dryer.

condenses to provide heat for drying the material being conveyed. Additional heat transfer can be provided by jacketing the trough or cylindrical shell. A wide range of materials can be dried including slurries, solutions, and solvent-laden solids. The boiling moisture can be purged with a small amount of inert gas. Standard conveyor dryers are as large as 3 ft in diameter by 20 ft long. More drying time can be provided by arranging a number of units in series, with one unit above another to save floor space. The last unit can be a cooler. Overall heat-transfer coefficients are comparable to, but somewhat less than, those for indirect-heat steam-tube rotary dryers. Major applications include removal of solvents from solids and drying of fine and sticky materials.

Fluidized-Bed Dryers

Free-flowing, moist particles can be dried continuously with a residence time of a few minutes by contact with hot gases in a fluidized-bed dryer, such as that shown in Figure 18.10a. This dryer consists of a cylindrical or rectangular fluidizing chamber to which wet particles are fed from a bin through a star valve or by a screw conveyor and fluidized by hot gases that are blown through a heater and into a plenum chamber

below the bed, from where they pass into the fluidizing chamber through a perforated distributor plate. The hot gases pass up through the bed, transferring heat for evaporation of the moisture, and pass out the top of the fluidizing chamber and into a cyclone for dust removal. The solids are circulated by the action of the hot gases in the bed, but eventually pass out of the chamber through an overflow duct, which also serves to establish the height of the fluidized bed.

At low gas velocities, solids are not fluidized, but form a fixed bed through which the gas flows upward with a decrease in pressure due to friction and drag of the particles. As the gas velocity is increased, the gas pressure drop across the bed increases until the minimum fluidization velocity is reached where the pressure drop is equal to the weight of the solids per unit cross-sectional area of bed normal to gas flow. At this point, the pressure drop is sufficient to support the weight of the bed. Further increases in gas velocity cause the bed to expand with little or no increase in gas pressure drop. Typically, fluidized-bed dryers are designed for gas velocities of about twice the minimum required for fluidization. That value depends mainly on the particle size and density, and gas density and viscosity. Superficial gas velocities in fluidized-bed dryers generally are in the range of 0.5–5.0 ft/s, which

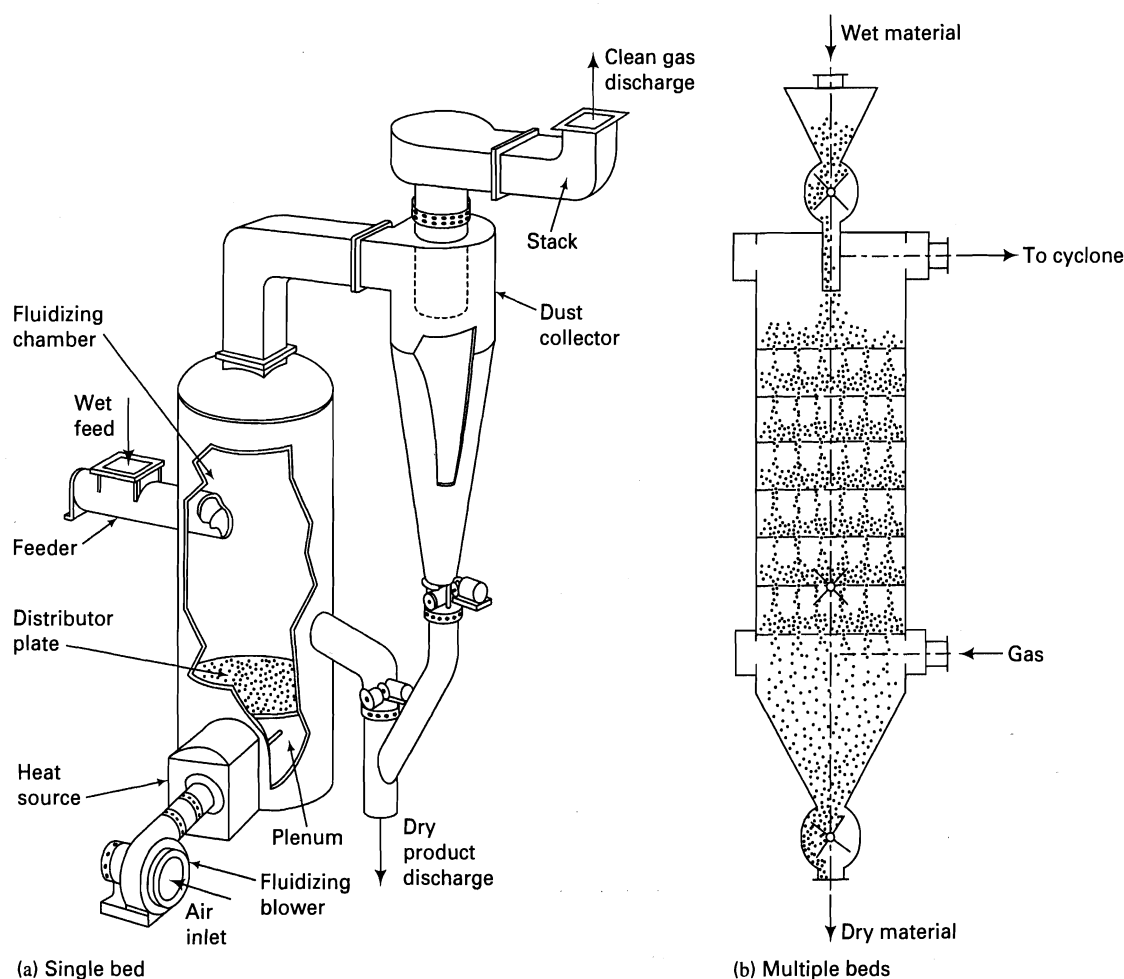


Figure 18.10 Fluidized-bed dryers.

[From W.L. McCabe, J.C. Smith, and P. Harriott, *Unit Operations of Chemical Engineering*, 5th ed., McGraw-Hill, New York (1993) with permission.]

provides stable, bubbling fluidization. Higher velocities can lead to undesirable slugging of large gas bubbles through the bed.

The capital and operating cost of a blower to provide sufficient gas pressure for the pressure drops across the distributor plate and the bed can be substantial. Therefore, the required solids residence time for drying is achieved by a shallow bed height and a large chamber cross-sectional area. Typical fluidized-bed heights range from 0.5 to 2.0 ft, with chamber diameters from 3 to 10 ft. However, chamber heights are much greater than fluidized-bed heights, because it is desirable to provide at least 6 ft of free-board height above the top surface of the fluidized bed so that the larger dust particles can settle back into the bed rather than being carried by the gas into the cyclone.

Because of intense mixing, the temperatures of the gas and solids in a fluidized bed are equal and uniform at the temperature of the discharged gas and solids. However, there is a substantial residence-time distribution for the particles in the bed. A fraction of the particles short-circuit from the feed inlet to the discharge duct with little residence time and opportunity to dry. Another fraction of the particles spends much more than the necessary residence time for complete drying. Thus, the nonuniform moisture content of the product solids may not meet specifications. When the final moisture content is critical, it may be advisable to smooth out the residence-time distribution by using a more elaborate multiple fluidized-bed dryer of the type shown in Figure 18.10b. Materials that are successfully dried in fluidized-bed dryers include coal, sand, limestone, iron ore, clay granules, granular fertilizer, granular desiccant, salt, sodium perborate, and polyvinylchloride (PVC). Large fluidized-bed dryers for coal and iron ore produce more than 0.5 million lb/h of dried material.

Spouted-Bed Dryers

When wet, free-flowing particles are larger than 1 mm in diameter, but are uniform in size and of low density, as in the case of various grains, a spouted-bed dryer, shown in Figure 18.11, is a good choice, particularly when the required drying time is more than just a few minutes. A high-velocity, hot gas enters the bottom of the drying chamber, entrains particles, and flows upward through a draft tube, above which the cross-sectional area for gas flow is significantly increased in a conical section, causing the gas velocity to decrease such that the particles are released to an annular-downcomer region. A fraction of the circulated solids is discharged from a duct in the conical section. The gas exits from the top of the vessel, following passage through a free-board region above the bed. For the drying of grains, entrainment of dust in the exiting gas is minimal.

Pneumatic-Conveyor (Flash) Dryers

When only surface moisture must be evaporated from materials that can be reduced to particles by a pulverizer, disinte-

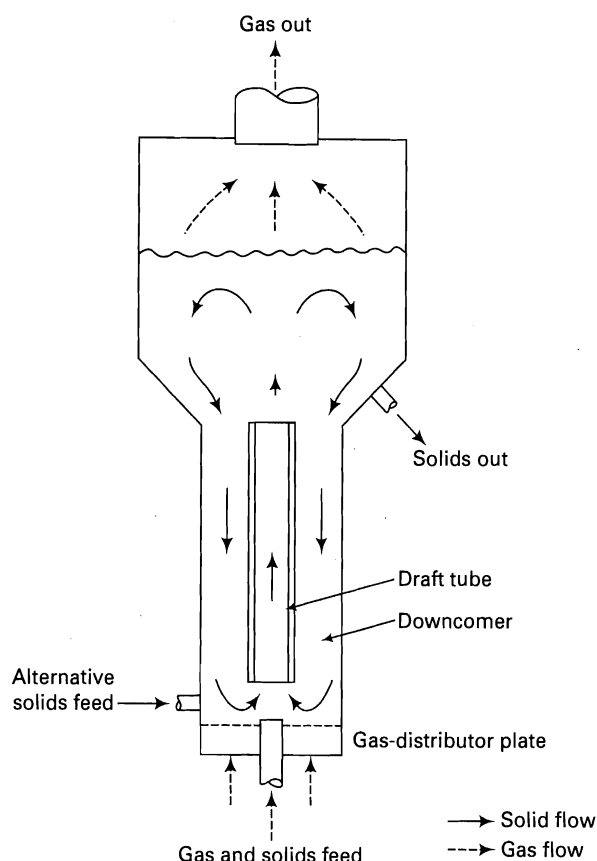


Figure 18.11 Spouted-bed dryer.

gration mill, or other deagglomeration device, a pneumatic-conveyor (gas-lift or flash) dryer is widely used and is particularly desirable when the material is temperature-sensitive, oxidizable, explosive, flammable, and/or non-erode. A common flash dryer configuration is shown in Figure 18.12. Wet solids are fed into a paddle-conveyor mixer and dropped into a hammer mill where the solids are disintegrated. Air is pulled, by an exhaust fan, through a furnace into the hammer mill, where the disintegrated solids are picked up and further deagglomerated into discrete particles while being pneumatically conveyed at upward high velocity in a duct where much of the drying takes place. Typically the holdup of solids in the gas is less than 2% by volume. The particle-gas mixture is separated in a cyclone separator, from which the solids are discharged. Because the particles travel upward in the drying duct at a velocity almost equal to that of the gas, a residence time of less than 5 s is provided. If additional time is needed, up to 30 s can be achieved by partial recycle of the solids leaving the cyclone separator. Recycle of solids is also useful for the disintegration of materials that are sticky or pasty. The deagglomerated particle sizes typically range from -30 to -300 mesh. If the particles are crystalline or friable, they may be subject to excessive breakage. Pneumatic conveying velocities range from 10 to 30 m/s, usually about 3 m/s greater than the terminal (free-fall) velocity of the largest particle to be conveyed out of the disintegrator. The distribution of remaining moisture in the product particles can be wide because of the

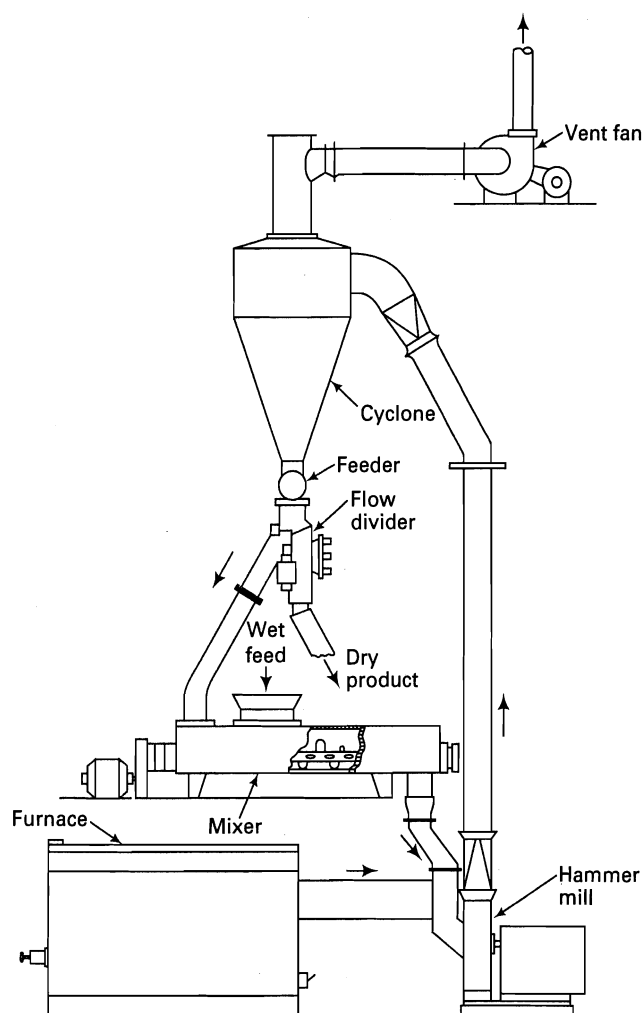


Figure 18.12 Pneumatic-conveyor (flash) dryer.

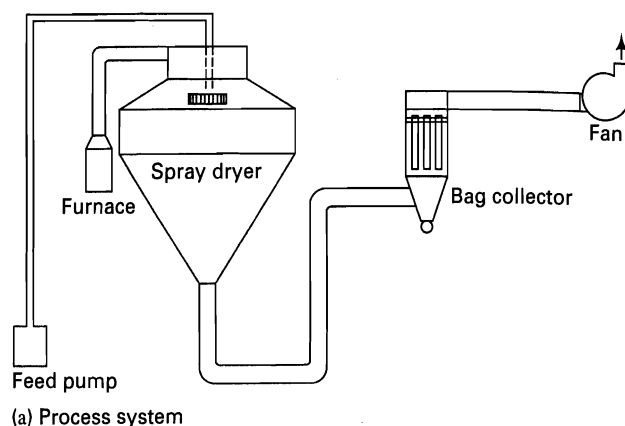
[From W.L. McCabe, J.C. Smith, and P. Harriott, *Unit Operations of Chemical Engineering*, 5th ed., McGraw-Hill, New York (1993) with permission.]

distribution of particle residence times. However, surface drying is rapid because inlet gas temperatures as high as 1500°F can be employed. Nevertheless, because: (1) the particles flow cocurrently with the gas, (2) the gas temperature decreases significantly, and (3) the particle residence time is short and evaporation of moisture is incomplete, the particles do not attain temperatures greater than about 200°F.

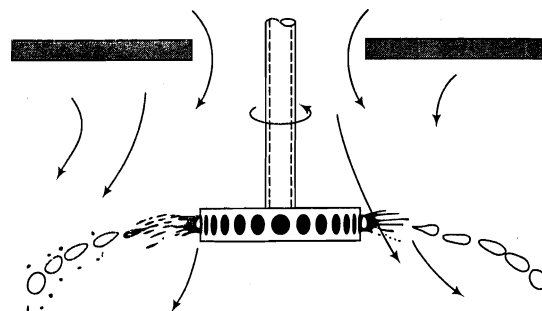
Large flash dryers are provided with pneumatic-conveying dryer ducts 1 m in diameter and 12 m high, with evaporation capacities of 20,000 lb/h of water. Compared to many other dryers, they have small floor-area requirements and are used for drying filter cakes, centrifuge cakes and slurries, yeast cake, whey, starch, sewage sludge, gypsum, fruit pulp, copper sulfate, clay, coal, chicken droppings, adipic acid, polystyrene beads, ammonium sulfate, and hexamethylene tetramine.

Spray Dryers

When solutions, slurries, or pumpable pastes, containing more than 50 wt% moisture at feed rates greater than



(a) Process system



(b) Centrifugal disk atomizer

Figure 18.13 Spray dryer.

1,000 lb/h are to be dried, a spray dryer should be considered. One of many spray-dryer configurations is shown in Figure 18.13a. The drying chamber frequently has a conical-shaped bottom section with a diameter at the top that may be nearly equal to the chamber height. The feed is pumped to the top center of the chamber where it is dispersed into small droplets or particles, ranging in size from 2 to 2,000 μm , by any of the following three types of atomizers: (1) single-fluid pressure nozzles, (2) pneumatic nozzles, and (3) centrifugal disks or spray wheels. Hot gas enters the chamber, after being heated in a furnace or heat exchanger, causing the moisture in the atomized feed to rapidly evaporate. The gas flows mainly cocurrently to the solids. The dried solids and gas are either partially separated in the chamber, followed by removal of dust from the gas by a cyclone separator, or, as shown in Figure 18.13a, are sent together to a cyclone separator, bag filter, or other gas-solid separator. The gas may be pushed or pulled through the unit by a fan.

In many respects, the spray dryer is similar in operating conditions to a pneumatic-conveyor dryer because the particles are small, the entering gas temperature can be very high, the residence time of the particles is short, mainly surface moisture is removed, and temperature-sensitive materials can be handled. However, a rather unique characteristic of a spray dryer is its ability, with some materials such as dyes, foods, and detergents, to produce, from a solution, rounded porous particles of fairly uniform size that can be rapidly dissolved or reacted in subsequent industrial or domestic applications. Although particle residence times are less than

5 s if only surface moisture is removed, residence times of up to 30 s can be provided for additional evaporation of internal moisture. Spray drying is also unique in that it combines, into one compact operation, the equivalent of some or all of the following sequence of operations: evaporation, crystallization or precipitation, filtration or centrifugation, size reduction, classification, and drying.

The most influential part of a spray dryer is the atomizer. Each of the three types has advantages and disadvantages and, therefore, the selection of the atomizer for a particular application is not straightforward. Pneumatic (two-fluid) nozzles impinge air, steam, or another gas on the feed at relatively low pressures of up to 100 psig, but are not efficient at high capacities. Consequently, they find applications only in pilot plants and commercial plants of low capacity. Exceptions are the dispersion of stringy and fibrous materials, thick pastes, certain filter cakes, and polymer solutions because with high atomizing gas-to-feed ratios, the smallest particles can be produced.

Pressure (single-fluid) nozzles, with orifice diameters of 0.012–0.15 in., require solution inlet pressures of 300–4,000 psig to achieve break-up of the feed stream. These nozzles can deliver the narrowest range of droplet sizes, but the droplet sizes are the largest delivered by the three types of atomizers, and multiple nozzles are required in large-diameter spray dryers. Also, orifice wear and plugging can be problems with some feeds. Because the spray is largely downward, chambers are relatively slender and tall, with height-to-diameter ratios of 4–5.

Centrifugal disks (spray wheels), of the type shown in Figure 18.13b, handle solutions or slurries, delivering thin sheets of feed that break up into small droplets in a nearly radial direction at high capacities. However, disks have the largest-diameter spray pattern and therefore require the largest-diameter drying chambers to prevent particles from striking the chamber wall while in a sticky state. Centrifugal disks range in diameter from a few inches in small units to up to 32 in. in very large units. The disks spin at 3,000–50,000 rpm, but a given disk can operate over a wide range of feed rate and rotation rate, without significantly affecting the particle-size distribution that occurs with variation of the feed pressure or by enlargement of or other damage to the orifice of a pressure nozzle.

Spray-dryer diameters for industrial applications are large, with 8–30 ft being common. Evaporation rates of up to 2,600 lb/h have been achieved in an 18-ft-diameter by 18-ft-high spray dryer equipped with a centrifugal-disk atomizer and supplied with an aqueous feed solution of 7 wt% dissolved solids and 11,000 cfm of air at 600°F. Larger spray dryers can evaporate up to 15,000 lb/h. Solutions, slurries, and pastes of the following materials are spray dried: detergents, blood, milk, eggs, starch, yeast, zinc sulfate, lignin, aluminum hydroxide, silica gel, magnesium chloride, manganese sulfate, aluminum sulfate, urea resin, sodium sulfide, coffee extract, tanning extract, color pigments, tea, tomato juice, polymer resins, and ceramics.

Drum Dryers

Approximately 100 years ago, and before the 1920s when spray dryers were introduced with applications for the drying of milk and detergent solutions, drum dryers were introduced to process solutions, slurries, and pastes with indirect heat. The first such dryer was the double-drum dryer, shown in Figure 18.14a, which is still the most versatile and widely applied drum dryer. It consists of two metal, cylindrical drums of identical size (1–5 ft diameter by 1.5–12 ft long), mounted side-by-side. One drum is movable so that the smallest distance between the two drums (nip) can be adjusted. The drums are heated on the inside by condensing steam at pressures as high as 12 atm. The feed to be dried enters at the top from a perforated pipe that runs the length of the two drums or from a pipe that swings like a pendulum from end-to-end of the drums. The drums are rotated in opposite directions toward each other at the top, as shown, causing the feed to form, on the hot surface of the drums, a clinging film or coating whose thickness is controlled by the nip. As the drums rotate, heat is transferred from the drum surfaces to the coating, causing it to dry. If the moisture is water, it exits as steam through a vapor hood mounted above the drums. If the moisture is a solvent or if the solid is dustable, a dryer enclosure can be provided. When the coating has made about three-quarters of a complete rotation, it is scraped off the drum surfaces by doctor blades that run the length of the drums. The dried material falls off as surface powder, chips, or more commonly as flakes, which typically are 1–3 mm thick, into conveyors. By adjusting (1) drum rotation rate, usually from 1 to 30 rpm, (2) drum-surface temperature, which is usually just a few degrees below the inside, condensing-steam temperature; (3) feed temperature; and (4) coating thickness, the moisture content of the dried material can be controlled. Drying times are in the range of 3–20 s. Performance data given by Walas [3] show capacities of double-drum dryers to be in the range of 1–60 kg dried product/h-m² of drum surface for feed moisture contents ranging from 10 to 90 wt%.

For drum dryers to be effective, the coating must adhere to the drum surface, which is often chrome plated. When necessary, other drum and feeding arrangements can be employed. For example, the twin-drum dryer with top feed shown in Figure 18.14b is not influenced by drum spacing because the drums rotate away from each other at the top. Much thicker coatings can then be formed and materials, like inorganic crystals, that might score or cause other damage to closely spaced drums can be processed. To improve the likelihood of the feed adhering to the drums, the feed may be splashed onto the surfaces of the twin-drum dryer as shown in Figure 18.14c.

For very viscous solutions or pastes that might cause undue pressure to be put on the surfaces of a double-drum dryer, a single-drum dryer can be used. The coating can be applied by using a top feed with applicator rolls, as shown in Figure 18.14d. If a porous product (e.g., malted milk) is desired or if the material is temperature-sensitive, a

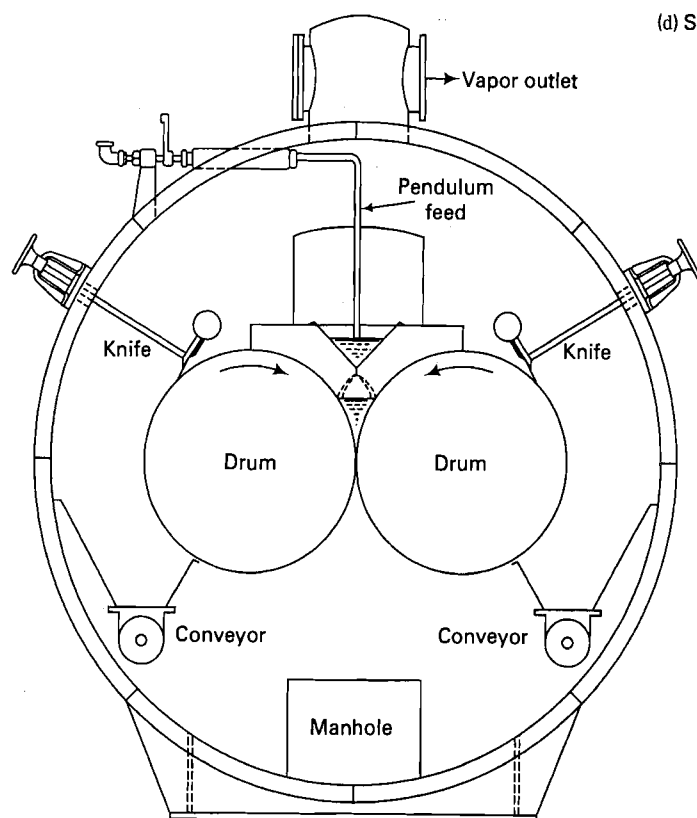
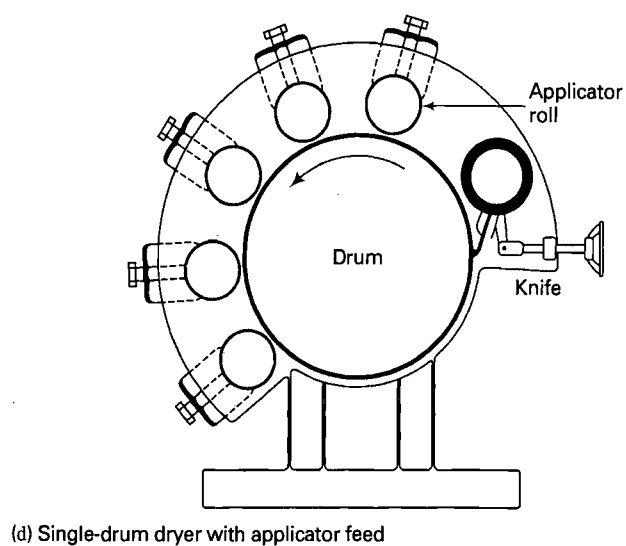
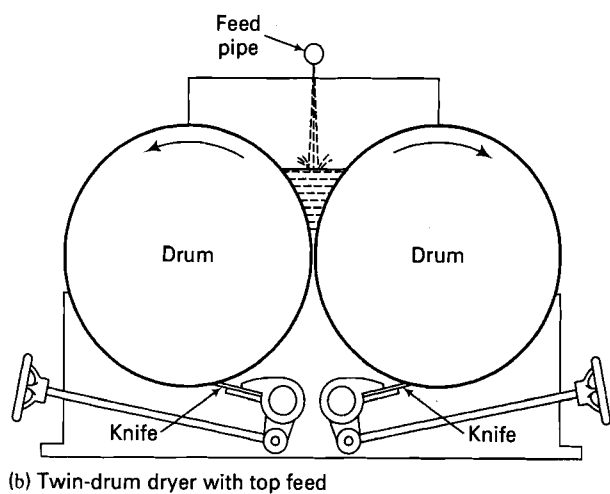
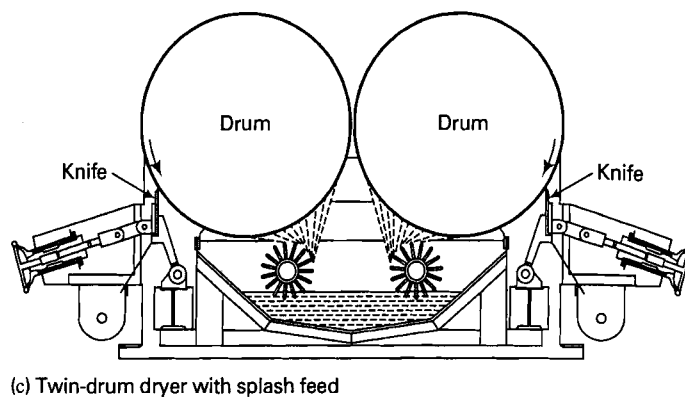
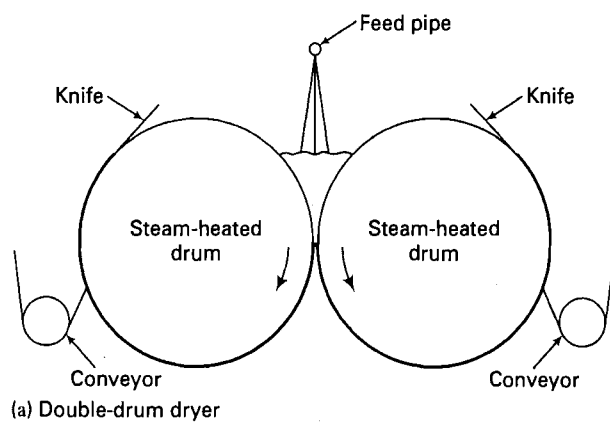


Figure 18.14 Drum dryers.

single-drum dryer or double-drum dryer, as shown in Figure 18.14e, can be enclosed so that a vacuum can be pulled to reduce the boiling point of the moisture.

Drum dryers are used to dry a wide variety of materials, including starch solutions, brewers yeast, potatoes, skim milk, malted milk, coffee, tanning extract, and vegetable glue; slurries of $\text{Mg}(\text{OH})_2$, $\text{Fe}(\text{OH})_2$, and CaCO_3 ; and solutions of sodium acetate, Na_2SO_4 , Na_2HPO_4 , CrSO_4 , and various organic compounds. However, many of the applications of drum dryers are now handled with spray dryers. Drum dryers belong to a class of hot-cylinder dryers. Units with large numbers of cylinders in series and parallel are used to dry continuous sheets of woven fabrics and paper pulp at evaporation rates of about $10 \text{ kg/h}\cdot\text{m}^2$.

Infrared Drying

In direct-heat dryers, the transfer of heat by convection from hot gases to the wet material is often supplemented by thermal radiation from hot surfaces that surround the material. However, this radiant-heat contribution is usually a minor one, such that it is often ignored. For the drying of certain films, sheets, and coatings, however, the use of thermal radiation as the major source of heat offers distinct advantages.

Radiant energy is released from all matter as a result of oscillations and transitions of its electrons. For gases and transparent solids and liquids, the radiation can be emitted from throughout the volume of the matter. For opaque solids and liquids, this internal radiation is quickly absorbed by adjoining molecules so that the net transfer of energy by radiation is only from the surface. Of greatest importance in radiation heat transfer for the drying of solids is the transfer of radiation from a hot, opaque surface through a nonabsorbing gas or vacuum to the material being dried. This transfer can be viewed as the propagation of discrete photons (quanta) and/or as the propagation of electromagnetic waves, consisting, as shown in Figure 18.15a, of electric and magnetic fields that oscillate at right angles to each other and to the direction in which the radiation travels. As shown in the electromagnetic spectrum of Figure 18.15b, the wavelength, λ , of the radiation, which depends on the manner in which it is generated, covers an exceedingly wide range from gamma rays of $10^{-8} \mu\text{m}$ to long radio waves of $10^{10} \mu\text{m}$. Regardless of the wavelength, all radiation waves travel at the speed of light, c , in the particular medium, which for a vacuum is $2.998 \times 10^8 \text{ m/s}$. Accordingly, the following relationship exists between the frequency of the wave, ν , and the wavelength, λ :

$$\nu = c/\lambda \quad (18-1)$$

The frequency is usually expressed in Hz, which is one cycle/s. The energy transmitted by the wave depends on its frequency and is expressed in terms of the energy, E , of a photon by

$$E = h\nu \quad (18-2)$$

where h = Planck's constant = $6.62608 \times 10^{-34} \text{ J}\cdot\text{s}$.

A solid, opaque surface can emit infrared radiation by virtue of its temperature. This type of radiation is invisible

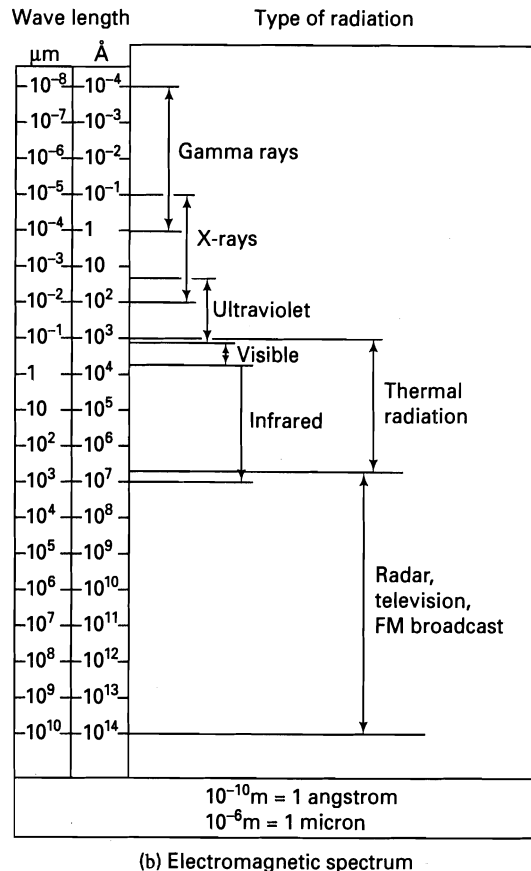
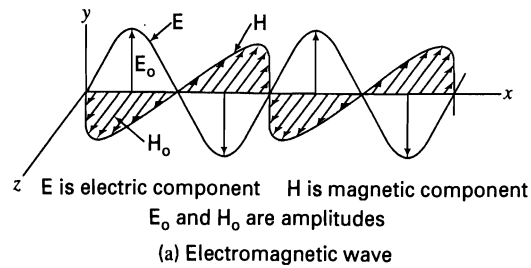


Figure 18.15 Radiation.

and has a wavelength, as shown in Figure 18.15b, in the range of $0.75\text{--}100 \mu\text{m}$. If the surface emitting the radiation is a so-called blackbody, b , such that the maximum amount of radiation will be emitted, that amount will be distributed over a range of wavelengths, depending on the temperature, as governed by the Planck distribution, which, in terms of radiant heat leaving diffusely from a unit area of surface, is

$$E_{\lambda,b} = C_1 / \{\lambda^5 [\exp(C_2/\lambda T) - 1]\} \quad (18-3)$$

where the units of $E_{\lambda,b}$ are $\text{W/m}^2\cdot\mu\text{m}$, $C_1 = 3.742 \times 10^8$, $C_2 = 1.439 \times 10^4$, T is in K, and λ is in μm . When (18-3) is integrated over the entire range of wavelengths, the result is the Stefan-Boltzmann equation

$$E_b = \sigma T^4 \quad (18-4)$$

where $\sigma = 5.67051 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4$ and T is in K. Thus, as the temperature of the infrared heat source is increased, the rate of heat transfer increases exponentially.

Sources of infrared radiant heat at surface temperatures in the range of 600–2,500 K are electrically heated metal-sheath rods, quartz tubes, and quartz lamps; and ceramic-enclosed gas burners. When the radiant energy reaches the material being dried, it is usually absorbed at the surface, from where it is transferred into the interior by conduction. In this respect, infrared radiant heat transfer to the surface is much like convective heat transfer to the surface. However, if the effective thermal conductivity of the material is low, the surface temperature may rise to an undesirable value, particularly if high-temperature, infrared-radiation sources are used. For that reason, infrared drying must be used with caution. Applications of growing interest include the drying of paper, paints, enamels, inks, glue-on flaps, and textiles. Continuous infrared dryers are more common than batch infrared dryers.

Dielectric and Microwave Drying

In contrast to infrared drying, dielectric and microwave drying involves the low-frequency, long-wavelength end of the electromagnetic spectrum of Figure 18.15b, where radio waves and microwaves reside. With nonelectrically conducting materials, heat is not absorbed at the surface but is generated throughout the material, reducing the need for heat conduction within the material and, thus, making this type of drying unique. Furthermore, the rate of energy dissipation in the material can be controlled over a very wide range. Other advantages over the more conventional drying methods include: (1) efficiency of energy usage because the energy dissipation occurs mainly in the moisture rather than in the solid substrate of other parts of the dryer; (2) operation at low temperatures, thus avoiding high material surface temperatures; and (3) more rapid drying. Dielectric and microwave drying is particularly useful for preheating materials and for removing the final traces of internal moisture. Dielectric drying is confined to frequencies between 1 and 150 MHz, while microwave drying utilizes frequencies from 300 MHz to 300 GHz. However, by international agreement, most microwave drying is done at only 915 and 2,450 MHz, as discussed by Mujumdar [1].

Equipment for dielectric and microwave drying consists of an energy generator and an applicator. The primary source of energy is the common alternating current of 50–60 Hz. A generator is used to boost this frequency to the much higher values quoted above. A negative-grid triode tube is used with dielectric systems, while magnetron or klystron tubes are used with microwave systems. Dielectric energy is usually applied by electrodes of various types and shapes, between which is placed the material to be dried. Microwave systems often use hollow, rectangular, metallic waveguides.

Dielectric systems are used to dry bulky materials such as lumber, ceramic monoliths, foam rubber, breakfast cereals, dog biscuits, crackers, biscuits, and cookies, as well as films, coatings, and other materials, such as paper, inks, adhesives, textiles, and penicillin, where high surface temperatures

must be avoided. Microwave systems are used to dry pasta, onions, seaweed, baseball bats, potato chips, pharmaceuticals, ceramic filters, and sand casting molds.

Freeze Drying

In freeze drying (also called lyophilization), the moisture in the feed material is first frozen, by cooling the material, and then sublimed by conductive, convective, and/or radiant heating. Because the structure and properties of the material to be dried are hardly altered by freeze drying, it has been widely applied industrially to the drying of biological materials, pharmaceuticals, and foodstuffs. Products of freeze drying are porous and nonshrunk. When foodstuffs are dehydrated by freeze drying and then stored under a dry, inert gas, they evade deterioration almost indefinitely and can be rehydrated almost perfectly to their original state for later consumption. The first major application of freeze drying was for the preservation of blood plasma during World War II.

When the moisture is water, the material must be cooled to at least 0°C to freeze the water if it is free, and even lower if the water is dissolved in the material. Most freeze drying is conducted with the material at –10°C or lower. At this temperature, ice has a vapor pressure of only 2 torr. Therefore, freeze drying must be conducted under a high vacuum. The feed material is usually in the form of a slab or particles. In both cases, heat for sublimation must be transferred from the heat source to the material under controlled conditions so that the moisture does not reach the melting point. In some cases, an even lower temperature, called the scorch point, must not be exceeded to avoid degradation of the material. During the drying period, which may take up to 20 hours, the resistance to heat transfer increases because an interface develops, between the porous freeze-dried layer and the frozen material, and gradually recedes into the material. The thermal resistance of the porous structure is greater than that of the frozen material.

For small quantities of biological and pharmaceutical materials, freeze drying is generally conducted batchwise on trays in vacuum cabinets, where the drying step follows the freezing step. The sublimed ice is desublimed on a cold metal plate that resides either inside the cabinet and adjacent to the trays or in a separate adjoining vessel. During the sublimation step, heat transfer is usually by conduction from the bottom and side surfaces of the tray, which contains coils of passages through which a heating fluid, e.g., vacuum steam, passes. For large quantities of foodstuffs, continuous freeze drying can be employed, as shown in Figure 18.16, using trays of prefrozen materials that are transported through a tunnel, past fixed heating platens, with vacuum locks at either end. With granular materials, drying times of less than 1 h can be achieved. Continuous freeze drying of small particles can also be accomplished rapidly in a fluidized bed, where heat transfer for sublimation is by convection and radiation. However, the resulting agitation of

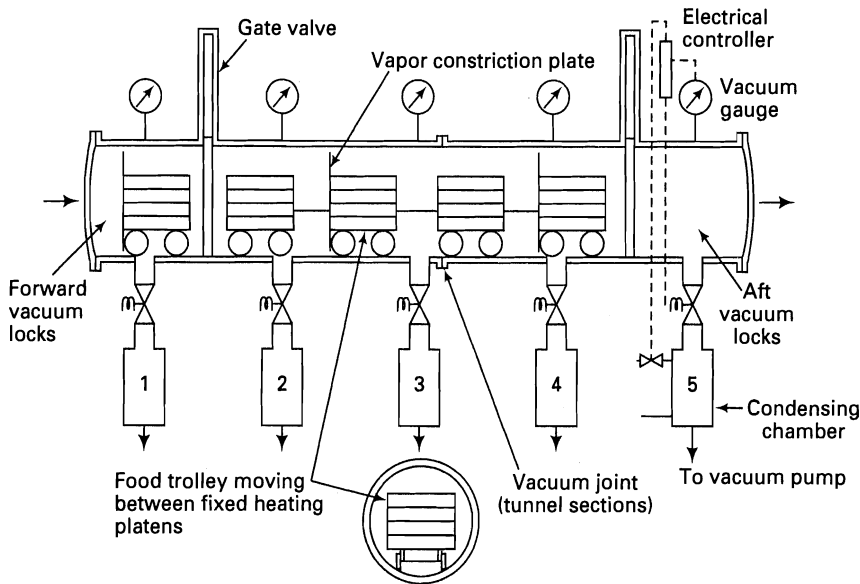


Figure 18.16 Tunnel freeze dryer.

the particles can cause breakage and dusting. In some cases, freeze drying can utilize infrared and microwave heating. Freeze drying is used for the sublimation of moisture from seafood, meat, vegetables, fruits, coffee, concentrated beverages, pharmaceuticals, blood plasma, and biological materials.

18.2 PSYCHROMETRY

For moisture to be evaporated from a wet solid, it must be heated to a temperature at which the vapor pressure of the moisture exceeds the partial pressure of the moisture in the gas in contact with the wet solid. In an indirect-heat dryer,

Table 18.4 Definitions of quantities useful in psychrometry: A = moisture, B = moisture-free gas, ideal-gas conditions

Quantity	Definition	Relationship	
Absolute, mass humidity	Moisture content of a gas by mass	$\mathcal{H} = \frac{M_A p_A}{M_B (P - p_A)}$	(18-5)
Molal humidity	Moisture content of a gas by mols	$\mathcal{H}_m = \frac{p_A}{P - p_A}$	(18-6)
Saturation humidity	Humidity at saturation	$\mathcal{H}_s = \frac{M_A P_A^s}{M_B (P - P_A^s)}$	(18-7)
Relative humidity (relative saturation as a percent)	Ratio of partial pressure of moisture to partial pressure of moisture at saturation	$\mathcal{H}_R = 100\% \times \frac{p_A}{P_A^s}$	(18-8)
Percentage humidity (percent saturation)	Ratio of humidity to humidity at saturation	$\mathcal{H}_P = 100\% \times \frac{\mathcal{H}}{\mathcal{H}_s}$	(18-9)
Humid volume	Volume of moisture-gas mixture per unit mass of moisture-free gas	$v_H = \frac{RT}{P} \left(\frac{1}{M_B} + \frac{\mathcal{H}}{M_A} \right)$	(18-10)
Humid heat	Specific heat of moisture-gas mixture per unit mass of moisture-free gas	$C_s = (C_P)_B + (C_P)_A \mathcal{H}$	(18-11)
Total enthalpy	Enthalpy of moisture-gas mixture per unit mass of moisture-free gas referred to temperature, T_o	$H = C_s (T - T_o) + \Delta H_o^{\text{vap}} \mathcal{H}$	(18-12)
Dew-point temperature	Temperature at which moisture begins to condense when mixture is cooled	T_{dew}	
Dry-bulb temperature	Temperature of mixture	T_d	
Wet-bulb temperature	Steady-state temperature attained by a wet-bulb thermometer	T_w	
Adiabatic-saturation temperature	Temperature attained when a gas is saturated with moisture in an adiabatic process	T_s	

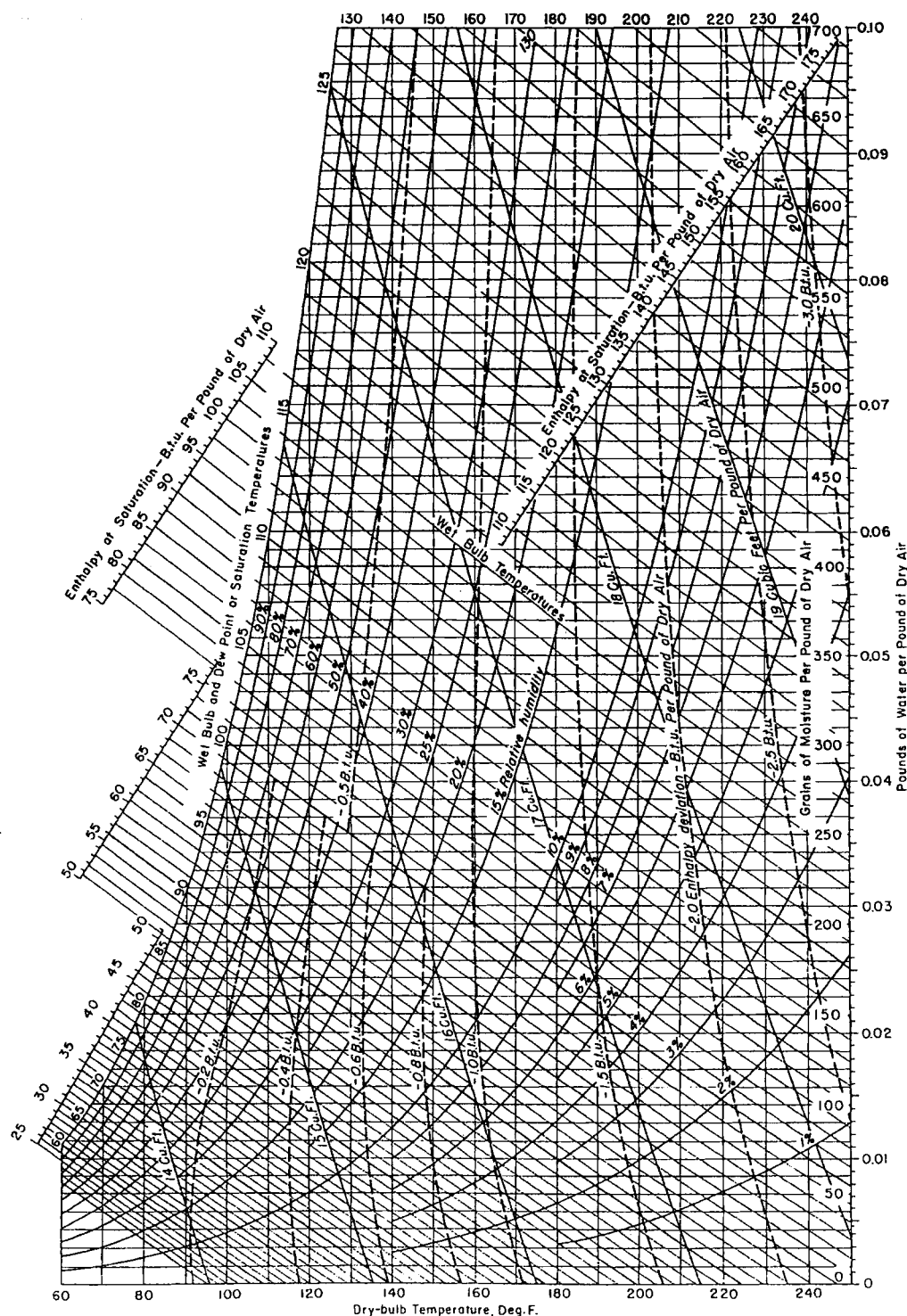


Figure 18.17 Psychrometric (humidity) chart for air-water at 1 atm.

where little or no gas is used to carry away the moisture as vapor, the partial pressure of the moisture approaches the total pressure and the temperature of evaporation approaches the boiling point of the moisture at the prevailing pressure as long as the moisture is free liquid at the surface of the solid. If the moisture interface recedes into the solid, a temperature higher than the boiling point is necessary at the solid-gas interface to transfer the heat for evaporation from that interface to the liquid-gas interface. In addition, if the moisture level drops to a point where it is all sorbed by the solid, the vapor pressure of this sorbed moisture is less than the

pure vapor pressure and an even higher temperature is required to evaporate the moisture. In a direct-heat dryer, similar situations can occur except that the temperature at which moisture evaporation occurs is dependent on the moisture content of the gas. Calculations involving the properties of moisture-gas mixtures for application to drying are most conveniently carried out with psychrometric charts.

A typical chart, given in Figure 18.17, is that for air-water vapor mixtures at 1-atm total pressure. Included in this chart are properties that are included in the list of definitions in Table 18.4, which applies to general moisture-gas mixtures

that obey the ideal-gas law. The definitions given by (18-5)–(18-9) for humidity and by (18-10)–(18-12) for humid volume, humid heat, and enthalpy are in terms of a unit mass or mole of moisture-free gas. In the case of a water vapor–air mixture, the term dry air is used. The use of Figure 18.17 and its basis in the relationships of Table 18.4 is illustrated in the following example.

EXAMPLE 18.1

Air at 131°F and 1 atm enters a direct-heat dryer with a humidity, $\mathcal{H}$, of 0.03 lb H₂O/lb H₂O-free air. Determine the following from the psychrometric chart of Figure 18.17 and/or relationships of Table 18.4:

- Molal humidity, $\mathcal{H}_m$
- Saturation humidity, $\mathcal{H}_s$
- Relative humidity, $\mathcal{H}_R$
- Percentage humidity, $\mathcal{H}_P$
- Humid volume, v_H
- Humid (specific) heat, C_s
- Enthalpy, H

SOLUTION

At 131°F, the vapor pressure of water is 118 torr = 0.155 atm. A = H₂O, B = air, and $\mathcal{H} = 0.03$ lb H₂O/lb H₂O-free air.

- (a) Combining (18-5) and (18-6),

$$\mathcal{H}_m = \frac{M_B}{M_A} \mathcal{H} = \frac{M_{\text{air}}}{M_{\text{H}_2\text{O}}} \mathcal{H} = \frac{28.97}{18.02} (0.03) = 0.048 \frac{\text{lb mol H}_2\text{O}}{\text{lb mol dry air}}$$

- (b) From (18-7),

$$\mathcal{H}_s = \frac{18.02}{28.97} \left(\frac{0.155}{1 - 0.155} \right) = 0.114 \frac{\text{lb H}_2\text{O}}{\text{lb dry air}}$$

- (c) From a rearrangement of (18-6),

$$p_{\text{H}_2\text{O}} = \frac{P \mathcal{H}_m}{1 + \mathcal{H}_m} = \frac{(1)(0.048)}{1 + 0.048} = 0.0458 \text{ atm}$$

From (18-8),

$$\mathcal{H}_R = 100 \left(\frac{0.0458}{0.155} \right) = 29.5\%$$

The same result is obtained from Figure 18.17.

- (d) From (18-9),

$$\mathcal{H}_P = 100 \left(\frac{0.03}{0.114} \right) = 26.3\%$$

- (e) From (18-10), for $R = 0.730 \frac{\text{atm} \cdot \text{ft}^3}{\text{lbmol} \cdot ^\circ\text{R}}$, $T = 131 + 460$
- $$= 591^\circ\text{R},$$

$$v_H = \frac{0.730(591)}{1} \left(\frac{1}{28.97} + \frac{0.03}{18.02} \right) = 15.6 \text{ ft}^3/\text{lb dry air}$$

which agrees with Figure 18.17.

- (f) From (18-11), using

$$(C_P)_{\text{air}} = 0.24 \text{ Btu/lb} \cdot ^\circ\text{F}$$

$$(C_P)_{\text{steam}} = 0.45 \text{ Btu/lb} \cdot ^\circ\text{F}$$

$$C_s = 0.24 + (0.45)(0.03) = 0.254 \frac{\text{Btu}}{\text{lb dry air}}$$

- (g) Equation (18-12) assumes that the enthalpy datum refers to air as a gas and water as a liquid. Taking $T_o = 32^\circ\text{F}$ and $\Delta H_o^{\text{vap}} = 1075 \text{ Btu/lb}$, (18-12) gives

$$H = 0.254(131 - 32) + 1,075(0.03) = 57.4 \text{ Btu/lb dry air}$$

EXAMPLE 18.2

At a certain location in a dryer where benzene is being evaporated from a solid, nitrogen gas at 50°F and 1.2 atm has a relative humidity for benzene of 35%.

Determine:

- Partial pressure of the benzene if the vapor pressure of benzene at 50°F = 45.6 torr.
- Humidity of the nitrogen-benzene mixture.
- Saturation humidity of the mixture.
- Percentage humidity of the mixture.

SOLUTION

$$\mathcal{H}_R = 35\%, P = 1.2 \text{ atm} = 912 \text{ torr}$$

$$A = \text{benzene}, B = \text{N}_2$$

$$M_A = 78.1, M_B = 28, p_{\text{benzene}}^s = 45.6 \text{ torr}$$

- (a) From (18-8),

$$p_{\text{benzene}} = \frac{p_{\text{benzene}}^s \mathcal{H}_R}{100} = \frac{(45.6)(35)}{100} = 16 \text{ torr}$$

- (b) From (18-5),

$$\mathcal{H} = \left(\frac{78.1}{28} \right) \left(\frac{16}{912 - 16} \right) = 0.050 \frac{\text{lb benzene}}{\text{lb dry nitrogen}}$$

- (c) From (18-7),

$$\mathcal{H}_s = \left(\frac{78.1}{28} \right) \left(\frac{45.6}{912 - 45.6} \right) = 0.147 \frac{\text{lb benzene}}{\text{lb dry nitrogen}}$$

- (d) From (18-9),

$$\mathcal{H}_P = 100 \left(\frac{0.050}{0.147} \right) = 34\%$$

Wet-bulb Temperature

In general, the temperature at which moisture evaporates in a direct-heat dryer is complex to determine and can vary from the inlet to the outlet of the dryer. However, when the dryer operates isobarically and adiabatically, with all of the energy for moisture evaporation supplied from the hot gas by convective heat transfer and with no energy required for heating the wet solid to the evaporation temperature, application of the following simplified heat- and mass-transfer

equations leads to an expression for the temperature of evaporation at a particular location in a dryer operating under continuous, steady-state conditions, or at a particular point in time for a dryer operating batchwise.

Because it is further assumed that the moisture being evaporated is free liquid, exerting its full vapor pressure, at the surface of the solid, this temperature of evaporation is referred to as the wet-bulb temperature, T_w , because it can be measured by covering the bulb of a thermometer with a wick saturated with the moisture to be evaporated, and passing a large quantity of partially saturated gas past the wick, as indicated in Figure 18.18a.

Refer to Figure 18.18b, where the wetted wick is replaced by an incremental amount of wet solid. Assume that the area for heat transfer = area for mass transfer = A . At steady-state conditions, the rate of convective heat transfer from the gas to the wet solid is given by Newton's law of cooling:

$$Q = h(T - T_w)A \quad (18-13)$$

The molar rate of mass transfer of the evaporated moisture from the wet surface of the solid, A , is

$$n_A = \frac{k_y(y_{Aw} - y_A)A}{(1 - y_A)_{LM}} \quad (18-14)$$

An enthalpy balance on the moisture being evaporated and heated to the gas temperature couples the heat and mass transfer to give

$$Q = n_A M_A [\Delta H_w^{\text{vap}} + (C_P)_A (T - T_w)] \quad (18-15)$$

To obtain a simplified relationship for the coupling in terms of T and $\mathcal{H}$, assume that the mole fraction of moisture in the gas is small in the bulk gas and at the wet solid-gas interface. Then, the bulk-flow effect in (18-14) becomes $(1 - y_A)_{LM} \sim 1.0$. Also, from (18-5), replacing p_A with $y_A P$

and solving for y_A , we obtain

$$y_A = \frac{\mathcal{H} M_A}{\frac{1}{M_B} + \frac{\mathcal{H}}{M_A}} \approx \frac{\mathcal{H} M_B}{M_A} \quad (18-16)$$

In addition, assume that the latent-heat effect in (18-15) is much greater than the sensible-heat effect. Then

$$\Delta H_w^{\text{vap}} + (C_P)_A (T - T_w) \approx \Delta H_w^{\text{vap}} \quad (18-17)$$

Combining (18-13) to (18-16) after making all these simplifications, we obtain

$$T_w = T - \frac{k_y M_B \Delta H_w^{\text{vap}}}{h} (\mathcal{H}_w - \mathcal{H}) \quad (18-18)$$

Assume that the Chilton-Colburn analogy for mass and heat transfer applies. Then, from (3-165) and (3-228),

$$\frac{k_y M}{G} N_{Sc}^{2/3} = \frac{h}{C_P G} N_{Pr}^{2/3} \quad (18-19)$$

If $M \approx M_B$, (18-19) simplifies to

$$\frac{k_y M_B}{h} = \frac{1}{(C_P)_B} \left(\frac{1}{N_{Le}} \right)^{2/3} \quad (18-20)$$

where

$$N_{Le} = \text{Lewis number} = N_{Sc}/N_{Pr} \quad (18-21)$$

The reciprocal of the Lewis number is sometimes referred to as the Luikov number, N_{Lu} . Substitution of (18-20) into (18-18) gives

$$T_w = T - \frac{\Delta H_w^{\text{vap}}}{C_P} \left(\frac{1}{N_{Le}} \right)^{2/3} (\mathcal{H}_w - \mathcal{H}) \quad (18-22)$$

In Equation (18-18), which defines the wet-bulb temperature, or (18-22), it is important to note that $\mathcal{H}_w$ refers to the saturation humidity at temperature T_w .

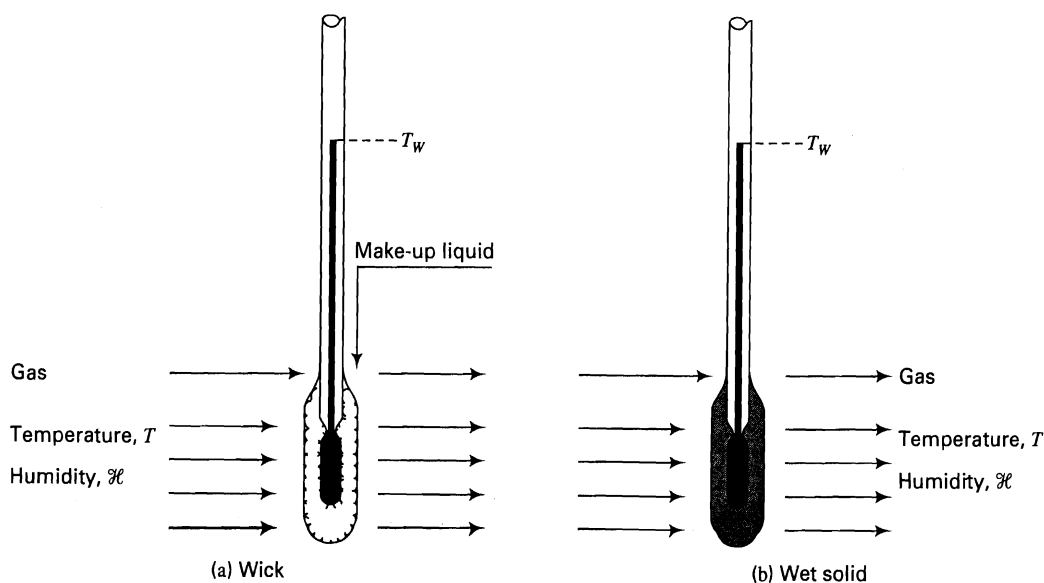


Figure 18.18 Wet-bulb temperature.

[From W.L. McCabe, J.C. Smith, and P. Harriott, *Unit Operations of Chemical Engineering*, 5th ed., McGraw-Hill, New York (1993) with permission.]

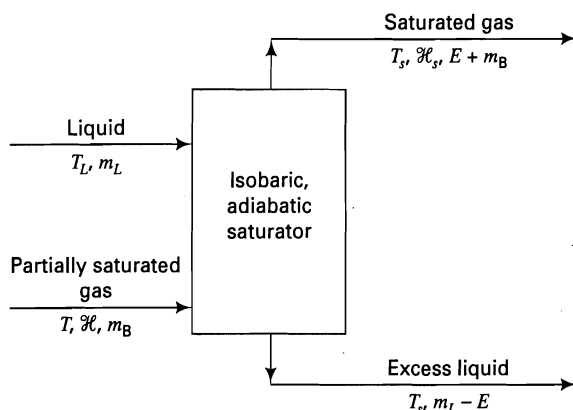
Table 18.5 Lewis Number for Liquids Evaporating into Air at 25°C

Liquid	N_{Le}	$(N_{Le})^{2/3} =$ Psychrometric Ratio
Benzene	2.44	1.812
Carbon tetrachloride	2.67	1.923
Chloroform	3.08	2.114
Ethyl acetate	2.58	1.880
Ethylene tetrachloride	3.05	2.101
Metaxylene	3.18	2.165
Methanol	1.37	1.233
Propanol	1.85	1.506
Toluene	2.64	1.908
Water	0.855	0.901

By coincidence, $(N_{Le})^{2/3}$ for the air-water vapor system at 25°C is close to 1.0 (actually 0.901). For air-organic vapor systems, as listed in Table 18.5, which is taken from Keey [7], $(N_{Le})^{2/3}$ is much less than 1.0. As will be shown in the next section, the value of $(N_{Le})^{2/3}$ has a great impact on the variation with location or time of the temperature of moisture evaporation in a direct-heat dryer.

Adiabatic-Saturation Temperature

In order to determine the possible change in the wet-bulb temperature of the wet solid, as it dries with location or time in the dryer, it is necessary to consider changes in the temperature and humidity of the gas as it cools because of the transfer of heat to the wet solid, and the transfer of moisture from the wet solid. A simplified relationship between gas temperature and humidity can be derived by considering the adiabatic saturation of a gas with an excess of moisture as pure liquid. Referring to the system of Figure 18.19, partially saturated gas at temperature T , humidity $\mathcal{H}$, and mass flow rate m_B (dry basis), together with liquid entering at temperature T_L and mass flow rate m_L , enters an isobaric and adiabatic chamber where a fraction of the liquid, (E/m_L) , is

**Figure 18.19** Adiabatic saturation of a gas with a liquid.

vaporized to saturate the gas. The gas and excess liquid leave the chamber in equilibrium at temperature T_s . An enthalpy balance using (18-12) for the enthalpy of the gas phase, but with the reference temperature $T_o = T_s$ (so as to simplify the balance), gives

$$m_L(C_{PA})_L(T_L - T_s) + m_B[C_{s,in}(T - T_s) + \Delta H_s^{\text{vap}}\mathcal{H}] = m_B[C_{s,out}(T_s - T_s) + \Delta H_s^{\text{vap}}\mathcal{H}_s] + (m_L - E)(C_{PA})_L(T_s - T_s) \quad (18-23)$$

Assume that the sensible heat required to heat the liquid from T_L to T_s is negligible. Then (18-23) simplifies to an equation for the adiabatic saturation temperature:

$$T_s = T - \frac{\Delta H_s^{\text{vap}}}{(C_s)_{in}}(\mathcal{H}_s - \mathcal{H}) \quad (18-24)$$

Equation (18-24) also applies for the determination of a gas temperature and humidity intermediate between T and T_s and $\mathcal{H}$ and $\mathcal{H}_s$, respectively, if sensible heat for the liquid is ignored. If (18-24) is compared to (18-18), it is seen that the wet-bulb and adiabatic-saturation temperatures would be equal if the $h/k_y M_B$ were equal to $(C_s)_{in}$. For the air-water system, as shown by Lewis [8] and referred to as the Lewis relation, this is almost the case, with

$$\frac{h}{k_y M_B} \approx 0.216 \text{ Btu/lb} \cdot ^\circ\text{F}$$

compared to $C_s \approx 0.24 \text{ Btu/lb} \cdot ^\circ\text{F}$. However, typically, a small amount of radiation heat transfer to the wet solid supplements the convective heat transfer. If h in the psychrometric ratio is replaced by $(h_c + h_r)$, the corrected ratio is almost identical to the humid heat at low humidities. Accordingly, for the air-water system, the wet-bulb temperature is taken to be equal to the adiabatic-saturation temperature and only one family of lines is shown for these two temperatures on the psychrometric chart. From (18-24), these lines have a negative slope of $\Delta H_s^{\text{vap}}/(C_s)_m$. The following example illustrates the use of the chart in Figure 18.17 to determine the relationship between gas (dry-bulb) temperature, wet-bulb temperature, and humidity. The example also illustrates the use of (18-24).

EXAMPLE 18.3

For the conditions of Example 18.1, determine the wet-bulb temperature, assuming it is equal to the adiabatic-saturation temperature, by using (a) the psychrometric chart and (b) (18-24).

SOLUTION

- (a) In Figure 18.17, the point $T = T_d = 131^\circ\text{F}$ and $\mathcal{H} = 0.03 \text{ lb H}_2\text{O/lb dry air}$ is plotted. This point lies just above the adiabatic-saturation-temperature line of 95°F at about 96.5°F .
- (b) Equation (18-24) involves an iterative calculation to determine $T_w = T_s$ because ΔH_s^{vap} and $\mathcal{H}_s$ are unknown.

Assume

$$T_s = 95^\circ\text{F}$$

$$\Delta H_s^{\text{vap}} = 1039.8 \text{ Btu/lb from steam tables}$$

$$(C_s)_{\text{in}} = 0.254 \text{ Btu/lb from Example 18.1}$$

$$P_A^s = 42.2 \text{ torr}$$

$$= \text{vapor pressure of water from Example 18.1}$$

From (18-7),

$$\mathcal{H}_s = \frac{18.02}{28.97} \left(\frac{42.2}{760 - 42.2} \right) = 0.0366 \frac{\text{lb H}_2\text{O}}{\text{lb dry air}}$$

From (18-24),

$$T_s = 131 - \frac{1039.8}{0.254} (0.0366 - 0.03) = 104^\circ\text{F}$$

Assume

$$T_s = 97^\circ\text{F}$$

$$\Delta H_s^{\text{vap}} = 1038.7 \text{ Btu/lb}$$

$$P_A^s = 45 \text{ torr}$$

$$\mathcal{H}_s = \frac{18.02}{28.97} \left(\frac{45}{760 - 45} \right) = 0.0391 \frac{\text{lb H}_2\text{O}}{\text{lb dry air}}$$

$$T_s = 131 - \frac{1038.7}{0.254} (0.0391 - 0.03) = 94^\circ\text{F}$$

As can be seen, this iterative calculation is very sensitive. By interpolation, T_s is almost identical to the value of 96.5°F read from Figure 18.17.

For systems other than air–water, the Lewis relation does not hold and it is common to indicate this by defining a psychrometric ratio as $h/(k_y C_s M_B)$. Calculated values of this ratio, using the Chilton–Colburn analogy, are included in Table 18.5 for several systems. Comparing (18-18) for the wet-bulb temperature to (18-24) for the adiabatic-saturation temperature, it can be noted that since the inverse of the psychrometric ratio for all air–organic vapor systems in Table 18.5 is less than one, such systems will have slopes of adiabatic-saturation lines that are less than slopes of wet-bulb lines. An example of this is the psychrometric chart for the air–toluene system at 1 atm shown in Figure 18.20. The use of this chart is illustrated in the next example.

EXAMPLE 18.4

Air is used to dry a solid wet with toluene at 1 atm. At a location where the air has a temperature of 140°F and a relative humidity of 10%, determine the humidity, the adiabatic-saturation temperature, and the wet-bulb temperature.

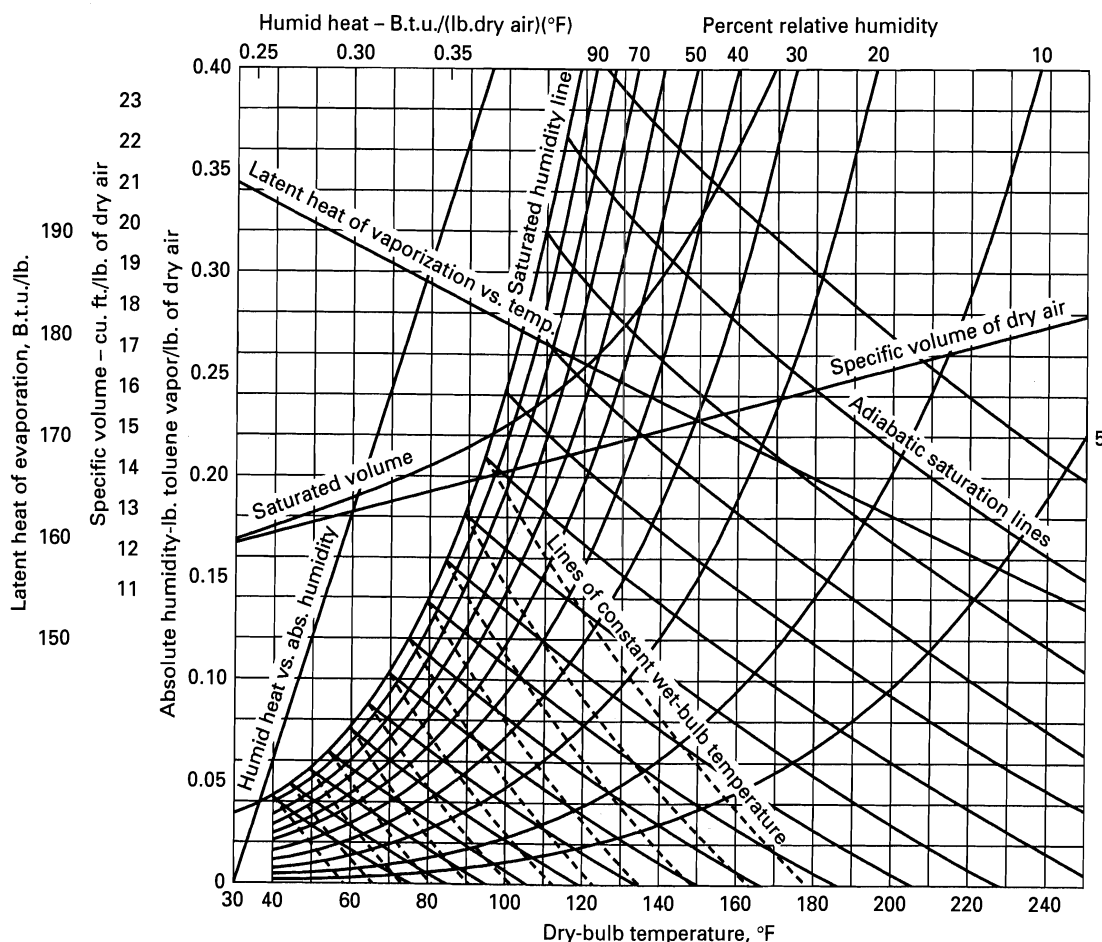


Figure 18.20 Psychrometric (humidity) chart for air–toluene at 1 atm.

[From *Perry's Chemical Engineers' Handbook*, 6th ed., R.H. Perry, D.W. Green, and J.O. Maloney, Eds., McGraw-Hill, New York (1984) with permission.]

SOLUTION

From Figure 18.20, the humidity is 0.062 lb toluene/lb dry air at the intersection point of a vertical temperature line and a curved percent relative-humidity line.

By following an interpolated adiabatic-saturation line from that intersection point to the saturated-humidity line, $T_s = 83^\circ\text{F}$.

By following an interpolated wet-bulb line from the intersection point to the saturated-humidity line, $T_w = 92^\circ\text{F}$.

Thus, the wet-bulb temperature is higher than the adiabatic-saturation temperature. In the next section, this causes a reduction in the driving force available for heat transfer during the drying of solids wet with organic moisture.

Moisture-Evaporation Temperature

The combination of the concepts of the adiabatic-saturation temperature and the wet-bulb temperature can be used to track the gas temperature and the moisture-evaporation (solid) temperature, T_v , with respect to location or time when removing surface moisture from a wet solid. The accuracy of the tracking is subject to the degree to which the assumptions of the two concepts are valid. If the moisture is water, T_v will be constant and equal to the constant T_w of the gas. If the moisture is an organic compound with properties similar to those in Table 18.5, T_v will still be equal to T_w , but, as shown next, that temperature will not be constant, but will decrease as the gas temperature decreases.

Let T_o and $\mathcal{H}_o$ be hypothetical entering conditions of a gas being used to dry a wet solid in an adiabatic, direct-heat dryer. At any point in the dryer, the conditions of the gas are given from (18-24) as

$$T_o - T_g = \frac{\Delta H_s^{\text{vap}}}{(C_s)_o} (\mathcal{H}_g - \mathcal{H}_o) \quad (18-25)$$

Assuming a quasi-steady-state transport condition at any point, the gas-liquid moisture interface conditions of T_v and $\mathcal{H}_v$ are related by the following form of (18-18):

$$T_v - T_g = \frac{k_y M_B \Delta H_s^{\text{vap}}}{h} (\mathcal{H}_g - \mathcal{H}_v) \quad (18-26)$$

For a continuous dryer, (18-26) holds regardless of whether the flows of gas and wet solid are countercurrent or cocurrent. Equations (18-25) and (18-26) can be combined to give

$$\frac{T_v - T_g}{T_o - T_g} = \left(\frac{k_y M_B (C_s)_o}{h} \right) \left(\frac{\mathcal{H}_g - \mathcal{H}_v}{\mathcal{H}_g - \mathcal{H}_o} \right) \quad (18-27)$$

The coefficient of the right-hand side of (18-27) is the inverse of the psychrometric ratio. For the air-water system, that ratio can be assumed to be 1.0, giving

$$\frac{T_v - T_g}{T_o - T_g} = \frac{\mathcal{H}_g - \mathcal{H}_v}{\mathcal{H}_g - \mathcal{H}_o} \quad (18-28)$$

Now assume that the wet solid in contact with the initial or entering gas is at the wet-bulb temperature, T_w , of the gas.

Then, at that point,

$$\frac{T_w - T_g}{T_o - T_g} = \frac{\mathcal{H}_g - \mathcal{H}_w}{\mathcal{H}_g - \mathcal{H}_o} \quad (18-29)$$

But, since T_o is fixed, T_v must remain at the value T_w regardless of the value of T_g . The result, which is independent of the direction of flow of the wet solid, is shown for the outcome of Example 18.5 below and in Figure 18.21 for air-water and air-toluene systems.

In the general case, (18-27) applies. Assume that the wet solid is dried batchwise, or flows in cocurrent flow to the gas in a continuous dryer, with an initial temperature equal to the wet-bulb temperature of the gas. Because the inverse of the psychrometric ratio is less than 1.0, T_w will now decrease as T_g decreases, as shown in the following example.

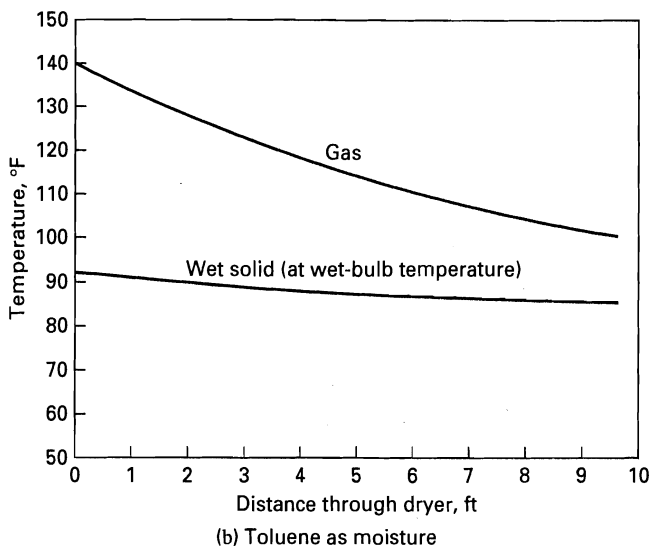
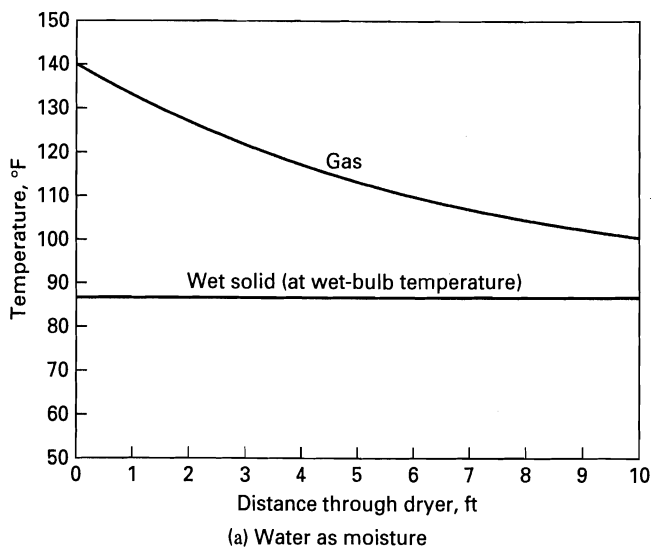


Figure 18.21 Temperature profiles for dryer of Example 18.5.

EXAMPLE 18.5

Air enters a continuous, adiabatic, direct-heat dryer at 140°F and 1 atm with the relative humidity below, and exits at 100°F. The wet solid enters in cocurrent flow at the wet-bulb temperature of the entering gas. Determine and plot the variation of the moisture-evaporation temperature, T_v , as a function of the distance through the dryer, for an exponential decrease of T_g according to the relation

$$T_g - T_s = (T_o - T_s) \exp(-0.1377z) \quad (1)$$

where z is distance through the dryer in feet and temperatures are in °F for:

- (a) water moisture with entering air of 12.5% $\mathcal{H}_R$
 (b) toluene moisture with entering air of 10% $\mathcal{H}_R$

SOLUTION

$$T_o = 140^\circ\text{F}, \quad \mathcal{H}_R = 12.5\%$$

- (a) From Figure 18.17, $\mathcal{H}_o = 0.015$ lb H₂O/lb dry air. $T_w = 86.5^\circ\text{F} = T_s$. As the gas cools, its humidity will follow the adiabatic-saturation line.

Using (1) and Figure 18.17,

z , ft	T_g , °F	$\mathcal{H}_g$, $\frac{\text{lb H}_2\text{O}}{\text{lb dry air}}$
0	140.0	0.015
2	127.1	0.018
4	117.3	0.020
6	109.9	0.022
8	104.3	0.0235
10	100.0	0.0245

Equation (18-28) is only satisfied for

$$T_v = T_w = 86.5^\circ\text{F}$$

$$\mathcal{H}_v = \mathcal{H}_w = 0.0275 \text{ lb H}_2\text{O/lb dry air}$$

For example, take $T_g = 109.9^\circ\text{F}$, $\mathcal{H}_g = 0.022$ lb H₂O/lb dry air. Using (18-28), take values of $T_v = 80^\circ\text{F}$ and 90°F , and compute the temperature and humidity ratios:

T_v , °F	$\mathcal{H}_v$, $\frac{\text{lb H}_2\text{O}}{\text{lb dry air}}$	$\frac{T_v - T_g}{T_o - T_g}$	$\frac{\mathcal{H}_g - \mathcal{H}_v}{\mathcal{H}_g - \mathcal{H}_o}$
86.5	0.0275	-0.777	-0.786
80.0	0.0223	-0.993	-0.043
90.0	0.0310	-0.661	-1.300

- (b) From Figure 18-20, for $T_o = 140^\circ\text{F}$, $\mathcal{H}_R = 10\%$,

$$\mathcal{H}_o = 0.062 \text{ lb toluene/lb dry air}$$

$$T_w \text{ for entering air} = 92^\circ\text{F}$$

$$T_s = 83^\circ\text{F}$$

From Table 18.4, $1/(\text{psychrometric ratio}) = 1/1.908 = 0.524$.
 From (18-27),

$$\frac{T_v - T_g}{140 - T_g} = 0.524 \left(\frac{\mathcal{H}_g - \mathcal{H}_v}{\mathcal{H}_g - 0.062} \right) \quad (2)$$

From (1),

$$T_g = T_v + (140 - T_v) \exp(-0.1377z) \quad (3)$$

Equation (2) is solved iteratively for T_v for each value of T_g , where $\mathcal{H}_v$ is the saturation humidity at T_v , as determined from Figure 18.20. For example, for $T_g = 115.9^\circ\text{F}$, with the humidity following the adiabatic saturation line, $\mathcal{H}_v = 0.095$, and (2) becomes:

$$\frac{T_v - 115.9}{140 - 115.9} = 0.524 \left(\frac{0.095 - \mathcal{H}_v}{0.095 - 0.062} \right)$$

or

$$T_v = 115.9 + 382.7(0.095 - \mathcal{H}_v) \quad (4)$$

This equation is solved by assuming T_v , determining $\mathcal{H}_v$ and then computing T_v .

Assumed T_v , °F	$\mathcal{H}_v$ from Fig. 18.20, lb toluene/lb dry air	T_v , °F from (4)
90	0.180	83.4
88	0.173	86.0
87	0.165	89.0

By interpolation, $T_v = 87.5^\circ\text{F}$.

This result can also be obtained graphically from Figure 18.20 using the construction shown in Figure 18.22, with an essentially identical result. Calculations for the other values of T_g give

T_g , °F	T_v , °F
140.0	92.0
126.3	90.0
115.9	87.5
107.9	86.5
100.0	85.5

From (3), the following values of z are computed:

T_g , °F	z , ft
140.0	0.0
126.3	2.3
115.9	4.5
107.9	6.7
100.0	9.6

Thus, the moisture-evaporation temperature, which is plotted in Figure 18.21b, decreases with decreasing gas temperature as discussed above.

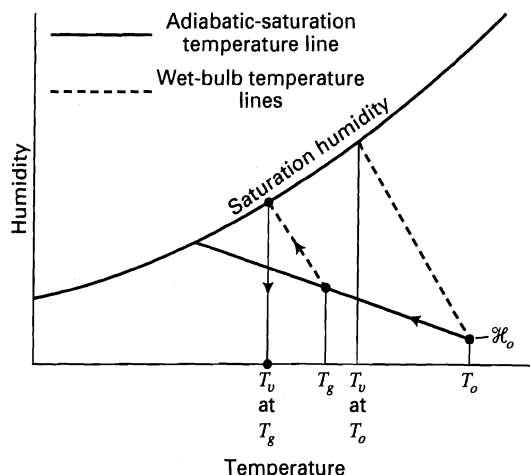


Figure 18.22 Adiabatic drying path for general vapor-moisture mixtures.

18.3 EQUILIBRIUM-MOISTURE CONTENT OF SOLIDS

As discussed by Faust et al. [9], wet solids can be grouped into the two following categories according to their drying behavior:

1. *Granular or crystalline solids that hold moisture in open pores between particles.* These are mainly inorganic materials, examples of which are crushed rocks, sand, catalysts, titanium dioxide, zinc sulfate, and sodium phosphates. During the drying process, the solid is unaffected by the removal of the moisture such that the selection of drying conditions and rate of drying are not critical to the final properties and appearance of the dried product. Materials in this category can be dried rapidly to very low moisture contents.
2. *Fibrous, amorphous, and gel-like materials that dissolve moisture or trap moisture in fibers or very fine pores.* These are mainly organic solids, including tree, plant, vegetable, and animal materials, such as wood, leather, soap, wood, eggs, glues, cereals, starch, cotton, and wool. These materials are significantly affected by moisture removal, often shrinking when dried and swelling when wetted. With this class of materials, drying in the later stages can be slow. If the surface is caused to dry too rapidly, moisture and temperature gradients can cause checking, warping, case hardening, and/or cracking. Therefore, the selection of drying conditions is a critical factor. Drying to low moisture contents is only possible when using a gas of low humidity.

In a direct-heat drying process, the extent to which moisture can be removed from a solid is limited, particularly for the second category of materials, by the *equilibrium-moisture content* of the solid, which depends on several factors including temperature, pressure, and moisture content of the gas. Even if the drying conditions produce a completely dry solid, subsequent exposure of the solid to an atmosphere of different humidity conditions can result in resorption of moisture.

A number of terms have been defined to describe and utilize the equilibrium-moisture content. These are shown in Figure 18.23 with reference to a hypothetical equilibrium isotherm. The moisture content, X , is expressed as mass of moisture per 100 mass units of bone-dry solid. This is the most commonly used way to express moisture content of a solid and is equivalent to weight % moisture on a dry-solid basis. This is analogous to expressions for humidity and is most convenient in drying calculations where the mass of bone-dry solid and bone-dry gas remain constant while moisture is transferred from the solid to the gas. Less common is weight % moisture on a wet-solid basis, W . The two moisture contents are related by the expression

$$X = \frac{100W}{100 - W} \quad (18-30)$$

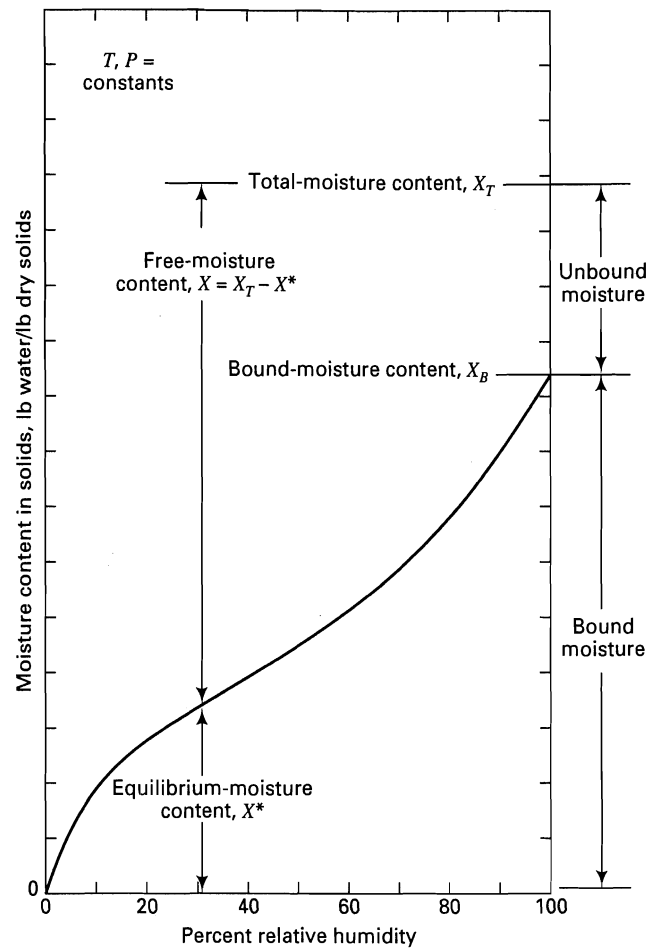


Figure 18.23 Typical isotherm for equilibrium moisture content.

or

$$W = \frac{100X}{100 + X} \quad (18-31)$$

Rarely used is moisture content on a volume basis because the volume occupied by wet solids of the second category shrinks during drying. Also, moisture content is almost never expressed on a mole basis because the molecular weight of the dry solid may not be known.

In Figure 18.23, the equilibrium-moisture content, X^* , is plotted for a typical solid of the second category, for a given temperature and pressure against the relative humidity, $\mathcal{H}_R$, where the limit is 100%. In some cases, humidity, $\mathcal{H}$, is used with a limit of the saturation humidity, $\mathcal{H}_s$, 1. At $\mathcal{H}_R = 100\%$, the equilibrium-moisture content is called *bound moisture*, X_B . If the wet solid has a total moisture content, X_T , greater than X_B , the excess, $X_T - X_B$, is called *unbound moisture*. At an intermediate relative humidity less than 100%, the excess of X_T over the equilibrium-moisture content, i.e., $X_T - X^*$, is called the *free-moisture content*.

In the presence of a saturated gas, only the unbound moisture can be removed during the drying process. For a partially saturated gas, only the free moisture can be removed. But if $\mathcal{H}_R = 0$, all solids, given enough time, may be dried to a bone-dry state. Solid materials that can contain bound moisture are *hygroscopic*. Bound moisture exhibits a vapor

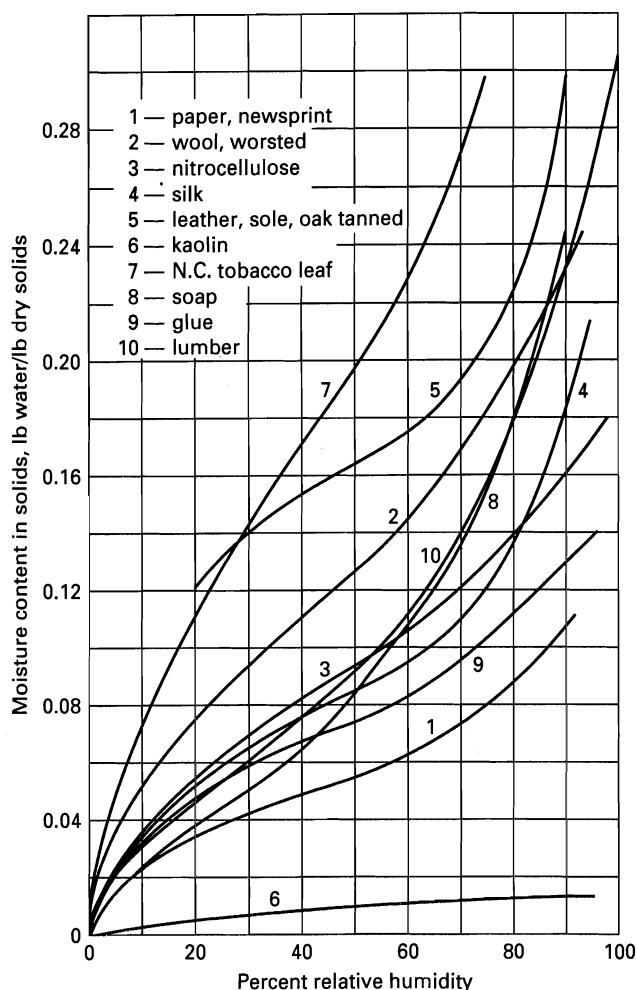


Figure 18.24 Equilibrium-moisture content at 25°C and 1 atm.

pressure less than the normal vapor pressure. The bound-moisture content of cellular materials such as wood is often referred to as the fiber-saturation point.

Equilibrium-moisture isotherms at 25°C and 1 atm are shown for several materials of the second category in Figure 18.24. Such curves must be determined experimentally. At low values of $\mathcal{H}_R$, e.g., < 10%, moisture is bound to the solid on its surfaces as an adsorbed monomolecular layer. Such bound moisture can also be present on solids of the first category. At intermediate values of $\mathcal{H}_R$, e.g., 20–60%, multimolecular layers may be built up successively on the monolayer. At large values of $\mathcal{H}_R$, e.g., > 60%, moisture is held in micropores so small (e.g., < 1 μm is radius) that a lowering of the vapor pressure occurs according to the Kelvin equation (15-14). In cellular materials, such as plant and tree matter, some of the moisture is held osmotically in fibers behind semipermeable membranes of cell walls.

Temperature has a significant effect on the equilibrium-moisture content, an example of which is shown in Figure 18.25 for cotton over a temperature range of 96–302°F. At an $\mathcal{H}_R$ of 20%, the equilibrium-moisture content decreases from 0.037 to 0.012 lb H_2O /lb dry cotton. The experimental determination of equilibrium-moisture isotherms is complicated by an apparent hysteresis effect, as shown in

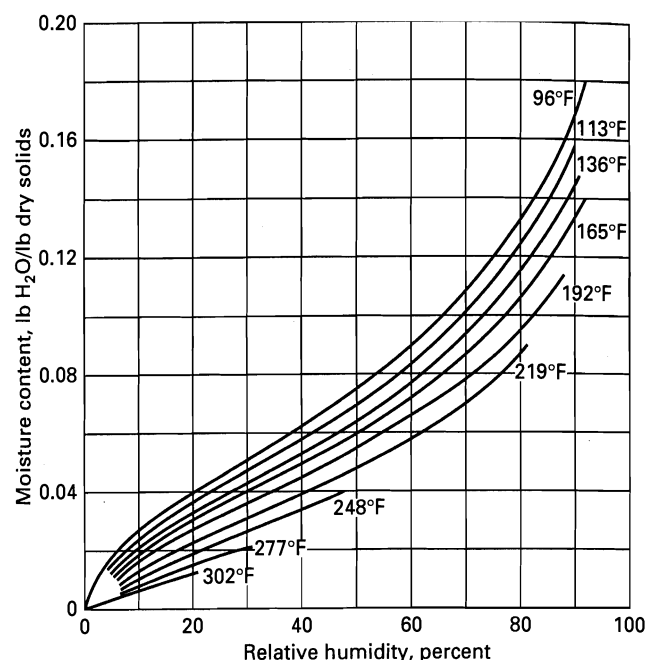


Figure 18.25 Effect of temperature on the equilibrium-moisture content of raw cotton at 1 atm.

Figure 18.26 for sulfite pulp. The sorption and desorption curves were obtained by wetting and drying the solid, respectively. The equilibrium-moisture content measured in drying experiments is always somewhat higher, particularly in the percent relative-humidity range of 30% to 80%. According to Luikov [10] the hysteresis effect may be due to either: (1) a failure to achieve true equilibrium or (2) irreversibility of evaporation and condensation in capillaries. For the latter, a possible explanation is based on the representations of moisture in necked capillaries, as shown in Figure 18.26. For drying (desorption), the capillary contains more moisture than for wetting (sorption). Thus, for a given relative humidity, the equilibrium moisture content for drying is less than for wetting.

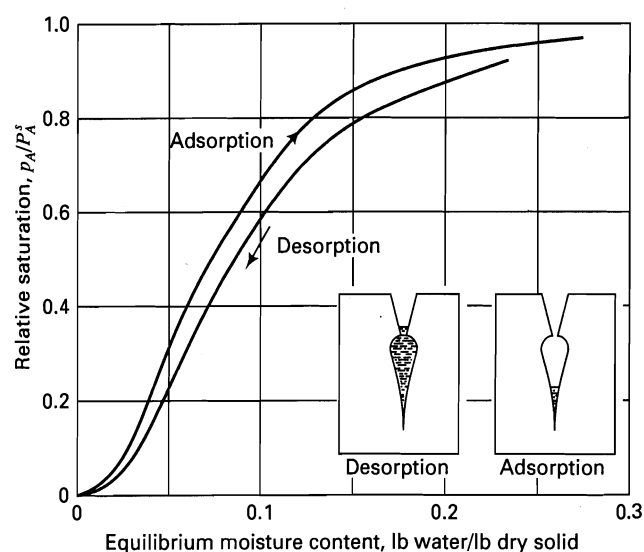


Figure 18.26 Effect of hysteresis on equilibrium-moisture content of sulfite pulp.

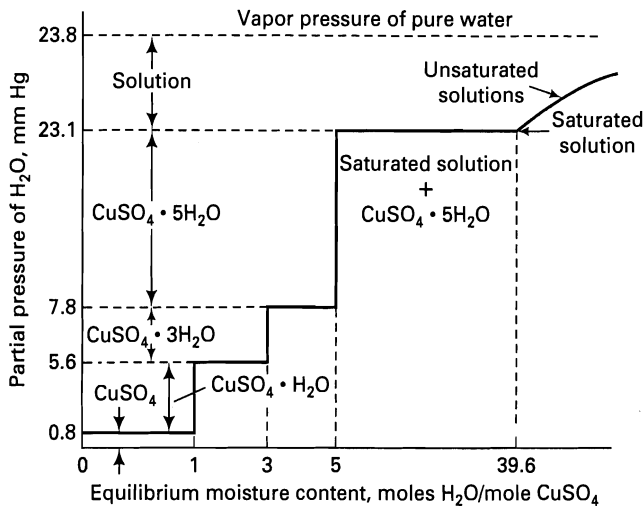


Figure 18.27 Equilibrium-moisture content for CuSO_4 at 25°C .

Bound moisture can also be defined as moisture that is held chemically as, e.g., water of hydration of inorganic crystals. This is one example of bound moisture dissolved in the solid. The vapor pressure of such moisture is lowered significantly below the true vapor pressure. For inorganic salts that form one or more hydrates, the hydrated form of the product will depend not only on the temperature, but also on the relative humidity of the gas in contact with the crystals. The effect of the latter for CuSO_4 , in terms of the partial pressure of water, is shown in Figure 18.27. At 25°C , the stable hydrate is $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$. However, if the partial pressure of H_2O is between 5.6 and 7.8 torr, the trihydrate tends to form. From 5.6 to 0.8 torr, the monohydrate is favored. Below 0.8 torr, CuSO_4 crystals are completely free of water.

EXAMPLE 18.6

One-kilogram blocks of wet Borax laundry soap with an initial water content of 20.2 wt% on the dry basis are dried with air in a tunnel dryer at 1 atm. In the limit, if the soap were brought to equilibrium with the air at 25°C and a relative humidity of 20%, determine the kg of moisture evaporated from each block.

SOLUTION

The initial moisture content of the soap on a wet basis is obtained from a rearrangement of (18-30):

$$W = \frac{100X}{100 + X} = \frac{100(20.2)}{100 + 20.2} = 16.8 \text{ wt\%}$$

Initial weight of moisture = $0.168(1.0) = 0.168 \text{ kg H}_2\text{O}$

Initial weight of dry soap = $1 - 0.168 = 0.832 \text{ kg dry soap}$

From Figure 18.24, for soap at $\mathcal{H}_R = 0.20$, $X^* = 0.037$

Final weight of moisture = $0.037(0.832) = 0.031 \text{ kg}$

Moisture evaporated = $0.168 - 0.031 = 0.137 \text{ kg H}_2\text{O/kg soap block}$

18.4 DRYING PERIODS

During drying of either category of wet solids, the decrease in average moisture content, X , as a function of time, t , for fixed gas conditions in a direct-heat dryer was observed experimentally by Sherwood [11, 12] to exhibit generally the type of relationship shown in Figure 18.28a provided that the exposed surface of the solid is initially covered with observable moisture. If that curve is differentiated with respect to time and multiplied by the ratio of the mass of dry solid to the interfacial area of contact between the mass of wet solid and the gas, a plot can be made of drying-rate flux, R ,

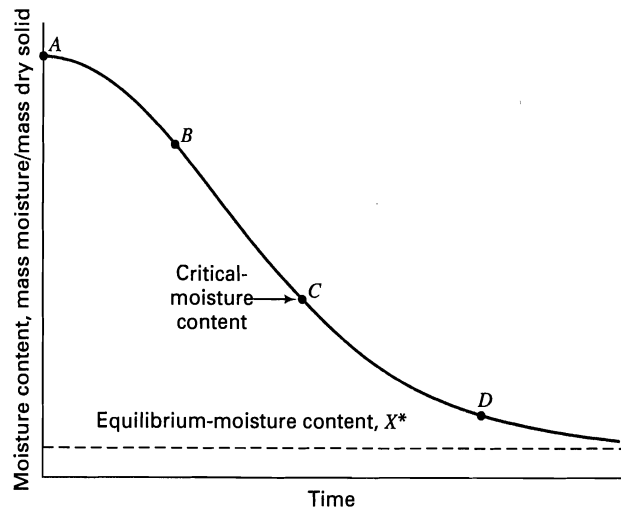
$$R = \frac{dm_v}{A dt} = -\frac{m_s}{A} \frac{dX}{dt} \quad (18-32)$$

where

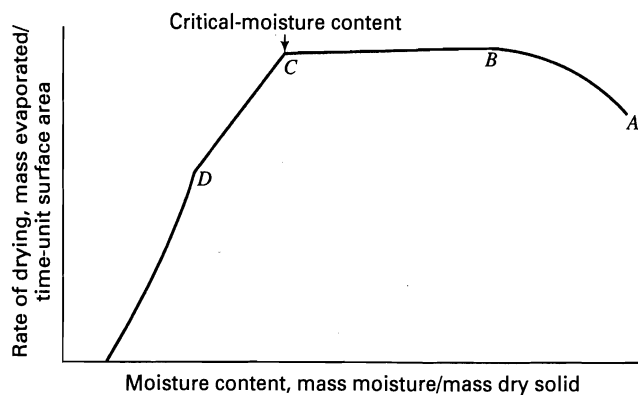
m_v = mass of moisture evaporated

m_s = mass of bone-dry solid

as a function of moisture content, as shown in Figure 18.28b. In both plots, the final equilibrium-moisture content is X^* . Although both plots can exhibit four drying periods, the periods are more distinct in the drying-rate curve. For some wet materials and/or some hot gas conditions, less than four



(a) Moisture content



(b) Drying rate

Figure 18.28 Drying curves for constant drying conditions.

drying periods may be observed. In the period from A to B, the wet solid is being preheated to an exposed-surface temperature equal to the wet-bulb temperature of the gas. Some moisture is evaporated in this preheat period, at an increasing rate, as the surface temperature increases.

At the end of the preheat period, if the wet solid is of the granular character of the first category, a cross section has the appearance of Figure 18.29a, where the exposed surface is still covered by a film of moisture. A wet solid of the second category is covered on the exposed surface by free moisture. The drying rate now becomes constant during the period from B to C, which prevails as long as free moisture still covers the exposed surface.

This surface moisture may be part of the original moisture that covered the surface or it may be moisture brought to the surface by capillary action in the case of wet solids of the first category or by liquid diffusion in the case of wet solids of the second category. In either case, the rate of drying is controlled by external mass and heat transfer between the exposed surface of the wet solid and the bulk

gas. The migration of moisture from the interior of the wet solid to the exposed surface is not a rate-affecting factor. This period is referred to as the *constant-rate drying period*. It terminates at point C, referred to as the *critical moisture content*. When drying wet solids of the first category under agitated conditions, as in a direct-heat rotary dryer, fluidized-bed dryer, flash dryer, or agitated batch dryer, such that all particle surfaces are in direct contact with the gas, the constant-rate drying period may extend all the way to X^* .

At the beginning of the period from C to D, the moisture just barely covers the exposed surface. From then until point D is reached, as shown in Figure 18.29b, the surface tends to a dry state because the rate of liquid travel by diffusion or capillary action to the exposed surface is not sufficiently fast. In this period, the exposed-surface temperature remains at the wet-bulb temperature if heat conduction is adequate, but the wetted exposed area for mass transfer decreases. Consequently, the rate of drying decreases linearly with decreasing average moisture content. This period is referred to as the *first falling-rate drying period*. It is not always observed with wet solids of the second category.

During the period from C to D, the liquid in the pores of wet solids of the first category begins to recede from the exposed surface. In the final period from D to E, as shown in Figure 18.29c, evaporation occurs from liquid surfaces in the pores, where the wet-bulb temperature prevails. However, the temperature of the exposed surface in the solid rises to approach the dry-bulb temperature of the gas. During this period, called the *second falling-rate drying period*, the rate of drying may be controlled by vapor diffusion for wet solids of the first category and by liquid diffusion for wet solids of the second category. The rate falls exponentially with decreasing moisture content.

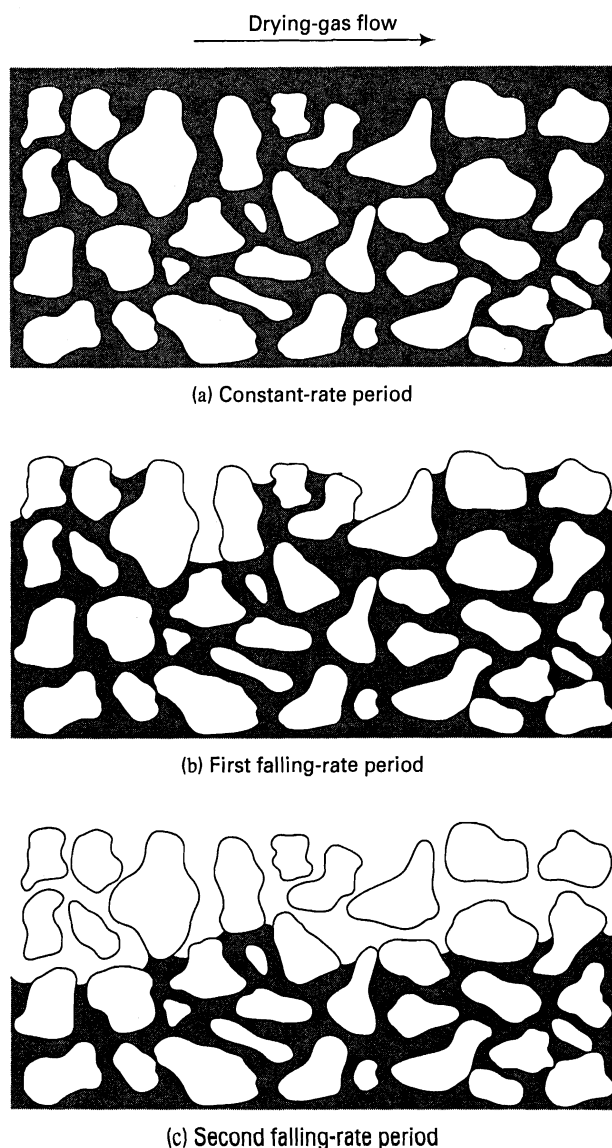


Figure 18.29 Drying stages for granular solids.

Constant-Rate Drying Period

In direct-heat equipment, drying involves the transfer of heat from the gas to the surface and interior of the wet solid, and mass transfer of moisture from the interior and surface of the solid to the gas. During the constant-rate period, the rate of mass transfer is determined by the resistance of the boundary layer or film of the gas phase in contact with the exterior wet surface of the solid. The wet solid is assumed to be at a uniform temperature so that the only resistance to convective heat transfer is also in the gas phase. The rate of moisture evaporation can then be based on convective heat transfer or mass transfer, according to the following conventional, but simplified, transport relationships, where thermal radiation, the bulk-flow effect for mass transfer, and the sensible-heat effect for the evaporated moisture are ignored:

$$\frac{dm_v}{dt} = \frac{h(T_g - T_i)A}{\Delta H_i^{\text{vap}}} = M_A k_y (y_i - y_g) A \quad (18-33)$$

where the subscript, i , refers to the gas at the interface with the solid.

As discussed in the previous section, the interface at these conditions is at the wet-bulb temperature, T_w . Although drying-rate calculations could be based on mass transfer using (18-33), it is more common to use the heat-transfer relation of (18-33) when air is the gas and water is the moisture because of the wide availability of the psychrometric chart for that system, the equality of the wet-bulb and adiabatic-saturation temperatures, and a wider availability of correlations for convective heat transfer than for mass transfer, although transport analogies can often be used to derive one from the other. Combining (18-32) and (18-33), the drying-rate flux for the constant-rate drying period, R_c , becomes

$$R_c = \frac{h(T_g - T_w)}{\Delta H_w^{\text{vap}}} \quad (18-34)$$

while a less-useful equivalent mass-transfer form is obtained by combining (18-16), (18-32), and (18-33):

$$R_c = M_B k_y (\mathcal{H}_w - \mathcal{H}_d) \quad (18-35)$$

where d and w refer to gas dry-bulb and wet-bulb conditions, respectively.

For some dryers, it is preferable to use a volumetric heat-transfer coefficient, (ha) , defined by

$$\frac{dm_v}{dt} = \frac{(ha)(T_g - T_i)V}{\Delta H_w^{\text{vap}}} \quad (18-36)$$

where

a = external surface area of wet solids per unit volume of dryer

V = volume of dryer

Then, the drying rate per unit dryer volume during the constant-rate drying period is

$$(R_c)_V = \frac{(ha)(T_d - T_w)}{\Delta H_w^{\text{vap}}} \quad (18-37)$$

Equations for estimating interphase heat-transfer coefficients were discussed for several geometries in Chapter 3. Empirical equations that are particularly useful for drying-rate calculations are summarized in Mujumbar [1]. Representative equations, when the gas is air, are listed here in Table 18.6. There, in (1), G is the mass velocity of air in the flow channel that passes over the wet surface. In (2), G is the mass velocity of the air impinging on the wet surface. In (3) to (8), d_p is the particle diameter and G is the superficial mass velocity.

The dramatic effect of exposed surface area of wet solids in drying operations was shown by Marshall and Hougen [13] and is illustrated in the following two examples that deal with batch drying. In Example 18.7, cross-circulation, batch tray drying is used to dry slabs of filter cake. In Example 18.8, the filter cake is extruded and then dried by through-circulation. The difference in the two drying times for the constant-rate drying period is very significant.

Table 18.6 Empirical Equations for Interphase Heat-transfer Coefficients for Application to Dryers (h in $\text{W/m}^2\text{-K}$, G in kg/hr-m^2 , d_p in m)

Geometry	Equation
Flat-plate, parallel flow	$h = 0.0204 G^{0.8}$ ($T_d = 45 - 150^\circ\text{C}$, $G = 2,450 - 29,300$) (1)
Flat-plate, perpendicular, impingement flow	$h = 1.17 G^{0.37}$ ($G = 3,900 - 19,500$) (2)
Packed beds, through-circulation	$h = 0.151 G^{0.59} / d_p^{0.41}$, ($N_{\text{Re}} > 350$) (3) $h = 0.214 G^{0.49} / d_p^{0.51}$ ($N_{\text{Re}} < 350$) (4)
Fluidized beds	$N_{\text{Nu}} = 0.0133 N_{\text{Re}}^{1.6}$ ($0 < N_{\text{Re}} < 80$) (5)
Pneumatic conveyors	$N_{\text{Nu}} = 0.316 N_{\text{Re}}^{0.8}$ ($8 < N_{\text{Re}} < 500$) (6)
Droplets in spray dryers	$N_{\text{Nu}} = 2 + 1.05 N_{\text{Re}}^{0.5} N_{\text{Pr}}^{1/3} N_{\text{Gu}}^{0.175}$ ($N_{\text{Re}} < 1000$) (7)
Spouted beds	$N_{\text{Nu}} = 0.0005 N_{\text{Re}_s}^{1.46} (u/u_s)^{1/3}$ (8)

$$N_{\text{Re}} = d_p G / \mu$$

$$N_{\text{Nu}} = h d_p / k$$

$$N_{\text{Pr}} = C_P \mu / k$$

$$N_{\text{Re}_s} = d_p G_s / \mu$$

$$G_s = \text{mass velocity for incipient spouting}$$

$$u = \text{velocity}$$

$$u_s = \text{incipient spouting velocity}$$

$$N_{\text{Gu}} = (T_d - T_w) / T_d \text{ in absolute temperature}$$

$$d_p = \text{particle size}$$

$$C_P = \text{specific heat of gas}$$

$$\mu = \text{viscosity of gas}$$

$$k = \text{thermal conductivity of gas}$$

EXAMPLE 18.7

A filter cake of calcium carbonate contained in a tray is to be dried by cross-circulation from the top surface. Each tray is 2.5 cm high with an area of 1.5 m^2 and is completely filled with 73 kg of wet filter cake having a water content of 30% on the dry basis. The heating medium is air at 1 atm and 170°F with a relative humidity of 10%. The average velocity of the air passing across the wet solid is 4 m/s. Estimate the time in hours to reach the experimentally determined, critical-moisture content (end of the constant-rate period) of 10% on the dry basis, if the preheat period is neglected.

SOLUTION

$$\text{H}_2\text{O in wet cake} = \left(\frac{30}{130} \right) (73) = 16.8 \text{ kg}$$

$$\text{H}_2\text{O in cake at } X_c = 0.10(73 - 16.8) = 5.6 \text{ kg}$$

$$m_v = \text{H}_2\text{O evaporated} = 16.8 - 5.6 = 11.2 \text{ kg}$$

For the constant-rate drying period, the heat-transfer form of (18-33) applies, which upon integration gives

$$t_c = \frac{m_v \Delta H_w^{\text{vap}}}{h(T_d - T_w)A}$$

where t_c is the time to reach the critical moisture content. From the humidity chart of Figure 18.17,

$$T_w = 100^\circ\text{F and } \mathcal{H} = 0.026 \frac{\text{lb H}_2\text{O}}{\text{lb dry air}}$$

$$T_d = 170^\circ\text{F} = 76.7^\circ\text{C}$$

$$T_d - T_w = 170 - 100 = 70^\circ\text{F} = 38.9\text{ K}$$

$$\text{At } T_w = 100^\circ\text{F}, \Delta H_w^{\text{vap}} = 1037.2 \text{ Btu/lb} = 2,413 \text{ kJ/kg}$$

$$G = u_{\text{avg}} \rho$$

From (18-10), Table 18.4,

$$v_H = 0.730(170 + 460) \left(\frac{1}{28.97} + \frac{0.026}{18.02} \right) = 16.5 \text{ ft}^3/\text{lb dry air}$$

or

$$\begin{aligned} 16.5/(1 + 0.026) &= 16.1 \text{ ft}^3/\text{lb moist air} \\ &= 1.004 \text{ m}^3/\text{kg moist air} = 1/\rho \end{aligned}$$

$$G = 4/1.004 = 3.98 \text{ kg/m}^2\text{-s} = 14,300 \text{ kg/m}^2\text{-h}$$

$$A = 1.5 \text{ m}^2$$

From Table 18.6, (1) applies for turbulent flow with T_d and G within the allowable range.

$$h = 0.0204(14,300)^{0.8} = 43 \text{ W/m}^2\text{-K} = 43 \text{ J/s-m}^2\text{-K}$$

From (1), using SI units

$$t_c = \frac{(11.2)[(2,413)(1,000)]}{(43)(38.9)(1.5)} = 10,800 \text{ s} = 2.99 \text{ h}$$

EXAMPLE 18.8

The filter cake of Example 18.7 is extruded into cylindrical-shaped pieces of 1/4-in. diameter and 1/2-in. length to form a bed that is 1.5 m² in cross-sectional area and 5 cm high, with an external porosity of 50%. Air at 170°F and 10% relative humidity passes through the bed at a superficial velocity of 2 m/s (average interstitial velocity of 4 m/s). Estimate the time in hours to reach the critical-moisture content, if the preheat period is neglected. Compare this time to that estimated in Example 18.7.

SOLUTION

Compared to the tray of Example 18.7, the bed is twice as high with the same cross-sectional area. Therefore, for a porosity of 50%, the bed contains the same amount of wet solids. Thus, as in Example 18.7,

$$m_{\text{wet cake}} = 73 \text{ kg}$$

$$m_v = 11.2 \text{ kg H}_2\text{O evaporated}$$

Also,

$$\Delta H_w^{\text{vap}} = 2,413 \text{ kJ/kg}$$

$$T_d - T_w = 38.9 \text{ K}$$

Assume that the density of the extrusions equals the density of the filter cake.

$$\rho_{\text{filter cake}} = \frac{73}{1.5 \left(\frac{2.5}{100} \right)} = 1,950 \text{ kg/m}^3.$$

$$\begin{aligned} \text{Volume of one extrusion} &= \pi D^2 L / 4 = \frac{3.14(0.25)^2(0.5)}{4} \\ &= 0.0245 \text{ in}^3 = 4.01 \times 10^{-7} \text{ m}^3. \end{aligned}$$

$$\begin{aligned} \text{Number of extrusions} &= 1.5 \left(\frac{2.5}{100} \right) (0.5) / 4.01 \times 10^{-7} \\ &= 46,800. \end{aligned}$$

$$\text{Surface area/extrusion} = \pi DL + \pi D^2/2$$

$$\begin{aligned} &= 3.14 \left[(0.25)(0.50) + \frac{(0.25)^2}{2} \right] \\ &= 0.49 \text{ in}^2 = 0.000316 \text{ m}^2 \end{aligned}$$

$$A = 46,800(0.000316) = 14.8 \text{ m}^2$$

Thus, the transport area is 14.8/1.5 = 9.9 times that for Example 18.7. From Table 18.6, (3) or (4) applies for estimating h , depending on N_{Re} . From Example 18.7, but with a superficial bed velocity of 50% of the crossflow velocity,

$$G = 3.98/2 = 1.96 \text{ kg/m}^2\text{-s}$$

Equations (3) and (4) refer to the work of Gamson, Thodos, and Hougen [14] for $N_{\text{Re}} > 350$ and Wilke and Hougen [15] for $N_{\text{Re}} < 350$, respectively. For both correlations, d_p is taken as the diameter of a sphere of the same surface area as the particle. For the extrusions of this example with $L = 2D$,

$$\pi d_p^2 = \frac{\pi D^2}{2} + 2\pi D^2 = 2.5\pi D^2$$

Solving (1),

$$d_p = D\sqrt{2.5} = 0.25\sqrt{2.5} = 0.395 \text{ in.} = 0.010 \text{ m}$$

$$\mu \approx 0.02 \text{ cP} = 2 \times 10^{-5} \text{ kg/m-s}$$

$$N_{\text{Re}} = \frac{d_p G}{\mu} = \frac{(0.010)(1.96)}{2 \times 10^{-5}} = 980$$

Therefore, (3) applies and

$$h = 0.151(14,300/2)^{0.59}/(0.010)^{0.41} = 188 \frac{\text{J}}{\text{s-m}^2\text{-K}}$$

The h is 188/43 = 4.4 times greater than in Example 18.7. From (1) in that example,

$$t_c = \frac{(11.2)[(2,413)(1,000)]}{(188)(38.9)(14.8)} = 250 \text{ s} = 4.16 \text{ min.}$$

This example and the preceding example show that cross-circulation drying can take hours, while through-circulation drying may require only minutes.

Falling-rate Drying Period

When the rate of drying in the constant-rate period is high and/or the distance that interior moisture must travel to reach the surface is large, the moisture may eventually fail to reach the surface fast enough to maintain a constant rate of drying. Then, a transition to the falling-rate period will occur. In Examples 18.7 and 18.8, the constant rates of drying are, respectively, from (18-34):

$$R_c = \frac{43(38.9)(3,600)}{(2,413)(1,000)} = 2.50 \text{ kg/h-m}^2$$

and

$$R_c = \frac{(188)(38.9)(3,600)}{(2,413)(1,000)} = 10.9 \text{ kg/h-m}^2$$

However, in Example 18.7, the moisture may have to travel from as far away as 25 mm to reach the exposed surface,

while in Example 18.8, the distance is only 3.2 mm. Therefore, as a first approximation, it might be expected that the critical-moisture contents for the two examples might not be the same. The value of 10% on the dry basis was taken from through-circulation drying experiments.

During drying, when moisture travels from the interior of a wet solid to the surface, a moisture profile develops in the wet solid. The shape of this profile depends on the nature of the moisture movement, as discussed by Hougen, McCauley, and Marshall [16]. If the wet solid is of the first category, where the moisture is not held in solution or in fibers, but is held as free moisture in the interstices of powders and granular solids such as paint pigments, minerals, clays, soil, and sand, or is moisture above the fiber-saturation point in textiles, paper, wood, and leather, then moisture movement occurs by capillary action. For wet solids of the second category, the internal moisture is bound moisture, as in the last stages of the drying of clay, starch, flour, textiles, paper and wood, or soluble moisture, as in soap, glue, gelatin, and paste. This type of moisture migrates to the surface by liquid diffusion. Moisture can also migrate by gravity, external pressure, and by vaporization-condensation sequences in the presence of a temperature gradient. In addition, vapor diffusion through the solid can occur in indirect-heat dryers when heating and vaporization occur at opposed surfaces.

A typical moisture profile for capillary flow is shown in Figure 18.30a. The profile is concave upward near the exposed surface, concave downward near the opposed surface, and with a point of inflection in between. For flow of moisture by diffusion, as shown in Figure 18.30b, the profile is concave downward throughout. If the diffusivity is independent of the moisture content, the solid curve applies. If, as is often the case, the diffusivity decreases with decreasing moisture content, due mainly to shrinkage, the dashed profile applies.

During the falling-rate period of drying, idealized theories for capillary flow and diffusion can be applied to estimate drying rates. Alternatively, the estimate could be made by a strictly empirical approach that ignores the mechanism of moisture movement, but instead relies on the experimental determination of drying rate as a function of average moisture content for a particular set of drying conditions. The empirical approach is examined first.

Empirical Approach

The empirical approach ignores the preheat period and is usually applied to experimental data in the form of Figure 18.31a (Case 1), but can be modified to be applied to data of Figures 18.31b (Case 2) and 18.31c (Case 3). In these plots, the abscissa is the free-moisture content, $X = X_T - X$, as shown in Figure 18.23, for the particular drying conditions. This allows all three plots to be extended to the origin. However, for a given application, all free moisture may not be removed.

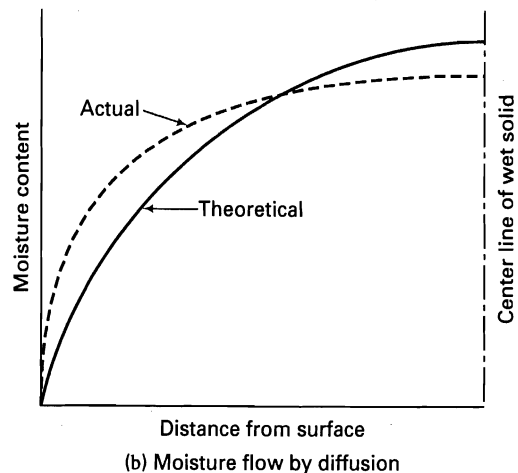
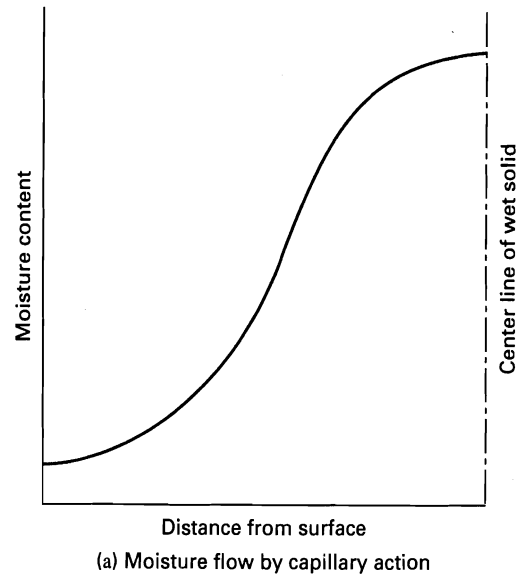


Figure 18.30 Moisture distribution in wet solids during drying.

[From W.L. McCabe, J.C. Smith, and P. Harriott, *Unit Operations of Chemical Engineering*, 5th ed., McGraw-Hill, New York (1993) with permission.]

From (18-32),

$$\int dt = -\frac{m_s}{A} \int \frac{dX}{R} \quad (18-38)$$

For the constant-rate period, and ignoring the preheat period, $R = R_c = \text{constant}$. Starting from an initial free-moisture content of X_o at time $t = 0$, the time to reach the critical free-moisture content, X_c , at time $t = t_c$ is obtained by integrating (18-38):

$$t_c = \frac{m_s(X_o - X_c)}{AR_c} \quad (18-39)$$

For Case 1 (Figure 18.31a) of the falling-rate period, the rate of drying is linear with X and terminates at the origin, according to

$$R = R_c X / X_c \quad (18-40)$$

Substituting (18-40) into (18-38) and integrating t from t_c to $t > 0$ and X from X_c to $X > 0$ gives the following

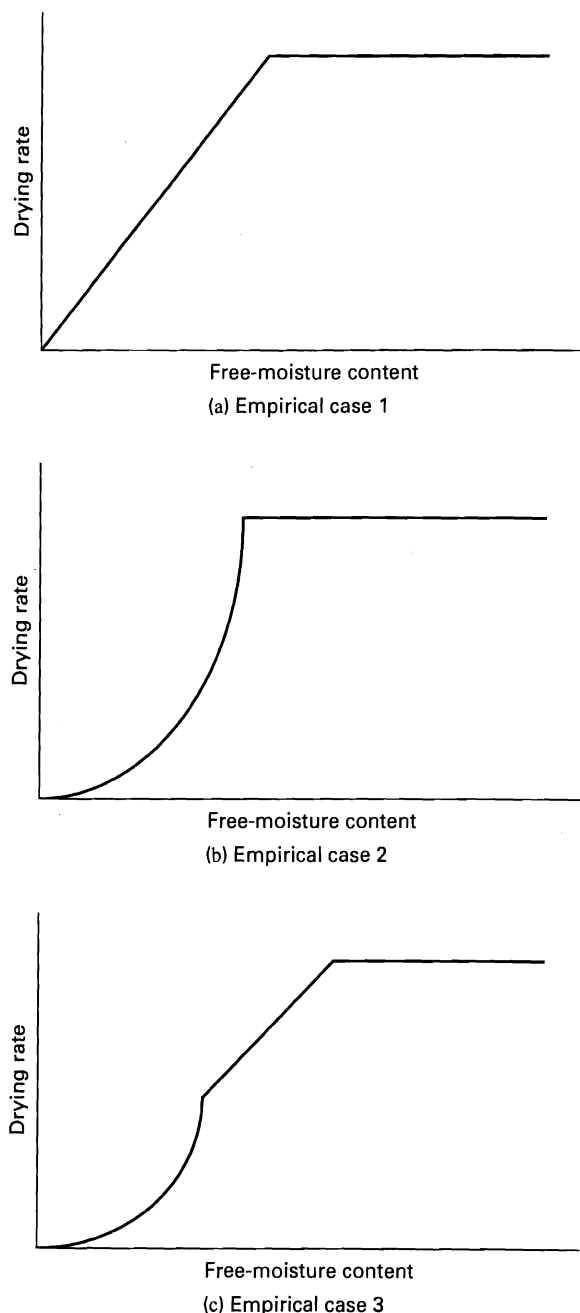


Figure 18.31 Drying-rate curves.

expression for the drying time in the falling-rate period, t_f :

$$t_f = t - t_c = \frac{m_s X_c}{A R_c} \ln \left(\frac{X_c}{X} \right) = \frac{m_s X_c}{A R_c} \ln \left(\frac{R_c}{R} \right) \quad (18-41)$$

The total drying time, t_T , is the sum of (18-39) and (18-41):

$$t_T = t_c + t_f = \frac{m_s}{A R_c} \left[(X_o - X_c) + X_c \ln \left(\frac{X_c}{X} \right) \right] \quad (18-42)$$

EXAMPLE 18.9

Marshall and Hougen [13] present experimental data for the through-circulation drying of 5/16-in. extrusions of ZnO in a bed of 1 ft² cross section by 1-in. high, using air of $T_d = 158^\circ\text{F}$ and

$T_w = 100^\circ\text{F}$ at a flow rate of 340 ft³/min. The data show a constant-rate period from $X_o = 33\%$ to $X_c = 13\%$, with $R_c = 1.42$ lb H₂O/h-lb bone-dry solid, followed by a falling-rate period that approximates Case 1 in Figure 18.31a. Calculate the drying time for the constant-rate period and the additional time in the falling-rate period to reach a free moisture content, X , of 1%.

SOLUTION

In (18-38), the drying rate, R , corresponds to mass of moisture evaporated per unit time per unit of exposed area of the wet material. In this example, the drying rate is not given per unit area, but per mass of bone-dry solid, with some associated exposed area. Equations (18-38) and (18-42) can be rewritten in terms of $R' = RA/m_s$, respectively, as

$$\int dt = - \int \frac{dX}{R'} \quad (1)$$

and

$$t_T = t_c + t_f = \frac{1}{R'_c} \left[(X_o - X_c) + X_c \ln \left(\frac{X_c}{X} \right) \right] \quad (2)$$

From (2), for just the constant-rate period,

$$t_c = \frac{1}{1.42} [0.33 - 0.13] = 0.141 \text{ h} = 8.45 \text{ min}$$

From (2), for just the falling-rate period,

$$t_f = \frac{1}{1.42} \left[0.13 \ln \left(\frac{0.13}{0.01} \right) \right] = 0.235 \text{ h} = 14.09 \text{ min}$$

The total drying time, ignoring the preheat period, is

$$t_T = t_c + t_f = 8.45 + 14.09 = 22.5 \text{ min}$$

For Case 2 (Figure 18.31b), R in the falling-rate period can be expressed as a parabolic function.

$$R = aX + bX^2 \quad (18-43)$$

The values of the parameters a and b are obtained by fitting (18-43) to the experimental drying-rate plot, subject to the constraint that $R = R_c$ at $X = X_c$. If (18-43) is substituted into (18-38) and the result is integrated for the falling-rate period from $X = X_c$ down to some final value of X_f and corresponding R_f , the time for the falling-rate period is found to be

$$t_f = t - t_c = \frac{m_s}{aA} \ln \left[\frac{X_c(a + bX_f)}{X_f(a + bX_c)} \right] = \frac{m_s}{aA} \ln \left[\frac{X_c^2 R_f}{X_f^2 R_c} \right] \quad (18-44)$$

EXAMPLE 18.10

Experimental data for through-circulation drying of 1/4-in.-diameter spherical pellets of a nonhygroscopic carburizing compound exhibit constant-rate drying of 1.9 lb H₂O/h-lb dry solid from $X_o = 30\%$ to $X_c = 21\%$, followed by a falling-rate period to $X_f = 4\%$ that fits (18-43) with $a = 3.23$ and $b = 27.7$ (both in lb H₂O/h-lb dry solid) for X as a fraction and R replaced by R' in lb H₂O/h-lb dry solid. Calculate the time for drying in the falling-rate period. Note that the values of a and b satisfy the constraint of $R'_c = 1.9$ at $X_c = 0.21$.

SOLUTION

For R' in the given units, (18-44) becomes

$$t_f = \frac{1}{a} \ln \left[\frac{X_c(a + bX_f)}{X_f(a + bX_c)} \right] \quad (1)$$

Thus,

$$t_f = \frac{1}{3.23} \ln \left\{ \frac{0.21}{0.04} \left[\frac{3.23 + 27.7(0.04)}{3.23 + 27.7(0.21)} \right] \right\} = 0.286 \text{ h} = 17.1 \text{ min}$$

For Case 3 (Figure 18.31c), the falling-rate period consists of two subregions. In the first subregion, which is linear,

$$R = \alpha X + \beta \quad (18-45)$$

with the constraints that $R = R_{c1}$ at X_{c1} , and $R = R_{c2}$ at X_{c2} . In the second subregion, (18-44) applies, but with the constraint that $R = R_{c2}$ at X_{c2} .

EXAMPLE 18.11

Experimental data of Sherwood [12] for the surface drying of a 3.18-cm-thick $\times$ 6.6-cm² cross-sectional area slab of a thick paste of CaCO_3 (whiting) from both sides by air at $T_d = 39.8^\circ\text{C}$ and $T_w = 23.5^\circ\text{C}$ and a cross-circulation velocity of 1 m/s exhibit the complex type of drying-rate curve shown in Figure 18.31c, with the following constants:

Constant-rate period:

$$X_o = 10.8\%$$

$$X_{c1} = 8.3\%$$

$$R_{c1} = 0.053 \text{ g H}_2\text{O/h-cm}^2$$

First falling-rate period:

$$X_{c2} = 3.7\%$$

$$R_{c2} = 0.038 \text{ g H}_2\text{O/h-cm}^2$$

Second falling-rate period to $X = 2.2\%$:

$$R = 29.03 X^2 - 0.048 X \quad (1)$$

Determine the time to dry a slab of the same dimensions at the same drying conditions, but from $X_o = 0.14$ to $X = 0.01$, ignoring the preheat period. Assume that the initial weight of the slab is 46.4 g. Drying is from both sides.

SOLUTION

Constant-rate period:

$$X_o = 0.14, X_{c1} = 0.083, R_{c1} = 0.053 \text{ g/h-cm}^2$$

$$m_s = 46.4 \left(\frac{1}{1.14} \right) = 40.7 \text{ g of moisture-free solid}$$

$$A = 2(6.6) = 13.2 \text{ cm}^2$$

From (18-39),

$$t_c = \frac{40.7(0.14 - 0.083)}{13.2(0.053)} = 3.32 \text{ h}$$

First falling-rate period:

In this period, R is linear with end points of ($R_{c1} = 0.053$, $X_{c1} = 0.083$) and ($R_{c2} = 0.038$, $X_{c2} = 0.037$). This gives for (18-45),

$$R = 0.0259 + 0.326X \quad (2)$$

Substitution of (2) into (18-38) and integration gives

$$\begin{aligned} t_{f1} &= -\frac{m_s}{A} \int_{X_{c1}}^{X_{c2}} \frac{dX}{0.0259 + 0.326X} \\ &= \frac{m_s}{A} \frac{1}{0.326} \ln \left(\frac{0.0259 + 0.326X_{c1}}{0.0259 + 0.326X_{c2}} \right) \\ &= \frac{m_s}{A} \frac{1}{0.326} \ln \left(\frac{R_{c1}}{R_{c2}} \right) \\ &= \frac{40.7}{13.2(0.326)} \ln \left(\frac{0.053}{0.038} \right) = 3.15 \text{ h} \end{aligned}$$

Second falling-rate period:

This period extends from $X_{c2} = 0.037$ to $X = 0.022$, with R given by (1) for (18-43), with $a = -0.048$ and $b = 29.03$.

From (18-44),

$$\begin{aligned} t_{f2} &= \frac{40.7}{(-0.048)(13.2)} \ln \left\{ \frac{0.037[-0.048 + 29.03(0.022)]}{0.022[-0.048 + 29.03(0.037)]} \right\} \\ &= 2.08 \text{ h} \end{aligned}$$

the total drying time is

$$t_T = 3.32 + 3.15 + 2.08 = 8.6 \text{ h}$$

For drying-rate curves of shapes other than those of Figure 18.31, the time for drying from any X_o to any X can be determined by numerical or graphical integration of (18-38) or (1) in Example 18.9, as illustrated in the following example.

EXAMPLE 18.12

Marshall and Hougen [13] present the following experimental data for the through-circulation drying of rayon waste. Determine the drying time if $X_o = 100\%$ and the final X is 10%. Assume that all moisture is free moisture.

X , lb H ₂ O/lb dry solid	R' , $\frac{\text{lb H}_2\text{O}}{\text{h-lb dry solid}}$
1.40	24
1.00	24
0.75	24
0.73	21
0.70	18
0.65	15.3
0.55	13
0.475	12.3
0.44	12.2
0.40	11
0.20	5.5
0	0

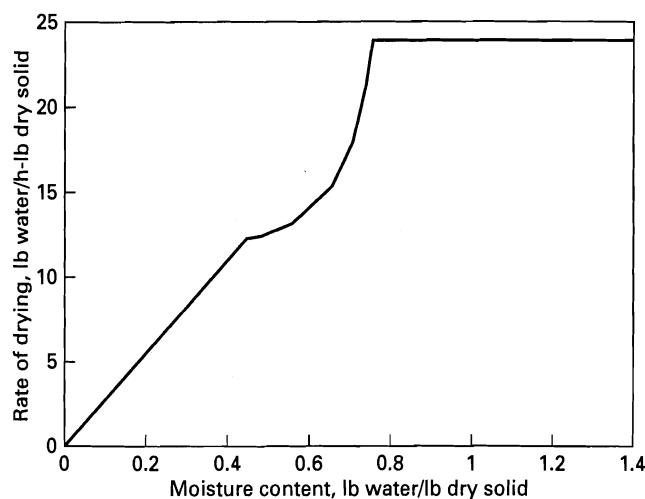


Figure 18.32 Experimental data for through-circulation drying of rayon waste.

SOLUTION

The data are plotted in Figure 18.32, where three distinct drying-rate periods are seen, but the two falling-rate periods are in the reverse order of Figure 18.31c. By numerical integration of (1) in Example 18.9 with a spreadsheet, the following drying times are obtained, noting that $R' = 2.75$ lb H₂O/h-lb dry solid at $X = 0.10$, $R_{c1} = 24$ at $X_{c1} = 0.75$, and $R_{c2} = 12.2$ at $X_{c2} = 0.44$.

$$t_c = 0.027 \text{ h} = 1.63 \text{ min}$$

$$t_{f1} = 1.28 \text{ min}$$

$$t_{f2} = 3.21 \text{ min}$$

$$t_T = 1.63 + 1.28 + 3.21 = 6.12 \text{ min}$$

Liquid-Diffusion Theory

The application of the empirical approach, in the preceding section, for determining drying time in the falling-rate period is limited to the particular conditions for which the experimental drying-rate curve is established. A more general approach, particularly for nonporous wet solids of the second category, discussed in Section 18.3, is the use of Fick's laws of diffusion. Once the moisture diffusion coefficient is established from experimental data for the particular wet solid, Fick's laws can be used to predict drying rates and moisture profiles for wet solids of other sizes and shapes and drying conditions during the falling-rate period.

Mathematical formulations of liquid diffusion in solids are readily obtained by analogy to the many solutions available for transient heat conduction in solids, as summarized, for example, by Carslaw and Jaeger [17] and discussed in Section 3.3. The following two solutions are of particular interest for the drying of slabs in the falling-rate period, where the area of the edges is small compared to the area of the two faces, or the edges are sealed to prevent escape of moisture. As in heat-conduction calculations, the equations also apply when one of the two faces is sealed so

that only the other face is exposed to gas:

Case 1: Initially uniform moisture profile in the wet solid with negligible resistance to mass transfer of moisture in the gas phase.

Case 2: Initially parabolic moisture profile in the wet solid with negligible resistance to mass transfer in the gas phase.

Although the equations for these two cases are developed here only for a slab with sealed edges, solutions are available in Carslaw and Jaeger [17] for a sphere and a cylinder with sealed edges. When edges of slabs and cylinders are not sealed, the method of Newman [18] can be applied as discussed in Section 3.3.

Case 1 This case applies to slow-drying materials for which the rate of drying is controlled by the internal diffusion of moisture to the exposed surface. This can occur if initially the wet solid has no surface liquid film and the external resistance to mass transfer is negligible, such that there is no constant-rate drying period. Alternatively, the wet solid can have a surface liquid film, but during the evaporation of that film in a constant-rate drying period controlled by gas-phase mass transfer, no moisture diffuses to the surface and following the completion of evaporation of that film, the resistance to mass transfer is entirely due to internal diffusion in a falling-rate period.

The slab, of thickness $2a$, is pictured in Figure 3.7a, where the edges at $x = \pm c$ and $y = \pm b$ are sealed to mass transfer. Internal diffusion of moisture is in the z direction only toward exposed faces at $z = \pm a$. Alternatively, the slab may be of thickness a with the face at $z = 0$ sealed to mass transfer. Initially, the moisture content throughout the slab, not counting any surface liquid film, is uniform at X_0 . At the beginning of the falling-rate period, $t = 0$, the exposed faces are (face is) brought to the equilibrium moisture content, X^* . For constant moisture diffusivity, D_{AB} , Fick's second law, as discussed in Section 3.3, applies:

$$\frac{\partial X}{\partial t} = D_{AB} \frac{\partial^2 X}{\partial z^2} \quad (18-46)$$

for $t \geq 0$ in the region $-a \leq z \leq a$, where the boundary conditions are

$$X = X_0 \quad \text{at} \quad t = 0 \quad \text{for} \quad -a < z < a$$

and

$$X = X^* \quad \text{at} \quad z = \pm a \quad \text{for} \quad t \geq 0$$

The solution to (18-46) for the moisture profile as a function of time under these boundary conditions, as discussed in Section 3.3, and first proposed for drying applications by Sherwood [11], is in terms of the unaccomplished free-

moisture change, the following modification of (3-80),

$$E = \frac{X - X^*}{X_o - X^*} = \frac{4}{\pi} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)} \times \exp \left[-\frac{\pi^2(2n+1)^2}{4} \left(\frac{D_{AB}t}{a^2} \right) \right] \cos \left[\frac{\pi(2n+1)}{2} \left(\frac{z}{a} \right) \right] \quad (18-47)$$

Thus, E is a function of two dimensionless groups, the Fourier number for diffusion, $D_{AB}t/a^2$, and the position ratio, z/a . This solution is plotted as $(1 - E)$ in terms of these two groups in Figure 3.8.

The rate of mass transfer from one face is also given in Chapter 3 as (3-82), which in terms of R , the rate of drying expressed as the mass of moisture evaporated per unit time per unit area, is

$$R = \frac{2D_{AB}(X_o - X^*)\rho_s}{a} \times \sum_{n=0}^{\infty} \exp \left[-\frac{\pi^2(2n+1)^2}{4} \left(\frac{D_{AB}t}{a^2} \right) \right] \quad (18-48)$$

where ρ_s = mass of dry solid/volume of slab.

Also of interest is the average moisture content of the slab during drying. From (3-85), which is derived in Chapter 3,

$$E_{\text{avg}} = \frac{X_{\text{avg}} - X^*}{X_o - X^*} = \frac{8}{\pi^2} \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} \exp \left[-\frac{\pi^2(2n+1)^2}{4} \left(\frac{D_{AB}t}{a^2} \right) \right] \quad (18-49)$$

Equations (18-47)–(18-49) can be used to determine the moisture diffusivity, D_{AB} , from experimental data, and then that value is used to estimate drying rates for other conditions as illustrated in the following example. However, such calculations must be made with caution because often the diffusivity is not constant, as shown by Sherwood [11] for the drying of slabs of soap, but decreases with decreasing moisture content because of shrinkage and/or case-hardening. In that case, numerical solutions of (18-46) are necessary.

EXAMPLE 18.13

A piece of poplar wood 15.2 cm long $\times$ 15.2 cm wide $\times$ 1.9 cm thick, with the edges sealed with a waterproofing cement was dried from both faces in a tunnel dryer using cross-circulation of air at 1 m/s. The initial moisture content was 39.7% on the dry basis. The total initial weight of the wet piece was 264 g, and no shrinkage occurred during drying. The direction of moisture diffusion was perpendicular to the grain. The equilibrium moisture content was 5% on the dry basis. The following experimental data were obtained for the average moisture content as a function of time. Included are values of E_{avg} , computed from its definition, given in (18-49), and values of t/a^2 , where $a = 0.5$ (1.90) = 0.95 cm.

t, h	$X_{\text{avg}},$ g H ₂ O/g dry wood	$t/a^2,$ h/cm ²	E_{avg}
0.36	0.362	0.40	0.900
0.90	0.328	1.00	0.800
1.53	0.303	1.70	0.730
1.94	0.291	2.15	0.694
2.89	0.267	3.20	0.626
3.47	0.255	3.85	0.591
4.02	0.245	4.45	0.562
4.92	0.230	5.45	0.520
5.82	0.218	6.45	0.483
6.95	0.204	7.70	0.443
8.03	0.192	8.90	0.409
8.98	0.183	9.95	0.382

Using the data, determine the average value of the diffusivity by nonlinear regression of (18-49) and use that value to determine the drying time from $X_o = 45\%$ to $X = 10\%$ with $X^* = 6\%$ for a piece of poplar measuring 72 in. long $\times$ 12 in. wide $\times$ 1 in. thick, neglecting mass transfer from the edges and assuming that all drying takes place in the falling-rate period with negligible resistance in the gas phase.

SOLUTION

$$m_s = 264 \left(\frac{1}{1 + 0.397} \right) = 189 \text{ g dry wood}$$

$$A \text{ for two faces} = 2(15.2)^2 = 462 \text{ cm}^2$$

At any instant, from (18-38),

$$R = -\frac{m_s}{A} \frac{dX_{\text{avg}}}{dt} \quad (1)$$

From a plot of the data, approximate values of R as a function of X_{avg} are computed to be

$R, \text{g H}_2\text{O/h}\cdot\text{cm}^2$	$X_{\text{avg}}, \text{g H}_2\text{O/g dry solid}$
0.02622	0.345
0.01573	0.315
0.01258	0.297
0.01019	0.279
0.00847	0.261
0.00760	0.250
0.00661	0.238
0.00582	0.224
0.00503	0.211
0.00446	0.198
0.00404	0.187

These results are plotted in Figure 18.33, where it appears that all of the drying takes place in the falling-rate period. Thus, the data may be consistent with the Case 1 diffusion theory.

To determine the average moisture diffusivity, a spreadsheet is used to prepare a semi-log plot of the data points as E_{avg} against t/a^2 , as shown in Figure 18.34. Equation (18-49) is then evaluated on the spreadsheet for different values of the moisture diffusivity until the best fit of the data is obtained, based on minimizing the error sum of squares (ESS) of the differences between E_{avg} of the data points and corresponding E_{avg} values calculated from (18-49).

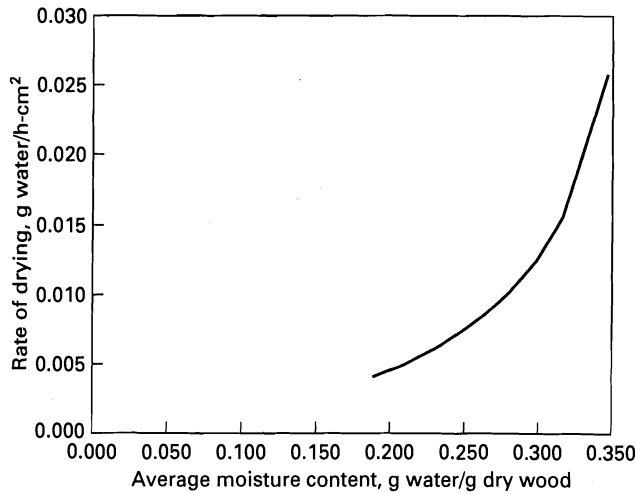


Figure 18.33 Experimental data for drying of poplar wood.

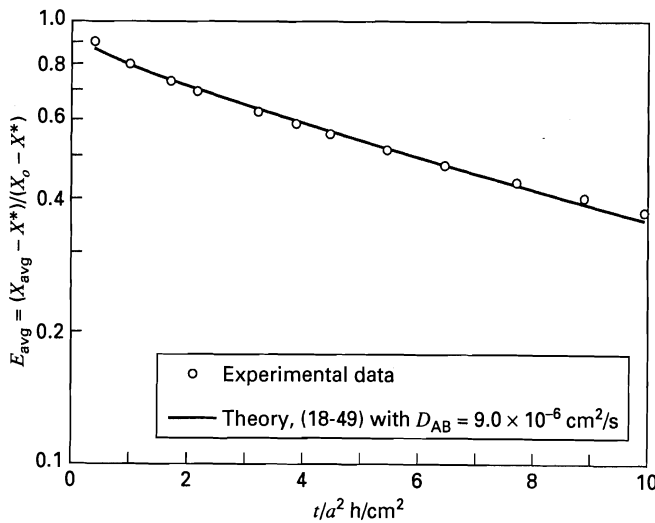


Figure 18.34 Best fit by diffusion theory of experimental data for drying of poplar wood.

The best fit is found for $D_{AB} = 9.0 \times 10^{-6} \text{ cm}^2/\text{s}$ with an ESS = 0.001669. The best fit of (18-49) is included as a line in Figure 18.34. It may be noted that for values of $N_{FOM} > 0.1$, only the first term in the infinite series of (18-49) is significant and therefore (18-49) approaches a straight line on a semi-log plot, as can be observed for the theoretical line in Figure 18.34, when $t/a^2 > 3.2 \text{ h/cm}^2$.

To determine the drying time for the 72-in. $\times$ 12-in. $\times$ 1-in. piece of poplar, assume that all drying takes place in the diffusion-controlled, falling-rate period and that mass transfer from the edges is negligible. Further assume that the drying time will be long enough that $N_{FOM} > 0.1$. Then, as just mentioned, (18-49) reduces to

$$\ln \left(\frac{X_{avg} - X^*}{X_o - X^*} \right) = \ln \left(\frac{8}{\pi^2} \right) - \frac{\pi^2}{4} \left(\frac{D_{AB} t}{a^2} \right) \quad (2)$$

Solving (2) for $(D_{AB} t/a^2)$,

$$N_{FOM} = \frac{D_{AB} t}{a^2} = \frac{4}{\pi^2} \ln \left[\frac{8}{\pi^2} \left(\frac{X_o - X^*}{X_{avg} - X^*} \right) \right] \quad (3)$$

$$X_{avg} = 0.10, X_o = 0.45, X^* = 0.06$$

From (3),

$$N_{FOM} = \frac{4}{(3.14)^2} \ln \left[\frac{8}{(3.14)^2} \left(\frac{0.45 - 0.06}{0.10 - 0.06} \right) \right] = 0.839$$

Since $N_{FOM} > 0.1$, (2) and (3) are valid,

$$a = 0.5 \text{ in.} = 1.27 \text{ cm}$$

$$D_{AB} = 9.0 \times 10^{-6} \text{ cm}^2/\text{s} \text{ from the above experiments}$$

$$t = \frac{a^2 N_{FOM}}{D_{AB}} = \frac{(1.27)^2 (0.839)}{(9.0 \times 10^{-6}) (3600)} = 41.8 \text{ h}$$

Case 2 When a liquid-diffusion-controlled, falling-rate drying period is preceded by a constant-rate period, that rate of drying is determined by external mass transfer in the gas phase, as discussed earlier, but diffusional resistance to the flow of moisture in the solid causes a moisture profile to be established in the solid. This moisture profile approaches a parabolic distribution, as discussed by Sherwood [19] and Gilliland and Sherwood [20].

For the slab of Figure 3.7a, Fick's second law, as given by (18-46), still applies, with $X = X_o$ at $t = 0$ for $-a < z < a$. However, during the constant-rate drying period, the slab-gas interface boundary conditions are changed from Case 1 to the conditions $\partial X/\partial z = 0$ at $z = 0$ for $t \geq 0$ and $R_c = -D_{AB} \rho_s (\partial X/\partial z)$ at $z = \pm a$ for $t \geq 0$. This latter boundary condition is more conveniently expressed in the form

$$\frac{\partial X}{\partial z} = -\frac{R_c}{\rho_s D_{AB}} \quad (18-50)$$

where the term on the right-hand side is a constant during the constant-rate period. This is analogous to a constant-heat-flux boundary condition in heat transfer. The solution to this case for the moisture profile as a function of time during the constant-rate drying period is given from Walker, et al. [21] as

$$X = X_o - \frac{R_c a}{D_{AB} \rho_s} \left\{ \frac{1}{2} \left(\frac{z}{a} \right)^2 - \frac{1}{6} + \frac{D_{AB} t}{a^2} - \frac{2}{\pi^2} \sum_{m=1}^{\infty} \frac{(-1)^m}{m^2} \exp \left[-m^2 \pi^2 \left(\frac{D_{AB} t}{a^2} \right) \right] \cos \left(\frac{\pi m z}{a} \right) \right\} \quad (18-51)$$

where for small values of $D_{AB} t/z^2$, the infinite series term is significant and converges very slowly.

The average moisture content in the slab at any time during the constant-rate period is defined by

$$X_{avg} = \frac{1}{a} \int_0^a X dz \quad (18-52)$$

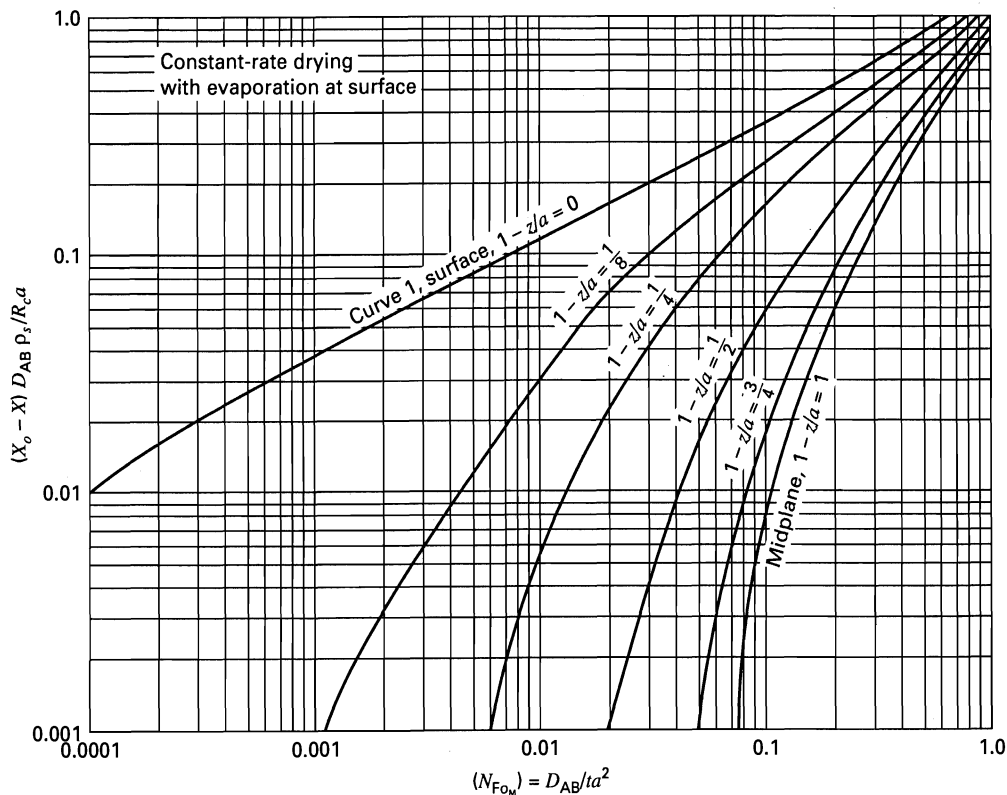
If (18-52) is integrated after substitution of X from (18-51), the result is

$$(X_o - X_{avg}) \frac{D_{AB} \rho_s}{R_c a} = \frac{D_{AB} t}{a^2} = N_{FOM} \quad (18-53)$$

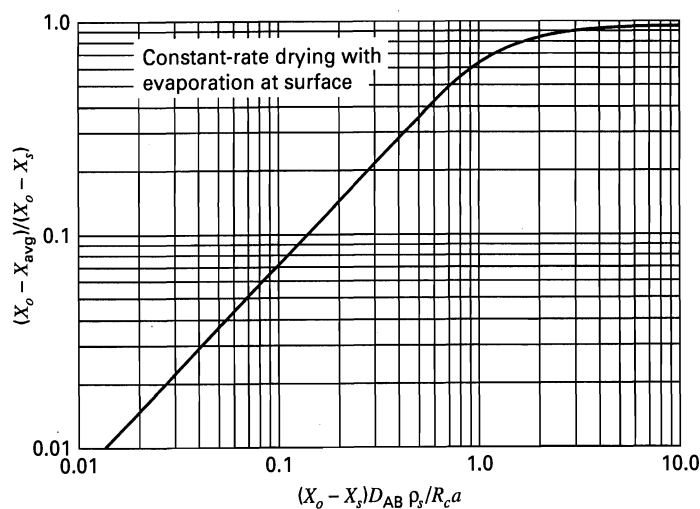
From (18-51), it is seen that the generalized moisture profile during the constant-rate drying period, $(X_o - X)D_{AB}\rho_s/R_c a$, is a function of the dimensionless position ratio, z/a , and N_{FoM} , where the latter is equal to the generalized, average moisture content, as given by (18-53). A plot of (18-51) for six different position ratios, is given in Figure 18.35a.

Equation (18-51) is based on the assumption that during the constant-rate drying period, moisture will be supplied to

the surface by liquid diffusion at a rate sufficient to maintain the constant moisture-evaporation rate. As discussed above, the average moisture content at which the constant-rate period ends and the falling-rate period begins is called the critical moisture content, X_c . In the empirical approach to the falling-rate period, X_c must be known from experiment for the particular conditions being evaluated because X_c is not a constant for a given material, but also depends on



(a) Moisture profile change



(b) Surface moisture change

Figure 18.35 Changes in moisture concentration during constant-rate period while diffusion in the solid occurs.

[From W.H. Walker, W.K. Lewis, W.H. McAdams, and E.R. Gilliland, *Principles of Chemical Engineering*, 3rd ed., McGraw-Hill, New York (1937) with permission.]

a number of other factors, including moisture diffusivity, slab thickness, initial and equilibrium moisture contents, and all factors that influence the moisture-evaporation flux in the constant-rate drying period. A very useful aspect of (18-51) is that it can be used to predict X_c . The basis for the prediction is the assumption that the falling-rate period will begin when the moisture content at the surface reaches the equilibrium-moisture content corresponding to the conditions of the surrounding gas. This prediction is facilitated, as described by Walker, et al. [21], by replotting an extension of Curve 1 in Figure 18.35a for the moisture content at the surface, X_s , in the form shown in Figure 18.35b. The use of Figure 18.35b and the predicted influence of several variables on the value of X_c is illustrated in the following example.

EXAMPLE 18.14

Experiments by Gilliland and Sherwood [20] with brick clay mix show that for certain drying conditions, the moisture profiles conform reasonably well to the Case 2 diffusion theory. Use Figure 18.35b to predict the critical-moisture content for the drying of clay slabs from the two faces only under three different sets of conditions. For all three sets, $X_o = 0.30$, $X^* = 0.05$, $\rho_s = 1.6 \text{ g/cm}^3$, and $D_{AB} = 0.3 \text{ cm}^2/\text{h}$. The other conditions are

	Set 1	Set 2	Set 3
a , slab half-thickness, cm	0.5	0.5	1.0
R_c , drying rate in constant-rate drying period, $\text{g/cm}^2\text{-h}$	0.2	0.4	0.2

SOLUTION

For Set 1, using $X_s = X^* = 0.05$,

$$(X_o - X_s) \frac{D_{AB} \rho_s}{R_c a} = (0.30 - 0.05) \frac{(0.3)(1.6)}{(0.2)(0.5)} = 1.20$$

From Figure 18.35b,

$$\frac{X_o - X_{\text{avg}}}{X_o - X_s} = 0.7$$

Solving,

$$X_{\text{avg}} = X_c = 0.25 - 0.7(0.25 - 0.05) = 0.11$$

In a similar manner,

$$X_c \text{ for Set 2} = 0.16$$

$$X_c \text{ for Set 3} = 0.16$$

These results show that doubling the rate of drying in the constant-rate period or doubling the slab thickness substantially increases the critical-moisture content.

For sufficiently large values of time, corresponding to a Fourier number for diffusion $N_{\text{FOM}} = \frac{D_{AB} t}{a^2} > 0.5$, the term for the infinite series in (18-51) approaches a value of zero, and, at all locations in the slab, X becomes a parabolic function of z .

A simple equation for the parabolic distribution can be formulated as follows from (18-51) in terms of the moisture contents at the surface and midplane of the slab. At the surface $z = \pm a$, the long-time form is

$$X_o - X_s = \frac{R_c a}{D_{AB} \rho_s} \left[\frac{1}{3} + N_{\text{FOM}} \right] \quad (18-54)$$

Similarly, at the midplane, $z = 0$, where $X = X_m$,

$$X_o - X_m = \frac{R_c a}{D_{AB} \rho_s} \left[-\frac{1}{6} + N_{\text{FOM}} \right] \quad (18-55)$$

Combining (18-51), (18-54), and (18-55), the dimensionless moisture-content profile becomes

$$\frac{X_m - X}{X_m - X_s} = \left(\frac{z}{a} \right)^2 \quad (18-56)$$

EXAMPLE 18.15

For the conditions of Example 18.14, determine the drying time for the constant-rate drying period and whether the parabolic moisture-content profile will be closely approached by the end of that period.

SOLUTION

From (18-39),

$$t_c = \frac{m_s (X_o - X_c)}{A R_c} \quad (1)$$

For a half-slab of thickness a ,

$$m_s = \rho_s a A \quad (2)$$

Combining (1) and (2),

$$t_c = \frac{\rho_s a}{R_c} (X_o - X_c) \quad (3)$$

For Set 1 of Example 18.14,

$$t_c = \frac{(1.6)(0.5)}{(0.2)} (0.30 - 0.11) = 0.76 \text{ h}$$

$$N_{\text{FOM}} = \frac{D_{AB} t_c}{a^2} = \frac{(0.3)(0.76)}{(0.5)^2} = 0.91$$

Because $N_{\text{FOM}} > 0.5$, the parabolic profile will be closely approached. In a similar manner, the following results are obtained for Sets 2 and 3:

	Set 2	Set 3
t_c , h	0.28	1.12
N_{FOM}	0.34	0.34
Parabolic profile closely approached	no	no

For Sets 2 and 3, the parabolic moisture-content profiles are not closely approached. However, the absolute errors in $X_o - X$ at the surface and midplane are determined from (18-51) to be only 1.1% and 4.3%, respectively.

An approximate theoretical estimate of the additional drying time required for the falling-rate period is derived as follows from the development by Walker, et al. [21]. At the end of the constant-rate period, the rate of flow of moisture

by Fickian diffusion to the surface of the slab, where it is then evaporated, may be equated to the reduction in average moisture content of the slab. Thus,

$$R = -\frac{\rho_s a A}{A} \frac{dX_{\text{avg}}}{dt} = -D_{\text{AB}} \rho_s \left. \frac{dX}{dz} \right|_s \quad (18-57)$$

From the parabolic moisture profile of (18-56), at the surface, $z = +a$,

$$-\left. \frac{dX}{dz} \right|_{z=+a} = \frac{2}{a}(X_m - X_s) \quad (18-58)$$

However, it is more desirable to convert this expression from one in terms of X_m to one in terms of X_{avg} . To do this, (18-56) can be substituted into (18-52) for the definition of X_{avg} , followed by integration to give

$$X_{\text{avg}} = \frac{2}{3}X_m + \frac{1}{3}X_s \quad (18-59)$$

which can be rewritten as

$$X_m - X_s = \frac{3}{2}(X_{\text{avg}} - X_s) \quad (18-60)$$

Substitution of (18-60) in (18-58), followed by substitution of the result into (18-57), gives

$$R = -a\rho_s \frac{dX_{\text{avg}}}{dt} = \frac{3D_{\text{AB}}\rho_s}{a}(X_{\text{avg}} - X_s) \quad (18-61)$$

The falling-rate period is assumed to begin, as discussed above, with $X_s = X^*$. If the parabolic moisture profile is maintained during the falling-rate period and if $X_s = X^*$ remains constant, then (18-61) applies during that period and a straight-line falling-rate period of the type shown in Figure 18.31a will be obtained. Integration of (18-61) from the start of the falling-rate period when $X_{\text{avg}} = X_c$, gives

$$t_f = \frac{a^2}{3D_{\text{AB}}} \ln \left[\frac{X_c - X^*}{X_{\text{avg}} - X^*} \right] \quad (18-62)$$

Thus, the duration of the falling-rate period is predicted to be directly proportional to the square of the slab half-thickness and inversely proportional to the moisture liquid diffusivity. Equation (18-62) gives reasonable predictions for nonporous slabs of materials such as wood, clay, and soap when the slabs are thick and D_{AB} is low. However, serious deviations can occur when D_{AB} depends strongly on X and/or temperature. In that case, an average value of D_{AB} can be used to obtain an approximate result. A summary of experimental average moisture liquid diffusivities for a wide range of water-wet solids is tabulated in Chapter 4 of Mujumdar [1].

EXAMPLE 18.16

Gilliland and Sherwood [20] obtained data of the drying of water-wet $7 \times 7 \times 2.54$ -cm slabs of 193.9 g (bone-dry) brick clay mix for direct-heat convective air drying from the two faces in both the constant-rate and falling-rate periods. For $X_o = 0.273$, $X^* = 0.03$, the rate of drying in the constant-rate period to $X_c = 0.165$ was

0.157 g/h-cm². The air velocity past the two faces was 15.2 m/s with $T_d = 25^\circ\text{C}$ and $T_w = 17^\circ\text{C}$. During the falling-rate period, experimental average slab moisture contents were as follows:

Time from start of the constant-drying rate period, minutes	X_{avg}
67	0.165 (critical value)
87	0.145
102	0.134
119	0.124
138	0.114
162	0.106
183	0.099
205	0.095
216	0.090

At the end of the constant-drying-rate period, the moisture profile was shown to be very nearly parabolic. From other experiments, $D_{\text{AB}} = 0.72 \times 10^{-4}$ cm²/s.

Use (18-62) to predict values of X_{avg} during the falling-rate period and compare the predicted values to the experimental values.

SOLUTION

Solving (18-62) for X_{avg} gives

$$X_{\text{avg}} = X^* + (X_c - X^*) \exp(-3D_{\text{AB}}t_f/a^2) \quad (1)$$

where t_f is the time from the start of the falling-rate period.

For $t_f = 87 - 67 = 20$ min, from (1),

$$X_{\text{avg}} = 0.03 + (0.165 - 0.03) \times \exp[-3(0.72 \times 10^{-4})(20)(60)/(1.27)^2] = 0.145 \text{ cm}^2/\text{s}$$

Calculations for other values of time give the following results:

t_f , Time from start of falling-rate period, min	Experimental X_{avg}	Predicted X_{avg}
0	0.165	0.165
20	0.145	0.145
35	0.134	0.132
52	0.124	0.119
71	0.114	0.106
95	0.106	0.093
116	0.099	0.083
138	0.095	0.075
149	0.090	0.071

Comparing the predicted values of X_{avg} with the experimental values, it is seen that the deviation increases with increasing time. If the value of D_{AB} is reduced to 0.53×10^{-4} cm²/s, much better agreement is obtained with the ESS decreasing from 0.0013 to 0.000154 cm⁴/s².

Capillary-Flow Theory

For wet solids of the first category as discussed in Section 18.3, moisture is held, e.g., as free moisture in the interstices of powders and granular solids. Movement of moisture from the interior to the surface can occur by capillary action in the interstices, but may be opposed by gravity.

Forces holding liquid molecules together are cohesive forces. Additionally, liquid molecules may be attracted to a solid surface by adhesive forces. Thus, water held in a glass tube will tend to creep up the side of the tube until the adhesive forces are balanced by the weight of the liquid. For an ideal case of a capillary tube of small diameter partially immersed in a vertical orientation in a liquid reservoir, the liquid will rise in the tube to a height above the surface of the liquid in the reservoir. At equilibrium the height, h , will be

$$h = 2\sigma/\rho_L g r \quad (18-63)$$

where σ is the surface tension of the liquid and r is the radius of the capillary. This equation shows that the smaller the radius of the capillary, the larger the capillary effect. Unlike mass transfer by diffusion, which causes moisture to move from a region of higher concentration to one of lower concentration, moisture in interstices flows to regions of highest capillary regardless of concentration.

For capillary flow in granular beds of wet solids, the variable size and shape of the particles make it extremely difficult to develop a usable theory for predicting the rate of drying in the falling-rate period in terms of permeability and capillarity. However, interesting discussions and idealized theories are presented by Keey [7, 23] and Ceaglske and Kiesling [22]. For practical calculations, it appears that, despite pleas to the contrary, it is common to apply diffusion theory with effective diffusivities determined from experiment. In general, these diffusivities are lower than those for true diffusion of moisture in nonporous materials. Some values are included in a tabulation in Chapter 4 of Mujumdar [1]. For example, values for the effective diffusivity of water in beds of sand particles are given that cover a range of 1.0×10^{-2} to 8.0×10^{-4} cm²/s.

18.5 DRYER MODELS

In previous sections, general mathematical models for estimating drying rates and moisture profiles have been developed and discussed, with applications to batch tray dryers of the cross-circulation and through-circulation type. More specific models have been developed over the years for a number of continuous dryers. In this section, models are presented for three such dryers: (1) belt dryer with through-circulation, (2) direct-heat rotary dryer, and (3) fluidized-bed dryer, all of which are direct-heat dryers. Other models are considered in a special issue of "Drying Technology," edited by Genskow [24], and in Mujumdar [1].

Material and Energy Balances for Direct-heat Dryers

Consider the general case of a continuous, direct-heat dryer, as shown in Figure 18.36. Although countercurrent flow is shown, the following development of energy balances applies equally well to other flow configurations such as cocurrent flow and crossflow. Assume that the dryer is perfectly insulated such that the operation is adiabatic. As the solid is dried, moisture is transferred to the gas. Further assume that no solid is entrained into the gas and that changes in kinetic energy and potential energy are negligible. The flow rates of dry solid, m_s , and dry gas, m_g , do not change as drying proceeds. Therefore, at steady state, a material balance on the moisture is given by

$$X_{ws}m_s + \mathcal{H}_{gi}m_g = X_{ds}m_s + \mathcal{H}_{go}m_g \quad (18-64)$$

The rate of moisture evaporation, m_v , is given by a rearrangement of (18-64):

$$m_v = m_s(X_{ws} - X_{ds}) = m_g(\mathcal{H}_{go} - \mathcal{H}_{gi}) \quad (18-65)$$

where the subscripts are *ws* (wet solid), *ds* (dry solid), *gi* (gas in), and *go* (gas out).

A steady-state energy balance can be written in terms of enthalpies or in terms of specific heat and heat of vaporization. In either case, it is convenient to treat the dry gas, dry solid, and moisture (liquid and vapor) separately, and assume ideal mixtures. In terms of enthalpies, the steady-state energy balance is as follows, where *s*, *g*, and *m* refer, respectively, to dry solid, dry gas, and moisture:

$$\begin{aligned} m_s(H_s)_{ws} + X_{ws}m_s(H_m)_{ws} + m_g(H_g)_{gi} + \mathcal{H}_{gi}m_g(H_m)_{gi} \\ = m_s(H_s)_{ds} + X_{ds}m_s(H_m)_{ds} \\ + m_g(H_g)_{go} + \mathcal{H}_{go}m_g(H_m)_{go} \end{aligned} \quad (18-66)$$

A factored rearrangement of (18-66) is

$$\begin{aligned} m_s[(H_s)_{ds} - (H_s)_{ws} + X_{ds}(H_m)_{ds} - X_{ws}(H_m)_{ws}] \\ = m_g[(H_g)_{gi} - (H_g)_{go} + \mathcal{H}_{gi}(H_m)_{gi} \\ - \mathcal{H}_{go}(H_m)_{go}] \end{aligned} \quad (18-67)$$

where any convenient reference temperatures can be used to determine the enthalpies.

When the system is air, water, and a solid, a more convenient form of (18-67) can be obtained by evaluating the enthalpies of the solid and the air from specific heats, and obtaining moisture enthalpies from the steam tables. Often, the specific heat of the solid is almost constant over the temperature range of interest, and in the range from 25°C (78°F) to 400°C (752°F), the specific heat of dry air increases by

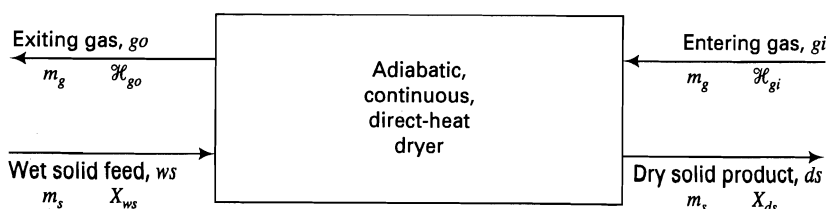


Figure 18.36 General configuration for a continuous, direct-heat dryer.

less than 3%, so that the use of a constant value of 0.242 Btu/lb-°F introduces little error. If the enthalpy reference temperature of the water is taken as T_o (usually 0°C or 32°F for liquid water when using the steam tables), (18-67) can be rewritten as

$$\begin{aligned} m_s[(C_P)_s(T_{ds} - T_{ws}) + X_{ds}(H_{H_2O})_{ds} - X_{ws}(H_{H_2O})_{ws}] \\ = m_g[(C_P)_{\text{air}}(T_{gi} - T_{go}) + \mathcal{H}_{gi}(H_{H_2O})_{gi} \\ - \mathcal{H}_{go}(H_{H_2O})_{go}] \end{aligned} \quad (18-68)$$

A further simplification in the energy balance for the air–water–solid system can be made by replacing enthalpies for water by their equivalents in terms of specific heats for liquid water and steam, and the heat of vaporization. In the range from 25°C (78°F) to 100°C (212°F), the specific heat of liquid water and steam are almost constant at 1 Btu/lb-°F and 0.447 Btu/lb-°F, respectively. The heat of vaporization of water over this same range decreases from 1049.8 to 970.3 Btu/lb, a change of almost 8%. Combining (18-65) with (18-68), and taking a thermodynamic path of water evaporation at the moisture-evaporation temperature, denoted T_v , the simplified energy balance is

$$\begin{aligned} Q = m_s\{(C_P)_s(T_{ds} - T_{ws}) + X_{ws}(C_P)_{H_2O(l)}(T_v - T_{ws}) \\ + X_{ds}(C_P)_{H_2O(l)}(T_{ds} - T_v) \\ + (X_{ws} - X_{ds})[\Delta H_v^{\text{vap}} + (C_P)_{H_2O(g)}(T_{go} - T_v)]\} \\ = m_g\{[(C_P)_{\text{air}} + \mathcal{H}_{gi}(C_P)_{H_2O(v)}](T_{gi} - T_{go})\} \end{aligned} \quad (18-69)$$

Equations (18-64) to (18-69) are useful for determining the required gas flow rate for drying a given flow rate of wet solids, as illustrated in the example below. Also of interest for sizing the dryer is the required heat-transfer rate, Q from the gas to the solid. For the air–water–solid system, this rate is equal to either the left-hand side or right-hand side of (18-69), as indicated. In the general case,

$$\begin{aligned} Q = m_g\{(H_g)_{gi} - (H_g)_{go} + \mathcal{H}_{gi}[(H_m)_{gi} - (H_m)_{go}]\} \\ = m_s\{(H_s)_{ds} - (H_s)_{ws} + X_{ds}(H_m)_{ds} - X_{ws}(H_m)_{ws} \\ + m_g[(H_m)_{go}(\mathcal{H}_{go} - \mathcal{H}_{gi})]\} \end{aligned} \quad (18-70)$$

EXAMPLE 18.17

A continuous, cocurrent-flow, direct-heat dryer is to be used to dry crystals of Epsom salt (magnesium sulfate heptahydrate). The feed to the dryer, a filter cake from a rotary, vacuum filter, consists of 2,854 lb/h of crystals (dry basis) with a moisture content of 25.8 wt% (dry basis) at 85°F and 14.7 psia. Air enters the dryer at 14.7 psia, with dry-bulb and wet-bulb temperatures of 250°F and 117°F, respectively. The final moisture content of the dried crystals is to be 1.5 wt% (dry basis), at a temperature of no more than 118°F to prevent decomposition to the hexahydrate (see Figure 17.2). Determine:

- The rate of moisture evaporation.
- The outlet temperature of the air.
- The rate of heat transfer.
- The entering air flow rate.

The average specific heat of Epsom salt is 0.361 Btu/lb-°F in the temperature range of 70–120°F.

SOLUTION

$$m_s = 2854 \text{ lb/h}, X_{ws} = 0.258, X_{ds} = 0.015$$

$$T_{ws} = 85^\circ\text{F}, T_{ds} = 118^\circ\text{F}, T_{gi} = 250^\circ\text{F}, T_v = 117^\circ\text{F}$$

From Figure 18.17, for $T_{db} = 250^\circ\text{F}$ and $T_{wb} = 117^\circ\text{F}$, $\mathcal{H}_{gi} = 0.0405$

- (a) From (18-65),

$$m_v = 2,854(0.258 - 0.015) = 694 \text{ lb/h}$$

- (b) Because the dryer operates cocurrently, the outlet temperature of the gas must be greater than the outlet temperature of the dry solid, which is taken as 118°F. The best value for T_{go} is obtained by optimizing the cost of the drying operation. A reasonable value for T_{go} can be estimated by using the concept of the number of heat-transfer units, which is analogous to the number of transfer units for mass transfer, as developed in Chapter 6. For heat transfer in a dryer, where the solids temperature throughout most of the dryer will be at T_v , the number of heat-transfer units is defined by

$$N_T = \ln \left[\frac{T_{gi} - T_v}{T_{go} - T_v} \right] \quad (1)$$

where economical values of N_T are usually in the range of 1.0–2.5. Assume a value here of 2.0. From (1),

$$2 = \ln \left[\frac{250 - 117}{T_{go} - 117} \right]$$

from which $T_{go} = 135^\circ\text{F}$

- (c) The rate of heat transfer is obtained from (18-69) using the conditions for the solid flow.

$$\begin{aligned} Q &= 2,854\{0.361(118 - 85) + 0.258(1)(117 - 85) \\ &\quad + 0.015(1)(118 - 117) + (0.258 - 0.015) \\ &\quad \times [1,027.5 + (0.447)(135 - 117)]\} \\ &= 2,854[11.9 + 8.3 + 0.02 + 249.7 + 2.0] \\ &= 2,854(271.9) = 776,000 \text{ Btu/h} \end{aligned}$$

It should be noted that the heat required to vaporize the 694 lb/h of moisture at 117°F is $(249.7/271.9) \times 100\% = 91.8\%$ of the total heat load.

- (d) The entering air flow rate is obtained from (18-69) using the far right-hand side of that equation with the above value of Q .

$$m_g = \frac{776,000}{[(0.242) + (0.0405)(0.447)](250 - 135)} = 25,940 \text{ lb/h}$$

The total entering air, including the humidity, is

$$25,940(1 + 0.0405) = 27,000 \text{ lb/h}$$

Belt Dryer with Through-Circulation

Consider the continuous, two-zone, through-circulation belt dryer shown in Figure 18.37a. A bed of wet-solid particles is conveyed continuously into Zone 1, where contact is made

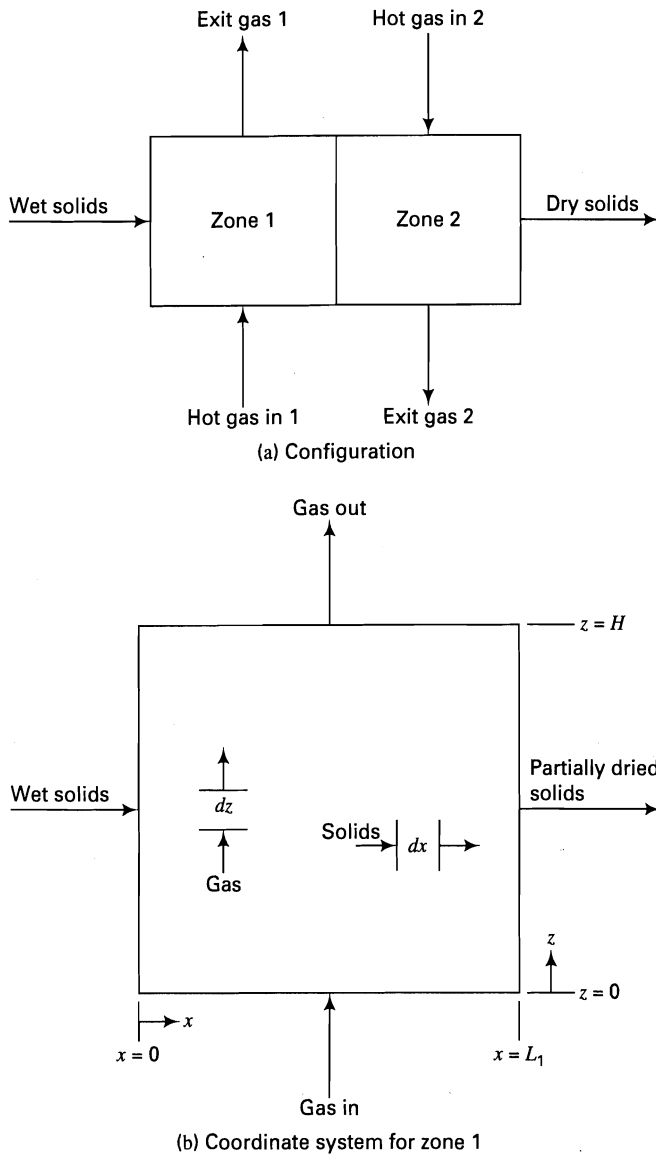


Figure 18.37 Continuous, two-zone, through-circulation belt dryer.

with hot gas passing upward through the bed. Because the temperature of the gas decreases as it passes through the bed, the temperature-driving force decreases so that the moisture content of solids near the bottom of the moving bed decreases more rapidly than for solids near the top. To obtain a dried solid of more uniform moisture content, the gas flow direction through the bed is reversed in Zone 2.

Based on the work of Thygeson and Grossmann [25], a mathematical model for Zone 1 can be developed as follows, using the coordinate system shown in Figure 18.37b. The model is based on the following assumptions:

1. Wet solids enter Zone 1 with a uniform moisture content of X_o on the dry basis.
2. Gas passes up through the moving bed in plug flow with no mass transfer in the vertical direction (i.e., no axial dispersion).
3. Drying takes place in the constant-rate period, controlled by the rate of heat transfer from the gas to the

surfaces of the solid particles, where the temperature is the adiabatic-saturation temperature.

4. Sensible-heat effects are negligible compared to latent-heat effects.
5. The void fraction of the bed is uniform and constant, and no mixing of solid particles occurs.
6. The solids are conveyed at a uniform linear speed, S .

Based on these assumptions, the temperature of the gas decreases with increasing distance z from the bottom of the bed, and is independent of the distance, x , in the direction in which the solids are conveyed, i.e., $T = T\{z\}$. The moisture content of the solids, however, varies in both the z and x directions, decreasing more rapidly near the bottom of the bed where the gas temperature is higher, i.e., $X = X\{z, x\}$.

Zone 1 With no mixing of solids, a material balance on the moisture in the solids at any vertical location, z , is given, for Zone 1, by

$$\frac{dX_1}{dt} = S \frac{dX_1}{dx} = -\frac{(ha)(T_1 - T_v)}{\Delta H_v^{\text{vap}}(\rho_b)_{ds}} \quad (18-71)$$

where a = surface area of solid particles per unit volume of bed, T_1 = bulk temperature of the gas in Zone 1, which depends on z , and $(\rho_b)_{ds}$ is the bulk density of solids when dry. The initial condition is $X_1 = X_o$ for $x = 0$.

An energy balance for the gas phase at any location, x , is given by

$$\rho_g(C_P)_g u_s \frac{dT_1}{dz} = -(ha)(T_1 - T_v) \quad (18-72)$$

where ρ_g = density of the gas and u_s = superficial velocity of gas through the bed. The initial condition is $T_1 = T_{gi}$ for $z = 0$.

Equation (18-71) is coupled to (18-72), which is independent of (18-71). Therefore, we can separate variables and integrate (18-72) to obtain

$$T_1 = T_v + (T_{gi} - T_v) \exp\left(-\frac{haz}{\rho_g(C_P)_g u_s}\right) \quad (18-73)$$

At $z = H$ at the top of the bed,

$$T_{go1} = T_v + (T_{gi} - T_v) \exp\left(-\frac{haH}{\rho_g(C_P)_g u_s}\right) \quad (18-74)$$

Equation (18-71) can now be solved by combining it with (18-73) to eliminate T_1 , followed by separation of variables and integration. The result is

$$\frac{X_1}{X_o} = 1 - \frac{xa(T_{gi} - T_v) \exp\left(-\frac{haz}{\rho_g(C_P)_g u_s}\right)}{S \Delta H_v^{\text{vap}}(\rho_b)_{ds}} \quad (18-75)$$

The moisture content $(X_1)_{L1}$ at $x = L_1$ is obtained by replacing x with L_1 .

If desired, X_{avg} at $x = L_1$ can be determined from

$$X_{\text{avg}} = \int_0^H (X_1)_{L_1} dz \quad (18-76)$$

Zone 2 In Zone 2, (18-71) still applies with X_1 and T_1 replaced by X_2 and T_2 , but the initial condition for X_o is $(X_1)_{L_1}$ from (18-75) for $x = L_1$, which depends on z . Equation (18-72) also applies with T_1 replaced by T_2 , but the initial condition is $T_2 = T_{gi}$ at $z = H$. The resulting integrated equations are

$$T_2 = T_v + (T_{gi} - T_v) \exp \left[-\frac{h a (H - z)}{\rho_g (C_P)_g u_s} \right] \quad (18-77)$$

with T_{go2} given by (18-74), where T_{go1} is replaced by T_{go2} , and

$$\frac{X_2}{(X_1)_{L_1}} = 1 - \frac{x h a (T_{gi} - T_v) \exp \left(-\frac{h a (H - z)}{\rho_g (C_P)_g u_s} \right)}{S \Delta H_v^{\text{vap}} (\rho_b)_{ds}} \quad (18-78)$$

where $(X_1)_{L_1}$ is the value from (18-75) for $x = L_1$ and the value of z in (18-78). The value of x in (18-78) is the distance from the start of Zone 2. Values of $(X_2)_{L_2}$ at any z are obtained from (18-78) for $x = L_2$. The average moisture content over the height of the moving bed leaving Zone 2 is then obtained from (18-76) with $(X_1)_{L_1}$ replaced by $(X_2)_{L_2}$. The application of the above relationships is illustrated in the following example.

EXAMPLE 18.18

The extruded filter cake of calcium carbonate in Example 18.8 is to be dried continuously with a belt dryer using through-circulation. The dryer is 6 ft wide and has a belt speed of 1 ft/min. The dryer consists of two drying zones, each 12 ft long. Air at 170°F and 10% relative humidity enters both zones, passing upward through the moving bed in the first zone, and downward through the second zone, at a superficial velocity of 2 m/s. The bed height on the belt is constant at 2 in. Predict the moisture-content distribution with height at the end of each zone, and the average moisture content at the end of Zone 2. Assume all drying is in the constant-rate period and neglect the preheat period.

SOLUTION

From data in Examples 18.7 and 18.8,

$$\begin{aligned} X_o &= 0.30 \\ (\rho_b)_{ds} &= \frac{1.00}{1.30} (1,950) = 1,500 \text{ kg/m}^3 \\ \epsilon_b &= 0.50 \\ T_v &= 37.8^\circ\text{C} = 311 \text{ K}, T_{gi} = 76.7^\circ\text{C} = 350 \text{ K} \\ \Delta H_v^{\text{vap}} &= 2,413 \text{ kJ/kg} \end{aligned}$$

From extrusion area and volume in Example 18.8,

$$a = \frac{(3.16 \times 10^{-4})(0.5)}{(4.01 \times 10^{-7})} = 395 \text{ m}^2/\text{m}^3 \text{ bed}$$

For $u_s = 2 \text{ m/s}$, $h = 0.188 (\text{kJ/s} \cdot \text{m}^2 \cdot \text{K}^2)$ from Example 18.8.

$$\begin{aligned} (C_P)_g &= 1.09 \text{ kJ/kg} \cdot \text{K} \\ \rho_g \text{ at 1 atm} &= 0.942 \text{ kg/m}^3 \\ S &= 1 \text{ ft/min} = 0.00508 \text{ m/s} \end{aligned}$$

The cross-sectional area of the moving bed normal to the conveying direction is $6(2/12) = 1 \text{ ft}^2 = 0.0929 \text{ m}^2$. For a belt speed of $1 \text{ ft/min} = 0.305 \text{ m/min}$, the volumetric flow of solids is $(0.0929)(0.305) = 0.0283 \text{ m}^3/\text{min}$. The mass rate of flow is $0.0283(1,500) = 42.5 \text{ kg/min}$ (dry basis).

Zone 1

$$\begin{aligned} H &= 0.167 \text{ ft} = 0.0508 \text{ m} \\ L_1 &= 12 \text{ ft} = 3.66 \text{ m} \end{aligned}$$

From (18-74), the gas temperature leaving the bed is

$$\begin{aligned} T_{go1} &= 37.8 + (76.7 - 37.8) \exp \left[-\frac{(0.188)(395)(0.0508)}{(0.942)(1.09)(2)} \right] \\ &= 44^\circ\text{C} = 317 \text{ K} \end{aligned}$$

The moisture-content distribution at $x = L_1$ is obtained from (18-75). For $z = 0$,

$$X_1 = 0.30 \left[1 - \frac{(3.66)(0.188)(395)(76.7 - 37.8)}{(0.00508)(2,413)(1,500)} \right] = 0.127$$

For other values of z , using a spreadsheet, the following values are obtained:

$z, \text{ m}$	$(X_1)_{L_1}$
0	0.127
0.0127	0.191
0.0254	0.231
0.0381	0.257
0.0508	0.273

Because of the considerable decrease in bulk-gas temperature as it passes upward through the bed, the moisture content varies considerably over the bed depth.

Zone 2 The flow of air is now reversed to further the drying and smooth out the moisture-content distribution. The value of X_o is replaced by values of $(X_1)_{L_1}$ above for corresponding values of z . Using (18-78) with a spreadsheet, the following distribution is obtained at $L_2 = 3.66 \text{ m}$ for a total length of both zones = $24 \text{ ft} = 7.32 \text{ m}$:

$z, \text{ m}$	$(X_2)_{L_2}$
0	0.116
0.0127	0.163
0.0254	0.178
0.0381	0.163
0.0508	0.116

A much more uniform moisture distribution is achieved. From (18-76) for Zone 2, using numerical integration with a spreadsheet, $(X_2)_{\text{avg}} = 0.155$.

Direct-Heat Rotary Dryer

As discussed by Kelly in Mujumdar [1], the design of a direct-heat rotary dryer, of the type shown in Figure 18.7, for drying a specified solids feed rate of initial moisture content X_{ws} to a final moisture content X_{ds} , involves the selection and determination of a number of factors, including heating gas inlet and outlet conditions, dryer-cylinder slope and rotation rate, number and type of lifting flights, heating-gas velocity and flow direction, and dryer cylinder diameter and length. Also of interest are the solids holdup, as a percent of dryer-cylinder volume, and solids residence time.

Ideally, a commercial-size direct-heat rotary dryer should be scaled up from pilot-plant data obtained in a laboratory unit. However, if a representative sample of the wet solid is not available for testing, the following procedure and model, based on test results with several materials in both pilot-plant-size and commercial-size dryers, is useful for a preliminary design.

The hot gas can flow countercurrently or cocurrently to the flow of the solids. In general, cocurrent flow is used for very wet, sticky solids, with high inlet-gas temperatures, and for nonhygroscopic solids. Countercurrent flow is preferred for low-to-moderate inlet-gas temperatures, where thermal efficiency becomes a factor. When solids are not subject to thermal degradation, melting, or sublimation, a high-inlet gas temperature up to approximately 1000°F can be used. The exit-gas temperature is determined largely from economics, as discussed in Example 18.17, where (1) can be used with N_T in the range of 1.5–2.5. The allowable gas velocity is determined from the dusting characteristics of the dried solid particles, and can vary widely with particle-size distribution and particle density. Some typical values for allowable gas velocity are as follows:

Material	Particle Density, ρ_p , lb/ft ³	Average Particle Size, d_p , μ m	Allowable Gas Velocity, u_{all} , ft/s
Plastic granules	69	920	3.5
Ammonium nitrate	104	900	4.5
Sand	164	110	1.0
Sand	164	215	2.0
Sand	164	510	5.0
Sawdust	27.5	640	1.0

Using an appropriate allowable gas velocity, u_{all} , the mass flow rate of gas at the gas discharge end, $(m_g)_{exit}$, of the dryer, and the gas density $(\rho_g)_{exit}$ at that end, the dryer diameter, D , can be estimated by the continuity equation

$$D = \left[\frac{4(m_g)_{exit}}{\pi u_{all}(\rho_g)_{exit}} \right]^{0.5} \quad (18-79)$$

The residence time of the solids in the dryer, θ , is related to the fractional volume holdup of solids, V_H , by the relation

$$\theta = \frac{LV_H}{F_V} \quad (18-80)$$

where L = length of dryer cylinder and F_V = solids volumetric velocity in volume/unit cross-sectional area-unit time. A conservative estimate of the holdup, including the effect of gas velocity, is obtained by combining (18-80) with a relation in [2]:

$$V_H = \frac{0.23F_V}{SN^{0.9}D} \pm 0.6 \frac{G(5/d_p^{0.5})}{\rho_p} \quad (18-81)$$

where

F_V = ft³ solids/(ft² cross section)-h

S = dryer cylinder slope, ft/ft

N = dryer cylinder rate of rotation, rpm

D = dryer diameter, ft

G = gas superficial mass velocity, lb/h-ft²

d_p = mass-average particle size, μ m

The plus (+) sign on the second term corresponds to countercurrent flow that tends to increase the holdup, while the minus (−) sign corresponds to cocurrent flow. Equation (18-81) holds for dryers with lifting flights that have lips, but is limited to gas velocities less than 3.5 ft/s. An improved, but more complex, model by Matchett and Sheikh [26] is valid for gas velocities to 10 ft/s. Optimal solids holdup is in the range of 10–18% of dryer volume so that flights run full and all or most of the solids are showered during each revolution.

When drying is in the constant-rate period such that the rate can be determined from the rate of heat transfer from the gas to the wet surface of the solids at the wet-bulb temperature, a volumetric heat-transfer coefficient, ha , can be used, which is defined by

$$Q = (ha)V\Delta T_{LM} \quad (18-82)$$

where V = volume of dryer cylinder = $\pi D^2 L/4$

$$\Delta T_{LM} = \frac{(T_g)_{in} - (T_g)_{out}}{\ln \left[\frac{(T_g)_{in} - T_v}{(T_g)_{out} - T_v} \right]} \quad (18-83)$$

and ha = volumetric heat-transfer coefficient based on dryer cylinder volume as given by the empirical correlation of McCormick [27], when the heating gas is air:

$$ha = KG^{0.67}/D \quad (18-84)$$

where

ha is in Btu/h-ft³-°F

G is in lb/h-ft²

D is in ft

A value of $K = 0.5$ is recommended in [2] for dryers operating at a peripheral cylinder speed of 1.0–1.25 ft/s and with a flight count of $2.4D$ – $3.0D$ per circle. When pilot-plant data are available, (18-84) can be used for scale-up to a larger diameter and a different value of G , where K is determined from the pilot-plant data.

It might be expected that a correlation for the volumetric heat-transfer coefficient, ha , would take into account the particle diameter because the solids are lifted and showered through the gas. However, the solids shower as curtains of some thickness, with the gas passing between curtains. Thus, particles inside the curtains do not receive significant exposure to the gas and the effective heat transfer area is more likely determined by the areas of the curtains. Nevertheless, (18-84) accounts for only two of the many possible variables and the inverse relation with dryer diameter is not well-supported by experimental data. A more satisfactory, but more complex, model for heat transfer is that of Schofield and Glikin [28], as modified by Langrish, Bahu, and Reay [29]. That model treats h and a separately.

EXAMPLE 18.19

Ammonium nitrate, with a moisture content of 15 wt% (dry basis), is to be fed into a direct-heat rotary dryer at a feed rate of 700 lb/min (dry basis) and a temperature of 70°F. Air at 250°F and 1 atm, with a humidity of 0.02 lb H₂O/lb dry air enters the dryer at the feed end and passes cocurrently with the solid through the dryer. The final moisture content of the solid is to be 1 wt% (dry basis). It may be assumed that all drying will take place in the constant-rate period. Make a preliminary estimate of the dryer diameter and length, assuming that such dryers are available in: (1) diameters from 1 to 5 ft in increments of 0.5 ft and from 5 to 20 ft in increments of 1.0 ft, and (2) lengths from 5 to 150 ft in increments of 5 ft.

SOLUTION

From the psychrometric chart (Figure 18.17), $T_{wb} = 107^\circ\text{F}$. Assume that all drying takes place at this temperature of the solid. A reasonable outlet temperature for the air can be estimated from (1) of Example 18.17, assuming $N_T = 1.5$. From that equation,

$$1.5 = \ln \left[\frac{250 - 107}{T_{go} - 107} \right]$$

Solving,

$$T_{go} = 140^\circ\text{F}$$

Assume solids outlet temperature = $T_{ds} = 135^\circ\text{F}$

Heat-transfer rate

$$m_s = 700(60) = 42,000 \text{ lb/h of solids (dry basis)}$$

$$(C_p)_s = 0.4 \text{ Btu/lb}\cdot^\circ\text{F}$$

$$T_{ws} = 70^\circ\text{F}$$

$$X_{ws} = 0.15$$

$$T_v = T_{wb} = 107^\circ\text{F}$$

$$X_{ds} = 0.01$$

$$\Delta H_v^{\text{vap}} = 1,033 \text{ Btu/lb}$$

From (18-65),

$$m_v = 42,000(0.15 - 0.01) = 5,880 \text{ lb/h H}_2\text{O evaporated}$$

From (18-69),

$$Q = 42,000\{(0.4)(135 - 70) + (0.15)(1)(107 - 70) + (0.01)(1)(135 - 70) + (0.15 - 0.01)[1,033 + (0.447)(140 - 107)]\} = 7,510,000 \text{ Btu/h}$$

Air flow rate:

$$m_g = \frac{7,510,000}{[(0.242) + (0.02)(0.447)](250 - 135)} = 260,000 \text{ lb/h entering dry air}$$

Dryer diameter:

Assume an allowable gas velocity at the exit end of the dryer of 4.5 ft/s.

$$(m_g)_{\text{exit}} = 260,000(1 + 0.02) + 5,880 = 271,000 \text{ lb/h total gas}$$

$$(\rho_g)_{\text{exit}} = \frac{PM}{RT_{go}}$$

$$M = \frac{271,000}{\frac{260,000}{29} + \frac{11,000}{18}} = 28.3$$

$$(\rho_g)_{\text{exit}} = \frac{(1)(28.3)}{(0.730)(600)} = 0.0646 \text{ lb/ft}^3$$

From (18-79),

$$D = \left[\frac{4(271,000)}{(3.14)(4.5)(3,600)(0.0646)} \right]^{0.5} = 18 \text{ ft}$$

Dryer length:

$$G = G_{\text{exit}} = \frac{(271,000)(4)}{(3.14)(18)^2} = 1,070 \text{ lb/h}\cdot\text{ft}^2$$

From (18-84),

$$ha = 0.5(1,070)^{0.67}/18 = 3 \text{ Btu/h}\cdot\text{ft}^3\cdot^\circ\text{F}$$

From (18-83), neglecting the periods of wet solids heating up to 107°F and the dry solids heating up to 135°F,

$$\Delta T_{LM} = \frac{250 - 140}{\ln \left[\frac{250 - 107}{140 - 107} \right]} = 75^\circ\text{F}$$

From (18-82)

$$V = \frac{7,510,000}{(3)(75)} = 33,400 \text{ ft}^3$$

$$\text{Cross-sectional area} = (3.14)(18)^2/4 = 254 \text{ ft}^2$$

$$L = \frac{33,400}{254} = 130 \text{ ft}$$

Fluidized-Bed Dryer

Consider the behavior of a bed of solid particles when a gas is passed up through the bed, as shown in Figure 18.38. At a very low gas velocity, the bed remains fixed. At a very high gas velocity, the bed disappears because the particles are pneumatically transported away by the gas when its local velocity exceeds the terminal settling velocity of the particles. At an

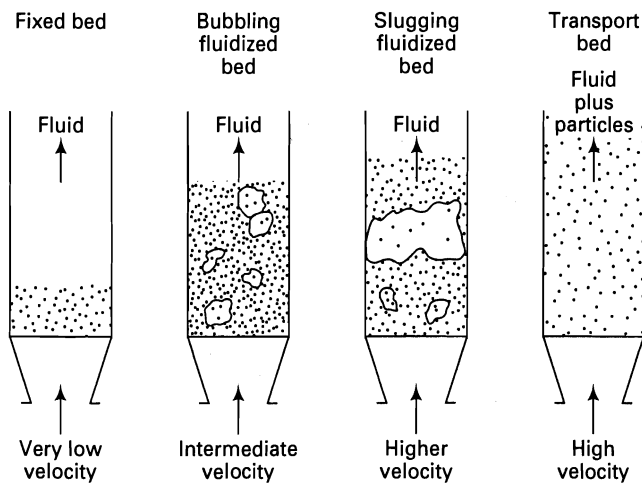


Figure 18.38 Regimes of fluidization of a bed of particles by a gas.

intermediate gas velocity, the bed is expanded, but particles are not carried out by the gas. Such a bed is said to be fluidized, because the bed of solids takes on some of the properties of a fluid. *Fluidization* is initiated when the gas velocity reaches the point where all the particles are just suspended by the gas. As the gas velocity is increased further, the bed expands and bubbles of gas are observed to pass up through the bed. This regime of fluidization is referred to as *bubbling fluidization* and is the most desirable regime for most fluidized-bed operations, including drying. If the gas velocity is increased further, a transition to *slugging fluidization* eventually occurs where bubbles coalesce and spread to a size that approximates the diameter of the vessel containing the bed.

Before fluidization occurs, when the bed of solids is fixed, the pressure drop across the bed for gas flow, ΔP_b , is predicted well by the Ergun [30] equation, discussed in Chapters 6 and 14:

$$\frac{\Delta P_b}{L_b} = 150 \frac{(1 - \epsilon_b)^2}{\epsilon_b^3} \frac{\mu u_s}{(\phi_s d_p)^2} + 1.75 \frac{(1 - \epsilon_b)}{\epsilon_b^3} \frac{\rho_g u_s^2}{\phi_s d_p} \quad (18-85)$$

where L_b = bed height, u_s = superficial gas velocity, and ϕ_s = particle sphericity. The first term on the right-hand side is dominant at low-particle Reynolds numbers where streamline flow exists and the second term dominates at high-particle Reynolds numbers where turbulent flow exists.

The onset of fluidization occurs when the drag force on the particles by the upward-flowing gas becomes equal to the weight of the particles (accounting for displaced gas):

$$\left(\frac{\Delta P}{\text{across bed}} \right) \left(\begin{array}{c} \text{cross-sectional} \\ \text{area} \\ \text{of bed} \end{array} \right) = \left(\begin{array}{c} \text{Volume} \\ \text{of} \\ \text{bed} \end{array} \right) \times \left(\begin{array}{c} \text{Volume} \\ \text{fraction} \\ \text{of solid} \\ \text{particles} \end{array} \right) \left[\left(\begin{array}{c} \text{Density} \\ \text{of} \\ \text{solid} \\ \text{particles} \end{array} \right) - \left(\begin{array}{c} \text{Density} \\ \text{of} \\ \text{displaced} \\ \text{gas} \end{array} \right) \right]$$

Thus,

$$\Delta P_b A_b = A_b L_b (1 - \epsilon_b) [(\rho_p - \rho_g)g] \quad (18-86)$$

The minimum gas-fluidization superficial velocity, u_{mf} , is obtained by solving (18-85) and (18-86) simultaneously for $u = u_{mf}$. For $N_{Re,p} = d_p u_{mf} \rho_g / \mu < 20$, the turbulent-flow contribution to (18-85) is negligible and the result is

$$u_{mf} = \frac{d_p^2 (\rho_p - \rho_g) g}{150 \mu} \left(\frac{\epsilon_b^3 \phi_s^2}{1 - \epsilon_b} \right) \quad (18-87)$$

For desirable operation in the bubbling fluidization regime, a superficial gas velocity of $u_s = 2u_{mf}$ is a reasonable choice. At this velocity, the bed will be expanded by about 10%, with no further increase in pressure drop across the bed. In this regime, the solid particles are well mixed and the bed temperature is uniform. Consequently, if the fluidized bed is operated continuously at steady-state conditions rather than batchwise with respect to the particles, the particles will have a residence-time distribution like that of a fluid in a continuous-stirred-tank reactor (CSTR). Some particles will be in the dryer for only a very short period of time and will experience almost no decrease in moisture content. Other particles will be in the dryer for a long time and may come to equilibrium before that time has elapsed. Thus, the dried solids will have a distribution of moisture content. This is in contrast to a batch-fluidization process where all particles have the same residence time and, therefore, a uniform final moisture content. This is an important distinction because continuous, fluidized-bed dryers are usually scaled up from data obtained in small-batch, fluidized-bed dryers. Therefore, it is important to have a relationship between batch drying time and continuous drying time.

The distribution of residence times for the effluent from a perfectly mixed vessel operating at continuous, steady-state conditions is given by Fogler [31] as

$$E\{t\} = \exp(-t/\tau)/\tau \quad (18-88)$$

where τ is the average residence time and $E\{t\}$ is defined such that $E\{t\}dt$ = the fraction of effluent with a residence time between t and $t + dt$. Thus, $\int_0^{t_1} E\{t\}dt$ = fraction of the effluent with a residence time less than t_1 . For example, if the average particle residence time is 10 min, 63.2% of the particles will have a residence time of less than 10 min. If the particles are small and porous such that all drying takes place in the constant-rate period and θ is the time for complete drying, then

$$\frac{X}{X_o} = 1 - \frac{t}{\theta}, \quad t \leq \theta \quad (18-89)$$

The average moisture content of the dried solids leaving the fluidized-bed reactor is obtained by integrating the following expression from 0 to only θ because $X = 0$ for $t > \theta$.

$$\bar{X}_{ds} = \int_0^\theta X E\{t\}dt = \int_0^\theta X_o \left(1 - \frac{t}{\theta}\right) E\{t\}dt \quad (18-90)$$

Combining (18-88) and (18-90) and integrating gives

$$\bar{X}_{ds} = X_o \left[1 - \frac{1 - \exp(-\theta/\tau)}{\theta/\tau} \right] \quad (18-91)$$

If the particles are nonporous and without surface moisture, such that all drying takes place in the falling-rate period, diffusion theory may apply such that the following empirical exponential expression may be used for the moisture content as a function of time:

$$\frac{X}{X_o} = \exp(-Bt) \quad (18-92)$$

In this case, the combination of (18-92) with (18-90), followed by integration from $t = 0$ to $t = \infty$ gives

$$\bar{X}_{ds} = 1/(1 + B\tau) \quad (18-93)$$

Values of θ and B must be determined from experiments with laboratory batch fluidized-bed dryers for scale-up to large commercial dryers operating under the same conditions.

EXAMPLE 18.20

10,000 lb/h of wet sand at 70°F with a moisture content of 20% (dry basis) is to be dried to a moisture content of 5% (dry basis) in a continuous, fluidized-bed dryer operating at a pressure of 1 atm in the freeboard region above the bed. The sand has a narrow size range with an average particle size of 500 μm and a sphericity, ϕ_s , of 0.67. The particle density is 2.6 g/cm³. When the sand is dry, the void fraction, ϵ_b , of a bed of the sand is 0.55. Fluidizing air will enter the bed at a temperature of 1,000°F with a humidity of 0.01 lb H₂O/lb dry air. The adiabatic-saturation temperature is estimated to be 145°F. Batch pilot-plant tests with a fluidization velocity of twice the minimum value show that drying takes place in the constant-rate period and all moisture can be removed in 8 minutes using air at the same conditions and with a bed temperature of 145°F. Determine the bed height and diameter for the large, continuous unit.

SOLUTION

$$d_p = 500 \mu\text{m} = 0.0500 \text{ cm}$$

Heat-transfer rate:

$$(C_p)_s = 0.20 \text{ Btu/lb}^\circ\text{F}$$

$$m_s = 10,000/(1 + 0.2) = 8,330 \text{ lb/h dry sand}$$

$$T_v = 145^\circ\text{F} = T_{go} = T_{ds}$$

$$\Delta H_v^{\text{vap}} = 1,011 \text{ Btu/lb}$$

$$T_{ws} = 70^\circ\text{F}$$

From (18-69),

$$Q = 8,330\{0.20(145 - 70) + 0.20(1)(145 - 70) + (0.20 - 0.05)(1,011)\} = 1,510,000 \text{ Btu/h}$$

Air rate:

$$m_g = \frac{1,510,000}{[(0.242) + (0.01)(0.447)](1,000 - 145)} = 7,170 \text{ lb/h dry air}$$

From (18-65),

$$m_v = 8,330(0.20 - 0.05) = 1,250 \text{ lb/h evaporated moisture}$$

$$\mathcal{H}_{go} = \frac{(7,170)(0.01) + 1,250}{7,170} = 0.184 \text{ lb H}_2\text{O/lb dry air}$$

Total exiting gas flow rate = 7,170(1 + 0.184) = 8,500 lb/h

Minimum fluidization velocity:

$$M \text{ of exiting gas} = \frac{8,500}{\frac{7,170}{29} + \frac{1,330}{18}} = 26.5$$

$$(\rho_g)_{\text{exit}} = \frac{PM}{RT_g} = \frac{(1)(26.5)}{(0.730)(605)} = 0.060 \text{ lb/ft}^3 = 0.00096 \text{ g/cm}^3$$

$$\mu = 0.048 \text{ lb/ft} \cdot \text{h} = 0.00020 \text{ g/cm} \cdot \text{s}$$

For small particles, assume streamline flow at u_{mf} so that (18-87) applies, but check to see if $N_{Re,p} < 20$, using cgs units,

$$u_{mf} = \frac{(0.0500)^2(2.6 - 0.00096)(980)(0.55)^3(0.67)^2}{150(0.00020)(1 - 0.55)} = 35.3 \text{ cm/s}$$

$$N_{Re,p} = \frac{d_p u_{mf} \rho_g}{\mu} = \frac{(0.0500)(35.3)(0.00096)}{0.00020} = 8.5$$

Since $N_{Re,p} < 20$, (18-87) does apply.

Use an actual superficial-gas velocity of $2(35.3) = 70.6 \text{ cm/s} = 8,340 \text{ ft/h}$

Bed diameter:

Equation (18-79) applies:

$$D = \left[\frac{4(8,500)}{3.14(8,340)(0.060)} \right]^{0.5} = 4.7 \text{ ft}$$

Bed density:

$$\text{Fixed-bed density} = \rho_s(1 - \epsilon_b) = 2.6(1 - 0.55)(62.4) = 73.0 \text{ lb/ft}^3$$

Assume bed expands by 10% upon fluidization to $u = 2u_{mf}$:

$$\rho_b = \frac{73.0}{1.10} = 66 \text{ lb/ft}^3 \text{ (dry basis)}$$

Average particle residence time:

For constant-rate drying in a batch dryer, all particles have the same residence time. From pilot-plant data, $\theta = 8 \text{ min}$ for complete drying.

For the large, continuous operation, (18-91) applies with $\bar{X}_{ds} = 0.05$, $X_o = 0.20$. Thus,

$$0.05 = 0.20 \left[1 - \frac{1 - \exp\left(-\frac{8}{\tau}\right)}{(8/\tau)} \right]$$

Solving this nonlinear equation, $\tau = 13.2$ minutes average residence time for particles. Only

$$\frac{(0.20 - 0.05)}{(0.20 - 0.0)} (8) = 6 \text{ min}$$

residence time would be required in a batch dryer to dry to 5% moisture. Therefore, more than double the residence time is needed in the continuous unit.

SUMMARY

1. Drying, in this chapter refers, to the removal of moisture from wet solids, solutions, slurries, and pastes. The moisture may be water or some other volatile liquid.
2. The two most common modes of drying are direct by heat transfer from a hot gas and indirect by heat transfer from a hot wall. The hot gas is frequently air, but can be a combustion gas, steam, nitrogen, or any other nonreactive gas.
3. Industrial drying equipment can be classified by operation (batch or continuous), mode (direct or indirect), and the degree to which the material being dried is agitated. Batch dryers include tray dryers and agitated dryers. Continuous dryers include tunnel dryers, belt or band dryers, turbo-tray tower dryers, rotary dryers, screw-conveyor dryers, fluidized-bed dryers, spouted-bed dryers, pneumatic-conveyor dryers, spray dryers, and drum dryers. Drying can also be accomplished with infrared radiation, dielectric and microwave radiation, and also from the frozen state by freeze drying.
4. Psychrometry, which deals with the properties of air-water mixtures and other gas-moisture mixtures, is useful in making drying calculations. Psychrometric (humidity) charts are particularly useful for determining the temperature at which surface moisture will be evaporated.
5. For the air-water system, the adiabatic-saturation temperature and the wet-bulb temperature are, by coincidence, almost identical. Thus, surface moisture is evaporated at the wet-bulb temperature. This greatly simplifies drying calculations.
6. Most solids can be grouped into one of two categories. Granular or crystalline solids that hold moisture in open pores between particles can be dried to very low moisture contents. Fibrous, amorphous, and gel-like materials that dissolve moisture or trap moisture in fibers or very fine pores can be dried to low moisture contents only with a gas of low humidity. The second category of materials can exhibit a significant equilibrium-moisture content that depends on the temperature, pressure, and humidity of the gas in contact with the material.
7. For drying calculations, the moisture content of a solid and a gas is usually based on the bone-dry solid and bone-dry gas, respectively. The bound-moisture content of a material in contact with a gas is the equilibrium-moisture content when the gas is saturated with the moisture. The excess-moisture content is called the

Bed height:

To achieve the average residence time of 13.2 min = 0.22 h, the expanded-bed volume must be

$$V_b = \frac{m_s \tau}{\rho_b} = \frac{8,330(0.22)}{66} = 27.8 \text{ ft}^3$$

$$H_b = \frac{V_b}{\pi D^2/4} = \frac{27.8(4)}{3.14(4.7)^2} = 1.6 \text{ ft}$$

unbound-moisture content. When the gas is not saturated, the excess-moisture content above the equilibrium-moisture content is called the free-moisture content. Solid materials that can contain bound moisture are hygroscopic. Bound moisture can be held chemically as water of hydration.

8. Drying by direct heat often takes place in four periods. The first is a preheat period accompanied by a rise in temperature but with little moisture removal. This is followed by a constant-rate period during which surface moisture is evaporated at the wet-bulb temperature. This moisture may be originally on the surface or moisture brought rapidly to the surface by diffusion or capillary action. The third period is a falling-rate period during which the rate of drying decreases linearly with time with little change in temperature. A fourth period may occur when the rate of drying falls off exponentially with time and the temperature rises.

9. The drying rate in the constant-rate period is governed by the rate of heat transfer from the gas to the surface of the solid. Empirical expressions for the heat-transfer coefficient have been formulated for a number of different types of direct-heat dryers.

10. The drying rate in the falling-rate period can be determined by using empirical expressions with experimental data. Diffusion theory can also be applied in some cases when moisture diffusivity is available or can be measured.

11. For direct-heat dryer models, material and energy balances can be applied to determine the rate of heat transfer from the gas to the wet solid and the gas flow rate.

12. A useful model for a two-zone belt dryer with through-circulation takes into account the changes in solids moisture content both vertically through the bed and in the direction of travel of the belt.

13. A model for preliminary sizing of a direct-heat rotary dryer is based on the use of a volumetric heat-transfer coefficient, assuming that the gas flows through curtains of cascading solids.

14. A model for sizing a large fluidized-bed dryer is based on the assumption of perfect mixing of the solids in the dryer when operating in the bubbling-fluidization regime. The procedure involves taking drying-time data from batch operation of a laboratory fluidized-bed dryer and correcting it for the expected solid particle residence-time distribution in the large dryer.

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EXERCISES

Section 18.1 (Use of the Internet is encouraged for the exercises of this section)

18.1 The surface moisture of crystals of NaCl of 0.5-mm average particle size is to be removed in a continuous, direct-heat dryer without a significant change to the particle size. What types of dryers would be suitable? How high could the gas feed temperature be?

18.2 A batch dryer is to be selected to dry 100 kg/h of a toxic, temperature-sensitive material (maximum of 50°C) of an average particle size of 350 μm . What dryers are suitable?

18.3 A thin, milk-like liquid is to be dried to produce a fine powder. What types of continuous, direct-heat dryers would be suitable? The material should not be heated above 200°C.

18.4 The selection of a batch or continuous dryer is determined largely by the condition of the feed, the temperature-sensitivity of the dried material, and the form of the dried product. What types of batch and continuous dryers would be suitable for the following cases:

- (a) A temperature-insensitive paste that must be maintained in slab form.
- (b) A temperature-insensitive paste that can be extruded.
- (c) A temperature-insensitive slurry.
- (d) A thin liquid from which flakes are to be produced.
- (e) Pieces of lumber.
- (f) Pieces of pottery.
- (g) Temperature-insensitive inorganic crystals where particle size is to be maintained and only surface moisture is to be removed.
- (h) Orange juice to produce a powder.

18.5 Solar drying has been used for centuries to dry and, thus, preserve fish, fruit, meat, plants, seeds, and wood. What are the advantages and disadvantages of this type of drying? What other types of dryers can be used to dry such materials? What type of dryer would you select to continuously dry beans?

18.6 Fluidized-bed dryers are used to dry a variety of vegetables, including potato granules, peas, diced carrots, and onion flakes. What are the advantages of this type of dryer for these types of materials?

18.7 Powdered milk can be produced from liquid milk in a three-stage process: (1) vacuum evaporation in a falling-film evaporator to a high-viscosity liquid of less than 50 wt% water; (2) spray drying to 7 wt% moisture; and (3) fluidized-bed drying to 3.6 wt% moisture. Give reasons why this three-stage process is preferable to a single-stage process involving just spray drying.

18.8 Deterioration must be strictly avoided when drying pharmaceutical products. Furthermore, such products are often produced from a nonaqueous solvent such as ethanol, methanol, acetone, etc. Explain why a closed-cycle spray dryer using nitrogen is frequently a good choice of dryer.

18.9 Paper is made from a suspension of fibers in water. The process begins by draining the fibers to a water-to-fiber ratio of 6:1, followed by pressing to a 2:1 ratio. What type of dryer could then be used to dry a continuous sheet to an equilibrium-moisture content of 8 wt% (dry basis)?

18.10 Green wood contains from 40 to 110 wt% moisture (dry basis) and must be dried before use to just under its equilibrium-moisture content when in the final environment. This moisture content is usually in the range from 6 to 15 wt% (dry basis). Why is it

important to dry the wood, and what is the best way to do it so as to avoid distortion, cracks, splits, and checks?

18.11 Wet coal is usually dried to a moisture content of less than 20 wt% (dry basis) before being transported, briquetted, coked, gasified, carbonized, or burned. What types of direct-heat dryers are suitable for drying coal? Can a spouted-bed dryer be used? If air is used as the heating medium, is there a fire and explosion hazard? Could superheated steam be used as the heating medium?

18.12 Drying is widely used to remove solvents from coated webs, which include coated paper and cardboard, coated plastic films and tapes (e.g., photographic films and magnetic tapes), and coated metallic sheets. The coatings may be water-based or other solvent-based. Solid coatings are also used. Typical coatings are 0.1 mm in wet thickness. Much of the drying time is usually spent in the falling-rate period where the rate of drying decreases in an exponentially decaying fashion with time. What types of dryers can be used with coated webs? Are infrared dryers a possibility? Why?

18.13 A number of polymers, including polyvinylchloride, polystyrene, and polymethylmethacrylate are made by suspension or emulsion polymerization, in which the product of polymerization reaction is finely divided solvent- or water-wet beads. For large production rates, direct-heat dryers are commonly used with air, inert gas, or superheated steam as the heating medium. Why are rotary dryers, fluidized-bed dryers, and spouted-bed dryers popular choices for the drying operation?

18.14 What are the advantages and disadvantages of superheated steam compared to air as the heating medium? Why might superheated steam be superior to air for the drying of lumber?

Section 18.2

18.15. A direct-heat dryer is to operate with air entering at 250°F and 1 atm with a wet-bulb temperature of 105°F. Determine from the psychrometric chart and/or relationships of Table 18.3 the following:

- Humidity.
- Molal humidity.
- Percentage humidity.
- Relative humidity.
- Saturation humidity.
- Humid volume.
- Humid heat.
- Enthalpy.
- Adiabatic-saturation temperature.
- Mole fraction of water in the air.

18.16 Air at 1 atm, 200°F, and a relative humidity of 15% enters a direct-heat dryer. Determine the following from the psychrometric chart and/or relationships of Table 18.3.

- Wet-bulb temperature.
- Adiabatic-saturation temperature.
- Humidity.
- Percentage humidity.
- Saturation humidity.
- Humid volume.
- Humid heat.
- Enthalpy.
- Partial pressure of water in the air.

18.17 Repeat Example 18.1 if the air is at 1.5 atm instead of 1.0 atm.

18.18 *n*-Hexane is being evaporated from a solid with nitrogen gas. At a point in the dryer where the gas is at 70°F and 1.1 atm, with a relative humidity for hexane of 25%, determine:

- Partial pressure of hexane at that point.
- Humidity of the nitrogen-hexane mixture.
- Percentage humidity of the nitrogen-hexane mixture.
- Mole fraction of hexane in the gas.

18.19 At a location in a dryer for evaporating toluene from a solid with air, the air is at 180°F, 1 atm, and a relative humidity of 15%. Determine the humidity, the adiabatic-saturation temperature, the wet-bulb temperature, and the psychrometric ratio.

18.20 Repeat Example 18.5 for water only, with air entering at 180°F and 1 atm, with a relative humidity of 15%, for an exit temperature of 120°F. In addition, plot temperature through the dryer.

18.21 Air enters a dryer at 1,000°F with a humidity of 0.01 kg H₂O/kg dry air. Determine the wet-bulb temperature if the air pressure is

- 1 atm.
- 0.8 atm.
- 1.2 atm.

18.22 Paper is being dried with recirculating air in a two-stage drying system operating at 1 atm. The air enters the first dryer at 180°F, where the air is adiabatically saturated with moisture. The air is then reheated in a heat exchanger to 174°F before entering the second dryer, where the air is adiabatically humidified to 80% relative humidity. The air is then cooled to 60°F in a second heat exchanger, causing some of the moisture to be condensed. This is followed by a third heater to heat the air to 180°F before it returns to the first dryer.

- Draw a process-flow diagram of the system and enter all of the given data.
- Determine the lb H₂O evaporated in each dryer per lb of dry air being circulated.
- Determine the lb H₂O condensed in the second heat exchanger per lb of dry air circulated.

18.23 Before being recirculated to a dryer, air at 96°F, 1 atm, and 70% relative humidity is to be dehumidified to 10% relative humidity. Cooling water is available at 50°F. Determine a method for carrying out the dehumidification, draw a labeled flow diagram of your process, and calculate the cooling-water requirement in lb H₂O per lb of dry air being circulated.

Section 18.3

18.24 Nitrocellulose fibers with an initial total water content of 40 wt% (dry basis) are dried in trays in a tunnel dryer operating at 1 atm. If the fibers are brought to equilibrium with air at 25°C and a relative humidity of 30%, determine the kg of moisture evaporated per kg of bone-dry fibers. The equilibrium-moisture content of the fibers is given in Figure 18.24.

18.25 Wet lumber of the type in Figure 18.24 is slowly dried from an initial total moisture content of 50 wt% to a moisture content in equilibrium with atmospheric air at 25°C and 40% relative humidity. Determine:

- The unbound moisture in the wet lumber before drying in lb water/lb bone-dry lumber.

- (b) The bound moisture in the wet lumber before drying in lb water/lb bone-dry lumber.
- (c) The free moisture in the wet lumber before drying, referred to as the final dried lumber, in lb water/lb bone-dry lumber.
- (d) The lb of moisture evaporated per lb of bone-dry wood.

18.26 Fifty pounds of cotton cloth containing 20% total moisture content (dry basis) are hung in a closed room containing 4,000 ft³ of air at 1 atm. Initially, the air is at 100°F at a wet-bulb temperature of 69°F. If the air is kept at 100°F, without admitting new air or venting the air, and the air is brought to equilibrium with the cotton cloth, determine the moisture content of the cotton cloth and the relative humidity of the air. Assume that the equilibrium-moisture content for cotton cloth at 100°F is the same as that of glue at 25°C as shown in Figure 18.24. Neglect the effect of the increase in air pressure, but calculate the final air pressure.

Section 18.4

18.27 Raw cotton having an initial total moisture content of 95% (dry basis) and a dry density 43.7 lb/ft³ is to be dried batchwise to a final moisture content of 10% (dry basis) in a cross-circulation tray dryer. The trays, which are insulated on the bottom, are each 3 cm high with an area of 1.5 m² and are completely filled. The heating medium, which is air at 160°F and 1 atm with a relative humidity of 30%, flows across the top surface of the tray at a mass velocity of 500 lb/h-ft². Equilibrium-moisture content isotherms for the cotton are given in Figure 18.25. Experiments have shown that under the given conditions, the critical-moisture content will be 0.4 lb water/lb bone-dry cotton and that the falling-rate drying period will be like that of Figure 18.31a, based on the free-moisture content. Determine:

- (a) The amount of raw cotton in pounds (wet basis) that can be dried in one batch if the dryer contains 16 trays.
- (b) The drying time for the constant-rate period.
- (c) The drying time for the falling-rate period.
- (d) The total drying time if the preheat period is 1 h.

18.28 Slabs of filter cake with a bone-dry density of 1,600 kg/m³ are to be dried from an initial free-moisture content of 110% (dry basis) to a final free-moisture content of 5% (dry basis) batchwise in trays that are 1 m long by 0.5 m wide with a depth of 3 cm. Drying will take place only from the top surface. The drying air conditions are 1 atm, 160°C, and a 60°C wet-bulb temperature. The air velocity across the trays is 3.5 m/s. Experiments under these drying conditions show a critical free-moisture content of 70% (dry basis), with a falling-rate period like that of Figure 18.31a, based on free-moisture content. Determine:

- (a) The drying time for the constant-rate period.
- (b) The drying time for the falling-rate period.

18.29 The filter cake of Exercise 18.28 is extruded into cylindrical-shaped pieces measuring 1/4 in. in diameter by 3/8 in. long that are placed in trays that are 6 cm high × 1 m long × 0.5 m wide and through which the air passes. The external porosity is 50%. If the superficial velocity of the air, which is at the same conditions as in Exercise 18.28, is 1.75 m/s, determine:

- (a) The drying time for the constant-rate period.
- (b) The drying time for the falling-rate period.

18.30 It takes 5 h to dry a wet solid, contained in a tray, from 36 to 8% moisture content, using air at constant conditions. Additional experiments give critical- and equilibrium-moisture contents of

15% and 5%, respectively. If the length of the preheat period is negligible and the falling-rate period is like that of Figure 18.31a, determine, for the same conditions, the drying time if the initial moisture content is 40% and a final moisture content of 7% is desired. All moisture contents are on the dry basis.

18.31 A tunnel dryer is to be designed to dry, by crossflow with air, a wet solid that will be placed in trays measuring 1.5 m long × 1.2 m wide × 25 cm deep. Drying will be from both sides. The initial total moisture content is 116% (dry basis) and the desired final average total moisture content is 10% (dry basis). The air conditions are 90°F and 1 atm with a relative humidity of 15%. The following laboratory data were obtained under the same conditions:

Equilibrium Moisture Content

% relative humidity	10	20	30	40	50	60	70	90
Moisture content, % (dry basis)	3.0	3.2	4.1	4.8	5.4	6.1	7.2	10.7

Drying Test		Drying Test	
Time, min	Moisture content, % (dry basis)	Time, min	Moisture content, % (dry basis)
0	116	362	31.4
36	106	415	28.6
125	81	465	24.8
194	61.8	506	22.8
211	57.4	601	15.4
242	49.6	635	13.5
277	42.8	785	11.4
313	37.1	822	10.2

Determine the total drying time to dry the same material from 110% to 10% moisture content if the air conditions are changed to 125°F and 20% relative humidity. Assume that the critical moisture content will not change and that the drying rate is proportional to the difference between the dry-bulb and wet-bulb temperatures of the air. All moisture contents are on the dry basis.

18.32 A piece of hemlock wood measuring 15.15 × 14.8 × 0.75 cm is to be dried from the two large faces from an initial total moisture content of 90% to a final average total moisture content of 10%, with drying taking place in the falling-rate period with liquid-diffusion controlling. The moisture diffusivity has been experimentally determined as 1.7×10^{-6} cm²/s. Estimate the drying time if bone-dry air is used. All moisture contents are on a dry basis.

18.33 Gilliland and Sherwood [20] obtained data for the drying of a water-wet piece of Hemlock wood measuring 15.15 × 14.8 × 0.75 cm, where only the two largest faces were exposed to the drying air, which was at a temperature of 25°C and passed over the faces at 3.7 m/s. The wet-bulb temperature of the air and the pressure may be assumed to be 17°C and 1 atm, respectively. The data below were obtained for the average moisture content (dry basis) of the wood as a function of time. From these data, determine whether Case 1 or Case 2 for the diffusion of moisture in solids applies. If Case 1 applies, determine the effective diffusivity. If Case 2 applies, determine:

- (a) The drying rate in g/h-cm² for the constant-rate period, assuming a wood density of 0.5 g/cm³ (dry basis) and no shrinkage upon drying.

- (b) The critical moisture content.
 (c) The predicted parabolic moisture content profile at the beginning of the falling-rate period.
 (d) The effective diffusivity during the falling-rate period.

In addition, for either case, describe what else could be determined from the data and explain how it could be determined.

Avg. Moisture		Avg. Moisture	
Time, h	Content, % (dry basis)	Time, h	Content, % (dry basis)
0	127	9	41.8
1	112	10	38.5
2	96.8	12	30.8
3	83.5	14	26.4
4	73.6	16	20.9
5	64.9	18	16.5
6	57.2	20	14.3
7	51.7	22	12.1
8	46.1	∞	6.6

18.34 When Case 1 of liquid diffusion is controlling during the falling-rate period, the time for drying can be determined from (3) under Example 18.13. Using that equation, derive an equation for the rate of drying to show that it varies inversely with the square of the thickness of the solid. If capillary movement controls the falling-rate period, an equation for the rate of drying can be derived by assuming the laminar flow of moisture takes place from the interior of the solid to the surface such that the rate of drying varies linearly with the average free-moisture content. If so, derive equations for the rate of drying and the time for drying in the falling-rate period to show that the rate of drying is inversely proportional to just the thickness of the solid. Outline an experimental procedure that could be used to determine whether diffusion or capillary flow governed in a given material.

18.35 In a cross-circulation tray dryer, the equations for the constant-rate period neglect radiation and assume that the bottoms of the trays are insulated so that heat transfer takes place only by convection from the gas to the surface of the solid where evaporation takes place. Under these conditions, evaporation occurs at the wet-bulb temperature of the gas when the moisture is water. In actual tray dryers, the bottoms of the trays are not insulated and heat transfer to the evaporating surface can also take place by convection from the gas to the tray bottom and thence by conduction through the tray and then through the wet solid. Derive an equation similar to (18-34) for the case where heat transfer by convection and conduction from the bottom side is taken into account. However, the conduction resistance of the tray bottom can be neglected. Show by combining your equation with the mass-transfer equation (18-35) that evaporation will now take place at a temperature higher than the wet-bulb temperature of the gas. What effect would heat transfer by radiation from the bottom surface of a tray to the tray below have on the temperature of evaporation?

Section 18.5

18.36 A tunnel dryer is to be used to dry 30 lb/h of raw cotton (dry basis) with a countercurrent flow of 1,800 lb/h of air (dry basis). The cotton enters at 70°F with a moisture content of 100% (dry basis) and exits at 150°F with 10% moisture (dry basis). The

air enters at 200°F and 1 atm with a relative humidity of 10%. The specific heat of dry cotton can be taken constant at 0.35 Btu/lb-°F. Calculate:

- (a) The rate of evaporation of moisture.
 (b) The outlet temperature of the air.
 (c) The rate of heat transfer.

18.37 A 25 wt% solution of coffee in water at 70°F is spray dried to a final moisture content of 5% (dry basis) with air that enters at 450°F and 1 atm with a humidity of 0.01 lb/lb (dry basis) and exits at 200°F. Assuming a reasonable value for the specific heat of coffee, calculate:

- (a) The air rate in lb dry air/lb coffee solution.
 (b) The temperature of evaporation.
 (c) The heat-transfer rate in Btu/lb coffee solution.

18.38 7,000 lb/h of wet, pulverized, clay particles with 27% moisture (dry basis) at 15°C and 1 atm enter a flash dryer where they are dried to a moisture content of 5% (dry basis) with a cocurrent flow of air that enters at 525°C. The dried solids exit at the air wet-bulb temperature of 50°C, while the air exits at 75°C. Assuming a reasonable value for the specific heat of clay, calculate:

- (a) The flow rate of air in lb/h (dry basis).
 (b) The rate of evaporation of moisture.
 (c) The heat-transfer rate in Btu/h.

18.39 5,000 lb/h of wet isophthalic acid crystals with 30 wt% moisture (wet basis) at 30°C and 1 atm enter an indirect-heat, steam-tube rotary dryer, where they are dried to a moisture content of 2 wt% (wet basis) by 25 psig steam (14 psia barometer) condensing inside the tubes. Evaporation takes place at 100°C, which is also the exit temperature of the crystals. The specific heat of isophthalic acid can be taken as 0.2 cal/g-°C. Determine:

- (a) The rate of evaporation of moisture.
 (b) The rate of heat transfer.
 (c) The quantity of steam required in lb/h.

18.40 The extruded filter cake of Examples 18.8 and 18.18 is to be dried under the same conditions as in Example 18.18 except that three drying zones 8 ft long each will be used, with flow upward in the first and third zones and downward in the second zone. Predict the moisture-content distribution with height at the end of each zone and the final average moisture content.

18.41 Repeat the calculations of Example 18.18 if the extrusions are 3/8 in. in diameter $\times$ 1/2 in. long. Compare your results with those of Example 18.18 and comment.

18.42 A direct-heat, countercurrent-flow, rotary dryer with a 6-ft diameter and 60-ft length is available to dry titanium dioxide particles at 70°F and 1 atm with a moisture content of 30% (dry basis) to a moisture content of 2% (dry basis). Hot air is available at 400°F with a humidity of 0.015 lb/lb dry air. Experiments show that an air-mass velocity of 500 lb/h-ft² will not cause serious dusting. The specific heat of solid titanium dioxide is 0.165 Btu/lb-°F, and its true density is 240 lb/ft³. Determine:

- (a) A reasonable production rate in lb/h of dry titanium dioxide (dry basis).
 (b) The heat-transfer rate in Btu/h.
 (c) A reasonable air rate in lb/h (dry basis).
 (d) Reasonable values for the exit air and exit solids temperatures.

18.43 A fluidized-bed dryer is to be sized to dry 5,000 kg/h (dry basis) of spherical polymer beads that are closely sized to 1 mm in diameter. The beads will enter the dryer at 25°C with a moisture content of 80% (dry basis). The drying medium will be superheated steam, which will enter the bed at 250°C. The pressure in the vapor space above the bed will be 1 atm. A fluidization velocity of twice the minimum will be used to obtain bubbling fluidization. The bed temperature, exit solids temperature, and exit vapor temperature can all be assumed to be 100°C. The beads are to be dried to a moisture content of 10% (dry basis). The void fraction of the bed before

fluidization is 0.47. The specific heat of the dry polymer is 1.15 kJ/kg-K, while the density is 1,500 kg/m³. Batch fluidization experiments show that drying will all take place in the falling-rate period, as governed by diffusion according to (18-92), where 50% of the moisture is evaporated in 150 s. Bed expansion is expected to be about 20%. Determine the dryer diameter, average bead residence time, and expanded bed height. Is the dryer size reasonable? If not, what changes in operation could be made to make the size reasonable? In addition, calculate the entering superheated-steam flow rate and the necessary heat-transfer rate.

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Physical Constants

Universal (ideal) gas law constant, R

$$1.987 \text{ cal/mol} \cdot \text{K} \text{ or } \text{Btu/lbmol} \cdot ^\circ\text{F}$$

$$8315 \text{ J/kmol} \cdot \text{K} \text{ or } \text{Pa} \cdot \text{m}^3/\text{kmol} \cdot \text{K}$$

$$8.315 \text{ kPa} \cdot \text{m}^3/\text{kmol} \cdot \text{K}$$

$$0.08315 \text{ bar} \cdot \text{L/mol} \cdot \text{K}$$

$$82.06 \text{ atm} \cdot \text{cm}^3/\text{mol} \cdot \text{K}$$

$$0.08206 \text{ atm} \cdot \text{L/mol} \cdot \text{K}$$

$$0.7302 \text{ atm} \cdot \text{ft}^3/\text{lbmol} \cdot ^\circ\text{R}$$

$$10.73 \text{ psia} \cdot \text{ft}^3/\text{lbmol} \cdot ^\circ\text{R}$$

$$1544 \text{ ft} \cdot \text{lbf/lbmol} \cdot ^\circ\text{R}$$

$$62.36 \text{ mmHg} \cdot \text{L/mol} \cdot \text{K}$$

$$21.9 \text{ in. Hg} \cdot \text{ft}^3/\text{lbmol} \cdot ^\circ\text{R}$$

Atmospheric pressure (sea level)

$$101.3 \text{ kPa} = 101,300 \text{ Pa} = 1.013 \text{ bar}$$

$$760 \text{ torr} = 29.92 \text{ in. Hg}$$

$$1 \text{ atm} = 14.696 \text{ psia}$$

Avogadro's number

$$6.022 \times 10^{23} \text{ molecules/mol}$$

Boltzmann constant

$$1.381 \times 10^{-23} \text{ J/K} \cdot \text{molecule}$$

Faraday's constant

$$96490 \text{ charge/g-equivalent}$$

Gravitational acceleration (sea level)

$$9.807 \text{ m/s}^2 = 32.174 \text{ ft/s}^2$$

Joule's constant (mechanical equivalent of heat)

$$4.184 \text{ J/cal}$$

$$778.2 \text{ ft} \cdot \text{lbf/Btu}$$

Planck's constant

$$6.626 \times 10^{-34} \text{ J} \cdot \text{s/molecule}$$

Speed of light in vacuum

$$2.998 \times 10^8 \text{ m/s}$$

Stefan-Boltzmann constant

$$5.671 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4$$

$$0.1712 \times 10^{-8} \text{ Btu/h} \cdot \text{ft}^2 \cdot ^\circ\text{R}^4$$

